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A Modern Approach to Advanced Calculus

To Connie

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MATHEMATICAL ANALYSIS

A Modern Approach to Advanced Calculus

by

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PREFACE

A glance at the Table of Contents will reveal that most of the topics which usually fall under the heading "Advanced Calculus" are treated in this book. The author's aim has been to provide a development of the subject matter which is honest, rigorous, up-to-date, and, at the same time, not too pedantic. Most of the "hard" theorems which are either omitted or treated rather skimpily in many texts on advanced calculus are proved here with great care. Some of them are ordinarily considered too difficult for a standard course in advanced calculus but too elementary for a course in real or complex function theory. With the inclusion of these theorems, the book helps to fill the gap between elementary calculus and advanced courses in analysis. More important than this, it introduces the reader to some of the abstract thinking that pervades modern mathematics.

Some of the features to be found in the book include:

A chapter on abstract set theory which contains a precise formulation of the function concept.

Point set topology in n-dimensional Euclidean space.

A satisfactory treatment of differentials.

An elementary discussion of connectedness.

An extensive treatment of the Riemann-Stieltjes integral, including complex integration and a discussion of the winding number.

A development of Jordan content and outer Lebesgue measure.

A proof of Green's theorem for plane regions bounded by arbitrary rectifiable Jordan curves.

A careful treatment of surfaces and surface integrals.

A thorough discussion of the interchange of limit processes.

A chapter on Fourier series and Fourier integrals, including the Fourier integral theorem, the convolution theorem for Fourier transforms, and the complex inversion formula for Laplace transforms.

An introduction to the theory of functions of a complex variable.

There is ample material here for a year's course, but many parts can easily be omitted without disturbing the continuity of the presentation. Nearly 500 exercises are included; many of these are designed to illustrate the general theory or to show where it can break down.

For various reasons, one of which is simply lack of space, there is a minimum of emphasis on applications and physical motivation in this book. It is a fairly easy matter for a lecturer to give a leisurely heuristic discussion that motivates a difficult concept, but in many instances the very same discussion may appear somewhat ridiculous when set down in print Furthermore, the approach to motivation is largely a matter of personal taste and hence is best provided in the form of classroom lectures as the individual instructor sees fit to introduce it

A clear understanding of the basic concepts of calculus is indispensable for anyone who wishes to learn about more advanced concepts in analysis Therefore it is hoped that this book, although designed primarily for mathematicians, may also be of interest to students of the physical and engineering sciences

While preparing the manuscript the author has had the best assistance one could wish for He is particularly grateful to Professors Paul R Halmos of the University of Chicago and M E Munroe of the University of Illinois Their thorough criticism of the preliminary version of the manuscript had a profound effect on the final form of the book Special thanks are also due to Dr Basil Gordon of the California Institute of Technology, who read the final draft and pointed out a number of errors Through the generosity of the California Institute, the expert services of Miss Rosemarie Stampfel were made available for typing of the manuscript The author also owes much to the students of Caltech, who provided the original incentive for this work

Finally, thanks are due to the Addison-Wesley Publishing Company, first, for their sincere interest in producing a book of this kind, and second, for the excellent help and cooperation which they have constantly proffered

TOM M APOSTOL

January, 1957

CONTENTS

CHAPTER 1

THE REAL AND COMPLEX NUMBER SYSTEMS

1–1 Introduction. The real number system is one of the fundamental concepts of mathematics. A thorough and exhaustive study of mathematical analysis would have to include a careful definition of what is meant by a real number, a discussion of how real numbers are constructed (starting, for example, with the integers), and a derivation of the principal properties of real numbers. Although these elements form a very interesting part of the foundations of mathematics, they will not be treated in detail here. As a matter of fact, in most phases of analysis it is only the *properties* of real numbers that concern us, rather than the methods used to construct the real number system. Therefore we shall simply list a set of axioms from which all the properties of real numbers can be derived. For discussions of the methods used to construct real numbers the reader should consult the references at the end of this chapter.

We shall assume that the reader is familiar with most of the properties of real numbers discussed in the next few pages, as well as some of their elementary consequences. We shall also assume that the reader has some knowledge of the elementary functions of calculus, such as the trigonometric, exponential, and logarithmic functions. Although infinite series will be treated in detail in a later chapter, a few basic facts about series (with which the reader is probably already familiar from his study of elementary calculus) will be used in the early part of this chapter.

1–2 Arithmetical properties of real numbers. Given any two real numbers x and y, we can form their sum $x + y$ and their product xy, and these satisfy the following axioms:

AXIOM 1. $x + y = y + x,\ xy = yx$ (*commutative laws*).

AXIOM 2. $x + (y + z) = (x + y) + z,$
$x(yz) = (xy)z$ (*associative laws*).

AXIOM 3. $x(y + z) = xy + xz$ (*distributive law*).

AXIOM 4. Given any two real numbers x and y, there exists a real number z such that $x + z = y$. This z is denoted by $y - x$; the number $x - x$ is denoted by 0. (It can be proved that 0 is independent of x.) We write $-x$ for $0 - x$.

AXIOM 5 There exists at least one real number $x \neq 0$. If x and y are two real numbers with $x \neq 0$, then there exists a real number z such that $xz = y$. This z is denoted by y/x, the number x/x is denoted by 1 and can be shown to be independent of x. We write x^{-1} for $1/x$ if $x \neq 0$.

From these axioms we can derive all the usual laws of arithmetic, for example, $-(-x) = x$, $(x^{-1})^{-1} = x$, $-(x - y) = y - x$, $x - y = x + (-y)$, etc. (For a more detailed explanation see Ref. 1–4.)

1–3 Order properties of real numbers. We also have a relation $<$ which establishes an ordering among the real numbers and which satisfies the following axioms:

AXIOM 6 Exactly one of the relations $x = y$, $x < y$, $x > y$ holds.

NOTE. $x > y$ means the same thing as $y < x$.

AXIOM 7 If $x < y$, then for every z we have $x + z < y + z$.

AXIOM 8 If $x > 0$ and $y > 0$, then $xy > 0$.

AXIOM 9 If $x > y$ and $y > z$, then $x > z$.

From these axioms we can derive the usual rules for operating with inequalities. For example, if we have $x < y$, then $xz < yz$ if $z > 0$, whereas $xz > yz$ if $z < 0$. Also, if $x > y > 0$ and $z > w > 0$, then $xz > yw$. (For a complete discussion of these rules see Ref. 1–1.) A tenth axiom is given in Section 1–9.

NOTE. The symbolism $x \leq y$ is used as an abbreviation for the statement: "$x < y$ or $x = y$." Thus we have $2 \leq 3$ since $2 < 3$, and $2 \leq 2$ since $2 = 2$. The symbol $\geq$ is similarly used.

1–4 Geometrical representation of real numbers. The real numbers can be represented geometrically as points on a line (the *real axis*). A point is selected to represent 0 and another point to represent 1, and these points determine the scale. Then each point on the real axis corresponds to one and only one real number and, conversely, each real number is represented by a single point.

1–5 Decimal representation of real numbers. Every real number x has a decimal expansion of the form

$$x = \pm N \cdot a_1 a_2 \cdots a_n \cdots,$$

where N is either 0 or a positive integer and each a_i is one of the digits from 0 to 9. The notation $N \cdot a_1 a_2 \cdots a_n \cdots$ is really an abbreviation for the infinite series

$$N + a_1(10)^{-1} + a_2(10)^{-2} + \cdots + a_n(10)^{-n} + \cdots.$$

When N is 0 and all the a_i are equal (say, to a), then the infinite series becomes a geometric series with ratio 1/10 and sum

$$a\left(\frac{1/10}{1 - 1/10}\right) = \frac{a}{9}.$$

If $a = 9$, this series has the sum 1, which means that the number 1 has the *two* decimal expansions

$$1 = 1.000... = 0.999....$$

More generally, if a number has a decimal expansion which ends in zeros, for example, $1/8 = 0.125000...$, then this number can also be written as a decimal expansion which ends in nines by decreasing the last nonzero digit by one unit. Thus we have $1/8 = 0.124999....$ Except for situations like this, decimal expansions are unique.

1–6 Rational numbers. Real numbers are classified into two main types, rational and irrational. Rational numbers are those real numbers which are the quotient of two integers; for example, $1/2$, $7/5$, -5. Irrational numbers are those real numbers which are not rational; for example, π, $\sqrt{2}$, e. Rational and irrational numbers can also be distinguished by their decimal representations. A rational number has a decimal expansion which eventually repeats itself: $1/2 = 0.5\underline{0}$, $2/3 = 0.\underline{6}$, $1/7 = 0.\underline{142857}$, $13/22 = 0.59\underline{09}$, $1/8 = 0.124\underline{9}$. That portion of the decimal which is underscored is understood to be repeated indefinitely in each case. (The reader should have no difficulty in showing that such repetitions must always occur when a rational number is written in its decimal expansion.) Conversely, every repeating decimal represents a rational number. This fact is not so obvious, but it can easily be proved by observing that a decimal which repeats forms a geometric series of the form

$$a + b(1 + r + r^2 + \cdots + r^n + \cdots),$$

where a, b, and r are rational, r being some power of 1/10. Such a series has the sum $a + b/(1 - r)$, which is rational. As a case in point, take the repeating decimal $x = 0.3\underline{14}$. We can write this out as a series as follows:

$$\begin{aligned} x &= \frac{3}{10} + 14(10^{-3}) + 14(10^{-5}) + 14(10^{-7}) + \cdots \\ &= \frac{3}{10} + 14(10^{-3})[1 + (10^{-2}) + (10^{-4}) + \cdots] \\ &= \frac{3}{10} + 14(10^{-3})\left[\frac{100}{99}\right] = \frac{311}{990}. \end{aligned}$$

Given two rational numbers, say a and b, their average $(a + b)/2$ is also rational. Hence between any two rational numbers there lies another rational number. It then follows that between any two rational numbers there must be *infinitely many* rational numbers (Why?), which implies that if we are given a certain rational number x, we cannot speak of the "next largest" rational number.

1-7 Some irrational numbers. Ordinarily it is not too easy to show that some particular number is irrational. There is no simple proof, for example, of the irrationality of e^π. However, the irrationality of certain numbers such as $\sqrt{2}$ and $\sqrt{3}$ is not too difficult to establish and, in fact, we can easily prove the following:

1-1 THEOREM. *If n is an integer which is not a perfect square, then $\sqrt{n}$ is irrational.*

Proof. Suppose first that n contains no square factor > 1. We assume that $\sqrt{n}$ is rational and obtain a contradiction. Let $\sqrt{n} = a/b$, where a and b are integers having no factor in common. Then $nb^2 = a^2$ and, since the left side of this equation is a multiple of n, so too is a^2. However, if a^2 is a multiple of n, a itself must be a multiple of n, since n has no square factors > 1. (This is easily seen by examining the factorization of a into its prime factors.) This means that $a = cn$, where c is some integer. Then the equation $nb^2 = a^2$ becomes $nb^2 = c^2n^2$, or $b^2 = nc^2$. The same argument shows that b must also be a multiple of n. Thus a and b are both multiples of n, which contradicts the fact that they have no factor in common. This completes the proof if n has no square factor > 1.

If n has a square factor, we can write $n = m^2k$, where $k > 1$ and k has no square factor > 1. Then $\sqrt{n} = m\sqrt{k}$, and if $\sqrt{n}$ were rational, the number $\sqrt{k}$ would also be rational, contradicting that which was just proved.

A different type of argument is needed to prove that the number e is irrational.

1–2 THEOREM. *If* $e^x = 1 + x + x^2/2! + x^3/3! + \cdots + x^n/n! + \cdots$, *then the number e is irrational.*

Proof. We shall prove that e^{-1} is irrational. The series for e^{-1} is an alternating series with terms which decrease steadily in absolute value. In such an alternating series the error made by stopping at the nth term has the algebraic sign of the first neglected term and is less in absolute value than the first neglected term. Hence, if $s_n = \sum_{k=0}^{n} (-1)^k/k!$, we have the inequality

$$0 < e^{-1} - s_{2k-1} < \frac{1}{(2k)!},$$

from which we obtain

$$0 < (2k - 1)!\,(e^{-1} - s_{2k-1}) < \frac{1}{2k} \leq \frac{1}{2}$$

for any integer $k \geq 1$. Now $(2k - 1)!\,s_{2k-1}$ is always an integer. If e^{-1} were rational, then we could choose k so large that $(2k - 1)!\,e^{-1}$ would also be an integer. But the last inequality says that the difference of these two integers would be a number between 0 and $\frac{1}{2}$, which is impossible. Thus e^{-1} cannot be rational, and hence e cannot be rational.

The ancient Greeks were aware of the existence of irrational numbers as early as 500 B.C. However, a satisfactory theory of such numbers was not developed until late in the nineteenth century, at which time three different theories were introduced by Cantor, Dedekind, and Weierstrass. For an account of the theories of Dedekind and Cantor and their equivalence, see Chapter I of E. W. HOBSON, *The Theory of Functions of a Real Variable and the Theory of Fourier's Series.* Vol. 1, 3rd ed. Cambridge: University Press, 1927.

1–8 Some fundamental inequalities. Calculations with inequalities arise quite frequently in analysis. They are of particular importance in dealing with the notion of absolute value. If x is any real number, then we define the absolute value of x, denoted by $|x|$, as follows:

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x \leq 0. \end{cases}$$

A fundamental inequality concerning absolute values is given in the following:

1–3 THEOREM. *If* $a \geq 0$, *then we have the inequality* $|x| \leq a$ *if, and only if,* $-a \leq x \leq a$.

Proof From the definition of $|x|$, we have the inequality $-|x| \leq x \leq |x|$, since $x = |x|$ or $x = -|x|$ If we assume that $|x| \leq a$, then we can write $-a \leq -|x| \leq x \leq |x| \leq a$ and thus half of the theorem is proved Conversely, let us assume $-a \leq x \leq a$ Then if $x > 0$, we have $|x| = x \leq a$, whereas if $x < 0$, we have $|x| = -x \leq a$ In either case we have $|x| \leq a$ and the theorem is proved.

As an important consequence we have

1-4 THEOREM $|x + y| \leq |x| + |y|$.

Proof We have $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$ Addition gives us $-(|x| + |y|) \leq x + y \leq |x| + |y|$, and from Theorem 1-3 we conclude that $|x + y| \leq |x| + |y|$

Since $|-x| = |x|$, Theorem 1-4 also yields the inequality

$$|x - y| \leq |x| + |y|$$

Replacing x by $y - x$, this can also be written in the form

$$|x - y| \geq |x| - |y|,$$

and by induction, we can prove the generalizations

$$|x_1 + x_2 + \cdots + x_n| \leq |x_1| + |x_2| + \cdots + |x_n|$$

and

$$|x_1 + x_2 + \cdots + x_n| \geq |x_1| - |x_2| - \cdots - |x_n|.$$

We shall now derive a very useful result known as the *Cauchy-Schwarz inequality*.

1-5 THEOREM (*Cauchy-Schwarz inequality*) *If $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ are arbitrary real numbers, we have*

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 \leq \left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} b_k^2\right).$$

Proof A sum of squares can never be negative Hence we have

$$\sum_{k=1}^{n} (a_k x + b_k)^2 \geq 0$$

for every real x This inequality can be written in the form

$$Ax^2 + 2Bx + C \geq 0,$$

where

$$A = \sum_{k=1}^{n} a_k^2, \qquad B = \sum_{k=1}^{n} a_k b_k, \qquad C = \sum_{k=1}^{n} b_k^2.$$

If $A > 0$, put $x = -B/A$ to obtain $B^2 - AC \leq 0$, which is the desired inequality. If $A = 0$, the proof is trivial.

NOTE. For an alternative proof of this theorem, see Exercise 1–15.

1–9 Infimum and supremum. Let S denote a collection of real numbers. The notation $x \in S$ means that the real number x is in the collection S, and we write $x \notin S$ to indicate that x is not in S.

The notation $\{x \mid x \text{ satisfies } P\}$ will be used to designate the collection of all real numbers x which satisfy the property P.

1–6 DEFINITION. *Let A be a set of real numbers. If there is a real number x such that $a \in A$ implies $a \leq x$, then x is called an upper bound for the set A and we say that A is bounded above.* [Lower bound is similarly defined.]

1–7 DEFINITION. *Let A be a set of real numbers bounded above. Suppose there is a real number x satisfying the following two conditions:*

(i) *x is an upper bound for A, and*

(ii) *if y is any upper bound for A, then $x \leq y$.*

Such a number x is called a least upper bound, or a supremum, of the set A. [The abbreviation lub is used for least upper bound and the abbreviation sup is used for supremum. The concept of greatest lower bound (glb), or infimum (inf), is similarly defined if A is bounded below.*]

It is easy to see that the sup and inf of a set are uniquely determined, whenever they exist (see Exercise 1–10).

Our final axiom for the real number system involves the notion of supremum:

AXIOM 10. If A is a nonempty set of real numbers which is bounded above, then A has a supremum.

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* We shall use the term "supremum" rather than "least upper bound" and the term "infimum" rather than "greatest lower bound."

As a consequence of this axiom, it follows that every nonempty set of real numbers which is bounded below has an infimum (The term "nonempty" means that the set must contain at least one real number)

NOTATIONS If x is the supremum of a set A and if y is the infimum of A, we write $x = \sup A$, $y = \inf A$

EXAMPLES

1 Let A be the set of all numbers of the form $1/n$, where n takes all positive integral values $n = 1, 2, 3,$ This set is bounded above by 1 and bounded below by 0 In fact, 1 is the sup of the set and 0 is the inf. In this example the sup is also one of the members of the set A, whereas the inf is not in A.

2 $A = \{x \mid 0 < x < 1\}$ In this case, $\sup A = 1$, $\inf A = 0$. However, neither 0 nor 1 belongs to A.

3 $A = \{x \mid 0 < x^2 < 2\}$. Since the inequality $0 < x^2 < 2$ is equivalent to the two inequalities $0 < x < \sqrt{2}$ and $-\sqrt{2} < x < 0$, the sup of this set is $\sqrt{2}$ and the inf is $-\sqrt{2}$

4 $A = \{x \mid x^2 > 2\}$ Here there is no sup and no inf This set is not bounded above or below

5 $A = \{x \mid x > 0 \text{ and } x^2 > 2\}$ In this example there is no sup but the inf is $\sqrt{2}$

The following property of the supremum will prove to be quite useful in some of the later work in this text

1-8 THEOREM *Let A and B be two bounded sets of real numbers with $a = \sup A$, $b = \sup B$ Let C denote the set*

$$C = \{x + y \mid x \in A, y \in B\}.$$

Then $a + b = \sup C$

Proof If $z \in C$, then $z = x + y \le a + b$, so that $a + b$ is an upper bound for C Let c be any upper bound for C We must show that $a + b \le c$ For this purpose let ϵ be a given positive number There exists a number x in A and a number y in B such that

$$a - \epsilon < x, \qquad b - \epsilon < y$$

By adding these inequalities, we find

$$a + b - 2\epsilon < x + y \le c$$

That is, $a + b \le c + 2\epsilon$ But since ϵ is arbitrary, this implies $a + b \le c$.

1–10 Complex numbers. It follows from the axioms governing the relation $<$ that the square of a real number is never negative. Thus, for example, the elementary quadratic equation $x^2 = -1$ has no solution among the real numbers. New types of numbers, called *complex numbers,* have been introduced to provide solutions to such equations. It turns out that the introduction of complex numbers provides, at the same time, solutions to general algebraic equations of the form

$$a_0 + a_1x + \cdots + a_nx^n = 0,$$

where the coefficients $a_0, a_1, \ldots, a_n$ are arbitrary real numbers. (This fact is known as the *Fundamental Theorem of Algebra.*)

We shall now define complex numbers and discuss them in further detail.

1–9 DEFINITION. *By a complex number we shall mean an ordered pair of real numbers which we denote by (x_1, x_2). The first member, x_1, is called the real part of the complex number; the second member, x_2, is called the imaginary part. Two complex numbers $x = (x_1, x_2)$ and $y = (y_1, y_2)$ are called equal, and we write $x = y$, if, and only if, $x_1 = y_1$ and $x_2 = y_2$. We define the sum $x + y$ and the product xy by the equations*

$$x + y = (x_1 + y_1, x_2 + y_2), \qquad xy = (x_1y_1 - x_2y_2, x_1y_2 + x_2y_1).$$

1–10 THEOREM. *The operations of addition and multiplication just defined satisfy the commutative, associative, and distributive laws.*

Proof. We shall illustrate only the distributive law, since it is the most difficult. If $x = (x_1, x_2)$, $y = (y_1, y_2)$, and $z = (z_1, z_2)$, then we have

$$\begin{aligned} x(y + z) &= (x_1, x_2)(y_1 + z_1, y_2 + z_2) \\ &= (x_1y_1 + x_1z_1 - x_2y_2 - x_2z_2, x_1y_2 + x_1z_2 + x_2y_1 + x_2z_1) \\ &= (x_1y_1 - x_2y_2, x_1y_2 + x_2y_1) + (x_1z_1 - x_2z_2, x_1z_2 + x_2z_1) \\ &= xy + xz. \end{aligned}$$

1–11 THEOREM.

$$(x_1, x_2) + (0, 0) = (x_1, x_2), \qquad (x_1, x_2)(0, 0) = (0, 0),$$
$$(x_1, x_2)(1, 0) = (x_1, x_2), \qquad (x_1, x_2) + (-x_1, -x_2) = (0, 0).$$

Proof. The proofs here are immediate from the definition, as are the proofs of Theorems 1–12, 1–13, 1–15, and 1–16.

1-12 THEOREM. *Given two complex numbers $x = (x_1, x_2)$ and $y = (y_1, y_2)$, there exists a complex number z such that $x + z = y$. In fact, $z = (y_1 - x_1, y_2 - x_2)$ This z is denoted by $y - x$. The complex number $(-x_1, -x_2)$ is denoted by $-x$*

1-13 THEOREM. *For any two complex numbers x and y, we have*

$$(-x)y = x(-y) = -(xy) = (-1, 0)(xy)$$

1-14 DEFINITION *If $x = (x_1, x_2) \neq (0, 0)$ and y are complex numbers, we define $x^{-1} = [x_1/(x_1^2 + x_2^2), -x_2/(x_1^2 + x_2^2)]$, and $y/x = yx^{-1}$*

1-15 THEOREM *If x and y are complex numbers with $x \neq (0, 0)$, there exists a complex number z such that $xz = y$, namely, $z = yx^{-1}$*

Of special interest are operations with complex numbers whose imaginary part is 0

1-16 THEOREM $(x_1, 0) + (y_1, 0) = (x_1 + y_1, 0)$,
$(x_1, 0)(y_1, 0) = (x_1 y_1, 0)$,
$(x_1, 0)/(y_1, 0) = (x_1/y_1, 0)$, *if* $y_1 \neq 0$.

NOTE It is evident from Theorem 1-16 that we can perform arithmetic operations on complex numbers with zero imaginary part by performing the usual real-number operations on the real parts alone Hence the complex numbers of the form $(x, 0)$ have the same arithmetic properties as the real numbers. For this reason it is convenient to think of the real number system as being a special case of the complex number system, and we agree to identify the complex number $(x, 0)$ and the real number x Therefore, we write $x = (x, 0)$ In particular, $0 = (0, 0)$ and $1 = (1, 0)$

1-11 Geometric representation of complex numbers. Just as real numbers are represented geometrically by points on a line, so complex numbers are represented by points in a plane The complex number $x = (x_1, x_2)$ can be thought of as the "point" with coordinates (x_1, x_2) When this is done, the definition of addition amounts to addition by the *parallelogram law* (See Fig 1-1)

The idea of expressing complex numbers geometrically as points on a plane was formulated by Gauss in his dissertation in 1799 and, independently, by Argand in 1806 Gauss later coined the somewhat unfortunate phrase "complex number." Other geometric interpretations of

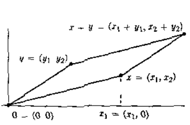

FIG 1-1. Addition of complex numbers

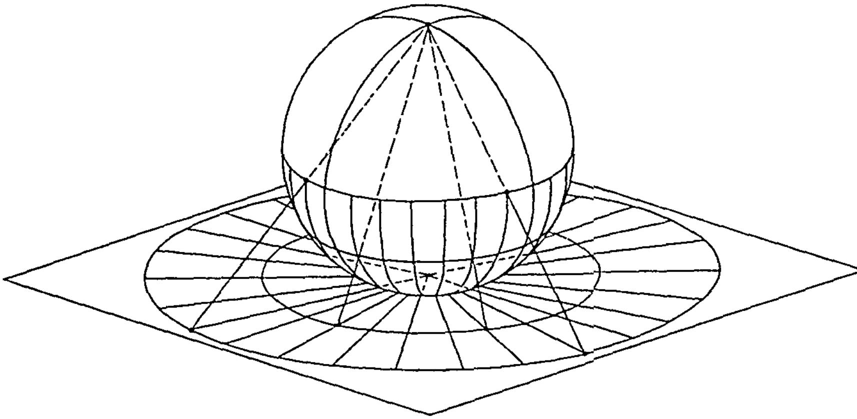

FIG. 1-2. Stereographic projection.

complex numbers are possible. Instead of using points on a plane, we can use points on other surfaces. Riemann found the sphere particularly convenient for this purpose. Points of the sphere are projected from the North Pole onto the tangent plane at the South Pole and thus there corresponds to each point of the plane a definite point of the sphere. With the exception of the North Pole itself, each point of the sphere corresponds to exactly one point of the plane. This correspondence is called a *stereographic projection.* (See Fig. 1-2.)

1-12 The imaginary unit. It is convenient to think of the complex number (x_1, x_2) as a two-dimensional vector with components x_1 and x_2. Adding two complex numbers by means of Definition 1-9 is then the same as adding two vectors component by component. The complex number $1 = (1, 0)$ plays the same role as a unit vector in the horizontal direction. The analog of a unit vector in the vertical direction will now be introduced.

1-17 DEFINITION. *The complex number* $(0, 1)$ *is denoted by* i *and is called the imaginary unit.*

1-18 THEOREM. *Every complex number* $x = (x_1, x_2)$ *can be represented in the form* $x = x_1 + ix_2$.

Proof. $x_1 = (x_1, 0), \qquad ix_2 = (0, 1)(x_2, 0) = (0, x_2),$

$$x_1 + ix_2 = (x_1, 0) + (0, x_2) = (x_1, x_2).$$

The next theorem tells us that the complex number i provides us with a solution to the equation $x^2 = -1$.

1-19 THEOREM $i^2 = -1$

Proof $i^2 = (0, 1)(0, 1) = (-1, 0) = -1$

1-13 Absolute value of a complex number. We now generalize the concept of absolute value to the complex number system

1-20 DEFINITION *If $x = (x_1, x_2)$, we define the modulus, or absolute value, of x to be the non-negative real number $|x|$ given by*

$$|x| = \sqrt{x_1^2 + x_2^2}\,.$$

1-21 THEOREM (i) $|(0, 0)| = 0$, *and* $|x| > 0$ *if* $x \neq 0$

(ii) $|xy| = |x|\,|y|$

(iii) $|x/y| = |x|/|y|$, *if* $y \neq 0$.

(iv) $|(x_1, 0)| = |x_1|$

Proof Statements (i) and (iv) are immediate To prove (ii), we write $x = x_1 + ix_2$, $y = y_1 + iy_2$, so that $xy = x_1y_1 - x_2y_2 + i(x_1y_2 + x_2y_1)$. Observing that we have

$$|xy|^2 = x_1^2y_1^2 + x_2^2y_2^2 + x_1^2y_2^2 + x_2^2y_1^2 = (x_1^2 + x_2^2)(y_1^2 + y_2^2) = |x|^2|y|^2,$$

statement (ii) follows Equation (iii) can be derived from (ii) by writing it in the form $|x| = |y|\,|x/y|$

Geometrically, $|x|$ represents the length of the segment joining the origin to the point x More generally, $|x - y|$ is the distance between the points x and y. Using this geometric interpretation, the following theorem states that one side of a triangle is less than the sum of the other two sides.

1-22 THEOREM *If x and y are complex numbers, then we have*

$$|x + y| \leq |x| + |y| \qquad \textit{(triangle inequality)}$$

The proof (without using geometry) is left as an exercise

1-14 Impossibility of ordering the complex numbers. As yet we have not defined a relation of the form $x < y$ if x and y are arbitrary complex numbers, for the reason that it is impossible to give a definition of $<$ for complex numbers which will have all the properties in Axioms 6 through 8 To illustrate, suppose we were able to define an order relation $<$ satisfying Axioms 6, 7, and 8 Then, since $i \neq 0$, we must have either $i > 0$ or $i < 0$, by Axiom 6 Let us assume $i > 0$ Then, taking $x = y = i$ in Axiom 8, we get $i^2 > 0$, or $-1 > 0$ Adding 1 to both sides (Axiom 7), we get $0 > 1$. On the other hand, applying Axiom 8 to $-1 > 0$ yields

$1 > 0$. Thus we have both $0 > 1$ and $1 > 0$, which, by Axiom 6, is impossible. Hence the assumption $i > 0$ leads us to a contradiction. [Why was the inequality $-1 > 0$ not already a contradiction?] A similar argument shows that we cannot have $i < 0$. Hence the complex numbers cannot be ordered in such a way that Axioms 6, 7, and 8 will be satisfied.

1–15 Complex exponentials. The exponential e^x (x real) was mentioned in Theorem 1–2. We now wish to define e^z, when z is a complex number. We shall do this in such a way that the principal properties of the real exponential function will be preserved. The main properties of e^x for x real are given by the law of exponents, $e^{x_1}e^{x_2} = e^{x_1+x_2}$, and the fact that $e^0 = 1$. We shall give a definition of e^z for complex z which will preserve these properties and which will reduce to the ordinary exponential when z is real.

If we write $z = x + iy$ (x, y real), then in order for the law of exponents to hold we want $e^{x+iy} = e^x e^{iy}$. It therefore remains to define what we shall mean by e^{iy}.

1–23 DEFINITION. *If $z = x + iy$, we define $e^z = e^{x+iy}$ to be the complex number $e^z = e^x (\cos y + i \sin y)$.*

This definition* clearly agrees with the usual exponential function when z is real (that is, $y = 0$). We shall now prove that the law of exponents still holds.

1–24 THEOREM. *If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ are two complex numbers, then we have*

$$e^{z_1} e^{z_2} = e^{z_1+z_2}.$$

Proof.

$$e^{z_1} = e^{x_1} (\cos y_1 + i \sin y_1), \qquad e^{z_2} = e^{x_2} (\cos y_2 + i \sin y_2),$$

$$e^{z_1}e^{z_2} = e^{x_1}e^{x_2} [\cos y_1 \cos y_2 - \sin y_1 \sin y_2 + i (\cos y_1 \sin y_2 + \sin y_1 \cos y_2)].$$

* Several plausible arguments can be given to justify the equation $e^{iy} = \cos y + i \sin y$. For example, let us write $e^{iy} = f(y) + ig(y)$ and try to determine the real-valued functions f and g so that the usual rules of operating with real exponentials will also apply to complex exponentials. Formal differentiation yields $e^{iy} = g'(y) - if'(y)$, if we assume that $(e^{iy})' = ie^{iy}$. Comparing the two expressions for e^{iy}, we see that f and g must satisfy the equations $f(y) = g'(y)$, $f'(y) = -g(y)$. Eliminating g yields $f(y) = -f''(y)$. Since we want $e^0 = 1$, we must have $f(0) = 1$ and $f'(0) = 0$. It follows that $f(y) = \cos y$ and $g(y) = -f'(y) = \sin y$. Of course, this argument *proves* nothing, but it strongly suggests that the definition $e^{iy} = \cos y + i \sin y$ is reasonable.

Now $e^{x_1}e^{x_2} = e^{x_1+x_2}$, since x_1 and x_2 are both real Also,

$$\cos y_1 \cos y_2 - \sin y_1 \sin y_2 = \cos (y_1 + y_2)$$

and

$$\cos y_1 \sin y_2 + \sin y_1 \cos y_2 = \sin (y_1 + y_2),$$

and hence

$$e^{z_1}e^{z_2} = e^{x_1+x_2} [\cos (y_1 + y_2) + i \sin (y_1 + y_2)] = e^{z_1+z_2}$$

We now derive some further properties of the complex exponential (In the following theorems, z, z_1, z_2 denote complex numbers)

1-25 THEOREM *e^z is never zero*

Proof $e^z e^{-z} = e^0 = 1$. Hence e^z cannot be zero

1-26 THEOREM *If x is real, then $|e^{ix}| = 1$.*

Proof $|e^{ix}|^2 = \cos^2 x + \sin^2 x = 1$, and $|e^{ix}| > 0$

1-27 THEOREM. *$e^z = 1$ if, and only if, z is an integral multiple of $2\pi i$.*

Proof If $z = 2\pi i n$, where n is an integer, then

$$e^z = \cos (2\pi n) + i \sin (2\pi n) = 1.$$

Conversely, suppose that $e^z = 1$ This means that $e^x \cos y = 1$ and $e^x \sin y = 0$. Since $e^x \neq 0$, we must have $\sin y = 0$, $y = k\pi$, where k is an integer But $\cos(k\pi) = (-1)^k$ Hence $e^x = (-1)^k$, since $e^x \cos (k\pi) = 1$. Since $e^x > 0$, k must be even Therefore $e^x = 1$ and hence $x = 0$ This proves the theorem

1-28 THEOREM *$e^{z_1} = e^{z_2}$ if, and only if, $z_1 - z_2 = 2\pi i n$ (where n is an integer)*

Proof $e^{z_1} = e^{z_2}$ if, and only if, $e^{z_1-z_2} = 1$

1-16 The argument of a complex number. If the point $z = (x, y) = x + iy$ is represented by polar coordinates r and θ, we can write $x = r \cos \theta$ and $y = r \sin \theta$, so that $z = r \cos \theta + ir \sin \theta = re^{i\theta}$ The two numbers r and θ uniquely determine z. Conversely, the positive number r is uniquely determined by z; in fact, $r = |z|$ However, z determines the angle θ only up to multiples of 2π. There are infinitely many values of θ which satisfy the equations $x = |z| \cos \theta$, $y = |z| \sin \theta$ but, of course, any two of them differ by some multiple of 2π. Each such θ is called an *argument* of z but *one* of these values is singled out and is called the *principal argument* of z.

1–29 DEFINITION. *Let $z = x + iy$ be a nonzero complex number. The unique real number θ which satisfies the conditions*

$$x = |z| \cos \theta, \qquad y = |z| \sin \theta, \qquad -\pi < \theta \leq +\pi$$

is called the principal argument of z, denoted by $\theta = \arg(z)$.

The above discussion immediately yields the following theorem:

1–30 THEOREM. *Every complex number $z \neq 0$ can be represented in the form $z = re^{i\theta}$, where $r = |z|$ and $\theta = \arg(z) + 2\pi n$, n being any integer.*

NOTE. This method of representing complex numbers is particularly useful in connection with multiplication and division, since we have

$$(r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1+\theta_2)} \qquad \text{and} \qquad \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)}.$$

1–31 THEOREM. *If $z_1 z_2 \neq 0$, then $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2) + 2\pi n(z_1, z_2)$, where*

$$n(z_1, z_2) = \begin{cases} 0, & \text{if } -\pi < \arg(z_1) + \arg(z_2) \leq +\pi, \\ +1, & \text{if } -2\pi < \arg(z_1) + \arg(z_2) \leq -\pi, \\ -1, & \text{if } \pi < \arg(z_1) + \arg(z_2) \leq 2\pi. \end{cases}$$

Proof. Write $z_1 = |z_1|e^{i\theta_1}$, $z_2 = |z_2|e^{i\theta_2}$, where $\theta_1 = \arg(z_1)$ and $\theta_2 = \arg(z_2)$. Then $z_1 z_2 = |z_1 z_2| e^{i(\theta_1+\theta_2)}$. Since $-\pi < \theta_1 \leq +\pi$ and $-\pi < \theta_2 \leq +\pi$, we have $-2\pi < \theta_1 + \theta_2 \leq 2\pi$. Hence there is an integer n such that $-\pi < \theta_1 + \theta_2 + 2n\pi \leq \pi$. This n is the same as the integer $n(z_1, z_2)$ given in the theorem, and for this n we have $\arg(z_1 z_2) = \theta_1 + \theta_2 + 2\pi n$. This proves the theorem.

1–17 Integral powers and roots of complex numbers.

1–32 DEFINITION. *Given a complex number z and an integer n, we define the nth power of z as follows:*

$$z^0 = 1, \qquad z^{n+1} = z^n z, \qquad \text{if } n \geq 0,$$
$$z^{-n} = (z^{-1})^n, \qquad \text{if } z \neq 0 \text{ and } n > 0.$$

Theorem 1–33, which states that the usual laws of exponents hold, can be proved by mathematical induction. The proof is left as an exercise.

1–33 THEOREM. *Given two integers m and n, we have*

$$z^n z^m = z^{n+m} \qquad \text{and} \qquad (z_1 z_2)^n = z_1^n z_2^n.$$

1-34 THEOREM *If $z \neq 0$, and if n is a positive integer, then there are exactly n distinct complex numbers $z_0, z_1, \ldots, z_{n-1}$ (called the* nth *roots of z), such that*

$$z_k^n = z, \qquad \text{for each } k = 0, 1, 2, \ldots, n-1$$

Furthermore, these roots can be obtained by use of the formulas

$$z_k = Re^{i\phi_k}, \qquad \text{where} \qquad R = |z|^{1/n},$$

and

$$\phi_k = \frac{\arg(z)}{n} + \frac{2\pi k}{n} \qquad (k = 0, 1, 2, \ldots, n-1).$$

NOTE The n nth roots of z are equally spaced on the circle of radius $R = |z|^{1/n}$, center at the origin

Proof The n complex numbers $Re^{i\phi_k}$, $0 \leq k \leq n-1$, are distinct and each is an nth root of z, since

$$(Re^{i\phi_k})^n = R^n e^{in\phi_k} = |z|e^{i[\arg(z)+2\pi k]} = z$$

We must now show that there are no other nth roots of z Suppose $w = Ae^{i\alpha}$ is a complex number such that $w^n = z$ Then $|w|^n = |z|$, and hence $A^n = |z|$, $A = |z|^{1/n}$. Therefore, $w^n = z$ can be written $e^{in\alpha} = e^{i[\arg(z)]}$, which implies $n\alpha - \arg(z) = 2\pi k$ for some integer k Hence $\alpha = [\arg(z) + 2\pi k]/n$ But when k runs through all integral values, w takes only the distinct values $z_0, \ldots, z_{n-1}$ (See Fig 1-3)

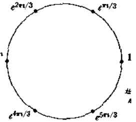

FIG 1-3 The six sixth roots of unity

1-18 Complex logarithms. By Theorem 1-25, e^z is never zero It is natural to ask if there are other values that e^z cannot assume The next theorem shows that zero is the only exceptional value

1-35 THEOREM *If z is a complex number $\neq 0$, then there exist complex numbers w such that $e^w = z$ One such w is the complex number*

$$\log|z| + i\arg(z),$$

and any other such w must have the form

$$\log|z| + i\arg(z) + 2n\pi i \qquad (n \text{ is an integer}).$$

Proof. Since $e^{\log|z|+i\arg(z)} = e^{\log|z|}e^{i\arg(z)} = |z|e^{i\arg(z)} = z$, we see that $w = \log|z| + i\arg(z)$ is a solution of the equation $e^w = z$. But if w_1 is any other solution, then $e^w = e^{w_1}$ and hence $w - w_1 = 2n\pi i$.

1–36 DEFINITION. *Let $z \neq 0$ be a given complex number. If w is a complex number such that $e^w = z$, then w is called a logarithm of z. The particular value of w given by*

$$w = \log|z| + i\arg(z)$$

is called the principal logarithm of z, and for this w we write

$$w = \operatorname{Log} z.$$

EXAMPLES.

1. Since $|i| = 1$ and $\arg(i) = \pi/2$, $\operatorname{Log}(i) = i\pi/2$.
2. Since $|-i| = 1$ and $\arg(-i) = -\pi/2$, $\operatorname{Log}(-i) = -i\pi/2$.
3. Since $|-1| = 1$ and $\arg(-1) = \pi$, $\operatorname{Log}(-1) = \pi i$.
4. If $x > 0$, $\operatorname{Log}(x) = \log x$, since $|x| = x$ and $\arg(x) = 0$.
5. Since $|1+i| = \sqrt{2}$ and $\arg(1+i) = \pi/4$, $\operatorname{Log}(1+i) = \log\sqrt{2} + i\pi/4$.

1–37 THEOREM. *If $z_1z_2 \neq 0$, then $\operatorname{Log}(z_1z_2) = \operatorname{Log} z_1 + \operatorname{Log} z_2 + 2\pi i n(z_1, z_2)$, where $n(z_1, z_2)$ is the integer defined in Theorem 1–31.*

Proof.
$$\begin{aligned}\operatorname{Log}(z_1z_2) &= \log|z_1z_2| + i\arg(z_1z_2)\\ &= \log|z_1| + \log|z_2| + i\,[\arg(z_1) + \arg(z_2)\\ &\qquad + 2\pi n(z_1, z_2)].\end{aligned}$$

1–19 Complex powers. Using complex logarithms, we can now give a definition of complex powers of complex numbers.

1–38 DEFINITION. *If $z \neq 0$ and if w is any complex number, we define*

$$z^w = e^{w\operatorname{Log} z}.$$

EXAMPLES.

1. $i^i = e^{i\operatorname{Log} i} = e^{i(i\pi/2)} = e^{-\pi/2}$.
2. $(-1)^i = e^{i\operatorname{Log}(-1)} = e^{i(i\pi)} = e^{-\pi}$.
3. If n is an integer, then $z^{n+1} = e^{(n+1)\operatorname{Log} z} = e^{n\operatorname{Log} z}e^{\operatorname{Log} z} = z^n z$, so that Definition 1–38 does not conflict with Definition 1–32.

The next two theorems give us rules for calculating with complex powers:

1–39 THEOREM. $z^{w_1}z^{w_2} = z^{w_1+w_2}$.

Proof. $z^{w_1+w_2} = e^{(w_1+w_2)\operatorname{Log} z} = e^{w_1\operatorname{Log} z}e^{w_2\operatorname{Log} z} = z^{w_1}z^{w_2}$.

1-40 THEOREM $(z_1z_2)^w = z_1^w z_2^w e^{2\pi i w\, n(z_1,\, z_2)}$, *where* $n(z_1, z_2)$ *is the integer defined in Theorem* 1-31

Proof. $(z_1z_2)^w = e^{w \operatorname{Log}(z_1z_2)} = e^{w[\operatorname{Log} z_1 + \operatorname{Log} z_2 + 2\pi i\, n(z_1,\, z_2)]}$.

1-20 Complex sines and cosines.

1-41 DEFINITION *Given a complex number* z, *we define*

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

NOTE When z is real, these equations are consistent with Definition 1-23

1-42 THEOREM *If* $z = x + iy$, *then we have*

$$\cos z = \cos x \cosh y - i \sin x \sinh y,$$

$$\sin z = \sin x \cosh y + i \cos x \sinh y.$$

Proof
$$\begin{aligned} 2\cos z &= e^{iz} + e^{-iz} \\ &= e^{-y}(\cos x + i \sin x) + e^{y}(\cos x - i \sin x) \\ &= \cos x(e^{y} + e^{-y}) - i \sin x(e^{y} - e^{-y}) \\ &= 2\cos x \cosh y - 2i \sin x \sinh y \end{aligned}$$

The proof for $\sin z$ is similar.

Further properties of sines and cosines are given in the exercises.

Exercises

Rational and irrational numbers.

1–1. Find the rational number whose decimal expansion is 0.3344444. . . .

1–2. Show that the decimal expansion of x will end in zeros (or in nines) if, and only if, x is a rational number whose denominator is of the form $2^n 5^m$, where m and n are positive integers or zero.

1–3. Show that $\sqrt{2} + \sqrt{3}$ is irrational.

1–4. If a, b, c, d are rational and if x is irrational, show that $(ax + b)/(cx + d)$ is usually irrational. When do exceptions occur?

1–5. Given any real $x > 0$, show how to find an irrational number between 0 and x.

1–6. If $a/b < c/d$ with $b > 0$, $d > 0$, show that $(a + c)/(b + d)$ lies between a/b and c/d.

1–7. Let a and b be positive integers. Show that $\sqrt{2}$ always lies between the two fractions a/b and $(a + 2b)/(a + b)$. Which fraction is closer to $\sqrt{2}$?

1–8. Let a and b denote the roots of the quadratic equation $x^2 - x - 1 = 0$, and let $x_n = (a^n - b^n)/(a - b)$. Show that $x_1 = 1$, $x_2 = 1$, $x_3 = 2, \ldots,$ $x_{n+1} = x_n + x_{n-1}$.

1–9. Determine for which values of the integer $n \geq 1$ the number $\sqrt{n - 1} + \sqrt{n + 1}$ is rational, and for which values it is irrational.

Upper bounds.

1–10. Show that the sup and inf of a set are uniquely determined whenever they exist.

1–11. Find the sup and inf of each of the following sets of real numbers:

(a) All numbers of the form $2^{-p} + 3^{-q} + 5^{-r}$, where p, q, and r take on all positive integral values.

(b) $S = \{x \mid 3x^2 - 10x + 3 < 0\}$.

(c) $S = \{x \mid (x - a)(x - b)(x - c)(x - d) < 0\}$, where $a < b < c < d$.

1–12. Let S be a set of real numbers bounded above and let a be the supremum of S. Let ϵ be a positive number. Show that there is at least one x in S, such that $a - \epsilon < x \leq a$.

1–13. Let A and B be two sets of real numbers bounded above. Let $a = \sup(A)$ and $b = \sup(B)$. Let C be the set of real numbers formed by considering all products of the form xy, where $x \in A$ and $y \in B$. Show that, in general, $ab \neq \sup(C)$.

1-14. Let x be a real number > 0, and let k be a positive integer > 1 Let a_0 denote the largest integer $\leq x$ and, assuming that $a_0, a_1, \ldots, a_{n-1}$ have been defined, let a_n denote the largest integer such that

$$a_0 + \frac{a_1}{k} + \frac{a_2}{k^2} + \cdots + \frac{a_n}{k^n} \leq x$$

(a) Show that $0 \leq a_i \leq k - 1$ for each $i = 1, 2, \ldots$.

(b) Explain how the numbers $a_0, a_1, a_2, \ldots$ can be obtained geometrically.

(c) Show that the infinite series $a_0 + a_1/k + a_2/k^2 + \cdots$ converges to the sum x

(d) Show that x is the sup of the set of partial sums of the series in (c).

NOTE The series in (c) yields a decimal expansion of x in the scale of k

Inequalities.

1-15. Prove Lagrange's identity for real numbers

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 = \left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} b_k^2\right) - \sum_{1 \leq k < j \leq n} (a_k b_j - a_j b_k)^2.$$

Use this identity to derive the Cauchy-Schwarz inequality

1-16. If $a_1 \geq a_2 \geq \cdots \geq a_n \geq 0$ and $b_1 \geq b_2 \geq \cdots \geq b_n \geq 0$, show that

$$\left(\sum_{k=1}^{n} a_k\right)\left(\sum_{k=1}^{n} b_k\right) \leq n \sum_{k=1}^{n} a_k b_k$$

1-17. Prove Minkowski's inequality

$$\left[\sum_{k=1}^{n} (a_k + b_k)^2\right]^{1/2} \leq \left(\sum_{k=1}^{n} a_k^2\right)^{1/2} + \left(\sum_{k=1}^{n} b_k^2\right)^{1/2}$$

1-18 Show that

$$\left(\sum_{k=1}^{n} a_k b_k c_k\right)^4 \leq \left(\sum_{k=1}^{n} a_k^4\right)\left(\sum_{k=1}^{n} b_k^2\right)^2\left(\sum_{k=1}^{n} c_k^4\right)$$

Complex numbers

1-19. Prove the associative law for multiplication of complex numbers. $x(yz) = (xy)z$

1-20. Prove the triangle inequality (Theorem 1-22) for complex numbers,

$$|x + y| \leq |x| + |y|,$$

by analytic methods (i.e., without using geometry). Deduce from this the following inequality:

$$|x - y| \geq ||x| - |y||.$$

1–21. If $z = x + iy$, then the complex conjugate of z is the complex number $\bar{z} = x - iy$ (x and y are real). Show that we have

(a) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$, (b) $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$, (c) $z\bar{z} = |z|^2$,

(d) $z + \bar{z} =$ twice the real part of z,

(e) $(z - \bar{z})/i =$ twice the imaginary part of z.

1–22. If z_1 and z_2 are complex numbers, show that $|z_1 - z_2|$ represents the distance between them.

1–23. Describe (geometrically) the sets of complex numbers z which satisfy the following conditions:

(a) $|z| = 1$, (b) $|z| < 1$, (c) $|z| \leq 1$,

(d) $z + \bar{z} = 1$, (e) $z - \bar{z} = i$, (f) $\bar{z} + z = |z|^2$.

1–24. Given three complex numbers z_1, z_2, z_3 such that $|z_1| = |z_2| = |z_3| = 1$ and $z_1 + z_2 + z_3 = 0$. Show that these numbers are vertices of an equilateral triangle inscribed in the unit circle with center at the origin.

1–25. If a and c are real constants, b complex, show that the equation

$$a\, z\bar{z} + b\bar{z} + \bar{b}z + c = 0 \qquad (a \neq 0, z = x + iy)$$

represents a circle in the xy-plane.

1–26. Recall the definition of the inverse tangent: given a real number t, $\tan^{-1}(t)$ is the unique real number θ which satisfies the two conditions

$$-\frac{\pi}{2} < \theta < +\frac{\pi}{2}, \qquad \tan\theta = t.$$

If $z = x + iy$, show that

(a) $\arg(z) = \tan^{-1}\left(\dfrac{y}{x}\right)$, if $x > 0$,

(b) $\arg(z) = \tan^{-1}\left(\dfrac{y}{x}\right) + \pi$, if $x < 0, y \geq 0$,

(c) $\arg(z) = \tan^{-1}\left(\dfrac{y}{x}\right) - \pi$, if $x < 0, y < 0$,

(d) $\arg(z) = \dfrac{\pi}{2}$ if $x = 0, y > 0$,

and

$$\arg(z) = -\frac{\pi}{2} \text{ if } x = 0, y < 0.$$

1-27. Define the following "pseudo-ordering" of the complex numbers: we say $z_1 < z_2$ if we have either

(i) $|z_1| < |z_2|$ or (ii) $|z_1| = |z_2|$ and $\arg(z_1) < \arg(z_2)$.

Which of Axioms 6, 7, 8, 9 are satisfied by this relation?

1-28. Which of Axioms 6, 7, 8, 9 are satisfied if the pseudo-ordering is defined as follows? We say $(x_1, y_1) < (x_2, y_2)$ if we have either

(i) $x_1 < x_2$ or (ii) $x_1 = x_2$ and $y_1 < y_2$.

1-29. State and prove a theorem analogous to Theorem 1-31, expressing $\arg(z_1/z_2)$ in terms of $\arg(z_1)$ and $\arg(z_2)$.

1-30. State and prove a theorem analogous to Theorem 1-37, expressing $\operatorname{Log}(z_1/z_2)$ in terms of $\operatorname{Log}(z_1)$ and $\operatorname{Log}(z_2)$.

1-31. Prove Theorem 1-33.

1-32. Show that the nth roots of 1 (also called the nth roots of unity) are given by $\alpha, \alpha^2, \ldots, \alpha^n$, where $\alpha = e^{2\pi i/n}$, and show that the roots $\neq 1$ satisfy the equation $1 + x + x^2 + \cdots + x^{n-1} = 0$.

1-33. (a) Show that $|e^z| \leq e^{|z|}$ for all complex $z \neq 0$.

(b) Show that there is no constant $M > 0$ such that $|\cos z| < M$ for all complex z.

1-34. If $w = u + iv$ (u, v real), show that

$$z^w = e^{u \log|z| - v \arg(z)} e^{i[v \log|z| + u \arg(z)]}.$$

1-35. (a) Show that $\operatorname{Log}(z^w) = w \operatorname{Log} z + 2\pi i n$, ($n$ is an integer).

(b) Show that $(z^w)^\alpha = z^{w\alpha} e^{2\pi i n\alpha}$, ($n$ is an integer).

1-36. (i) If θ and a are real numbers, $-\pi < \theta \leq +\pi$, show that

$$(\cos\theta + i\sin\theta)^a = \cos(a\theta) + i\sin(a\theta).$$

(ii) Show that, in general, the restriction $-\pi < \theta \leq +\pi$ is necessary in (i) by taking $\theta = -\pi$, $a = \frac{1}{2}$.

(iii) If a is an integer, show that the formula in (i) holds without any restriction on θ. In this case it is known as DeMoivre's theorem.

1-37. Use DeMoivre's theorem (Exercise 1-36) to derive the trigonometric identities

$$\sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta,$$
$$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta,$$

valid for real θ. Are these valid when θ is complex?

1-38. Define $\tan z = (\sin z)/(\cos z)$ and show that for $z = x + iy$, we have

$$\tan z = \frac{\sin 2x + i\sinh 2y}{\cos 2x + \cosh 2y}.$$

1–39. Let w be a given complex number. If $w \neq \pm 1$, show that there exist two values of $z = x + iy$ satisfying the conditions $\cos z = w$ and $-\pi < x \leq +\pi$. Find these values when $w = i$ and when $w = 2$.

1–40. Prove Lagrange's identity for complex numbers:

$$\left|\sum_{k=1}^{n} a_k b_k\right|^2 = \sum_{k=1}^{n} |a_k|^2 \sum_{k=1}^{n} |b_k|^2 - \sum_{1 \leq k < j \leq n} |a_k \bar{b}_j - a_j \bar{b}_k|^2.$$

Use this to deduce the Cauchy-Schwarz inequality for complex numbers.

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CHAPTER 2

SOME BASIC NOTIONS OF SET THEORY

2–1 Fundamentals of set theory. Since the latter part of the nineteenth century, the theory of sets as developed by G Cantor has not only influenced and enriched almost every branch of mathematics but has also helped to clarify the relationship between mathematics and philosophy We shall not attempt an exhaustive or rigorous treatment of the theory of sets, but rather we shall confine ourselves to a study of some of the more basic ideas involved (The reader who wishes to explore the subject further may consult the bibliography at the end of the chapter)

A collection of objects viewed as a single entity will be referred to as a *set* The objects in the collection will be called *elements* or *members* of the set, and they will be said to *belong to* or to be *contained in* the set The set, in turn, will be said to *contain* or to be *composed of* its elements For the most part we shall be interested in sets of mathematical objects, that is, sets of numbers, points, functions, curves, etc However, since much of the theory of sets does not depend on the nature of the individual objects in the collection, we gain a great economy of thought by discussing sets whose elements may be objects of any kind It is because of this quality of generality that the theory of sets has had such a strong effect in furthering the development of mathematics

2–2 Notations. *Sets* will usually be denoted by capital letters

$$A, B, C, \ldots, X, Y, Z,$$

and *elements* by lower-case letters $a, b, c, \ldots, x, y, z$ We write $x \in S$ to mean "x is an element of S," or "x belongs to S" If x does not belong to S, we write $x \notin S$ We sometimes designate sets by displaying the elements in braces, for example, the set of positive even integers less than 10 is denoted by $\{2, 4, 6, 8\}$

From a given set we may form new sets, called *subsets* of the given set For example, the set consisting of all positive integers less than 10 which are divisible by 4, namely, $\{4, 8\}$, is a subset of the set of even integers less than 10 In general, we say that a set A is a subset of B, and we write $A \subset B$ whenever every element of A also belongs to B The statement $A \subset B$ does not rule out the possibility that $B \subset A$. In fact, we have both $A \subset B$ and $B \subset A$ if, and only if, A and B have the same elements In this case we call the sets A and B equal and we write $A = B$ If A

and B are not equal, we write $A \neq B$. If $A \subset B$ but $A \neq B$, then we say that A is a *proper subset* of B.

It is convenient to consider the possibility of a set which contains no elements whatever; this set is called the *empty set* and is a subset of every set. The reader may find it helpful to picture a set as a box containing certain objects, its elements. The empty set is then an empty box.

2–3 Ordered pairs. Suppose we have a set consisting of two elements a and b; that is, the set $\{a, b\}$. By our definition of equality this set is the same as the set $\{b, a\}$, since no question of order is involved. However, it is also necessary to consider sets of two elements in which order *is* important. For example, in analytic geometry of the plane, the coordinates (x, y) of a point represent an *ordered pair* of numbers. The *point* $(3, 4)$ is different from the point $(4, 3)$, whereas the *set* $\{3, 4\}$ is the same as the set $\{4, 3\}$. When we wish to consider a set of two elements a and b as being *ordered*, we shall enclose the elements in parentheses: (a, b). Then a is called the first element and b the second. It is possible to give a purely set-theoretic definition of the concept of an ordered pair of objects (a, b). One such definition is the following:

2–1 DEFINITION. $(a, b) = \{\{a\}, \{a, b\}\}$.

This definition states that (a, b) is a set containing two elements, $\{a\}$ and $\{a, b\}$. Using this definition, we can prove the following theorem:

2–2 THEOREM. *$(a, b) = (c, d)$ if, and only if, $a = c$ and $b = d$.*

This theorem shows that Definition 2–1 is a "reasonable" definition of an ordered pair, in the sense that the object a has been distinguished from the object b. The proof of Theorem 2–2 will be an instructive exercise for the reader. (See Exercise 2–5.)

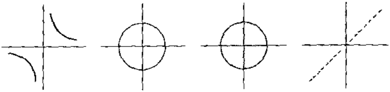
2–4 Cartesian product of two sets.

2–3 DEFINITION. *Given two sets A and B, the set of all ordered pairs (a, b) such that $a \in A$ and $b \in B$ is called the cartesian product of A and B, and is denoted by $A \times B$.*

EXAMPLE. If R denotes the set of all real numbers, then $R \times R$ is the set of all complex numbers.

2–5 Relations and functions in the plane. Let x and y denote real numbers, so that the ordered pair (x, y) can be thought of as representing the rectangular coordinates of a point in the xy-plane (or a complex number). We frequently encounter such expressions as

(a) $\qquad xy = 1, \qquad x^2 + y^2 = 1, \qquad x^2 + y^2 \leq 1, \qquad x < y.$



$xy = 1$ $x^2 + y^2 = 1$ $x^2 + y^2 \leq 1$ $x < y$

FIG 2–1 Relations in the plane.

Each of these expressions defines a certain set of ordered pairs (x, y) of real numbers, namely, the set of all pairs (x, y) for which the expression is satisfied. Such a set of ordered pairs is called a *plane relation*. The corresponding set of points plotted in the xy-plane is called the *graph* of the relation. The graphs of the relations described in (a) are shown in Fig 2–1.

Every set of ordered pairs (x, y) constitutes a plane relation. Certain special kinds of relations are known as *functions*. A function is a relation whose graph has the property that every vertical line intersects it at most once. In other words, whenever two points (x, y_1) and (x, y_2) in the graph have the same x-coordinate, then they must also have the same y-coordinate; i e , they must have $y_1 = y_2$. Thus, for each point (x, y) of the graph of a function, the x-coordinate uniquely determines the y-coordinate. For example, the relation defined by the equation $xy = 1$ is a function because for each x there is exactly one y such that $xy = 1$, namely, $y = 1/x$. On the other hand, the relation defined by the equation $x^2 + y^2 = 1$ is *not* a function because there are points of the graph which have the same x-coordinate but different y-coordinates. Two such points are $(0, 1)$ and $(0, -1)$. Similarly, the other examples given above are not functions.

If F is a set of ordered pairs which constitutes a function, then instead of writing $(x, y) \in F$ to indicate that the pair (x, y) is in the set F, it is customary to write $y = F(x)$. This is consistent with the fact that for each pair (x, y) in F, the second member y is uniquely determined by x.

If each point on the graph of a function F is projected vertically on the x-axis, we obtain a set of points on the x-axis known as the *domain* of the function, denoted by $D(F)$. Similarly, projecting the graph horizontally on the y-axis gives us another set called the *range* of the function, denoted by $R(F)$. For example, consider the function F defined as follows

$$\begin{aligned} F = \{(x, y) \mid y &= x^3, && \text{if } -1 \leq x \leq 1, \\ y &= \tfrac{3}{2}, && \text{if } \quad x = \tfrac{3}{2}, \\ y &= 2, && \text{if } 2 \leq x \leq 3\}. \end{aligned}$$

The graph of F is illustrated by Fig. 2-2. The domain $D(F)$ consists of the interval $-1 \leq x \leq +1$, the point $x = \frac{3}{2}$, and the interval $2 \leq x \leq 3$. The range $R(F)$ consists of the interval $-1 \leq y \leq +1$, the point $y = \frac{3}{2}$, and the point $y = 2$.

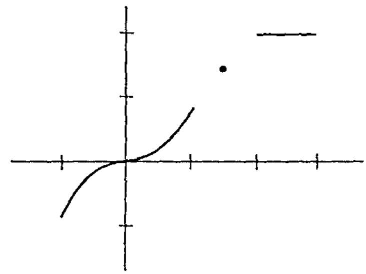


FIGURE 2-2.

2-6 General definition of relation. At this point we shall introduce a very general definition of function in which the "variables" need not be numbers at all but may be objects of any kind. Our discussion will be guided by the special case just considered, that of the notion of function in a plane. We first need the concept of an ordered pair of objects, which has been stated in Definition 2-1. We then define a *relation* as follows:

2-4 DEFINITION. *By a relation we shall mean a set of ordered pairs. If S is a set of ordered pairs, and if the pair $(a, b) \in S$, then we also say "a is in the relation S to b," and we sometimes write aSb.*

Formulas (a) in Section 2-5 give examples of relations in the plane. Further examples are:

1. $S =$ the set of ordered pairs of congruent triangles in a given plane; that is, S is a set of pairs (a, b) where the symbols a and b denote triangles in a given plane. The pair $(a, b) \in S$ if, and only if, the triangles a and b are congruent. (It is customary in plane geometry to write $a \cong b$ instead of aSb.)

2. $S =$ the set of ordered pairs (a, b) of people such that a is a parent of b. Thus aSb means "a is a parent of b."

3. $S =$ the set of ordered pairs (a, b) of people such that a is a child of b. Then aSb means "a is a child of b." [What is the connection between this relation and the relation in Example 2?]

Given a relation S, we can form two new sets, the *domain* of the relation, $D(S)$, and the *range* $R(S)$:

2-5 DEFINITION. *The domain of a relation S is defined to be the set of those x for which there exists a y such that $(x, y) \in S$. The range of S is the set of y for which there exists an x such that $(x, y) \in S$.*

2-7 General definition of function.

2-6 DEFINITION. *A relation F is called a function provided that*

$$(x, y) \in F \quad \text{and} \quad (x, z) \in F \quad \text{implies} \quad y = z.$$

A function, then, is a set of ordered pairs which has the special property that whenever two pairs (x, y) and (x, z) in the set have the same first element, they must also have the same second element Intuitively, a function can be thought of as a table consisting of two columns Each entry in the table is a pair of the form (x, y), with the column of x's the domain and the column of y's the range The elements of the range are known as *function values* If two entries (x, y) and (x, z) appear in the table with the same x-value, then for the table to be a function, it is necessary that $y = z$ Hence, for every x in the domain of a function F, there exists exactly one value of y which corresponds to this x, that is, exactly one y such that $(x, y) \varepsilon F$ Since this y is uniquely determined once x is known, we can introduce a symbol for it We write $y = F(x)$ to mean $(x, y) \varepsilon F$.

As an alternative to describing a function F by explicitly exhibiting the pairs it contains, it is usually preferable to describe the domain of F, and then, for each x in the domain, to describe how the function value $F(x)$ is obtained The reader should verify that this does, indeed, characterize F. (See Exercise 2–3)

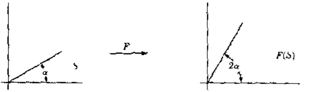


FIG 2–3. The mapping defined by $F(z) = z^2$

When the domain of F consists of real numbers, F is called a function of one real variable If the domain of F is a set of points in the plane (or a set of complex numbers), then F is called a function of two real variables (or a function of a complex variable) If S is a subset of the domain of F, we say that F is *defined on* S In this case, the set of $F(x)$ such that $x \varepsilon S$ is called the *image* of S under F and is denoted by $F(S)$ If $F(S) \subset T$, then F is called a *mapping* of S *into* T If $F(S) = T$, the mapping is said to be *onto* T A mapping of S into itself is also referred to as a *transformation*.

Consider, for example, the function of a complex variable defined by the equation $F(z) = z^2$. This function "maps" every sector S of the form $0 \leq \arg(z) \leq \alpha \leq \pi/2$ of the complex z-plane onto a sector $F(S)$ described by the inequalities $0 \leq \arg[F(z)] \leq 2\alpha$ (See Fig 2–3)

2-8 One-to-one functions and inverses.

2-7 DEFINITION. *Let F be a function defined on S. We say F is one-to-one on S if, and only if, for every x and y in S,*

$$F(x) = F(y) \qquad \textit{implies} \qquad x = y.$$

This is the same as saying that a function which is one-to-one on S assigns distinct function values to distinct members of S. Such functions are important because, as we shall presently see, they possess *inverses.* However, before stating the definition of the inverse of a function, it is convenient to introduce a more general notion, that of the *converse* of a relation.

2-8 DEFINITION. *Given a relation S, the new relation $\breve{S}$ defined by*

$$\breve{S} = \{(a, b) \mid (b, a) \in S\}$$

is called the converse of S.

Thus an ordered pair (a, b) belongs to $\breve{S}$ if, and only if, the pair (b, a), with elements interchanged, belongs to S. When the relation S is a *plane relation,* this simply means that the graph of $\breve{S}$ is the reflection of the graph of S with respect to the line $y = x$. In the relation defined by $x < y$, the converse relation is defined by $y < x$. The converse of "is a parent of" is the relation "is a child of."

2-9 DEFINITION. *Suppose that the relation F is a function. Consider the converse relation $\breve{F}$, which may or may not be a function. If $\breve{F}$ is also a function, then $\breve{F}$ is called the inverse of F and is denoted by F^{-1}.*

Figure 2-4 (a) illustrates an example of a function F for which $\breve{F}$ is not a function. In Fig. 2-4 (b) both F and its converse are functions.

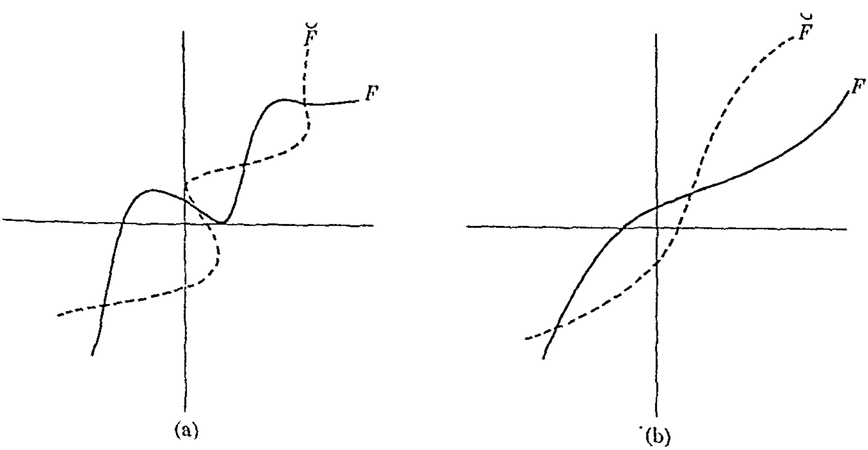


FIGURE 2-4.

The next theorem tells us that a function which is one-to-one on its domain always has an inverse

2-10 THEOREM *If the function F is one-to-one on its domain, then $\breve{F}$ is also a function*

Proof To show that $\breve{F}$ is a function, we must show that if $(x, y) \in \breve{F}$ and $(x, z) \in \breve{F}$, then $y = z$ But $(x, y) \in \breve{F}$ means that $(y, x) \in F$, that is, $x = F(y)$ Similarly, $(x, z) \in \breve{F}$ means that $x = F(z)$ Thus $F(y) = F(z)$ and, since we are assuming that F is one-to-one, this implies $y = z$ Hence, $\breve{F}$ is a function

2-9 Composite functions.

2-11 DEFINITION *Given two functions F and G such that $R(F) \subset D(G)$, we can form a new function, the composite GF of F and G, defined as follows for every x in the domain of F, $GF(x) = G[F(x)]$*

Since $R(F) \subset D(G)$, the element $F(x)$ is in the domain of G, and therefore it makes sense to consider $G[F(x)]$ In general, it is not true that $GF = FG$ In fact, FG may be meaningless unless the range of G is contained in the domain of F However, the associative law, $H(GF) = (HG)F$, always holds whenever each side of the equation has a meaning (Verification will be an interesting exercise for the reader See Exercise 2-4)

2-10 Sequences. Among the important examples of functions are those defined on subsets of the integers

2-12 DEFINITION *By a finite sequence of n terms we shall understand a function F whose domain is the set of numbers $\{1, 2, \ldots, n\}$*

The range is then the set $\{F(1), F(2), F(3), \ldots, F(n)\}$, customarily written $\{F_1, F_2, F_3, \ldots, F_n\}$ The elements of the range are called *terms* of the sequence and, of course, they may be arbitrary objects of any kind

2-13 DEFINITION *By an infinite sequence we shall mean a function F whose domain is the set $\{1, 2, 3, \ldots\}$ of all positive integers The range of F, that is, the set $\{F(1), F(2), F(3), \ldots\}$, is also written $\{F_1, F_2, F_3, \ldots\}$, and the function value F_n is called the nth term of the sequence*

For brevity, we shall occasionally use the notation $\{F_n\}$ to denote the infinite sequence whose nth term is F_n

Let $s = \{s_n\}$ be an infinite sequence, and let k be a function whose domain is the set of positive integers and whose range is a subset of the positive integers Assume that k is "order-preserving," that is, assume that

$$k(m) < k(n), \quad \text{if } m < n$$

Then the composite function sk is defined for all integers $n \geq 1$ and for every such n, we have

$$sk(n) = s_{k(n)} .$$

Such a composite function sk is said to be a *subsequence* of s. Again, for brevity, we often use the notation $\{s_{k(n)}\}$ or $\{s_{k_n}\}$ to denote the subsequence of $\{s_n\}$ whose nth term is $s_{k(n)}$.

EXAMPLE. Let $s = \{1/n\}$ and let k be defined by $k(n) = 2^n$. Then $sk = \{1/2^n\}$.

2-11 The number of elements in a set.

2-14 DEFINITION. *Two sets A and B are called equivalent, and we write $A \sim B$ if, and only if, there exists a one-to-one function F whose domain is the set A and whose range is the set B.*

We also say that F establishes a *one-to-one correspondence* between the sets A and B. Clearly, every set A is equivalent to itself (take F to be the "identity" function for which $F(x) = x$, if $x \in A$). Furthermore, if $A \sim B$ then $B \sim A$, because if F is a one-to-one function which makes A equivalent to B, then F^{-1} will make B equivalent to A. Also, if $A \sim B$ and if $B \sim C$, then $A \sim C$. (The proof is left to the reader. See Exercise 2-6.)

A set S is called *finite* and is said to contain n elements if

$$S \sim \{1, 2, \ldots, n\}.$$

The empty set is also considered finite. Sets which are not finite are called *infinite sets*. The chief difference between the two is that an infinite set must be equivalent to some proper subset of itself, whereas a finite set cannot be equivalent to any proper subset of itself. (See Exercise 2-7.) It is easy to see, for example, that the set $A = \{1, 2, 3, \ldots\}$ of all positive integers is equivalent to the proper subset $\{2, 4, 8, 16, \ldots\}$ consisting of powers of 2. The one-to-one function F which makes them equivalent is defined by $F(x) = 2^x$ for each x in A.

2-15 DEFINITION. *If a set A is equivalent to the set of all positive integers or to some (finite or infinite) subset of the positive integers, then A is said to be countable (or denumerable). A set which is not countable is called uncountable (or nondenumerable).*

Therefore, for every countable set A, there corresponds a (finite or infinite) sequence $s = \{s_n\}$ having distinct terms, these terms being the elements of A. The set A is thus the same as the range of s.

2-16 THEOREM. *Every subset of a countable set is countable.*

Proof Let S be the given countable set and let $A \subset S$ If A is finite, there is nothing to prove, so we can assume that A is infinite (which means S is also infinite). Let $s = \{s_n\}$ be an infinite sequence of distinct terms such that $S = \{s_1, s_2, \quad\}$ Define a function on the positive integers as follows

Let $k(1)$ be the smallest positive integer m such that $s_m \in A$ Assuming that $k(1), k(2), \quad, k(n-1)$ have been defined, let $k(n)$ be the smallest positive integer $m > k(n-1)$ such that $s_m \in A$ Then k is order-preserving $m > n$ implies $k(m) > k(n)$ Form the composite function sk The domain of sk is the set of positive integers and the range of sk is A Furthermore, sk is one-to-one, since $sk(n) = sk(m)$ implies $s_{k(n)} = s_{k(m)}$, which implies $k(n) = k(m)$, and this implies $n = m$ This proves the theorem

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Roughly speaking, a countable set can have no more elements than the set of integers There are sets, however, which are not countable

2-17 THEOREM *The set of all real numbers is uncountable*

Proof It suffices to show that the set of x satisfying $0 < x < 1$ is uncountable If the real numbers in this interval were countable, there would be a sequence $s = \{s_n\}$ whose terms would constitute the whole interval We shall show that this is impossible by constructing, in the interval, a real number which is not a term of this sequence Write each s_n as a decimal

$$s_n = 0\ u_{n\ 1}u_{n\ 2}u_{n\ 3} \quad ,$$

where each $u_{n\ ,}$ is 0, 1, , or 9 Consider the real number y which has the decimal expansion

$$y = 0\ v_1v_2v_3 \quad ,$$

where

$$v_n = \begin{cases} 1, & \text{if } u_{n\ n} \neq 1, \\ 2, & \text{if } u_{n\ n} = 1 \end{cases}$$

Then no term of the sequence $\{s_n\}$ can be equal to y, since y differs from s_1 in the first decimal place, differs from s_2 in the second decimal place, . , from s_n in the nth decimal place (A situation like $s_n = 0\ 1999$ and $y = 0\ 2000$ cannot occur here because of the way the v_n are chosen.) Since $0 < y < 1$, the theorem is proved

2-18 THEOREM *Let P denote the set of all positive integers and let $P \times P$ be the cartesian product of P with itself Then $P \times P$ is countable.*

Proof. Define a function f on $P \times P$ as follows:

$$f(m, n) = 2^m 3^n, \qquad \text{if } (m, n) \in P \times P.$$

Then f is one-to-one on $P \times P$ [Why?] and the range of f is a subset of P. This completes the proof.

2–12 Set algebra. Given two sets A_1 and A_2, we define a new set, called the *union* of A_1 and A_2, denoted by $A_1 \cup A_2$, as follows:

2–19 DEFINITION. *The union $A_1 \cup A_2$ is the set of those elements which belong either to A_1 or to A_2 or to both.*

This is the same as saying that $A_1 \cup A_2$ consists of those elements which belong to at least one of the sets A_1, A_2. Since there is no question of order involved in this definition, the union $A_1 \cup A_2$ is the same as $A_2 \cup A_1$; that is, set addition is commutative. The definition is also phrased in such a way that set addition is associative:

$$A_1 \cup (A_2 \cup A_3) = (A_1 \cup A_2) \cup A_3.$$

The definition of union can be extended to any finite or infinite collection of sets:

2–20 DEFINITION. *If F is an arbitrary collection of sets, then the union of all the sets in F is defined to be the set of those elements which belong to at least one of the sets in F, and is denoted by*

$$\bigcup_{A \in F} A.$$

If F is a finite collection of sets, $F = \{A_1, \ldots, A_n\}$, we write

$$\bigcup_{A \in F} A = \bigcup_{k=1}^{n} A_k = A_1 \cup A_2 \cup \cdots \cup A_n.$$

If F is a countable collection, $F = \{A_1, A_2, \ldots\}$, we write

$$\bigcup_{A \in F} A = \bigcup_{k=1}^{\infty} A_k = A_1 \cup A_2 \cup \cdots.$$

2–21 DEFINITION. *If F is an arbitrary collection of sets, then the intersection of all sets in F is defined to be the set of those elements which belong to every one of the sets in F, and is denoted by*

$$\bigcap_{A \in F} A.$$

The intersection of two sets A_1 and A_2 is denoted by $A_1 \cap A_2$ and consists of those elements common to both sets. If A_1 and A_2 have no elements in common, then $A_1 \cap A_2$ is the empty set and A_1 and A_2 are said to be *disjoint*. If F is a finite collection (as above), we write

$$\bigcap_{A \varepsilon F} A = \bigcap_{k=1}^{n} A_k = A_1 \cap A_2 \cap \cdots \cap A_n$$

and if F is a countable collection, we write

$$\bigcap_{A \varepsilon F} A = \bigcap_{k=1}^{\infty} A_k = A_1 \cap A_2 \cap \cdots.$$

If the sets in the collection have no elements in common, their intersection is the empty set. Our definitions of union and intersection apply, of course, even when F is not countable. Because of the way we have defined unions and intersections, the commutative and associative laws are automatically satisfied.

2–22 Definition. *The complement of A relative to B, denoted by $B - A$, is defined to be the set*

$$B - A = \{x \mid x \varepsilon B, \text{ but } x \not\varepsilon A\}$$

Note that $B - (B - A) = A$ whenever $A \subset B$. Also note that $B - A = B$ if $B \cap A$ is empty.

The notions of union, intersection, and complement are illustrated in Fig. 2–5.

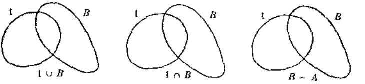

Fig. 2–5. Operations on sets.

2–23 Theorem. *Let F be a collection of sets. Then for any set B, we have*

$$B - \bigcup_{A \varepsilon F} A = \bigcap_{A \varepsilon F} (B - A)$$

and

$$B - \bigcap_{A \varepsilon F} A = \bigcup_{A \varepsilon F} (B - A)$$

Proof. Let $S = \bigcup_{A \varepsilon F} A$, $T = \bigcap_{A \varepsilon F} (B - A)$. If $x \varepsilon B - S$, then $x \varepsilon B$, but $x \not\varepsilon S$. Hence, it is not true that x belongs to at least one A in F; therefore x belongs to no A in F. Hence, for every A in F, $x \varepsilon B - A$. But this implies $x \varepsilon T$, so that $B - S \subset T$. Reversing the steps, we obtain $T \subset B - S$, and this proves that $B - S = T$. To prove the second statement, use a similar argument.

2-24 DEFINITION. *If F is a collection of sets such that every two distinct sets in F are disjoint, then F is said to be a collection of disjoint sets.*

2-25 THEOREM. *If F is a countable collection of disjoint sets, say $F = \{A_1, A_2, \ldots\}$, such that each set A_n is countable, then the union $\bigcup_{k=1}^{\infty} A_k$ is also countable.*

Proof. Let $A_n = \{a_{1,n}, a_{2,n}, a_{3,n} \ldots\}$, $n = 1, 2, \ldots$, and let $S = \bigcup_{k=1}^{\infty} A_k$. Then every element x of S is in at least one of the sets in F and hence $x = a_{m,n}$ for some pair of integers (m, n). The pair (m, n) is uniquely determined by x, since F is a collection of disjoint sets. Hence the function f defined by $f(x) = (m, n)$ if $x = a_{m,n}$, $x \varepsilon S$, has domain S. The range $f(S)$ is a subset of $P \times P$ (where P is the set of positive integers) and hence is countable. But f is one-to-one and therefore $S \sim f(S)$, which means that S is also countable.

2-26 THEOREM. *If $F = \{A_1, A_2, \ldots\}$ is a countable collection of sets, let $G = \{B_1, B_2, \ldots\}$, where $B_1 = A_1$ and, for $n > 1$,*

$$B_n = A_n - \bigcup_{k=1}^{n-1} A_k.$$

Then G is a collection of disjoint sets, and we have

$$\bigcup_{k=1}^{\infty} A_k = \bigcup_{k=1}^{\infty} B_k.$$

Proof. Each set B_n is constructed so that it has no elements in common with the earlier sets $B_1, B_2, \ldots, B_{n-1}$. Hence G is a collection of disjoint sets. Let $A = \bigcup_{k=1}^{\infty} A_k$ and $B = \bigcup_{k=1}^{\infty} B_k$. We shall show that $A = B$. First of all, if $x \varepsilon A$, then $x \varepsilon A_k$ for some k. If n is the smallest such k, then $x \varepsilon A_n$ but $x \not\varepsilon \bigcup_{k=1}^{n-1} A_k$, which means that $x \varepsilon B_n$, and therefore $x \varepsilon B$. Hence $A \subset B$. Conversely, if $x \varepsilon B$, then $x \varepsilon B_n$ for some n, and therefore $x \varepsilon A_n$ for this same n. Thus $x \varepsilon A$ and this proves that $B \subset A$.

Using Theorems 2-25 and 2-26, we immediately obtain

2-27 THEOREM *If F is a countable collection of countable sets, then the union of all sets in F is also a countable set*

EXAMPLES

1 The set S of all rational numbers is a countable set

Proof Let A_n denote the set of all positive rational numbers having denominator n The set of all positive rational numbers is equal to $\bigcup_{k=1}^{\infty} A_k$ From this it follows that S is countable, since each A_n is countable

2 The set S of intervals with rational endpoints is a countable set

Proof Let $\{x_1, x_2, \quad\}$ denote the set of rational numbers and let A_n be the set of all intervals whose left endpoint is x_n and whose right endpoint is rational Then A_n is countable and $S = \bigcup_{k=1}^{\infty} A_k$

Exercises

2-1. Let S be a relation and let $D(S)$ be its domain. The relation S is said to be

(i) *reflexive* if $a \in D(S)$ implies $(a, a) \in S$,
(ii) *symmetric* if $(a, b) \in S$ implies $(b, a) \in S$,
(iii) *transitive* if $(a, b) \in S$ and $(b, c) \in S$ implies $(a, c) \in S$.

A relation which is symmetric, reflexive, and transitive is called an *equivalence relation*. Determine which of these properties is possessed by S, if S is the set of all pairs of real numbers (x, y) such that

(a) $x \leq y$, (b) $x < y$, (c) $x < |y|$,
(d) $x^2 + y^2 = 1$, (e) $x^2 + y^2 < 0$, (f) $x^2 + x = y^2 + y$.

2-2. The following functions F and G are defined for all real x by the equations given. In each case where the composite function GF can be formed, give the domain of GF and a formula (or formulas) for $GF(x)$.

(a) $F(x) = 1 - x$, $G(x) = x^2 + 2x$.
(b) $F(x) = x + 5$, $G(x) = |x|/x$, if $x \neq 0$, $G(0) = 1$.
(c) $F(x) = 2x$ if $0 \leq x \leq 1$; $F(x) = 1$, otherwise.
$G(x) = x^2$, if $0 \leq x \leq 1$; $G(x) = 0$, otherwise.

Find $F(x)$ if $G(x)$ and $GF(x)$ are given as follows:

(d) $G(x) = x^3$, $GF(x) = x^3 - 3x^2 + 3x - 1$.
(e) $G(x) = 3 + x + x^2$, $GF(x) = x^2 - 3x + 5$.

2-3. Show that two functions F and G are equal if, and only if, the following two conditions hold:

(a) F and G have the same domain.
(b) For every x in the domain of F, $F(x) = G(x)$.

2-4. Given three functions F, G, H, what restrictions must be placed on their domains so that the following four composite functions can be defined?

$$GF, \qquad HG, \qquad H(GF), \qquad (HG)F.$$

Assuming that $H(GF)$ and $(HG)F$ can be defined, prove the associative law:

$$H(GF) = (HG)F.$$

2-5. Prove Theorem 2-2. [*Hint:* $(a, b) = (c, d)$ means $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$. Now appeal to the definition of set equality.]

2-6. Prove that if $A \sim B$ and $B \sim C$, then $A \sim C$.

2-7. Prove that a set is finite if, and only if, it contains no equivalent proper subset.

2-8. Prove the following set-theoretic identities for union and intersection

(a) $A \cup (B \cup C) = (A \cup B) \cup C, \qquad A \cap (B \cap C) = (A \cap B) \cap C$
(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
(c) $(A \cup B) \cap (A \cup C) = A \cup (B \cap C)$
(d) $(A \cup B) \cap (B \cup C) \cap (C \cup A) = (A \cap B) \cup (A \cap C) \cup (B \cap C)$
(e) $A \cap (B - C) = (A \cap B) - (A \cap C)$
(f) $(A - C) \cap (B - C) = (A \cap B) - C$
(g) $(A - B) \cup B = A$ if, and only if, $B \subset A$

2-9. Let S be a finite set consisting of n elements and let T be the collection of all subsets of S. Show that T is a finite set and find the number of elements in T.

2-10. Let R denote the set of real numbers and let S denote the set of all real-valued functions whose domain is R. Show that S and R are not equivalent. [*Hint.* Assume $S \sim R$ and let f be a one-to-one function such that $f(R) = S$. If $a \in R$, let $g_a = f(a)$ be the real-valued function in S which corresponds to the real number a. Now define h by the equation $h(x) = 1 + g_x(x)$ if $x \in R$, and show that $h \notin S$.]

2-11. Let S be the collection of all sequences whose terms are the integers 0 and 1. Show that S is uncountable.

2-12. Show that the following sets are countable: (a) the set of circles in the complex plane having rational radii and centers with rational coordinates, (b) any collection of disjoint intervals of positive length.

2-13. Let f be a real-valued function defined for every x in the interval $0 \le x \le 1$. Suppose there is a positive number M having the following property: for every choice of a finite number of points $x_1, x_2, \ldots, x_n$ in the interval $0 \le x \le 1$, the sum

$$|f(x_1) + \cdots + f(x_n)| \le M.$$

Let S be the set of those x in $0 \le x \le 1$ for which $f(x) \ne 0$. Prove that S is countable.

2-14. What is wrong with the following "proof" that the set of all intervals of positive length is countable?

Let $\{x_1, x_2, \ldots\}$ denote the countable set of rational numbers and let I be any interval of positive length. Then I contains infinitely many rational points x_n, but among these there will be one with *smallest index n*. Define a function F by means of the equation $F(I) = n$, if x_n is the rational number with smallest index in the interval I. This function establishes a one-to-one correspondence between the set of all intervals and a subset of the positive integers. Hence the set of all intervals is countable.

2-15. (i) Define an *ordered triple* (a, b, c) as follows:

$$(a, b, c) = ((a, b), c).$$

Show that $(a_1, b_1, c_1) = (a_2, b_2, c_2)$ if, and only if, $a_1 = a_2$, $b_1 = b_2$, $c_1 = c_2$.

(ii) Define the cartesian product of three sets A, B, and C to be the set

$$A \times B \times C = \{(a, b, c) \mid a \in A, b \in B, c \in C\}.$$

Show that $A \times B \times C = (A \times B) \times C = A \times (B \times C)$.

(iii) Give similar definitions for ordered n-tuples $(a_1, a_2, \ldots, a_n)$ and for n-fold cartesian products $A_1 \times A_2 \times \cdots \times A_n$.

Suggested References

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CHAPTER 3

ELEMENTS OF POINT SET THEORY

3–1 Introduction. A large part of the previous chapter dealt with "abstract" sets, that is, sets of arbitrary objects whose nature is immaterial In this chapter we specialize our sets to be sets of real numbers or sets of complex numbers and study such collections in greater detail

In this area of study it is convenient and helpful to use geometrical terminology to a large extent. Thus, we shall talk about sets of points on the real axis or sets of points in the plane Later in the chapter we shall see how most of the concepts introduced can be extended to three or more dimensions For brevity, we denote by E_1 the set of all real numbers and by E_2 the set of all complex numbers; E_1 is also called the *real line* and E_2 the *complex plane.*

3–2 Intervals and open sets in E_1. The simplest types of sets encountered on the real line are intervals. It is sometimes important to distinguish between intervals which include their endpoints and intervals which do not.

3–1 DEFINITION *Assume $a < b$ The open interval (a, b) is defined to be the set*

$$(a, b) = \{x \mid a < x < b\}$$

The closed interval $[a, b]$ is the set $\{x \mid a \leq x \leq b\}$ The half-open intervals $(a, b]$ and $[a, b)$ are similarly defined, using the inequalities $a < x \leq b$ and $a \leq x < b$, respectively Infinite intervals are defined as follows·

$$(a, +\infty) = \{x \mid a < x\}, \qquad [a, +\infty) = \{x \mid a \leq x\},$$

$$(-\infty, a) = \{x \mid x < a\}, \qquad (-\infty, a] = \{x \mid x \leq a\}.$$

The real line E_1 is sometimes referred to as the open interval $(-\infty, +\infty)$ A single point is also considered as a "degenerate" closed interval

NOTE. The symbols $+\infty$ and $-\infty$ are used here purely for convenience in notation and are not to be considered as being real numbers Later we shall extend the real number system to include these two symbols, but until this is done, the reader should understand that all real numbers are "finite" and satisfy the axioms listed in Chapter 1

We now introduce the following definitions:

3-2 DEFINITION. *Let $h > 0$, and let x be a given point. The open interval $(x - h, x + h)$ is called a neighborhood with x as center and of radius h. We denote the neighborhood by $N(x; h)$, or simply by $N(x)$ if the radius is unimportant.*

3-3 DEFINITION. *Let S be a set in E_1 and assume $x \varepsilon S$. Then x is called an interior point of S if there is some neighborhood $N(x)$ all of whose points belong to S.*

3-4 DEFINITION. *Let S be a set in E_1. Then S is called an open set if every point of S is an interior point of S.*

Thus an open set is such that each of its points can be enclosed in a neighborhood which is completely contained in the set. The simplest kind of an open set is an open interval. The empty set is also open [Why?], as is the real line E_1.

The next two theorems show how additional open sets can be constructed from given open sets.

3-5 THEOREM. *The union of any collection of open sets is an open set.*

Proof. Let F be a collection of open sets and let S denote the union $S = \bigcup_{A \varepsilon F} A$. Assume $x \varepsilon S$. Then x must belong to at least one of the sets in F, say $x \varepsilon A$. Since A is open, there exists a neighborhood $N(x) \subset A$. But $A \subset S$, so that $N(x) \subset S$ and hence x is an interior point of S. This means that S is open.

3-6 THEOREM. *The intersection of a finite collection of open sets is open.*

Proof. Let $S = \bigcap_{k=1}^{n} A_k$ where each A_k is open. Assume $x \varepsilon S$. (If S is empty, there is nothing to prove.) Then $x \varepsilon A_k$ for every $k = 1, 2, \ldots, n$ and hence there is a neighborhood $N(x; r_k) \subset A_k$. Let r be the smallest of the positive numbers $r_1, r_2, \ldots, r_n$. Then $x \varepsilon N(x; r) \subset S$. That is, S is open.

Thus we see that from given open sets, new open sets can be formed by taking arbitrary unions or finite intersections. Arbitrary intersections, on the other hand, will not always lead to open sets. For example, the intersection of all open intervals of the form $(-1/n, 1/n)$, $n = 1, 2, 3, \ldots$ is the set consisting of 0 alone.

The union of a countable collection of disjoint open intervals is an open set and, remarkably enough, every open set on the real line can be obtained in this way. The next section is devoted to a proof of this statement.

3-3 The structure of open sets in E_1

3-7 Definition *A set of points in E_1 is said to be bounded if it is a subset of some finite interval*

3-8 Definition *Let S be an open set in E_1 and let (a, b) be an open interval which is contained in S, but whose endpoints are not in S Then (a, b) is called a component interval of S*

3-9 Theorem *Let S be a bounded open set Then we have.*

(i) *Each point of S belongs to a uniquely determined component interval of S*

(ii) *The component intervals of S form a countable collection of disjoint sets whose union is S*

Proof Assume $x \in S$ Then x is contained in some open interval I which consists entirely of points of S There are many such intervals I, but the "largest" of these will be the desired component interval To show that such a largest interval exists, we proceed as follows Let a be the inf of the left endpoints of all open intervals $I \subset S$ which contain x, and let b be the sup of the right endpoints of these intervals (The boundedness of S is used here) Then (a, b) contains all the open intervals $I \subset S$ which contain x, and hence (a, b) contains x Furthermore, from the way (a, b) was constructed, it is easy to see that $(a, b) \subset S$ [Why?] The endpoint b, however, cannot belong to S, because then for some neighborhood $N(b)$ the interval $(a, b) \cup N(b)$ would be contained in S and would contain x, thus contradicting the definition of b Similarly, the endpoint $a \not\in S$. Therefore (a, b) is a component interval of S which contains x Thus, with each x in S we have associated at least one component interval I_x If two of these intervals, I_x and I_y, have interior points in common, they must be identical, since their endpoints do not belong to S This proves (i)

The union of all such intervals I_x is clearly S To prove (ii) it remains to show that they form a countable collection For this purpose, let $\{x_1, x_2, x_3, \quad \}$ denote the countable set of rational numbers In each component interval I_x there will be infinitely many x_n, but among these there will be exactly one *with smallest index n* We then define a function F by means of the equation $F(I_x) = n$, if x_n is the rational number in I_x with smallest index n This function F is one-to-one since $F(I_x) = F(I_y) = n$ implies that I_x and I_y have x_n in common and this implies $I_x = I_y$ Therefore F establishes a one-to-one correspondence between the intervals I_x and a subset of the positive integers This proves (ii) (Compare this part of the proof with Exercise 2-14)

Note It follows from Theorem 3-9 that an open interval in E_1 cannot be expressed as the union of two disjoint open sets when neither set is empty. [Why?]

3–10 THEOREM. *Every open set in E_1 is the union of a countable collection of disjoint open intervals.*

Proof. Let S be the given open set and let J_n denote the interval $(-n, n)$. Then the set $S_n = S \cap J_n$ is a bounded open set and $S = \bigcup_{k=1}^{\infty} S_k$. Since each S_n is the union of a countable collection of disjoint open intervals, the same is true of S.

3–4 Accumulation points and the Bolzano-Weierstrass theorem in E_1. One of the fundamental concepts in the theory of point sets is the notion of an accumulation point of a set.

3–11 DEFINITION. *Let S be a set in E_1 and x a point in E_1, x not necessarily in S. Then x is called an accumulation point of S, provided every neighborhood of x contains at least one point of S distinct from x.*

3–12 THEOREM. *If x is an accumulation point of S, then every neighborhood $N(x)$ contains infinitely many points of S.*

Proof. Assume the contrary; that is, suppose a neighborhood $N(x)$ exists which contains only a finite number of points of S distinct from x, say $x_1, x_2, \ldots, x_n$. If r denotes the smallest of the positive numbers $|x - x_1|, |x - x_2|, \ldots, |x - x_n|$, then $N(x; r/2)$ will be a neighborhood of x which contains no points of S distinct from x. This is a contradiction.

This theorem implies, in particular, that a set cannot have an accumulation point unless it contains infinitely many points to begin with. The converse, however, is not true. The set of integers $\{1, 2, 3, \ldots\}$ is an example of an infinite set with no accumulation points. But if an infinite set is *bounded*, then it must have an accumulation point. This is a very important result in analysis known as the *Bolzano-Weierstrass theorem.*

3–13 THEOREM (*Bolzano-Weierstrass*). *If a bounded set S in E_1 contains infinitely many points, then there is at least one point in E_1 which is an accumulation point of S.*

Proof. Let S be contained in the finite interval $[a, b]$. Subdivide $[a, b]$ into two equal subintervals. At least one of these subintervals contains an infinite subset of S. Call one such subinterval $[a_1, b_1]$. Bisect $[a_1, b_1]$ and obtain a subinterval $[a_2, b_2]$ containing an infinite subset of S, and continue this process. In this way a countable collection of intervals is obtained, the nth interval $[a_n, b_n]$ being of length $b_n - a_n = (b - a)/2^n$. Clearly, the sup of the left endpoints a_n and the inf of the right endpoints b_n must be equal, say to x. [Why are they equal?] The point x will be an accumulation point of S because, if r is any positive number, the interval $[a_n, b_n]$ will be contained in $N(x; r)$ as soon as n is large enough so that $b_n - a_n < r/2$. The neighborhood $N(x; r)$ contains a point of S distinct from x and hence x is an accumulation point of S. This proves the theorem. (Observe that the accumulation point x may or may not belong to S)

EXAMPLES

1. The set of numbers of the form $1/n$, $n = 1, 2, 3,$, has 0 as an accumulation point

2 The set of rational numbers has every real number as an accumulation point

3 Every point of the closed interval $[a, b]$ is an accumulation point of the set of numbers in the open interval (a, b)

3-5 Closed sets in E_1.

3-14 DEFINITION *A set is called closed if it contains all its accumulation points*

Thus a closed interval is a closed set An open interval, however, is not a closed set because it does not contain its endpoints, both of which are accumulation points of the set Of course, by Definition 3-14 a set which has no accumulation points is automatically closed and, in particular, every finite set is closed The empty set is also closed, as is the whole real line E_1 (These last two sets are also open!) A set which is not closed need not be open, as for example, the half-open interval $(a, b]$, which is neither open nor closed

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The next theorem expresses a relation between open and closed sets.

3-15 THEOREM *If S is open, then the complement $E_1 - S$ is closed Conversely, if S is closed, then $E_1 - S$ is open*

Proof Assume that S is open and let x be an accumulation point of $E_1 - S$ If $x \not\in E_1 - S$, then $x \in S$ and hence some neighborhood $N(x) \subset S$ But, being a subset of S, this neighborhood can contain no points of $E_1 - S$, contradicting the fact that x is an accumulation point of $E_1 - S$ Therefore, $x \in E_1 - S$ and $E_1 - S$ is closed

Next, assume that S is closed and let $x \in E_1 - S$ Then $x \not\in S$ and hence x cannot be an accumulation point of S, since S is closed Therefore some neighborhood $N(x)$ has no points of S and must consist only of points of $E_1 - S$ That is, $N(x) \subset E_1 - S$ and $E_1 - S$ must therefore be open

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As a consequence of Theorems 3-15, 3-6, and 3-5, we obtain

3-16 THEOREM. *The union of a finite collection of closed sets is closed and the intersection of an arbitrary collection of closed sets is closed.*

Theorem 3–15 can be generalized as follows:

3–17 THEOREM. *If S is closed, then the complement of S (relative to any open set containing S) is open. If S is open, then the complement of S (relative to any closed set containing S) is closed.*

Proof. Assume $S \subset A$. Then $A - S = E_1 - [S \cup (E_1 - A)]$. (The reader should verify this equation.) If S is closed and A is open, then $E_1 - A$ is closed, $S \cup (E_1 - A)$ is closed, $A - S$ is open. The converse is similarly proved.

3–6 Extensions to higher dimensions. Most of what we have discussed in this chapter can be generalized to the complex plane. The starting point of the foregoing discussion in E_1 was the concept of a neighborhood. A neighborhood of a point x_0 in E_1 was defined to be an open interval of the form $(x_0 - h, x_0 + h)$. However, such an interval can also be described by the inequality $|x - x_0| < h$. If x and x_0 are complex numbers, this inequality still makes sense and, in fact, describes the interior of a circle with center at x_0 and radius h. Such a circle is called a two-dimensional neighborhood. Having the notion of neighborhood, we can define open and closed sets and accumulation points in E_2 exactly as they were defined earlier for E_1. Moreover, there is no reason why we should restrict ourselves to E_2, since no additional effort is required to extend all these ideas to three-dimensional space and higher.

Recall that a "point" in two-dimensional space was defined to be an ordered pair of real numbers (x_1, x_2). Similarly, a point in three-dimensional space is an ordered triple of real numbers (x_1, x_2, x_3). It is just as easy to consider an ordered set of n real numbers $(x_1, x_2, \ldots, x_n)$ and refer to this as a point in n-dimensional space.

3–18 DEFINITION. *Let $n > 0$ be an integer. An ordered set of n real numbers $(x_1, x_2, \ldots, x_n)$ is called an n-dimensional point or a vector with n components. Points or vectors will be denoted by single boldfaced letters; for example, $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ or $\mathbf{y} = (y_1, y_2, \ldots, y_n)$. The number x_k is called the* kth *coordinate of the point* $\mathbf{x}$ *or the* kth *component of the vector* $\mathbf{x}$. *The set of all n-dimensional points is called n-dimensional Euclidean space and is denoted by E_n.*

The reader may wonder whether there is any advantage in discussing spaces of dimension greater than three. Actually, the language of n-dimensional space makes many complicated situations much easier to comprehend. The reader is probably familiar enough with three-dimensional vector analysis to realize the advantage of writing the equations of motion of a system having three degrees of freedom as a single vector equation rather than as three scalar equations. There is a similar advantage if the system has n degrees of freedom Higher dimensional spaces arise quite naturally in such fields as relativity, and statistical and quantum mechanics In fact, even *infinite*-dimensional spaces are quite common in quantum mechanics

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We shall now define algebraic operations on n-dimensional points

3-19 DEFINITION *Let* $\mathbf{x} = (x_1, \ldots, x_n)$ *and* $\mathbf{y} = (y_1, \ldots, y_n)$ *be in* E_n *We define:*

(a) *Equality*:

$$\mathbf{x} = \mathbf{y} \textit{ if, and only if, } x_1 = y_1, \ldots, x_n = y_n$$

(b) *Sum.*

$$\mathbf{x} + \mathbf{y} = (x_1 + y_1, \ldots, x_n + y_n)$$

(c) *Multiplication by scalars (scalar = a real number).*

$$a\mathbf{x} = (ax_1, \ldots, ax_n) \qquad (a \text{ real})$$

(d) *Difference*

$$\mathbf{x} - \mathbf{y} = \mathbf{x} + (-1)\mathbf{y}$$

(e) *Zero vector or origin*

$$\mathbf{0} = (0, \ldots, 0)$$

(f) *Unit coordinate vectors*:

$$\mathbf{u}_k = (\delta_{k,1}, \delta_{k,2}, \ldots, \delta_{k,n}) \qquad (k = 1, 2, \ldots, n),$$

where $\delta_{k,j}$ *is the "Kronecker delta," defined by* $\delta_{k,j} = 0$, *if* $k \neq j$, $\delta_{k,k} = 1$

NOTE If $\mathbf{x} = (x_1, \ldots, x_n)$, then $\mathbf{x} = x_1\mathbf{u}_1 + \cdots + x_n\mathbf{u}_n$ and, because of (a), this representation is unique

The basic idea which really enabled us to generalize the concept of neighborhood from E_1 to E_2 was the notion of absolute value This can now be extended to higher dimensions in such a way that its fundamental properties will be preserved

3-20 DEFINITION *Let* $\mathbf{x} = (x_1, \ldots, x_n)$ *and* $\mathbf{y} = (y_1, \ldots, y_n)$ *be in* E_n *We define*:

(a) *The absolute value, or length, or norm of* $\mathbf{x}$

$$|\mathbf{x}| = \sqrt{x_1^2 + \cdots + x_n^2}.$$

(b) *The distance between* $\mathbf{x}$ *and* $\mathbf{y}$:

$$|\mathbf{x} - \mathbf{y}| = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}.$$

It is now easy to establish the essential properties of absolute value.

3–21 THEOREM. *Let* $\mathbf{x}$ *and* $\mathbf{y}$ *denote points in* E_n. *Then we have*

(a) $|\mathbf{x}| \geq 0$, *and* $|\mathbf{x}| = 0$ *if, and only if,* $\mathbf{x} = \mathbf{0}$.
(b) $|\mathbf{x} - \mathbf{y}| = |\mathbf{y} - \mathbf{x}|$.
(c) $|\mathbf{x} + \mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|$.

Proof. Statements (a) and (b) are immediate from the definition. To prove (c) we make use of the Cauchy-Schwarz inequality (Theorem 1–5), which can now be written as

$$\left(\sum_{k=1}^{n} x_k y_k\right)^2 \leq |\mathbf{x}|^2 |\mathbf{y}|^2.$$

Since we have

$$\begin{aligned} |\mathbf{x} + \mathbf{y}|^2 &= \sum_{k=1}^{n} (x_k + y_k)^2 = \sum_{k=1}^{n} (x_k^2 + 2x_k y_k + y_k^2) \\ &= |\mathbf{x}|^2 + |\mathbf{y}|^2 + 2\sum_{k=1}^{n} x_k y_k \\ &\leq |\mathbf{x}|^2 + |\mathbf{y}|^2 + 2|\mathbf{x}||\mathbf{y}| = (|\mathbf{x}| + |\mathbf{y}|)^2, \end{aligned}$$

property (c) follows at once.

With the notion of distance, we can now proceed to define what is meant by a neighborhood in E_n.

3–22 DEFINITION. *By an open sphere of radius* $r > 0$ *and center at the point* $\mathbf{x}_0$ *in* E_n, *we mean the set of all* $\mathbf{x}$ *in* E_n *such that* $|\mathbf{x} - \mathbf{x}_0| < r$. *Those* $\mathbf{x}$ *such that* $|\mathbf{x} - \mathbf{x}_0| \leq r$ *form a closed sphere, the boundary of the sphere being the set of* $\mathbf{x}$ *with* $|\mathbf{x} - \mathbf{x}_0| = r$. *An open sphere with center at* $\mathbf{x}_0$ *is called a neighborhood of* $\mathbf{x}_0$ *and is denoted by* $N(\mathbf{x}_0)$ *or by* $N(\mathbf{x}_0; r)$, *if* r *is its radius. The corresponding closed sphere is denoted by* $\overline{N}(\mathbf{x}_0)$. *The open sphere with its center removed is called a deleted neighborhood of* $\mathbf{x}_0$ *and is denoted by* $N'(\mathbf{x}_0)$.

With this generalization of the notion of neighborhood we can proceed to define open and closed sets in E_n exactly as they were defined in E_1. It should be pointed out, however, that this method of defining n-dimensional neighborhoods is only one of many possible definitions which would serve equally well For example, instead of using circles in E_2, spheres in E_3, and n-dimensional spheres in E_n, we could use rectangles in E_2, rectangular parallelepipeds in E_3, and n-dimensional parallelepipeds in E_n These can be defined as follows.

3–23 DEFINITION *Let* $\mathbf{a} = (a_1, \ldots, a_n)$ *and* $\mathbf{b} = (b_1, \ldots, b_n)$ *be two distinct points in* E_n, *such that* $a_k \leq b_k$ *for each* $k = 1, 2, \ldots, n$ *The n-dimensional closed interval* $[\mathbf{a}, \mathbf{b}]$ *is defined to be the set*

$$[\mathbf{a}, \mathbf{b}] = \{(x_1, \ldots, x_n) \mid a_k \leq x_k \leq b_k, k = 1, 2, \ldots, n\}.$$

If $a_k < b_k$ *for every* k, *the n-dimensional open interval* $(\mathbf{a}, \mathbf{b})$ *is the set*

$$(\mathbf{a}, \mathbf{b}) = \{(x_1, \ldots, x_n) \mid a_k < x_k < b_k, k = 1, 2, \ldots, n\}.$$

Thus, for example, $(\mathbf{a}, \mathbf{b})$ can be considered as the "cartesian product"

$$(\mathbf{a}, \mathbf{b}) = (a_1, b_1) \times (a_2, b_2) \times \cdots \times (a_n, b_n)$$

of the n one-dimensional open intervals (a_k, b_k) An open interval in E_n could then be called a neighborhood of any of its points, and in the theorems which follow it really makes no difference whether a neighborhood is taken to be a "sphere" or an "interval" This is because each sphere about a point $\mathbf{x}_0$ includes an interval which contains $\mathbf{x}_0$ and, conversely, each interval containing $\mathbf{x}_0$ contains a sphere about $\mathbf{x}_0$ Nevertheless, in the discussion which follows, we shall use spherical neighborhoods, in accordance with Definition 3–22 In E_2, spheres are called *disks* and intervals are called *rectangles* Although all definitions and theorems which follow are stated without recourse to "intuitive" reasoning, the reader should make an effort to visualize the ideas geometrically whenever possible It is often helpful to illustrate the meaning of certain concepts by drawing a figure in the plane

3–24 DEFINITION *Let* S *be a set of points in* E_n *and assume* $\mathbf{x} \in S$ *Then* $\mathbf{x}$ *is called an interior point of* S *if there exists a neighborhood* $N(\mathbf{x}) \subset S$ *The set* S *is said to be open if each of its points is an interior point The interior of* S *is the collection of its interior points*

Examples of open sets in the plane are the interior of a disk, the first quadrant, that is, the set $\{(x, y) \mid x > 0, y > 0\}$, the whole space Any n-dimensional open sphere and any n-dimensional open interval is an open set in E_n (See Exercise 3–8) The reader should be cautioned that an open interval (a, b) in E_1 is no longer an open set when it is considered as a subset of the plane. In fact, *no* subset of E_1 (except the empty set) can be open in E_2, because such a set can contain no two-dimensional neighborhoods.

3–25 THEOREM. *The union of an arbitrary collection of open sets in E_n is an open set in E_n, and the intersection of a finite collection of open sets in E_n is an open set in E_n.*

Proof. The proofs are exactly the same as stated for the one-dimensional case (Theorems 3–5 and 3–6).

3–26 DEFINITION. *Assume $S \subset E_n$, $\mathbf{x} \in E_n$. Then $\mathbf{x}$ is called an accumulation point of S if every neighborhood $N(\mathbf{x})$ contains at least one point of S distinct from $\mathbf{x}$; that is, if $N'(\mathbf{x}) \cap S$ is not empty.*

3–27 THEOREM. *Assume $S \subset E_n$, $\mathbf{x} \in E_n$. If $\mathbf{x}$ is an accumulation point of S, then every neighborhood $N(\mathbf{x})$ contains infinitely many points of S.*

Proof. The proof is identical to that of Theorem 3–12.

3–28 DEFINITION. *A set S in E_n is said to be bounded if S lies entirely within some sphere; that is, if for some $r > 0$ we have $S \subset N(\mathbf{0}; r)$.*

3–29 THEOREM (*Bolzano-Weierstrass*). *If a bounded set S in E_n contains infinitely many points, then there exists at least one point in E_n which is an accumulation point of S.*

Proof. Since S is bounded, S lies within some n-dimensional sphere $N(\mathbf{0}; a)$, $a > 0$, and therefore within the n-dimensional interval J_1 defined by the inequalities

$$-a \le x_k \le a \qquad (k = 1, 2, \ldots, n).$$

The proof of the theorem is an extension of the ideas used in treating the one-dimensional case. (The reader may find it helpful to visualize the proof in the case of E_2 by referring to Fig. 3–1.) Denote J_1 symbolically as a cartesian product

$$J_1 = I_1^{(1)} \times I_2^{(1)} \times \cdots \times I_n^{(1)};$$

that is, as a set of points $(x_1, \ldots, x_n)$, where $x_k \in I_k^{(1)}$ and where each $I_k^{(1)}$ is a one-dimensional interval $-a \le x_k \le a$. Each interval $I_k^{(1)}$ can be bisected to form two subintervals $I_{k,1}^{(1)}$ and $I_{k,2}^{(1)}$, defined by the inequalities

$$I_{k,1}^{(1)} : -a \le x_k \le 0; \qquad I_{k,2}^{(1)} : 0 \le x_k \le a.$$

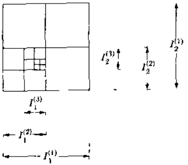

FIG 3-1 Proof of the Bolzano-Weierstrass theorem in E_2.

Next, we consider all possible cartesian products of the form

$$\text{(a)} \qquad I_{1\,k_1}^{(1)} \times I_{2\,k_2}^{(1)} \times \cdots \times I_{n\,k_n}^{(1)},$$

where each $k_i = 1$ or 2 There are exactly 2^n such products and, of course, each such product is an n-dimensional interval The union of these 2^n intervals is the original interval J_1, which contains S, and hence at least one of the 2^n intervals in (a) must contain infinitely many points of S. One of these we denote by J_2, which can then be expressed as

$$J_2 = I_1^{(2)} \times I_2^{(2)} \times \cdots \times I_n^{(2)},$$

where each $I_k^{(2)}$ is one of the subintervals of $I_k^{(1)}$ of length $a/2$. We now proceed with J_2 as we did with J_1, bisecting each interval $I_k^{(2)}$ and arriving at an n-dimensional interval J_3 containing an infinite subset of S. If we continue the process, we obtain a countable collection of n-dimensional intervals $J_1, J_2, J_3, \ldots$, where the mth interval J_m has the property that it contains an infinite subset of S and can be expressed in the form

$$J_m = I_1^{(m)} \times I_2^{(m)} \times \cdots \times I_n^{(m)}, \qquad \text{where } I_k^{(m)} \subset I_k^{(1)}$$

Writing

$$I_k^{(m)} = [a_k^{(m)}, b_k^{(m)}],$$

we have

$$b_k^{(m)} - a_k^{(m)} = \frac{a}{2^{m-1}} \qquad (k = 1, 2, \ldots, n)$$

For each fixed k, the sup of all left endpoints $a_k^{(m)} (m = 1, 2, \ldots)$ must therefore be equal to the inf of all right endpoints $b_k^{(m)} (m = 1, 2, \ldots)$, and their common value we denote by t_k. We now assert that the point $\mathbf{t} = (t_1, t_2, \ldots, t_n)$ is an accumulation point of S. To see this, take any neighborhood $N(\mathbf{t}; r)$. The point $\mathbf{t}$, of course, belongs to each of the intervals $J_1, J_2, \ldots$ constructed above, and when m is such that $a/2^{m-1} < r/2$, this neighborhood will include J_m. But since J_m contains infinitely many points of S, so will $N(\mathbf{t}; r)$, which proves that $\mathbf{t}$ is indeed an accumulation point of S.

3–30 DEFINITION. *A set S in E_n is said to be closed if it contains all its accumulation points.*

EXAMPLES. A closed sphere in E_n is a closed set. An n-dimensional closed interval is a closed set. A set which is closed in E_1 is also closed in E_n for $n > 1$.

3–31 THEOREM. *A set S in E_n is closed if, and only if, $E_n - S$ is open.*

Proof. The proof is identical to that of Theorem 3–15.

3–32 THEOREM. *The union of a finite collection of closed sets in E_n is closed and the intersection of an arbitrary collection of closed sets in E_n is closed.*

Proof. Theorems 3–25 and 3–31.

3–33 THEOREM. *Assume $S \subset A \subset E_n$. If A is open and S is closed, then $A - S$ is open. If A is closed and S is open, then $A - S$ is closed.*

Proof. $A - S = E_n - [S \cup (E_n - A)]$.

3–7 The Heine-Borel covering theorem.

Our purpose in this section is to use the Bolzano-Weierstrass theorem to prove an important result known as the *Heine-Borel covering theorem*. We begin by defining a *covering* of a set.

3–34 DEFINITION. *A collection F of sets is said to be a covering of a given set S if $S \subset \bigcup_{A \in F} A$. The collection F is also said to cover S. If F is a collection of open sets, then F is called an open covering of S.*

EXAMPLES.

1. The collection of all intervals of the form $1/n < x < 2/n$ $(n = 2, 3, 4, \ldots)$ is an open covering of the interval $0 < x < 1$. This is an example of a countable covering.
2. The real axis E_1 is covered by the collection of all open intervals (a, b). This covering is not countable. However, it contains a countable covering of E_1, namely, all intervals of the form $(n, n + 2)$, where n runs through the integers.

3 Let $S = \{(x, y) \mid x > 0, y > 0\}$. The collection F of all circles with centers at (x, x) and with radius x, $x > 0$, is a covering of S. This covering is not countable. However, it contains a countable covering of S, namely, all those circles in which x is rational. (See Exercise 3–14.)

Our first "covering theorem" tells us that every open covering of a set S in E_n contains a countable subcollection which also covers S. We first prove the following preliminary result.

3–35 THEOREM. *Let $G = \{A_1, A_2, \ldots\}$ denote the countable collection of neighborhoods in E_n having rational radii and centers at points with rational coordinates. Assume $\mathbf{x} \in E_n$ and let S be an open set in E_n which contains $\mathbf{x}$. Then at least one of the neighborhoods in G contains $\mathbf{x}$ and is contained in S. That is, we have*

$$\mathbf{x} \in A_k \subset S \qquad \textit{for some } A_k \textit{ in } G$$

Proof. The collection G is countable because of Theorem 2–27. If $\mathbf{x} \in E_n$ and if S is an open set containing $\mathbf{x}$, then there is some neighborhood $N(\mathbf{x}; r) \subset S$. We shall find a point $\mathbf{y}$ in S with rational coordinates that is "near" $\mathbf{x}$ and, using this point as center, will then find a neighborhood in G which lies within $N(\mathbf{x}, r)$ and which contains $\mathbf{x}$. Writing

$$\mathbf{x} = (x_1, x_2, \ldots, x_n),$$

find a rational number y_k such that $|y_k - x_k| < r/(4n)$ for each $k = 1, 2, \ldots, n$. Then

$$|\mathbf{y} - \mathbf{x}| \leq |y_1 - x_1| + \cdots + |y_n - x_n| < \frac{r}{4}.$$

Next, find a rational number q such that $r/4 < q < r/2$. Then $\mathbf{x} \in N(\mathbf{y}; q)$ and $N(\mathbf{y}, q) \subset N(\mathbf{x}, r) \subset S$. But $N(\mathbf{y}, q) \in G$ and hence the theorem is proved. (See Fig. 3–2 for the situation in E_2.)

3–36 THEOREM. *Assume $A \subset E_n$ and let F be an open covering of A. Then there is a countable subcollection of F which also covers A.*

Proof. Let $G = \{A_1, A_2, \ldots\}$ denote the countable collection of all neighborhoods in E_n having rational centers and rational radii. This set G will be used to help us extract a countable subcollection of F which covers A.

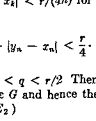

FIG. 3-2. Theorem 3–35 in E_2.

Assume $\mathbf{x} \in A$. Then there is an open set S in F such that $\mathbf{x} \in S$. By Theorem 3–35 there is a neighborhood A_k in G such that $\mathbf{x} \in A_k \subset S$. There are, of course, infinitely many such A_k corresponding to each S, but we choose only one of these, for example, the one of smallest index, say $m = m(\mathbf{x})$. Then we have $\mathbf{x} \in A_{m(\mathbf{x})} \subset S$. The set of all neighborhoods $A_{m(\mathbf{x})}$ obtained as $\mathbf{x}$ varies over all elements of A is a countable collection of open sets which covers A. To get a countable subcollection of F which covers A, we simply correlate to each set $A_{k(\mathbf{x})}$ one of the sets S of F which contained $A_{k(\mathbf{x})}$. This completes the proof.

One more result is needed as a preliminary to the Heine-Borel theorem. It is here that the Bolzano-Weierstrass theorem is used.

3–37 THEOREM. *Let $\{Q_1, Q_2, \ldots\}$ be a countable collection of nonempty sets in E_n such that:*

(i) $Q_{k+1} \subset Q_k \qquad (k = 1, 2, 3, \ldots)$.

(ii) *Each set Q_k is closed and Q_1 is bounded.*

Then the intersection $\bigcap_{k=1}^{\infty} Q_k$ is closed and nonempty.

Proof. Let $S = \bigcap_{k=1}^{\infty} Q_k$. Then S is closed because of Theorem 3–32. To show that S is nonempty we exhibit a point $\mathbf{x}$ in S. We can assume that each Q_k contains infinitely many points; otherwise the proof is trivial. Now form a collection of distinct points $A = \{\mathbf{x}_1, \mathbf{x}_2, \ldots\}$ where $\mathbf{x}_k \in Q_k$. Since A is an infinite set contained in the bounded set Q_1, it has an accumulation point, say $\mathbf{x}$. We shall show that $\mathbf{x} \in S$ by verifying that $\mathbf{x} \in Q_k$ for each k. It will suffice to show that $\mathbf{x}$ is an accumulation point of each Q_k, since they are all closed sets. But every neighborhood of $\mathbf{x}$ contains infinitely many points of A, and since all except (possibly) a finite number of the points of A belong to Q_k, this neighborhood also contains infinitely many points of Q_k. Therefore $\mathbf{x}$ is an accumulation point of Q_k and the theorem is proved.

Theorem 3–36 states that from any open covering of an arbitrary set A in E_n we can extract a countable covering. The Heine-Borel theorem tells us that if, in addition, we know that A is *closed* and *bounded*, we can reduce the covering to a *finite* covering.

3–38 THEOREM (*Heine-Borel*). *Let F be an open covering of a closed and bounded set A in E_n. Then a finite subcollection of F also covers A.*

Proof A countable subcollection of F, say $\{I_1, I_2, \quad\}$, covers A by Theorem 3-36 Consider, for $m \geq 1$, the finite union

$$S_m = \bigcup_{k=1}^{m} I_k$$

This is open, since it is the union of open sets We shall show that for some value of m the union S_m covers A

For this purpose we consider the complement $E_n - S_m$, which is closed Define a countable collection of sets $\{Q_1, Q_2, \quad\}$ as follows $Q_1 = A$, and for $m > 1$, $Q_m = A \cap (E_n - S_m)$ That is, Q_m consists of those points of A which lie outside of S_m If we can show that for some value of m the set Q_m is empty, then we will have shown that for this m no point of A lies outside S_m, in other words, we will have shown that some S_m covers A.

Observe the following properties of the sets Q_m Each set Q_m is closed, since it is the intersection of the closed set A and the closed set $E_n - S_m$ The sets Q_m are decreasing, since the S_m are increasing, that is, $Q_{m+1} \subset Q_m$ The sets Q_m, being subsets of A, are all bounded Therefore, if no set Q_m is empty, we can apply Theorem 3-37 to conclude that their intersection $\bigcap_{k=1}^{\infty} Q_k$ is also not empty This means that there is some point in A which is in all the sets Q_m, or, what is the same thing, outside all the sets S_m But this is impossible, since $A \subset \bigcup_{k=1}^{\infty} S_k$ Therefore some Q_m must be empty, and this completes the proof

---

The Heine-Borel theorem will prove to be a useful tool in many of the later chapters The basic idea contained in the theorem appeared in a paper by E Heine (*Crelles Journal* **74**, 1872) in which he proved that every function which is continuous on a closed interval is uniformly continuous on the interval (See Theorem 4-24) In 1895 E. Borel (*Ann de l'Ecole Normal*, (3), **12**) explicitly stated and proved Theorem 3-38 for the case of a closed interval in E_1 covered by a countable set of open intervals Lebesgue later extended the result and some authors refer to Theorem 3-38 as the Borel-Lebesgue theorem E Lindelof proved Theorem 3-36 somewhat later (*Comptes Rendus*, **137**, 1903) W H Young (*Proc London Math Soc*, **35** 1902) had discovered a result essentially equivalent to Theorem 3-36 (for E_1) before Lindelof s paper appeared, but the theorem seems to have been first stated explicitly by Lindelof and it is now usually referred to as the Lindelof covering theorem

3-8 Compactness. We have just seen that if a set S in E_n is closed and bounded, then any open covering of S can be reduced to a finite covering It is natural to inquire whether there might be sets other than closed and bounded sets which also have this property Such sets will be called *compact*

3–39 DEFINITION. *A set S in E_n is said to be compact if, and only if, every open covering of S contains a finite subcollection which also covers S.*

The Heine-Borel theorem states that every closed and bounded set in E_n is compact. We now prove the converse result.

3–40 THEOREM. *Let S be a compact set in E_n. Then we have:*

(i) *Every infinite subset of S has an accumulation point in S.*
(ii) *The set S is closed and bounded.*

Proof. If S is finite, both statements hold trivially. Let S be infinite and assume that (i) does not hold. Then S has an infinite subset T such that if $\mathbf{x} \in S$, then $\mathbf{x}$ is not an accumulation point of T. Therefore, for each $\mathbf{x}$ in S there exists a neighborhood $N(\mathbf{x}; r_{\mathbf{x}})$ such that $N'(\mathbf{x}; r_{\mathbf{x}}) \cap T$ is empty. The collection

$$F = \{N(\mathbf{x}; r_{\mathbf{x}}/2) \mid N'(\mathbf{x}; r_{\mathbf{x}}) \cap T \text{ is empty}\}$$

is an open covering of S. Since S is compact, a finite number of these neighborhoods cover S, say

$$S \subset N(\mathbf{x}_1; r_1/2) \cup \cdots \cup N(\mathbf{x}_p; r_p/2),$$

where $N'(\mathbf{x}_k; r_k) \cap T$ is empty for each $k = 1, 2, \ldots, p$. Consider the set $\overline{N}$ obtained by taking the union of the closed spheres

$$\overline{N} = \overline{N}(\mathbf{x}_1; r_1/2) \cup \cdots \cup \overline{N}(\mathbf{x}_p; r_p/2).$$

This is a closed set and we have $T \subset S \subset \overline{N}$. Hence T is a bounded set (since $\overline{N}$ is obviously bounded) and, since T is infinite, it must have an accumulation point, say $\mathbf{x}_0$. The point $\mathbf{x}_0$ is also an accumulation point of $\overline{N}$ and hence $\mathbf{x}_0 \in \overline{N}$, since $\overline{N}$ is closed. This means that

$$\mathbf{x}_0 \in \overline{N}(\mathbf{x}_k; r_k/2) \subset N(\mathbf{x}_k; r_k)$$

for some k, $1 \leq k \leq p$. Therefore the neighborhood

$$N(\mathbf{x}_0; r_k/4) \subset N(\mathbf{x}_k; r_k).$$

(See Fig. 3–3 for the case of E_2.) But every neighborhood of $\mathbf{x}_0$ must contain infinitely many points of T, and this contradicts the fact that $N'(\mathbf{x}_k; r_k) \cap T$ is empty. Thus part (i) of the theorem is proved.

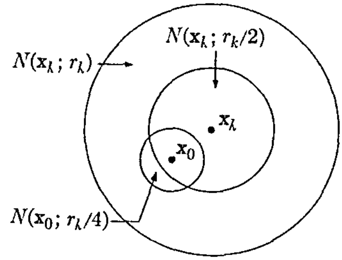

FIG. 3–3. Theorem 3–40 in E_2.

In the course of the proof of (i) we have also shown that S is bounded. To complete the proof of (ii) we must show that S is closed. Let $\mathbf{x}$ be an accumulation point of S. Since every neighborhood of $\mathbf{x}$ contains infinitely many points of S, we can consider the neighborhoods $N(\mathbf{x}, 1/k)$, $k = 1, 2, \ldots$, and obtain a countable set of distinct points, say $T = \{\mathbf{x}_1, \mathbf{x}_2, \ldots\}$, contained in S, such that $\mathbf{x}_k \in N(\mathbf{x}; 1/k)$. The point $\mathbf{x}$ is also an accumulation point of T. Since T is an infinite subset of S, part (i) of the theorem tells us that T must have an accumulation point in S. The theorem will then be proved if we show that $\mathbf{x}$ is the only accumulation point of T.

To do this, let $\mathbf{y} \neq \mathbf{x}$. Then we have

$$|\mathbf{y} - \mathbf{x}| \leq |\mathbf{y} - \mathbf{x}_k| + |\mathbf{x}_k - \mathbf{x}| < |\mathbf{y} - \mathbf{x}_k| + 1/k, \qquad \text{if } \mathbf{x}_k \in T$$

If k_0 is taken so large that $1/k < \frac{1}{2}|\mathbf{y} - \mathbf{x}|$ whenever $k \geq k_0$, the last inequality leads to $\frac{1}{2}|\mathbf{y} - \mathbf{x}| < |\mathbf{y} - \mathbf{x}_k|$. This shows that $\mathbf{x}_k \notin N(\mathbf{y}; r)$ when $k \geq k_0$, if $r = \frac{1}{2}|\mathbf{y} - \mathbf{x}|$. Hence $\mathbf{y}$ cannot be an accumulation point of T. This completes the proof of (ii).

3–9 Infinity in the real number system. We now adjoin to the real number system two "ideal points," denoted by the symbols $+\infty$ and $-\infty$ ("plus infinity" and "minus infinity").

3–41 DEFINITION *By the extended real number system E_1^* we shall mean the union of the set of real numbers E_1 with two symbols $+\infty$ and $-\infty$ which satisfy the following properties:*

(i) *If $x \in E_1$, then we have*
$$x + (+\infty) = +\infty, \quad x + (-\infty) = -\infty,$$
$$x - (+\infty) = -\infty, \quad x - (-\infty) = +\infty,$$
$$x/(+\infty) = x/(-\infty) = 0$$

(ii) *If $x > 0$, then we have*
$$x(+\infty) = +\infty, \quad x(-\infty) = -\infty$$

(iii) *If $x < 0$, then we have*
$$x(+\infty) = -\infty, \quad x(-\infty) = +\infty$$

(iv) $(+\infty) + (+\infty) = (+\infty)(+\infty) = (-\infty)(-\infty) = +\infty,$
$(-\infty) + (-\infty) = (+\infty)(-\infty) = -\infty$

(v) *If $x \in E_1$, then we have $-\infty < x < +\infty$.*

3–42 DEFINITION. *Every open interval $(a, +\infty)$ is called a neighborhood of $+\infty$ and every open interval $(-\infty, a)$ is called a neighborhood of $-\infty$.*

NOTATION. We denote E_1 by $(-\infty, +\infty)$ and E_1^* by $[-\infty, +\infty]$.

The principal reason for introducing the symbols $+\infty$ and $-\infty$ is one of convenience. For example, if we define $+\infty$ to be the sup of any set in E_1^* which is not bounded above, then *every* set in E_1^* has a sup. The sup is finite if the set is bounded above and $+\infty$ if it is not bounded above.

Warning: E_1^* does not satisfy all the axioms for the real number system listed in Chapter 1.

3–10 Infinity in the complex plane.

We shall now adjoin to the complex number system one ideal point denoted by the symbol ∞.

3–43 DEFINITION. *By the extended complex number system E_2^* we shall mean the union of the complex plane E_2 with a symbol ∞ which satisfies the following properties:*

(i) *If $z \in E_2$, then we have $z + \infty = z - \infty = \infty$, $z/\infty = 0$.*

(ii) *If $z \in E_2$, but $z \neq 0$, then $z(\infty) = \infty$ and $z/0 = \infty$.*

(iii) $\infty + \infty = (\infty)(\infty) = \infty$.

3–44 DEFINITION. *Every open set in E_2 of the form $\{z \mid |z| > r \geq 0\}$ is called a neighborhood of ∞.*

The reader may wonder why *two* symbols, $+\infty$ and $-\infty$, are adjoined to E_1 but only *one* symbol, ∞, is adjoined to E_2. The answer lies in the fact that there is an ordering relation $<$ among the real numbers, but no such relation occurs among the complex numbers. In order that certain properties of real numbers involving the relation $<$ hold *without exception*, we need two symbols, $+\infty$ and $-\infty$, as defined above. We have already mentioned that in E_1^* *every* set has a sup, for example.

In E_2 it turns out to be more convenient to have just one ideal point. By way of illustration, let us recall the stereographic projection which establishes a one-to-one correspondence between the points of the complex plane and those points on the surface of the sphere distinct from the North Pole. The apparent exception at the North Pole can be removed by regarding it as the geometric representative of the ideal point ∞. We then get a one-to-one correspondence between the extended complex plane E_2^* and the total surface of the sphere. It is geometrically evident that if the South Pole is placed on the origin of the complex plane, the exterior of a "large" circle in the plane will correspond, by stereographic projection, to a "small" spherical cap about the North Pole. This illustrates vividly why we have defined a neighborhood of ∞ by an inequality of the form $|z| > r$.

Open and closed sets in E_1 and E_2

3–1. Prove that an open interval in E_1 is an open set and that a closed interval is a closed set

3–2. Determine all the accumulation points of the following sets in E_1 and decide whether the sets are open or closed (or neither)

(a) All integers
(b) The interval $(a, b]$
(c) All numbers of the form $1/n$, $(n = 1, 2, 3, \ldots)$
(d) All rational numbers
(e) All numbers of the form $2^{-n} + 5^{-m}$, $(m, n = 1, 2, \ldots)$
(f) All numbers of the form $(-1)^n + (1/m)$, $(m, n = 1, 2, \ldots)$
(g) All numbers of the form $(1/n) + (1/m)$, $(m, n = 1, 2, \ldots)$.
(h) All numbers of the form $(-1)^n/[1 + (1/n)]$, $(n = 1, 2, \ldots)$

3–3. The same as Exercise 3–2 for the following sets in E_2

(a) All complex z such that $|z| > 1$
(b) All complex z such that $|z| \geq 1$.
(c) All complex numbers of the form $(1/n) + (i/m)$, $(m, n = 1, 2, \ldots)$
(d) All points (x, y) such that $x^2 - y^2 < 1$
(e) All points (x, y) such that $x > 0$
(f) All points (x, y) such that $x \geq 0$

3–4. Show that the set of rational numbers in the interval $(0, 1)$ cannot be expressed as the intersection of a countable collection of open sets

3–5. Prove that the only sets in E_1 which are both open and closed are the empty set and E_1 itself Is a similar statement true for E_2?

3–6. Show that every closed set in E_1 is the intersection of a countable collection of open sets

3–7. Show that a nonempty, bounded closed set S in E_1 is either a closed interval, or that S can be obtained from a closed interval by removing a countable disjoint collection of open intervals whose endpoints belong to S.

Open and closed sets in E_n

3–8. Show that n-dimensional open spheres and n-dimensional open intervals are open sets in E_n

3–9. Show that the interior of a set S in E_n is an open set.

3–10. If $\mathbf{x} \in S$ but if $\mathbf{x}$ is not an accumulation point of S, then $\mathbf{x}$ is called an *isolated point* of S. Show that the collection of isolated points of S is countable.

3–11. A set S in E_n is called *convex* if, for every $\mathbf{x}$ in S, $\mathbf{y}$ in S, and for each real θ satisfying $0 < \theta < 1$, we have $\theta\mathbf{x} + (1 - \theta)\mathbf{y} \in S$. Interpret this result geometrically (in E_2 and E_3) and show that both open and closed spheres in E_n are convex.

3–12. If S is a set in E_n, let S' denote the set of accumulation points of S, and let $\bar{S} = S \cup S'$. (The set S' is called the *derived set* of S and $\bar{S}$ is called the *closure* of S.) Show that:

(a) S' is a closed set, that is, $(S')' \subset S'$.
(b) If $S \subset T$, then $S' \subset T'$.
(c) $(S \cup T)' = S' \cup T'$.
(d) $(\bar{S})' = S'$.
(e) $\bar{S}$ is a closed set.
(f) S is closed if, and only if, $S = \bar{S}$.

3–13. Let $\{Q_1, Q_2, \ldots\}$ be a collection of closed and bounded sets in E_n. Define $S_m = \bigcap_{k=1}^{m} Q_k$. Find a condition on the sets $S_1, S_2, \ldots$, which is necessary and sufficient for the intersection $\bigcap_{k=1}^{\infty} Q_k$ to be nonempty.

Covering theorems.

3–14. Show that the set of circles in the xy-plane with center at (x, x) and radius $x > 0$, x rational, is a countable covering of the set

$$\{(x, y) \mid x > 0, y > 0\}.$$

3–15. The collection F of intervals of the form

$$\frac{1}{n} < x < \frac{2}{n} \qquad (n = 2, 3, 4, \ldots),$$

is an open covering of the interval $0 < x < 1$. Show (without using Theorem 3–40) that no finite subcollection of F covers the interval $0 < x < 1$.

3–16. Give an example of a set S which is closed but not bounded and exhibit a countable open covering F such that no finite subset of F covers S.

3–17. Given a set S in E_n with the property that for every $\mathbf{x}$ in S there exists a neighborhood $N(\mathbf{x})$ such that $N(\mathbf{x}) \cap S$ is countable. Show that S is countable.

·**3–18.** Show that a collection of disjoint open sets in E_n is necessarily countable. Give an example of a collection of disjoint closed sets which is not countable.

3–19. Assume that $S \subset E_n$. A point $\mathbf{x}$ in E_n is said to be a *condensation point* of S if every neighborhood $N(\mathbf{x})$ has the property that $N(\mathbf{x}) \cap S$ is not countable. Show that if S is not countable, then there exists a point $\mathbf{x}$ *in* S such that $\mathbf{x}$ is a condensation point of S.

3-20. Assume that $S \subset E_n$ and assume that S is not countable. Let T denote the set of condensation points of S. Show that

(a) $S - T$ is countable, (b) $S \cap T$ is not countable,
(c) T is a closed set, (d) T contains no isolated points.

Note that Exercise 3-19 is a very special case of (b).

3-21. A set S in E_n is said to be *perfect* if $S = S'$, that is, if S is a closed set which contains no isolated points. Show that if F is an uncountable closed set in E_n, then F can be expressed in the form $F = A \cup B$, where A is a perfect set and B is a countable set (Cantor-Bendixon theorem). [*Hint*. Use Exercise 3-20.]

Suggested References

3-1 Burkill, J. C., *The Lebesgue Integral*. Cambridge University Press, 1951.

3-2 Newman, M. H. A., *Elements of the Topology of Plane Sets of Points*. 2nd ed. Cambridge University Press, 1951.

3-3 Rudin, W., *Principles of Mathematical Analysis*. New York: McGraw-Hill, 1953.

CHAPTER 4

THE LIMIT CONCEPT AND CONTINUITY

4–1 The definition of limit. The reader is already familiar with the notion of limit as introduced in elementary calculus, where, in fact, several kinds of limits are usually presented. For example, we have the idea of the *limit of a sequence,* which describes what we write symbolically in the form

$$\lim_{n\to\infty} x_n = A.$$

There is also the *limit of a function,* indicated by notation such as

$$\lim_{x\to x_0} f(x) = A.$$

Two-dimensional limits, introduced for functions of two variables, are usually denoted by expressions of the form

$$\lim_{\substack{x\to x_0\\ y\to y_0}} f(x, y) = A,$$

or by similar expressions. Each of these limit concepts is given a separate definition and each has associated with it its own body of theorems. Nevertheless, the same underlying idea exists in all these limits and if we analyze the separate definitions carefully, we can formulate a general concept of limit which will include all the foregoing limits as special cases.

Let us begin with the equation $\lim_{x\to x_0} f(x) = A$, which is meant to convey the idea that when x is sufficiently near to x_0, then $f(x)$ will be as near to A as desired. The statements "x is sufficiently near to x_0" and "$f(x)$ will be as near to A as desired" are made mathematically precise by the following type of definition.

4–1 DEFINITION. *The symbolism*

$$\lim_{x\to x_0} f(x) = A$$

means that for every number $\epsilon > 0$, *there is another number* $\delta > 0$ *such that whenever* $0 < |x - x_0| < \delta$, *then* $|f(x) - A| < \epsilon$.

In the case of sequences, $\lim_{n\to\infty} x_n = A$ means, intuitively, that the terms x_n of our sequence will be as near to A as desired, provided only that n is sufficiently large A more rigorous definition takes the form

4-2 DEFINITION *We write*

$$\lim_{n\to\infty} x_n = A$$

to mean that for every $\epsilon > 0$, there is an integer N such that whenever $n > N$, then $|x_n - A| < \epsilon$

Let us examine these definitions more closely to see what they have in common Essentially, both begin "For every $\epsilon > 0$," and both end with a certain absolute value $< \epsilon$ Definition 4-1 deals with a function f, Definition 4-2 with a sequence $\{x_n\}$ However, a sequence is also a function—one whose domain of definition is the set of integers $\{1, 2, 3, \quad \}$ This much, then, the two definitions have in common The essential difference seems to be that the first definition involves a number δ which measures the "nearness" of x to x_0, and the second deals with a number N which measures the "largeness" of n However, if we use neighborhood terminology, we see that both definitions really involve the same principle To say that $0 < |x - x_0| < \delta$ means that x is in a neighborhood of x_0, but $x \neq x_0$ That is, x is in a deleted neighborhood $N'(x_0)$ Also, to say that $n > N$ is the same as saying that n is in a deleted neighborhood of $+\infty$ Finally, the parts of these two definitions concerned with ϵ can also be rephrased in neighborhood terminology For example, the inequality $|f(x) - A| < \epsilon$ states that $f(x) \in N(A, \epsilon)$ and, similarly, $|x_n - A| < \epsilon$ means that $x_n \in N(A, \epsilon)$ Therefore, the basic principle involved in both definitions of limit is that for every neighborhood $N(A)$ there must exist a neighborhood $N(x_0)$ such that $x \in N'(x_0)$ implies $f(x) \in N(A)$ This principle will be expressed formally in Definition 4-3, but first we wish to make some remarks concerning notation and terminology that will be used in this and later chapters

If f is a real-valued function defined at a point $\mathbf{x} = (x_1, \quad , x_m)$ in E_m, we use either $f(x_1, \quad , x_m)$ or $f(\mathbf{x})$ to denote the value of f at that point. Such a function is sometimes called a function of the m real "variables" $x_1, \quad , x_m$, and sometimes a function of the *one* m-dimensional "variable" $\mathbf{x}$ If we have several real-valued functions, say $f_1, \quad , f_k$, defined on a common subset S of E_m, it is extremely convenient to introduce a *vector-valued function* $\mathbf{f}$, defined by the equation

$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \quad , f_k(\mathbf{x})), \qquad \text{if } \mathbf{x} \in S$$

Each value $\mathbf{f}(\mathbf{x})$ is then a vector in E_k The real-valued functions $f_1, \quad , f_k$ are referred to as the components of $\mathbf{f}$, and we write $\mathbf{f} = (f_1, \ldots, f_k)$ to

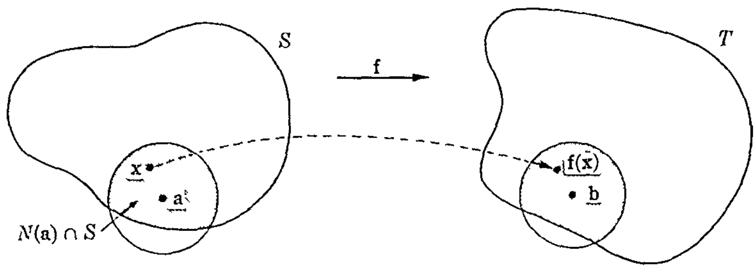

FIG. 4-1. Definition of limit.

indicate this relationship. Vector-valued functions will usually be denoted by bold-faced type.

Our definition of limit (Fig. 4-1) now assumes the following form:

4-3 DEFINITION. *Let* **f** *be a function defined on a set S in E_m and let the range of* **f** *be a subset T of E_k. If* **a** *is an accumulation point of S and if* **b** ε E_k, *then the symbolism*

$$\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{b}$$

is defined to mean the following: For every neighborhood $N(\mathbf{b}) \subset E_k$, there exists a neighborhood $N(\mathbf{a}) \subset E_m$ such that

$$\mathbf{x} \;\varepsilon\; N'(\mathbf{a}) \cap S \quad \textit{implies} \quad \mathbf{f}(\mathbf{x}) \;\varepsilon\; N(\mathbf{b}).$$

NOTE. We write $\mathbf{x} \;\varepsilon\; N'(\mathbf{a}) \cap S$ rather than $\mathbf{x} \;\varepsilon\; N'(\mathbf{a})$ in order to make certain that **x** is in the domain of **f**. Also, we require that **a** be an accumulation point of S in order to make certain that the intersection $N'(\mathbf{a}) \cap S$ will never be the empty set.

The symbolism $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{b}$ is read "the limit of **f**(**x**), as **x** tends to **a**, is **b**" or, "**f**(**x**) approaches **b** as **x** approaches **a**." We sometimes indicate this by writing $\mathbf{f}(\mathbf{x}) \to \mathbf{b}$ as $\mathbf{x} \to \mathbf{a}$. Observe that the definition still makes sense when $\mathbf{a} = +\infty$ or when $\mathbf{a} = -\infty$ (when $S \subset E_1$), or when $\mathbf{a} = \infty$ (when $S \subset E_2$). It also makes sense when $\mathbf{b} = +\infty$ or $-\infty$ (when **f** is a real-valued function), or when $\mathbf{b} = \infty$ (when **f** is complex-valued). When **f** is a sequence whose nth term is $\mathbf{x}_n$, it is customary to write $\lim_{n\to\infty} \mathbf{x}_n$ instead of $\lim_{x\to+\infty} \mathbf{f}(x)$.

The terminology "$\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x})$ exists" is also used. This means that there is a point **b** (in the space under consideration) such that $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{b}$. When we use this terminology for real- or complex-valued functions, it will be understood that **b** is a "finite" point. When the limit is "infinite," we will explicitly write $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = +\infty$ (or $-\infty$, or ∞, as the case may be).

NOTE If $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x})$ exists (finite or infinite), its value is uniquely determined [Why?]

Strictly speaking, we should somehow indicate the fact that the limit just defined depends on the set S through which $\mathbf{x}$ is allowed to range. This will usually be clear from the context but, if necessary, we will write

$$\lim_{\substack{\mathbf{x}\to\mathbf{a}\\ \mathbf{x}\in S}} \mathbf{f}(\mathbf{x}) = \mathbf{b}$$

to emphasize this fact more explicitly An important special case of this occurs when S is an interval in E_1 having a as its left endpoint We then write

$$\lim_{\substack{x\to a\\ x\in S}} \mathbf{f}(x) = \lim_{x\to a+} \mathbf{f}(x) = \mathbf{b},$$

and $\mathbf{b}$ is called the *right-hand* limit at a Left-hand limits, denoted by $\lim_{x\to a-} \mathbf{f}(x)$, are similarly defined when S is an interval in E_1 having a as its right endpoint

4-2 Some basic theorems on limits.

4-4 THEOREM *Let* $\mathbf{a}$ *be an accumulation point of a set* S *in* E_m *Then there exists an infinite sequence* $\{\mathbf{x}_n\}$ *whose terms are distinct points of* S, *such that* $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{a}$

Proof There exists a point $\mathbf{x}_1$ of S, $\mathbf{x}_1 \neq \mathbf{a}$, in the neighborhood $N(\mathbf{a}; r_1)$ of radius $r_1 = 1$ about $\mathbf{a}$ as center Now consider the neighborhood $N(\mathbf{a}, r_2)$ of radius $r_2 = |\mathbf{a} - \mathbf{x}_1|/2$ There is a point $\mathbf{x}_2$ of S in this neighborhood distinct from $\mathbf{x}_1$ and distinct from $\mathbf{a}$ Similarly, in the neighborhood $N(\mathbf{a}; r_3)$, of radius $r_3 = |\mathbf{a} - \mathbf{x}_2|/2$, there is a point $\mathbf{x}_3$ of S distinct from $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{a}$ Having chosen $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{n-1}$ in this way, we find that there is a point $\mathbf{x}_n$ of S in the neighborhood $N(\mathbf{a}, r_n)$ of radius $r_n = |\mathbf{a} - \mathbf{x}_{n-1}|/2 < 2^{-n+1}$ such that $\mathbf{x}_n$ is distinct from the previous $\mathbf{x}_k$'s The sequence $\{\mathbf{x}_n\}$ so defined satisfies all the conditions in the theorem

NOTE. Suppose a sequence $\{\mathbf{x}_n\}$ has a limit, say $\mathbf{a} = \lim_{n\to\infty} \mathbf{x}_n$, and let $S = \{\mathbf{x}_1, \mathbf{x}_2, \ldots\}$ denote the range of the sequence If S is infinite, it follows at once from the definition of limit that a is an accumulation point of S. Theorem 4-4 tells us that S can have no further accumulation points Therefore, a sequence $\{\mathbf{x}_n\}$ whose range S is infinite has a limit if, and only if, S has exactly one accumulation point, in which case the accumulation point is also the limit of the sequence

4–5 THEOREM. *Let* $\mathbf{f}$ *be defined on a set* S *in* E_m *with function values in* E_k, *and let* $\mathbf{a}$ *be an accumulation point of* S. *Let* $\{\mathbf{x}_n\}$ *be an infinite sequence whose terms are points of* S, *such that each term* $\mathbf{x}_n \neq \mathbf{a}$ *but such that*

$$\lim_{n\to\infty} \mathbf{x}_n = \mathbf{a}.$$

Then we have:

(i) *If* $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{b}$, *then* $\lim_{n\to\infty} \mathbf{f}(\mathbf{x}_n) = \mathbf{b}$.

(ii) *Conversely, if for every such sequence we know that* $\lim_{n\to\infty} \mathbf{f}(\mathbf{x}_n)$ *exists, then all these sequences have the same limit (call it* $\mathbf{y}$*) and also* $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x})$ *exists and equals* $\mathbf{y}$.

Proof. (i) For every neighborhood $N(\mathbf{b})$ in E_k, there is a neighborhood $N(\mathbf{a})$ in E_m such that $\mathbf{x} \in N'(\mathbf{a}) \cap S$ implies $\mathbf{f}(\mathbf{x}) \in N(\mathbf{b})$. But for the neighborhood $N(\mathbf{a})$, there exists an integer M such that $n > M$ implies $\mathbf{x}_n \in N'(\mathbf{a})$. These two statements together imply that $\lim_{n\to\infty} \mathbf{f}(\mathbf{x}_n) = \mathbf{b}$.

(ii) Let $\{\mathbf{p}_n\}$ and $\{\mathbf{q}_n\}$ be two sequences such that $\mathbf{p}_n \neq \mathbf{a}$, $\mathbf{q}_n \neq \mathbf{a}$ for each n, and such that $\lim_{n\to\infty} \mathbf{p}_n = \lim_{n\to\infty} \mathbf{q}_n = \mathbf{a}$. Let $\mathbf{p} = \lim_{n\to\infty} \mathbf{f}(\mathbf{p}_n)$, $\mathbf{q} = \lim_{n\to\infty} \mathbf{f}(\mathbf{q}_n)$. It is easy to prove that $\mathbf{p} = \mathbf{q}$. In fact, if $\mathbf{p} \neq \mathbf{q}$ we consider the new sequence $\{\mathbf{x}_n\}$ defined as follows:

$$\mathbf{x}_n = \begin{cases} \mathbf{p}_n, & \text{if } n \text{ is even,} \\ \mathbf{q}_n, & \text{if } n \text{ is odd.} \end{cases}$$

Then $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{a}$ and hence $\lim_{n\to\infty} \mathbf{f}(\mathbf{x}_n)$ exists. But this is impossible, since the terms of $\{\mathbf{f}(\mathbf{x}_n)\}$ are ultimately near both $\mathbf{p}$ and $\mathbf{q}$. Hence $\mathbf{p} = \mathbf{q}$. Call their common value $\mathbf{y}$. We will show next that $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x})$ exists and equals $\mathbf{y}$. If this were not so, there would be at least one neighborhood $N(\mathbf{y})$ such that every deleted neighborhood $N'(\mathbf{a}; r)$ would contain some $\mathbf{x}$ in S for which $\mathbf{f}(\mathbf{x}) \notin N(\mathbf{y})$. For each $n = 1, 2, \ldots$, let $\mathbf{x}_n$ be a point in $N'(\mathbf{a}; 1/n) \cap S$ for which $\mathbf{f}(\mathbf{x}_n) \notin N(\mathbf{y})$. This defines a sequence $\{\mathbf{x}_n\}$ such that $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{a}$, $\mathbf{x}_n \neq \mathbf{a}$. But for this sequence it is not true that $\lim_{n\to\infty} \mathbf{f}(\mathbf{x}_n) = \mathbf{y}$, contradicting what was just proved. Therefore, $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{y}$, and this completes the proof.

4–3 The Cauchy condition. In the definition of the statement $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{b}$, it was assumed that the limiting value $\mathbf{b}$ was given. The following theorem gives us a condition (called the *Cauchy condition*) which enables us to determine, without knowing its value in advance, whether such a point $\mathbf{b}$ exists.

We first prove such a theorem for sequences and then use the result to obtain a similar criterion for more general functions. In effect, the Cauchy condition states that if the terms of a sequence are getting close to *some-*

thing, then they must be getting close to each other The converse also holds true

4-6 THEOREM (*Cauchy condition for sequences*) *Let* $\{\mathbf{x}_n\}$ *be an infinite sequence whose terms are points in* E_k *There exists a point* $\mathbf{y}$ *in* E_k *such that* $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{y}$ *if, and only if, the following condition is satisfied: For every* $\epsilon > 0$, *there exists an integer* N *such that* $n > N$ *and* $m > N$ *implies* $|\mathbf{x}_n - \mathbf{x}_m| < \epsilon$

Proof Assume first that $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{y}$ Then, given $\epsilon > 0$, there is an integer N such that $n > N$ implies $|\mathbf{x}_n - \mathbf{y}| < \epsilon/2$ But we have

$$|\mathbf{x}_n - \mathbf{x}_m| = |(\mathbf{x}_n - \mathbf{y}) + (\mathbf{y} - \mathbf{x}_m)| \leq |\mathbf{x}_n - \mathbf{y}| + |\mathbf{y} - \mathbf{x}_m|,$$

so that if n and m are both $>N$, it follows that $|\mathbf{x}_n - \mathbf{x}_m| < \epsilon$

To prove the converse, suppose the Cauchy condition holds Let $S = \{\mathbf{x}_1, \mathbf{x}_2, \quad\}$ If S is finite, then all except a finite number of the terms $\mathbf{x}_n$ must be equal and $\lim_{n\to\infty} \mathbf{x}_n$ will then exist and be equal to this common value If S is infinite, the Cauchy condition implies that S is bounded

In fact, if we take $\epsilon = 1$, an integer N_1 exists such that $n > N_1$ implies $|\mathbf{x}_n - \mathbf{x}_{N_1}| < 1$ This means that all the points $\mathbf{x}_n$ with $n > N_1$ lie inside a sphere of radius 1 about $\mathbf{x}_{N_1}$ as center If M denotes the largest of the numbers $|\mathbf{x}_1|, \quad , |\mathbf{x}_{N_1}|$, then S lies inside a sphere of radius $M + 1$ about the origin By the Bolzano-Weierstrass theorem, the set S must have an accumulation point, say $\mathbf{y}$ Because of the Cauchy condition, S cannot have more than one accumulation point Hence $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{y}$ and the proof is complete

4-7 THEOREM (*Cauchy condition for functions*) *Let* $\mathbf{f}$ *be defined on a set* S *in* E_m, *the function values being in* E_k *Let* $\mathbf{a}$ *be an accumulation point of* S *There exists a point* $\mathbf{b}$ *in* E_k *such that* $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{b}$ *if, and only if, the following condition holds. For every* $\epsilon > 0$, *there is a neighborhood* $N(\mathbf{a})$ *such that* $\mathbf{x}$ *and* $\mathbf{y}$ *in* $N'(\mathbf{a}) \cap S$ *implies* $|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})| < \epsilon$

Proof Assume $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{b}$, and let $\epsilon > 0$ be given There is a neighborhood $N(\mathbf{a})$ such that $\mathbf{x} \in N'(\mathbf{a}) \cap S$ implies $|\mathbf{f}(\mathbf{x}) - \mathbf{b}| < \epsilon/2$ If $\mathbf{y}$ is also in $N'(\mathbf{a}) \cap S$, then $|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})| \leq |\mathbf{f}(\mathbf{x}) - \mathbf{b}| + |\mathbf{b} - \mathbf{f}(\mathbf{y})| < \epsilon$

Conversely, suppose the condition of the theorem holds, and let $N(\mathbf{a})$ be the neighborhood which corresponds to ϵ Let $\{\mathbf{x}_n\}$ be a sequence whose terms are distinct points of S such that $\mathbf{x}_n \neq \mathbf{a}$ and such that $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{a}$ Then for some n_0, we have $\mathbf{x}_n \in N'(\mathbf{a})$ for all $n > n_0$, and therefore $|\mathbf{f}(\mathbf{x}_n) - \mathbf{f}(\mathbf{x}_m)| < \epsilon$ for all $m > n_0$, $n > n_0$ By the Cauchy condition for sequences, this means that $\lim_{n\to\infty} \mathbf{f}(\mathbf{x}_n)$ exists Part (ii) of Theorem 4-5 implies that $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x})$ exists This completes the proof

4-4 Algebra of limits. The usual rules for calculating with limits are given in the next theorem.

4-8 THEOREM. *Let f and g be two functions, each defined on a set S in E_n, with function values in E_1 or in E_2. Let* $\mathbf{a}$ *be an accumulation point of S and assume we have*

$$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = A, \qquad \lim_{\mathbf{x}\to\mathbf{a}} g(\mathbf{x}) = B.$$

Then we also have

(i) $\lim_{\mathbf{x}\to\mathbf{a}} [f(\mathbf{x}) \pm g(\mathbf{x})] = A \pm B,$

(ii) $\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x})g(\mathbf{x}) = AB,$

(iii) $\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x})/g(\mathbf{x}) = A/B$ *if* $B \neq 0$.

Proof. We will prove only (ii), leaving the other parts as exercises. Let ϵ, $0 < \epsilon < 1$, be given and let ϵ' be a second number, $0 < \epsilon' < 1$, which will be made to depend on ϵ in a way to be described later. Choose a neighborhood $N(\mathbf{a})$ such that if $\mathbf{x} \in N'(\mathbf{a}) \cap S$, we have both

$$|f(\mathbf{x}) - A| < \epsilon' \qquad \text{and} \qquad |g(\mathbf{x}) - B| < \epsilon'.$$

Then

$$|f(\mathbf{x})| = |A + (f(\mathbf{x}) - A)| < |A| + \epsilon' < |A| + 1.$$

Writing $f(\mathbf{x})g(\mathbf{x}) - AB = f(\mathbf{x})g(\mathbf{x}) - Bf(\mathbf{x}) + Bf(\mathbf{x}) - AB$, we have

$$\begin{aligned} |f(\mathbf{x})g(\mathbf{x}) - AB| &\leq |f(\mathbf{x})||g(\mathbf{x}) - B| + |B||f(\mathbf{x}) - A| \\ &\leq (|A| + 1)\epsilon' + |B|\epsilon' = \epsilon'(|A| + |B| + 1). \end{aligned}$$

If we choose $\epsilon' = \epsilon/(|A| + |B| + 1)$, we see that $|f(\mathbf{x})g(\mathbf{x}) - AB| < \epsilon$ whenever $\mathbf{x} \in N'(\mathbf{a}) \cap S$, and this proves (ii).

4-5 Continuity. In the definition of the symbol $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{b}$, it is not necessary that the function be defined *at* $\mathbf{x} = \mathbf{a}$. Moreover, in case $\mathbf{f}$ *is* defined at $\mathbf{x} = \mathbf{a}$, the value $\mathbf{f}(\mathbf{a})$ need not be equal to $\mathbf{b}$; when we do have $\mathbf{f}(\mathbf{a}) = \mathbf{b}$, the function is called *continuous* at $\mathbf{a}$.

4-9 DEFINITION. *Let* $\mathbf{f}$ *be defined on a set S in E_n with function values in E_m, and let* $\mathbf{a}$ *be an accumulation point of S. We say that* $\mathbf{f}$ *is continuous at the point* $\mathbf{a}$ *provided that*

(i) $\mathbf{f}$ *is defined at* $\mathbf{a}$,

(ii) $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{f}(\mathbf{a})$.

If $\mathbf{a}$ *is not an accumulation point of* S, *we say* $\mathbf{f}$ *is continuous at* $\mathbf{a}$ *provided only* (i) *holds. If* $\mathbf{f}$ *is continuous at every point of* S, *we say* $\mathbf{f}$ *is continuous on the set* S.

It is convenient to note that whenever $\mathbf{f}$ is continuous at $\mathbf{a}$, we can write (ii) of Definition 4–9 as follows:

$$\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{f}\,(\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{x})$$

Thus, when we deal with continuous functions, the limit symbol may be interchanged with the function symbol. Observe also that continuity at $\mathbf{a}$ means that for every neighborhood $N(\mathbf{f}(\mathbf{a}))$, there exists a neighborhood $N(\mathbf{a})$ such that $\mathbf{f}[N(\mathbf{a}) \cap S] \subset N[\mathbf{f}(\mathbf{a})]$.

From the algebra of limits we obtain the following theorem at once:

4–10 THEOREM. *Let f and g be continuous at the point* $\mathbf{a}$ *in* E_n *and assume that f and g have function values in* E_1 *or* E_2. *Then* $f + g$, $f - g$, *and* $f \cdot g$ *are each continuous at* $\mathbf{a}$. *The quotient* f/g *is also continuous at* $\mathbf{a}$, *provided that* $g(\mathbf{a}) \neq 0$.

NOTE. We denote by $f + g$, $f - g$, $f \cdot g$, and f/g the function whose value at $\mathbf{x}$ is, respectively, $f(\mathbf{x}) + g(\mathbf{x})$, $f(\mathbf{x}) - g(\mathbf{x})$, $f(\mathbf{x})g(\mathbf{x})$, and $f(\mathbf{x})/g(\mathbf{x})$. The product $f \cdot g$ should not be confused with the composition fg, defined by $fg(\mathbf{x}) = f[g(\mathbf{x})]$.

The analog of Theorem 4–10 for the composition of f and g is a much deeper result.

4–11 THEOREM. *Let* $\mathbf{f}$ *be a function defined on a set* S *in* E_n *and assume* $\mathbf{f}(S) \subset E_m$. *Let* $\mathbf{g}$ *be defined on* $\mathbf{f}(S)$ *with values in* E_k, *and let* $\mathbf{gf}$ *be the composite function defined on* S *by the equation* $\mathbf{gf}(\mathbf{x}) = \mathbf{g}[\mathbf{f}(\mathbf{x})]$. *Suppose that we have*

(i) *The point* $\mathbf{a}$ *is an accumulation point of* S *and* $\mathbf{f}$ *is continuous at* $\mathbf{a}$.

(ii) *The point* $\mathbf{f}(\mathbf{a})$ *is an accumulation point of* $\mathbf{f}(S)$ *and* $\mathbf{g}$ *is continuous at* $\mathbf{f}(\mathbf{a})$.

Then the composite function $\mathbf{gf}$ *is also continuous at* $\mathbf{a}$; *that is,*

$$\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{g}[\mathbf{f}(\mathbf{x})] = \mathbf{g}[\mathbf{f}(\mathbf{a})].$$

Proof. Let $\mathbf{b} = \mathbf{f}(\mathbf{a})$. Then by (ii), for every neighborhood $N[\mathbf{g}(\mathbf{b})]$ in E_k, there is a neighborhood $N(\mathbf{b})$ in E_m such that $\mathbf{y} \in N(\mathbf{b}) \cap \mathbf{f}(S)$ implies $\mathbf{g}(\mathbf{y}) \in N[\mathbf{g}(\mathbf{b})]$. For the neighborhood $N(\mathbf{b})$ we can find a neighborhood

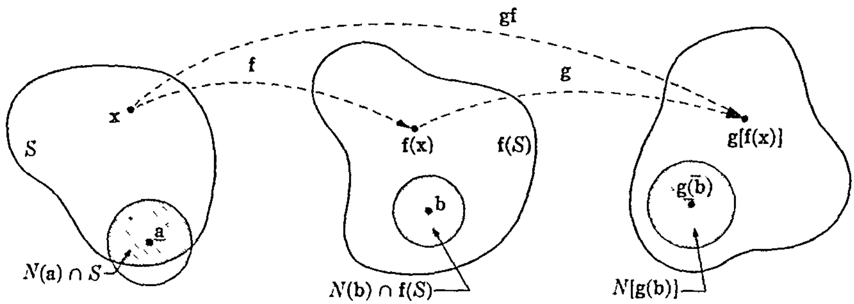

FIG. 4–2. Theorem 4–11 when $E_n = E_m = E_k = E_2$.

$N(\mathbf{a})$ in E_n such that $\mathbf{x} \in N(\mathbf{a}) \cap S$ implies $\mathbf{f}(\mathbf{x}) \in N(\mathbf{b})$. Combining these two statements, we see that for every neighborhood of $\mathbf{g}(\mathbf{b}) = \mathbf{g}[\mathbf{f}(\mathbf{a})] = \mathbf{gf}(\mathbf{a})$ in E_k, there exists a neighborhood $N(\mathbf{a})$ in E_n such that $\mathbf{x} \in N(\mathbf{a}) \cap S$ implies $\mathbf{gf}(\mathbf{x}) \in N[\mathbf{gf}(\mathbf{a})]$. This proves the theorem. (See Fig. 4–2 for the case $E_m = E_n = E_k = E_2$.)

4–6 Examples of continuous functions. Let $S = E_2$. It is a trivial exercise to show that the following complex-valued functions are continuous on E_2: (a) constant functions, defined by $f(x) = c$ for every x in E_2, and (b) the identity function defined by $f(x) = x$ for every x in E_2. Repeated application of the algebra of continuous functions then establishes the continuity of every polynomial:

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

the a_i being complex numbers.

If S is a set in E_2 where the polynomial f does not vanish, then $1/f$ is also continuous on S. Therefore a rational function g/f, where g and f are polynomials, is continuous in E_2 wherever the denominator does not vanish.

The familiar real-valued functions of elementary calculus, such as the exponential, trigonometric, and logarithmic functions, are all continuous wherever they are defined. The continuity of these elementary functions justifies the common practice of evaluating certain limits by substituting the limiting value of the "independent variable"; for example,

$$\lim_{x \to 0} e^x = e^0 = 1.$$

The continuity of the complex exponential and trigonometric functions is a consequence of the continuity of the corresponding real-valued functions. (See Exercise 4–6.)

4-7 Functions continuous on open or closed sets. Functions which are continuous on open or closed sets have special properties of particular significance. Before discussing these in detail, it will be convenient to introduce some further terminology.

4-12 DEFINITION. *Let* $\mathbf{f}$ *be a function whose domain is* S *and whose range is* T, *and let* Y *be a subset of* T. *By the inverse image of* Y *under* $\mathbf{f}$, *denoted by* $\mathbf{f}^{-1}(Y)$, *we shall mean the largest subset of* S *which* $\mathbf{f}$ *maps onto* Y, *that is to say,*

$$\mathbf{f}^{-1}(Y) = \{\mathbf{x} \mid \mathbf{x} \in S, \mathbf{f}(\mathbf{x}) \in Y\}.$$

NOTE. If $\mathbf{f}$ has an inverse function $\mathbf{f}^{-1}$, the inverse image of Y under $\mathbf{f}$ is the same as the image of Y under $\mathbf{f}^{-1}$, and in this case there is no ambiguity in the notation $\mathbf{f}^{-1}(Y)$.

4-13 THEOREM. *Let* $\mathbf{f}$ *be a function with domain* S *and range* T. *Assume* $X \subset S$ *and* $Y \subset T$. *Then we have*

(i) *If* $X = \mathbf{f}^{-1}(Y)$, *then* $Y = \mathbf{f}(X)$.

(ii) *If* $Y = \mathbf{f}(X)$, *then* $X \subset \mathbf{f}^{-1}(Y)$.

The proof of Theorem 4-13 is a direct translation of the definition of the symbols $\mathbf{f}^{-1}(Y)$ and $\mathbf{f}(X)$, and is left to the reader. It should be observed that, in general, we cannot conclude that $Y = \mathbf{f}(X)$ implies $X = \mathbf{f}^{-1}(Y)$. (See Fig. 4-3.)

Note that the statements in Theorem 4-13 can also be expressed as follows:

$$Y = \mathbf{f}[\mathbf{f}^{-1}(Y)], \qquad X \subset \mathbf{f}^{-1}[\mathbf{f}(X)].$$

4-14 THEOREM. *Let* $\mathbf{f}$ *be a function which is continuous on a closed set* S *in* E_m, *and let the range of* $\mathbf{f}$ *be a set* T *in* E_k. *Then if* Y *is a closed subset of* T, *the inverse image* $\mathbf{f}^{-1}(Y)$ *will be a closed subset of* S.

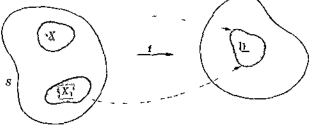

FIG. 4-3. $Y = \mathbf{f}(X) = \mathbf{f}(X_1)$, $\mathbf{f}^{-1}(Y) = X \cup X_1$.

Proof. Let $\mathbf{x}$ be an accumulation point of $\mathbf{f}^{-1}(Y)$. Then $\mathbf{x}$ is also an accumulation point of S and hence $\mathbf{x} \in S$, since S is closed. We must show that $\mathbf{x} \in \mathbf{f}^{-1}(Y)$ or, equivalently, that $\mathbf{f}(\mathbf{x}) \in Y$. There exists an infinite sequence $\{\mathbf{x}_n\}$, whose terms are points in $\mathbf{f}^{-1}(Y)$, such that $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{x}$. By continuity at $\mathbf{x}$, we have $\lim_{n\to\infty} \mathbf{f}(\mathbf{x}_n) = \mathbf{f}(\mathbf{x})$. But each point $\mathbf{f}(\mathbf{x}_n) \in Y$ and, since Y is closed, the limit $\mathbf{f}(\mathbf{x}) \in Y$. [Why?] This proves that $\mathbf{f}^{-1}(Y)$ is closed.

Note that the hypothesis that S is closed is necessary in Theorem 4–14. For example, if S is an open set and if $\mathbf{f}(\mathbf{x}) = \mathbf{a}$ for every $\mathbf{x}$ in S, then the range of $\mathbf{f}$ is the closed set $\{\mathbf{a}\}$. In this case the inverse image of a closed set is open.

4–15 THEOREM. *Let* $\mathbf{f}$ *be a function which is continuous on an open set* S *in* E_m, *and let the range of* $\mathbf{f}$ *be a set* T *in* E_k. *Then if* Y *is an open subset of* T, *the inverse image* $\mathbf{f}^{-1}(Y)$ *will be an open subset of* S.

Proof. We shall show that $\mathbf{f}^{-1}(Y)$ is a union of open sets. Assume that

$$\mathbf{x}_0 \in \mathbf{f}^{-1}(Y).$$

Then there exists a point $\mathbf{y}_0$ in Y such that $\mathbf{y}_0 = \mathbf{f}(\mathbf{x}_0)$. Since Y is open, there is a neighborhood $N(\mathbf{y}_0) \subset Y$. If the radius of this neighborhood is ϵ, there exists a $\delta > 0$ such that for all $\mathbf{x}$ in $N(\mathbf{x}_0; \delta) \cap S$ we have $|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}_0)| < \epsilon$, which means that $\mathbf{f}(\mathbf{x}) \in N(\mathbf{y}_0)$. The open set $N(\mathbf{x}_0; \delta) \cap S$ is therefore a subset of $\mathbf{f}^{-1}(Y)$. But the union of all such open sets $N(\mathbf{x}_0; \delta) \cap S$ obtained as $\mathbf{x}_0$ ranges through the set $\mathbf{f}^{-1}(Y)$ is exactly $\mathbf{f}^{-1}(Y)$ itself. Hence $\mathbf{f}^{-1}(Y)$ is open, and the proof is complete.

We have already seen that the image of an open set under a continuous mapping is not always open. The example in E_1, defined by the equation $f(x) = \tan^{-1}(x)$, shows that the image of a closed set under a continuous mapping need not be closed, since f maps E_1 onto the open interval $(-\pi/2, \pi/2)$. However, the image of a *compact* set under a continuous mapping is always compact. This will be proved in the next theorem.

4–8 Functions continuous on compact sets.

4–16 THEOREM. *Let* $\mathbf{f}$ *be a function which is continuous on a compact set* S *in* E_m, *and assume that* $\mathbf{f}(S) \subset E_k$. *Then* $\mathbf{f}(S)$ *is a compact set.*

Proof. Let F be an open covering of $\mathbf{f}(S)$, so that $\mathbf{f}(S) \subset \bigcup_{A \in F} A$. Take a point $\mathbf{x}$ in S. Then $\mathbf{f}(\mathbf{x})$ belongs to at least one of the open sets in F. Denote one of these sets by $A_{\mathbf{x}}$. Since $A_{\mathbf{x}}$ is open, there is a neighborhood $N[\mathbf{f}(\mathbf{x})] \subset A_{\mathbf{x}}$. By continuity, there is a neighborhood $N(\mathbf{x})$ such that

$f[N(\mathbf{x}) \cap S] \subset N[f(\mathbf{x})]$. The collection of all such neighborhoods $N(\mathbf{x})$ obtained as $\mathbf{x}$ varies through S is an open covering of S. Since S is compact, a finite number of neighborhoods cover S, say $S \subset N_1(\mathbf{x}_1) \cup \cdots \cup N_p(\mathbf{x}_p)$. Now we must have

$$\mathbf{f}[N_q(\mathbf{x}_q) \cap S] \subset N[\mathbf{f}(\mathbf{x}_q)] \subset A_{\mathbf{x}_q} \qquad (q = 1, 2, \ldots, p),$$

where $A_{\mathbf{x}_q} \in F$. It follows that $\mathbf{f}(S) \subset A_{\mathbf{x}_1} \cup \cdots \cup A_{\mathbf{x}_p}$, which proves that $\mathbf{f}(S)$ is compact.

4-17 THEOREM. *Let* $\mathbf{f}$ *be a function which is continuous on a compact set* S *in* E_m, *and assume that* $\mathbf{f}(S) \subset E_k$. *Suppose further that* $\mathbf{f}$ *is one-to-one on* S *so that the inverse function* $\mathbf{f}^{-1}$ *exists. Then* $\mathbf{f}^{-1}$ *is continuous on* $\mathbf{f}(S)$.

Proof. Let $\mathbf{y}$ be an accumulation point of $\mathbf{f}(S)$ and write $\mathbf{y} = \lim_{n\to\infty} \mathbf{y}_n$, where each $\mathbf{y}_n \in \mathbf{f}(S)$. Let $\mathbf{x}_n = \mathbf{f}^{-1}(\mathbf{y}_n)$. Then the set $\{\mathbf{x}_1, \mathbf{x}_2, \ldots\}$ is an infinite subset of the compact set S and hence has an accumulation point, say $\mathbf{x}$, in S. A subsequence of $\{\mathbf{x}_n\}$, say $\{\mathbf{x}_n'\}$, can then be chosen so that $\lim_{n\to\infty} \mathbf{x}_n' = \mathbf{x}$. By continuity of $\mathbf{f}$, we have $\lim_{n\to\infty} \mathbf{f}(\mathbf{x}_n') = \mathbf{f}(\mathbf{x})$. On the other hand, $\lim_{n\to\infty} \mathbf{f}(\mathbf{x}_n') = \mathbf{y}$, since the sequence $\{\mathbf{f}(\mathbf{x}_n')\}$ is a subsequence of $\{\mathbf{y}_n\}$. Therefore $\mathbf{x} = \mathbf{f}^{-1}(\mathbf{y})$, which means that $\mathbf{x}$ is the only accumulation point of the set $\{\mathbf{x}_1, \mathbf{x}_2, \ldots\}$. Hence we can write $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{x}$ or, what is the same thing, $\lim_{n\to\infty} \mathbf{f}^{-1}(\mathbf{y}_n) = \mathbf{f}^{-1}(\mathbf{y})$, which implies continuity of $\mathbf{f}^{-1}$ at $\mathbf{y}$.

4-9 Topological mappings.

4-18 DEFINITION. *Let* $\mathbf{f}$ *be continuous on* S *and assume that* $\mathbf{f}$ *is one-to-one on* S *so that the inverse function* $\mathbf{f}^{-1}$ *exists. Then* $\mathbf{f}$ *is called a topological mapping or a homeomorphism if, in addition,* $\mathbf{f}^{-1}$ *is continuous on* $\mathbf{f}(S)$.

A property of a set which remains invariant under every topological mapping is called a *topological property*. Thus, for example, Theorem 4-16 tells us that compactness is a topological property.

Topological mappings are particularly important in the theory of space curves. For example, a *simple arc* is the topological image of an interval and a *simple closed curve* is the topological image of a circle. These ideas will be considered in more detail in Chapter 8.

4-10 Properties of real-valued continuous functions.

4-19 DEFINITION. *Let* f *be a real-valued function defined on a set* S *in* E_n. *Then* f *is said to have an absolute maximum on the set* S *if there exists a point* $\mathbf{a}$ *in* S *such that*

$$f(\mathrm{x}) \leq f(\mathrm{a}), \qquad \text{for all } \mathrm{x} \text{ in } S.$$

If $\mathrm{a} \in S$ *and if there is a neighborhood* $N(\mathrm{a})$ *such that*

$$f(\mathrm{x}) \leq f(\mathrm{a}), \qquad \text{for all } \mathrm{x} \text{ in } N(\mathrm{a}) \cap S,$$

then f *is said to have a relative maximum at the point* a. *(Absolute minimum and relative minimum are similarly defined, using* $f(\mathrm{x}) \geq f(\mathrm{a})$.*)*

The terms "local" and "global" are also used instead of "relative" and "absolute." In general, a property of a set S which is described in terms of a neighborhood of a point in S is referred to as a *local property* of S; continuity at a point and the notion of relative maximum are local properties. On the other hand, a property which involves the set as a whole is called a *global property*. Hence, continuity on a set is a global property, as is the possession of an absolute maximum on a set.

4–20 THEOREM. *Let* f *be a real-valued function which is continuous on a compact set* S *in* E_n. *Then* f *has an absolute maximum and an absolute minimum on* S.

Proof. In Theorem 4–16 we proved that $f(S)$ is compact. Also, we have $\inf f(S) \leq f(\mathrm{x}) \leq \sup f(S)$ for every x in S. To prove the present theorem it suffices to show that every compact set of real numbers contains its inf and its sup. For finite sets this is obvious.

Let A be a closed and bounded infinite set in E_1 and let $a = \sup A$. Then, if $a \notin A$, for every $\epsilon > 0$, there exists a point x in A such that $a - \epsilon \leq x < a$. This means that every neighborhood of a contains points of A distinct from a. Hence a is an accumulation point of A. Since A is closed, this contradicts the assumption $a \notin A$. Therefore A contains its sup. (A similar argument is valid for the inf.)

The next two theorems deal with properties of real-valued continuous functions defined on closed and bounded intervals in E_1. The first of these is a very important and intuitively evident statement which says that if the graph of a continuous function lies above the x-axis at one end of an interval and below the x-axis at the other end, then it must cross the axis somewhere in between.

4–21 THEOREM (*Bolzano*). *Let* f *be real-valued and continuous on a closed interval* $[a, b]$ *in* E_1, *and suppose that* $f(a)$ *and* $f(b)$ *have different signs; that is, assume* $f(a)f(b) < 0$. *Then there is at least one point* x *in the open interval* (a, b) *such that* $f(x) = 0$.

Proof. For definiteness, assume $f(a) > 0$ and $f(b) < 0$. Let

$$A = \{x \mid x \in [a, b], f(x) \geq 0\}.$$

Then A is a nonempty subset of $[a, b]$ (since $a \in A$) Let $c = \sup A$ Then $a < c < b$ [Why?] We shall show that $f(c) = 0$.

If $f(c) \neq 0$, there exists a neighborhood $N(c)$ such that at each point x in $N(c)$ the numbers $f(x)$ and $f(c)$ have the same sign [Why?] This contradicts the definition of c [Why?] It follows that we must have $f(c) = 0$.

An immediate consequence of Bolzano's theorem is the *intermediate value theorem* for continuous functions

4-22 THEOREM *If f is real-valued and continuous on a closed interval S in E_1, then f assumes every value between its maximum, $\sup f(S)$, and its minimum, $\inf f(S)$*

Proof Let $f(x_1) = \inf f(S)$ and $f(x_2) = \sup f(S)$, and assume $\inf f(S) < c < \sup f(S)$ Apply Theorem 4-21 to the function g, defined (on the interval joining x_1 and x_2) by the equation $g(x) = f(x) - c$

It follows from this theorem that the set of values assumed by a continuous function f on a closed interval S is another closed interval, namely, $[\inf f(S), \sup f(S)]$ (If f is constant on S, this will be a degenerate interval)

4-11 Uniform continuity. Suppose we have a function $\mathbf{f}$ continuous on a set S, so that for each accumulation point $\mathbf{a}$ of S, we have $\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{f}(\mathbf{a})$ In the ϵ and δ terminology this means that, given the accumulation point $\mathbf{a}$ and given ϵ, there is a number $\delta > 0$ (depending on $\mathbf{a}$ and on ϵ) such that $0 < |\mathbf{x} - \mathbf{a}| < \delta$ implies $|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{a})| < \epsilon$ In general, we cannot expect that for a fixed ϵ the same value of δ will serve equally well for *every* such point $\mathbf{a}$ This might happen, however, when it does, the function is called *uniformly continuous* on the set S

4-23 DEFINITION *Let $\mathbf{f}$ be defined on a set S in E_n with function values in E_m Then $\mathbf{f}$ is said to be uniformly continuous on S if the following statement holds.*

For every $\epsilon > 0$, there exists a $\delta > 0$ (depending only on ϵ) such that if $\mathbf{x} \in S$ and $\mathbf{y} \in S$ and $|\mathbf{x} - \mathbf{y}| < \delta$, then $|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})| < \epsilon$

Observe that uniform continuity is a property of the whole set S, that is, a global property To emphasize the difference between *continuity on a set S* and *uniform continuity on a set S*, we consider the following examples

1 Let S be the half-open interval $0 < x \leq 1$, and let f be defined for each x in S by the formula $f(x) = 1/x$ This function is continuous on the set S However, we shall prove that f is *not uniformly continuous* on S Let $\epsilon = 10$, and suppose we could find a δ, $0 < \delta < 1$, to satisfy the

condition of the definition. Taking $x = \delta$, $y = \delta/11$, we obtain

$$|x - y| = \frac{10}{11}\delta < \delta$$

and

$$|f(x) - f(y)| = \frac{11}{\delta} - \frac{1}{\delta} = \frac{10}{\delta} > 10.$$

Hence for these two points we would always have $|f(x) - f(y)| > 10$, contradicting the definition of uniform continuity.

2. Now let $f(x) = x^2$ if $x \in S$, where S is the same set as above. This function *is* uniformly continuous on S. To prove this, observe that we have

$$|f(x) - f(y)| = |x^2 - y^2| = |(x - y)(x + y)| < 2|x - y|.$$

If $|x - y| < \delta$, then $|f(x) - f(y)| < 2\delta$. Hence, if ϵ is given, we need only take $\delta = \epsilon/2$ to guarantee that $|f(x) - f(y)| < \epsilon$ for *every pair* x, y with $|x - y| < \delta$. Thus f is uniformly continuous on S.

An instructive exercise is to show that in the second example f is not uniformly continuous on the whole real axis E_1.

It follows from Definition 4–23 that uniform continuity on a set implies continuity on the set. (The reader should verify this. See Exercise 4–21.) The converse of this statement is also true when the set is compact.

4–24 THEOREM (*Heine*). *Let S be a compact set in E_n. If $\mathbf{f}$ is continuous on S, then $\mathbf{f}$ is also uniformly continuous on S.*

Proof. Let $\epsilon > 0$ be given. Then each point $\mathbf{a}$ in S has associated with it a neighborhood $N(\mathbf{a}; r)$, with r depending on $\mathbf{a}$, such that

$$\mathbf{x} \in N(\mathbf{a}; r) \cap S \quad \text{implies} \quad |\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{a})| < \frac{\epsilon}{2}\cdot$$

Consider the collection of all halfsize neighborhoods $N(\mathbf{a}; r/2)$. These cover S. Since S is compact, a finite number of them also cover S, say

$$S \subset N\left(\mathbf{a}_1; \frac{r_1}{2}\right) \cup \cdots \cup N\left(\mathbf{a}_m; \frac{r_m}{2}\right)\cdot$$

In any neighborhood of twice the size, $N(\mathbf{a}_k; r_k)$, it is true that

$$\mathbf{x} \in N(\mathbf{a}_k; r_k) \cap S \quad \text{implies} \quad |\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{a}_k)| < \frac{\epsilon}{2}\cdot$$

Let 2δ be the smallest of the numbers $r_1, \ldots, r_m$. We shall show that this δ works in the definition of uniform continuity.

For this purpose, consider two points of S, say $\mathbf{x}$ and $\mathbf{y}$, with $|\mathbf{x} - \mathbf{y}| < \delta$. By the above discussion, there is some neighborhood $N(\mathbf{a}_k, r_k/2)$ containing $\mathbf{x}$. Then $|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{a}_k)| < \epsilon/2$. But we also have

$$|\mathbf{y} - \mathbf{a}_k| = |(\mathbf{y} - \mathbf{x}) + (\mathbf{x} - \mathbf{a}_k)| < \delta + \frac{r_k}{2} \leq \frac{r_k}{2} + \frac{r_k}{2} = r_k$$

That is, $\mathbf{y} \in N(\mathbf{a}_k; r_k) \cap S$. Hence we also have $|\mathbf{f}(\mathbf{y}) - \mathbf{f}(\mathbf{a}_k)| < \epsilon/2$, and therefore $|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})| < \epsilon$. This proves the theorem.

4–12 Discontinuities of real-valued functions.

4–25 DEFINITION *If f is a real-valued function defined on an open interval (a, b) in E_1 and if c is a point in $[a, b)$, we write $f(c+)$ to denote the right-hand limit, $\lim_{x \to c+} f(x)$, whenever the limit exists. If $c \in (a, b]$, we define $f(c-) = \lim_{x \to c-} f(x)$.*

It is clear that we have $f(x+) = f(x-) = f(x)$ if, and only if, f is continuous at x. A point at which f is not continuous is called a *discontinuity* of f. Of course, a point x will be a discontinuity of f if f is not defined at x. If f is defined at x, then x can be a discontinuity of f only under one of the following conditions:

(a) If either $f(x+)$ or $f(x-)$ does not exist
(b) If both $f(x+)$ and $f(x-)$ exist but have different values
(c) If $f(x+) = f(x-) \neq f(x)$.

4–26 DEFINITION *Let f be a real-valued function defined on an interval $[a, b]$. If $f(x+)$ and $f(x-)$ both exist at some interior point x, then the difference $f(x) - f(x-)$ is called the left-hand jump of f at x, the difference $f(x+) - f(x)$ is called the right-hand jump of f at x, and their sum, $f(x+) - f(x-)$, is called the jump of f at x. If any one of these three numbers is different from 0, then x is called a jump discontinuity of f. (For the endpoints a and b, only one-sided jumps are considered.) We say that f is continuous from the right at x if the right-hand jump is equal to 0, whereas f is discontinuous from the right if either $f(x+)$ does not exist or the right-hand jump is $\neq 0$. (Similar terminology applies from the left.)*

In the case that $f(x+) = f(x-) \neq f(x)$, the point x is also said to be a *removable discontinuity*, since the discontinuity can be removed by redefining f at x to have the value $f(x+) = f(x-)$.

If either $\lim_{x \to c+} f(x)$ or $\lim_{x \to c-} f(x)$ is infinite ($+\infty$ or $-\infty$), then f is sometimes said to have an *infinite discontinuity* at c.

EXAMPLES

1. The function f defined by $f(x) = x/|x|$ if $x \neq 0$, $f(0) = A$, has a jump discontinuity at $x = 0$, regardless of the value of A. (See Fig. 4–4.)

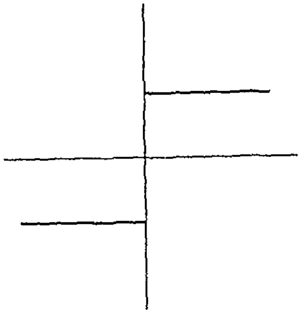

FIG. 4-4. Jump discontinuity.

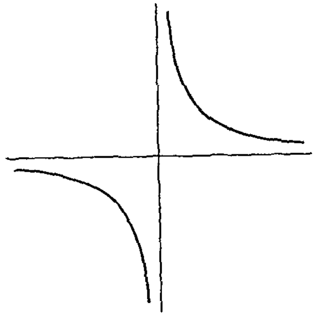

FIG. 4-5. Infinite discontinuity.

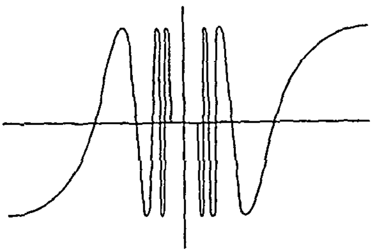

FIG. 4-6. A case where $f(0+)$ and $f(0-)$ do not exist.

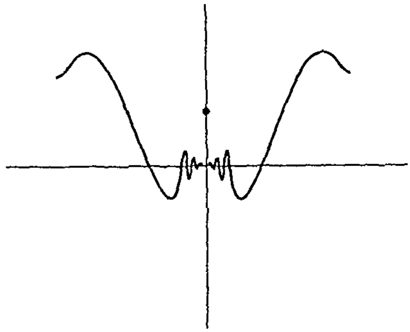

FIG. 4-7. Removable discontinuity.

2. The function f defined by $f(x) = 1$ if $x \neq 0$, $f(0) = 0$, has a removable discontinuity at $x = 0$.

3. The function f defined by $f(x) = 1/x$ if $x \neq 0$, $f(0) = A$, has an infinite discontinuity at $x = 0$. (See Fig. 4-5.)

4. The function f defined by $f(x) = \sin(1/x)$ if $x \neq 0$, $f(0) = A$, has a discontinuity at $x = 0$, regardless of the value of A. In this case, neither $f(0+)$ nor $f(0-)$ exists. (See Fig. 4-6.)

5. The function f defined by $f(x) = x \sin(1/x)$ if $x \neq 0$, $f(0) = 1$, has a removable discontinuity at $x = 0$, since $f(0+) = f(0-) = 0$. (See Fig. 4-7.)

4-13 Monotonic functions.

4-27 DEFINITION. *Let f be a real-valued function defined on a set S in E_1. If, for every pair of points x and y in S, $x < y$ implies $f(x) \leq f(y)$, then f is said to be increasing (or nondecreasing) on S. If $x < y$ implies $f(x) < f(y)$, then f is said to be strictly increasing on S. (Decreasing* *functions are similarly defined) A function is called monotonic on S if it is increasing on S or decreasing on S*

If f is an increasing function, then $-f$ is a decreasing function Because of this simple fact, in many situations involving monotonic functions it suffices to consider only the case of increasing functions

We shall prove that monotonic functions always have finite right- and left-hand limits, and hence their discontinuities (if any) must be jump discontinuities

4-28 THEOREM *If f is increasing on $[a, b]$, then $f(x_0+)$ and $f(x_0-)$ both exist for each x_0 in (a, b) and we have*

$$f(x_0-) \leq f(x_0) \leq f(x_0+)$$

At the endpoints we have $f(a) \leq f(a+)$ and $f(b-) \leq f(b)$

Proof Let $A = \{f(x) \mid a < x < x_0\}$ Since f is increasing, this set is bounded above by $f(x_0)$. If we put $c = \sup A$, then $c \leq f(x_0)$ We shall show that $f(x_0-)$ exists and that it equals c To do this, we must show that for every $\epsilon > 0$, there exists a $\delta > 0$ such that $x_0 - \delta < x < x_0$ implies $|f(x) - c| < \epsilon$ But since $c = \sup A$, there are elements $f(x)$ of A such that $c - \epsilon < f(x) \leq c$ Let $x_1 < x_0$ be such that $c - \epsilon < f(x_1) \leq c$. Then for every x in (x_1, x_0) we must also have $c - \epsilon < f(x) \leq c$, and hence $|f(x) - c| < \epsilon$ Therefore the number $\delta = x_0 - x_1$ has the required property (The proof that $f(x_0+)$ exists and is $\geq f(x_0)$ is similar, and only trivial modifications are needed for the endpoints)

4-29 THEOREM *Let f be strictly increasing on $[a, b]$ Then we have.*

(i) *The inverse function f^{-1} exists and is strictly increasing on its domain*

(ii) *If f is continuous on $[a, b]$, then f^{-1} is continuous on $[f(a), f(b)]$*

Proof We need only prove (i), since (ii) follows immediately from the more general result in Theorem 4-17 To prove (i), we first observe that f is one-to-one, since it is strictly increasing, and hence f^{-1} exists To see that f^{-1} is also strictly increasing, let $y_1 < y_2$ be two points in the domain of f^{-1} Then there are points x_1, x_2 in $[a, b]$ such that $y_1 = f(x_1)$, $y_2 = f(x_2)$ We cannot have $x_1 \geq x_2$, for then we would also have $y_1 \geq y_2$ The only alternative is $x_1 < x_2$, and this means that f^{-1} is strictly increasing

Theorem 4-29 tells us that a continuous, strictly increasing function is a topological mapping Conversely, every topological mapping of an interval $[a, b]$ onto an interval $[c, d]$ must be a strictly monotonic function The verification of this fact will be an instructive exercise for the reader. (See Exercise 4-31)

4-14 Necessary and sufficient conditions for continuity. Continuous functions can be characterized in a very elegant fashion if we introduce a slight generalization of the notions of open and closed sets.

4-30 DEFINITION. *Let A and B be two sets in E_n, with $A \subset B$. We say that A is closed relative to B if those accumulation points of A which lie in B are also in A. We say that A is open relative to B if the complement $B - A$ is closed relative to B.*

It is clear that every closed set in E_n is closed relative to E_n, and every open set in E_n is open relative to E_n. It is also clear that every set is both open and closed relative to itself. A set which is open relative to B need not, of course, be an open set (relative to E_n). However, such a set must be the intersection of B with an open set. This statement, which will be proved in Theorem 4-31, is illustrated by Fig. 4-8. The shaded region (which includes the arc abc, except for the endpoints a and c) is open relative to B.

4-31 THEOREM. *Assume that $A \subset B \subset E_n$. Then A is open relative to B if, and only if, A is the intersection of B with a set which is open relative to E_n.*

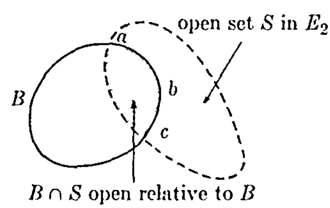

FIG. 4-8. Theorem 4-31.

Proof. Let $A = B \cap S$, where S is open. We shall show that $B - A$ is closed relative to B. Let $\mathbf{x}$ be an accumulation point of $B - A$ which lies in B. We must show that $\mathbf{x} \in B - A$, that is, that $\mathbf{x} \notin A$. Since $\mathbf{x} \in B$ and $A = B \cap S$, it suffices to show that $\mathbf{x} \notin S$. But $\mathbf{x}$ is also an accumulation point of $E_n - S$, since $B - A = B - S \subset E_n - S$. Now $E_n - S$ is closed, so $\mathbf{x} \in E_n - S$, $\mathbf{x} \notin S$. This proves that A is open relative to B.

Conversely, let A be open relative to B. Then $A \subset B$ and we must find an open set S such that $A = B \cap S$. Now $B - A$ is closed relative to B, which means that the accumulation points of $B - A$ which lie in B must also lie in $B - A$. That is, the accumulation points of $B - A$ cannot lie in A. Therefore, if $\mathbf{x} \in A$, then $\mathbf{x}$ is not an accumulation point of $B - A$, and hence there is some neighborhood $N(\mathbf{x})$ such that $N(\mathbf{x}) \cap (B - A)$ is empty. Let S be the union of all such neighborhoods $N(\mathbf{x})$ obtained as $\mathbf{x}$ ranges through A. Then S is an open set containing A (by construction). Hence $A \subset B \cap S$. But if $\mathbf{x} \in B \cap S$ there is a neighborhood $N(\mathbf{y})$ such that $N(\mathbf{y}) \cap (B - A)$ is empty, where $\mathbf{y} \in A$ and $\mathbf{x} \in N(\mathbf{y})$. Therefore $\mathbf{x} \notin (B - A)$. But since $\mathbf{x} \in B$, we must have $\mathbf{x} \in B - (B - A) = A$. This shows that $A = B \cap S$, and the proof is complete.

Of course, Theorem 4-31 also holds with "open" replaced by "closed" throughout. Using these concepts, we can give the following characterization of continuous functions.

4-32 THEOREM. *Let* $\mathbf{f}$ *be a function defined on a set* S *in* E_n, *and let* $T = \mathbf{f}(S)$ *be a subset of* E_m. *Then the following three statements are equivalent:*

(a) *The function* $\mathbf{f}$ *is continuous on* S.

(b) *If* Y *is closed relative to* T, *then* $\mathbf{f}^{-1}(Y)$ *is closed relative to* S.

(c) *If* Y *is open relative to* T, *then* $\mathbf{f}^{-1}(Y)$ *is open relative to* S.

Proof. Assume that (a) holds and let Y be closed relative to T. If $\mathbf{x}$ is an accumulation point of $\mathbf{f}^{-1}(Y)$ which is also in S, we will show that $\mathbf{x} \in \mathbf{f}^{-1}(Y)$ or, what is the same thing, that $\mathbf{f}(\mathbf{x}) \in Y$. There exists a sequence $\{\mathbf{x}_k\}$, whose terms are points in $\mathbf{f}^{-1}(Y)$, such that $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{x}$. Since $\mathbf{x} \in S$, by continuity we have $\lim_{k\to\infty} \mathbf{f}(\mathbf{x}_k) = \mathbf{f}(\mathbf{x})$. But each $\mathbf{f}(\mathbf{x}_k) \in Y$ and the limit $\mathbf{f}(\mathbf{x}) \in T$, hence $\mathbf{f}(\mathbf{x}) \in Y$, since Y is closed relative to T. Therefore, (a) implies (b).

Next, assume that (b) holds and let Y be open relative to T. Then $T - Y$ is closed relative to T, and hence $\mathbf{f}^{-1}(T - Y)$ is closed relative to S. But $\mathbf{f}^{-1}(T - Y) = S - \mathbf{f}^{-1}(Y)$. (The reader should verify this equation.) Therefore, $\mathbf{f}^{-1}(Y)$ must be open relative to S, and hence (b) implies (c).

Finally, assume that (c) holds. To show that $\mathbf{f}$ is continuous on S, we take a point $\mathbf{a}$ in S and let $\mathbf{b} = \mathbf{f}(\mathbf{a})$ be the corresponding point in T. We must find, then, for every neighborhood $N(\mathbf{b})$, a neighborhood $N(\mathbf{a})$ such that

$$\mathbf{x} \in N(\mathbf{a}) \cap S \quad \text{implies} \quad \mathbf{f}(\mathbf{x}) \in N(\mathbf{b})$$

Given $N(\mathbf{b})$, the set $T_1 = N(\mathbf{b}) \cap T$ is open relative to T, and hence, by (c), $\mathbf{f}^{-1}(T_1)$ is open relative to S. Therefore, $\mathbf{f}^{-1}(T_1) = S \cap U$, where U is open. But $\mathbf{a} \in \mathbf{f}^{-1}(T_1)$ and hence $\mathbf{a} \in U$. Therefore there exists a neighborhood $N(\mathbf{a}) \subset U$ *such that* $N(\mathbf{a}) \cap S \subset \mathbf{f}^{-1}(T_1)$. *(See Fig. 4-9.) But* this means that if $\mathbf{x} \in N(\mathbf{a}) \cap S$, then $\mathbf{x} \in \mathbf{f}^{-1}(T_1)$, or $\mathbf{f}(\mathbf{x}) \in T_1 \subset N(\mathbf{b})$. Therefore $\mathbf{f}$ is continuous at the point $\mathbf{a}$, and this completes the proof.

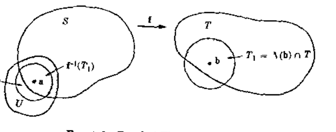

FIG. 4-9. Proof of Theorem 4-32.

Exercises

Note. Unless otherwise stated, all functions in the following exercises are real-valued.

Limits.

4–1. Prove statements (i) and (iii) of Theorem 4–8.

4–2. Compute the following limits (whenever they exist):

$$\text{(a)} \quad \lim_{\substack{(x,y)\to(0,0)\\(x,y)\in S}} \frac{xy}{x^2+y^2}, \qquad \text{(b)} \quad \lim_{\substack{(x,y)\to(0,0)\\(x,y)\in S}} \frac{x^2-y^2}{x^2+y^2},$$

when S is each of the following sets in E_2:

(i) $S = \{(x, y) \mid y = ax\}$ $(a \neq 0)$.
(ii) $S = \{(x, y) \mid y = ax^2\}$ $(a \neq 0)$.
(iii) $S = \{(x, y) \mid y^2 = ax\}$ $(a \neq 0)$.
(iv) $S = E_2$.

4–3. Let f be defined on the interval (a, b) and assume $x \in (a, b)$. Consider the two statements

(i) $\lim_{h\to 0} |f(x + h) - f(x)| = 0.$

(ii) $\lim_{h\to 0} |f(x + h) - f(x - h)| = 0.$

(a) Show that (i) always implies (ii).
(b) Give an example in which (ii) holds but (i) does not hold.

4–4. Consider the functions f defined in E_2 as follows:

(a) $f(x, y) = \dfrac{x^2 - y^2}{x^2 + y^2}$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$.

(b) $f(x, y) = \dfrac{(xy)^2}{(xy)^2 + (x - y)^2}$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$.

(c) $f(x, y) = \dfrac{1}{x} \sin (xy)$ if $x \neq 0$, $f(0, y) = y$.

$$\text{(d)} \quad f(x, y) = \begin{cases} (x + y) \sin (1/x) \sin (1/y) & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } x = 0 \text{ or } y = 0. \end{cases}$$

(e) $f(x, y) = \dfrac{\sin x - \sin y}{\tan x - \tan y}$ if $\tan x \neq \tan y$,

$f(x, y) = \cos^3 x$ if $\tan x = \tan y$

In each of the preceding examples, determine whether the following limits exist and evaluate those limits that do exist

$$\lim_{x\to 0} [\lim_{y\to 0} f(x, y)], \qquad \lim_{y\to 0} [\lim_{x\to 0} f(x, y)], \qquad \lim_{(x, y)\to(0, 0)} f(x, y)$$

4–5. If x is any real number in $[0, 1]$, show that the following limit exists

$$\lim_{m\to\infty} [\lim_{n\to\infty} \cos^{2n} (m!\pi x)],$$

and that its value is 0 or 1, according to whether x is irrational or rational

4–6. Let $f_1, \ldots, f_n$ be n real-valued functions defined on a set S in E_m, and consider the vector-valued function $\mathbf{f}$ defined for each $\mathbf{x}$ in S by the equation $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_n(\mathbf{x}))$. Let $\mathbf{a}$ be an accumulation point of S and assume $\mathbf{b} \in E_n$. Show that the equation

$$\lim_{\mathbf{x}\to\mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{b}, \qquad \mathbf{b} = (b_1, \ldots, b_n)$$

is equivalent to the n equations

$$\lim_{\mathbf{x}\to\mathbf{a}} f_k(\mathbf{x}) = b_k \qquad (k = 1, 2, \ldots, n)$$

$$\left[\textit{Hint}\quad |f_k(\mathbf{x}) - b_k| \le |\mathbf{f}(\mathbf{x}) - \mathbf{b}| \le \sum_{k=1}^{n} |f_k(\mathbf{x}) - b_k|\right]$$

Continuity of real-valued functions

4–7. Let f be continuous on $[a, b]$ and let $f(x) = 0$ when x is rational. Show that $f(x) = 0$ for every x in $[a, b]$

4–8 Let f be continuous at the point $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ in E_n. Keep $a_2, a_3, \ldots, a_n$ fixed and define a new function g of one real variable by the equation

$$g(x) = f(x, a_2, \ldots, a_n)$$

Show that g is continuous at the point $x = a_1$. (This is sometimes stated as follows "A continuous function of n variables is continuous in each variable separately")

4–9. Show by an example that the converse of the statement in Exercise 4–8 is not true in general

4–10. Let f, g, and h be defined on $[0, 1]$ as follows:

$$\begin{aligned} f(x) &= g(x) = h(x) = 0, && \text{whenever } x \text{ is irrational;} \\ f(x) &= 1 \text{ and } g(x) = x, && \text{whenever } x \text{ is rational;} \\ h(x) &= 1/n, \text{ if } x \text{ is the rational number } m/n \text{ (in lowest terms);} \\ h(0) &= 1. \end{aligned}$$

Show that f is not continuous anywhere in $[0, 1]$, that g is continuous only at $x = 0$, and that h is continuous only at the irrational points in $[0, 1]$.

4–11. For each x in $[0, 1]$, let $f(x) = x$ if x is rational, and let $f(x) = 1 - x$ if x is irrational. Show that

(a) f is continuous only at the point $x = \frac{1}{2}$.
(b) f assumes every value between 0 and 1.

4–12. Let f be defined on E_1 and assume that there exists at least one point x_0 in E_1 at which f is continuous. Suppose also that, for every x and y in E_1, f satisfies the equation

$$f(x + y) = f(x) + f(y).$$

Show that there exists a constant a such that $f(x) = ax$ for all x.

4–13. Let $f_1, \ldots, f_m$ be m real-valued functions defined on a set S in E_n. Assume that each f_k is continuous at the point $\mathbf{a}$ of S. Define a new function f as follows: For each $\mathbf{x}$ in S, $f(\mathbf{x})$ is the largest of the m numbers $f_1(\mathbf{x}), \ldots, f_m(\mathbf{x})$. Discuss the continuity of f at $\mathbf{a}$.

4–14. Let f be defined and bounded on the closed interval $S = [a, b]$. (That is, assume that $f(S)$ is a bounded set.) If T is a subset of S, the number

$$\Omega_f(T) = \sup \{f(x) - f(y) \mid x \in T, y \in T\}$$

is called the *oscillation* of f on T. If $x \in S$, the oscillation of f at x is defined to be the number

$$\omega_f(x) = \lim_{h \to 0+} \Omega_f(N(x; h) \cap S).$$

Show that this limit always exists and that $\omega_f(x) = 0$ if, and only if, f is continuous at x.

4–15. Let f be continuous on $[a, b]$ and define g as follows: $g(a) = f(a)$ and, for $a < x \le b$, let $g(x)$ be the maximum value of f in the subinterval $[a, x]$. Show that g is continuous on $[a, b]$.

4–16. Let f be continuous on $[a, b]$ and let S be the square in the xy-plane given by $a \le x \le b$, $a \le y \le b$. Define g on S as follows: $g(x, x) = f(x)$; for every point (x, y) in S with $a \le x < y \le b$, let $g(x, y)$ be the maximum value of f in the subinterval $[x, y]$ of $[a, b]$. If $x > y$, $g(x, y) = g(y, x)$. Show that g is continuous on S.

4–17. Let f be defined and continuous on a closed set S in E_n. Let

$$A = \{\mathbf{x} \mid \mathbf{x} \in S, f(\mathbf{x}) = 0\}$$

Show that A is a closed set

4–18. Let f be continuous on an open set S in E_n, assume that $\mathbf{x}_0 \in S$, and assume that $f(\mathbf{x}_0) > 0$ Show that there is a neighborhood $N(\mathbf{x}_0)$ such that $f(\mathbf{x}) > 0$ for every $\mathbf{x}$ in this neighborhood

4–19. Let f be continuous on a closed interval $[a, b]$ Suppose that f has a local maximum at x_1 and a local maximum at x_2 Show that there must be a third point between x_1 and x_2 where f has a local minimum.

4–20. Let f be a function defined on $[0, 1]$ with the following property: For every real number y, either there is no x in $[0, 1]$ for which $f(x) = y$ or there are exactly *two* values of x in $[0, 1]$ for which $f(x) = y$

(a) Show that f cannot be continuous on $[0, 1]$

(b) Construct a function f which has the above property.

Uniform continuity

4–21. Show that a function which is uniformly continuous on a set S is also continuous on S.

4–22. Show that the function f defined by $f(x) = x^2$ is not uniformly continuous on E_1.

4–23. Assume that f is uniformly continuous on a bounded set S in E_n. Show that f must be bounded on S, that is, show that there is a constant $M > 0$ such that $|f(\mathbf{x})| \leq M$ for all $\mathbf{x}$ in S.

4–24. Let $\mathbf{f}$ be defined on a set S in E_n with function values in E_m. Instead of using Definition 4–23, define uniform continuity as follows

DEFINITION. *Let $\mathbf{f}$ be continuous on S Then $\mathbf{f}$ is called uniformly continuous on S if, given $\epsilon > 0$, there exists a $\delta > 0$ (depending only on ϵ) such that whenever $\mathbf{x}$ and $\mathbf{y}$ are two accumulation points of S with $|\mathbf{x} - \mathbf{y}| < \delta$, then*

$$|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})| < \epsilon$$

Show that Heine's theorem (Theorem 4–24) is still valid with this definition of uniform continuity.

4–25. Let $\mathbf{f}$ be a function defined on a set S in E_n and assume that $\mathbf{f}(S) \subset E_m$ Let $\mathbf{g}$ be defined on $\mathbf{f}(S)$ with values in E_k, and let $\mathbf{gf}$ denote the composite function defined by $\mathbf{gf}(\mathbf{x}) = \mathbf{g}[\mathbf{f}(\mathbf{x})]$, if $\mathbf{x} \in S$. If $\mathbf{f}$ is uniformly continuous on S and if $\mathbf{g}$ is uniformly continuous on $\mathbf{f}(S)$, show that $\mathbf{gf}$ is uniformly continuous on S

4–26. Prove by direct application of the definition (without using the Heine-Borel covering theorem) that a polynomial is uniformly continuous on every finite closed interval in E_1.

Discontinuities.

4–27. Locate and classify the discontinuities of the functions f defined on E_1 by the following equations:

(a) $f(x) = (\sin x)/x$ if $x \neq 0, f(0) = 0.$
(b) $f(x) = e^{1/x}$ if $x \neq 0, f(0) = 0.$
(c) $f(x) = e^{1/x} + \sin(1/x)$ if $x \neq 0, f(0) = 0.$
(d) $f(x) = 1/(1 - e^{1/x})$ if $x \neq 0, f(0) = 0.$

4–28. Locate the points in E_2 at which each of the functions in Exercise 4–4 is not continuous.

Monotonic functions.

4–29. Let f be defined in the open interval (a, b) and assume that for each interior point x of (a, b) there exists a neighborhood $N(x)$ in which f is increasing. Show that f is an increasing function throughout (a, b).

4–30. Let f be continuous on the closed interval $[a, b]$ and assume that f does not have a local maximum or a local minimum at any interior point. Show that f must be monotonic on $[a, b]$.

4–31. If f is one-to-one and continuous on $[a, b]$, show that f must be strictly monotonic on $[a, b]$. That is, prove that every topological mapping of $[a, b]$ onto an interval $[c, d]$ must be strictly monotonic.

4–32. Let f be an increasing function defined on $[a, b]$ and let $x_1, \ldots, x_n$ be n points in the interior such that $a < x_1 < x_2 < \cdots < x_n < b$.

(a) Show that $\sum_{k=1}^{n} [f(x_k+) - f(x_k-)] \leq f(b-) - f(a+)$.
(b) Deduce from part (a) that the set of discontinuities of f is countable.

CHAPTER 5

DIFFERENTIATION OF FUNCTIONS OF ONE REAL VARIABLE

5–1 Introduction. That branch of mathematics known as differential calculus centers around a special limit process, *differentiation*, which will be considered in detail in this chapter. Two different types of problems—the physical problem of finding the instantaneous velocity of a moving particle, and the geometrical problem of finding the tangent line to a curve at a given point—both lead quite naturally to the same basic idea which underlies the notion of derivative. Here, we shall not be concerned so much with the applications of differentiation to either mechanics or geometry, but instead will confine ourselves to a study of the general mathematical properties of the derivative. This chapter deals with differentiation of *real-valued functions* defined on open intervals in E_1 and Chapter 6 will treat generalizations to E_n.

5–2 Definition of derivative. If f is defined on an open interval (a, b), then for two distinct points x and x_0 in (a, b) we can form the *difference quotient*

$$\frac{f(x) - f(x_0)}{x - x_0}.$$

We keep x_0 fixed and study the behavior of this quotient as x assumes values arbitrarily close to x_0.

5–1 DEFINITION. *Let f be defined on the open interval (a, b), and assume that $x_0 \in (a, b)$. Then f is said to have a derivative at x_0 whenever the limit*

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists. This limit, denoted by $f'(x_0)$, is called the derivative of f at x_0.

We can think of the above limit process as defining, from a given function f, a new function f', whose domain consists of those points in (a, b) where f has a derivative. The function f' is called the *first derivative* of f. Similarly, the nth derivative of f, denoted by $f^{(n)}$, is defined to be the first derivative of $f^{(n-1)}$, $n = 2, 3, \ldots$. (By our definition, it does not make sense to consider $f^{(n)}$ unless $f^{(n-1)}$ is defined on an open interval.) Other notations with which the reader may be familiar are

$$f'(x) = Df(x) = \frac{df(x)}{dx} = \frac{dy}{dx} \qquad [\text{where } y = f(x)],$$

or similar notations. The function f itself is sometimes written $f^{(0)}$

5–2 THEOREM. *If f has a derivative at a point x_0 in (a, b), then f is continuous at x_0.*

Proof. If $x \in (a, b)$, $x \neq x_0$, we can write

$$f(x) - f(x_0) = \frac{f(x) - f(x_0)}{x - x_0}(x - x_0).$$

Applying Theorem 4–8 (ii), we find $\lim_{x\to x_0} f(x) = f(x_0)$. This proves the assertion.

The next theorem gives us somewhat more information about the behavior of f in the immediate vicinity of a point where the derivative exists.

5–3 THEOREM. *If f has a derivative at an interior point x_0 of (a, b), then there is a neighborhood $N(x_0)$ and a positive number M such that $x \in N'(x_0)$ implies $|f(x) - f(x_0)| < M|x - x_0|$.*

Proof. Given $\epsilon > 0$, there is a neighborhood $N(x_0) \subset (a, b)$ such that

$$x \in N'(x_0) \quad \text{implies} \quad \left|\frac{f(x) - f(x_0)}{x - x_0} - f'(x_0)\right| < \epsilon.$$

If we take the neighborhood corresponding to $\epsilon = 1$, we can write

$$\left|\frac{f(x) - f(x_0)}{x - x_0}\right| < |f'(x_0)| + 1$$

whenever $x \in N'(x_0)$. The conclusion of the theorem follows by taking $M = 1 + |f'(x_0)|$.

Functions for which the conclusion of Theorem 5–3 holds are said to satisfy a *Lipschitz condition* at x_0. Geometrically, this means that the graph of the function must lie between the graphs of the two straight lines $y - f(x_0) = M(x - x_0)$ and $y - f(x_0) = -M(x - x_0)$ whenever $x \in N'(x_0)$. (See Fig. 5–1.) The number M, of course, depends on

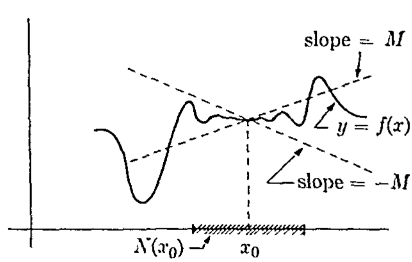

FIG. 5–1. Lipschitz condition at x_0.

x_0. Functions which satisfy a Lipschitz condition at x_0 are automatically continuous at x_0, but the converse is not true. (See Exercise 5-1.)

5-3 Algebra of derivatives. The reader should have no difficulty in applying the usual proofs of elementary calculus to derive the following theorem.

5-4 THEOREM. *If f and g are defined on (a, b), then at those points where f and g have derivatives, the functions $f + g$, $f - g$, and $f \cdot g$ also have derivatives. This is also true of f/g at points x where $g(x) \neq 0$. These derivatives are given by the following formulas:*

$$(f \pm g)' = f' \pm g',$$
$$(f \cdot g)' = f \cdot g' + f' \cdot g,$$
$$(f/g)' = [g \cdot f' - f \cdot g']/g^2 \qquad (\textit{at points } x \textit{ where } g(x) \neq 0).$$

From the definition we see at once that if $f(x)$ is constant for all x, then $f'(x)$ is always 0. Also, if $f(x) = x$, then $f'(x) = 1$ for all x. Repeated application of Theorem 5-4 tells us that if $f(x) = x^n$ (n a positive integer), then $f'(x) = nx^{n-1}$ for all x. Applying Theorem 5-4 again, we see that every polynomial has a derivative everywhere in E_1 and every rational function has a derivative wherever it is defined.

5-4 The chain rule. A much deeper result concerning derivatives is the so-called *chain rule* for differentiating a composite function.

5-5 THEOREM (*Chain rule*). *Let f be continuous on a closed interval S and let $f(S)$ be the image of S under f. Let g be defined on $f(S)$ and consider the composite function gf defined for each x in S by $gf(x) = g[f(x)]$. Assume that x_0 is an interior point of S such that $y_0 = f(x_0)$ is an interior point of $f(S)$. Assume, further, that both $f'(x_0)$ and $g'(y_0)$ exist. Then gf has a derivative at x_0 and its value is $f'(x_0)g'(y_0)$. That is,*

$$(gf)'(x_0) = g'[f(x_0)]f'(x_0)$$

Proof. We must consider the limit of the quotient

$$\frac{gf(x) - gf(x_0)}{x - x_0}$$

as x tends to x_0. As a first attempt toward the proof, it seems natural to write

$$\frac{gf(x) - gf(x_0)}{x - x_0} = \frac{g[f(x)] - g[f(x_0)]}{f(x) - f(x_0)} \cdot \frac{f(x) - f(x_0)}{x - x_0}.$$

If we let $x \to x_0$, then the second factor on the right has $f'(x_0)$ as a limit and the first factor should have $g'[f(x_0)]$ as a limit. However, this argument fails because the denominator $f(x) - f(x_0)$ of the first factor might be zero in every neighborhood of x_0, and hence the first factor would be undefined at such x's. Therefore a more involved argument is necessary.

We write $y_0 = f(x_0)$ and define a new function h as follows:

$$h(y) = \frac{g(y) - g(y_0)}{y - y_0} - g'(y_0) \qquad \text{if } y \neq y_0,\, y \in f(S), \qquad h(y_0) = 0.$$

Then we have $\lim_{y \to y_0} h(y) = 0$ because $g'(y_0)$ is assumed to exist. Hence h is continuous at y_0. Now h is defined for every point of $f(S)$ and hence it makes sense to consider the composite function hf. Since f is continuous at x_0 and since h is continuous at $y_0 = f(x_0)$, the theorem on continuity of composite functions (Theorem 4-11) tells us that we have

$$\lim_{x \to x_0} hf(x) = h[f(x_0)] = h(y_0) = 0.$$

Taking $y = f(x)$ in the definition of h, we can write

$$hf(x) = \frac{g[f(x)] - g(y_0)}{f(x) - y_0} - g'(y_0), \qquad \text{if } f(x) \neq y_0,$$

or

$$gf(x) - g(y_0) = [hf(x) + g'(y_0)][f(x) - y_0],$$

and this equation holds even if $f(x) = y_0$. Now keep $x \neq x_0$, and divide both sides by $x - x_0$ to obtain

$$\frac{gf(x) - gf(x_0)}{x - x_0} = [hf(x) + g'(y_0)] \frac{f(x) - f(x_0)}{x - x_0}.$$

We are now ready to consider the limit as $x \to x_0$. The second factor on the right has the limit $f'(x_0)$ and we have just seen that $\lim_{x \to x_0} hf(x) = 0$. Hence the left side must have the limit $g'(y_0)f'(x_0)$, and this completes the proof.

5-5 One-sided derivatives and infinite derivatives. Up to this point, the statement that f has a derivative at x_0 has meant that x_0 was *interior* to an interval in which f was defined and that the limit defining $f'(x_0)$ was *finite*. It is convenient to extend the scope of our ideas somewhat in order to discuss derivatives at endpoints of intervals. It is also desirable to introduce *infinite* derivatives, so that the usual geometric interpretation of a derivative as the slope of a tangent line will still be valid in case the tangent line happens to be vertical. Because in such a case we cannot prove that f is continuous at x_0, we explicitly require it to be so.

5-6 DEFINITION *Let f be defined on a closed interval S and assume that f is continuous at the point x_0 in S. Then f is said to have a right-hand derivative at x_0 if the right-hand limit*

$$\lim_{x \to x_0 +} \frac{f(x) - f(x_0)}{x - x_0}$$

exists as a finite value, or if the limit is $+\infty$ or $-\infty$. This limit will be denoted by $f'_+(x_0)$. Left-hand derivatives, denoted by $f'_-(x_0)$, are similarly defined. In addition, if x_0 is an interior point of S, then we say that f has the derivative $f'(x_0) = +\infty$ if both the right- and left-hand derivatives at x_0 are $+\infty$. (The derivative $f'(x_0) = -\infty$ is similarly defined.)

It is clear that f has a derivative at x_0 if, and only if, $f'_+(x_0) = f'_-(x_0) = f'(x_0)$.

Figure 5-2 illustrates some of these concepts. At the point x_1 we have $f'_+(x_1) = -\infty$. At x_2 the left-hand derivative is 0 and the right-hand derivative is -1. Also, $f'(x_3) = -\infty$, $f'_-(x_4) = -1$, $f'_+(x_4) = +1$, $f'(x_6) = +\infty$, and $f'_-(x_7) = 2$. There is no derivative (one-sided or otherwise) at x_5, since f is not continuous there.

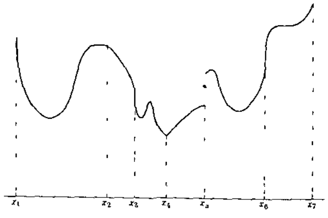

FIG. 5-2 One-sided derivatives and infinite derivatives

5–6 Functions with nonzero derivative.

5–7 THEOREM. *Let f be defined on the open interval (a, b) and suppose that for some point x_0 in (a, b) we have $f'(x_0) > 0$ or $f'(x_0) = +\infty$. Then there is a neighborhood $N(x_0) \subset (a, b)$ such that for each x in $N'(x_0)$ we have $f(x) > f(x_0)$ if $x > x_0$, and $f(x) < f(x_0)$ if $x < x_0$.*

Proof. Suppose $f'(x_0)$ is finite and positive. Then for every $\epsilon > 0$ there is a neighborhood $N(x_0) \subset (a, b)$ such that

$$x \in N'(x_0) \quad \text{implies} \quad \left|\frac{f(x) - f(x_0)}{x - x_0} - f'(x_0)\right| < \epsilon.$$

Take the neighborhood corresponding to $\epsilon = \frac{1}{2}f'(x_0)$. Then if $x \in N'(x_0)$, we have

$$-\frac{1}{2}f'(x_0) < \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) < \frac{1}{2}f'(x_0),$$

or

$$0 < \frac{1}{2}f'(x_0) < \frac{f(x) - f(x_0)}{x - x_0} < \frac{3}{2}f'(x_0).$$

Hence in this neighborhood the quotient $[f(x) - f(x_0)]/(x - x_0)$ is positive. But this implies that $f(x) - f(x_0)$ has the same sign as $x - x_0$.

If $f'(x_0) = +\infty$, there will be a neighborhood $N(x_0)$ such that

$$\frac{f(x) - f(x_0)}{x - x_0} > 1, \qquad \text{whenever } x \in N'(x_0).$$

In this neighborhood the quotient is again positive and the conclusion follows as before.

A result similar to Theorem 5–7 holds, of course, if $f'(x_0) < 0$ or $f'(x_0) = -\infty$ at some interior point of (a, b).

5–7 Functions with zero derivative. We can use Theorem 5–7 to obtain the following important result which shows a connection between derivatives and local maxima and minima.

5–8 THEOREM. *Let f be defined on an open interval (a, b) and assume that f has a local maximum or a local minimum at an interior point x_0 of (a, b). If f has a derivative at x_0, then $f'(x_0)$ must be 0.*

Proof. If $f'(x_0)$ is positive or $+\infty$, then f cannot have a local maximum or a local minimum at x_0 because of Theorem 5–7. Similarly, $f'(x_0)$ cannot be negative or $-\infty$. However, because there *is* a derivative at x_0, the only other possibility is $f'(x_0) = 0$.

The converse of Theorem 5–8 is not true In general, knowing that $f'(x_0) = 0$ is not enough to determine whether f has a maximum or minimum at x_0 In fact, it may have neither, as can be verified by the example in which $f(x) = x^3$ and $x_0 = 0$ In this case, $f'(0) = 0$ but f is increasing in every neighborhood of 0

Furthermore, it should be emphasized that f can have a local maximum or minimum at x_0 without $f'(x_0)$ being zero The example in which $f(x) = |x|$ illustrates a case where the function has a minimum at $x = 0$ but, of course, there is no derivative at 0 The theorem just proved (5–8) assumes that f must have a derivative (finite or infinite) at x_0 The theorem also assumes that x_0 must be an *interior* point of (a, b) In the example $f(x) = x$, $a \le x \le b$, f takes on its maximum and minimum at the endpoints but $f'(x)$ is never zero in $[a, b]$

5–8 Rolle's theorem. It is geometrically evident that a sufficiently "smooth" curve which crosses the x-axis at both endpoints of an interval $[a, b]$ must have a "turning point" somewhere between a and b The precise statement of this fact is a theorem of great importance in calculus, known as *Rolle's theorem.* It is contained in part (i) of the following theorem.

5–9 THEOREM *Let f be defined in the open interval (a, b) and assume that f has a derivative (finite or infinite) at each interior point. Assume also that both limits $f(a+) = \lim_{x\to a+} f(x)$ and $f(b-) = \lim_{x\to b-} f(x)$ exist (and are finite) Then we have:*

(i) *If $f(a+) = f(b-)$, there exists at least one interior point x_0 of (a, b) such that $f'(x_0) = 0$*

(ii) *If $f'(x) \neq 0$ for every x in (a, b), then f is monotonic in (a, b). In fact, f is strictly increasing if $f(a+) < f(b-)$ and strictly decreasing if $f(a+) > f(b-)$.*

Proof The hypothesis implies that f is continuous on the open interval (a, b) Define a new function g as follows

$$g(x) = f(x) \quad \text{if } x \in (a, b), \qquad g(a) = f(a+), \qquad g(b) = f(b-).$$

Then g is continuous on the closed interval $[a, b]$ and hence attains its maximum M and its minimum m somewhere in $[a, b]$. If $f'(x)$ is never zero in (a, b), Theorem 5–8 tells us that the extreme values of g cannot be attained at any interior points

To prove (i), assume that $f(a+) = f(b-)$ If $f'(x) \neq 0$ for every x in (a, b), we must have $m = g(a) = g(b) = M$ (by the preceding remarks). This implies that f is constant on (a, b). Therefore, the nonvanishing of f' in (a, b) is incompatible with the equation $f(a+) = f(b-)$. This proves (i)

To prove (ii), assume that $f'(x) \neq 0$ for every x in (a, b) and assume that $f(a+) < f(b-)$. Then $m = g(a) < g(b) = M$, and therefore $g(a) < g(x) < g(b)$ for every x in (a, b). Taking a fixed value of x, say $x = x_1$, $a < x_1 < b$, and applying the same argument to the closed interval $[x_1, b]$, we find that

$$g(x_1) < g(x) < g(b), \qquad \text{if } x_1 < x < b.$$

But this means that $f(x_1) < f(x)$ whenever $a < x_1 < x < b$. Therefore f is strictly increasing on (a, b). (If $f(a+) > f(b-)$, the argument is similar.) This proves (ii).

5–9 The Mean Value Theorem of differential calculus. One of the most useful tools of differential calculus is the *Mean Value Theorem.*

5–10 THEOREM (*The Mean Value Theorem of differential calculus*). *Let f be defined on the closed interval $[a, b]$ and assume that f has a derivative (finite or infinite) at each interior point. Assume further that the limits $f(a+)$ and $f(b-)$ both exist and satisfy the condition*

$$f(a) - f(a+) = f(b) - f(b-).$$

Then there is at least one interior point x_0 of (a, b) such that

$$f(b) - f(a) = f'(x_0)(b - a).$$

NOTE. The condition $f(a) - f(a+) = f(b) - f(b-)$ states that the right-hand jump at a and the left-hand jump at b must have the same magnitude but opposite signs. In particular, this condition is satisfied if f is continuous at both endpoints.

Geometrically, Theorem 5–10 states that on a sufficiently "smooth" curve joining two points A and B there is a tangent line to the curve having the same slope as the chord AB.

The Mean Value Theorem will be derived as a corollary of the following more general result.

5–11 THEOREM (*Generalized Mean Value Theorem*). *Let f and g be two functions defined on the closed interval $[a, b]$ and assume that each function has a derivative, finite or infinite, at each interior point. At the endpoints, assume that the limits $f(a+)$, $f(b-)$, $g(a+)$, and $g(b-)$ exist and satisfy the relation*

(i) $[f(a+) - f(b-)][g(a) - g(b)] = [f(a) - f(b)][g(a+) - g(b-)]$.

Then there is at least one interior point x_0 of (a, b) such that

(ii) $f'(x_0)[g(b) - g(a)] = g'(x_0)[f(b) - f(a)]$.

NOTE Hypothesis (i) is automatically satisfied if both f and g are continuous at the endpoints of $[a, b]$

Proof Define a new function h as follows.

$$h(x) = f(x)[g(a) - g(b)] - g(x)[f(a) - f(b)], \qquad \text{if } x \in [a, b]$$

Hypothesis (i) implies that $h(a+) = h(b-)$ Since h satisfies all the conditions of Rolle's theorem, we must have $h'(x_0) = 0$ for some interior point x_0 But this is equivalent to statement (ii) Theorem 5–10 is obtained when $g(x) = x$

NOTE The reader should interpret Theorem 5–11 geometrically by referring to the curve in the xy-plane described by the parametric equations $x = g(t)$, $y = f(t)$, $a \le t \le b$

We have already seen in Theorem 5–9 that f must be monotonic in (a, b) if f' never vanishes there Essentially the same result can be obtained as an immediate consequence of the Mean Value Theorem

5–12 THEOREM *If f is continuous on $[a, b]$ and if f has a derivative which takes on only positive values (finite or infinite) in the interior, then f is strictly increasing on $[a, b]$ If f' takes on only negative values (finite or infinite) in (a, b), then f is strictly decreasing on $[a, b]$ If $f'(x) = 0$ for each x in (a, b), then f is constant throughout $[a, b]$.*

Proof If we apply the Mean Value Theorem to an arbitrary subinterval $[x_1, x_2]$ of $[a, b]$, we find

$$f(x_2) - f(x_1) = f'(x_0)(x_2 - x_1), \qquad \text{where } x_0 \in (x_1, x_2)$$

All the statements of the theorem follow at once from this equation

By applying Theorem 5–12 to the difference $f - g$, we see that when two functions f and g are continuous in $[a, b]$ and have equal finite derivatives at each interior point, then f and g differ by a constant throughout $[a, b]$

5–10 Intermediate value theorem for derivatives. In Theorem 4–22 of the last chapter we proved that a function f which is continuous on a closed interval $[a, b]$ assumes every value between its maximum and its minimum on the interval In particular, f assumes every value between $f(a)$ and $f(b)$. A similar result will now be proved for functions which are derivatives

5–13 THEOREM (*Intermediate value theorem for derivatives*) *Assume that f is defined on the closed interval $[a, b]$ and that f has a derivative (finite or* *infinite) at each interior point. Assume also that f has finite one-sided derivatives $f'_+(a)$ and $f'_-(b)$ at the endpoints, with $f'_+(a) \neq f'_-(b)$. Then, if c is a real number between $f'_+(a)$ and $f'_-(b)$, there exists at least one interior point x such that $f'(x) = c$.*

Proof. Define a new function g as follows:

$$g(x) = \frac{f(x) - f(a)}{x - a} \qquad \text{if } x \neq a, \qquad g(a) = f'_+(a).$$

Then g is continuous on the closed interval $[a, b]$. By the intermediate value theorem for continuous functions, g takes on every value between $f'_+(a)$ and $[f(b) - f(a)]/(b - a)$ in the interior (a, b). By the Mean Value Theorem, we have $g(x) = f'(x_0)$ for some x_0 in (a, x) whenever $x \in (a, b)$. Therefore f' takes on every value between $f'_+(a)$ and $[f(b) - f(a)]/(b - a)$ in the interior (a, b). A similar argument applied to the function h, defined by

$$h(x) = \frac{f(x) - f(b)}{x - b} \qquad \text{if } x \neq b, \qquad h(b) = f'_-(b),$$

shows that f' takes on every value between $[f(b) - f(a)]/(b - a)$ and $f'_-(b)$ in the interior (a, b). Combining these results, we see that f' takes on every value between $f'_+(a)$ and $f'_-(b)$ in the interior (a, b), and this proves the theorem.

NOTE. Theorem 5–13 is still valid if one or both of the one-sided derivatives $f'_+(a), f'_-(b)$, is infinite. The proof in this case can be given by considering the auxiliary function g defined by the equation $g(x) = f(x) - cx$, if $x \in [a, b]$. Details are left to the reader. (See Exercise 5–22.)

5–11 Taylor's formula with remainder. The Mean Value Theorem has already been given a geometrical interpretation, but there is another way of looking at the theorem which also helps us gain an insight into its meaning.

If f is continuous on $[a, b]$ and has a derivative at each interior point, then, given x in $(a, b]$, we can write

$$f(x) = f(a) + f'(x_0)(x - a), \qquad \text{where } a < x_0 < x.$$

This equation says that the quantity $f'(x_0)(x - a)$ measures the error made when $f(x)$ is approximated by $f(a)$. Unfortunately, the Mean Value Theorem does not tell us how to find x_0; it simply says that $a < x_0 < x$.

If x is not too far from a, then $[f(x) - f(a)]/(x - a)$ will be approximately $f'(a)$. That is, $f'(x_0)$ will be nearly equal to $f'(a)$, and hence the equation

$$f(x) = f(a) + f'(a)(x - a)$$

should be approximately correct when $x - a$ is small. This means that f is approximately a linear function near a. Taylor's theorem tells us that, more generally, f can be approximated by a polynomial of degree $n - 1$ if $f^{(n)}$ exists in (a, b). The value of this theorem lies in the fact that it gives us a useful expression for the error made by this approximation.

5–14 THEOREM (*Taylor*). *Let f be a function having a finite* nth *derivative $f^{(n)}$ everywhere in the open interval (a, b) and assume that $f^{(n-1)}$ is continuous on the closed interval $[a, b]$. Assume that $x_0 \in [a, b]$. Then, for every x in $[a, b]$, $x \neq x_0$, there exists a point x_1 interior to the interval joining x and x_0 such that*

$$f(x) = f(x_0) + \sum_{k=1}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \frac{f^{(n)}(x_1)}{n!}(x - x_0)^n.$$

Taylor's theorem will be obtained as a consequence of a more general result that is a direct extension of the generalized Mean Value Theorem.

5–15 THEOREM. *Let f and g be two functions having finite* nth *derivatives $f^{(n)}$ and $g^{(n)}$ in the open interval (a, b) and continuous $(n - 1)$st derivatives in the closed interval $[a, b]$. Assume that $x_0 \in [a, b]$. Then, for every x in $[a, b]$, $x \neq x_0$, there exists a point x_1 interior to the interval joining x and x_0 such that*

$$\left[f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k\right] g^{(n)}(x_1)$$
$$= f^{(n)}(x_1)\left[g(x) - \sum_{k=0}^{n-1} \frac{g^{(k)}(x_0)}{k!}(x - x_0)^k\right].$$

NOTE. For the special case in which $g(x) = (x - x_0)^n$, we have $g^{(k)}(x_0) = 0$ for $0 \leq k \leq n - 1$ and $g^{(n)}(x) = n!$. This theorem then reduces to Taylor's theorem.

Proof. For simplicity, assume that $x_0 < b$ and that $x > x_0$. Keep x fixed and define new functions F and G as follows:

$$F(t) = f(t) + \sum_{k=1}^{n-1} \frac{f^{(k)}(t)}{k!} (x - t)^k,$$

$$G(t) = g(t) + \sum_{k=1}^{n-1} \frac{g^{(k)}(t)}{k!} (x - t)^k,$$

for each t in $[x_0, x]$. Then F and G are continuous on the closed interval $[x_0, x]$ and have finite derivatives in the open interval (a, x). Therefore, Theorem 5–11 is applicable and we can write

$$F'(x_1)[G(x) - G(x_0)] = G'(x_1)[F(x) - F(x_0)], \qquad \text{where } x_1 \in (x_0, x).$$

This reduces to the equation

$$\text{(a)} \qquad F'(x_1)[g(x) - G(x_0)] = G'(x_1)[f(x) - F(x_0)],$$

since $G(x) = g(x)$ and $F(x) = f(x)$. If, now, we compute the derivative of the sum defining $F(t)$, keeping in mind that each term of the sum is a product, we find that all terms cancel but one, and we are left with

$$F'(t) = \frac{(x - t)^{n-1}}{(n - 1)!} f^{(n)}(t).$$

Similarly, we obtain

$$G'(t) = \frac{(x - t)^{n-1}}{(n - 1)!} g^{(n)}(t).$$

If we put $t = x_1$ and substitute into (a), we obtain the formula of the theorem.

Exercises

In the following exercises assume, where necessary, a knowledge of the formulas for differentiating the elementary trigonometric, exponential, and logarithmic functions.

5–1. A function f is said to satisfy a Lipschitz condition of order α at x_0 if there exists a positive number M (which may depend on x_0) and a neighborhood $N(x_0)$ such that

$$x \in N'(x_0) \quad \text{implies} \quad |f(x) - f(x_0)| < M|x - x_0|^\alpha$$

(a) Show that a function which satisfies a Lipschitz condition of order α is continuous at x_0 if $\alpha > 0$, and has a derivative at x_0 if $\alpha > 1$.

(b) Give an example of a function satisfying a Lipschitz condition of order 1 at x_0 for which $f'(x_0)$ does not exist.

5–2. In each of the following cases, determine the intervals in which the function f is increasing or decreasing and find the maxima and minima (if any) in the set where each f is defined.

(a) $f(x) = x^3 + ax + b$, $\quad x \in E_1$

(b) $f(x) = \log (x^2 - 9)$, $\quad |x| > 3$.

(c) $f(x) = x^{2/3}(x - 1)^4$, $\quad 0 \le x \le 1$.

(d) $f(x) = (\sin x)/x$ if $x \ne 0$, $f(0) = 0$, $\quad 0 \le x \le \pi/2$

5–3. Find a polynomial f of lowest possible degree such that

$$f(x_1) = a_1, \qquad f(x_2) = a_2, \qquad f'(x_1) = b_1, \qquad f'(x_2) = b_2,$$

where $x_1 \ne x_2$ and a_1, a_2, b_1, b_2 are given real numbers.

5–4. Define f as follows: $f(x) = e^{-1/x^2}$ if $x \ne 0$, $f(0) = 0$. Show that

(a) f is continuous for all x.

(b) $f^{(n)}$ is continuous for all x, and that $f^{(n)}(0) = 0$, $(n = 1, 2, \ldots)$

5–5. Define f, g, and h as follows $f(0) = g(0) = h(0) = 0$ and, if $x \ne 0$, $f(x) = \sin (1/x)$, $g(x) = x \sin (1/x)$, $h(x) = x^2 \sin (1/x)$. Show that

(a) $f'(x) = -1/x^2 \cos (1/x)$, if $x \ne 0$, $\quad f'(0)$ does not exist.

(b) $g'(x) = \sin (1/x) - 1/x \cos (1/x)$, if $x \ne 0$; $\quad g'(0)$ does not exist.

(c) $h'(x) = 2x \sin (1/x) - \cos (1/x)$, if $x \ne 0$; $\quad h'(0) = 0$;

$\lim_{x\to 0} h'(x)$ does not exist.

5–6. Derive Leibnitz's formula for the nth derivative of the product h of two functions f and g:

$$h^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x), \qquad \text{where } \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

5–7. Let f and g be two functions defined and having finite third order derivatives $f'''(x)$ and $g'''(x)$ for all x in E_1. If $f(x)g(x) = 1$ for all x, show that the relations in (a), (b), (c), and (d) hold at those points where the denominators are not zero:

(a) $f'(x)/f(x) + g'(x)/g(x) = 0.$

(b) $f''(x)/f'(x) - 2f'(x)/f(x) - g''(x)/g'(x) = 0.$

(c) $\dfrac{f'''(x)}{f'(x)} - 3\dfrac{f'(x)g''(x)}{f(x)g'(x)} - 3\dfrac{f''(x)}{f(x)} - \dfrac{g'''(x)}{g'(x)} = 0.$

(d) $\dfrac{f'''(x)}{f'(x)} - \dfrac{3}{2}\left(\dfrac{f''(x)}{f'(x)}\right)^2 = \dfrac{g'''(x)}{g'(x)} - \dfrac{3}{2}\left(\dfrac{g''(x)}{g'(x)}\right)^2.$

NOTE. The expression which appears on the left side of (d) is called the *Schwarzian derivative* of f at x.

(e) Show that f and g have the same Schwarzian derivative if

$$g(x) = [af(x) + b]/[cf(x) + d], \text{ where } ad - bc \neq 0.$$

[*Hint:* If $c \neq 0$, write $(af + b)/(cf + d) = (a/c) + (bc - ad)/[c(cf + d)]$, and apply part (d).]

5–8. Let f_1, f_2, g_1, g_2 be four functions having derivatives in (a, b). Define F by means of the determinant

$$F(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{vmatrix}, \qquad \text{if } x \in (a, b).$$

(a) Show that $F'(x)$ exists for each x in (a, b) and that

$$F'(x) = \begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & f_2(x) \\ g_1'(x) & g_2'(x) \end{vmatrix}.$$

(b) State and prove a more general result for nth order determinants.

5–9. Given n functions $f_1, \ldots, f_n$, each having nth order derivatives in (a, b). A function W, called the *Wronskian* of $f_1, \ldots, f_n$, is defined as follows: For each x in (a, b), $W(x)$ is the value of the determinant of order n whose element in the kth row and mth column is $f_m^{(k-1)}(x)$, where $k = 1, 2, \ldots, n$ and $m = 1, 2, \ldots, n$. [The expression $f_m^{(0)}(x)$ is written for $f_m(x)$.]

(a) Show that $W'(x)$ can be obtained by replacing the last row of the determinant defining $W(x)$ by the nth derivatives $f_1^{(n)}(x), \ldots, f_n^{(n)}(x)$.

(b) Assuming the existence of n constants $c_1, \ldots, c_n$, not all zero, such that $c_1 f_1(x) + \cdots + c_n f_n(x) = 0$ for every x in (a, b), show that $W(x) = 0$ for each x in (a, b).

NOTE. A set of functions satisfying such a relation is said to be a *linearly dependent set* on (a, b).

(c) The vanishing of the Wronskian throughout (a, b) is necessary, but not sufficient, for linear dependence of $f_1, \ldots, f_n$. Show that in the case of two functions, if the Wronskian vanishes throughout (a, b) and if one of the functions does not vanish in (a, b), then they form a linearly dependent set in (a, b).

5-10. Given a function f defined and having a finite derivative in (a, b) and such that $\lim_{x \to b-} f(x) = +\infty$. Prove that $\lim_{x \to b-} f'(x)$ either fails to exist or is infinite.

5-11. Show that the formula in the Mean Value Theorem can be written as follows:

$$\frac{f(x+h) - f(x)}{h} = f'(x + \theta h), \qquad \text{where } 0 < \theta < 1.$$

Determine θ as a function of x and h when

(a) $f(x) = x^2$, (b) $f(x) = x^3$,

(c) $f(x) = e^x$, (d) $f(x) = \log x$, $x > 0$.

Keep $x \neq 0$ fixed, and find $\lim_{h \to 0} \theta$ in each case.

5-12. Take $f(x) = 3x^4 - 2x^3 - x^2 + 1$ and $g(x) = 4x^3 - 3x^2 - 2x$ in Theorem 5-15. Show that $f'(x)/g'(x)$ is never equal to the quotient $[f(1) - f(0)]/[g(1) - g(0)]$ if $0 < x \leq 1$. How do you reconcile this with the equation

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(x_1)}{g'(x_1)}, \qquad a < x_1 < b,$$

obtainable from Theorem 5-15 when $n = 1$?

5-13. In each of the following special cases of Theorem 5-15, take $n = 1$, $x_0 = a$, $x = b$, and show that $x_1 = (a + b)/2$.

(a) $f(x) = \sin x$, $g(x) = \cos x$; (b) $f(x) = e^x$, $g(x) = e^{-x}$.

Can you find a general class of such pairs of functions f and g for which x_1 will always be $(a + b)/2$ and such that both examples (a) and (b) are in this class?

5–14. Given a function f defined and having a finite derivative f' in the half-open interval $0 < x \leq 1$ and such that $|f'(x)| < 1$. Define $a_n = f(1/n)$ for $n = 1, 2, 3, \ldots$, and show that $\lim_{n\to\infty} a_n$ exists. [*Hint*: Use the Cauchy condition.]

5–15. Assume that f has a finite derivative at each point of the open interval (a, b). Assume also that $\lim_{x\to x_0} f'(x)$ exists and is finite for some interior point x_0. Prove that the value of this limit must be $f'(x_0)$.

5–16. Let f be continuous on (a, b) with a finite derivative f' everywhere in (a, b), except possibly at x_0. If $\lim_{x\to x_0} f'(x)$ exists and has the value A, show that $f'(x_0)$ must also exist and have the value A.

5–17. Let f be continuous on $[0, 1]$, $f(0) = 0$, $f'(x)$ finite for each x in $(0, 1)$. Prove that if f' is an increasing function on $(0, 1)$, then so too is the function g defined by the equation $g(x) = f(x)/x$.

5–18. Let h be a fixed positive number. Show that there is no function f satisfying the following three conditions: $f'(x)$ exists for $x \geq 0$, $f'(0) = 0$, $f'(x) \geq h$ for $x > 0$.

5–19. Assuming that f and g have continuous derivatives, use the Mean Value Theorem to derive the chain rule for differentiating the composite function gf.

5–20. If $h > 0$ and if $f'(x)$ exists (and is finite) for every x in $(a - h, a + h)$, and if f is continuous on $[a - h, a + h]$, show that we have:

(a) $$\frac{f(a + h) - f(a - h)}{h} = f'(a + \theta h) + f'(a - \theta h), \qquad 0 < \theta < 1;$$

(b) $$\frac{f(a + h) - 2f(a) + f(a - h)}{h} = f'(a + \lambda h) - f'(a - \lambda h), \qquad 0 < \lambda < 1.$$

(c) If $f''(a)$ exists, show that

$$f''(a) = \lim_{h\to 0} \frac{f(a + h) - 2f(a) + f(a - h)}{h^2}.$$

(d) Give an example where the limit of the quotient in (c) exists but where $f''(a)$ does not exist.

5–21. Let f have a finite derivative in (a, b) and assume that $x_0 \in (a, b)$. Consider the following condition: *For every $\epsilon > 0$ there exists a neighborhood $N'(x_0; \delta)$, whose radius δ depends only on ϵ and not on x_0, such that if $x \in N'(x_0; \delta)$, then*

$$\left|\frac{f(x) - f(x_0)}{x - x_0} - f'(x_0)\right| < \epsilon.$$

Show that f' is continuous on (a, b) if this condition holds throughout (a, b).

5–22. Prove Theorem 5–13 if one or both of $f'_+(a)$, $f'_-(b)$ is infinite.

5-23. Give an example of a pair of functions f and g having finite derivatives in $(0, 1)$ such that

$$\lim_{x \to 0} \frac{f(x)}{g(x)} = 0$$

but such that $\lim_{x \to 0} f'(x)/g'(x)$ does not exist, choosing g so that $g'(x)$ is never zero

5-24. Prove the following theorem

Let f and g be two functions having finite nth derivatives in (a, b) For some interior point x_1 in (a, b), assume that $f(x_1) = f'(x_1) = \cdots = f^{(n-1)}(x_1) = 0$, and that $g(x_1) = g'(x_1) = \cdots = g^{(n-1)}(x_1) = 0$, but that $g^{(n)}(x)$ is never zero in (a, b). Show that

$$\lim_{x \to x_1} \frac{f(x)}{g(x)} = \frac{f^{(n)}(x_1)}{g^{(n)}(x_1)}$$

NOTE $f^{(n)}$ and $g^{(n)}$ are not assumed to be continuous at x_1 [*Hint.* Let $F(x) = f(x) - (x - x_1)^{n-1}f^{(n)}(x_1)/(n - 1)!$, define G similarly, and apply Theorem 5-15 to the functions F and G]

5-25 Show that the formula in Taylor's theorem can also be written as follows.

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{(x - x_0)(x - x_1)^{n-1}}{(n - 1)!} f^{(n)}(x_1),$$

where x_1 is interior to the interval joining x and x_0 Let $1 - \theta = (x - x_1)/(x - x_0)$ Show that $0 < \theta < 1$ and deduce the following form of the remainder term (due to Cauchy)

$$\frac{(1 - \theta)^{n-1}(x - x_0)^n}{(n - 1)!} f^{(n)}[\theta x + (1 - \theta)x_0]$$

[*Hint* Take $G(t) = g(t) = t$ in the proof of Theorem 5-15]

CHAPTER 6

DIFFERENTIATION OF FUNCTIONS OF SEVERAL VARIABLES

6–1 Introduction. In Chapter 5 we considered derivatives of functions defined on subsets of the real line E_1. We now wish to discuss differentiation of real-valued functions of several variables. Perhaps the simplest way to proceed is to reduce the discussion to the one-dimensional case by treating a function of several variables as a function of one variable at a time, holding the others fixed. This leads us to the concept of *partial derivative*, with which the reader is already somewhat familiar from his knowledge of elementary calculus.

If $\mathbf{x} = (x_1, \ldots, x_n)$ is a point in E_n, and if $\mathbf{y} = (y_1, \ldots, y_n)$ is another point all of whose coordinates except the kth are the same as those of $\mathbf{x}$, that is, $y_i = x_i$ if $i \neq k$ and $y_k \neq x_k$, then we can consider the limit

$$\lim_{y_k \to x_k} \frac{f(\mathbf{y}) - f(\mathbf{x})}{y_k - x_k}.$$

When this limit exists, it is called the partial derivative of f with respect to the kth coordinate and is denoted by $D_k f(\mathbf{x})$, or $f_k(\mathbf{x})$, or $\partial f(\mathbf{x})/\partial x_k$, or by a similar expression. We shall adhere to the notation $D_k f(\mathbf{x})$.

This process then yields, from a given function f, n further functions $D_1 f, D_2 f, \ldots, D_n f$ defined at those points in E_n where the corresponding limits exist. One-sided and infinite partial derivatives could be defined as in the one-dimensional case, but we shall be interested only in finite derivatives which exist at interior points of certain open sets in E_n.

In generalizing a concept from E_1 to E_n, we seek to preserve what we consider to be the important properties in the one-dimensional case. For example, the existence of the derivative at x implies continuity at x in the one-dimensional case. Therefore it seems desirable to have a notion of derivative for functions of several variables which will imply continuity. Partial derivatives do *not* do this. A function of n variables can have partial derivatives at a point with respect to each of the variables and yet not be continuous at the point. Consider the following example of a function of two variables:

$$f(x, y) = \begin{cases} x + y, & \text{if } x = 0 \text{ or } y = 0, \\ 1, & \text{otherwise.} \end{cases}$$

The partial derivatives $D_1f(0, 0)$ and $D_2f(0,0)$ both exist. In fact,

$$D_1f(0, 0) = \lim_{x\to 0} \frac{f(x, 0) - f(0, 0)}{x - 0} = \lim_{x\to 0} \frac{x}{x} = 1$$

and, similarly, $D_2f(0, 0) = 1$ On the other hand, it is clear that this function is not continuous at $(0, 0)$

The existence of the partial derivatives with respect to each variable separately *implies continuity in each variable separately*, but, as we have just seen, this does not necessarily imply continuity in all the variables simultaneously. The difficulty with partial derivatives is that by their very definition we are forced to consider only one variable at a time Partial derivatives give us the rate of change of a function in the direction of each coordinate axis It is natural to seek a more general concept of derivative which does not restrict our considerations to the special directions of the coordinate axes, but which allows us to study the rate of change in an arbitrary direction The *directional derivative* serves this purpose

Before we introduce the directional derivative, we wish to remark that in this chapter we shall restrict ourselves to functions defined on *open sets* S in E_n, so that with each point $\mathbf{x}$ in S there will be a neighborhood $N(\mathbf{x}) \subset S$. Every point $\mathbf{y}$ in $N(\mathbf{x})$ can then be expressed in the form $\mathbf{y} = \mathbf{x} + \lambda\mathbf{u}$, where $\mathbf{u}$ is a unit vector; that is to say, $\mathbf{u} \in E_n$ and $|\mathbf{u}| = 1$. The number λ has an absolute value not exceeding the radius of the sphere $N(\mathbf{x})$.

6-2 The directional derivative.

6-1 DEFINITION *Let f be a real-valued function defined on an open set S in E_n and assume $\mathbf{x} \in S$ Let $\mathbf{u}$ be a unit vector in E_n We define the directional derivative of f at $\mathbf{x}$ in the direction $\mathbf{u}$ to be the number*

$$D_\mathbf{u}f(\mathbf{x}) = \lim_{\lambda\to 0} \frac{f(\mathbf{x} + \lambda\mathbf{u}) - f(\mathbf{x})}{\lambda}$$

whenever the limit exists

Observe that in the one-dimensional case, this reduces to the definition of $f'(\mathbf{x})$ if we take $\mathbf{u} = 1$. Also, this definition includes the kth partial derivative as a special case when the unit vector $\mathbf{u}$ is taken to be the kth unit coordinate vector $\mathbf{u}_k$ (that is, the vector having all components zero except the kth, which has the value 1) We then write D_k instead of $D_{\mathbf{u}_k}$. Observe also that if we introduce $F(\lambda) = f(\mathbf{x} + \lambda\mathbf{u})$, we have $D_\mathbf{u}f(\mathbf{x}) = F'(0)$.

If a function f defined in E_n has a directional derivative in *every* direction $\mathbf{u}$ at a point $\mathbf{x}$, then, in particular, all partial derivatives $D_1f, \ldots, D_nf$ exist at $\mathbf{x}$. The converse is not true, however. For example, the function f considered above, which has the value $x + y$ at (x, y) if $x = 0$ or $y = 0$ and has the value 1 otherwise, has finite partials $D_1f(0, 0)$ and $D_2f(0, 0)$. Nevertheless, if we consider any *other* direction $\mathbf{u} = (a_1, a_2)$, $a_1 \neq 0$, $a_2 \neq 0$, we have

$$\frac{f(\lambda a_1, \lambda a_2) - f(0, 0)}{\lambda} = \frac{1}{\lambda}$$

and this does not tend to a finite limit as $\lambda \to 0$.

A rather surprising fact is that a function may have a finite directional derivative in every direction at some point but may fail to be continuous at that point. For example, consider the function of two variables defined by the formulas

$$f(x, y) = \begin{cases} xy^2/(x^2 + y^4), & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$$

Let $\mathbf{u} = (a_1, a_2)$ be an arbitrary unit vector in E_2. Then we have

$$\frac{f(\lambda a_1, \lambda a_2) - f(0, 0)}{\lambda} = \frac{a_1 a_2^2}{a_1^2 + \lambda^2 a_2^4},$$

and hence $D_{\mathbf{u}}f(0, 0) = a_2^2/a_1$ if $a_1 \neq 0$. If $a_1 = 0$, we find $D_{\mathbf{u}}f(0, 0) = 0$. Therefore, $D_{\mathbf{u}}f(0, 0)$ is finite for all directions $\mathbf{u}$. On the other hand, the function f takes on the value $\frac{1}{2}$ at each point of the parabola $x = y^2$ (except at the origin), so that f is clearly not continuous at $(0, 0)$, since $f(0, 0) = 0$.

Thus we see that even the existence of *all* directional derivatives at a point fails to imply continuity at the point. For this reason directional, like partial derivatives, are a somewhat unsatisfactory extension of the one-dimensional concept of derivative. We now introduce a more suitable generalization which does imply continuity and, at the same time, permits us to extend the principal theorems of one-dimensional derivative theory to functions of several variables. The concept which seems to serve this purpose best is the notion of *differential*. We shall first discuss the one-dimensional case in detail before we define differentials in n-dimensions.

6–3 Differentials of functions of one real variable.

6–2 DEFINITION. *Let f be a real-valued function defined on an open interval S in E_1. Construct a new function g of two real variables as follows: For every point x in S such that $f'(x)$ exists (finite), and for every real number t, let*

$$g(x; t) = f'(x)t.$$

The function g so defined is called the differential of f.

NOTE We write $g(x, t)$ rather than $g(x, t)$ to place further emphasis on the different roles of x and t. The first point, x, must be a point where $f'(x)$ exists, whereas the second point, t, is an arbitrary point in E_1. We sometimes say that $g(x, t)$ is the differential of f at x with *increment* t.

The differential can be given a geometric interpretation as indicated in Fig 6–1. Observe in the figure that when t is "small," the difference $f(x + t) - f(x)$ and the differential $f'(x)t$ are nearly equal. This fact, which is of fundamental importance, is described precisely in the next theorem.

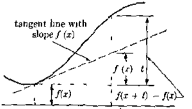

FIG 6–1. Geometric interpretation of the differential in E_1

6-3 THEOREM *Let f have a finite derivative at x, and let $g(x; t) = f'(x)t$. Then for every $\epsilon > 0$, there exists a neighborhood $N(x)$ such that for every y in $N'(x)$ we have the inequality*

$$|f(y) - f(x) - g(x, y - x)| < \epsilon|y - x|$$

Proof Given $\epsilon > 0$, there is a neighborhood $N(x)$ such that $y \in N'(x)$ implies

$$\left|\frac{f(y) - f(x)}{y - x} - f'(x)\right| < \epsilon$$

Multiplication by $|y - x|$ gives the result.

The next property of differentials is an immediate consequence of the definition.

6-4 THEOREM *If $f'(x)$ exists and if $g(x, t) = f'(x)t$, then for all real numbers t, t', α, and α', we have*

$$g(x, \alpha t + \alpha' t') = \alpha g(x, t) + \alpha' g(x; t')$$

Proof $g(x; \alpha t + \alpha' t') = f'(x)(\alpha t + \alpha' t') = \alpha[f'(x)t] + \alpha'[f'(x)t']$.

NOTE We describe the property just proved by saying that g is *linear* in the second variable.

6–4 Differentials of functions of several variables. Instead of defining a differential of a function of several variables by a *formula*, as we did in E_1, we choose to define a differential in E_n by the *properties* we wish it to possess. The properties we want are the natural extensions of those embodied in Theorems 6–3 and 6–4.

6–5 DEFINITION. *Let f be a real-valued function defined on an open set S in E_n, and assume $\mathbf{x} \in S$. We say that f has a differential at $\mathbf{x}$ if there exists another function g which satisfies the following conditions:*

(a) *g is a real-valued function of two n-dimensional variables, and the function values, denoted by $g(\mathbf{x};\ \mathbf{t})$, are defined for the given point $\mathbf{x}$ in S and for every point $\mathbf{t}$ in E_n.*

(b) *g is linear in the second variable; that is, for every pair of points $\mathbf{t}$ and $\mathbf{t}'$ in E_n and for every pair of real numbers α and α', we have*

$$g(\mathbf{x}; \alpha\mathbf{t} + \alpha'\mathbf{t}') = \alpha g(\mathbf{x}; \mathbf{t}) + \alpha' g(\mathbf{x}; \mathbf{t}').$$

(c) *For every $\epsilon > 0$, there exists a neighborhood $N(\mathbf{x})$ such that $\mathbf{y} \in N'(\mathbf{x})$ implies*

$$|f(\mathbf{y}) - f(\mathbf{x}) - g(\mathbf{x}; \mathbf{y} - \mathbf{x})| < \epsilon|\mathbf{y} - \mathbf{x}|.$$

NOTE. It is important to observe that we have no guarantee in advance that, for given f, *any* such function g exists. We shall prove later that when f is suitably restricted, there will be one and only one such function g.

Before we deal with the question of the existence of differentials, we shall prove some theorems on the assumption that a differential *does* exist. The first of these is a consequence of the linearity property and tells us that the value of a differential must be a linear combination of the components of the second variable.

6–6 THEOREM. *Assume that f has a differential $g(\mathbf{x}; \mathbf{t})$ at $\mathbf{x}$, and write $\mathbf{t} = (t_1, \ldots, t_n)$. Then there exist n real numbers $a_1(\mathbf{x}), \ldots, a_n(\mathbf{x})$ (depending on $\mathbf{x}$ but independent of $\mathbf{t}$) such that*

$$g(\mathbf{x}; \mathbf{t}) = \sum_{k=1}^{n} a_k(\mathbf{x}) t_k.$$

Proof. We can write $\mathbf{t} = t_1\mathbf{u}_1 + \cdots + t_n\mathbf{u}_n$, where $\mathbf{u}_k$ is the kth unit coordinate vector. From the linearity of g in the second variable, we then have

$$g(\mathbf{x}, \mathbf{t}) = g(\mathbf{x}, t_1\mathbf{u}_1 + \cdots + t_n\mathbf{u}_n) = g(\mathbf{x}; \mathbf{u}_1)t_1 + \cdots + g(\mathbf{x}; \mathbf{u}_n)t_n.$$

This proves the theorem, if we take $a_k(\mathbf{x}) = g(\mathbf{x}, \mathbf{u}_k)$, $k = 1, 2, \ldots, n$.

Theorem 6–6 will now be used to prove that if the differential exists at all, it is uniquely determined. In fact, we will show that the n numbers $a_1(\mathbf{x}), \ldots, a_n(\mathbf{x})$ of Theorem 6–6 are simply the n partial derivatives $D_1f(\mathbf{x}), \ldots, D_nf(\mathbf{x})$.

6–7 THEOREM (*Uniqueness theorem*). *Assume that f has a differential $g(\mathbf{x}, \mathbf{t})$ at $\mathbf{x}$ and write*

$$g(\mathbf{x}, \mathbf{t}) = \sum_{k=1}^{n} a_k(\mathbf{x})t_k,$$

in accordance with Theorem 6–6. Then each partial derivative $D_kf(\mathbf{x})$ exists and we have

$$a_k(\mathbf{x}) = D_kf(\mathbf{x}), \qquad k = 1, 2, \ldots, n.$$

Proof. By hypothesis, for every $\epsilon > 0$ there is a neighborhood $N(\mathbf{x})$ such that $\mathbf{y} \in N'(\mathbf{x})$ implies $|f(\mathbf{y}) - f(\mathbf{x}) - g(\mathbf{x}, \mathbf{y} - \mathbf{x})| < \epsilon|\mathbf{y} - \mathbf{x}|$. By Theorem 6–6, we also have

$$g(\mathbf{x}, \mathbf{y} - \mathbf{x}) = \sum_{r=1}^{n} a_r(\mathbf{x})(y_r - x_r).$$

Writing $\mathbf{y} = \mathbf{x} + \lambda\mathbf{u}_k$, where $|\lambda|$ is less than the radius of $N(\mathbf{x})$ and $\mathbf{u}_k$ is the kth unit coordinate vector, we have

$$0 < |\mathbf{y} - \mathbf{x}| = |\lambda|, \qquad y_k - x_k = \lambda, \qquad y_r - x_r = 0, \qquad \text{if } r \neq k.$$

Therefore, the fundamental inequality becomes

$$|f(\mathbf{x} + \lambda\mathbf{u}_k) - f(\mathbf{x}) - \lambda a_k(\mathbf{x})| < \epsilon|\lambda|.$$

Dividing by $|\lambda|$, we find

$$\left|\frac{f(\mathbf{x} + \lambda\mathbf{u}_k) - f(\mathbf{x})}{\lambda} - a_k(\mathbf{x})\right| < \epsilon,$$

and this implies that $D_kf(\mathbf{x})$ exists and has the value $a_k(\mathbf{x})$.

We have therefore proved that *if* a function f has a differential $g(\mathbf{x}; \mathbf{t})$ at $\mathbf{x}$, this differential is uniquely determined and must necessarily have the form

$$g(\mathbf{x}; \mathbf{t}) = \sum_{k=1}^{n} D_k f(\mathbf{x}) t_k, \qquad \text{if } \mathbf{t} = (t_1, \ldots, t_n).$$

It is customary to use the symbol df instead of g for the differential. It is also customary to use the symbols $dx_1, \ldots, dx_n$ instead of $t_1, \ldots, t_n$ for the components of $\mathbf{t}$, in which case the symbol $d\mathbf{x}$ is used in place of the vector $\mathbf{t}$. Theorem 6–7 then states that

$$df(\mathbf{x}; d\mathbf{x}) = \sum_{k=1}^{n} D_k f(\mathbf{x}) dx_k, \qquad \text{if } d\mathbf{x} = (dx_1, \ldots, dx_n).$$

This is sometimes expressed more briefly in the following notation:

$$df = \frac{\partial f}{\partial x_1} dx_1 + \cdots + \frac{\partial f}{\partial x_n} dx_n.$$

Indeed this last formula is often used as the definition of df and the properties given in our definition are then proved as theorems.

It is quite easy to see that the directional derivative $D_{\mathbf{u}} f(\mathbf{x})$ will exist in every direction $\mathbf{u}$ if f has a differential at $\mathbf{x}$. In fact, the directional derivative is merely a special case of the differential.

6–8 THEOREM. *Let f have a differential at a point $\mathbf{x}$ of an open set S in E_n, and let $\mathbf{u}$ be a unit vector in E_n. Then the directional derivative $D_{\mathbf{u}} f(\mathbf{x})$ exists and we have*

$$D_{\mathbf{u}} f(\mathbf{x}) = df(\mathbf{x}; \mathbf{u}).$$

Proof. Given $\epsilon > 0$, there exists a neighborhood $N(\mathbf{x}; \delta)$ such that

$$|f(\mathbf{y}) - f(\mathbf{x}) - df(\mathbf{x}; \mathbf{y} - \mathbf{x})| < \epsilon |\mathbf{y} - \mathbf{x}|, \qquad \text{if } \mathbf{y} \in N'(\mathbf{x}; \delta).$$

Given the unit vector $\mathbf{u}$, for every real $\lambda \neq 0$ such that $|\lambda| < \delta$, the point $\mathbf{x} + \lambda\mathbf{u}$ will be in $N'(\mathbf{x}; \delta)$. Taking $\mathbf{y} = \mathbf{x} + \lambda\mathbf{u}$ in the inequality, we obtain the relation

$$\left| \frac{f(\mathbf{x} + \lambda\mathbf{u}) - f(\mathbf{x})}{\lambda} - df(\mathbf{x}; \mathbf{u}) \right| < \epsilon, \qquad \text{if } 0 < |\lambda| < \delta.$$

But this means that $D_{\mathbf{u}} f(\mathbf{x})$ exists and has the value $df(\mathbf{x}; \mathbf{u})$.

Next, we prove that the existence of the differential df at $\mathbf{x}$ implies continuity of f at $\mathbf{x}$ In fact, we can prove a little more—namely, that f satisfies a Lipschitz condition at $\mathbf{x}$ This, in turn, will yield the continuity

6–9 THEOREM *Let f be defined on an open set S in E_n and assume that f has a differential at a point $\mathbf{x}$ in S Then f satisfies a Lipschitz condition at $\mathbf{x}$ That is to say, there exists a positive number M and a neighborhood $N(\mathbf{x})$ such that*

$$\mathbf{y} \in N'(\mathbf{x}) \quad \textit{implies} \quad |f(\mathbf{y}) - f(\mathbf{x})| < M|\mathbf{y} - \mathbf{x}|$$

Proof. Taking $\epsilon = 1$ in part (c) of Definition 6–5, there exists a neighborhood $N(\mathbf{x})$ such that $\mathbf{y} \in N'(\mathbf{x})$ implies

$$|f(\mathbf{y}) - f(\mathbf{x}) - df(\mathbf{x}, \mathbf{y} - \mathbf{x})| < |\mathbf{y} - \mathbf{x}|$$

This implies the inequality

$$|f(\mathbf{y}) - f(\mathbf{x})| < |df(\mathbf{x}, \mathbf{y} - \mathbf{x})| + |\mathbf{y} - \mathbf{x}|$$

But, by Theorem 6–7, we have

$$|df(\mathbf{x}, \mathbf{y} - \mathbf{x})| = \left|\sum_{k=1}^{n} D_k f(\mathbf{x})(y_k - x_k)\right| \leq \sum_{k=1}^{n} |D_k f(\mathbf{x})|\,|\mathbf{y} - \mathbf{x}|.$$

The theorem follows if we take $M = 1 + \sum_{k=1}^{n} |D_k f(\mathbf{x})|$

6–10 THEOREM *If f has a differential at $\mathbf{x}$, then f is continuous at $\mathbf{x}$*

Proof By Theorem 6–9 there is a positive number M and a neighborhood $N(\mathbf{x})$ such that $\mathbf{y} \in N'(\mathbf{x})$ implies $|f(\mathbf{y}) - f(\mathbf{x})| < M|\mathbf{y} - \mathbf{x}|$ Given $\epsilon > 0$, choose $\delta > 0$ so that $\delta < \epsilon/M$, and also so that δ is less than the radius of $N(\mathbf{x})$ Then $|\mathbf{y} - \mathbf{x}| < \delta$ implies $|f(\mathbf{y}) - f(\mathbf{x})| < \epsilon$, and this establishes the continuity

NOTE Actually, we have proved that a function which satisfies a Lipschitz condition at $\mathbf{x}$ must also be continuous at $\mathbf{x}$

6–5 The gradient vector. Before we develop the theory of differentials any further, we shall reformulate some of our results in vector terminology.

6–11 DEFINITION *Given a function f for which the n partial derivatives $D_1 f, \ldots, D_n f$ exist at a point $\mathbf{x}$ in E_n The vector-valued function denoted by ∇f and defined by the equation*

$$\nabla f(\mathbf{x}) = (D_1 f(\mathbf{x}), \ldots, D_n f(\mathbf{x}))$$

is called the gradient of f.

We sometimes write the symbolic equation

$$\nabla f = (D_1 f, \ldots, D_n f)$$

and think of ∇f as being an idealized "vector" whose kth "component" is the function $D_k f$. Instead of writing $(\nabla f)_k$ for the kth component, we write $D_k f$. (The symbol ∇ is pronounced "del.")

6–12 DEFINITION. *Given two vectors* $\mathbf{a} = (a_1, \ldots, a_n)$ *and* $\mathbf{b} = (b_1, \ldots, b_n)$ *in* E_n, *the real number denoted by* $\mathbf{a} \cdot \mathbf{b}$ *and defined by the equation*

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + \cdots + a_n b_n$$

is called the inner product (or dot product) of $\mathbf{a}$ *and* $\mathbf{b}$.

It follows at once from this definition that we have

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}, \qquad (\lambda \mathbf{a}) \cdot \mathbf{b} = \lambda(\mathbf{a} \cdot \mathbf{b}) \quad \text{and} \quad \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$$

for all vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, in E_n and for every real number λ. Using inner products, the Cauchy-Schwarz inequality (Theorem 1–5) assumes the form

$$|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}|\,|\mathbf{b}|.$$

By employing these vectorial concepts, we can restate some of the previous results in a very simple and elegant form.

6–13 THEOREM. *Let f have a differential at a point* $\mathbf{x}$ *in* E_n. *Then the gradient vector* $\nabla f(\mathbf{x})$ *exists and we have*

$$df(\mathbf{x}; \mathbf{t}) = \nabla f(\mathbf{x}) \cdot \mathbf{t}, \qquad \textit{if } \mathbf{t} \in E_n.$$

The directional derivative $D_{\mathbf{u}} f(\mathbf{x})$ *also exists and is given by*

$$D_{\mathbf{u}} f(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot \mathbf{u}, \qquad \textit{if } \mathbf{u} \in E_n, \qquad |\mathbf{u}| = 1.$$

Furthermore, if there exists a vector $\mathbf{F}(\mathbf{x})$ *such that*

$$df(\mathbf{x}; \mathbf{t}) = \mathbf{F}(\mathbf{x}) \cdot \mathbf{t}$$

for all $\mathbf{t}$ *in* E_n, *then* $\mathbf{F}(\mathbf{x})$ *must be the gradient vector* $\nabla f(\mathbf{x})$.

Proof The first statement is Theorem 6–7 and the second is Theorem 6–8 The last statement also follows at once from Theorem 6–7

Observe that the formula $df(\mathbf{x}; \mathbf{t}) = \nabla f(\mathbf{x}) \cdot \mathbf{t}$ bears a strong resemblance to the equation $df(x, t) = f'(x)t$, which holds in the one-dimensional case This suggests that the gradient vector ∇f plays the same role in E_n as the derivative f' in E_1 We shall see further evidence of this in some of the later work For example, if f has a local maximum or a local minimum at an interior point $\mathbf{x}$ of an open set S in E_n, and if $\nabla f(\mathbf{x})$ exists, then we must have $\nabla f(\mathbf{x}) = \mathbf{0}$ (See Exercise 6–1 and compare with Theorem 5–8)

6–6 Differentials of composite functions and the chain rule. We come now to a very important theorem which provides us with an extension of the "chain rule" (Theorem 5–5) of the previous chapter In one dimension, the chain rule gives us a formula for differentiating a composite function $h = gf$ in terms of the derivatives of g and f The formula states that

$$h'(z) = g'[f(z)]f'(z) \tag{1}$$

When we proceed to functions of several variables, the simplest type of composite function arises as follows We are given a function g of n real variables, say $g(\mathbf{x}) = g(x_1, \quad, x_n)$ Each of the variables $x_1, \quad ., x_n$ is then replaced by a function of another real variable z, say $x_k = f_k(z)$, and this transforms g into a new function h, where $h(z) = g[f_1(z), . \quad , f_n(z)]$. The extended chain rule in this case tells us how to compute the derivative $h'(z)$ in terms of the partial derivatives of g and the derivatives of $f_1, \ldots, f_n$. The formula turns out to be a sum of terms resembling (1), namely,

$$h'(z) = \sum_{k=1}^{n} D_k g[\mathbf{f}(z)] f_k'(z), \tag{2}$$

where we have written $\mathbf{f}(z)$ for the vector $(f_1(z), \quad , f_n(z))$ The sum which appears in this equation can also be expressed as an *inner product* of two vectors, and formula (2) takes the form

$$h'(z) = \nabla g[\mathbf{f}(z)] \quad \mathbf{f}'(z), \tag{3}$$

where $\mathbf{f}'(z)$ denotes the vector $(f_1'(z), \quad . \quad , f_n'(z))$ In this form, the extended chain rule resembles the result in the one-dimensional case more closely, except that now the gradient ∇g plays the role of the derivative g'.

The situation is quite similar if the new variable $\mathbf{z}$ is an m-dimensional variable, say $\mathbf{z} = (z_1, \ldots, z_m)$. The composite function h so formed is a function of the m variables $z_1, \ldots, z_m$, and the chain rule in this case consists of a set of m formulas, one for each partial derivative $D_1h, \ldots, D_mh$. The formula for $D_rh(\mathbf{z})$ is again a sum, namely,

$$(4)\qquad D_rh(\mathbf{z}) = \sum_{k=1}^{n} D_kg[\mathbf{f}(\mathbf{z})]D_rf_k(\mathbf{z}) \qquad (r = 1, 2, \ldots, m).$$

If $D_r\mathbf{f}(\mathbf{z})$ is written for the vector $(D_rf_1(\mathbf{z}), \ldots, D_rf_n(\mathbf{z}))$, then this sum can also be expressed as an inner product and formula (4) assumes a form which again resembles the one-dimensional case:

$$(5)\qquad D_rh(\mathbf{z}) = \nabla g[\mathbf{f}(\mathbf{z})] \cdot D_r\mathbf{f}(\mathbf{z}) \qquad (r = 1, 2, \ldots, m).$$

We will now prove these formulas under the hypothesis that the functions $f_1, \ldots, f_n$ and g have differentials. Of course, we need prove only (4), since the others are merely special cases.

6–14 THEOREM. *Let $\mathbf{f} = (f_1, \ldots, f_n)$ be a vector-valued function defined on an open set Z in E_m with function values in E_n, and let g be a real-valued function defined on an open set X in E_n which contains $\mathbf{f}(Z)$. Let $\mathbf{z}$ be a point in Z at which each function f_k has a differential and suppose g has a differential at $\mathbf{f}(\mathbf{z})$. Then the composite function $h = g\mathbf{f}$ (defined on Z by the equation $h(\mathbf{z}) = g[\mathbf{f}(\mathbf{z})]$) also has a differential at $\mathbf{z}$. Moreover, the gradient vector $\nabla h(\mathbf{z})$ is given by the formula*

$$(6)\qquad \nabla h(\mathbf{z}) = \sum_{k=1}^{n} D_kg[\mathbf{f}(\mathbf{z})]\nabla f_k(\mathbf{z}).$$

NOTE. By equating the rth components in this vector equation, we find that each partial derivative $D_rh(\mathbf{z})$ is given by formulas (4) and (5). The formula for D_rh is sometimes written as follows:

$$(7)\qquad \frac{\partial h}{\partial z_r} = \frac{\partial g}{\partial x_1}\frac{\partial f_1}{\partial z_r} + \cdots + \frac{\partial g}{\partial x_n}\frac{\partial f_n}{\partial z_r} \qquad (r = 1, 2, \ldots, m).$$

Proof. If we prove that $dh(\mathbf{z}; \mathbf{t})$ exists and has the value

$$dh(\mathbf{z}; \mathbf{t}) = \left(\sum_{k=1}^{n} D_kg[\mathbf{f}(\mathbf{z})]\nabla f_k(\mathbf{z})\right) \cdot \mathbf{t},$$

then (6) will follow from the uniqueness theorem (Theorem 6–7). For this purpose we must find, for every $\epsilon > 0$, a neighborhood $N(\mathbf{z})$ such that $\mathbf{z} + \mathbf{t} \in N'(\mathbf{z})$ implies

$$\left|h(\mathbf{z}+\mathbf{t}) - h(\mathbf{z}) - \left(\sum_{k=1}^{n} D_k g[\mathbf{f}(\mathbf{z})]\nabla f_k(\mathbf{z})\right)\cdot \mathbf{t}\right| < \epsilon|\mathbf{t}|. \tag{8}$$

Write $\mathbf{x} = \mathbf{f}(\mathbf{z})$ and $\mathbf{y} = \mathbf{f}(\mathbf{z}+\mathbf{t})$ Then $h(\mathbf{z}+\mathbf{t}) - h(\mathbf{z}) = g(\mathbf{y}) - g(\mathbf{x})$ Since g has a differential at $\mathbf{x}$, given $\epsilon' > 0$ there exists a neighborhood $N(\mathbf{x}, \delta)$ such that

$$|g(\mathbf{y}) - g(\mathbf{x}) - \nabla g(\mathbf{x})\cdot(\mathbf{y}-\mathbf{x})| \le \epsilon'|\mathbf{y}-\mathbf{x}|, \qquad \text{if } \mathbf{y} \in N(\mathbf{x};\delta) \tag{9}$$

Now $\mathbf{f}$ satisfies a Lipschitz condition at $\mathbf{z}$, and hence there is a neighborhood $N(\mathbf{z}, r_0)$ and a constant $M > 0$ such that $|\mathbf{y} - \mathbf{x}| < M|\mathbf{t}|$ if $\mathbf{z} + \mathbf{t} \in N'(\mathbf{z}; r_0)$ If we take $r_0 < \delta/M$ and $\mathbf{z}+\mathbf{t}$ in $N'(\mathbf{z}; r_0)$, then $\mathbf{y} \in N(\mathbf{x}, \delta)$ and (9) becomes

$$|h(\mathbf{z}+\mathbf{t}) - h(\mathbf{z}) - \nabla g(\mathbf{x})\cdot(\mathbf{y}-\mathbf{x})| < \epsilon' M|\mathbf{t}| \tag{10}$$

Next, we shall prove that there is a constant $A \ge 0$ and a neighborhood $N(\mathbf{z}, r)$ such that

$$\left|\nabla g(\mathbf{x})\cdot(\mathbf{y}-\mathbf{x}) - \left(\sum_{k=1}^{n} D_k g[\mathbf{f}(\mathbf{z})]\nabla f_k(\mathbf{z})\right)\cdot \mathbf{t}\right| \le \epsilon' nA|\mathbf{t}| \tag{11}$$

whenever $\mathbf{z}+\mathbf{t} \in N'(\mathbf{z}, r)$ This, together with (10), implies (8) if we take $\epsilon' = \epsilon/(M + nA)$ and $N(\mathbf{z}) = N(\mathbf{z}, r) \cap N(\mathbf{z}, r_0)$

The left member of (11) can be written as follows:

$$\left|\sum_{k=1}^{n} D_k g[\mathbf{f}(\mathbf{z})]\{f_k(\mathbf{z}+\mathbf{t}) - f_k(\mathbf{z}) - \nabla f_k(\mathbf{z})\cdot\mathbf{t}\}\right|. \tag{12}$$

Since each function f_k has a differential at $\mathbf{z}$, there exists a neighborhood $N(\mathbf{z}; r_k)$ such that

$$|f_k(\mathbf{z}+\mathbf{t}) - f_k(\mathbf{z}) - \nabla f_k(\mathbf{z})\cdot\mathbf{t}| < \epsilon'|\mathbf{t}|, \qquad \text{if } \mathbf{z}+\mathbf{t} \in N'(\mathbf{z}; r_k).$$

If A denotes the largest of the n numbers $|D_k g[\mathbf{f}(\mathbf{z})]|$, $k = 1, 2, \ldots, n$, and if r denotes the smallest of the radii $r_1, \ldots, r_n$, we find that the expression in (12) is bounded by $nA\epsilon'|\mathbf{t}|$ whenever $\mathbf{z}+\mathbf{t} \in N'(\mathbf{z}, r)$. This proves (11) and hence (8).

6-7 Cauchy's invariant rule. An important theorem concerning differentials of composite functions (known as *Cauchy's invariant rule*) is an immediate consequence of the chain rule just discussed

Under the hypotheses of Theorem 6–14, let us write the differential of g in the following notation:

$$dg(\mathbf{x}; d\mathbf{x}) = \nabla g(\mathbf{x}) \cdot d\mathbf{x}, \qquad \text{where } d\mathbf{x} = (dx_1, \ldots, dx_n) \in E_n,$$

and let us also write

$$dh(\mathbf{z}; d\mathbf{z}) = \nabla h(\mathbf{z}) \cdot d\mathbf{z}, \qquad \text{where } d\mathbf{z} = (dz_1, \ldots, dz_m) \in E_m.$$

Since $h = g\mathbf{f}$, the function value $h(\mathbf{z})$ can be obtained from the function value $g(\mathbf{x})$ by making the formal replacement $\mathbf{x} = \mathbf{f}(\mathbf{z})$. Cauchy's invariant rule tells us that the value of the differential $dh(\mathbf{z}; d\mathbf{z})$ can also be obtained from the value of the differential $dg(\mathbf{x}; d\mathbf{x})$ by making a formal replacement: $\mathbf{x}$ must be replaced by $\mathbf{f}(\mathbf{z})$ and the kth component of $d\mathbf{x}$ (namely, dx_k) must be replaced by the differential $df_k(\mathbf{z}; d\mathbf{z})$. This type of formal manipulation is particularly useful in performing computations with differentials.

6–15 THEOREM. *Under the hypotheses of Theorem* 6–14, *we have*

$$dh(\mathbf{z}; \mathbf{t}) = dg[\mathbf{f}(\mathbf{z}); df_1(\mathbf{z}; \mathbf{t}), \ldots, df_n(\mathbf{z}; \mathbf{t})]$$

for every $\mathbf{t}$ *in* E_m.

Proof. $dg[\mathbf{f}(\mathbf{z}); df_1(\mathbf{z}; \mathbf{t}), \ldots, df_n(\mathbf{z}; \mathbf{t})]$

$$= \sum_{k=1}^{n} D_k g[\mathbf{f}(\mathbf{z})] df_k(\mathbf{z}; \mathbf{t}) = \sum_{k=1}^{n} D_k g[\mathbf{f}(\mathbf{z})] \sum_{r=1}^{m} D_r f_k(\mathbf{z}) t_r$$

$$= \sum_{r=1}^{m} \left(\sum_{k=1}^{n} D_k g[\mathbf{f}(\mathbf{z})] D_r f_k(\mathbf{z}) \right) t_r$$

$$= \sum_{r=1}^{m} D_r h(\mathbf{z}) t_r = dh(\mathbf{z}; \mathbf{t}).$$

In practice, certain notational conveniences are adopted which tend to simplify computations with differentials. For example, let f be a given function of two variables and introduce the equation

$$z = f(x, y).$$

Then it is customary to use the symbol dz to denote the differential $df(x, y; dx, dy)$. Thus, for example, from the equation

$$z = e^{2xy} \tag{13}$$

we write

$$dz = 2ye^{2xy}\,dx + 2xe^{2xy}\,dy \tag{14}$$

When this is done, it is understood that the expression equated to dz in Eq (14) is the differential of the function f defined by

$$f(x, y) = e^{2xy}. \tag{15}$$

Suppose that a "change of variable" is now introduced in (13), say by means of the equations

$$x = r\cos\theta, \qquad y = r\sin\theta \tag{16}$$

The formal replacement of x and y in (13) by the expressions in (16) transforms Eq (13) into

$$z = e^{2r^2\sin\theta\cos\theta} = e^{r^2\sin 2\theta} \tag{17}$$

Adopting the notational convenience mentioned above, we obtain, from (17),

$$dz = 2r\sin 2\theta\, e^{r^2\sin 2\theta}\,dr + 2r^2\cos 2\theta\, e^{r^2\sin 2\theta}\,d\theta, \tag{18}$$

where dz now represents the differential of the function h defined by

$$h(r,\theta) = e^{r^2\sin 2\theta} \tag{19}$$

We are now faced with a conflict. the same symbol dz is being used for the differential of the function f defined in Eq (15) and also for the differential of the function h defined in (19) Cauchy's invariant rule allows us to resolve this conflict and to justify the double use of the symbol dz In fact, the function h defined in (19) is, in reality, the composite of f with the function defined in (16), that is to say,

$$h(r, \theta) = f(r\cos\theta, r\sin\theta)$$

If we now apply our convention to the equations in (16), we can write

$$dx = \cos\theta\,dr - r\sin\theta\,d\theta, \qquad dy = \sin\theta\,dr + r\cos\theta\,d\theta. \tag{20}$$

Cauchy's invariant rule then tells us that when the expressions for x, y, dx, and dy in Eqs. (16) and (20) are substituted in the right member of (14), we will obtain the right member of (18). This is easily verified in this example

6–8 The Mean Value Theorem for functions of several variables. The extension of the Mean Value Theorem of differential calculus to functions of several variables will now be obtained. Before stating the theorem, however, we first define what is meant by a line segment in E_n.

6–16 DEFINITION. *If* $\mathbf{x}$ *and* $\mathbf{y}$ *are two distinct points in* E_n, *then by the line segment* $L(\mathbf{x}, \mathbf{y})$ *joining* $\mathbf{x}$ *and* $\mathbf{y}$, *we mean the set*

$$L(\mathbf{x}, \mathbf{y}) = \{\mathbf{z} \mid \mathbf{z} = \theta\mathbf{x} + (1 - \theta)\mathbf{y}, 0 < \theta < 1\}.$$

NOTE. $L(\mathbf{x}, \mathbf{y})$ is also called the *open* line segment. The *closed* line segment, $L[\mathbf{x}, \mathbf{y}]$, is similarly defined, using $0 \leq \theta \leq 1$. An open line segment is not an open set if $n > 1$, but a closed line segment is always a closed set.

6–17 THEOREM. (*Mean Value Theorem in* E_n). *Assume that* f *has a differential at each point of an open set* S *in* E_n. *Let* $\mathbf{x}$ *and* $\mathbf{y}$ *be two points of* S *such that the line segment* $L(\mathbf{x}, \mathbf{y}) \subset S$. *Then there exists a point* $\mathbf{z}$ *of* $L(\mathbf{x}, \mathbf{y})$ *such that*

$$f(\mathbf{y}) - f(\mathbf{x}) = \nabla f(\mathbf{z}) \cdot (\mathbf{y} - \mathbf{x}).$$

Proof. Keep $\mathbf{x}$ and $\mathbf{y}$ fixed, $\mathbf{x} \neq \mathbf{y}$, and introduce a new function h by means of the equation

$$h(\lambda) = f(\mathbf{x} + \lambda\mathbf{u}), \qquad \text{if } 0 \leq \lambda \leq \rho,$$

where $\mathbf{u}$ is the unit vector given by $\mathbf{u} = (\mathbf{y} - \mathbf{x})/\rho$, $\rho = |\mathbf{y} - \mathbf{x}|$. Then $\mathbf{x} + \rho\mathbf{u} = \mathbf{y}$ and we have $f(\mathbf{y}) - f(\mathbf{x}) = h(\rho) - h(0)$. The existence of the differential of f in S implies the existence of the derivative h' in the closed interval $0 \leq \lambda \leq \rho$. Therefore we can apply the one-dimensional Mean Value Theorem to write

$$h(\rho) - h(0) = \rho h'(\bar{\rho}), \qquad \text{where } 0 < \bar{\rho} < \rho.$$

In order to compute $h'(\bar{\rho})$, we keep λ fixed, $0 < \lambda < \rho$, and introduce a new function g by the equation

$$g(\alpha) = f(\mathbf{x} + \lambda\mathbf{u} + \alpha\mathbf{u}) = h(\lambda + \alpha), \qquad \text{if } 0 \leq \alpha \leq \rho - \lambda.$$

Then $g'(0) = D_{\mathbf{u}}f(\mathbf{x} + \lambda\mathbf{u}) = h'(\lambda)$. Putting $\lambda = \bar{\rho}$, we obtain

$$h'(\bar{\rho}) = D_{\mathbf{u}}f(\mathbf{x} + \bar{\rho}\mathbf{u}) = \nabla f(\mathbf{x} + \bar{\rho}\mathbf{u}) \cdot \mathbf{u} = \nabla f(\mathbf{z}) \cdot \mathbf{u},$$

where $\mathbf{z} = \mathbf{x} + \bar{\rho}\mathbf{u}$. This proves the theorem, since $\mathbf{z} \in L(\mathbf{x}, \mathbf{y})$.

6-9 A sufficient condition for existence of the differential. Up to now we have been deriving consequences of the hypothesis that the differential of a function exists We have also seen that neither the existence of all partial derivatives nor the existence of all directional derivatives suffices to establish the existence of the differential (since neither implies continuity) However, we shall now prove that the existence of *continuous* partials does imply the existence of the differential

6-18 THEOREM *Let f be defined on an open set S in E_n and assume that the n partial derivatives $D_1f, \ldots, D_nf$ exist and are continuous at a point $\mathbf{x}$ in S Then f has a differential at $\mathbf{x}$*

Proof The only possible candidate for $df(\mathbf{x}, \mathbf{t})$ is, of course, the inner product $\nabla f(\mathbf{x}) \cdot \mathbf{t} = \sum_{k=1}^{n} D_kf(\mathbf{x})t_k$ This is clearly linear in $\mathbf{t}$ The whole difficulty in the proof lies in verifying that it also satisfies the fundamental inequality (c) of Definition 6-5 For this purpose we shall express the difference $f(\mathbf{y}) - f(\mathbf{x})$ as a sum of n terms, where the kth term will be an "approximation" to $D_kf(\mathbf{x})(y_k - x_k)$

First of all, the continuity of the partials $D_1f, \ldots, D_nf$ tells us that, given $\epsilon > 0$, there exists a neighborhood $N(\mathbf{x}, \delta)$ such that for every $k = 1, 2, \ldots, n$, we have

$$(21) \qquad |D_kf(\mathbf{y}) - D_kf(\mathbf{x})| < \frac{\epsilon}{n}, \qquad \text{if } \mathbf{y} \in N(\mathbf{x}, \delta)$$

(Actually, there is a different neighborhood for each k, but we take δ to be the radius of the smallest of these) In the remainder of the proof we assume that $\mathbf{y}$ is in the neighborhood $N'(\mathbf{x}, \delta/2)$ Then $\mathbf{y} = \mathbf{x} + \lambda\mathbf{u}$, where $\mathbf{u}$ is a unit vector and $|\lambda| = |\mathbf{y} - \mathbf{x}| < \delta/2$ We can also write $\mathbf{u} = a_1\mathbf{u}_1 + \cdots + a_n\mathbf{u}_n$, where $\mathbf{u}_k$ is the kth unit coordinate vector. The difference $f(\mathbf{y}) - f(\mathbf{x})$ is now expressed as a sum by means of the following identity.

$$(22) \qquad f(\mathbf{y}) - f(\mathbf{x}) = f(\mathbf{x} + \lambda\mathbf{u}) - f(\mathbf{x})$$
$$= \sum_{k=1}^{n} [f(\mathbf{x} + \lambda\mathbf{v}_k) - f(\mathbf{x} + \lambda\mathbf{v}_{k-1})],$$

where $\mathbf{v}_0, \mathbf{v}_1, \ldots, \mathbf{v}_n$ are any vectors in E_n such that $\mathbf{v}_0 = \mathbf{0}$ and $\mathbf{v}_n = \mathbf{u}$ If the vectors $\mathbf{v}_k$ are defined so that they satisfy the recurrence relation $\mathbf{v}_k = \mathbf{v}_{k-1} + a_k\mathbf{u}_k$, then we will have

$$\mathbf{v}_0 = \mathbf{0}, \quad \mathbf{v}_1 = a_1\mathbf{u}_1, \quad \mathbf{v}_2 = a_1\mathbf{u}_1 + a_2\mathbf{u}_2, \quad \ldots, \quad \mathbf{v}_n = \mathbf{u},$$

and the general term of the sum in (22) becomes

$$f(\mathbf{x} + \lambda\mathbf{v}_k) - f(\mathbf{x} + \lambda\mathbf{v}_{k-1}) = f(\mathbf{x} + \lambda\mathbf{v}_{k-1} + \lambda a_k\mathbf{u}_k) - f(\mathbf{x} + \lambda\mathbf{v}_{k-1})$$

This is the difference of the function values at two points which differ only in their kth coordinate. Furthermore, each of the two points is in the neighborhood $N'(\mathbf{x}; \delta/2)$, since $|\mathbf{v}_k|^2 = a_1^2 + \cdots + a_k^2 \leq |\mathbf{u}|^2 = 1$ and $|\lambda \mathbf{v}_k| \leq |\lambda| < \delta/2$.

We now keep λ fixed and introduce a new function g, where

$$g(\alpha) = f(\mathbf{x} + \lambda\mathbf{v}_{k-1} + \alpha\mathbf{u}_k), \qquad \text{if } -|\lambda a_k| \leq \alpha \leq |\lambda a_k|.$$

The previous equation can then be written in the form

$$f(\mathbf{x} + \lambda\mathbf{v}_k) - f(\mathbf{x} + \lambda\mathbf{v}_{k-1}) = g(\lambda a_k) - g(0).$$

Furthermore, the derivative $g'(\alpha)$ exists for each α in the closed interval $-|\lambda a_k| \leq \alpha \leq |\lambda a_k|$. In fact, we have

$$\frac{g(\alpha + h) - g(\alpha)}{h} = \frac{f(\mathbf{x} + \lambda\mathbf{v}_{k-1} + \alpha\mathbf{u}_k + h\mathbf{u}_k) - f(\mathbf{x} + \lambda\mathbf{v}_{k-1} + \alpha\mathbf{u}_k)}{h},$$

so that when $h \to 0$ we find $g'(\alpha) = D_kf(\mathbf{x} + \lambda\mathbf{v}_{k-1} + \alpha\mathbf{u}_k)$. The one-dimensional Mean Value Theorem can now be applied and we obtain

$$g(\lambda a_k) - g(0) = \lambda a_k g'(\alpha_k), \qquad \text{where } -|\lambda a_k| < \alpha_k < |\lambda a_k|.$$

Equation (22) then becomes

$$\begin{aligned} f(\mathbf{y}) - f(\mathbf{x}) &= \lambda \sum_{k=1}^{n} a_k D_k f(\mathbf{x} + \lambda\mathbf{v}_{k-1} + \alpha_k\mathbf{u}_k) \\ &= \nabla f(\mathbf{x}) \cdot (\mathbf{y} - \mathbf{x}) + \lambda \sum_{k=1}^{n} a_k[D_k f(\mathbf{x} + \lambda\mathbf{v}_{k-1} + \alpha_k\mathbf{u}_k) \\ &\qquad - D_k f(\mathbf{x})]. \end{aligned}$$

Since we have $|\lambda\mathbf{v}_{k-1} + \alpha_k\mathbf{u}_k| \leq |\lambda| + |\lambda a_k| \leq 2|\lambda| < \delta$, we may apply inequality (21) to each term of the sum in modified Eq. (22), and we find

$$|f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x}) \cdot (\mathbf{y} - \mathbf{x})| < |\lambda|\epsilon = \epsilon|\mathbf{y} - \mathbf{x}|.$$

Since $\mathbf{y}$ was an arbitrary point in $N'(\mathbf{x}; \delta/2)$, this proves the theorem.

NOTE. The condition that the n partials be continuous at $\mathbf{x}$, although sufficient, is by no means a necessary condition for the existence of the differential at $\mathbf{x}$. In fact, consider the special case in which f can be expressed as a sum of n functions, say

$$f(\mathbf{x}) = f_1(x_1) + \cdots + f_n(x_n), \qquad \text{if } \mathbf{x} = (x_1, \ldots, x_n),$$

where the kth function f_k is a function of the one-dimensional variable x_k

alone Then the mere existence of the partials $D_1f, \ldots, D_nf$ will suffice to establish the existence of df. This is rather easy to prove and is left as an exercise (See Exercise 6–3) A similar, but slightly more general result is contained in Exercise 6–4.

6–10 Partial derivatives of higher order. If a function f, defined on an open set S in E_n, has finite partial derivatives, then the derivatives $D_1f, \ldots, D_nf$ are new functions also defined on S and we can consider their partial derivatives This gives rise to *second-order* partial derivatives

6–19 DEFINITION. *Let f be defined on an open set S in E_n, and assume that the partial derivative D_kf exists on S Then $D_{r,k}f$ will denote the partial derivative of D_kf with respect to the rth variable, that is, $D_{r,k}f = D_r(D_kf)$ whenever this derivative exists Higher-order partial derivatives are similarly defined.*

Other notations are

$$D_{r,k}f = \frac{\partial^2 f}{\partial x_r \partial x_k}, \qquad D_{p,q,r}f = \frac{\partial^3 f}{\partial x_p \partial x_q \partial x_r}.$$

The example

$$f(x, y) = \begin{cases} xy(x^2 - y^2)/(x^2 + y^2), & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0), \end{cases}$$

shows that $D_{1,2}f(x, y)$ is not necessarily the same as $D_{2,1}f(x, y)$ In fact, in this example we have

$$D_1f(x, y) = \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^2}, \qquad \text{if } (x, y) \neq (0, 0),$$

and $D_1f(0, 0) = 0$ Hence, $D_1f(0, y) = -y$ for all y and therefore $D_{2,1}f(0, y) = -1$, $D_{2,1}f(0, 0) = -1$ On the other hand, we have

$$D_2f(x, y) = \frac{x(x^4 - 4x^2y^2 - y^4)}{(x^2 + y^2)^2}, \qquad \text{if } (x, y) \neq (0, 0),$$

and $D_2f(0, 0) = 0$, so that $D_2f(x, 0) = x$ for all x Therefore, $D_{1,2}f(x, 0) = 1$, $D_{1,2}f(0, 0) = 1$, and we see that $D_{2,1}f(0, 0) \neq D_{1,2}f(0, 0)$.

The next theorem gives us a criterion for determining when the two mixed partials $D_{1,2}f$ and $D_{2,1}f$ will be equal.

6–20 THEOREM. *If D_1f, D_2f, and $D_{2,1}f$ are continuous in a neighborhood of the point (x_0, y_0) in E_2, then $D_{1,2}f(x_0,y_0)$ exists and we have*

$$D_{1,2}f(x_0, y_0) = D_{2,1}f(x_0, y_0).$$

Proof. We are to prove that the limit

$$\lim_{h\to 0} \frac{D_2f(x_0 + h, y_0) - D_2f(x_0, y_0)}{h}$$

exists and has the value $D_{2,1}f(x_0, y_0)$. Let (x, y) be a point in the neighborhood of (x_0, y_0) given in the theorem. Then we have

$$D_2f(x, y) = \lim_{k\to 0} \frac{f(x, y + k) - f(x, y)}{k}.$$

Hence the numerator of the quotient under consideration can be written in the form

$$D_2f(x_0 + h, y_0) - D_2f(x_0, y_0) = \lim_{k\to 0} \frac{f(x_0 + h, y_0 + k) - f(x_0 + h, y_0)}{k} - \lim_{k\to 0} \frac{f(x_0, y_0 + k) - f(x_0, y_0)}{k}.$$

Keeping k fixed, we introduce a new function g_k by means of the relation

$$g_k(t) = f(x_0 + t, y_0 + k) - f(x_0 + t, y_0).$$

Thus we can write

$$D_2f(x_0 + h, y_0) - D_2f(x_0, y_0) = \lim_{k\to 0} \frac{g_k(h) - g_k(0)}{k} = \lim_{k\to 0} h\, \frac{g_k'(\bar{h})}{k},$$

where $\bar{h}$ lies between 0 and h. (Note that $\bar{h}$ also depends on k, since g_k depends on k.) The derivative of g_k can be computed as follows:

$$g_k'(t) = D_1f(x_0 + t, y_0 + k) - D_1f(x_0 + t, y_0)$$

and hence we have

$$\begin{aligned}\frac{D_2f(x_0 + h, y_0) - D_2f(x_0, y_0)}{h} &\\ &= \lim_{k\to 0} \frac{D_1f(x_0 + \bar{h}, y_0 + k) - D_1f(x_0 + \bar{h}, y_0)}{k} \\ &= \lim_{k\to 0} D_{2,1}f(x_0 + \bar{h}, \bar{y}),\end{aligned}$$

where $\bar{y}$ lies between y_0 and $y_0 + k$.

It remains to prove that we have

$$\lim_{h \to 0} \lim_{k \to 0} D_{2,1}f(x_0 + \bar{h}, \bar{y}) = D_{2,1}f(x_0, y_0)$$

When $k \to 0$, then $\bar{y} \to y_0$, but the behavior of $\bar{h}$ as a function of k is not known. If $\bar{h}$ were *independent* of k, then we could write

$$\lim_{k \to 0} D_{2,1}f(x_0 + \bar{h}, \bar{y}) = D_{2,1}f(x_0 + \bar{h}, y_0)$$

because $D_{2,1}f$ is continuous in a neighborhood of (x_0, y_0). Since $\bar{h} \to 0$ when $h \to 0$, we would obtain

$$\lim_{h \to 0} D_{2,1}f(x_0 + \bar{h}, y_0) = D_{2,1}f(x_0, y_0).$$

However, the fact that $\bar{h}$ depends on k in an unknown manner makes necessary a slightly more involved argument.

Let us define $F(h)$ to be the limit

$$F(h) = \lim_{k \to 0} D_{2,1}f(x_0 + \bar{h}, \bar{y}),$$

known to exist by what we have already proved. The theorem will be proved if we can show that $\lim_{h \to 0} F(h) = D_{2,1}f(x_0, y_0)$.

Let $\epsilon > 0$ be given. Then there is a neighborhood $N(x_0, y_0)$, of radius δ, say, such that

$$|D_{2,1}f(x, y) - D_{2,1}f(x_0, y_0)| < \frac{\epsilon}{2}, \qquad \text{if } (x, y) \in N(x_0, y_0)$$

The point $(x_0 + \bar{h}, \bar{y})$ will be in the neighborhood $N(x_0, y_0)$ if h and k are chosen so that $|h| < \delta/2$ and $|k| < \delta/2$. Keeping h fixed, with $|h| < \delta/2$, we will have

$$0 \leq |D_{2,1}f(x_0 + \bar{h}, \bar{y}) - D_{2,1}f(x_0, y_0)| < \frac{\epsilon}{2}, \qquad \text{if } |k| < \frac{\delta}{2}$$

If we let $k \to 0$ in this inequality, the term $D_{2,1}f(x_0 + \bar{h}, \bar{y})$ tends to $F(h)$ and the other terms are independent of k, so that we can write

$$0 \leq |F(h) - D_{2,1}f(x_0, y_0)| \leq \frac{\epsilon}{2} < \epsilon,$$

provided that $|h| < \delta/2$. But this is precisely the meaning of the statement

$$\lim_{h \to 0} F(h) = D_{2,1}f(x_0, y_0)$$

and hence the theorem is proved.

NOTE. It should be observed that the theorem still holds if the roles of $D_{1,2}f$ and $D_{2,1}f$ are interchanged.

If f is a function of two variables, then there are four second-order partial derivatives to consider; namely, $D_{1,1}f$, $D_{1,2}f$, $D_{2,1}f$, and $D_{2,2}f$. We have just shown that only three of these are distinct if f is suitably restricted.

The number of partial derivatives of order k which can be formed is 2^k. If all these derivatives are continuous in a neighborhood of the point (x_0, y_0), then certain of the mixed partials will be equal. Each mixed partial is of the form $D_{r_1,\ldots,r_k}f$, where each r_j is either 1 or 2. If we have two such mixed partials, $D_{r_1,\ldots,r_k}f$ and $D_{p_1,\ldots,p_k}f$, where the k-tuple $(r_1, \ldots, r_k)$ is a permutation of the k-tuple $(p_1, \ldots, p_k)$, then the two partials will be equal at (x_0, y_0) if all 2^k partials are continuous in a neighborhood of (x_0, y_0). This statement can be easily proved by mathematical induction, using Theorem 6-20 (which is the case $k = 2$). We omit the proof for general k. From this it follows that among the 2^k partial derivatives of order k, there are only $k + 1$ *distinct* partials in general, namely, those of the form $D_{r_1,\ldots,r_k}f$, where the k-tuple $(r_1, \ldots, r_k)$ assumes the following $k + 1$ forms:

$$(2, 2, \ldots, 2), \qquad (1, 2, 2, \ldots, 2), \qquad (1, 1, 2, \ldots, 2), \ldots, \qquad (1, 1, \ldots, 1, 2), \qquad (1, \ldots, 1).$$

Similar statements hold, of course, for functions of n variables. In this case, there are n^k partial derivatives of order k that can be formed. Continuity of all these partials at a point $\mathbf{x}$ implies that $D_{r_1,\ldots,r_k}f(\mathbf{x})$ is unchanged when the indices $r_1, \ldots, r_k$ are permuted. Each r_i is now a positive integer $\leq n$.

6-11 Taylor's formula for functions of several variables. Taylor's formula (Theorem 5-14) will now be extended to functions of several variables. In order to state the general theorem in a form which resembles the one-dimensional case, we first introduce higher-order differentials.

6-21 DEFINITION. *Let f be a real-valued function defined on a subset of E_n. The second-order differential d^2f is a function of two n-dimensional variables defined, for those points $\mathbf{x}$ in E_n where f has second-order partial derivatives and for every $\mathbf{t}$ in E_n, by the equation*

$$d^2f(\mathbf{x}; \mathbf{t}) = \sum_{i=1}^{n} \sum_{j=1}^{n} D_{i,j}f(\mathbf{x})t_j t_i, \qquad \text{if } \mathbf{t} = (t_1, \ldots, t_n).$$

The third-order differential d^3f is defined by the equation

$$d^3f(\mathbf{x}, \mathbf{t}) = \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{k=1}^{n} D_{i,j,k}f(\mathbf{x})t_k t_j t_i,$$

and the mth *order differential* $d^m f$ *is similarly defined (when all* mth *order partials exist)*

NOTE Higher-order differentials do not enjoy all the properties of first-order differentials For example, they are not linear in the second variable, nor do they satisfy Cauchy's invariant rule (See Exercise 6–25)

6–22 THEOREM *Let* f *have continuous partial derivatives of order* m *at each point of an open set* S *in* E_n *If* $\mathbf{a} \in S$, $\mathbf{b} \in S$, $\mathbf{a} \neq \mathbf{b}$, *and if* $L(\mathbf{a}, \mathbf{b}) \subset S$, *then there exists a point* $\mathbf{z}$ *on the line segment* $L(\mathbf{a}, \mathbf{b})$ *such that*

$$f(\mathbf{b}) - f(\mathbf{a}) = \sum_{k=1}^{m-1} \frac{1}{k!} d^k f(\mathbf{a}; \mathbf{b} - \mathbf{a}) + \frac{1}{m!} d^m f(\mathbf{z}; \mathbf{b} - \mathbf{a}).$$

Proof Define a new function g of one real variable by the equation

$$g(t) = f[t\mathbf{b} + (1 - t)\mathbf{a}], \qquad \text{if } 0 \le t \le 1.$$

Then $g(1) = f(\mathbf{b})$ and $g(0) = f(\mathbf{a})$ We will prove the theorem by applying the one-dimensional Taylor formula to g, writing

$$(23) \qquad g(1) - g(0) = \sum_{k=1}^{m-1} \frac{g^{(k)}(0)}{k!} + \frac{1}{m!} g^{(m)}(\bar{t}), \qquad \text{where } 0 < \bar{t} < 1.$$

Now g is a composite function given by $g(t) = f[\mathbf{p}(t)]$, where $\mathbf{p}(t) = t\mathbf{b} + (1 - t)\mathbf{a}$ The kth component of $\mathbf{p}$ has a derivative given by $p_k'(t) = b_k - a_k$ Applying the chain rule (Theorem 6–14), we see that g' exists in $0 \le t \le 1$ and is given by the formula

$$g'(t) = \sum_{j=1}^{n} D_j f[\mathbf{p}(t)](b_j - a_j) = df(\mathbf{p}(t), \mathbf{b} - \mathbf{a})$$

Again applying the chain rule, we obtain

$$g''(t) = \sum_{i=1}^{n} \sum_{j=1}^{n} D_{i,j} f[\mathbf{p}(t)](b_j - a_j)(b_i - a_i) = d^2 f(\mathbf{p}(t), \mathbf{b} - \mathbf{a}).$$

Similarly, we find that $g^{(m)}(t) = d^m f(\mathbf{p}(t); \mathbf{b} - \mathbf{a})$. Substitution in (23) then yields the theorem, since $\bar{t}\mathbf{b} + (1 - \bar{t})\mathbf{a} \in L(\mathbf{a}, \mathbf{b})$.

6–12 Differentiation of functions of a complex variable. In this section we shall discuss briefly derivatives of complex-valued functions defined on subsets of the complex plane. Such functions are, of course, vector-valued functions whose domain and range are subsets of E_2. All the considerations of Chapter 4 concerning limits and continuity of vector-valued functions apply, in particular, to functions of a complex variable. There is, however, one essential difference between the set of complex numbers E_2 and the set of n-dimensional vectors E_n (when $n > 2$) that plays an important role here. In the complex number system we have the four algebraic operations of addition, subtraction, multiplication, and division, and these operations satisfy most of the "usual" laws of algebra that hold for the real number system. In particular, they satisfy the first five axioms for real numbers listed in Chapter 1. (Axioms 6 through 10 involve the ordering relation $<$, which cannot exist among the complex numbers.) Any algebraic system which satisfies Axioms 1 through 5 is said to form a *field*. (For a thorough discussion of fields, see Ref. 1–1.) Multiplication and division, it turns out, cannot be introduced in E_n (for $n > 2$) in such a way that E_n will become a field* which includes E_2. Since division *is* possible in E_2, however, we can form the fundamental difference quotient $[f(z) - f(z_0)]/(z - z_0)$ which was used to define the derivative in E_1, and it now becomes clear how the derivative should be defined in E_2.

6–23 DEFINITION. *Let f be a complex-valued function defined on an open set S in E_2, and assume $z_0 \in S$. Then f is said to have a derivative $f'(z_0)$ at z_0 if the limit*

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'(z_0)$$

exists.

By means of this limit process, a new complex-valued function f' is defined at those points z of S where $f'(z)$ exists. Higher-order derivatives $f'', f''', \ldots$ are, of course, similarly defined.

The following statements can now be proved for complex-valued functions by using exactly the same proofs as in the real case:

(a) *Existence of $f'(z_0)$ implies continuity of f at z_0.*

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*For example, if it were possible to define multiplication in E_3 so as to make E_3 a field including E_2, we could argue as follows: For every $\mathbf{x}$ in E_3, the vectors $1, \mathbf{x}, \mathbf{x}^2, \mathbf{x}^3$ would be linearly dependent (see Ref. 1–1, p. 172). Hence for each $\mathbf{x}$ in E_3, a relation of the form $a_0 + a_1\mathbf{x} + a_2\mathbf{x}^2 + a_3\mathbf{x}^3 = \mathbf{0}$ would hold, where a_0, a_1, a_2, a_3 are real numbers. But every polynomial of degree three with real coefficients is a product of a linear polynomial and a quadratic polynomial with real coefficients. The only roots such polynomials can have are either real numbers or complex numbers.

(b) *If two functions f and g have derivatives at z_0, then their sum, difference, product, and quotient also have derivatives at z_0 and are given by the usual formulas (as in Theorem 5–4). In the case of f/g, we must assume $g(z_0) \neq 0$.*

(c) *The chain rule is valid, that is to say, we have*

$$(gf)'(z_0) = g'[f(z_0)]f'(z_0)$$

if the domain of g contains a neighborhood of $f(z_0)$ and if $f'(z_0)$ and $g'[f(z_0)]$ both exist

When $f(z) = z$, Definition 6–23 immediately yields $f'(z) = 1$ for all z in E_2 Using (b) repeatedly, we find that $f'(z) = nz^{n-1}$ when $f(z) = z^n$ (n is a positive integer) This also holds when n is a negative integer, provided $z \neq 0$ Therefore, we may compute derivatives of complex polynomials and complex rational functions by the same techniques used in elementary differential calculus

6–13 The Cauchy-Riemann equations. If f is a complex-valued function of a complex variable, we can write each function value in the form

$$f(z) = u(z) + iv(z),$$

where u and v are real-valued functions of a complex variable We can, of course, also consider u and v to be real-valued functions of two real variables and then we write

$$f(z) = u(x, y) + iv(x, y), \qquad \text{if } z = x + iy.$$

In either case, we write $f = u + iv$ and we refer to u and v as the real and imaginary parts of f For example, in the case of the complex exponential function f, defined by

$$f(z) = e^z = e^x \cos y + ie^x \sin y$$

(see Definition 1–23), the real and imaginary parts are given by

$$u(x, y) = e^x \cos y, \qquad v(x, y) = e^x \sin y$$

Similarly, when $f(z) = z^2 = (x + iy)^2$, we find

$$u(x, y) = x^2 - y^2, \qquad v(x, y) = 2xy.$$

In the next theorem we shall see that the existence of the derivative f' places a rather severe restriction on the real and imaginary parts u and v

6–24 THEOREM. *Let f be a function of a complex variable defined on an open set S in E_2, and write $f = u + iv$. If f has a derivative $f'(z_0)$ at the point $z_0 = x_0 + iy_0$ of S, then u and v must have finite partial derivatives D_1u, D_2u, D_1v, D_2v at (x_0, y_0), and they are related to $f'(z_0)$ by the two equations*

$$f'(z_0) = D_1u(x_0, y_0) + iD_1v(x_0, y_0)$$

and

$$f'(z_0) = D_2v(x_0, y_0) - iD_2u(x_0, y_0).$$

This implies, in particular, that we must have

$$D_1u(x_0, y_0) = D_2v(x_0, y_0) \quad \textit{and} \quad D_1v(x_0, y_0) = -D_2u(x_0, y_0).$$

NOTE. These last two equations are known as the Cauchy-Riemann equations. They are usually seen in the form

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

Proof. The existence of $f'(z_0)$ implies that, if $\epsilon > 0$ is given, there exists a neighborhood $N(z_0; \delta)$ such that

$$\left|\frac{f(z) - f(z_0)}{z - z_0} - f'(z_0)\right| < \epsilon, \qquad \text{whenever } z \in N'(z_0; \delta).$$

Let us write $z - z_0 = x - x_0 + i(y - y_0)$ and $f'(z_0) = A(z_0) + iB(z_0)$, where $A(z_0)$ and $B(z_0)$ are real. If we consider only those z for which $y = y_0$ and consider real and imaginary parts separately, the foregoing inequality implies the two inequalities

$$\left|\frac{u(x, y_0) - u(x_0, y_0)}{x - x_0} - A(z_0)\right| < \epsilon$$

and

$$\left|\frac{v(x, y_0) - v(x_0, y_0)}{x - x_0} - B(z_0)\right| < \epsilon$$

whenever $0 < |x - x_0| < \delta$. But this means that

$$D_1u(x_0, y_0) = A(z_0) \quad \text{and} \quad D_1v(x_0, y_0) = B(z_0).$$

Similarly, if we take only those z for which $x = x_0$, we find

$$\left| \frac{v(x_0, y) - v(x_0, y_0)}{y - y_0} - A(z_0) \right| < \epsilon$$

and

$$\left| \frac{u(x_0, y) - u(x_0, y_0)}{y - y_0} + B(z_0) \right| < \epsilon$$

whenever $0 < |y - y_0| < \delta$ This implies

$$D_2v(x_0, y_0) = A(z_0) \quad \text{and} \quad D_2u(x_0, y_0) = -B(z_0)$$

and the theorem follows at once

This theorem tells us that a *necessary* condition for the function $f = u + iv$ to have a derivative at z_0 is that the four partials, D_1u, D_2u, D_1v, D_2v, must exist at z_0 and satisfy the Cauchy-Riemann equations This condition, however, is *not sufficient*, as we see by considering the example in which u and v are defined as follows

$$u(x, y) = \frac{x^3 - y^3}{x^2 + y^2} \quad \text{if } (x, y) \neq (0, 0), \quad u(0, 0) = 0,$$

$$v(x, y) = \frac{x^3 + y^3}{x^2 + y^2} \quad \text{if } (x, y) \neq (0, 0), \quad v(0, 0) = 0$$

It is easily seen that $D_1u(0, 0) = D_1v(0,0) = 1$ and that $D_2u(0,0) = -D_2v(0, 0) = -1$, so that the Cauchy-Riemann equations hold at $(0,0)$ Nevertheless, the function $f = u + iv$ cannot have a derivative at $z = 0$ In fact, for $x = 0$, the difference quotient becomes

$$\frac{f(z) - f(0)}{z - 0} = \frac{-y + iy}{iy} = 1 + i,$$

whereas for $x = y$, it becomes

$$\frac{f(z) - f(0)}{z - 0} = \frac{xi}{x + ix} = \frac{1 + i}{2},$$

and hence $f'(0)$ cannot exist

The Cauchy-Riemann equations suffice to establish the existence of the derivative of $f = u + iv$ at z_0, provided that u and v have *continuous* partials in some neighborhood of z_0 That is, we have

6–25 THEOREM *Let u and v be two real-valued functions defined on an open set S in E_2, and assume that the four partial derivatives D_1u, D_2u, D_1v, D_2v exist and are continuous on S. If for some point z_0 in S we have*

$$D_1u(z_0) = D_2v(z_0) \quad \textit{and} \quad D_2u(z_0) = -D_1v(z_0),$$

then the function $f = u + iv$ *has a derivative at* z_0.

Proof. The existence of continuous partials implies that u and v have differentials on S and hence the two-dimensional Mean Value Theorem is applicable. If $N(z_0)$ is a neighborhood of z_0 contained in S, then for any z in $N'(z_0)$ we can write

$$u(z) - u(z_0) = \nabla u(z_1) \cdot (z - z_0), \qquad \text{where } z_1 \in L(z, z_0),$$

and

$$v(z) - v(z_0) = \nabla v(z_2) \cdot (z - z_0), \qquad \text{where } z_2 \in L(z, z_0).$$

For simplicity we use vector terminology and, as usual, $a \cdot b$ denotes the inner product of a and b. Then we have

$$f(z) - f(z_0) = [\nabla u(z_0) + i\nabla v(z_0)] \cdot (z - z_0) + \epsilon_1 + i\epsilon_2,$$

where

$$\epsilon_1 = [\nabla u(z_1) - \nabla u(z_0)] \cdot (z - z_0), \qquad \epsilon_2 = [\nabla v(z_2) - \nabla v(z_0)] \cdot (z - z_0).$$

But now we can write

$$\begin{aligned}
&[\nabla u(z_0) + i\nabla v(z_0)] \cdot (z - z_0) \\
&\quad = D_1u(z_0)(x - x_0) + D_2u(z_0)(y - y_0) + iD_1v(z_0)(x - x_0) \\
&\qquad\qquad + iD_2v(z_0)(y - y_0) \\
&\quad = D_1u(z_0)[(x - x_0) + i(y - y_0)] + iD_1v(z_0)[(x - x_0) + i(y - y_0)]
\end{aligned}$$

because of the Cauchy-Riemann equations. Hence we have

$$\frac{f(z) - f(z_0)}{z - z_0} = D_1u(z_0) + iD_1v(z_0) + \frac{\epsilon_1 + i\epsilon_2}{z - z_0}.$$

But ϵ_1 is given by the expression

$$\epsilon_1 = [D_1u(z_1) - D_1u(z_0)](x - x_0) + [D_2u(z_1) - D_2u(z_0)](y - y_0),$$

so that we have

$$|\epsilon_1| \le |z - z_0|\{|D_1u(z_1) - D_1u(z_0)| + |D_2u(z_1) - D_2u(z_0)|\}.$$

Similarly,

$$|\epsilon_2| \leq |z - z_0|\{|D_1v(z_2) - D_1v(z_0)| + |D_2v(z_2) - D_2v(z_0)|\}$$

If $\epsilon > 0$ is given, the original neighborhood $N(z_0)$ can be chosen so that all four inequalities

$$|D_ku(z) - D_ku(z_0)| < \frac{\epsilon}{4}, \qquad |D_kv(z) - D_kv(z_0)| < \frac{\epsilon}{4} \qquad (k = 1, 2),$$

hold whenever $z \in N'(z_0)$ But if z is in $N'(z_0)$, so are the points z_1 and z_2, and hence we have

$$\left|\frac{\epsilon_1 + i\epsilon_2}{z - z_0}\right| < \epsilon, \qquad \text{whenever } z \in N'(z_0).$$

This proves that $f'(z_0)$ exists and equals $D_1u(z_0) + iD_1v(z_0)$

We can use Theorems 6–24 and 6–25 to obtain the derivative of the exponential function Writing $f = u + iv$, where $f(z) = e^z$, we have $u(x, y) = e^x \cos y$, $v(x, y) = e^x \sin y$, and hence

$$D_1u(x, y) = e^x \cos y = D_2v(x, y), \qquad D_2u(x, y) = -e^x \sin y = -D_1v(x,y).$$

Applying Theorem 6–25, we see that $f'(z)$ exists *everywhere* in E_2 In order to compute the derivative, we use Theorem 6–24 to obtain

$$f'(z) = e^x \cos y + ie^x \sin y = f(z).$$

That is, the exponential function is its own derivative (as in the real case) Further examples are given in Exercise 6–27

6–26 THEOREM *Let u and v be two real-valued functions defined on an open set S in E_2, and assume that the partial derivatives D_1u, D_2u, $D_{1,2}u$, D_1v, D_2v, $D_{1,2}v$ exist and are continuous on S If u and v satisfy the Cauchy-Riemann equations everywhere in S, then the partial derivatives $D_{1,1}u$, $D_{1,1}v$, $D_{2,2}u$, and $D_{2,2}v$ also exist in S and satisfy the equations*

$$D_{1,1}u + D_{2,2}u = 0, \qquad D_{1,1}v + D_{2,2}v = 0$$

everywhere in S

Proof. Since $D_1u = D_2v$, we have $D_{1,1}u = D_{1,2}v$ Similarly, $D_2u = -D_1v$ implies $D_{2,2}u = -D_{2,1}v$. The hypothesis of continuity implies $D_{1,2}v = D_{2,1}v$, and hence $D_{1,1}u = -D_{2,2}u$ In the same way, we find $D_{1,1}v = -D_{2,2}v$.

NOTE. The partial differential equation $D_{1,1}u + D_{2,2}u = 0$ is known as *Laplace's equation* and is usually seen in the form

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

The equation is also written more briefly as follows:

$$\nabla^2 u = 0 \quad \text{or} \quad \Delta u = 0.$$

Functions satisfying Laplace's equation on a set S are said to be *harmonic* on S.

Differentials and partial derivatives

6-1. Let f have finite partial derivatives $D_kf(\mathbf{x})$, $k = 1, 2, \ldots, n$, at each point $\mathbf{x}$ of an open set S in E_n. If f has a local maximum or a local minimum at the point $\mathbf{x}_0$ in S, show that $D_kf(\mathbf{x}_0) = 0$ for each $k = 1, 2, \ldots, n$.

6-2. (a) Show that every unit vector $\mathbf{u}$ in E_3 can be expressed in the form $\mathbf{u} = (\cos\alpha_1, \cos\alpha_2, \cos\alpha_3)$. What are the geometric meanings of $\alpha_1, \alpha_2, \alpha_3$?

(b) If f is defined on an open set S in E_3 and if f has a differential at the point $\mathbf{x}$ of S, show that

$$D_{\mathbf{u}}f(\mathbf{x}) = D_1f(\mathbf{x})\cos\alpha_1 + D_2f(\mathbf{x})\cos\alpha_2 + D_3f(\mathbf{x})\cos\alpha_3,$$

where $\mathbf{u}$ is the vector in part (a).

6-3. Given n real-valued functions $f_1, \ldots, f_n$, defined and having finite derivatives in the interval (a, b). For each $\mathbf{x}$ in the n-dimensional interval

$$S = \{(x_1, \ldots, x_n) \mid a < x_k < b, k = 1, 2, \ldots, n\},$$

define $f(\mathbf{x}) = f_1(x_1) + \cdots + f_n(x_n)$. Show that f has a differential at each point of S.

6-4. Given n functions $f_1, \ldots, f_n$ defined on an open set S in E_n. For each $\mathbf{x}$ in S, define $f(\mathbf{x}) = f_1(\mathbf{x}) + \cdots + f_n(\mathbf{x})$. If, for each $k = 1, 2, \ldots, n$, we have

$$\lim_{\substack{\mathbf{y}\to\mathbf{x}\\ y_k \neq x_k}} \frac{f_k(\mathbf{y}) - f_k(\mathbf{x})}{y_k - x_k} = a_k,$$

show that f has a differential at $\mathbf{x}$. (The number a_k may depend on $\mathbf{x}$.)

6-5. Let f have a differential $df(\mathbf{x}; \mathbf{t})$ at the point $\mathbf{x}$ in E_n, and assume that $f(\mathbf{x}) = 0$. Let g be continuous at $\mathbf{x}$. Show that the product function h defined by $h(\mathbf{x}) = f(\mathbf{x})g(\mathbf{x})$ also has a differential at $\mathbf{x}$ and that $dh(\mathbf{x}, \mathbf{t}) = g(\mathbf{x})\, df(\mathbf{x}, \mathbf{t})$, if $\mathbf{t} \in E_n$.

6-6. Consider the functions defined in E_2 by the following formulas.

(i) $f(x, y) = \dfrac{xy}{x^2 + y^2}$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$.

(ii) $f(x, y) = \dfrac{x^2y^2}{x^4 + y^4}$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$.

(a) In each case show that the partial derivatives $D_1f(x, y)$ and $D_2f(x,y)$ exist for every (x, y) in E_2 and evaluate these derivatives explicitly in terms of x and y.

(b) Explain why the functions described in (i) and (ii) do not have differentials at (0, 0).

Gradients and the chain rule.

6–7. Let $\mathbf{a}$ and $\mathbf{b}$ be two vectors in E_n. Show that the absolute value of their inner product never exceeds the product of their lengths; that is, $|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}|\,|\mathbf{b}|$. Show also that $|\mathbf{a} \cdot \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}|$ if, and only if $\mathbf{b} = \lambda\mathbf{a}$ or $\mathbf{a} = \lambda\mathbf{b}$ for some real number λ.

6–8. Assume that f has a differential at the point $\mathbf{x}$ in E_n and suppose that $|\nabla f(\mathbf{x})| \neq 0$. Show that there is one and only one direction $\mathbf{u}$ in E_n, $|\mathbf{u}| = 1$, such that $D_{\mathbf{u}}f(\mathbf{x}) = |\nabla f(\mathbf{x})|$, and that this is the direction for which $|D_{\mathbf{u}}f(\mathbf{x})|$ has its maximum value.

6–9. Compute the gradient vector $\nabla f(x, y)$ at those points (x, y) in E_2 where it exists:

(a) $f(x, y) = x^2y^2 \log (x^2 + y^2)$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$.

(b) $f(x, y) = xy \sin \dfrac{1}{x^2 + y^2}$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$.

6–10. Let f and g be real-valued functions defined on E_1 with continuous second derivatives f'' and g''. Define

$$F(x, y) = f[x + g(y)] \text{ for each } (x, y) \text{ in } E_2.$$

Find formulas for all partials of F of first and second order in terms of the derivatives of f and g. Verify the relation

$$(D_1F)(D_{1,2}F) = (D_2F)(D_{1,1}F).$$

6–11. Given a function f defined in E_2. Let

$$F(r, \theta) = f(r \cos \theta, r \sin \theta).$$

(a) Assume appropriate differentiability properties of f and show that

$$D_1F(r, \theta) = \cos \theta\, D_1f(x, y) + \sin \theta\, D_2f(x, y),$$

$$D_{1,1}F(r, \theta) = \cos^2 \theta D_{1,1}f(x, y) + 2 \sin \theta \cos \theta\, D_{1,2}f(x, y) + \sin^2 \theta D_{2,2}f(x, y),$$

where $x = r \cos \theta$, $y = r \sin \theta$.

(b) Find similar formulas for D_2F, $D_{1,2}F$, and $D_{2,2}F$.

(c) Verify the formula

$$|\nabla f(r\cos\theta, r\sin\theta)|^2 = [D_1F(r,\theta)]^2 + \frac{1}{r^2}[D_2F(r,\theta)]^2.$$

6–12. If f and g have gradient vectors $\nabla f(\mathbf{x})$ and $\nabla g(\mathbf{x})$ at a point $\mathbf{x}$ in E_n, show that the product function h defined by $h(\mathbf{x}) = f(\mathbf{x})g(\mathbf{x})$ also has a gradient vector at $\mathbf{x}$ and that

$$\nabla h(\mathbf{x}) = f(\mathbf{x})\nabla g(\mathbf{x}) + g(\mathbf{x})\nabla f(\mathbf{x})$$

State and prove a similar result for the quotient f/g

6–13. Let f be a function having a derivative f' at each point in E_1 and let g be defined on E_3 by the equation

$$g(x, y, z) = x^2 + y^2 + z^2$$

If h denotes the composite function $h = fg$, show that

$$|\nabla h(x, y, z)|^2 = 4g(x, y, z)\{f'[g(x, y, z)]\}^2.$$

6–14. Let f have a differential at each point (x, y) in E_2 Let g_1 and g_2 be defined on E_3 by the equations

$$g_1(x, y, z) = x^2 + y^2 + z^2, \qquad g_2(x, y, z) = x + y + z,$$

and let $\mathbf{g}$ be the vector-valued function whose values (in E_2) are given by

$$\mathbf{g}(x, y, z) = (g_1(x, y, z), g_2(x, y, z))$$

Let h be the composite function $h = f\mathbf{g}$ and show that

$$|\nabla h|^2 = 4(D_1f)^2g_1 + 4(D_1f)(D_2f)g_2 + 3(D_2f)^2$$

6–15. Let f be defined on an open set S in E_n We say that f is homogeneous of degree p over S if $f(\lambda\mathbf{x}) = \lambda^p f(\mathbf{x})$ for every real λ and for every $\mathbf{x}$ in S for which $\lambda\mathbf{x} \in S$ If such a function has a differential at $\mathbf{x}$, show that

$$\mathbf{x}\cdot\nabla f(\mathbf{x}) = pf(\mathbf{x})$$

NOTE. This is known as *Euler's theorem* for homogeneous functions [*Hint* For fixed $\mathbf{x}$, define $g(\lambda) = f(\lambda\mathbf{x})$ and compute $g'(1)$] Also prove the converse. That is, show that if $\mathbf{x}\cdot\nabla f(\mathbf{x}) = pf(\mathbf{x})$ for all $\mathbf{x}$ in an open set S, then f must be homogeneous of degree p over S.

Mean Value Theorems.

6–16. Given a function f of two variables having a differential everywhere in a neighborhood $N(\mathbf{x})$ of a point $\mathbf{x} = (x_1, x_2)$ in E_2. Show that if

$$\mathbf{y} = (y_1, y_2) \in N'(\mathbf{x}),$$

we have

$$f(\mathbf{y}) - f(\mathbf{x}) = (y_1 - x_1)D_1 f(z_1, y_2) + (y_2 - x_2)D_2 f(x_1, z_2),$$

where $z_1 \in L(x_1, y_1)$ and $z_2 \in L(x_2, y_2)$.

[*Hint*: Consider $g(t) = f[ty_1 + (1 - t)x_1, y_2] + f[x_1, ty_2 + (1 - t)x_2]$.]

6–17. Given a function f having a differential everywhere in a neighborhood $N(\mathbf{x})$ of a point $\mathbf{x} = (x_1, x_2, x_3)$ in E_3. Show that if $\mathbf{y} = (y_1, y_2, y_3) \in N'(\mathbf{x})$, we have

$$f(\mathbf{y}) - f(\mathbf{x}) = (y_1 - x_1)D_1 f(z_1, y_2, y_3) + (y_2 - x_2)D_2 f(x_1, z_2, y_3) + (y_3 - x_3)D_3 f(x_1, x_2, z_3),$$

where

$$z_1 \in L(x_1, y_1), \qquad z_2 \in L(x_2, y_2), \qquad z_3 \in L(x_3, y_3).$$

6–18. State and prove a generalization of the result in Exercise 6–17 for functions of n variables.

6–19. Do the partial derivatives $D_{1,2}f(0, 0)$ and $D_{2,1}f(0, 0)$ exist in either of the examples of Exercise 6–6?

6–20. Prove that mixed partials $D_{1,2}f$ and $D_{2,1}f$ will be equal at a point (x, y) in E_2 if each partial derivative $D_1 f$ and $D_2 f$ exists in some neighborhood of (x, y) and has a differential at (x, y).

6–21. Let f be a function of two variables. Use induction and Theorem 6–20 to prove that if the 2^k partial derivatives of f of order k are continuous in a neighborhood of a point (x_0, y_0), then all mixed partials of the form $D_{r_1,\ldots,r_k}f$ and $D_{p_1,\ldots,p_k}f$ will be equal at (x_0, y_0) if the k-tuple $(r_1, \ldots, r_k)$ contains the same number of ones as the k-tuple $(p_1, \ldots, p_k)$.

Differentials of higher order and Taylor's formula.

6–22. Define directional derivatives of higher order by the formula

$$D_{\mathbf{u}}^{k+1} f = D_{\mathbf{u}}(D_{\mathbf{u}}^k f), \qquad \text{if } |\mathbf{u}| = 1.$$

(a) Show that $D_{\mathbf{u}}^k f(\mathbf{x}) = d^k f(\mathbf{x}; \mathbf{u})$ when either side exists ($k > 1$).

(b) Show that Taylor's formula (Theorem 6–22) can be expressed as follows:

$$f(\mathbf{b}) - f(\mathbf{a}) = \sum_{k=1}^{m-1} \frac{1}{k!} D_{\mathbf{u}}^k f(\mathbf{a})|\mathbf{b} - \mathbf{a}|^k + \frac{1}{m!} D_{\mathbf{u}}^m f(\mathbf{z})|\mathbf{b} - \mathbf{a}|^m,$$

where $\mathbf{u} = (\mathbf{b} - \mathbf{a})/|\mathbf{b} - \mathbf{a}|$

6-23. If f is a function of two variables having continuous partials of order k on some open set S in E_2, show that

$$d^k f(\mathbf{x}, \mathbf{t}) = \sum_{r=0}^{k} \binom{k}{r} t_1^r t_2^{k-r} D_{p_1, \ldots, p_k} f(\mathbf{x}), \qquad \text{if } \mathbf{x} \in S, \qquad \mathbf{t} = (t_1, t_2),$$

where in the rth term we have $p_1 = \cdots = p_r = 1$ and $p_{r+1} = \cdots = p_k = 2$ Use this result to give an alternative expression for Taylor's formula (Theorem 6-22) in the case when $n = 2$ NOTE The symbol $\binom{k}{r}$ is the binomial coefficient $k!/[r!(k - r)!]$

6-24. Use Taylor's formula to expand the following in powers of $(x - 1)$ and $(y - 2)$

(a) $f(x, y) = x^3 + y^3 + xy^2$, (b) $f(x, y) = x^2 + xy + y^2$.

6-25. Let $h = g\mathbf{f}$, where $\mathbf{f}$, g, and h satisfy the hypotheses of Theorem 6-14. In addition, assume that the second order differentials $d^2 f_1, \ldots, d^2 f_n$ exist at $\mathbf{z}$ and that $d^2 g$ exists at $\mathbf{f}(\mathbf{z})$ Show that

(a)
$$D_{s,r} h(\mathbf{z}) = \sum_{k=1}^{n} \sum_{j=1}^{n} D_{j,k} g[\mathbf{f}(\mathbf{z})] D_s f_j(\mathbf{z}) D_r f_k(\mathbf{z})$$
$$+ \sum_{k=1}^{n} D_k g[\mathbf{f}(\mathbf{z})] D_{s,r} f_k(\mathbf{z}).$$

(b)
$$d^2 h(\mathbf{z}, \mathbf{t}) = d^2 g[\mathbf{f}(\mathbf{z}), v_1, \ldots, v_n] + dg[\mathbf{f}(\mathbf{z}), w_1, \ldots, w_n],$$

where $v_i = df_i(\mathbf{z}, \mathbf{t})$ and $w_i = d^2 f_i(\mathbf{z}, \mathbf{t})$, $i = 1, 2, \ldots, n$

NOTE. The presence of the second term on the right in (b) shows that there is no invariance theorem for second order differentials.

6-26. Let $f = u + iv$ and assume that $f'(z_0)$ exists. Write $z = z_0 + re^{i\alpha}$ (α is real, fixed) and let $r \to 0$ in the difference quotient $[f(z) - f(z_0)]/(z - z_0)$ to obtain

$$f'(z_0) = e^{-i\alpha}(D_{\mathbf{a}} u + i D_{\mathbf{a}} v),$$

where $\mathbf{a} = (\cos\alpha, \sin\alpha)$ is a unit vector and $D_{\mathbf{a}}u$, $D_{\mathbf{a}}v$ are directional derivatives. If $\mathbf{b} = (\cos\beta, \sin\beta)$ where $\beta = \alpha + \frac{1}{2}\pi$, show by a similar argument that

$$f'(z_0) = e^{-i\alpha}(D_{\mathbf{b}}v - iD_{\mathbf{b}}u).$$

Deduce that $D_{\mathbf{a}}u = D_{\mathbf{b}}v$, $D_{\mathbf{a}}v = -D_{\mathbf{b}}u$. (The Cauchy-Riemann equations are a special case.) The directional derivatives are to be evaluated at z_0.

6–27. (i) In each of the following examples write $f = u + iv$ and find explicit formulas for $u(x, y)$ and $v(x, y)$:

(a) $f(z) = \sin z$, (b) $f(z) = \cos z$,

(c) $f(z) = |z|$, (d) $f(z) = \bar{z}$,

(e) $f(z) = \arg z \quad (z \neq 0)$, (f) $f(z) = \operatorname{Log} z \quad (z \neq 0)$,

(g) $f(z) = e^{z^2}$, (h) $f(z) = z^{\alpha} \quad (\alpha \text{ complex}, z \neq 0)$.

(These functions are to be defined as indicated in Chapter 1.) (ii) Show that u and v satisfy the Cauchy-Riemann equations for the following values of z: All z in (a), (b), (g); no z in (c), (d), (e); all z except real $z \leq 0$ in (f), (h). (In part (h), the Cauchy-Riemann equations hold for all z if α is a nonnegative integer and they hold for all $z \neq 0$ if α is a negative integer.) (iii) Compute the derivative $f'(z)$ in (a), (b), (f), (g), (h), whenever it exists.

6–28. Write $f = u + iv$ and assume that f has a derivative at each point of an open set S in E_2. Assume also that u and v have continuous partials on S. Show that

(a) If u is constant on S, then f is constant on S.

(b) If $u^2 + v^2$ is constant on S, then f is constant on S.

(c) If $g(z) = \overline{f(\bar{z})}$, show that $g'(z)$ exists everywhere on $\bar{S}$, the mirror image of S in the real axis.

6–29. Assume that u and v satisfy the conditions of Theorem 6–26, let $f = u + iv$, $g = |f'|^2$, $h = 1 + |f|^2$, and show that

(a) $\nabla^2 h = 4g$, (b) $\nabla^2 h = h^2\nabla^2(\log h)$.

6–30. Let S be the open disk $\{z \mid |z| < 1\}$ and let $\bar{S} = \{z \mid |z| \leq 1\}$. Write $f = u + iv$, assume that f is continuous on $\bar{S}$, and suppose f has a continuous derivative everywhere on S. Assume further that $u(x, y) = x$ and $v(x, y) = y$ whenever $x^2 + y^2 = 1$, and that $f'(z) \neq 0$ if $z \in S$. Prove that $|f(z)| < 1$ if $z \in S$.

CHAPTER 7

APPLICATIONS OF PARTIAL DIFFERENTIATION

7–1 Introduction. This chapter comprises two principal parts The first part is devoted to proving a very important theorem of analysis, the *implicit function theorem*, the second part treats extremum problems

The implicit function theorem in its simplest form deals with a function f such that $y = f(x)$, this function being determined "implicitly" by an equation of the form $F(x, y) = 0$ The problem here is this. Given the equation $F(x, y) = 0$, when can we "solve" this equation for y and obtain an explicit solution $y = f(x)$? Also, if we know something about the continuity or differentiability of F, what can we conclude about the continuity or differentiability of f? The problem assumes a more general form when we have a function F of several (say, n) variables, and we ask whether we can solve an equation of the form $F(x_1, \ldots, x_n) = 0$ for one of the variables in terms of the others Even more generally, we will consider a system of several equations involving several variables and we will study the problem of solving these equations for some of the variables in terms of the remaining variables Under rather general conditions, a solution will always exist, and the exact statement of these conditions, with some conclusions about the solution, constitutes the implicit function theorem

An important special case of this theorem is the well-known problem in algebra of solving n linear equations of the form

$$(1) \qquad \sum_{j=1}^{n} a_{ij}x_j = b_i \qquad (i = 1, 2, \ldots, n)$$

where the a_{ij} and the b_i are considered as given numbers and $x_1, \ldots, x_n$ represent "unknowns" It is known that there exists a unique solution to this system if, and only if, the determinant of the coefficients is not zero In the general implicit function theorem, the nonvanishing of a certain determinant, known as a *Jacobian*, also comes into consideration. Jacobian determinants will be defined and their basic properties derived in the next section, but first we wish to make some remarks concerning notations and terminology

Most of the results in this chapter are concerned with collections of functions, where each function in the collection is a function of several variables Consequently, the statements of many of these results are usually rather complicated. Some simplification can be achieved by using vector terminology. For example, if we are dealing with a collection of n real-valued functions $f_1, \ldots, f_n$ defined on a set S, then we denote by $\mathbf{f}$ the vector-valued function whose value at the point $\mathbf{x}$ in S is $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_n(\mathbf{x}))$. We also write $\mathbf{f} = (f_1, \ldots, f_n)$ to indicate this relationship, and f_k is then called the kth component of $\mathbf{f}$. We will say that $\mathbf{f}$ *has a differential at the point* $\mathbf{x}$ if each component of $\mathbf{f}$ has a differential at $\mathbf{x}$. This will be abbreviated by writing $\mathbf{f} \in D$ at $\mathbf{x}$. We will further say that $\mathbf{f}$ is *continuously differentiable at* $\mathbf{x}$ if each component f_k of $\mathbf{f}$ has continuous first partials $D_j f_k$ $(j = 1, 2, \ldots, n)$ at $\mathbf{x}$. This will be denoted symbolically by writing $\mathbf{f} \in C'$ at $\mathbf{x}$. We also write $\mathbf{f} \in C'$ on S if $\mathbf{f} \in C'$ at each point of S; the notation $\mathbf{f} \in D$ on S has a similar meaning. Note that the statement $\mathbf{f} \in C'$ at $\mathbf{x}$ implies $\mathbf{f} \in D$ at $\mathbf{x}$, because of Theorem 6–18.

7–2 Jacobians.

7–1 DEFINITION. *Let $f_1, \ldots, f_n$ be n real-valued functions defined on an open set S in E_n, and let $\mathbf{f} = (f_1, \ldots, f_n)$. By the Jacobian of $\mathbf{f}$ we mean the real-valued function $J_{\mathbf{f}}$ whose values are given by the determinant*

$$J_{\mathbf{f}}(\mathbf{x}) = \begin{vmatrix} D_1 f_1(\mathbf{x}) & D_2 f_1(\mathbf{x}) \cdots D_n f_1(\mathbf{x}) \\ \vdots & \vdots \qquad\quad \vdots \\ D_1 f_n(\mathbf{x}) & D_2 f_n(\mathbf{x}) \cdots D_n f_n(\mathbf{x}) \end{vmatrix}$$

at those points $\mathbf{x}$ in S where all partials $D_j f_i(\mathbf{x})$ exist.

NOTE. The notation $\partial(f_1, \ldots, f_n)/\partial(x_1, \ldots, x_n)$ is also used instead of $J_{\mathbf{f}}(\mathbf{x})$ if $\mathbf{x} = (x_1, \ldots, x_n)$.

For brevity, a matrix A whose element in the ith row and jth column is a_{ij} will be written $A = [a_{ij}]$ and the determinant of this matrix will be denoted by $\det [A] = \det [a_{ij}]$. Thus, $J_{\mathbf{f}}(\mathbf{x}) = \det [D_j f_i(\mathbf{x})]$. The elements in the ith row of the matrix $[D_j f_i(\mathbf{x})]$ are the components of the gradient vector $\nabla f_i(\mathbf{x})$.

We recall from the elementary theory of n-rowed square matrices that the product AB of the matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ (in the order given) is the matrix $[c_{ij}]$, where

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

and that $\det [AB] = \det [A] \det [B]$. This fact will be used in the proof of the next theorem. (For an account of the theory of matrices and determinants, see Ref. 7–2.)

7-2 THEOREM (*Multiplication theorem for Jacobians*) *Let* $\mathbf{f} = (f_1, \dots, f_n)$ *be defined on an open set* S *in* E_n *and let* $\mathbf{g} = (g_1, \dots, g_n)$ *be defined on an open set* T *in* E_n, *where* $\mathbf{f}(S) \subset T$. *Let* $\mathbf{h} = (h_1, \dots, h_n)$ *be the composite function defined by the equation*

$$\mathbf{h}(\mathbf{x}) = \mathbf{g}[\mathbf{f}(\mathbf{x})], \qquad \text{if } \mathbf{x} \in S$$

Then, for every $\mathbf{x}$ *in* S *such that* $\mathbf{f} \in D$ *at* $\mathbf{x}$ *and* $\mathbf{g} \in D$ *at* $\mathbf{f}(\mathbf{x})$, *we have*

$$J_{\mathbf{h}}(\mathbf{x}) = J_{\mathbf{g}}[\mathbf{f}(\mathbf{x})]J_{\mathbf{f}}(\mathbf{x})$$

Proof. The assumptions $\mathbf{f} \in D$ at $\mathbf{x}$ and $\mathbf{g} \in D$ at $\mathbf{f}(\mathbf{x})$ assure us that $\mathbf{h} \in D$ at $\mathbf{x}$ because of the chain rule (Theorem 6-14). Multiplication of determinants then yields

$$J_{\mathbf{g}}[\mathbf{f}(\mathbf{x})]J_{\mathbf{f}}(\mathbf{x}) = \det[D_j g_i(\mathbf{f}(\mathbf{x}))] \det[D_j f_i(\mathbf{x})]$$

$$= \det\left[\sum_{k=1}^{n} D_k g_i(\mathbf{f}(\mathbf{x})) D_j f_k(\mathbf{x})\right]$$

But the chain rule also tells us that the last sum is $D_j h_i(\mathbf{x})$, and this proves the theorem

In particular, if $\mathbf{g} = \mathbf{f}^{-1}$, then $\mathbf{h}(\mathbf{x}) = \mathbf{x}$ for all $\mathbf{x}$ in S and $J_{\mathbf{h}}(\mathbf{x}) = 1$. Hence the last equation in the theorem becomes

$$J_{\mathbf{f}^{-1}}[\mathbf{f}(\mathbf{x})]J_{\mathbf{f}}(\mathbf{x}) = 1$$

This is sometimes written as follows

$$\frac{\partial(f_1, \dots, f_n)}{\partial(x_1, \dots, x_n)} \frac{\partial(x_1, \dots, x_n)}{\partial(f_1, \dots, f_n)} = 1.$$

The equation $J_{\mathbf{h}}(\mathbf{x}) = J_{\mathbf{g}}[\mathbf{f}(\mathbf{x})]J_{\mathbf{f}}(\mathbf{x})$ in Theorem 7-2 bears a strong resemblance to the chain rule $h'(x) = g'[f(x)]f'(x)$ of one-dimensional derivative theory and suggests that the Jacobian should perhaps be considered as a generalization (to vector-valued functions) of the notion of derivative Other reasons for adopting this point of view will be seen in some of the later chapters, in particular when we study the formula for changing variables in a multiple integral However, it should be pointed out that for complex-valued functions defined in E_2, the Jacobian $J_f(z)$ is not the same as the derivative $f'(z)$. In fact, if we write $f = u + iv$, then $f'(z)$ is given by the equation

$$f'(z) = D_1u(x, y) + iD_1v(x, y), \qquad z = x + iy,$$

whereas the Jacobian $J_f(z)$ is given by

$$J_f(z) = \begin{vmatrix} D_1u(x, y) & D_2u(x, y) \\ D_1v(x, y) & D_2v(x, y) \end{vmatrix} = D_1u(x, y)D_2v(x, y)$$
$$- D_1v(x, y)D_2u(x, y).$$

If the Cauchy-Riemann equations are also satisfied, we can write

$$J_f(z) = [D_1u(x, y)]^2 + [D_1v(x, y)]^2$$

and we see that, in this case, J_f and f' are related by the equation

$$J_f(z) = |f'(z)|^2.$$

7-3 Functions with nonzero Jacobian. In this section we obtain some properties of vector-valued functions having a nonzero Jacobian at certain points. These results will be used later in the proof of the implicit function theorem.

7-3 THEOREM. *Let K be an open sphere in E_n with center at $\mathbf{x}_0$ and let $\overline{K}$ be the corresponding closed sphere. Let $\mathbf{f} = (f_1, \ldots, f_n)$ be a vector-valued function which is continuous on $\overline{K}$ and assume that all the partial derivatives $D_jf_i(\mathbf{x})$ exist if $\mathbf{x} \in K$. Assume further that $\mathbf{f}$ is one-to-one on $\overline{K}$ and that the Jacobian $J_{\mathbf{f}}(\mathbf{x}) \neq 0$ for each $\mathbf{x}$ in K. Then $\mathbf{f}(K)$, the image of K under $\mathbf{f}$, contains a neighborhood of the point $\mathbf{f}(\mathbf{x}_0)$.*

Proof. Let $B(K)$ denote the boundary of K, that is,

$$B(K) = \{\mathbf{x} \mid |\mathbf{x} - \mathbf{x}_0| = r\},$$

where r is the radius of K. Then $B(K)$ is a closed and bounded set. Define a real-valued function g on $B(K)$ as follows:

$$g(x) = |\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}_0)|, \qquad \text{if } \mathbf{x} \in B(K).$$

Then $g(\mathbf{x}) > 0$ for each $\mathbf{x}$ in $B(K)$ because $\mathbf{f}$ is one-to-one on $\overline{K}$. Furthermore, g is continuous on $B(K)$, since $\mathbf{f}$ is continuous on $\overline{K}$. But $B(K)$ is a compact set and hence g must take on its absolute minimum m somewhere on $B(K)$. This minimum m must be positive, since $g(\mathbf{x}) > 0$ for all $\mathbf{x}$ in $B(K)$. Using this positive number m, let T be the open sphere with center

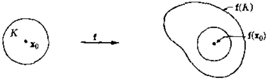

FIG. 7-1. Theorem 7-3 in E_2.

at $\mathbf{f}(\mathbf{x}_0)$ and radius $m/2$. We will prove that $T \subset \mathbf{f}(K)$ and this will prove the theorem. (See Fig. 7-1.)

To do this, we must show that $\mathbf{y} \in T$ implies $\mathbf{y} \in \mathbf{f}(K)$. Choose a point $\mathbf{y}$ in T, keep $\mathbf{y}$ fixed, and define a new real-valued function h on $\overline{K}$ as follows:

$$h(\mathbf{x}) = |\mathbf{f}(\mathbf{x}) - \mathbf{y}|, \quad \text{if } \mathbf{x} \in \overline{K}.$$

Then h is continuous on the compact set $\overline{K}$ and therefore has an absolute minimum on $\overline{K}$. We will show that h attains its minimum somewhere in the *open* sphere K. At the center we have $h(\mathbf{x}_0) = |\mathbf{f}(\mathbf{x}_0) - \mathbf{y}| < m/2$, since $\mathbf{y} \in T$. Hence the minimum value of h in $\overline{K}$ must also be $< m/2$. But at a boundary point $\mathbf{x}$ of $B(K)$, h has the value

$$\begin{aligned} h(\mathbf{x}) &= |\mathbf{f}(\mathbf{x}) - \mathbf{y}| = |\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}_0) - (\mathbf{y} - \mathbf{f}(\mathbf{x}_0))| \\ &\geq |\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}_0)| - |\mathbf{f}(\mathbf{x}_0) - \mathbf{y}| > g(\mathbf{x}) - \frac{m}{2} \geq \frac{m}{2}, \end{aligned}$$

so that the minimum cannot occur on the boundary of K. Hence there is an *interior point* $\mathbf{a}$ in K such that h attains its minimum at $\mathbf{a}$. At this point the square of h will also have a minimum. Since $h^2(\mathbf{x}) = |\mathbf{f}(\mathbf{x}) - \mathbf{y}|^2 = \sum_{r=1}^{n} [f_r(\mathbf{x}) - y_r]^2$ and since each partial derivative $D_k(h^2)$ must be zero at $\mathbf{a}$, we must have

$$\sum_{r=1}^{n} [f_r(\mathbf{a}) - y_r] D_k f_r(\mathbf{a}) = 0 \qquad (k = 1, 2, \ldots, n).$$

But this is a system of linear equations whose determinant $J_{\mathbf{f}}(\mathbf{a})$ is not zero, since $\mathbf{a} \in K$. This implies $f_r(\mathbf{a}) = y_r$ for each r, or $\mathbf{f}(\mathbf{a}) = \mathbf{y}$. That is, $\mathbf{y} \in \mathbf{f}(K)$. Hence $T \subset \mathbf{f}(K)$ and the theorem is proved.

Let us return now to the system of linear equations

$$\sum_{j=1}^{n} a_{ij} x_j = b_i \qquad (i = 1, 2, \ldots, n)$$

It is known from algebra that if $\det [a_{ij}] \neq 0$, this system has a unique solution. In fact, the solution can be obtained by Cramer's rule, which expresses each x_k as a quotient of two determinants, say $x_k = A_k/D$, where $D = \det [a_{ij}]$ and A_k is the determinant of the matrix obtained by replacing the kth column of $[a_{ij}]$ by $b_1, \ldots, b_n$. (For a proof of Cramer's rule, see Ref. 7-2 at the end of the chapter.) If we define $f_i(\mathbf{x}) = \sum_{j=1}^{n} a_{ij}x_j$, where $\mathbf{x} = (x_1, \ldots, x_n)$, then the vector-valued function $\mathbf{f} = (f_1, \ldots, f_n)$ has the Jacobian $J_{\mathbf{f}}(\mathbf{x}) = \det [a_{ij}]$. To say that the foregoing system of linear equations has a unique solution whenever $\det [a_{ij}] \neq 0$ is to say that this function $\mathbf{f}$ is one-to-one whenever $J_{\mathbf{f}}(\mathbf{x}) \neq 0$. [Why?] This result is analogous to Theorem 5-7, in which we proved that a real-valued function f is one-to-one in some neighborhood of a point x whenever the derivative $f'(x) \neq 0$. In the next theorem the result is extended to vector-valued functions which are continuously differentiable.

7-4 THEOREM. *Let $\mathbf{f} = (f_1, \ldots, f_n) \in C'$ on an open set S in E_n, and assume that the Jacobian $J_{\mathbf{f}}(\mathbf{x}_0) \neq 0$ for some point $\mathbf{x}_0$ in S. Then there exists a neighborhood $N(\mathbf{x}_0)$ such that $\mathbf{f}$ is one-to-one in this neighborhood.*

Proof. Let $\mathbf{Z}_1, \ldots, \mathbf{Z}_n$ be n points in S and let $\mathbf{Z} = (\mathbf{Z}_1; \ldots; \mathbf{Z}_n)$ denote that point in E_{n^2} whose first n components are the components of $\mathbf{Z}_1$, whose next n components are the components of $\mathbf{Z}_2$, and so on. Define a real-valued function h as follows:

$$h(\mathbf{Z}) = \det [D_j f_i(\mathbf{Z}_i)].$$

This function is continuous at those points $\mathbf{Z}$ in E_{n^2} where $h(\mathbf{Z})$ is defined, because each $D_j f_i$ is continuous on S and a determinant is a polynomial in its n^2 entries. Let $\mathbf{Z}_0$ be the special point in E_{n^2} obtained by putting $\mathbf{Z}_1 = \mathbf{Z}_2 = \cdots = \mathbf{Z}_n = \mathbf{x}_0$. Then $h(\mathbf{Z}_0) = J_{\mathbf{f}}(\mathbf{x}_0) \neq 0$ and hence, by continuity, there is some n-dimensional neighborhood $N(\mathbf{x}_0)$ such that $\det [D_j f_i(\mathbf{Z}_i)] \neq 0$ if each $\mathbf{Z}_i \in N(\mathbf{x}_0)$. We can now prove that $\mathbf{f}$ is one-to-one in this neighborhood. In fact, assume that $\mathbf{f}(\mathbf{x}) = \mathbf{f}(\mathbf{y})$, where $\mathbf{x} \in N(\mathbf{x}_0)$, $\mathbf{y} \in N(\mathbf{x}_0)$, $\mathbf{x} \neq \mathbf{y}$. Since $N(\mathbf{x}_0)$ is convex, the line segment $L(\mathbf{x}, \mathbf{y}) \subset N(\mathbf{x}_0)$ and we can apply the n-dimensional Mean Value Theorem to write

$$0 = f_i(\mathbf{y}) - f_i(\mathbf{x}) = \nabla f_i(\mathbf{Z}_i) \cdot (\mathbf{y} - \mathbf{x}) \qquad (i = 1, 2, \ldots, n),$$

where each $\mathbf{Z}_i \in L(\mathbf{x}, \mathbf{y})$ and hence $\mathbf{Z}_i \in N(\mathbf{x}_0)$. But this is a system of linear equations of the form

$$\sum_{k=1}^{n} (y_k - x_k)a_{ik} = 0, \qquad \text{with } a_{ik} = D_k f_i(\mathbf{Z}_i).$$

The determinant of this system is not zero, since $Z_i \in N(\mathbf{x}_0)$. Hence $y_k - x_k = 0$ for each k, and this contradicts the assumption that $\mathbf{x} \neq \mathbf{y}$. We have shown, therefore, that $\mathbf{x} \neq \mathbf{y}$ implies $\mathbf{f}(\mathbf{x}) \neq \mathbf{f}(\mathbf{y})$ and hence that $\mathbf{f}$ is one-to-one in $N(\mathbf{x}_0)$.

NOTE. The reader should be cautioned that Theorem 7–4 is a *local* theorem and not a global theorem. The nonvanishing of $J_{\mathbf{f}}(\mathbf{x}_0)$ guarantees the existence of $\mathbf{f}^{-1}$ in a neighborhood of $\mathbf{x}_0$. It does not follow that $\mathbf{f}$ has an inverse in S, even though $J_{\mathbf{f}}(\mathbf{x}) \neq 0$ for every $\mathbf{x}$ in S. The following example will illustrate this point. Let f be the complex-valued function defined by $f(z) = e^z$ if $z \in E_2$. Writing $z = x + iy$, we have

$$J_f(z) = |f'(z)|^2 = |e^z|^2 = e^{2x}$$

and we see that $J_f(z) \neq 0$ for every z in E_2. However, f is not one-to-one in E_2 because $f(z_1) = f(z_2)$ for every pair of points z_1 and z_2 which differ by $2\pi i$.

It follows from Theorem 7–4 that, under the conditions given, $\mathbf{f}$ has what might be called a "local inverse" in a neighborhood of a point $\mathbf{x}_0$ where $J_{\mathbf{f}}(\mathbf{x}_0) \neq 0$. In the next theorem we derive some local differentiability properties of this local inverse function.

7–4 The inverse function theorem.

7–5 THEOREM (*Inverse function theorem*). *Let $\mathbf{f} = (f_1, \ldots, f_n) \in C'$ on an open set S in E_n and let $T = \mathbf{f}(S)$. Assume that the Jacobian $J_{\mathbf{f}}(\mathbf{x}_0) \neq 0$ for some point $\mathbf{x}_0$ in S. Then there is a uniquely determined function $\mathbf{g}$ and there are two open sets $X \subset S$ and $Y \subset T$ such that*

(i) $\mathbf{x}_0 \in X$ *and* $\mathbf{f}(\mathbf{x}_0) \in Y$,

(ii) $Y = \mathbf{f}(X)$,

(iii) $\mathbf{f}$ *is one-to-one on* X,

(iv) $\mathbf{g}$ *is defined on* Y, $\mathbf{g}(Y) = X$, *and* $\mathbf{g}[\mathbf{f}(\mathbf{x})] = \mathbf{x}$ *for every* $\mathbf{x}$ *in* X,

(v) $\mathbf{g} \in C'$ *on* Y.

Proof. The Jacobian $J_{\mathbf{f}}$ is continuous on S and, since $J_{\mathbf{f}}(\mathbf{x}_0) \neq 0$, there is a neighborhood $N_1(\mathbf{x}_0)$ such that $J_{\mathbf{f}}(\mathbf{x}) \neq 0$ whenever $\mathbf{x} \in N_1(\mathbf{x}_0)$. By Theorem 7–4 there exists a neighborhood $N(\mathbf{x}_0) \subset N_1(\mathbf{x}_0)$ such that $\mathbf{f}$ is one-to-one in $N(\mathbf{x}_0)$. If $\overline{K}$ is a closed sphere contained in $N(\mathbf{x}_0)$ (with center at $\mathbf{x}_0$), and if K is the corresponding open sphere, then $\mathbf{f}(K)$ must contain a neighborhood of $\mathbf{f}(\mathbf{x}_0)$, by Theorem 7–3. Call this neighborhood Y, and let $X = \mathbf{f}^{-1}(Y) \cap K$. (See Fig. 7–2.) Then X is open, by Theorem 4–15. Applying Theorem 4–17 to $\overline{K}$, we find that there exists a function $\mathbf{g}$ (the

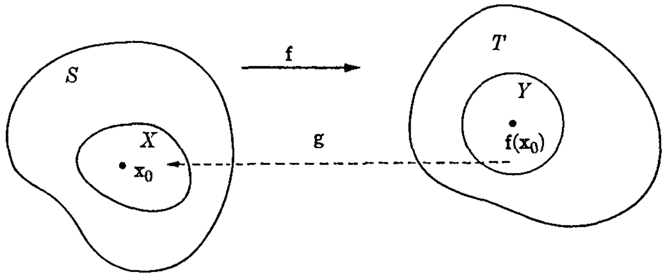

FIG. 7–2. The inverse-function theorem in E_2.

function $\mathbf{f}^{-1}$ of Theorem 4–17) defined on $\mathbf{f}(\overline{K})$ such that $\mathbf{g}[\mathbf{f}(\mathbf{x})] = \mathbf{x}$ for all $\mathbf{x}$ in $\overline{K}$. Moreover, since $\overline{K}$ is compact, $\mathbf{g}$ is continuous on $\mathbf{f}(\overline{K})$. Since $X \subset \overline{K}$ and $Y \subset \mathbf{f}(\overline{K})$, this proves (i), (ii), (iii), and (iv).

It remains to verify (v). For this purpose, define a real-valued function h by the equation $h(\mathbf{Z}) = \det [D_j f_i(\mathbf{Z}_i)]$, where $\mathbf{Z}_1, \ldots, \mathbf{Z}_n$ are n points in S, and $\mathbf{Z} = (\mathbf{Z}_1; \ldots; \mathbf{Z}_n)$ is the corresponding point in E_{n^2}. Then, arguing as in the proof of Theorem 7–4, there is a neighborhood $N_2(\mathbf{x}_0)$ such that $h(\mathbf{Z}) \neq 0$ if each $\mathbf{Z}_i \in N_2(\mathbf{x}_0)$. We can now assume that in the earlier part of the proof the neighborhood $N(\mathbf{x}_0)$ was chosen so that $N(\mathbf{x}_0) \subset N_2(\mathbf{x}_0)$. Then $\overline{K} \subset N_2(\mathbf{x}_0)$ and $h(\mathbf{Z}) \neq 0$ if each $\mathbf{Z}_i \in \overline{K}$.

To prove (v), write $\mathbf{g} = (g_1, \ldots, g_n)$. We must show that each $g_k \in C'$ on Y. In order to prove that $D_r g_k$ exists on Y, assume $\mathbf{y} \in Y$ and consider the difference quotient $[g_k(\mathbf{y} + \lambda\mathbf{u}_r) - g_k(\mathbf{y})]/\lambda$, where $\mathbf{u}_r$ is the rth unit coordinate vector. (Since Y is open, $\mathbf{y} + \lambda\mathbf{u}_r \in Y$ if $|\lambda|$ is sufficiently small.) Let $\mathbf{x} = \mathbf{g}(\mathbf{y})$ and $\mathbf{x}' = \mathbf{g}(\mathbf{y} + \lambda\mathbf{u}_r)$. Then both $\mathbf{x}$ and $\mathbf{x}'$ are in X and $\mathbf{f}(\mathbf{x}') - \mathbf{f}(\mathbf{x}) = \lambda\mathbf{u}_r$. Hence, $f_i(\mathbf{x}') - f_i(\mathbf{x}) = 0$ if $i \neq r$, and equals λ if $i = r$. By Theorem 6–17 we have

$$\frac{f_i(\mathbf{x}') - f_i(\mathbf{x})}{\lambda} = \nabla f_i(\mathbf{Z}_i) \cdot \frac{\mathbf{x}' - \mathbf{x}}{\lambda} \qquad (i = 1, 2, \ldots, n),$$

where each $\mathbf{Z}_i$ is on the segment joining $\mathbf{x}$ and $\mathbf{x}'$ and hence $\mathbf{Z}_i \in K$. The expression on the left is 1 or 0, according to whether $i = r$ or $i \neq r$. This is a system of n linear equations in n unknowns $(x_j' - x_j)/\lambda$ and has a unique solution, since $\det [D_j f_i(\mathbf{Z}_i)] = h(\mathbf{Z}) \neq 0$. Solving for the kth unknown by Cramer's rule, we obtain an expression for $[g_k(\mathbf{y} + \lambda\mathbf{u}_r) - g_k(\mathbf{y})]/\lambda$ as a quotient of determinants. As $\lambda \to 0$, the point $\mathbf{x}' \to \mathbf{x}$, since $\mathbf{g}$ is continuous, and hence each $\mathbf{Z}_i \to \mathbf{x}$, since $\mathbf{Z}_i$ is on the segment joining $\mathbf{x}$ and $\mathbf{x}'$. The determinant which appears in the denominator has for its limit the number $\det[D_j f_i(\mathbf{x})] = J_{\mathbf{f}}(\mathbf{x})$, and this is not 0, since $\mathbf{x} \in X$. Therefore, the following limit exists:

$$\lim_{\lambda\to 0} \frac{g_k(\mathbf{y} + \lambda \mathbf{u}_r) - g_k(\mathbf{y})}{\lambda} = D_r g_k(\mathbf{y})$$

This establishes the existence of $D_r g_k(\mathbf{y})$ for each $\mathbf{y}$ in Y and each $r = 1, 2, \ldots, n$. Moreover, this limit is a quotient of two determinants involving only the derivatives $D_j f_i(\mathbf{x})$. The continuity of the $D_j f_i$ then yields the continuity of each partial $D_r g_k$. This completes the proof.

Observe that the foregoing proof also provides us with a method for computing $D_r g_k(\mathbf{y})$. However, the derivatives $D_r g_k$ can be obtained more easily (without recourse to a limiting process) by using the multiplication theorem for Jacobians which, in this case, states that $J_f(\mathbf{x}) J_g(\mathbf{y}) = 1$ if $\mathbf{y} = \mathbf{f}(\mathbf{x})$, $\mathbf{x} \in X$. When the corresponding matrix equation is written out in detail, it is equivalent to the following system of n^2 equations:

$$\sum_{k=1}^{n} D_k g_i(\mathbf{y}) D_j f_k(\mathbf{x}) = \delta_{ij}, \qquad (\text{where } \mathbf{y} = \mathbf{f}(\mathbf{x}))$$

For each fixed i, we obtain n linear equations as j runs through the values $1, 2, \ldots, n$. These can then be solved for the n unknowns, $D_1 g_i(\mathbf{y}), \ldots, D_n g_i(\mathbf{y})$, by Cramer's rule.

7-5 The implicit function theorem. The reader knows that the equation of a curve in the xy-plane can be expressed either in an "explicit" form, such as $y = f(x)$, or in an "implicit" form, such as $F(x, y) = 0$. However, if we are given an equation of the form $F(x, y) = 0$, this does not necessarily represent a function. (Take, for example, $x^2 + y^2 - 5 = 0$.) The equation $F(x, y) = 0$ *does* always represent a *relation*, namely, that set of all pairs (x, y) which satisfy the equation. The following question therefore presents itself quite naturally: When is the relation defined by $F(x, y) = 0$ also a function? In other words, when can the equation $F(x, y) = 0$ be solved explicitly for y in terms of x, yielding a unique solution? The implicit function theorem deals with this question *locally*. It tells us that, given a point (x_0, y_0) such that $F(x_0, y_0) = 0$, under certain conditions there will be a neighborhood of (x_0, y_0) such that *in this neighborhood* the relation defined by $F(x, y) = 0$ is also a function. The conditions are that F and D_2F be continuous in some neighborhood of (x_0, y_0) and that $D_2F(x_0, y_0) \neq 0$. In its more general form, the theorem treats, instead of one equation in two variables, a system of n equations in $n + k$ variables:

$$f_r(x_1, \ldots, x_n; t_1, \ldots, t_k) = 0 \qquad (r = 1, 2, \ldots, n)$$

This system can be solved for $x_1, \ldots, x_n$ in terms of $t_1, \ldots, t_k$, provided that certain partial derivatives are continuous and provided that the $n \times n$ Jacobian determinant $\partial(f_1, \ldots, f_n)/\partial(x_1, \ldots, x_n)$ is not zero.

For brevity, we shall adopt the following notation in this theorem: Points in $(n + k)$-dimensional space E_{n+k} will be written in the form $(\mathbf{x}; \mathbf{t})$, where $\mathbf{x} = (x_1, \ldots, x_n) \in E_n$ and $\mathbf{t} = (t_1, \ldots, t_k) \in E_k$.

7–6 THEOREM (*Implicit function theorem*). *Let* $\mathbf{f} = (f_1, \ldots, f_n)$ *be a vector-valued function defined on an open set* S *in* E_{n+k} *with values in* E_n. *Suppose* $\mathbf{f} \in C'$ *on* S. *Let* $(\mathbf{x}_0; \mathbf{t}_0)$ *be a point in* S *for which* $\mathbf{f}(\mathbf{x}_0; \mathbf{t}_0) = \mathbf{0}$ *and for which the* $n \times n$ *determinant* $\det [D_j f_i(\mathbf{x}_0; \mathbf{t}_0)] \neq 0$. *Then there exists a k-dimensional neighborhood* T_0 *of* $\mathbf{t}_0$ *and one, and only one, vector-valued function* $\mathbf{g}$, *defined on* T_0 *and having values in* E_n, *such that*

(i) $\mathbf{g} \in C'$ *on* T_0,

(ii) $\mathbf{g}(\mathbf{t}_0) = \mathbf{x}_0$,

(iii) $\mathbf{f}(\mathbf{g}(\mathbf{t}); \mathbf{t}) = \mathbf{0}$ *for every* $\mathbf{t}$ *in* T_0.

Proof. We shall apply the inverse function theorem to a certain vector-valued function $\mathbf{F} = (F_1, \ldots, F_n; F_{n+1}, \ldots, F_{n+k})$ defined on S and having values in E_{n+k}. The function $\mathbf{F}$ is defined as follows: For $1 \leq m \leq n$, let $F_m(\mathbf{x}; \mathbf{t}) = f_m(\mathbf{x}; \mathbf{t})$, and for $1 \leq m \leq k$, let $F_{n+m}(\mathbf{x}; \mathbf{t}) = t_m$. We can then write $\mathbf{F} = (\mathbf{f}; \mathbf{I})$, where $\mathbf{f} = (f_1, \ldots, f_n)$ and where $\mathbf{I}$ is the identity function defined by $\mathbf{I}(\mathbf{t}) = \mathbf{t}$ for each $\mathbf{t}$ in E_k. The Jacobian $J_{\mathbf{F}}(\mathbf{x}; \mathbf{t})$ then has the same value as the $n \times n$ determinant $\det [D_j f_i(\mathbf{x}; \mathbf{t})]$ because the terms which appear in the last k rows and also in the last k columns of $J_{\mathbf{F}}(\mathbf{x}; \mathbf{t})$ form a $k \times k$ determinant with ones along the main diagonal and zeros elsewhere; the intersection of the first n rows and n columns consists of the $n \times n$ determinant $\det [D_j f_i(\mathbf{x}; \mathbf{t})]$, and the remaining terms are zeros. Hence the Jacobian $J_{\mathbf{F}}(\mathbf{x}_0; \mathbf{t}_0) \neq 0$. Also, $\mathbf{F}(\mathbf{x}_0; \mathbf{t}_0) = (\mathbf{0}; \mathbf{t}_0)$. Therefore, by Theorem 7–5, there exist open sets X and Y containing $(\mathbf{x}_0; \mathbf{t}_0)$ and $(\mathbf{0}; \mathbf{t}_0)$, respectively, such that $\mathbf{F}$ is one-to-one on X, and $Y = \mathbf{F}(X)$. Also, there exists a local inverse function $\mathbf{G}$, defined on Y and having values in X, such that $\mathbf{G}[\mathbf{F}(\mathbf{x}; \mathbf{t})] = (\mathbf{x}; \mathbf{t})$ and such that $\mathbf{G} \in C'$ on Y.

Now $\mathbf{G}$ can be reduced to components as follows: $\mathbf{G} = (\mathbf{v}; \mathbf{w})$ where

$(\mathbf{x};\mathbf{t}) = \mathbf{F}(\mathbf{x}',\mathbf{t}')$, we must have $\mathbf{t}' = \mathbf{t}$. Therefore, $\mathbf{v}(\mathbf{x},\mathbf{t}) = \mathbf{v}[\mathbf{F}(\mathbf{x}',\mathbf{t})] = \mathbf{x}'$ and $\mathbf{w}(\mathbf{x};\mathbf{t}) = \mathbf{w}[\mathbf{F}(\mathbf{x}',\mathbf{t})] = \mathbf{t}$. Hence the function $\mathbf{G}$ can be described as follows. Given a point $(\mathbf{x};\mathbf{t})$ in Y, we have $\mathbf{G}(\mathbf{x};\mathbf{t}) = (\mathbf{x}',\mathbf{t})$, where $\mathbf{x}'$ is that point in E_n such that $(\mathbf{x},\mathbf{t}) = \mathbf{F}(\mathbf{x}',\mathbf{t})$. This statement implies that $\mathbf{F}[\mathbf{v}(\mathbf{x};\mathbf{t}),\mathbf{t}] = (\mathbf{x};\mathbf{t})$, for every $(\mathbf{x},\mathbf{t})$ in Y.

Now we are ready to define the set T_0 and the function $\mathbf{g}$ in the theorem. Let $T_0 = \{\mathbf{t} \mid \mathbf{t} \in E_k,\ (\mathbf{0},\mathbf{t}) \in Y\}$, and for each $\mathbf{t}$ in T_0 define $\mathbf{g}(\mathbf{t}) = \mathbf{v}(\mathbf{0},\mathbf{t})$. The set T_0 is open [Why?]. Moreover, $\mathbf{g} \in C'$ on T_0 because $\mathbf{G} \in C'$ on Y and the components of $\mathbf{g}$ are taken from the components of $\mathbf{G}$. Also, $\mathbf{g}(\mathbf{t}_0) = \mathbf{v}(\mathbf{0};\mathbf{t}_0) = \mathbf{x}_0$ because $(\mathbf{0},\mathbf{t}_0) = \mathbf{F}(\mathbf{x}_0,\mathbf{t}_0)$. Finally, the equation $\mathbf{F}[\mathbf{v}(\mathbf{x},\mathbf{t}),\mathbf{t}] = (\mathbf{x},\mathbf{t})$, which holds for every $(\mathbf{x},\mathbf{t})$ in Y, yields (by considering the components in E_n) the equation $\mathbf{f}[\mathbf{v}(\mathbf{x},\mathbf{t}),\mathbf{t}] = \mathbf{x}$. Taking $\mathbf{x} = \mathbf{0}$, we see that for every $\mathbf{t}$ in T_0, we have $\mathbf{f}[\mathbf{g}(\mathbf{t}),\mathbf{t}] = \mathbf{0}$, and this completes the proof of statements (i), (ii), and (iii). It remains to prove that there is only one such function $\mathbf{g}$. But this follows at once from the one-to-one character of $\mathbf{f}$. If there were another function, say $\mathbf{h}$, which satisfied (iii), then we would have $\mathbf{f}[\mathbf{g}(\mathbf{t}),\mathbf{t}] = \mathbf{f}[\mathbf{h}(\mathbf{t}),\mathbf{t}]$, and this would imply $(\mathbf{g}(\mathbf{t});\mathbf{t}) = (\mathbf{h}(\mathbf{t}),\mathbf{t})$, or $\mathbf{g}(\mathbf{t}) = \mathbf{h}(\mathbf{t})$ for every $\mathbf{t}$ in T_0.

7–6 Extremum problems. In the remainder of this chapter we shall consider real-valued functions f with a view toward determining those points (if any) at which f has a local extremum, that is, either a local maximum or a local minimum.

We have already obtained one result (Theorem 5–8) in this connection for functions of one variable. In that theorem we found that a *necessary* condition for a function f to have a local extremum at an interior point x_0 of an interval is that $f'(x_0) = 0$, provided that $f'(x_0)$ exists. This condition, however, is not sufficient, as we can see by taking $f(x) = x^3$, $x_0 = 0$. We now derive a *sufficient* condition.

7–7 THEOREM. *For some integer $n \geq 1$, let f have a continuous nth derivative in the open interval (a, b). Suppose also that for some interior point x_0 in (a, b) we have*

$$f'(x_0) = f''(x_0) = \cdots = f^{(n-1)}(x_0) = 0, \quad \textit{but} \quad f^{(n)}(x_0) \neq 0.$$

Then for n even, f has a local minimum at x_0 if $f^{(n)}(x_0) > 0$, and a local maximum at x_0 if $f^{(n)}(x_0) < 0$. If n is odd, there is neither a local maximum nor a local minimum at x_0.

Proof. Since $f^{(n)}(x_0) \neq 0$, there exists a neighborhood $N(x_0)$ such that for every x in $N(x_0)$, the derivative $f^{(n)}(x)$ will have the same sign as $f^{(n)}(x_0)$. Now by Taylor's formula (Theorem 5–14), for every x in $N(x_0)$ we can write

$$f(x) - f(x_0) = \frac{f^{(n)}(x_1)}{n!}(x - x_0)^n, \qquad \text{where } x_1 \varepsilon N(x_0).$$

If n is even, this equation implies $f(x) \geq f(x_0)$ when $f^{(n)}(x_0) > 0$, and $f(x) \leq f(x_0)$ when $f^{(n)}(x_0) < 0$. If n is odd and $f^{(n)}(x_0) > 0$, then $f(x) > f(x_0)$ when $x > x_0$, but $f(x) < f(x_0)$ when $x < x_0$, and there can be no extremum at x_0. A similar statement holds if n is odd and $f^{(n)}(x_0) < 0$. This proves the theorem.

We turn now to functions of several variables. In Exercise 6–1 we have given a *necessary* condition for a function to have a local maximum or a local minimum at an interior point $\mathbf{x}$ of an open set. The condition is that each partial derivative $D_k f(\mathbf{x})$ must be zero at that point. We can also state this in terms of directional derivatives by saying that $D_{\mathbf{u}} f(\mathbf{x})$ must be zero for every direction $\mathbf{u}$.

The converse of this statement is not true, however. Consider the following example of a function of two real variables:

$$f(x, y) = (y - x^2)(y - 2x^2).$$

Here we have $D_1 f(0, 0) = D_2 f(0,0) = 0$. Now $f(0, 0) = 0$, but the function assumes both positive and negative values in every neighborhood of $(0, 0)$. (See Fig. 7–3.) This example illustrates another interesting phenomenon. If we take a fixed straight line through the origin and restrict the point (x, y) to move along this line toward $(0, 0)$, then the point will finally enter the region above the parabola $y = 2x^2$ (or below the parabola $y = x^2$) in which $f(x, y)$ becomes and stays positive for every $(x, y) \neq (0, 0)$. Therefore, along every such line, f has a minimum at $(0, 0)$, but the origin is not a local minimum in any two-dimensional neighborhood of $(0, 0)$.

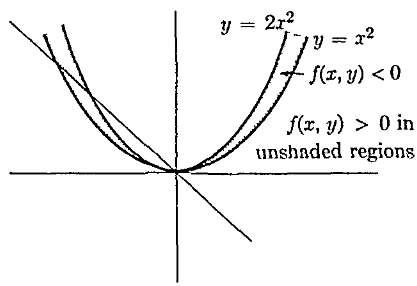

FIGURE 7–3.

7–7 Sufficient conditions for a local extremum. The next theorem contains a *sufficient* condition for a local extremum of a function of two variables.

7–8 THEOREM. *Let f have continuous second-order partials in an open set S in E_2. Assume that $(x_0, y_0) \varepsilon S$ and suppose that $D_1 f(x_0, y_0) = D_2 f(x_0, y_0) = 0$. Let Δ denote the value of the determinant*

$$\Delta = \begin{vmatrix} D_{1,1}f(x_0, y_0) & D_{1,2}f(x_0, y_0) \\ D_{2,1}f(x_0, y_0) & D_{2,2}f(x_0, y_0) \end{vmatrix}$$

and assume that $\Delta \neq 0$ If $\Delta > 0$ and $D_{1,1}f(x_0, y_0) > 0$, then f has a local minimum at (x_0, y_0), if $\Delta > 0$ and $D_{1,1}f(x_0,y_0) < 0$, then f has a local maximum at (x_0, y_0) If $\Delta < 0$, then f has neither a local maximum nor a local minimum at (x_0, y_0)

NOTE The determinant Δ is known as the *Hessian* of f

Proof If we apply the two-dimensional Taylor formula (Theorem 6-22) with $m = n = 2$, we find that for some point $(\bar{x}, \bar{y})$ on the segment joining (x, y) to (x_0, y_0), we have

$$\begin{aligned} f(x, y) - f(x_0, y_0) &= \frac{1}{2!} d^2f[(\bar{x}, \bar{y}), (x - x_0, y - y_0)] \\ &= \frac{1}{2} [(x - x_0)^2 D_{1,1}f(\bar{x}, \bar{y}) \\ &\quad + 2(x - x_0)(y - y_0)D_{1,2}f(\bar{x}, \bar{y}) \\ &\quad + (y - y_0)^2 D_{2,2}f(\bar{x}, \bar{y})] \\ &= \frac{1}{2} (ah^2 + 2bhk + ck^2), \end{aligned}$$

where $h = x - x_0$, $k = y - y_0$, $a = D_{1,1}f(\bar{x}, \bar{y})$, $b = D_{1,2}f(\bar{x}, \bar{y})$, and $c = D_{2,2}f(\bar{x}, \bar{y})$ By the assumed continuity of the partial derivatives, there will be a neighborhood in which a, b, and c will have the same signs as the numbers $D_{1,1}f(x_0, y_0)$, $D_{1,2}f(x_0, y_0)$, and $D_{2,2}f(x_0, y_0)$, respectively. We restrict ourselves to points (x, y) in this neighborhood in the remainder of the proof Write $A = D_{1,1}f(x_0, y_0)$, $B = D_{1,2}f(x_0, y_0)$, $C = D_{2,2}f(x_0,y_0)$, and $Q = \frac{1}{2}(Ah^2 + 2Bhk + Ck^2)$ Then $f(x, y) - f(x_0, y_0)$ has the same sign as Q

Suppose now that $\Delta > 0$ Then we must have $AC - B^2 > 0$ and hence $A \neq 0$ Then we can write

$$Q = \frac{1}{2A} [(Ah + Bk)^2 + (AC - B^2)k^2]$$

The expression in square brackets is positive and hence $f(x, y) - f(x_0, y_0)$ has the same sign as A Thus, in this case, f has a local maximum at (x_0, y_0) if $A < 0$ and a local minimum if $A > 0$

Suppose next that $\Delta < 0$ We will now exhibit points (x, y) arbitrarily close to (x_0, y_0) such that $f(x, y) - f(x_0, y_0)$ is both positive and negative. If A and C are both zero, then $Q = Bhk$ where $B \neq 0$ This can clearly be made to assume both positive and negative values for small h and k. On the other hand, if $A \neq 0$, we can write $Q = [(Ah + Bk)^2 + (AC - B^2)k^2]/(2A)$. Taking $k = 0$, we find that $f(x, y) - f(x_0, y_0)$ has the same sign as A. But taking $k \neq 0$ and $h = -Bk/A$, we find $Q = (AC - B^2)k^2/(2A)$, and this has the sign opposite to that of A. (A similar argument holds if $C \neq 0$.) This completes the proof.

The extension of Theorem 7–8 to functions of several variables depends on the theory of quadratic forms. A real-valued function ϕ defined on E_n by an equation of the type

$$\phi(\mathbf{x}) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}x_ix_j,$$

where $\mathbf{x} = (x_1, \ldots, x_n) \in E_n$ and the a_{ij} are real numbers, is called a *quadratic form.* The form ϕ is called *symmetric* if $a_{ij} = a_{ji}$ for all i and j, *positive definite* if $\mathbf{x} \neq \mathbf{0}$ implies $\phi(\mathbf{x}) > 0$, and *negative definite* if $\mathbf{x} \neq \mathbf{0}$ implies $\phi(\mathbf{x}) < 0$. Let $\Delta = \det [a_{ij}]$ and let Δ_{n-k} denote the determinant with $n - k$ rows, obtained by deleting the last k rows and columns of Δ. Also, put $\Delta_0 = 1$. From the theory of quadratic forms it is known that a necessary and sufficient condition for a symmetric form ϕ to be positive definite is that the $n + 1$ numbers $\Delta_0, \Delta_1, \ldots, \Delta_n$ be positive. The form is negative definite if, and only if, the same $n + 1$ numbers are alternately positive and negative. (See Ref. 7–2; in particular, Theorem 9–26, p. 197.)

Quadratic forms enter into the theory of partial differentiation by way of the second-order differential

$$d^2f(\mathbf{x}; \mathbf{t}) = \sum_{i=1}^{n} \sum_{j=1}^{n} D_{i,j}f(\mathbf{x})t_it_j,$$

when $\mathbf{x}$ is held fixed and d^2f is considered as a function of $\mathbf{t}$. If all the derivatives $D_{i,j}f$ are continuous at $\mathbf{x}$, then the mixed partials $D_{i,j}f(\mathbf{x})$ and $D_{j,i}f(\mathbf{x})$ are equal and d^2f is a *symmetric* form.

An argument similar to that given in Theorem 7–8 can now be employed to derive the following sufficient condition for an extremum.

7–9 THEOREM. *Let f have continuous second-order partial derivatives on an open set S in E_n. Let $\mathbf{x}_0$ be a point of S for which $D_1f(\mathbf{x}_0) = \cdots = D_nf(\mathbf{x}_0) = 0$. Assume that the determinant $\Delta = \det [D_{i,j}f(\mathbf{x}_0)] \neq 0$. Let $\Delta_0 = 1$ and let Δ_{n-k} be the determinant obtained from Δ by deleting* *the last k rows and columns If the* $n+1$ *numbers* $\Delta_0, \Delta_1, \ldots, \Delta_n$ *are all positive, then f has a local minimum at* $\mathbf{x}_0$ *If these numbers are alternately positive and negative, then f has a local maximum at* $\mathbf{x}_0$.

Proof Using Taylor's formula (Theorem 6–22) with $m = 2$, we find that

$$f(\mathbf{x}) - f(\mathbf{x}_0) = \frac{1}{2}\, d^2 f(\mathbf{z}, \mathbf{x} - \mathbf{x}_0) = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} D_{i,j} f(\mathbf{z}) t_i t_j,$$

where $\mathbf{z} \in L(\mathbf{x}, \mathbf{x}_0)$ and t_k is the kth component of $\mathbf{x} - \mathbf{x}_0$ By the continuity of the second partials, there will be a neighborhood $N(\mathbf{x}_0)$ in which this quadratic form has the same sign as the quadratic form

$$\phi(\mathbf{t}) = \sum_{i=1}^{n} \sum_{j=1}^{n} D_{i,j} f(\mathbf{x}_0) t_i t_j, \qquad \text{if } \mathbf{t} = (t_1, \ldots, t_n)$$

If $\mathbf{x}$ is restricted to $N'(\mathbf{x}_0)$, then the conclusion follows at once from the results on symmetric quadratic forms written above

7–8 Extremum problems with side conditions. Let us consider the following type of extremum problem. Suppose that $f(x, y, z)$ represents the temperature at the point (x, y, z) in space and we ask for the maximum or minimum value of the temperature on a certain surface If the equation of the surface is given explicitly in the form $z = h(x, y)$, then in the expression $f(x, y, z)$ we can replace z by $h(x, y)$ to obtain the temperature on the surface as a function of x and y alone, say $F(x, y) = f[x, y, h(x, y)]$ The problem is then reduced to finding the extreme values of F However, in practice, certain difficulties arise The equation of the surface might be given in an implicit form, say $g(x, y, z) = 0$, and it may be impossible, in practice, to solve this equation explicitly for z in terms of x and y, or even for x or y in terms of the remaining variables The problem might be further complicated by asking for the extreme values of the temperature at those points which lie on a given *curve* in space Such a curve is the intersection of two surfaces, say $g_1(x, y, z) = 0$ and $g_2(x, y, z) = 0$ If we could solve these two equations simultaneously, say for x and y in terms of z, then we could introduce these expressions into f and obtain a new function of z alone, whose extrema we would then seek In general, however, this procedure cannot be carried out and a more practicable method must be sought A very elegant and useful method for attacking such problems was developed by Lagrange.

Lagrange's method provides a *necessary* condition for an extremum and can be described as follows. Let $f(x_1, \ldots, x_n)$ be an expression whose extreme values are sought when the variables are restricted by a certain number of side conditions, say $g_1(x_1, \ldots, x_n) = 0, \ldots, g_m(x_1, \ldots, x_n) = 0$. We then form the linear combination

$$\phi(x_1, \ldots, x_n) = f(x_1, \ldots, x_n) + \lambda_1 g_1(x_1, \ldots, x_n) + \cdots + \lambda_m g_m(x_1, \ldots, x_n),$$

where $\lambda_1, \ldots, \lambda_m$ are m constants. We then differentiate ϕ with respect to each coordinate and consider the following system of $n + m$ equations:

$$\begin{aligned} D_r\phi(x_1, \ldots, x_n) &= 0, \qquad r = 1, 2, \ldots, n, \\ g_k(x_1, \ldots, x_n) &= 0, \qquad k = 1, 2, \ldots, m. \end{aligned}$$

Lagrange discovered that if the point $(x_1, \ldots, x_n)$ is a solution of the extremum problem, then it will also satisfy this system of $n + m$ equations. In practice, one attempts to solve this system for the $n + m$ "unknowns," $\lambda_1, \ldots, \lambda_m$, and $x_1, \ldots, x_n$. The points $(x_1, \ldots, x_n)$ so obtained must then be tested to determine whether they yield a maximum, a minimum, or neither. The numbers $\lambda_1, \ldots, \lambda_m$, which are introduced only to help solve the system for $x_1, \ldots, x_n$, are known as Lagrange's multipliers. One multiplier is introduced for each side condition.

A complicated analytic criterion exists for distinguishing between maxima and minima in such problems. (See, for example, Ref. 7–1.) However, this criterion is not very useful in practice and in any particular problem it is usually easier to rely on some other means (for example, physical or geometrical considerations) to make this distinction.

The following theorem establishes the validity of Lagrange's method:

7–10 THEOREM. *Let f be a real-valued function such that $f \in C'$ on an open set S in E_n. Let $g_1, \ldots, g_m$ be m real-valued functions such that $\mathbf{g} = (g_1, \ldots, g_m) \in C'$ on S, and assume that $m < n$. Let X_0 be that subset of S on which $\mathbf{g}$ vanishes, that is,*

$$X_0 = \{\mathbf{x} \mid \mathbf{x} \in S, \mathbf{g}(\mathbf{x}) = \mathbf{0}\}.$$

Assume that $\mathbf{x}_0 \in X_0$ and assume that there exists a neighborhood $N(\mathbf{x}_0)$ such that $f(\mathbf{x}) \leq f(\mathbf{x}_0)$ for all $\mathbf{x}$ in $X_0 \cap N(\mathbf{x}_0)$ or such that $f(\mathbf{x}) \geq f(\mathbf{x}_0)$ for all $\mathbf{x}$ in $X_0 \cap N(\mathbf{x}_0)$. Assume also that the m-rowed determinant $\det[D_j g_i(\mathbf{x}_0)] \neq 0$. Then there exist m real numbers $\lambda_1, \ldots, \lambda_m$ such that the following n equations are satisfied:

$$(2) \qquad D_r f(\mathbf{x}_0) + \sum_{k=1}^{m} \lambda_k D_r g_k(\mathbf{x}_0) = 0 \qquad (r = 1, 2, \ldots, n).$$

NOTE. The n equations in (2) are equivalent to the following vector equation:

$$\nabla f(\mathbf{x}_0) + \lambda_1 \nabla g_1(\mathbf{x}_0) + \cdots + \lambda_m \nabla g_m(\mathbf{x}_0) = \mathbf{0}$$

Proof. Consider the following system of m linear equations in the m unknowns $\lambda_1, \ldots, \lambda_m$

$$\sum_{k=1}^{m} \lambda_k D_r g_k(\mathbf{x}_0) = -D_r f(\mathbf{x}_0) \qquad (r = 1, 2, \ldots, m)$$

This system has a unique solution since, by hypothesis, the determinant of the system is not zero. Therefore, the *first m equations* in (2) are satisfied. We must now verify that for this choice of $\lambda_1, \ldots, \lambda_m$, the remaining $n - m$ equations in (2) are also satisfied.

To do this, we apply the implicit function theorem. Since $m < n$, every point $\mathbf{x}$ in S can be written in the form $\mathbf{x} = (\mathbf{x}', \mathbf{t})$, say, where $\mathbf{x}' \in E_m$ and $\mathbf{t} \in E_{n-m}$. In the remainder of this proof we will write $\mathbf{x}'$ for $(x_1, \ldots, x_m)$ and $\mathbf{t}$ for $(x_{m+1}, \ldots, x_n)$, so that $t_k = x_{m+k}$. In terms of the vector-valued function $\mathbf{g} = (g_1, \ldots, g_m)$, we can now write $\mathbf{g}(\mathbf{x}_0', \mathbf{t}_0) = \mathbf{0}$ if $\mathbf{x}_0 = (\mathbf{x}_0', \mathbf{t}_0)$. Since $\mathbf{g} \in C'$ on S, and since the determinant $\det[D_j g_i(\mathbf{x}_0', \mathbf{t}_0)] \neq 0$, all the conditions of the implicit function theorem are satisfied. Therefore, there exists an $(n - m)$-dimensional neighborhood T_0 of $\mathbf{t}_0$ and a unique vector-valued function $\mathbf{h} = (h_1, \ldots, h_m)$, defined on T_0 and having values in E_m, such that $\mathbf{h} \in C'$ on T_0, $\mathbf{h}(\mathbf{t}_0) = \mathbf{x}_0'$, and for every $\mathbf{t}$ in T_0, we have $\mathbf{g}[\mathbf{h}(\mathbf{t}); \mathbf{t}] = \mathbf{0}$. This amounts to saying that the system of m equations

$$g_1(x_1, \ldots, x_n) = 0, \ldots, g_m(x_1, \ldots, x_n) = 0$$

can be "solved" for $x_1, \ldots, x_m$ in terms of $x_{m+1}, \ldots, x_n$, giving the solutions in the form $x_r = h_r(x_{m+1}, \ldots, x_n)$, $r = 1, 2, \ldots, m$. We shall now "substitute" these expressions for $x_1, \ldots, x_m$ into the expression $f(x_1, \ldots, x_n)$ and also into each expression $g_p(x_1, \ldots, x_n)$. That is to say, we define a new function F as follows:

$$F(x_{m+1}, \ldots, x_n) = f[h_1(x_{m+1}, \ldots, x_n), \ldots, h_m(x_{m+1}, \ldots, x_n); x_{m+1}, \ldots, x_n],$$

and we define m new functions $G_1, \ldots, G_m$ as follows:

$$G_p(x_{m+1}, \ldots, x_n) = g_p[h_1(x_{m+1}, \ldots, x_n), \ldots, h_m(x_{m+1}, \ldots, x_n); x_{m+1}, \ldots, x_n]$$

More briefly, we can write $F(\mathbf{t}) = f[\mathbf{H}(\mathbf{t})]$ and $G_p(\mathbf{t}) = g_p[\mathbf{H}(\mathbf{t})]$, where $\mathbf{H}(\mathbf{t}) = (\mathbf{h}(\mathbf{t}); \mathbf{t})$. Here $\mathbf{t}$ is restricted to lie in the set T_0.

Each function G_p so defined is identically zero on the set T_0 by the implicit function theorem. Therefore, each derivative D_rG_p is also identically zero on T_0 and, in particular, $D_rG_p(\mathbf{t}_0) = 0$. But by the chain rule (Theorem 6–14), we can compute these derivatives as follows:

$$D_rG_p(\mathbf{t}_0) = \sum_{k=1}^{n} D_kg_p(\mathbf{x}_0)D_rH_k(\mathbf{t}_0) \qquad (r = 1, 2, \ldots, n - m).$$

But $H_k(\mathbf{t}) = h_k(\mathbf{t})$ if $1 \leq k \leq m$, and $H_k(\mathbf{t}) = x_k$ if $m + 1 \leq k \leq n$. Therefore, when $m + 1 \leq k \leq n$, we have $D_rH_k(\mathbf{t}) \equiv 0$ if $m + r \neq k$ and $D_rH_{m+r}(\mathbf{t}) = 1$ for every $\mathbf{t}$. Hence the above set of equations becomes

(3)
$$\sum_{k=1}^{m} D_kg_p(\mathbf{x}_0)D_rh_k(\mathbf{t}_0) + D_{m+r}g_p(\mathbf{x}_0) = 0 \qquad \begin{cases} p = 1, 2, \ldots, m, \\ r = 1, 2, \ldots, n - m. \end{cases}$$

By continuity of $\mathbf{h}$, there will be a neighborhood $N(\mathbf{t}_0) \subset T_0$ such that $\mathbf{t} \in N(\mathbf{t}_0)$ implies $(\mathbf{h}(\mathbf{t}); \mathbf{t}) \in N(\mathbf{x}_0)$, where $N(\mathbf{x}_0)$ is the neighborhood in the statement of the theorem. Hence, $\mathbf{t} \in N(\mathbf{t}_0)$ implies $(\mathbf{h}(\mathbf{t}); \mathbf{t}) \in X_0 \cap N(\mathbf{x}_0)$ and therefore, by hypothesis, we have either $F(\mathbf{t}) \leq F(\mathbf{t}_0)$ for all $\mathbf{t}$ in $N(\mathbf{t}_0)$ or else we have $F(\mathbf{t}) \geq F(\mathbf{t}_0)$ for all $\mathbf{t}$ in $N(\mathbf{t}_0)$. That is, F has a local maximum or a local minimum at the interior point $\mathbf{t}_0$. Each partial derivative $D_rF(\mathbf{t}_0)$ must therefore be zero. If we use the chain rule to compute these derivatives, we find

$$D_rF(\mathbf{t}_0) = \sum_{k=1}^{n} D_kf(\mathbf{x}_0)D_rH_k(\mathbf{t}_0) \qquad (r = 1, \ldots, n - m),$$

and hence we can write

(4)
$$\sum_{k=1}^{m} D_kf(\mathbf{x}_0)D_rh_k(\mathbf{t}_0) + D_{m+r}f(\mathbf{x}_0) = 0 \qquad (r = 1, \ldots, n - m).$$

If we now multiply (3) by λ_p, sum on p, and add the result to (4), we find

$$\sum_{k=1}^{m} \left[D_kf(\mathbf{x}_0) + \sum_{p=1}^{m} \lambda_p D_kg_p(\mathbf{x}_0) \right] D_rh_k(\mathbf{t}_0) + D_{m+r}f(\mathbf{x}_0)$$
$$+ \sum_{p=1}^{m} \lambda_p D_{m+r}g_p(\mathbf{x}_0) = 0,$$

for $r = 1, \ldots, n - m$. In the sum over k, the expression in square brackets vanishes because of the way $\lambda_1, \ldots, \lambda_m$ were defined. Thus we are left with

$$D_{m+r}f(\mathbf{x}_0) + \sum_{p=1}^{m} \lambda_p D_{m+r}g_p(\mathbf{x}_0) = 0 \qquad (r = 1, 2, \ldots, n - m),$$

and these are exactly the equations needed to complete the proof.

In attempting the solution of a particular extremum problem by Lagrange's method, it is usually very easy to determine the system of equations (2) but, in general, it is not a simple matter to actually *solve* the system. Special devices can often be employed to obtain the extreme values of f directly from (2) without first finding the particular points where these extremes are taken on The following example illustrates some of these devices:

EXAMPLE *A quadric surface with center at the origin has the equation*

$$Ax^2 + By^2 + Cz^2 + 2Dyz + 2Ezx + 2Fxy = 1$$

Find the length of its semi-axes.

Solution. Let us write (x_1, x_2, x_3) instead of (x, y, z), and introduce the quadratic form

$$q(\mathbf{x}) = \sum_{j=1}^{3} \sum_{i=1}^{3} a_{ij}x_ix_j, \tag{5}$$

where $\mathbf{x} = (x_1, x_2, x_3)$ and the $a_{ij} = a_{ji}$ are chosen so that the equation of the surface becomes $q(\mathbf{x}) = 1$ (Hence the quadratic form is symmetric and positive definite) The problem is equivalent to finding the extreme values of $f(\mathbf{x}) = |\mathbf{x}|^2 = x_1^2 + x_2^2 + x_3^2$ subject to the side condition $g(\mathbf{x}) = 0$, where $g(\mathbf{x}) = q(\mathbf{x}) - 1$ Using Lagrange's method, we introduce one multiplier and consider the vector equation

$$\nabla f(\mathbf{x}) + \lambda \nabla q(\mathbf{x}) = 0 \tag{6}$$

(since $\nabla g = \nabla q$). In this particular case, both f and q are homogeneous functions of degree 2 and we can apply Euler's theorem (see Exercise 6–15) in (6) to obtain

$$\mathbf{x} \cdot \nabla f(\mathbf{x}) + \lambda \mathbf{x} \cdot \nabla q(\mathbf{x}) = 2f(\mathbf{x}) + 2\lambda q(\mathbf{x}) = 0.$$

Since $q(\mathbf{x}) = 1$ on the surface we find $\lambda = -f(\mathbf{x})$, and (6) becomes

$$t\nabla f(\mathbf{x}) - \nabla q(\mathbf{x}) = \mathbf{0}, \tag{7}$$

where $t = 1/f(\mathbf{x})$. (We cannot have $f(\mathbf{x}) = 0$ in this problem.) The vector equation (7) then leads to the following three equations for x_1, x_2, x_3:

$$\begin{aligned} (a_{11} - t)x_1 + a_{12}x_2 + a_{13}x_3 &= 0, \\ a_{21}x_1 + (a_{22} - t)x_2 + a_{23}x_3 &= 0, \\ a_{31}x_1 + a_{32}x_2 + (a_{33} - t)x_3 &= 0. \end{aligned}$$

Since $\mathbf{x} = \mathbf{0}$ cannot yield a solution to our problem, the determinant of this system must vanish. That is, we must have

$$\begin{vmatrix} a_{11} - t & a_{12} & a_{13} \\ a_{21} & a_{22} - t & a_{23} \\ a_{31} & a_{32} & a_{33} - t \end{vmatrix} = 0. \tag{8}$$

Equation (8) is called the *characteristic equation* of the quadratic form in (5). In this case, the geometrical nature of the problem assures us that the three roots t_1, t_2, t_3 of this cubic must be real and positive. [Since $q(\mathbf{x})$ is symmetric and positive definite, the general theory of quadratic forms also guarantees that the roots of (8) are all real and positive. (See Ref. 7–2, Theorem 9–25.)] The semi-axes of the quadric surface are $t_1^{-1/2}$, $t_2^{-1/2}$, $t_3^{-1/2}$.

Exercises

Jacobians

7–1. Let f be the complex-valued function defined for each complex $z \neq 0$ by the equation $f(z) = 1/\bar{z}$. Show that $J_f(z) = -|z|^{-4}$. Show that f is one-to-one and compute f^{-1} explicitly.

7–2. Let $\mathbf{f} = (f_1, f_2, f_3)$ be the vector-valued function defined (for every point (x_1, x_2, x_3) in E_3 for which $x_1 + x_2 + x_3 \neq -1$) as follows

$$f_k(x_1, x_2, x_3) = \frac{x_k}{1 + x_1 + x_2 + x_3} \qquad (k = 1, 2, 3)$$

Show that $J_{\mathbf{f}}(x_1, x_2, x_3) = (1 + x_1 + x_2 + x_3)^{-4}$. Show that $\mathbf{f}$ is one-to-one and compute $\mathbf{f}^{-1}$ explicitly.

7–3. Let $\mathbf{f} = (f_1, \ldots, f_n)$ be a vector-valued function defined in E_n, suppose $\mathbf{f} \in C'$ on E_n, and let $J_{\mathbf{f}}(\mathbf{x})$ denote the Jacobian. Let $g_1, \ldots, g_n$ be n real-valued functions defined on E_1 and having continuous derivatives $g_1', \ldots, g_n'$. Let $h_k(\mathbf{x}) = f_k[g_1(x_1), \ldots, g_n(x_n)]$, $k = 1, 2, \ldots, n$, and put $\mathbf{h} = (h_1, \ldots, h_n)$. Show that

$$J_{\mathbf{h}}(\mathbf{x}) = J_{\mathbf{f}}[g_1(x_1), \ldots, g_n(x_n)]\, g_1'(x_1) \cdots g_n'(x_n)$$

7–4. (a) If $x(r, \theta) = r \cos \theta$, $y(r, \theta) = r \sin \theta$, show that

$$\frac{\partial(x, y)}{\partial(r, \theta)} = r$$

(b) If $x(r, \theta, \phi) = r \cos \theta \sin \phi$, $y(r, \theta, \phi) = r \sin \theta \sin \phi$, $z = r \cos \phi$, show that

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = -r^2 \sin \phi.$$

7–5. (a) State precise conditions on f and g which will ensure that the equations $x = f(u, v)$, $y = g(u, v)$ can be solved for u and v in a neighborhood of (x_0, y_0). If the solutions are $u = F(x, y)$, $v = G(x, y)$, and if $J = \partial(f, g)/\partial(u, v)$, show that

$$\frac{\partial F}{\partial x} = \frac{1}{J}\frac{\partial g}{\partial v}, \qquad \frac{\partial F}{\partial y} = -\frac{1}{J}\frac{\partial f}{\partial v}, \qquad \frac{\partial G}{\partial x} = -\frac{1}{J}\frac{\partial g}{\partial u}, \qquad \frac{\partial G}{\partial y} = \frac{1}{J}\frac{\partial f}{\partial u}.$$

(b) Compute J and the partial derivatives of F and G at $(x_0, y_0) = (1, 1)$ when $f(u, v) = u^2 - v^2$, $g(u, v) = 2uv$.

· **7–6.** Let $\mathbf{f}$ and $\mathbf{g}$ be related as in Theorem 7–5. Consider the case $n = 3$ and show that we have

$$J_{\mathbf{f}}(\mathbf{x})D_1g_i(\mathbf{y}) = \begin{vmatrix} \delta_{i,1} & D_1f_2(\mathbf{x}) & D_1f_3(\mathbf{x}) \\ \delta_{i,2} & D_2f_2(\mathbf{x}) & D_2f_3(\mathbf{x}) \\ \delta_{i,3} & D_3f_2(\mathbf{x}) & D_3f_3(\mathbf{x}) \end{vmatrix} \qquad (i = 1, 2, 3),$$

where $\mathbf{y} = \mathbf{f}(\mathbf{x})$. Use this to deduce the formula

$$D_1g_1 = \frac{\partial(f_2, f_3)}{\partial(x_2, x_3)} \Big/ \frac{\partial(f_1, f_2, f_3)}{\partial(x_1, x_2, x_3)}$$

and obtain similar expressions for the other eight derivatives D_kg_i.

7–7. Let $f = u + iv$ be a complex-valued function satisfying the following conditions: $u \in C'$ and $v \in C'$ on the open disk $A = \{z \mid |z| < 1\}$; f is continuous on the closed disk $\overline{A} = \{z \mid |z| \leq 1\}$; $u(x, y) = x$ and $v(x, y) = y$ whenever $x^2 + y^2 = 1$; the Jacobian $J_f(z) > 0$ if $z \in A$. Let $B = f(A)$ denote the image of A under f and prove that:

(a) If X is an open subset of A, then $f(X)$ is an open subset of B.

(b) B is an open disk of radius 1.

(c) For each point $u_0 + iv_0$ in B, there is only a finite number of points z in A such that $f(z) = u_0 + iv_0$.

Extremum problems.

7–8. Find and classify the extreme values (if any) of the functions defined by the following equations:

(a) $f(x, y) = y^2 + x^2y + x^4$,

(b) $f(x, y) = x^2 + y^2 + x + y + xy$,

(c) $f(x, y) = (x - 1)^4 + (x - y)^4$,

(d) $f(x, y) = y^2 - x^3$.

7–9. Find the shortest distance from the point $(0, b)$ on the y-axis to the parabola $x^2 - 4y = 0$. Solve this problem using Lagrange's method and also without using Lagrange's method.

7–10. Solve the following geometric problems by Lagrange's method:

(a) Find the shortest distance from the point (a_1, a_2, a_3) in E_3 to the plane whose equation is $b_1x_1 + b_2x_2 + b_3x_3 + b_0 = 0$.

(b) Find the point on the line of intersection of the two planes

$$a_1x_1 + a_2x_2 + a_3x_3 + a_0 = 0$$

and

$$b_1x_1 + b_2x_2 + b_3x_3 + b_0 = 0$$

which is nearest the origin

7-11. Find the maximum value of $|\sum_{k=1}^{n} a_kx_k|$, if $\sum_{k=1}^{n} x_k^2 = 1$, by using
(a) the Cauchy-Schwarz inequality.
(b) Lagrange's method

7-12. Find the maximum of $(x_1x_2 \cdots x_n)^2$ under the restriction

$$x_1^2 + \cdots + x_n^2 = 1$$

Use the result to derive the following inequality, valid for positive real numbers $a_1, \ldots, a_n$:

$$(a_1 \cdots a_n)^{1/n} \leq \frac{a_1 + \cdots + a_n}{n}$$

7-13. If $f(\mathbf{x}) = x_1^k + \cdots + x_n^k$, $\mathbf{x} = (x_1, \ldots, x_n)$, show that a local extreme of f, subject to the condition $x_1 + \cdots + x_n = a$, is a^kn^{1-k}.

7-14. Show that all points (x_1, x_2, x_3, x_4) where $x_1^2 + x_2^2$ has a local extremum subject to the two side conditions $x_1^2 + x_3^2 + x_4^2 = 4$, $x_2^2 + 2x_3^2 + 3x_4^2 = 9$, are found among

$$(0, 0, \pm\sqrt{3}, \pm 1),\ (0, \pm 1, \pm 2, 0),\ (\pm 1, 0, 0, \pm\sqrt{3}),\ (\pm 2, \pm 3, 0, 0).$$

Which of these yield a local maximum and which yield a local minimum? Give reasons for your conclusions.

7-15. Show that the extreme values of $f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2$, subject to the two side conditions

$$\sum_{j=1}^{3} \sum_{i=1}^{3} a_{ij}x_ix_j = 1 \qquad (a_{ij} = a_{ji})$$

and

$$b_1x_1 + b_2x_2 + b_3x_3 = 0, \qquad (b_1, b_2, b_3) \neq (0, 0, 0),$$

are $t_1^{-1/2}$, $t_2^{-1/2}$, where t_1 and t_2 are the roots of the equation

$$\begin{vmatrix} b_1 & b_2 & b_3 & 0 \\ a_{11} - t & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} - t & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} - t & b_3 \end{vmatrix} = 0$$

Show that this is a quadratic equation in t and give a geometric argument to explain why the roots t_1, t_2 are real and positive.

7–16. Let $\Delta = \det [x_{ij}]$ and let $\mathbf{X}_i = (x_{i1}, \ldots, x_{in})$. A famous theorem of Hadamard states that $|\Delta| \leq d_1 \cdots d_n$, if $d_1, \ldots, d_n$ are n positive constants such that $|\mathbf{X}_i|^2 = d_i^2 (i = 1, 2, \ldots, n)$. Prove this by treating Δ as a function of n^2 variables subject to n constraints, using Lagrange's method to show that, when Δ has an extreme under these conditions, we must have

$$\Delta^2 = \begin{vmatrix} d_1^2 & 0 & 0 \cdots 0 \\ 0 & d_2^2 & 0 \cdots 0 \\ \vdots & \vdots & \vdots \quad\ \vdots \\ 0 & 0 & 0 \cdots d_n^2 \end{vmatrix}.$$

References

7–1. Hancock, H., *Theory of Maxima and Minima.* Boston: Ginn, 1917.

7–2. Perlis, S., *Theory of Matrices.* Reading, Mass.: Addison-Wesley, 1952.

CHAPTER 8

FUNCTIONS OF BOUNDED VARIATION, RECTIFIABLE CURVES AND CONNECTED SETS

8–1 Introduction. Some of the basic properties of monotonic functions were derived in Chapter 4. In the first part of this chapter we shall discuss functions of bounded variation, a class of functions very closely related to monotonic functions. We shall find that these functions are intimately connected with curves having finite arc length (rectifiable curves). They also play a fundamental role in the theory of integration (which is developed in the next chapter). In the latter part of this chapter we shall deal with properties of connected sets.

8–2 Properties of monotonic functions.

8–1 THEOREM. *Let f be an increasing function defined on $[a, b]$ and let $x_0, x_1, \ldots, x_n$ be $n + 1$ points such that*

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b$$

Then we have the inequality

$$\sum_{k=1}^{n-1} [f(x_k+) - f(x_k-)] \leq f(b) - f(a)$$

Proof. Assume that $y_k \in (x_k, x_{k+1})$. For $1 \leq k \leq n - 1$, we have $f(x_k+) \leq f(y_k)$ and $f(y_{k-1}) \leq f(x_k-)$, so that $f(x_k+) - f(x_k-) \leq f(y_k) - f(y_{k-1})$. If we add these inequalities, the sum on the right telescopes and we obtain

$$\sum_{k=1}^{n-1} [f(x_k+) - f(x_k-)] \leq f(y_{n-1}) - f(y_0) \leq f(b) - f(a)$$

This is the inequality in the theorem.

The difference $f(x_k+) - f(x_k-)$ is, of course, the jump of f at x_k. The foregoing theorem tells us that for every finite collection of points x_k in (a, b), the sum of the jumps at these points is always bounded by $f(b) - f(a)$. This result can be used to prove the following theorem.

8-2 THEOREM. *If f is monotonic on $[a, b]$, then the set of discontinuities of f is countable.*

Proof. Assume that f is increasing and let S_m be the set of points in (a, b) at which the jump of f exceeds $1/m$, $m > 0$. If $x_1 < x_2 < \cdots < x_{n-1}$ are in S_m, Theorem 8-1 tells us that

$$\frac{n-1}{m} \leq f(b) - f(a).$$

This means that S_m must be a finite set. But the set of discontinuities of f in (a, b) is a subset of the union $\bigcup_{m=1}^{\infty} S_m$ and hence is countable. (If f is decreasing, the argument can be applied to $-f$.)

8-3 Functions of bounded variation.

8-3 DEFINITION. *If $[a, b]$ is a finite interval, then a set of points*

$$P = \{x_0, x_1, \ldots, x_n\}$$

satisfying the inequalities $a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$ is called a partition of $[a, b]$. The interval $[x_{k-1}, x_k]$ is called the kth subinterval of P and we write $\Delta x_k = x_k - x_{k-1}$, so that $\sum_{k=1}^{n} \Delta x_k = b - a$. The collection of all possible partitions of $[a, b]$ will be denoted by $\mathcal{P}[a, b]$.

8-4 DEFINITION. *Let f be defined on $[a, b]$. If $P = \{x_0, x_1, \ldots, x_n\}$ is a partition of $[a, b]$, write $\Delta f_k = f(x_k) - f(x_{k-1})$, $k = 1, 2, \ldots, n$. If there exists a positive number M such that*

$$\sum_{k=1}^{n} |\Delta f_k| \leq M$$

for all partitions of $[a, b]$, then f is said to be of bounded variation on $[a, b]$.

Examples of functions of bounded variation are provided by the next two theorems.

8-5 THEOREM. *If f is monotonic on $[a, b]$, then f is of bounded variation on $[a, b]$.*

Proof. Let f be increasing. Then for every partition of $[a, b]$ we have $\Delta f_k \geq 0$ and hence

$$\sum_{k=1}^{n} |\Delta f_k| = \sum_{k=1}^{n} \Delta f_k = \sum_{k=1}^{n} [f(x_k) - f(x_{k-1})] = f(b) - f(a).$$

8-6 THEOREM. *If f is continuous on $[a, b]$ and if f' exists and is bounded in the interior, say $|f'(x)| \le A$ for all x in (a, b), then f is of bounded variation on $[a, b]$*

Proof Applying the Mean Value Theorem, we can write

$$\Delta f_k = f(x_k) - f(x_{k-1}) = f'(t_k)(x_k - x_{k-1}), \text{ where } t_k \in (x_{k-1}, x_k).$$

This implies

$$\sum_{k=1}^{n} |\Delta f_k| = \sum_{k=1}^{n} |f'(t_k)|\, \Delta x_k \le A \sum_{k=1}^{n} \Delta x_k = A(b - a).$$

8-7 THEOREM *If f is of bounded variation on $[a, b]$, say $\sum|\Delta f_k| \le M$ for all partitions of $[a, b]$, then f is bounded on $[a, b]$ In fact, $|f(x)| \le |f(a)| + M$ if $x \in [a, b]$.*

Proof. Assume that $x \in (a, b)$ Using the special partition $P = \{a, x, b\}$, we find

$$|f(x) - f(a)| + |f(b) - f(x)| \le M.$$

This implies $|f(x) - f(a)| \le M$, $|f(x)| \le |f(a)| + M$, and the theorem follows

EXAMPLES.

1. It is easy to construct a continuous function which is not of bounded variation For example, let $f(x) = x \cos(\pi/2x)$ if $x \neq 0$, $f(0) = 0$. Then f is continuous on $[0, 1]$, but if we consider the partition

$$P = \left\{0, \frac{1}{2n}, \frac{1}{2n-1}, \quad , \frac{1}{3}, \frac{1}{2}, 1\right\},$$

an easy calculation shows that we have

$$\sum_{k=1}^{n} |\Delta f_k| = 1 + \frac{1}{2} + \frac{1}{3} + \quad + \frac{1}{n}.$$

This cannot be bounded for all n, since the series $\sum_{n=1}^{\infty} (1/n)$ diverges. In this example the derivative f' exists in $(0, 1)$ but f' is not bounded on $(0, 1)$. However, f' is bounded on any closed interval not containing the origin and hence f will be of bounded variation in such an interval

2 An example similar to the first is given by $f(x) = x^2 \cos(1/x)$ if $x \neq 0$, $f(0) = 0$. This f *is* of bounded variation on $[0, 1]$, since f' is bounded on

[0, 1]. In fact, $f'(0) = 0$ and, for $x \neq 0, f'(x) = \sin(1/x) + 2x\cos(1/x)$, so that $|f'(x)| \leq 3$ for all x in [0, 1].

3. Boundedness of f' is not necessary for f to be of bounded variation. For example, let $f(x) = x^{1/3}$. This function is monotonic (and hence of bounded variation) in every finite interval. However, $f'(x) \to +\infty$ as $x \to 0$.

8-4 Total variation.

8-8 DEFINITION. *Let f be of bounded variation on $[a, b]$, and let $\sum(P)$ denote the sum $\sum_{k=1}^{n} |\Delta f_k|$ corresponding to the partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$. The number*

$$V_f(a, b) = \sup \{\textstyle\sum(P) \mid P \in \mathcal{P}[a, b]\}$$

is called the total variation of f on the interval $[a, b]$.

NOTE. When there is no danger of misunderstanding, we will write V_f instead of $V_f(a, b)$.

Since f is of bounded variation on $[a, b]$, the number V_f is finite. Also, $V_f \geq 0$, since each sum $\sum(P) \geq 0$. Moreover, $V_f(a, b) = 0$ if, and only if, f is constant on $[a, b]$. [Why?]

8-9 THEOREM. *Assume that f and g are each of bounded variation on $[a, b]$. Then so are their sum, difference, and product. Also, we have*

$$V_{f \pm g} \leq V_f + V_g \quad \textit{and} \quad V_{f \cdot g} \leq AV_f + BV_g,$$

where $A = \sup \{|g(x)| \mid x \in [a, b]\}$, $B = \sup \{|f(x)| \mid x \in [a, b]\}$.

Proof. Let $h(x) = f(x)g(x)$. For every partition P of $[a, b]$, we have

$$\begin{aligned} |\Delta h_k| &= |f(x_k)g(x_k) - f(x_{k-1})g(x_{k-1})| \\ &= |[f(x_k)g(x_k) - f(x_{k-1})g(x_k)] \\ &\quad + [f(x_{k-1})g(x_k) - f(x_{k-1})g(x_{k-1})]| \leq A|\Delta f_k| + B|\Delta g_k|. \end{aligned}$$

This implies that h is of bounded variation and that $V_h \leq AV_f + BV_g$. The proofs for the sum and difference are simpler and will be omitted.

Quotients were not included in the foregoing theorem because the reciprocal of a function of bounded variation need not be of bounded variation. For example, if $f(x) \to 0$ as $x \to x_0$, then $1/f$ will not be bounded in any interval containing x_0 and (by Theorem 8-7) $1/f$ cannot be of bounded variation in such an interval. To extend Theorem 8-9 to quotients, it suffices to exclude functions whose values become arbitrarily close to zero.

8-10 THEOREM. *Let f be of bounded variation on $[a, b]$ and assume that f is bounded away from zero, that is, suppose that there exists a positive number m such that $0 < m \leq |f(x)|$ for all x in $[a, b]$. Then $g = 1/f$ is also of bounded variation on $[a, b]$, and $V_g \leq V_f/m^2$.*

Proof.

$$|\Delta g_k| = \left|\frac{1}{f(x_k)} - \frac{1}{f(x_{k-1})}\right| = \left|\frac{\Delta f_k}{f(x_k)f(x_{k-1})}\right| \leq \frac{|\Delta f_k|}{m^2}.$$

In the last two theorems the interval $[a, b]$ was kept fixed and $V_f(a, b)$ was considered as a function of f. If we keep f fixed and study the total variation as a function of the interval $[a, b]$, we can prove the following "additive" property.

8-11 THEOREM. *Let f be of bounded variation on $[a, b]$, and assume that $c \in (a, b)$. Then f is of bounded variation on $[a, c]$ and on $[c, b]$ and we have*

$$V_f(a, b) = V_f(a, c) + V_f(c, b).$$

Proof. We first prove that f is of bounded variation on $[a, c]$ and on $[c, b]$. Let P_1 be a partition of $[a, c]$ and let P_2 be a partition of $[c, b]$. Then $P_0 = P_1 \cup P_2$ is a partition of $[a, b]$. If $\sum(P)$ denotes the sum $\sum|\Delta f_k|$ corresponding to the partition P (of the appropriate interval), we can write

$$\sum(P_1) + \sum(P_2) = \sum(P_0) \leq V_f(a, b). \tag{1}$$

This shows that each sum $\sum(P_1)$ and $\sum(P_2)$ is bounded by $V_f(a, b)$ and this means that f is of bounded variation on $[a, c]$ and on $[c, b]$. From (1) we also obtain the inequality

$$V_f(a, c) + V_f(c, b) \leq V_f(a, b)$$

because of Theorem 1-8.

To obtain the reverse inequality, let $P = \{x_0, x_1, \ldots, x_n\} \in \mathcal{P}[a, b]$ and let $P_0 = P \cup \{c\}$ be the (possibly new) partition obtained by adjoining the point c. If $c \in [x_{k-1}, x_k]$, then we have

$$|f(x_k) - f(x_{k-1})| \leq |f(x_k) - f(c)| + |f(c) - f(x_{k-1})|$$

and hence $\sum(P) \leq \sum(P_0)$. Now the points of P_0 in $[a, c]$ determine a partition P_1 of $[a, c]$ and those in $[c, b]$ determine a partition P_2 of $[c, b]$. The corresponding sums for all these partitions are connected by the relation

$$\sum(P) \leq \sum(P_0) = \sum(P_1) + \sum(P_2) \leq V_f(a, c) + V_f(c, b).$$

Therefore, $V_f(a, c) + V_f(c, b)$ is an upper bound for every sum $\sum(P)$. Since this cannot be smaller than the least upper bound, we must have

$$V_f(a, b) \leq V_f(a, c) + V_f(c, b),$$

and this completes the proof.

Now we are in a position to keep the function f and the left endpoint of the interval fixed and study the total variation as a function of the right endpoint. The additive property just proved implies important consequences for this function.

8–12 THEOREM. *Let f be of bounded variation on $[a, b]$. Let V be defined on $[a, b]$ as follows: $V(x) = V_f(a, x)$ if $a < x \leq b$, $V(a) = 0$. Then:*

(i) *V is an increasing function on $[a, b]$.*

(ii) *$V - f$ is an increasing function on $[a, b]$.*

Proof. If $a < x < y \leq b$, we can write $V_f(a, y) = V_f(a, x) + V_f(x, y)$. This implies $V(y) - V(x) = V_f(x, y) \geq 0$. Hence $V(x) \leq V(y)$, and (i) holds.

To prove (ii), let $D(x) = V(x) - f(x)$ if $x \in [a, b]$. Then, if $a \leq x < y \leq b$, we have

$$D(y) - D(x) = V(y) - V(x) - [f(y) - f(x)] = V_f(x, y) - [f(y) - f(x)].$$

But from the definition of $V_f(x, y)$ it follows that we have

$$f(y) - f(x) \leq V_f(x, y).$$

This means that $D(y) - D(x) \geq 0$, and (ii) holds.

The following simple and elegant characterization of functions of bounded variation is a consequence of Theorem 8–12.

8-13 THEOREM *Let f be defined on $[a, b]$. Then f is of bounded variation on $[a, b]$ if, and only if, f can be expressed as the difference of two increasing functions.*

Proof If f is of bounded variation on $[a, b]$, we can write $f = V - D$, where V is the function of Theorem 8-12 and $D = V - f$. Both V and D are increasing functions on $[a, b]$

The converse follows at once from Theorems 8-5 and 8-9.

The representation of a function of bounded variation as a difference of two increasing functions is by no means unique If $f = f_1 - f_2$, where f_1 and f_2 are increasing, we also have $f = (f_1 + g) - (f_2 + g)$, where g is an arbitrary increasing function, and we get a new representation of f If g is *strictly* increasing, the same will be true of $f_1 + g$ and $f_2 + g$ Therefore, Theorem 8-13 also holds if "increasing" is replaced by "strictly increasing"

8-5 Continuous functions of bounded variation.

8-14 THEOREM *Let f be of bounded variation on $[a, b]$. If $x \in (a, b]$, let $V(x) = V_f(a, x)$ and put $V(a) = 0$. Then every point of continuity of f is also a point of continuity of V The converse also holds true.*

Proof. Since V is monotonic, the right- and left-hand limits $V(x+)$ and $V(x-)$ exist for each point x in (a, b) Because of Theorem 8-13, the same is true of $f(x+)$ and $f(x-)$

If $a < x < y \leq b$, then we have [by definition of $V_f(x, y)$]

$$0 \leq |f(y) - f(x)| \leq V(y) - V(x)$$

Letting $y \to x$, we find

$$0 \leq |f(x+) - f(x)| \leq V(x+) - V(x)$$

Similarly, $0 \leq |f(x) - f(x-)| \leq V(x) - V(x-)$. These inequalities imply that a point of continuity of V is also a point of continuity of f.

To prove the converse, let f be continuous at the point c in (a, b) Then, given $\epsilon > 0$, there exists a $\delta > 0$ such that $0 < |x - c| < \delta$ implies $|f(x) - f(c)| < \epsilon/2$ For this same ϵ, there also exists a partition P of $[c, b]$, say $P = \{x_0, x_1, \ldots, x_n\}$, $x_0 = c$, $x_n = b$, such that

$$V_f(c, b) - \frac{\epsilon}{2} < \sum_{k=1}^{n} |\Delta f_k|$$

Adding more points to P can only increase the sum $\sum |\Delta f_k|$ and hence we can assume that $0 < x_1 - x_0 < \delta$. This means that

$$|\Delta f_1| = |f(x_1) - f(c)| < \frac{\epsilon}{2},$$

and the foregoing inequality now becomes

$$V_f(c, b) - \frac{\epsilon}{2} < \frac{\epsilon}{2} + \sum_{k=2}^{n} |\Delta f_k| \leq \frac{\epsilon}{2} + V_f(x_1, b),$$

since $\{x_1, x_2, \ldots, x_n\}$ is a partition of $[x_1, b]$. We therefore have

$$V_f(c, b) - V_f(x_1, b) < \epsilon.$$

But $0 \leq V(x_1) - V(c) = V_f(a, x_1) - V_f(a, c) = V_f(c, x_1) = V_f(c, b) - V_f(x_1, b) < \epsilon$. Hence we have shown that $0 < x_1 - c < \delta$ implies $0 \leq V(x_1) - V(c) < \epsilon$. This proves that $V(c+) = V(c)$. A similar argument yields $V(c-) = V(c)$. The theorem is therefore proved for all interior points of $[a, b]$. (Trivial modifications are needed for the endpoints.)

Combining Theorem 8–14 with 8–13, we can state

8–15 THEOREM. *Let f be continuous on $[a, b]$. Then f is of bounded variation on $[a, b]$ if, and only if, f can be expressed as the difference of two increasing continuous functions.*

NOTE. The theorem also holds if "increasing" is replaced by "strictly increasing."

Of course, discontinuities of a function of bounded variation (if any) must be jump discontinuities because of Theorem 8–13. Moreover, Theorem 8–2 tells us that they form a countable set.

8–6 Curves. In his study of elementary calculus, the reader has become familiar with the method of representing a plane curve parametrically by a pair of equations of the form

$$x = \alpha_1(t), \qquad y = \alpha_2(t), \qquad a \leq t \leq b.$$

The interval $[a, b]$ on which the two functions α_1 and α_2 are defined is called the *parametric interval* and the functions α_1 and α_2 are usually assumed to be continuous and to have certain differentiability properties on this interval. This manner of describing a curve is particularly convenient, especially in physical problems dealing with the motion of a particle along a curved path. The parametric interval can then be interpreted as a "time interval" and the parametric equations specify the position of the particle at time t Similar considerations apply, of course, to curves in three-dimensional space In this case, three parametric equations are used to describe the curve Much can be gained in simplicity, however, by writing *one* vector equation rather than several parametric equations, and we will adopt this convenience in the work that follows Moreover, there is no need to restrict ourselves to E_2 or E_3, since it is just as easy to give a definition of what is meant by a curve in E_n

8-16 DEFINITION *Let $\alpha = (\alpha_1, \ldots, \alpha_n)$ be a continuous vector-valued function defined on a closed interval $[a, b]$ The image of $[a, b]$ under α is said to be a curve described by α and it is said to join the points $\alpha(a)$ and $\alpha(b)$ If $\alpha(a) = \alpha(b)$, the curve is called a closed curve If $\alpha(a) \neq \alpha(b)$, the curve is called an arc with endpoints $\alpha(a)$ and $\alpha(b)$ If α is one-to-one on $[a, b]$, the curve is called a simple arc, or a Jordan arc If α is one-to-one on the half-open interval $[a, b)$ and if $\alpha(a) = \alpha(b)$, the curve is called a simple closed curve, or a Jordan curve*

The distinctions between the various terms just defined should be carefully noted (See Fig 8-1)

NOTE If α is constant on $[a, b]$ the resulting curve is called a *point curve* Jordan arcs and Jordan curves are both referred to as *simple* curves

FIG 8-1 Curves in E_2

8-7 Equivalence of continuous vector-valued functions. Certain situations arise in which a description of a curve as a mere set of points is unsatisfactory This is particularly true in physical problems dealing with the motion of a particle, where it is important to know not only the set of points on the curve traced out by the particle, but also the order in which these points are visited during the motion For example, suppose a particle moves in the xy-plane in such a way that its position at time t is given by

$$\alpha(t) = e^{2\pi i t}, \quad \text{if } 0 \le t \le 1 \tag{2}$$

The curve so described is the unit circle $x^2 + y^2 = 1$. If, instead of (2), the equation of motion is

$$\alpha(t) = e^{-2\pi i t}, \qquad \text{if } 0 \leq t \leq 1, \tag{3}$$

the same set of points is visited by the particle but the motion takes place in a "direction" opposite to that in (2). Again, the equation

$$\alpha(t) = e^{2\pi i t^2}, \qquad \text{if } 0 \leq t \leq 1, \tag{4}$$

describes a motion similar to that in (2) but with a different "speed." The "direction" of motion is the same in (2) and (4), however. For the time being we shall not be concerned with the "speed" at which a curve is traversed, but rather with the actual set of points on the curve and the *order* in which these points are traced out. All the motions which trace out a given curve in the same "direction" will be called properly equivalent. The precise meaning of this idea is given in the following definition.

8–17 DEFINITION. *Let* $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)$ *and* $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_n)$ *be two vector-valued functions defined and continuous on* $[a, b]$ *and* $[c, d]$, *respectively. The function* $\boldsymbol{\beta}$ *is said to be properly equivalent to* $\boldsymbol{\alpha}$ *if there exists a real-valued function* h, *continuous and strictly increasing on* $[c, d]$, *such that* $h(c) = a$, $h(d) = b$, *and such that*

$$\boldsymbol{\beta}(t) = \boldsymbol{\alpha}[h(t)] \qquad \textit{for all } t \textit{ in } [c, d].$$

On the other hand, $\boldsymbol{\beta}$ *is called improperly equivalent to* $\boldsymbol{\alpha}$ *if*

$$\boldsymbol{\beta}(t) = \boldsymbol{\alpha}[k(t)] \qquad \textit{for all } t \textit{ in } [c, d],$$

where k *is continuous and strictly decreasing on* $[c, d]$, *with* $k(c) = b$ *and* $k(d) = a$. *Two functions* $\boldsymbol{\alpha}$ *and* $\boldsymbol{\beta}$ *are called equivalent if they are properly or improperly equivalent.*

NOTE. When $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ describe *closed* curves, the above definition requires $\boldsymbol{\alpha}(a) = \boldsymbol{\alpha}(b) = \boldsymbol{\beta}(c) = \boldsymbol{\beta}(d)$.

If we use physical terminology and think of $\boldsymbol{\alpha}$ as describing a motion from A to B, say, along a curve Γ, then all functions properly equivalent to $\boldsymbol{\alpha}$ also describe motions from A to B along Γ, whereas all functions improperly equivalent to $\boldsymbol{\alpha}$ describe motions from B to A along Γ.

Since continuous strictly increasing functions have continuous strictly increasing inverses (Theorem 4–29), it follows that $\boldsymbol{\beta}$ is properly equivalent to $\boldsymbol{\alpha}$ if, and only if, $\boldsymbol{\alpha}$ is properly equivalent to $\boldsymbol{\beta}$. Moreover, if $\boldsymbol{\alpha}$ is properly equivalent to $\boldsymbol{\beta}$ and if $\boldsymbol{\beta}$ is properly equivalent to $\boldsymbol{\gamma}$, then $\boldsymbol{\alpha}$ and $\boldsymbol{\gamma}$ are properly equivalent (Similar remarks hold, of course, for improper equivalence.) Also, it is clear that two equivalent functions always describe the same curve. For those functions which describe *simple arcs* the converse also holds That is to say, we have

8-18 THEOREM. *Two functions α and β which describe Jordan arcs are equivalent if, and only if, they describe the same arc*

Proof Equivalent functions necessarily describe the same curve The whole difficulty lies in the converse deduction

Let α be defined on $[a, b]$, β on $[c, d]$, and assume that they describe the same Jordan arc Γ Since β is one-to-one and continuous on the compact set $[c, d]$, Theorem 4-17 tells us that β^{-1} exists and is continuous on $\beta([c, d]) = \alpha([a, b]) = \Gamma$ Define $h(t) = \beta^{-1}[\alpha(t)]$ if $t \in [a, b]$ Then h is continuous on $[a, b]$ (by Theorem 4-11) and $\alpha(t) = \beta[h(t)]$ by the very definition of h The theorem will be proved if we can show that h is strictly monotonic on $[a, b]$

Since both α and β^{-1} are one-to-one, the same is true of h For definiteness, assume that $h(a) < h(b)$ We will show that h is strictly increasing on $[a, b]$ and that $h(a) = c$, $h(b) = d$. If $a < t_1 < t_2 < b$, then $h(t_1) \neq h(t_2)$. Suppose that $h(t_1) > h(t_2)$ If $h(a) < h(t_2)$, then in the interval $[a, t_1]$ all values between $h(a)$ and $h(t_1)$ will be attained by h, in particular $h(t_2)$ This contradicts one-to-oneness If $h(a) > h(t_2)$, then $h(a)$ would be assumed in the interval $[t_2, b]$, which is also impossible. Therefore the inequality $h(t_1) > h(t_2)$ cannot hold Hence $h(t_1) < h(t_2)$, and this means that h is strictly increasing in the open interval (a, b) By continuity, the same is true on $[a, b]$ From the definition of h it follows that $h(a) = c$ and $h(b) = d$ Therefore we have shown that α and β are properly equivalent (In case $h(a) > h(b)$, a similar argument shows that α and β are improperly equivalent)

In the second half of the foregoing proof we made essential use of the fact that α and β described Jordan *arcs* The theorem must be modified when dealing with Jordan *curves* In fact, two functions describing the same Jordan curve are not necessarily equivalent For example, if α and β are two such functions (defined on $[a, b]$ and $[c, d]$, respectively), we must have $\alpha(a) = \alpha(b)$ and $\beta(c) = \beta(d)$ If $\alpha(a) \neq \beta(c)$, the functions cannot be equivalent (according to Definition 8-17) However, the fact that they describe the same curve implies $\alpha(a) = \beta(e)$ for some point e in (c, d). In this case, we can make a "shift" of the parametric interval and obtain a new function γ, defined on $[e, d - c + e]$, which is equivalent to α and describes the same curve as β This idea of "shifting" the parametric interval is important enough to warrant a formal definition

8–19 DEFINITION. *Let β be a continuous function defined on* $[c, d]$, *with* $\beta(c) = \beta(d)$, *and assume that* $e \in (c, d)$. *The function* γ, *defined on* $[e, d - c + e]$ *by the equations*

$$\gamma(t) = \beta(t) \text{ if } t \in [e, d], \qquad \gamma(t) = \beta(t - d + c) \text{ if } t \in [d, d - c + e],$$

is said to be obtained from β *by shifting the interval* $[c, d]$ *to* $[e, d - c + e]$.

NOTE. It is clear that γ and β describe the same closed curve. However, γ and β are not equivalent.

The analog of Theorem 8–18 for closed curves can now be stated as follows:

8–20 THEOREM. *Assume that two continuous functions* α *and* β, *defined on* $[a, b]$ *and* $[c, d]$, *respectively, describe the same Jordan curve. If* $\alpha(a) = \beta(c)$, *then* α *and* β *are equivalent. If* $\alpha(a) \neq \beta(c)$, *then there is exactly one point* e *in* (c, d) *such that* $\alpha(a) = \beta(e)$. *In this case,* α *is equivalent to the function obtained from* β *by shifting* $[c, d]$ *to* $[e, d - c + e]$.

Proof. A sketch of the proof will be given. The details can easily be filled in by the reader.

Case 1. Assume that $\alpha(a) = \beta(c)$. For each t in (a, b), there is exactly one x in (c, d) such that $\alpha(t) = \beta(x)$. Define h on (a, b) as follows: $h(t) = x$ if $\alpha(t) = \beta(x)$. It is easy to prove that h is continuous and strictly monotonic on the open interval (a, b). By defining $h(a) = \lim_{t \to a+} h(t)$ and $h(b) = \lim_{t \to b-} h(t)$, the same is true on the closed interval. Moreover, $\alpha(t) = \beta[h(t)]$ for each t in $[a, b]$ and hence α and β are equivalent.

Case 2. If $\alpha(a) \neq \beta(c)$, let $\alpha(a) = \beta(e)$ and reduce this case to Case 1 by shifting the interval $[c, d]$ to $[e, d - c + e]$.

Theorem 8–20 suggests a way of extending the definition of the term *equivalence* so that Theorem 8–18 will also hold for Jordan curves.

8–21 DEFINITION. *Let* α *and* β *(defined on* $[a, b]$ *and* $[c, d]$, *respectively) describe two closed curves. We shall say that* α *and* β *are properly equivalent whenever the following conditions are satisfied:*

(i) *If* $\alpha(a) = \beta(c)$, *then* α *and* β *must be properly equivalent in the sense of Definition* 8–17.

(ii) *If* $\alpha(a) \neq \beta(c)$, *then there must exist a function* γ, *obtained from* β *by shifting* $[c, d]$ *to* $[e, d - c + e]$, *where* $e \in (c, d)$, *such that* α *and* γ *are properly equivalent in the sense of Definition* 8–17.

The extended definition of improper equivalence is similar. With this generalized meaning of the term equivalence, it follows that *two functions* *defining simple curves are equivalent if, and only if, they describe the same curve*

It is important to observe that the preceding statement does not hold if the curves are *not* simple For example, the complex-valued function α, defined on $[0, 1]$ by the equation $\alpha(t) = e^{2\pi it}$, describes the same curve as the function β defined on $[0,2]$ by the same formula $\beta(t) = e^{2\pi it}$ However, β does not describe a Jordan curve, since $e^{2\pi i(t+1)} = e^{2\pi it}$ and each point of the curve is obtained from two distinct values of t in $[0,2]$. It is easy to show in this case that α and β cannot be equivalent (See Exercise 8-8)

8-8 Directed paths. Using the concept of equivalence introduced in the previous section, we can now assign a precise mathematical meaning to the intuitive idea of "traversing" a curve in a given "direction " This is done by introducing the notion of a *directed path*

8-22 DEFINITION *Let Γ be a curve described by a continuous vector-valued function $\boldsymbol{\alpha}$ defined on $[a, b]$, and let $\mathbf{A} = \boldsymbol{\alpha}(a)$, $\mathbf{B} = \boldsymbol{\alpha}(b)$ Associated with Γ, we have two collections of functions, namely*

(i) *All functions properly equivalent to $\boldsymbol{\alpha}$.*

(ii) *All functions improperly equivalent to $\boldsymbol{\alpha}$*

These two collections, denoted by $\Gamma[\boldsymbol{\alpha}]$ and $-\Gamma[\boldsymbol{\alpha}]$, respectively, are called directed paths along Γ and they are said to have opposite directions If $\mathbf{A} \neq \mathbf{B}$, so that Γ is an arc, we also write $\Gamma(\mathbf{A}, \mathbf{B})$ instead of $\Gamma[\boldsymbol{\alpha}]$ to denote the collection (i), *and we say that $\Gamma(\mathbf{A}, \mathbf{B})$ is the path from $\mathbf{A}$ to $\mathbf{B}$ along Γ The point $\mathbf{A}$ is called the initial point of $\Gamma(\mathbf{A}, \mathbf{B})$ and $\mathbf{B}$ is its terminal point Similarly, $-\Gamma[\boldsymbol{\alpha}]$ is denoted by $\Gamma(\mathbf{B}, \mathbf{A})$*

If we use physical terminology and think of Γ as a collection of points visited by a moving particle, then a function in $\Gamma(\mathbf{A}, \mathbf{B})$ describes a motion from $\mathbf{A}$ to $\mathbf{B}$ along Γ, whereas a function in $\Gamma(\mathbf{B}, \mathbf{A})$ describes a motion from $\mathbf{B}$ to $\mathbf{A}$ along Γ.

For greater flexibility of language, we shall occasionally use the term *path* to refer to each of the functions in the sets $\Gamma[\boldsymbol{\alpha}]$ and $-\Gamma[\boldsymbol{\alpha}]$ Thus, if we speak of a path $\boldsymbol{\alpha}$ from $\mathbf{A}$ to $\mathbf{B}$, we shall mean that $\boldsymbol{\alpha} \in \Gamma(\mathbf{A}, \mathbf{B})$

An important example of a closed path in the plane is the *positively oriented circle* (See Fig 8-2) This is defined as follows:

8-23 DEFINITION. *Let z_0 be a point in E_2 and let r be a given positive number. Consider the complex-valued function α defined by the equation*

FIG. 8-2 Positively oriented circle.

$$\alpha(t) = z_0 + re^{2\pi i t}, \qquad \textit{where} \qquad 0 \le t \le 1.$$

The collection of functions properly equivalent to α *is called the positively oriented circle with center* z_0 *and radius* r, *and it is said to have a positive, or counterclockwise, direction. The path with opposite direction is said to have a negative, or clockwise, direction.*

8-9 Rectifiable curves. Recall that the closed line segment $L[\mathbf{x}, \mathbf{y}]$ joining two distinct points $\mathbf{x}$ and $\mathbf{y}$ in E_n is the set

$$L[\mathbf{x}, \mathbf{y}] = \{t\mathbf{x} + (1 - t)\mathbf{y} \mid 0 \le t \le 1\}.$$

Therefore, $L[\mathbf{x}, \mathbf{y}]$ is the curve described by the parametric equation

$$\boldsymbol{\alpha}(t) = t\mathbf{x} + (1 - t)\mathbf{y}, \qquad \text{if} \quad 0 \le t \le 1.$$

The length of $L[\mathbf{x}, \mathbf{y}]$ is, of course, $|\mathbf{x} - \mathbf{y}|$. The concept of length for more general curves will now be introduced. As usual, this is done by approximating the curve by means of inscribed polygons. (See Fig. 8-3.)

8-24 DEFINITION. *Consider a curve in* E_n *described by a continuous function* $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)$ *defined on* $[a, b]$. *For each partition*

$$P = \{t_0, t_1, \ldots, t_m\}$$

of $[a, b]$, *the set of points*

$$\pi(P) = \{\mathbf{x} \mid \mathbf{x} \in L[\boldsymbol{\alpha}(t_{k-1}), \boldsymbol{\alpha}(t_k)], k = 1, 2, \ldots, m\}$$

is called the inscribed polygon determined by P. *The length of* $\pi(P)$, *denoted by* $|\pi(P)|$, *is defined to be the sum*

$$|\pi(P)| = \sum_{k=1}^{m} |\boldsymbol{\alpha}(t_k) - \boldsymbol{\alpha}(t_{k-1})|.$$

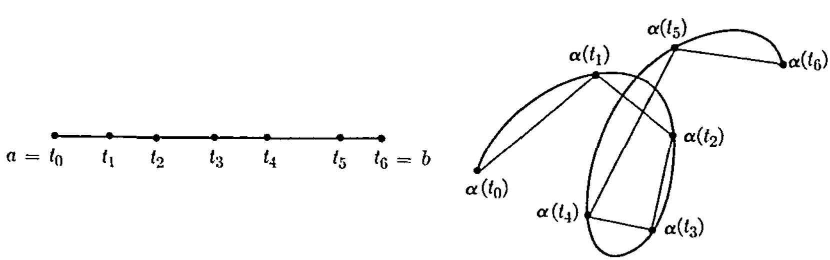

FIG. 8-3. A partition of $[a, b]$ into six subintervals and the corresponding inscribed polygon.

If there exists a positive number M such that $|\pi(P)| < M$ for all partitions of $[a, b]$, then the curve described by α is said to be rectifiable. In this case, the number

$$\sup \{|\pi(P)| \mid P \in \mathcal{P}[a, b]\}$$

is called the arc length of the curve and is denoted by $\Lambda_\alpha(a, b)$

NOTE If α and β are equivalent functions, defined on $[a, b]$ and $[c, d]$, respectively, every partition P of $[a, b]$ gives rise to a partition P' of $[c, d]$ such that $\pi(P) = \pi(P')$, and conversely Hence the curve described by α is rectifiable if, and only if, the curve described by β is rectifiable In this case, we have $\Lambda_\alpha(a, b) = \Lambda_\beta(c, d)$ It follows that for *simple* curves the arc length is completely determined by the set of points on the curve and does not depend on the particular function used to describe the curve. In such a case we will write $\Lambda(\Gamma)$ for the length of Γ

It is an easy matter to characterize all rectifiable curves

8-25 THEOREM. *A curve described by a function $\alpha = (\alpha_1, \ldots, \alpha_n)$ defined on $[a, b]$ is rectifiable if, and only if, each component α_k is of bounded variation on $[a, b]$ If the curve is rectifiable, we have the inequalities*

$$(5) \quad V_k(a, b) \leq \Lambda_\alpha(a, b) \leq V_1(a, b) + \cdots + V_n(a, b) \quad (k = 1, 2, \ldots, n)$$

where $V_k(a, b)$ denotes the total variation of α_k on $[a, b]$

Proof If $P = \{t_0, t_1, \ldots, t_m\}$ is a partition of $[a, b]$, we have

$$(6) \qquad \sum_{i=1}^{m} |\alpha_k(t_i) - \alpha_k(t_{i-1})| \leq |\pi(P)| \leq \sum_{i=1}^{m} \sum_{j=1}^{n} |\alpha_j(t_i) - \alpha_j(t_{i-1})|$$

for each $k = 1, 2, \ldots, n$ All assertions of the theorem follow easily from (6).

8-10 Properties of arc length. If a function α (defined on $[a, b]$) describes a rectifiable curve, each component α_k is of bounded variation on every subinterval $[x, y] \subset [a, b]$ In this section we keep α fixed and study the arc length $\Lambda_\alpha(x, y)$ as a function of the interval $[x, y]$

8-26 THEOREM *If a rectifiable curve is described by a function α defined on $[a, b]$ and if $c \in (a, b)$, then we have*

$$\Lambda_\alpha(a, b) = \Lambda_\alpha(a, c) + \Lambda_\alpha(c, b).$$

Proof. Adjoining the point c to a partition P of $[a, b]$, we get a partition P_1 of $[a, c]$ and a partition P_2 of $[c, b]$ such that

$$|\pi(P)| \leq |\pi(P_1)| + |\pi(P_2)| \leq \Lambda_{\alpha}(a, c) + \Lambda_{\alpha}(c, b).$$

This implies $\Lambda_{\alpha}(a, b) \leq \Lambda_{\alpha}(a, c) + \Lambda_{\alpha}(c, b)$. To obtain the reverse inequality, let P_1 and P_2 be arbitrary partitions of $[a, c]$ and $[c, b]$, respectively. Then $P = P_1 \cup P_2$ is a partition of $[a, b]$ for which we have

$$|\pi(P_1)| + |\pi(P_2)| = |\pi(P)| \leq \Lambda_{\alpha}(a, b).$$

Since the supremum of all sums $|\pi(P_1)| + |\pi(P_2)|$ is the sum $\Lambda_{\alpha}(a, c) + \Lambda_{\alpha}(c, b)$ (see Theorem 1–8), the theorem follows.

8–27 THEOREM. *Consider a rectifiable curve described by a function $\alpha = (\alpha_1, \ldots, \alpha_n)$ defined on $[a, b]$. If $x \in (a, b]$, let $s(x) = \Lambda_{\alpha}(a, x)$ and let $s(a) = 0$. Then we have:*

(i) *The function s so defined is increasing and continuous on $[a, b]$.*

(ii) *If there is no subinterval of $[a, b]$ on which α is constant, then s is strictly increasing on $[a, b]$. In particular, this holds when α describes a simple curve.*

Proof. If $a \leq x < y \leq b$, Theorem 8–26 implies $s(y) - s(x) = \Lambda_{\alpha}(x, y) \geq 0$. This proves that s is increasing on $[a, b]$. Furthermore, we will have $s(y) - s(x) > 0$ unless $\Lambda_{\alpha}(x, y) = 0$. But by inequality (5), $\Lambda_{\alpha}(x, y) = 0$ implies $V_k(x, y) = 0$ for each k and this, in turn, implies that α is constant on $[x, y]$. Hence (ii) holds.

To prove that s is continuous, we use inequality (5) again to write

$$0 \leq s(y) - s(x) = \Lambda_{\alpha}(x, y) \leq \sum_{k=1}^{n} V_k(x, y).$$

If we let $y \to x$, we find each term $V_k(x, y) \to 0$ and hence $s(x) = s(x+)$. Similarly, $s(x) = s(x-)$ and the proof is complete.

8–11 Connectedness. The remaining sections of this chapter are devoted to certain concepts in point set theory which will be needed in some of the later work, particularly in the study of multiple integrals and line integrals. One of the most important of these concepts is the notion of *connectedness.* Actually, there are a number of different "kinds" of connectedness that are useful in analysis, two of which will be discussed here. The first of these is called *arcwise connectedness* and is defined as follows:

8-28 DEFINITION *A set S in E_n is said to be arcwise connected if every pair of points in S can be joined by an arc lying in S*

Thus, for example, every convex set is arcwise connected, since the line segment joining two points of such a set lies in the set In particular, open spheres and closed spheres are arcwise connected The reader can easily verify the arcwise connectedness of the set in Fig 8-4 (a union of two tangent closed disks) The set in Fig 8-5 consists of those points on the curve described by $y = \sin (1/x)$, $0 < x \leq 1$, along with the points on the horizontal segment $-1 \leq x \leq 0$ This set is not arcwise connected [Why?]

Definition 8-28, although suggested by intuition, is often unwieldy when it comes to deriving useful properties of arcwise connected sets We now define a second type of connectedness which is more general than arcwise connectedness, and which has the advantage of being expressible in rather simple set-theoretic terms

8-29 DEFINITION *A set S in E_n is said to be disconnected whenever $S = A \cup B$, where A and B are nonempty disjoint sets, both open relative to S (and hence both closed relative to S) A set S is called connected if it is not disconnected.*

In other words, a connected set S is one that cannot be covered by two open sets (or by two closed sets) whose intersections with S are nonempty disjoint sets (The reader may wish to recall Definition 4-30 and Theorem 4-31)

As a consequence of Theorem 3-9 it follows that an open interval in E_1 is a connected set Examples of connected sets in E_2 are illustrated in Fig 8-6

The relation between connectedness and arcwise connectedness is given in the following theorem

8-30 THEOREM *Every arcwise connected set is connected*

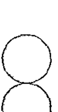

FIG 8-4. Arcwise connected

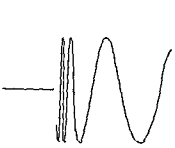

FIG 8-5 Not arcwise connected.

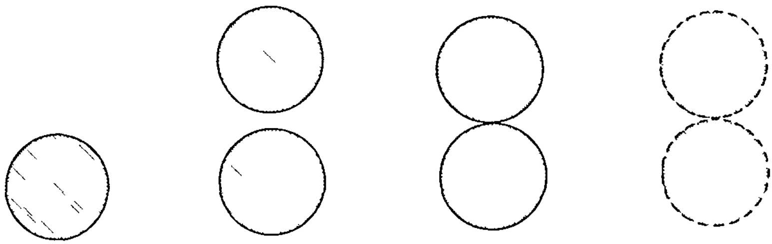

FIG. 8-6. Connected and disconnected sets in E_2.

Proof. Let S be arcwise connected and suppose that $S = A \cup B$, where A and B are disjoint nonempty sets, both open relative to S. We will show that this leads to a contradiction.

Choose $\mathbf{x}_0$ in A, $\mathbf{y}_0$ in B, and join them by an arc C in S described by a function α (defined on $[a, b]$), where $\alpha(a) = \mathbf{x}_0$, $\alpha(b) = \mathbf{y}_0$. Let I_A and I_B denote the following subsets of the open interval (a, b):

$$I_A = \{x \mid x \in (a, b), \alpha(x) \in A\}, \qquad I_B = \{x \mid x \in (a, b), \alpha(x) \in B\}.$$

Then $(a, b) = I_A \cup I_B$ and $I_A \cap I_B$ is empty. If I_B is empty, then $\alpha^{-1}(B \cap C) = \{b\}$. But this is impossible, since $\alpha^{-1}(B \cap C)$ must be open relative to $[a, b]$ (by Theorem 4-32). Hence I_B is nonempty and, similarly, I_A is nonempty. On the other hand, we have $I_A = \alpha^{-1}((A \cap C) - \{\mathbf{x}_0\})$ and $I_B = \alpha^{-1}((B \cap C) - \{\mathbf{y}_0\})$. But $(A \cap C) - \{\mathbf{x}_0\}$ and $(B \cap C) - \{\mathbf{y}_0\}$ are open relative to C [Why?] and hence, by continuity of α, I_A and I_B must be open relative to (a, b). This means that I_A and I_B are nonempty, disjoint open sets whose union is the open interval (a, b). This contradicts the fact that (a, b) is connected. [Connectedness of the open interval (a, b) is a consequence of Theorem 3-9.]

NOTE Theorem 8-30 implies, of course, that every curve is connected. In particular, a closed interval $[a, b]$ in E_1 is connected.

The converse of Theorem 8-30 is not true in general. There are connected sets which are not arcwise connected, an example of which is the set in Fig. 8-5. (See Exercise 8-21.) However, we can prove that these concepts are equivalent for *open sets.*

8-31 THEOREM. *Every open connected set is arcwise connected.*

Proof. Let S be an open connected set and assume that $\mathbf{x} \in S$. We will show that $\mathbf{x}$ can be joined to every point $\mathbf{y}$ in S by an arc lying in S. Let A

denote that subset of S which can be so joined to $\mathbf{x}$, and let $B = S - A$. Then $S = A \cup B$, where A and B are disjoint We will show that A and B are both open

Assume that $\mathbf{a} \varepsilon A$ and join $\mathbf{a}$ to $\mathbf{x}$ by an arc, say Γ, lying in S Since $\mathbf{a} \varepsilon S$, there is a neighborhood $N(\mathbf{a}) \subset S$ Every $\mathbf{y}$ in $N(\mathbf{a})$ can be joined to $\mathbf{a}$ by a line segment (in S) and thence to $\mathbf{x}$ by Γ. Thus $\mathbf{y} \varepsilon A$ if $\mathbf{y} \varepsilon N(\mathbf{a})$. That is $N(\mathbf{a}) \subset A$ and hence A is open

To see that B is also open, assume that $\mathbf{b} \varepsilon B$ Then there is a neighborhood $N(\mathbf{b}) \subset S$ But if a point $\mathbf{y}$ in $N(\mathbf{b})$ could be joined to $\mathbf{x}$ by an arc, say Γ', lying in S, the point $\mathbf{b}$ itself could also be so joined by first joining $\mathbf{b}$ to $\mathbf{y}$ [by a line segment in $N(\mathbf{b})$] and then using Γ' But since $\mathbf{b} \not\varepsilon A$, no point of $N(\mathbf{b})$ can be in A That is, $N(\mathbf{b}) \subset B$ and hence B is open.

Therefore we have the decomposition $S = A \cup B$, where A and B are disjoint open sets Moreover, A is not empty since $\mathbf{x} \varepsilon A$ Since S is connected, it follows that B must be empty But this means that $S = A$ Now A is clearly arcwise connected, because any two of its points can be suitably joined by first joining each of them to $\mathbf{x}$ Therefore, S is arcwise connected and the theorem is proved.

It is easy to prove that connectedness of a set is a property which remains invariant under continuous mappings. In particular, it is a topological property

8–32 THEOREM *Let $\mathbf{f}$ be a vector-valued function defined and continuous on a connected set S in E_n and assume that $\mathbf{f}(S) \subset E_m$ Then $\mathbf{f}(S)$ is a connected set*

Proof Assume that $\mathbf{f}(S)$ is disconnected and write $\mathbf{f}(S) = A_1 \cup B_1$, where A_1 and B_1 are nonempty disjoint sets, both open relative to $\mathbf{f}(S)$. By continuity, the sets $A = \mathbf{f}^{-1}(A_1)$ and $B = \mathbf{f}^{-1}(B_1)$ are open relative to S But

$$S = \mathbf{f}^{-1}(A_1 \cup B_1) = \mathbf{f}^{-1}(A_1) \cup \mathbf{f}^{-1}(B_1) = A \cup B$$

Also, A and B are disjoint because A_1 and B_1 are disjoint Furthermore, A and B are nonempty since A_1 and B_1 are nonempty. This contradicts the connectedness of S

In particular, this theorem gives us another proof of the fact that a curve is a connected set (being the continuous image of a closed interval). Furthermore, Theorem 4–16 tells us that a curve is also a *compact* set This simple observation enables us to prove that every open connected set is *polygonally connected* That is, we have the following theorem:

8–33 THEOREM *Let S be an open connected set Then every pair of points in S can be joined by a polygonal curve lying in S*

NOTE. A curve is said to be *polygonal* (or a *polygon*) if it can be expressed as the union of a finite number of line segments. Thus, for example, the polygons $\pi(P)$ of Definition 8-24 are polygonal curves.

Proof. If $\mathbf{x}$ and $\mathbf{y}$ are two points of S, there exists an arc Γ in S joining $\mathbf{x}$ and $\mathbf{y}$. We shall construct a corresponding inscribed polygon lying in S.

Let Γ be described by a continuous function α defined on $[a, b]$. Each point $\mathbf{z}$ on Γ can be covered by a neighborhood $N(\mathbf{z}; r_{\mathbf{z}}) \subset S$. By compactness of Γ, it is easy to find *one* radius $r > 0$ such that for every $\mathbf{z}$ on Γ we have $N(\mathbf{z}; r) \subset S$. By uniform continuity of α, for this r there exists a $\delta > 0$ such that for every partition $P = \{t_0, t_1, \ldots, t_m\}$ of $[a, b]$, with each $|t_k - t_{k-1}| < \delta$, we will have $|\alpha(t_k) - \alpha(t_{k-1})| < r$. But for such P the polygon $\pi(P) \subset S$. In fact, if $\mathbf{z} \in \pi(P)$ and if $\mathbf{z}$ is on the segment joining $\alpha(t_{k-1})$ to $\alpha(t_k)$, then $|\mathbf{z} - \alpha(t_k)| \leq |\alpha(t_{k-1}) - \alpha(t_k)| < r$. Hence, $\mathbf{z} \in N(\alpha(t_k); r) \subset S$. That is, $\pi(P) \subset S$. (See Fig. 8-7.)

NOTE. If $S \subset E_2$, each of the segments of the polygon $\pi(P)$ used in the above proof can be replaced by the union of a horizontal and a vertical segment as suggested by the dotted lines in Fig. 8-7. A polygonal curve in E_2 consisting entirely of horizontal and vertical segments is called a *step polygon*. Thus every pair of points in an open connected set S in E_2 can be joined by a step polygon lying in S. Moreover, this can always be done by using a *simple* step polygon, that is, one having no self-intersections. Furthermore, every pair of points can also be joined by using a polygonal curve containing *no* horizontal or vertical line segments.

8-34 THEOREM. *Let F be a collection of connected sets such that the intersection $T = \bigcap_{A \in F} A$ is not empty. Then the union $S = \bigcup_{A \in F} A$ is a connected set.*

Proof. Assume that S is disconnected. We will obtain a contradiction. Write $S \subset U \cup V$, where U and V are open sets such that $S \cap U$ and $S \cap V$ are nonempty disjoint sets. Assume that $\mathbf{x} \in T$. Then $\mathbf{x} \in S$ and hence one of the sets U or V contains $\mathbf{x}$. Let U be the set which contains $\mathbf{x}$. Since $S \cap V$ is not empty, there is a point $\mathbf{y}$ in $S \cap V$. Then $\mathbf{y} \in A$ for some A in F. Thus we have $\mathbf{x} \in A \cap U$ and $\mathbf{y} \in A \cap V$. Hence the sets $A \cap U$ and $A \cap V$ are nonempty. They are also disjoint, being subsets of $S \cap U$ and $S \cap V$, respectively. Since $A \subset S \subset U \cup V$, we conclude that A is disconnected. This is a contradiction.

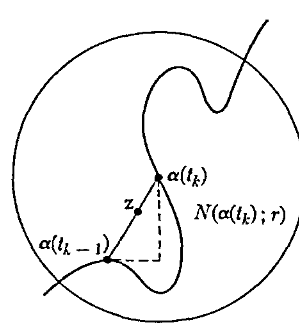

FIGURE 8-7.

8–12 Components of a set. We will prove next that an arbitrary set in E_n can be resolved in a unique fashion into connected "pieces," called *components*. These are defined as follows.

8–35 DEFINITION. *Assume that $S \subset E_n$. If T is a connected subset of S which is not contained in any other connected subset of S, then T is called a component of S.*

In other words, components of S are maximal connected subsets of S.

8–36 THEOREM. *Every point of a set S in E_n belongs to a uniquely determined component of S. If S is open, each component of S is an open connected set.*

Proof. If $\mathbf{x} \in S$, the union of all connected subsets of S which contain $\mathbf{x}$ will be a component of S containing $\mathbf{x}$. Two distinct components of S cannot contain $\mathbf{x}$, otherwise their union would be a larger connected set containing $\mathbf{x}$. This proves the first statement.

If S is open, there is a neighborhood $N(\mathbf{x}) \subset S$. If $\mathbf{x}$ belongs to the component T of S, then $N(\mathbf{x}) \subset T$, since neighborhoods are connected. Therefore, T is open. This proves the second statement.

Another way of stating Theorem 8–36 is to say that the components of S form a collection of disjoint sets whose union is S. In particular, every *open* set is a union of open connected sets. By the Lindelöf theorem (Theorem 3–36), the components of an open set form a *countable* collection. (Theorem 3–9 is a special case.)

We conclude this section with the following generalization of the intermediate value theorem for continuous functions (Theorem 4–22).

8–37 THEOREM. *Let f be a real-valued function defined and continuous on a connected set S in E_n. If f takes on two different values in S, say a and b, then for each number c between a and b there exists a point $\mathbf{x}$ in S such that $f(\mathbf{x}) = c$.*

The proof is an easy consequence of Theorem 8–32 and is left to the reader. (See Exercise 8–22.)

8–13 Regions.

8–38 DEFINITION. *Assume that $S \subset E_n$. A point $\mathbf{x}$ in E_n is called an exterior point of S if there exists a neighborhood of $\mathbf{x}$ containing no points of S. If, however, every neighborhood of $\mathbf{x}$ contains at least one point of S and at least one point not belonging to S, then $\mathbf{x}$ is called a boundary point of S. The collection of all boundary points of S is called the boundary of S and is denoted by ∂S.*

Thus the interior of S (see Definition 3-24), the boundary of S, and the set of exterior points of S are three pairwise disjoint sets whose union is the whole space E_n. Of course, no exterior point can belong to S and a boundary point may or may not belong to S. A boundary point of S need not be an accumulation point of S; however, a point *in* S which is not an accumulation point of S (called an *isolated point of* S) is necessarily a boundary point. A closed set contains all its boundary points; an open set contains none of its boundary points.

8-39 DEFINITION. *A set is called a region if it is the union of an open connected set with some, none, or all its boundary points. If none of the boundary points are included, the region is called an open region. If all the boundary points are included, the region is called a closed region.*

NOTE. Some authors use the term *domain* instead of *open region*, especially in E_2.

A curve C in E_2 is a closed and bounded set. Consequently, its complement $E_2 - C$ is an open set which (by Theorem 8-36) is the union of disjoint open regions. If we consider these open regions as subsets of the extended plane E_2^*, there will be exactly one component which contains the ideal point ∞. In other words, one and only one of the components of $E_2 - C$ is unbounded. (In Fig. 8-8 the unbounded region is the unshaded portion of the plane.)

8-14 Statement of the Jordan curve theorem and related results. A simple closed curve C in the plane divides $E_2 - C$ into *two* components having C as their common boundary. (See Fig. 8-9.) Although this statement may seem intuitively "evident" for certain familiar Jordan curves such as circles, ellipses, or polygons, the proof for an arbitrary Jordan curve is by

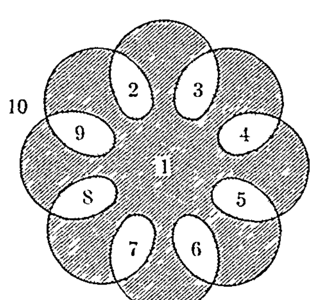

FIG. 8-8. A curve C for which $E_2 - C$ has ten components.

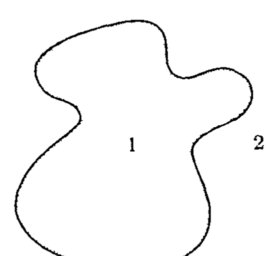

FIG. 8-9. A Jordan curve C and the two components of $E_2 - C$.

no means simple and will not be attempted here. Camille Jordan (1892) was the first to point out that this statement (now known as the *Jordan curve theorem*) requires proof. Although Jordan and others published incomplete proofs of this theorem, it was not until 1905 that a correct proof was given by Veblen. The precise statement of the Jordan curve theorem follows:

8-40 THEOREM. *Let C be a Jordan curve in E_2 and let $S = E_2 - C$. Then $S = A \cup B$ where A and B are two open regions, exactly one of which is bounded. Furthermore, $\partial A = \partial B = C$.*

The bounded component of $E_2 - C$ is called the *interior* (or the *inner region*) of C and is denoted by $I(C)$. The unbounded component of $E_2 - C$ is the *exterior* (or the *outer region*) of C. Subsets of the interior are said to be *inside* C and subsets of the exterior are said to be *outside* C.

The Jordan curve theorem (and some related theorems which are stated below) will be needed later in connection with the theory of line integrals. The results, all of them dealing with curves in E_2, are stated here (without proof) for easy reference.

8-41 THEOREM. *Let C be a Jordan curve in E_2. If $x \in C$ and if $y \in I(C)$, there exists an arc joining x and y which (except for the endpoint x) lies entirely inside C. (The arc need not be rectifiable.)*

8-42 THEOREM. *Let C and C' be two Jordan curves in E_2. If $C' \subset I(C)$, then $I(C') \subset I(C)$.*

NOTE. Proofs of the Jordan curve theorem can be found in References 8-1, 8-3, and 8-4. References 8-1 and 8-3 also contain proofs of Theorem 8-41. Reference 8-4 has a proof of Theorem 8-42 and also of the theorem which follows.

8-43 THEOREM. *Let Γ be a Jordan curve in E_2 and let C be a circle lying inside Γ. Let a and b be two distinct points of C. Then there exist two points α and β of Γ and two nonintersecting simple polygonal curves P and Q joining a to α and b to β, respectively, having the following properties.*

(i) *Except for endpoints, P and Q both lie inside Γ and outside C.*

(ii) *The points a and b decompose C into two arcs, C_1 and C_2, joining a and b. Similarly, α and β decompose Γ into two arcs, Γ_1 and Γ_2, joining α and β. Define*

$$G_1 = C_1 \cup \Gamma_1 \cup P \cup Q \quad \text{and} \quad G_2 = C_2 \cup \Gamma_2 \cup P \cup Q.$$

Then G_1 and G_2 are Jordan curves such that either the interior of C is inside both G_1 and G_2 or else the interior of C is outside both G_1 and G_2.

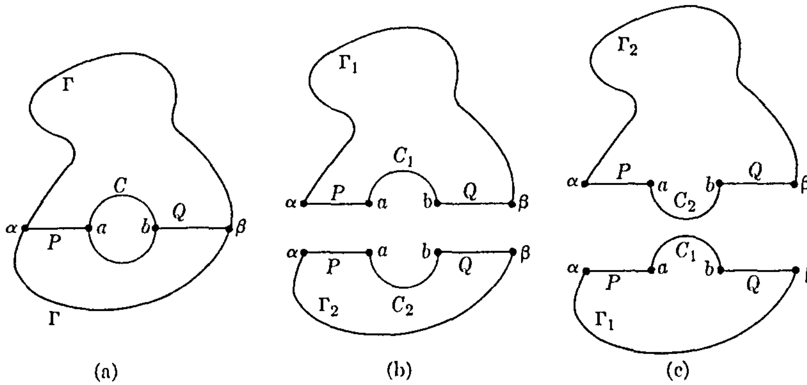

FIG. 8-10. Illustrating Theorem 8-43.

NOTE. By labeling C_1 and C_2 properly, we can arrange matters so that the interior of C is outside both G_1 and G_2.

This theorem is illustrated by Figure 8-10. In this case, P and Q are line segments. Figures 8-10 (b) and 8-10 (c) illustrate the two possibilities that may occur in (ii).

Monotonic functions

8-1. Let f be increasing on $[a, b]$ and let $D = \{x_1, x_2, \quad\}$ denote the set of discontinuities of f in the interior (a, b).

(a) Use Theorem 8-1 to show that the infinite series

$$\sum_{k=1}^{\infty} [f(x_k+) - f(x_k-)]$$

converges and has a sum not exceeding $f(b) - f(a)$

(b) Define a function j_f, called the *jump function* of f, as follows $j_f(a) = 0$, and if $a < x \leq b$, then

$$j_f(x) = f(a+) - f(a) + \sum_{x_k < x} [f(x_k+) - f(x_k-)] + f(x) - f(x-),$$

where the sum is taken over those points x_k in $D \cap (a, x)$ It is obvious that j_f is increasing on $[a, b]$ Derive the following further properties

(i) $j_f(y) - j_f(x) \leq f(y) - f(x)$ if $a \leq x < y < b$

(ii) $j_f(x+) - j_f(x) = f(x+) - f(x)$ and $j_f(x-) - j_f(x) = f(x-) - f(x)$.

(iii) $f(x) = j_f(x) + g(x)$, where g is *increasing* and *continuous* on $[a, b]$

Functions of bounded variation

8-2. Determine whether or not the following functions f are of bounded variation on $[0, 1]$

(a) $f(x) = x^2 \sin(1/x)$ if $x \neq 0, f(0) = 0$

(b) $f(x) = \sqrt{x} \sin(1/x)$ if $x \neq 0, f(0) = 0$

8-3. A function f, defined on $[a, b]$, is said to satisfy a uniform Lipschitz condition of order $\alpha > 0$ on $[a, b]$ if there exists a constant $M > 0$ such that $|f(x) - f(y)| < M|x - y|^\alpha$ for all x and y in $[a, b]$ (Compare with Exercise 5-1)

(a) If f is such a function, show that $\alpha > 1$ implies f is constant on $[a, b]$, whereas $\alpha = 1$ implies f is of bounded variation $[a, b]$.

(b) Give an example of a function f satisfying a uniform Lipschitz condition of order $\alpha < 1$ on $[a, b]$ such that f is not of bounded variation on $[a, b]$

(c) Give an example of a function f which is of bounded variation on $[a, b]$ but which satisfies no uniform Lipschitz condition on $[a, b]$

8–4. Show that a polynomial f is of bounded variation on every finite interval $[a, b]$. Describe a method for finding the total variation of f on $[a, b]$ if the zeros of the derivative f' are known.

8–5. If f is defined everywhere in E_1, then f is said to be of bounded variation on $(-\infty, +\infty)$ if f is of bounded variation on every finite interval and if there exists a positive number M such that $V_f(a, b) < M$ for all finite intervals $[a, b]$. The total variation of f on $(-\infty, +\infty)$ is then defined to be the sup of all numbers $V_f(a, b)$, $-\infty < a < b < +\infty$, and is denoted by $V_f(-\infty, +\infty)$. Similar definitions apply to half-open infinite intervals $[a, +\infty)$ and $(-\infty, b]$.

(a) State and prove theorems for the infinite interval $(-\infty, +\infty)$ analogous to Theorems 8–7, 8–9, 8–10, 8–11, and 8–12.

(b) Show that Theorem 8–5 is true for $(-\infty, +\infty)$ if "monotonic" is replaced by "bounded and monotonic." State and prove a similar modification of Theorem 8–13.

8–6. Assume that f is of bounded variation on $[a, b]$ and let

$$P = \{x_0, x_1, \ldots, x_n\} \in \mathcal{P}[a, b].$$

As usual, write $\Delta f_k = f(x_k) - f(x_{k-1})$, $k = 1, 2, \ldots, n$. Define

$$A(P) = \{k \mid \Delta f_k > 0\}, \qquad B(P) = \{k \mid \Delta f_k < 0\}.$$

The numbers

$$p_f(a, b) = \sup\left\{\sum_{k \in A(P)} \Delta f_k \mid P \in \mathcal{P}[a, b]\right\}$$

and

$$n_f(a, b) = \sup\left\{\sum_{k \in B(P)} |\Delta f_k| \mid P \in \mathcal{P}[a, b]\right\}$$

are called, respectively, the positive and negative variations of f on $[a, b]$. For each x in $(a, b]$, let $V(x) = V_f(a, x)$, $p(x) = p_f(a, x)$, $n(x) = n_f(a, x)$, and set $V(a) = p(a) = n(a) = 0$. Show that we have:

(a) $V(x) = p(x) + n(x)$.

(b) $0 \leq p(x) \leq V(x)$ and $0 \leq n(x) \leq V(x)$.

(c) p and n are increasing on $[a, b]$.

(d) $f(x) = f(a) + p(x) - n(x)$.

NOTE. Part (d) gives an alternative proof of Theorem 8–13.

(e) $2p(x) = V(x) + f(x) - f(a), \qquad 2n(x) = V(x) - f(x) + f(a)$.

(f) Every point of continuity of f is also a point of continuity of p and of n.

Curves

8–7. Fill in the details of the proof of Theorem 8–20

8–8. Define complex valued functions α and β as follows

$$\alpha(t) = e^{2\pi i t} \text{ if } t \in [0, 1], \qquad \beta(t) = e^{2\pi i t} \text{ if } t \in [0, 2].$$

(a) Show that α and β describe the same curve but that α and β are not equivalent (according to Definition 8–21)

(b) Show that $\Lambda_\beta[0, 2] = 2\Lambda_\alpha[0, 1]$

8–9. Prove that every Jordan curve is a topological image of the unit circle S in E_2 $S = \{x \mid x \in E_2, |x| = 1\}$

8–10. For a partition $P = \{t_0, t_1, \ldots, t_m\}$ of $[a, b]$, the number $|P|$ (called the *norm* of P) is defined to be the largest of the m positive numbers $t_k - t_{k-1}$, $k = 1, 2, \ldots, m$ Let $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)$ be a continuous function defined on $[a, b]$ Prove that the curve described by $\boldsymbol{\alpha}$ is rectifiable and has arc length L if, and only if, the following statement holds For every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|P| < \delta \text{ implies } |\,|\pi(P)| - L| < \epsilon$$

8–11. Let Γ be a rectifiable simple curve of length $\Lambda(\Gamma)$ described by a function $\boldsymbol{\alpha}$ defined on $[a, b]$ Let $s(x) = \Lambda_{\boldsymbol{\alpha}}(a, x)$ if $a < x \le b$, and let $s(a) = 0$

(a) Show that s^{-1} exists and is continuous on $[0, \Lambda(\Gamma)]$.

(b) Define $\boldsymbol{\beta}$ by the equation $\boldsymbol{\beta}(t) = \boldsymbol{\alpha}[s^{-1}(t)]$ if $t \in [0, \Lambda(\Gamma)]$, and show that $\boldsymbol{\beta}$ is properly equivalent to $\boldsymbol{\alpha}$

NOTE Since $\boldsymbol{\alpha}(t) = \boldsymbol{\beta}[s(t)]$, the function $\boldsymbol{\beta}$ is said to provide a representation of Γ with arc length as parameter The result of this exercise shows that such a representation always exists when Γ is a rectifiable simple curve

8–12. Let f and g be two real-valued continuous functions of bounded variation defined on $[a, b]$, with $0 < f(x) < g(x)$ for each x in (a, b), $f(a) = g(a)$, $f(b) = g(b)$ Let h be the complex-valued function defined on the interval $[a, 2b - a]$ as follows:

$$h(t) = t + i f(t), \qquad \text{if } a \le t \le b,$$
$$h(t) = t + i g(2b - t), \qquad \text{if } b \le t \le 2b - a.$$

(a) Show that h describes a rectifiable Jordan curve Γ.

(b) Explain, by means of a sketch, the geometric relationship between f, g, and h

(c) Show that the set of points

$$R = \{(x, y) \mid a \leq x \leq b,\ f(x) \leq y \leq g(x)\}$$

is a region in E_2 whose boundary is the curve Γ.

(d) Let H be the complex-valued function defined on $[a, 2b - a]$ as follows:

$$H(t) = t - \tfrac{1}{2} i[g(t) - f(t)], \qquad \text{if } a \leq t \leq b,$$
$$H(t) = t + \tfrac{1}{2} i[g(2b - t) - f(2b - t)], \qquad \text{if } b \leq t \leq 2b - a.$$

Show that H describes a rectifiable Jordan curve Γ_0 which is the boundary of the region

$$R_0 = \{(x, y) \mid a \leq x \leq b, f(x) - g(x) \leq 2y \leq g(x) - f(x)\}.$$

(e) Show that R_0 has the x-axis as a line of symmetry. (The region R_0 is called the *symmetrization* of R with respect to the *x-axis.*)

(f) Show that the length of Γ_0 does not exceed the length of Γ.

Connected sets.

8–13. Show that an open set S in E_n is disconnected if $S = A \cup B$, where A and B are disjoint open sets, neither of which is empty. Show that the same statement also holds with "open" replaced by "closed" throughout.

8–14. Prove that a set S in E_n is connected if, and only if, the only subsets of S which are both open and closed relative to S are S itself and the empty set.

8–15. Show that the union of two open spheres $N(\mathbf{x}; a)$ and $N(\mathbf{y}; b)$ in E_n is a connected set if, and only if, $|\mathbf{x} - \mathbf{y}| < a + b$. What can you say about the connectedness of the union if one or both of the spheres is closed?

8–16. Let S be a subset of E_n. A continuous real-valued function f defined on S is said to be *two-valued* on S if $f(S) \subset \{0, 1\}$ (in other words, a two-valued function is a continuous function whose only possible values are 0 and 1). Prove that S is connected if, and only if, every two-valued function on S is constant.

8–17. Use the result of Exercise 8–16 to give alternate proofs of Theorems 8–30, 8–32, and 8–34.

8–18. Let S be a connected set in E_n and let $\bar{S}$ denote its closure. (See Exercise 3–12.) If $S \subset T \subset \bar{S}$, show that T must also be connected. (In particular, this shows that the closure of a connected set is connected.)

8–19. Prove that the cartesian product of two connected sets is connected. [*Hint*: Use the result of Exercise 8–16.]

8–20. Show that the only connected subsets of E_1 which contain more than one point are the intervals (open, closed, or half-open)

8–21. Let

$$A = \{(x, y) \mid 0 < x \leq 1, y = \sin(1/x)\}, \; B = \{(x, y) \mid y = 0, -1 \leq x \leq 0\},$$

and let $S = A \cup B$ Show that S is connected but not arcwise connected (See Fig 8–5 following Definition 8–28)

8–22. Prove Theorem 8–37

8–23. Let x and y be two points of an open region S in E_2

(a) Prove that x and y can be joined by a simple step-polygon in S

(b) Prove that x and y can be joined by a simple polygon in S containing no horizontal or vertical line segments

8–24. Let $F = \{F_1, F_2, \ldots\}$ be a countable collection of connected compact sets such that $F_{n+1} \subset F_n (n = 1, 2, \ldots)$ Show that the intersection $\bigcap_{k=1}^{\infty} F_k$ is connected and closed

8–25. Let S be an open connected set in E_n Let T be a component of $E_n - S$ Show that $E_n - T$ is connected

Suggested References

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8–2 Lefschetz, S, *Introduction to Topology* Princeton University Press, 1949

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CHAPTER 9

THEORY OF RIEMANN-STIELTJES INTEGRATION

9–1 Introduction. The calculus can be considered as dealing principally with two fundamental problems: finding the tangent to a curve, and finding the area under a curve. The first problem is solved by the limit process known as *differentiation*; the second is solved by another limit process—*integration*—to which we now turn.

The reader will recall from elementary calculus that when we seek the area under the graph of a positive function f defined on $[a, b]$, we subdivide the interval $[a, b]$ into a finite number (say n) of subintervals, the kth subinterval having length Δx_k, and we consider sums of the form $\sum_{k=1}^{n} f(t_k)\, \Delta x_k$, where t_k is some point in the kth subinterval. Each such sum represents an approximation to the area we seek, by means of rectangles. If the function f is sufficiently well behaved in $[a, b]$ (continuous, for example), then there is some hope that these sums will tend to a limit as we let $n \to \infty$, making the successive subdivisions finer and finer. This, roughly speaking, is what is involved in Riemann's definition of the definite integral $\int_a^b f(x)\, dx$. (A precise definition will be given below.)

The two concepts, derivative and integral, arise in entirely different ways and it is a remarkable fact indeed that the two are intimately connected. If we consider the definite integral of a continuous function f as a function of its upper limit, say we write

$$F(x) = \int_a^x f(t)\, dt,$$

then F has a derivative and $F'(x) = f(x)$. This important result shows that differentiation and integration are, in a sense, inverse operations.

In this chapter we shall study the process of integration in some detail. We shall actually consider a more general concept than that of Riemann: this is the *Riemann-Stieltjes integral*, which involves two functions f and α, rather than one function. The symbol for such an integral is $\int_a^b f(x)\, d\alpha(x)$, or something similar, and the usual Riemann integral occurs as the special case in which $\alpha(x) = x$. When α has a continuous derivative, the definition is such that the symbol $d\alpha(x)$ can then be interpreted as the differential $\alpha'(x)\, dx$ and the Stieltjes integral $\int_a^b f(x)\, d\alpha(x)$ becomes the Riemann integral $\int_a^b f(x)\, \alpha'(x)\, dx$. However, the Stieltjes integral still makes sense when α is not differentiable or even when α is discontinuous In fact, it is in dealing with *discontinuous* α that the importance of the Stieltjes integral becomes apparent By a suitable choice of a discontinuous α, any finite or infinite sum can be expressed as a Stieltjes integral, and summation and ordinary Riemann integration then become special cases of this more general process Problems in physics which involve mass distributions that are partly discrete and partly continuous can also be treated by using Stieltjes integrals In the mathematical theory of probability this integral is a very useful tool that makes possible the simultaneous treatment of continuous and discrete random variables

9-2 Notations. For brevity, we shall make certain stipulations concerning notations and terminology to be used in this chapter We shall be working with a finite interval $[a, b]$ and, unless otherwise stated, all functions denoted by f, g, α, or β will be assumed to be real-valued functions defined and *bounded* on $[a, b]$ Complex-valued functions will be dealt with in the latter part of the chapter Extensions to unbounded functions and infinite intervals will be discussed in Chapter 14

As in Chapter 8, a partition P of $[a, b]$ is a finite set of points, say $P = \{x_0, x_1, \ldots, x_n\}$, such that $a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$ The norm of P is the largest of the n numbers $\Delta x_k = x_k - x_{k-1}$, and is denoted by $|P|$ The partition P' of $[a, b]$ is finer than P (or a refinement of P) if $P \subset P'$. Thus, $P \subset P'$ implies $|P'| \leq |P|$, but the converse does not hold. The symbol $\Delta\alpha_k$ denotes the difference $\Delta\alpha_k = \alpha(x_k) - \alpha(x_{k-1})$, so that $\sum_{k=1}^{n} \Delta\alpha_k = \alpha(b) - \alpha(a)$ Finally, the set of all possible partitions of $[a, b]$ is denoted by $\mathcal{P}[a, b]$

9-3 The definition of the Riemann-Stieltjes integral.

9-1 DEFINITION. *Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$ and let t_k be a point in the subinterval $[x_{k-1}, x_k]$ A sum of the form*

$$S(P, f, \alpha) = \sum_{k=1}^{n} f(t_k)\, \Delta\alpha_k$$

is called a Riemann-Stieltjes sum of f with respect to α We say f is Riemann-integrable with respect to α on $[a, b]$, and we write "$f \in R(\alpha)$ on $[a, b]$" if there exists a number A having the following property: For every $\epsilon > 0$, there exists a partition P_ϵ of $[a, b]$ such that for every partition P finer than P_ϵ and for every choice of the points t_k in $[x_{k-1}, x_k]$, we have $|S(P, f, \alpha) - A| < \epsilon$

When such a number A exists, it is uniquely determined and is denoted by $\int_a^b f\,d\alpha$ or by $\int_a^b f(x)\,d\alpha(x)$. We also say that the Riemann-Stieltjes integral $\int_a^b f\,d\alpha$ *exists.* The functions f and α are referred to as the *integrand* and the *integrator*, respectively. In the special case when $\alpha(x) = x$, we write $S(P, f)$ instead of $S(P, f, \alpha)$, and $f \in R$ instead of $f \in R(\alpha)$. The integral is then called a Riemann integral and is denoted by $\int_a^b f\,dx$ or by $\int_a^b f(x)\,dx$. The numerical value of $\int_a^b f(x)\,d\alpha(x)$ depends only on f, α, a, and b, and does not depend on the symbol x. The letter x is a "dummy variable" and may be replaced by any other convenient symbol.

NOTE. This is one of several accepted definitions of the Riemann-Stieltjes integral. An alternative (but not equivalent) definition is stated in Exercise 9–4.

9–4 Linearity properties. It is an easy matter to prove that the integral operates in a linear fashion on both the integrand and the integrator. This is the content of the next two theorems.

9–2 THEOREM. *If $f \in R(\alpha)$ and if $g \in R(\alpha)$ on $[a, b]$, then $c_1 f + c_2 g \in R(\alpha)$ on $[a, b]$ (for any two constants c_1 and c_2) and we have*

$$\int_a^b (c_1 f + c_2 g)\,d\alpha = c_1 \int_a^b f\,d\alpha + c_2 \int_a^b g\,d\alpha.$$

Proof. Let $h = c_1 f + c_2 g$. Given a partition P of $[a, b]$, we can write

$$\begin{aligned} S(P, h, \alpha) &= \sum_{k=1}^{n} h(t_k)\,\Delta\alpha_k = c_1 \sum_{k=1}^{n} f(t_k)\,\Delta\alpha_k + c_2 \sum_{k=1}^{n} g(t_k)\,\Delta\alpha_k \\ &= c_1 S(P, f, \alpha) + c_2 S(P, g, \alpha). \end{aligned}$$

Given $\epsilon > 0$, choose P'_ϵ so that $P'_\epsilon \subset P$ implies $|S(P, f, \alpha) - \int_a^b f\,d\alpha| < \epsilon$, and choose P''_ϵ so that $P''_\epsilon \subset P$ implies $|S(P, g, \alpha) - \int_a^b g\,d\alpha| < \epsilon$. If we take $P_\epsilon = P'_\epsilon \cup P''_\epsilon$, then, for P finer than P_ϵ, we have

$$\left| S(P, h, \alpha) - c_1 \int_a^b f\,d\alpha - c_2 \int_a^b g\,d\alpha \right| \le |c_1|\epsilon + |c_2|\epsilon,$$

and this proves the theorem.

9–3 THEOREM. *If $f \in R(\alpha)$ and $f \in R(\beta)$ on $[a, b]$, then $f \in R(c_1\alpha + c_2\beta)$ on $[a, b]$ (for any two constants c_1 and c_2) and we have*

$$\int_a^b f\,d(c_1\alpha + c_2\beta) = c_1 \int_a^b f\,d\alpha + c_2 \int_a^b f\,d\beta.$$

The proof is similar to that of Theorem 9-2 and is left as an exercise.

A result somewhat analogous to the previous two theorems tells us that the integral is also "additive" with respect to the interval of integration

9-4 THEOREM *Assume that $c \in (a, b)$. If two of the three integrals in* (1) *exist, then the third also exists and we have*

$$\int_a^c f\,d\alpha + \int_c^b f\,d\alpha = \int_a^b f\,d\alpha. \tag{1}$$

Proof If P is a partition of $[a, b]$ such that $c \in P$, let $P' = P \cap [a, c]$ and $P'' = P \cap [c, b]$ denote the corresponding partitions of $[a, c]$ and $[c, b]$, respectively. The Riemann-Stieltjes sums for these partitions are connected by the equation

$$S(P, f, \alpha) = S(P', f, \alpha) + S(P'', f, \alpha)$$

Assume that $\int_a^c f\,d\alpha$ and $\int_c^b f\,d\alpha$ exist. Then, given $\epsilon > 0$, there is a partition P'_ϵ of $[a, c]$ such that

$$\left|S(P', f, \alpha) - \int_a^c f\,d\alpha\right| < \epsilon/2 \qquad \text{whenever } P' \text{ is finer than } P'_\epsilon,$$

and a partition P''_ϵ of $[c, b]$ such that

$$\left|S(P'', f, \alpha) - \int_c^b f\,d\alpha\right| < \epsilon/2 \qquad \text{whenever } P'' \text{ is finer than } P''_\epsilon.$$

Then $P_\epsilon = P'_\epsilon \cup P''_\epsilon$ is a partition of $[a, b]$ such that P finer than P_ϵ implies $P'_\epsilon \subset P'$ and $P''_\epsilon \subset P''$. Hence, if P is finer than P_ϵ, we can combine the foregoing results to obtain the inequality

$$\left|S(P, f, \alpha) - \int_a^c f\,d\alpha - \int_c^b f\,d\alpha\right| < \epsilon$$

This proves that $\int_a^b f\,d\alpha$ exists and equals $\int_a^c f\,d\alpha + \int_c^b f\,d\alpha$. The reader can easily verify that a similar argument proves the theorem in the remaining cases

Using mathematical induction, we can prove a similar result for a decomposition of $[a, b]$ into a finite number of subintervals

NOTE. The preceding type of argument cannot be used to prove that the integral $\int_a^c f\,d\alpha$ exists whenever $\int_a^b f\,d\alpha$ exists. The conclusion is correct, however. (See Exercise 9–2.) For integrators α of bounded variation, this fact will later be proved in Theorem 9–21.

9–5 DEFINITION. *If $a < b$, we define $\int_b^a f\,d\alpha = -\int_a^b f\,d\alpha$ whenever $\int_a^b f\,d\alpha$ exists. We also define $\int_a^a f\,d\alpha = 0$.*

The equation in Theorem 9–4 can now be written as follows:

$$\int_a^b f\,d\alpha + \int_b^c f\,d\alpha + \int_c^a f\,d\alpha = 0.$$

9–5 Integration by parts. A remarkable connection exists between the integrand and the integrator in a Riemann-Stieltjes integral. The existence of $\int_a^b f\,d\alpha$ implies the existence of $\int_a^b \alpha\,df$, and the converse is also true. Moreover, a very simple relation holds between the two integrals.

9–6 THEOREM. *If $f \in R(\alpha)$ on $[a, b]$, then $\alpha \in R(f)$ on $[a, b]$ and we have*

$$\int_a^b f(x)\,d\alpha(x) + \int_a^b \alpha(x)\,df(x) = f(b)\alpha(b) - f(a)\alpha(a).$$

NOTE. This equation, which provides a kind of reciprocity law for the integral, is known as the *formula for integration by parts.*

Proof. Let $\epsilon > 0$ be given. Since $\int_a^b f\,d\alpha$ exists, there is a partition P_ϵ of $[a, b]$ such that for every P' finer than P_ϵ, we have

$$\left|S(P', f, \alpha) - \int_a^b f\,d\alpha\right| < \epsilon. \tag{2}$$

Consider an arbitrary Riemann-Stieltjes sum for the integral $\int_a^b \alpha\,df$, say

$$S(P, \alpha, f) = \sum_{k=1}^n \alpha(t_k)\,\Delta f_k = \sum_{k=1}^n \alpha(t_k)f(x_k) - \sum_{k=1}^n \alpha(t_k)f(x_{k-1}),$$

where P is finer than P_ϵ. Writing $A = f(b)\alpha(b) - f(a)\alpha(a)$, we have the identity

$$A = \sum_{k=1}^n f(x_k)\alpha(x_k) - \sum_{k=1}^n f(x_{k-1})\alpha(x_{k-1}).$$

Subtracting the last two displayed equations, we find

$$A - S(P, \alpha, f) = \sum_{k=1}^{n} f(x_k)[\alpha(x_k) - \alpha(t_k)]$$

$$+ \sum_{k=1}^{n} f(x_{k-1})[\alpha(t_k) - \alpha(x_{k-1})].$$

The two sums on the right can be combined into a single sum of the form $S(P', f, \alpha)$, where P' is that partition of $[a, b]$ obtained by taking the points x_k and t_k together Then P' is finer than P and hence finer than P_ϵ Therefore the inequality (2) is valid and this means that we have

$$\left| A - S(P, \alpha, f) - \int_a^b f\,d\alpha \right| < \epsilon$$

whenever P is finer than P_ϵ But this is exactly the statement that $\int_a^b \alpha\,df$ exists and equals $A - \int_a^b f\,d\alpha$

9-6 Change of variable in a Riemann-Stieltjes integral

9-7 THEOREM *Let $f \in R(\alpha)$ on $[a, b]$ and let g be a strictly monotonic continuous function defined on an interval S having endpoints c and d. Assume that $a = g(c)$, $b = g(d)$. Let h and β be the composite functions defined as follows:*

$$h(x) = f[g(x)], \qquad \beta(x) = \alpha[g(x)], \qquad \text{if } x \in S.$$

Then $h \in R(\beta)$ on S and we have

$$\int_a^b f\,d\alpha = \int_c^d h\,d\beta.$$

Proof For definiteness, assume that g is strictly increasing on S (This implies $c < d$) Then g is one-to-one and has a strictly increasing, continuous inverse g^{-1} defined on $[a, b]$ Therefore, for every partition $P = \{y_0, \ldots, y_n\}$ of $[c, d]$, there corresponds one and only one partition $P' = \{x_0, \ldots, x_n\}$ of $[a, b]$ with $x_k = g(y_k)$. In fact, we can write $P' = g(P)$ and $P = g^{-1}(P')$. Furthermore, a refinement of P produces a corresponding refinement of P', and the converse also holds

If $\epsilon > 0$ is given, there is a partition P'_ϵ of $[a, b]$ such that P' finer than P'_ϵ implies $|S(P', f, \alpha) - \int_a^b f\,d\alpha| < \epsilon$. Let $P_\epsilon = g^{-1}(P'_\epsilon)$ be the corresponding partition of $[c, d]$, and let $P = \{y_0, \ldots, y_n\}$ be a partition of $[c, d]$ finer than P_ϵ. Form a Riemann-Stieltjes sum

$$S(P, h, \beta) = \sum_{k=1}^{n} h(u_k)\, \Delta\beta_k,$$

where $u_k \in [y_{k-1}, y_k]$ and $\Delta\beta_k = \beta(y_k) - \beta(y_{k-1})$. If we put $t_k = g(u_k)$ and $x_k = g(y_k)$, then $P' = \{x_0, \ldots, x_n\}$ is a partition of $[a, b]$ finer than P'_ϵ. Moreover, we then have

$$\begin{aligned} S(P, h, \beta) &= \sum_{k=1}^{n} f[g(u_k)]\{\alpha[g(y_k)] - \alpha[g(y_{k-1})]\} \\ &= \sum_{k=1}^{n} f(t_k)\{\alpha(x_k) - \alpha(x_{k-1})\} = S(P', f, \alpha), \end{aligned}$$

since $t_k \in [x_{k-1}, x_k]$. Therefore, $|S(P, h, \beta) - \int_a^b f\, d\alpha| < \epsilon$ and the theorem is proved.

NOTE. This theorem applies, in particular, to Riemann integrals, that is, when $\alpha(x) = x$. Another theorem of this type, in which g is not required to be monotonic, will later be proved for Riemann integrals. (See Theorem 9-33.)

9-7 Reduction to a Riemann integral. The next theorem tells us that we are permitted to replace the symbol $d\alpha(x)$ by $\alpha'(x)\, dx$ in the integral $\int_a^b f(x)\, d\alpha(x)$ whenever α has a continuous derivative α'.

9-8 THEOREM. *Let $f \in R(\alpha)$ on $[a, b]$ and assume that α has a continuous derivative α' on $[a, b]$. Then the Riemann integral $\int_a^b f(x)\alpha'(x)\, dx$ exists and we have*

$$\int_a^b f(x)\, d\alpha(x) = \int_a^b f(x)\alpha'(x)\, dx.$$

Proof. Let $g(x) = f(x)\alpha'(x)$ and consider a Riemann sum

$$S(P, g) = \sum_{k=1}^{n} g(t_k)\, \Delta x_k = \sum_{k=1}^{n} f(t_k)\alpha'(t_k)\, \Delta x_k.$$

The same partition P and the same choice of the t_k can be used to form the Riemann-Stieltjes sum

$$S(P, f, \alpha) = \sum_{k=1}^{n} f(t_k)\, \Delta\alpha_k.$$

Applying the Mean Value Theorem, we can write

$$\Delta\alpha_k = \alpha'(v_k)\,\Delta x_k, \qquad \text{where} \qquad v_k \in (x_{k-1}, x_k),$$

and hence

$$S(P, f, \alpha) - S(P, g) = \sum_{k=1}^{n} f(t_k)[\alpha'(v_k) - \alpha'(t_k)]\,\Delta x_k$$

Since f is bounded, we can write $|f(x)| \leq M$ for all x in $[a, b]$, where $M > 0$. Continuity of α' on $[a, b]$ implies uniform continuity on $[a, b]$. Hence, if $\epsilon > 0$ is given, there exists a $\delta > 0$ (depending only on ϵ) such that

$$0 \leq |x - y| < \delta \qquad \text{implies} \qquad |\alpha'(x) - \alpha'(y)| < \frac{\epsilon}{2M(b - a)}.$$

If we take a partition P'_ϵ with norm $|P'_\epsilon| < \delta$, then for any finer partition P we will have $|\alpha'(v_k) - \alpha'(t_k)| < \epsilon/[2M(b - a)]$ in the preceding equation. For such P we therefore have

$$|S(P, f, \alpha) - S(P, g)| < \frac{\epsilon}{2}.$$

On the other hand, since $f \in R(\alpha)$ on $[a, b]$, there exists a partition P''_ϵ such that P finer than P''_ϵ implies

$$\left|S(P, f, \alpha) - \int_a^b f\,d\alpha\right| < \frac{\epsilon}{2}.$$

Combining the last two inequalities, we see that when P is finer than $P_\epsilon = P'_\epsilon \cup P''_\epsilon$, we will have $|S(P, g) - \int_a^b f\,d\alpha| < \epsilon$, and this proves the theorem.

9-8 Step functions as integrators. If α is constant throughout $[a, b]$, the integral $\int_a^b f\,d\alpha$ exists and has the value 0, since each sum $S(P, f, \alpha) = 0$. However, if α is constant except for a jump discontinuity at one point, the integral $\int_a^b f\,d\alpha$ need not exist and, if it does exist, its value need not be zero. The situation is described more fully in the following theorem.

9-9 THEOREM. *Given $a < c < b$. Define α on $[a, b]$ as follows: The values $\alpha(a)$, $\alpha(c)$, $\alpha(b)$ are arbitrary, $\alpha(x) = \alpha(a)$ if $a \leq x < c$, and $\alpha(x) = \alpha(b)$ if $c < x \leq b$. Let f be defined on $[a, b]$ in such a way that at least one of the functions f or α is continuous from the left at c and at least one is continuous from the right at c. Then $f \in R(\alpha)$ on $[a, b]$ and we have*

$$\int_a^b f\,d\alpha = f(c)[\alpha(c+) - \alpha(c-)].$$

NOTE. The result also holds if $c = a$, provided that we write $\alpha(c)$ for $\alpha(c-)$, and it holds for $c = b$ if we write $\alpha(c)$ for $\alpha(c+)$. We will prove later (Theorem 9–28) that the integral does not exist if both f and α are discontinuous from the right or from the left at c.

Proof. If $c \in P$, every term in the sum $S(P, f, \alpha)$ is zero except the two terms arising from the subinterval separated by c, say

$$S(P, f, \alpha) = f(t_{k-1})[\alpha(c) - \alpha(c-)] + f(t_k)[\alpha(c+) - \alpha(c)],$$

where $t_{k-1} \le c \le t_k$. This equation can also be written as follows:

$$\Delta = [f(t_{k-1}) - f(c)][\alpha(c) - \alpha(c-)] + [f(t_k) - f(c)][\alpha(c+) - \alpha(c)],$$

where $\Delta = S(P, f, \alpha) - f(c)[\alpha(c+) - \alpha(c-)]$. Hence we have

$$|\Delta| \le |f(t_{k-1}) - f(c)||\alpha(c) - \alpha(c-)| + |f(t_k) - f(c)||\alpha(c+) - \alpha(c)|.$$

If f is continuous at c, for every $\epsilon > 0$ there is a $\delta > 0$ such that $|P| < \delta$ implies

$$|f(t_{k-1}) - f(c)| < \epsilon \quad \text{and} \quad |f(t_k) - f(c)| < \epsilon.$$

In this case, we obtain the inequality

$$|\Delta| \le \epsilon|\alpha(c) - \alpha(c-)| + \epsilon|\alpha(c+) - \alpha(c)|.$$

But this inequality holds whether or not f is continuous at c. For example, if f is discontinuous both from the right and from the left at c, then $\alpha(c) = \alpha(c-)$ and $\alpha(c) = \alpha(c+)$ and we get $\Delta = 0$. On the other hand, if f is continuous from the left and discontinuous from the right at c, we must have $\alpha(c) = \alpha(c+)$ and we get $|\Delta| \le \epsilon|\alpha(c) - \alpha(c-)|$. Similarly, if f is continuous from the right and discontinuous from the left at c, we have $\alpha(c) = \alpha(c-)$ and $|\Delta| \le \epsilon|\alpha(c+) - \alpha(c)|$. Hence the last displayed inequality holds in every case. This proves the theorem.

Theorem 9–9 tells us that the value of a Riemann-Stieltjes integral can be altered by changing the value of f at a single point and it is easy to see that the *existence* of the integral can also be affected by such a change.

Consider the following example:

$$\alpha(x) = 0 \text{ if } x \neq 0, \quad \alpha(0) = -1,$$
$$f(x) = 1, \quad \text{if } -1 \leq x \leq +1.$$

In this case Theorem 9–9 implies $\int_{-1}^{1} f\,d\alpha = 0$. But if we re-define f so that $f(0) = 2$ and $f(x) = 1$ if $x \neq 0$, we can easily see that $\int_{-1}^{1} f\,d\alpha$ will not exist. In fact, when P is a partition which includes 0 as a point of subdivision, we find

$$S(P, f, \alpha) = f(t_k)[\alpha(x_k) - \alpha(0)] + f(t_{k-1})[\alpha(0) - \alpha(x_{k-2})]$$
$$= f(t_k) - f(t_{k-1}),$$

where $x_{k-2} \leq t_{k-1} \leq 0 \leq t_k \leq x_k$. The value of this sum is 0, 1, or -1, depending on the choice of t_k and t_{k-1}. Hence, $\int_{-1}^{1} f\,d\alpha$ does not exist in this case. However, in a Riemann integral $\int_a^b f(x)\,dx$, the values of f *can* be changed at a finite number of points without affecting either the existence or the value of the integral. To prove this, it suffices to consider the case where $f(x) = 0$ for all x in $[a, b]$ except for one point, say $x = c$. But for such a function it is obvious that $|S(P, f)| \leq |f(c)||P|$. Since $|P|$ can be made arbitrarily small, it follows that $\int_a^b f(x)\,dx = 0$.

The integrator α in Theorem 9–9 is a special case of an important class of functions known as *step functions*. These are functions which are constant throughout an interval except for a finite number of jump discontinuities.

9–10 DEFINITION (*Step function*). *Let α be defined on $[a, b]$ in such a way that α is discontinuous at a finite number of points c_k, where*

$$a \leq c_1 < c_2 < \cdots < c_n \leq b.$$

If α is constant in each open subinterval (c_{k-1}, c_k), then α is called a step function and the number $\alpha(c_k+) - \alpha(c_k-)$ is called the jump at c_k. If $c_1 = a$, the jump at c_1 is $\alpha(c_1+) - \alpha(c_1)$, and if $c_n = b$, the jump at c_n is $\alpha(c_n) - \alpha(c_n-)$.

Step functions provide the connecting link between Riemann-Stieltjes integrals and finite sums:

9–11 THEOREM (*Reduction of a Riemann-Stieltjes integral to a finite sum*). *Let α be a step function defined on $[a, b]$ with jump α_k at x_k, where $a \leq x_1 < x_2 < \cdots < x_n \leq b$. Let f be defined on $[a, b]$ in such a* *way that not both f and α are discontinuous from the right or from the left at each x_k. Then $\int_a^b f\, d\alpha$ exists and we have*

$$\int_a^b f(x)\, d\alpha(x) = \sum_{k=1}^{n} f(x_k)\alpha_k.$$

Proof. By Theorem 9–4, $\int_a^b f\, d\alpha$ can be written as a sum of integrals of the type considered in Theorem 9–9.

One of the simplest step functions is the *greatest-integer function.* Its value at x is the greatest integer which is less than or equal to x and is denoted by $[x]$. Thus, $[x]$ is the unique integer satisfying the inequalities $[x] \leq x < [x] + 1$.

9–12 THEOREM. *Every finite sum can be written as a Riemann-Stieltjes integral. In fact, given a sum $\sum_{k=1}^{n} a_k$, define f on $[0, n]$ as follows:*

$$f(x) = a_k \text{ if } k - 1 < x \leq k \qquad (k = 1, 2, \ldots, n),\ f(0) = 0.$$

Then

$$\sum_{k=1}^{n} a_k = \sum_{k=1}^{n} f(k) = \int_0^n f(x)\, d[x],$$

where $[x]$ is the greatest integer $\leq x$.

Proof. The greatest-integer function is a step function, continuous from the right and having jump 1 at each integer. The function f is continuous from the left at $1, 2, \ldots, n$. Now apply Theorem 9–11.

We shall illustrate the use of Riemann-Stieltjes integrals by deriving a remarkable formula known as *Euler's summation formula,* which relates the integral of a function over an interval $[a, b]$ with the sum of the function values at the *integers* in $[a, b]$. It can sometimes be used to approximate integrals by sums or, conversely, to estimate the values of certain sums by means of integrals.

9–13 THEOREM (*Euler's summation formula*). *If f has a continuous derivative f' on $[a, b]$, then we have*

$$\sum_{a<n\leq b} f(n) = \int_a^b f(x)\, dx + \int_a^b f'(x)((x))\, dx + f(a)((a)) - f(b)((b)),$$

where $((x)) = x - [x]$ *When a and b are integers, this becomes*

$$\sum_{n=a}^{b} f(n) = \int_a^b f(x)\,dx + \int_a^b f'(x)\left(x - [x] - \frac{1}{2}\right)dx + \frac{f(a) + f(b)}{2}.$$

NOTE $\sum_{a<n\leq b}$ means the sum from $n = [a] + 1$ to $n = [b]$

Proof Applying Theorem 9–6 (integration by parts), we can write

$$\int_a^b f(x)\,d(x - [x]) + \int_a^b (x - [x])\,df(x)$$
$$= f(b)(b - [b]) - f(a)(a - [a])$$

Since the greatest-integer function has unit jumps in $[a, b]$ at the integers $x = [a] + 1, [a] + 2, \ldots, [b]$, we can write

$$\int_a^b f(x)\,d[x] = \sum_{a<n\leq b} f(n)$$

If we combine this with the previous equation, the theorem follows at once.

9–9 Monotonically increasing integrators. Upper and lower integrals. The further theory of Riemann-Stieltjes integration will now be developed for monotonically increasing integrators, and we shall see later (in Theorem 9–20) that this is just as general as studying the theory for integrators which are of bounded variation

When α is increasing, the differences $\Delta\alpha_k$ which appear in the Riemann-Stieltjes sums are all non-negative This simple fact plays a vital role in the development of the theory For brevity, we shall use the abbreviation "$\alpha \nearrow$ on $[a, b]$" to mean that "α is increasing on $[a, b]$"

It was stated earlier that the problem of finding the area under the graph of a function f is approached by considering the Riemann sums $\sum f(t_k)\,\Delta x_k$ as approximations to the area by means of rectangles Such sums also arise quite naturally in certain physical problems requiring the use of integration for their solution Another approach to these problems is by means of *upper* and *lower* Riemann sums For example, in the case of areas, we can consider approximations from "above" and from "below" by means of the sums $\sum M_k\,\Delta x_k$ and $\sum m_k\,\Delta x_k$, where M_k and m_k denote, respectively, the sup and inf of the function values in the kth subinterval Our geometric intuition tells us that the upper sums are at least as big as the area we seek, whereas the lower sums cannot exceed this area (See Fig 9–1) Therefore it seems natural to ask· What is the smallest possible

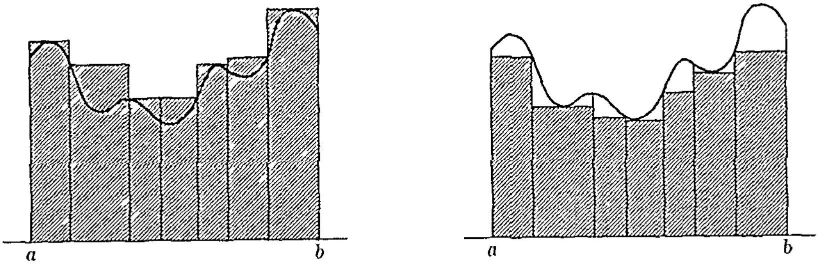

FIG. 9–1. Upper and lower Riemann sums as areas.

value of the upper sums? This leads us to consider the inf of all upper sums, a number called the *upper integral* of f. The *lower integral* is similarly defined to be the sup of all lower sums. For "reasonable" functions (for example, continuous functions) both these integrals will be equal to $\int_a^b f(x)\,dx$. However, in general, these integrals will be different and hence it becomes an important problem to find conditions on the function which will ensure that the upper and lower integrals will be the same. We shall now discuss this type of problem for Riemann-Stieltjes integrals.

9–14 DEFINITION. *Let P be a partition of $[a, b]$ and let*

$$M_k(f) = \sup \{f(x) \mid x \in [x_{k-1}, x_k]\},$$

$$m_k(f) = \inf \{f(x) \mid x \in [x_{k-1}, x_k]\}.$$

The numbers

$$U(P, f, \alpha) = \sum_{k=1}^{n} M_k(f)\,\Delta\alpha_k \quad \text{and} \quad L(P, f, \alpha) = \sum_{k=1}^{n} m_k(f)\,\Delta\alpha_k$$

are called, respectively, the upper and lower Stieltjes sums of f with respect to α for the partition P.

NOTE. We always have $m_k(f) \le M_k(f)$. If $\alpha \nearrow$ on $[a, b]$, then $\Delta\alpha_k \ge 0$ and we can also write $m_k(f)\,\Delta\alpha_k \le M_k(f)\,\Delta\alpha_k$, from which it follows that the lower sums do not exceed the upper sums. Furthermore, if $t_k \in [x_{k-1}, x_k]$, then $m_k(f) \le f(t_k) \le M_k(f)$. Therefore, when $\alpha \nearrow$, we have the inequalities

$$L(P, f, \alpha) \le S(P, f, \alpha) \le U(P, f, \alpha)$$

relating the upper and lower sums to the Riemann-Stieltjes sums. These inequalities, which are frequently used in the material that follows, do not necessarily hold when α is not an increasing function.

The next theorem shows that, for increasing α, refinement of the partition increases the lower sums and decreases the upper sums.

9–15 THEOREM *Assume that $\alpha \nearrow$ on $[a, b]$. Then:*

(i) *If P' is finer than P, we have*

$$U(P', f, \alpha) \leq U(P, f, \alpha) \quad \text{and} \quad L(P', f, \alpha) \geq L(P, f, \alpha)$$

(ii) *For any two partitions P_1 and P_2, we have*

$$L(P_1, f, \alpha) \leq U(P_2, f, \alpha)$$

Proof. It suffices to prove (i) when P' contains exactly one more point than P, say the point c. If c is in the ith subinterval of P, we can write

$$U(P', f, \alpha) = \sum_{\substack{k=1 \\ k \neq i}}^{n} M_k(f)\, \Delta\alpha_k + M'[\alpha(c) - \alpha(x_{i-1})] + M''[\alpha(x_i) - \alpha(c)],$$

where M' and M'' denote the sup of f in $[x_{i-1}, c]$ and $[c, x_i]$. But, since $M' \leq M_i(f)$ and $M'' \leq M_i(f)$, we have $U(P', f, \alpha) \leq U(P, f, \alpha)$. (The inequality for lower sums is proved in a similar fashion.)

To prove (ii), let $P = P_1 \cup P_2$. Then we have

$$L(P_1, f, \alpha) \leq L(P, f, \alpha) \leq U(P, f, \alpha) \leq U(P_2, f, \alpha).$$

NOTE. It follows from this theorem that we also have (for increasing α)

$$m[\alpha(b) - \alpha(a)] \leq L(P_1, f, \alpha) \leq U(P_2, f, \alpha) \leq M[\alpha(b) - \alpha(a)],$$

where M and m denote the sup and inf of f on $[a, b]$.

9–16 DEFINITION *Assume that $\alpha \nearrow$ on $[a, b]$. The upper Stieltjes integral of f with respect to α is defined as follows:*

$$\overline{\int_a^b} f\, d\alpha = \inf \{U(P, f, \alpha) \mid P \in \mathcal{P}[a, b]\}.$$

The lower Stieltjes integral is similarly defined:

$$\underline{\int_a^b} f\, d\alpha = \sup \{L(P, f, \alpha) \mid P \in \mathcal{P}[a, b]\}.$$

NOTE. We sometimes write $\bar{I}(f, \alpha)$ and $\underline{I}(f, \alpha)$ for the upper and lower integrals. In the special case where $\alpha(x) = x$, the upper and lower sums are denoted by $U(P, f)$ and $L(P, f)$ and are called upper and lower Riemann sums. The corresponding integrals, denoted by $\bar{\int}_a^b f(x)\,dx$ and by $\underline{\int}_a^b f(x)\,dx$, are called upper and lower Riemann integrals. They were first introduced by J. G. Darboux (1875).

9–17 THEOREM. *Assume that $\alpha \nearrow$ on $[a, b]$. Then $\underline{I}(f, \alpha) \le \bar{I}(f, \alpha)$.*

Proof. If $\epsilon > 0$ is given, there exists a partition P_1 such that

$$U(P_1, f, \alpha) < \bar{I}(f, \alpha) + \epsilon.$$

By Theorem 9–15, it follows that $\bar{I}(f, \alpha) + \epsilon$ is an upper bound to all lower sums $L(P, f, \alpha)$. Hence, $\underline{I}(f, \alpha) \le \bar{I}(f, \alpha) + \epsilon$, and, since ϵ is arbitrary, this implies $\underline{I}(f, \alpha) \le \bar{I}(f, \alpha)$.

It is an easy matter to give an example in which $\underline{I}(f, \alpha) < \bar{I}(f, \alpha)$.

EXAMPLE. Let $\alpha(x) = x$ and define f on $[a, b] = [0, 1]$ as follows:

$$f(x) = 1, \text{ if } x \text{ is rational}, \qquad f(x) = 0, \text{ if } x \text{ is irrational}.$$

Then for every partition P of $[a, b]$, we have $M_k(f) = 1$ and $m_k(f) = 0$, since every subinterval contains both rational and irrational numbers. Therefore, $U(P, f) = 1$ and $L(P, f) = 0$ for all P. It follows that we have

$$\bar{\int_a^b} f\,dx = 1 \qquad \text{and} \qquad \underline{\int_a^b} f\,dx = 0.$$

Observe that the same result holds if $f(x) = 0$ when x is rational, and $f(x) = 1$ when x is irrational.

Upper and lower integrals share many of the properties of the integral. For example, we have

$$\bar{\int_a^b} f\,d\alpha = \bar{\int_a^c} f\,d\alpha + \bar{\int_c^b} f\,d\alpha$$

if $a < c < b$, and the same equation holds for lower integrals. However, certain equations which hold for integrals must be replaced by inequalities when they are stated for upper and lower integrals. For example, we have

$$\overline{\int_a^b} (f + g)\, d\alpha \le \overline{\int_a^b} f\, d\alpha + \overline{\int_a^b} g\, d\alpha$$

and

$$\underline{\int_a^b} (f + g)\, d\alpha \ge \underline{\int_a^b} f\, d\alpha + \underline{\int_a^b} g\, d\alpha$$

These remarks can be easily verified by the reader (See Exercise 9-7.)

9-10 Riemann's condition. If we are to expect equality of the upper and lower integrals, then we must also expect the upper sums to become arbitrarily close to the lower sums. Hence it seems reasonable to seek those functions f for which the difference $U(P, f, \alpha) - L(P, f, \alpha)$ can be made arbitrarily small.

9-18 DEFINITION *We say that f satisfies Riemann's condition with respect to α on $[a, b]$ if, for every $\epsilon > 0$, there exists a partition P_ϵ such that P finer than P_ϵ implies*

$$0 \le U(P, f, \alpha) - L(P, f, \alpha) < \epsilon$$

9-19 THEOREM *Assume that $\alpha \nearrow$ on $[a, b]$. Then the following three statements are equivalent*

(i) $f \in R(\alpha)$ *on* $[a, b]$

(ii) *f satisfies Riemann's condition with respect to α on $[a, b]$*

(iii) $\underline{I}(f, \alpha) = \bar{I}(f, \alpha)$

Proof We will prove that part (i) implies (ii), part (ii) implies (iii), and part (iii) implies (i). Assume that (i) holds. If $\alpha(b) = \alpha(a)$, then (ii) holds trivially, so we can assume that $\alpha(a) < \alpha(b)$. Given $\epsilon > 0$, choose P_ϵ so that for any finer P and all choices of t_k and t'_k in $[x_{k-1}, x_k]$, we have

$$\left| \sum_{k=1}^{n} f(t_k)\, \Delta\alpha_k - A \right| < \frac{\epsilon}{3} \quad \text{and} \quad \left| \sum_{k=1}^{n} f(t'_k)\, \Delta\alpha_k - A \right| < \frac{\epsilon}{3},$$

where $A = \int_a^b f\, d\alpha$. Combining these inequalities, we find

$$\left| \sum_{k=1}^{n} [f(t_k) - f(t'_k)]\, \Delta\alpha_k \right| < \frac{2}{3}\,\epsilon$$

Since $M_k(f) - m_k(f) = \sup \{f(x) - f(x') \mid x, x' \text{ in } [x_{k-1}, x_k]\}$, it follows that for every $h > 0$ we can choose t_k and t_k' so that $f(t_k) - f(t_k') > M_k(f) - m_k(f) - h$. Making a choice corresponding to

$$h = \tfrac{1}{3}\epsilon/[\alpha(b) - \alpha(a)],$$

we can write

$$U(P, f, \alpha) - L(P, f, \alpha) = \sum_{k=1}^{n} [M_k(f) - m_k(f)] \, \Delta\alpha_k$$

$$< \sum_{k=1}^{n} [f(t_k) - f(t_k')] \, \Delta\alpha_k + h \sum_{k=1}^{n} \Delta\alpha_k < \epsilon.$$

Hence, (i) implies (ii).

Next, assume that (ii) holds. If $\epsilon > 0$ is given, there exists a partition P_ϵ such that P finer than P_ϵ implies $U(P, f, \alpha) < L(P, f, \alpha) + \epsilon$. Hence, for such P we have

$$\bar{I}(f, \alpha) \leq U(P, f, \alpha) < L(P, f, \alpha) + \epsilon \leq \underline{I}(f, \alpha) + \epsilon.$$

That is, $\bar{I}(f, \alpha) \leq \underline{I}(f, \alpha) + \epsilon$ for every $\epsilon > 0$. Therefore, $\bar{I}(f, \alpha) \leq \underline{I}(f, \alpha)$. But, by Theorem 9-17, we also have the opposite inequality. Hence (ii) implies (iii).

Finally, assume that $\bar{I}(f, \alpha) = \underline{I}(f, \alpha)$ and let A denote their common value. We will prove that $\int_a^b f \, d\alpha$ exists and equals A. Given $\epsilon > 0$, choose P_ϵ' so that $U(P, f, \alpha) < \bar{I}(f, \alpha) + \epsilon$ for all P finer than P_ϵ'. Also choose P_ϵ'' such that $L(P, f, \alpha) > \underline{I}(f, \alpha) - \epsilon$ for all P finer than P_ϵ''. If $P_\epsilon = P_\epsilon' \cup P_\epsilon''$, we can write

$$\underline{I}(f, \alpha) - \epsilon < L(P, f, \alpha) \leq S(P, f, \alpha) \leq U(P, f, \alpha) < \bar{I}(f, \alpha) + \epsilon$$

for every P finer than P_ϵ. But, since $\underline{I}(f, \alpha) = \bar{I}(f, \alpha) = A$, this means that $|S(P, f, \alpha) - A| < \epsilon$ whenever P is finer than P_ϵ. This proves that $\int_a^b f \, d\alpha$ exists and equals A, and the proof of the theorem is now complete.

9-11 Integrators of bounded variation. In Theorem 8-13 we found that every function α of bounded variation on $[a, b]$ can be expressed as the difference of two increasing functions. If $\alpha = \alpha_1 - \alpha_2$ is such a decomposition and if $f \in R(\alpha_1)$ and $f \in R(\alpha_2)$ on $[a, b]$, it follows by linearity that $f \in R(\alpha)$ on $[a, b]$. However, the converse is not always true. If $f \in R(\alpha)$ on $[a, b]$, it is quite possible to choose increasing functions α_1 and α_2 such that $\alpha = \alpha_1 - \alpha_2$, but such that neither integral $\int_a^b f \, d\alpha_1$, $\int_a^b f \, d\alpha_2$

exists The difficulty, of course, is due to the nonuniqueness of the decomposition $\alpha = \alpha_1 - \alpha_2$ However, we can prove that there is at least one decomposition for which the converse *is* true, namely, when α_1 is the total variation of α and $\alpha_2 = \alpha_1 - \alpha$ (Recall Definition 8-8)

9-20 THEOREM *Assume that α is of bounded variation on $[a, b]$ Let $V(x)$ denote the total variation of α on $[a, x]$ if $a < x \le b$, and let $V(a) = 0$ Let f be defined and bounded on $[a, b]$ If $f \in R(\alpha)$ on $[a, b]$, then $f \in R(V)$ on $[a, b]$*

Proof If $V(b) = 0$, then V is constant and the result is trivial Suppose, therefore, that $V(b) > 0$ Suppose also that $|f(x)| \le M$ if $x \in [a, b]$ Since V is increasing, we need only verify that f satisfies Riemann's condition with respect to V on $[a, b]$

Given $\epsilon > 0$, choose P_ϵ so that for any finer P and all choices of points t_k and t'_k in $[x_{k-1}, x_k]$ we have

$$\left| \sum_{k=1}^{n} [f(t_k) - f(t'_k)] \Delta\alpha_k \right| < \frac{\epsilon}{4} \qquad \text{and} \qquad V(b) < \sum_{k=1}^{n} |\Delta\alpha_k| + \frac{\epsilon}{4M} \cdot$$

For P finer than P_ϵ we will establish the two inequalities

$$\sum_{k=1}^{n} [M_k(f) - m_k(f)] (\Delta V_k - |\Delta\alpha_k|) < \frac{\epsilon}{2}$$

and

$$\sum_{k=1}^{n} [M_k(f) - m_k(f)] |\Delta\alpha_k| < \frac{\epsilon}{2},$$

which, by addition, yield $U(P, f, V) - L(P, f, V) < \epsilon$

To prove the first inequality, we note that $\Delta V_k - |\Delta\alpha_k| \ge 0$ and hence

$$\sum_{k=1}^{n} [M_k(f) - m_k(f)] (\Delta V_k - |\Delta\alpha_k|) \le 2M \sum_{k=1}^{n} (\Delta V_k - |\Delta\alpha_k|)$$

$$= 2M\left(V(b) - \sum_{k=1}^{n} |\Delta\alpha_k|\right) < \frac{\epsilon}{2} \cdot$$

To prove the second inequality, let $A(P) = \{k \mid \Delta\alpha_k \ge 0\}$, $B(P) = \{k \mid \Delta\alpha_k < 0\}$ and let $h = \frac{1}{4}\epsilon/V(b)$. If $k \in A(P)$, choose t_k and t'_k so that $f(t_k) - f(t'_k) > M_k(f) - m_k(f) - h$; but, if $k \in B(P)$, choose t_k and t'_k

so that $f(t_k') - f(t_k) > M_k(f) - m_k(f) - h$. Then

$$\begin{aligned}\sum_{k=1}^{n} [M_k(f) - m_k(f)] \, |\Delta\alpha_k| &< \sum_{k \varepsilon A(P)} [f(t_k) - f(t_k')] \, |\Delta\alpha_k| \\ &\quad + \sum_{k \varepsilon B(P)} [f(t_k') - f(t_k)] \, |\Delta\alpha_k| + h \sum_{k=1}^{n} |\Delta\alpha_k| \\ &= \sum_{k=1}^{n} [f(t_k) - f(t_k')] \, \Delta\alpha_k + h \sum_{k=1}^{n} |\Delta\alpha_k| \\ &< \frac{\epsilon}{4} + hV(b) = \frac{\epsilon}{4} + \frac{\epsilon}{4} = \frac{\epsilon}{2}.\end{aligned}$$

It follows that $f \varepsilon R(V)$ on $[a, b]$.

Note. This theorem (along with Theorem 8–12) enables us to reduce the theory of Riemann-Stieltjes integration for integrators of bounded variation to the case of *increasing integrators*. Riemann's condition then becomes available and it turns out to be a particularly useful tool in this work. As a first application we shall obtain a result which is closely related to Theorem 9–4.

9–21 Theorem. *Let α be of bounded variation on $[a, b]$ and assume that $f \varepsilon R(\alpha)$ on $[a, b]$. Then $f \varepsilon R(\alpha)$ on every subinterval $[c, d] \subset [a, b]$.*

Proof. It suffices to assume that $\alpha \nearrow$ on $[a, b]$. By Theorem 9–4, it also suffices to prove that $\int_a^c f \, d\alpha$ and $\int_a^d f \, d\alpha$ each exist. Assume that $a < c < b$. If P is a partition of $[a, x]$, let $\Delta(P, x)$ denote the difference

$$\Delta(P, x) = U(P, f, \alpha) - L(P, f, \alpha)$$

of the upper and lower sums associated with the interval $[a, x]$. Since $f \varepsilon R(\alpha)$ on $[a, b]$, Riemann's condition holds. Hence, if $\epsilon > 0$ is given, there exists a partition P_ϵ of $[a, b]$ such that $\Delta(P, b) < \epsilon$ if P is finer than P_ϵ. We can assume that $c \varepsilon P_\epsilon$. The points of P_ϵ in $[a, c]$ form a partition P_ϵ' of $[a, c]$. If P' is a partition of $[a, c]$ finer than P_ϵ', then $P = P' \cup P_\epsilon$ is a partition of $[a, b]$ composed of the points of P' along with those points of P_ϵ in $[c, b]$. Now the sum defining $\Delta(P', c)$ contains only part of the terms in the sum defining $\Delta(P, b)$. Since each term is ≥ 0 and since P is finer than P_ϵ, we have

$$\Delta(P', c) \leq \Delta(P, b) < \epsilon.$$

That is, P' finer than P_ϵ' implies $\Delta(P', c) < \epsilon$. Hence, f satisfies Riemann's condition on $[a, c]$ and $\int_a^c f\,d\alpha$ exists. The same argument, of course, shows that $\int_a^d f\,d\alpha$ exists, and by Theorem 9-4 it follows that $\int_c^d f\,d\alpha$ exists.

9-22 THEOREM *Assume that $\alpha \nearrow$ on $[a, b]$. If $f \in R(\alpha)$ and $g \in R(\alpha)$ on $[a, b]$ and if $f(x) \le g(x)$ for all x in $[a, b]$, then we have*

$$\int_a^b f(x)\,d\alpha(x) \le \int_a^b g(x)\,d\alpha(x).$$

Proof. For every partition P, the corresponding Riemann-Stieltjes sums satisfy

$$S(P, f, \alpha) = \sum_{k=1}^{n} f(t_k)\,\Delta\alpha_k \le \sum_{k=1}^{n} g(t_k)\,\Delta\alpha_k = S(P, g, \alpha)$$

since $\alpha \nearrow$ on $[a, b]$. From this the theorem follows easily.

In particular, this theorem implies that $\int_a^b g(x)\,d\alpha(x) \ge 0$ whenever $g(x) \ge 0$ and $\alpha \nearrow$ on $[a, b]$.

9-23 THEOREM *Assume that $\alpha \nearrow$ on $[a, b]$. If $f \in R(\alpha)$ on $[a, b]$, then $|f| \in R(\alpha)$ on $[a, b]$ and we have the inequality*

$$\left|\int_a^b f(x)\,d\alpha(x)\right| \le \int_a^b |f(x)|\,d\alpha(x).$$

Proof. Using the notation of Definition 9-14, we can write

$$M_k(f) - m_k(f) = \sup\{f(x) - f(y) \mid x, y \text{ in } [x_{k-1}, x_k]\}.$$

Since the inequality $||f(x)| - |f(y)|| \le |f(x) - f(y)|$ always holds, it follows that we have

$$M_k(|f|) - m_k(|f|) \le M_k(f) - m_k(f).$$

Multiplying by $\Delta\alpha_k$ and summing on k, we obtain

$$U(P, |f|, \alpha) - L(P, |f|, \alpha) \le U(P, f, \alpha) - L(P, f, \alpha)$$

for every partition P of $[a, b]$. By applying Riemann's condition, we find that $|f| \in R(\alpha)$ on $[a, b]$. The inequality in the theorem follows by taking $g = |f|$ in Theorem 9-22.

NOTE The converse of Theorem 9-23 is not true. (See Exercise 9-9.)

9-24 THEOREM. *Assume that* $\alpha\nearrow$ *on* $[a, b]$. *If* $f \in R(\alpha)$ *on* $[a, b]$, *then* $f^2 \in R(\alpha)$ *on* $[a, b]$.

Proof. Using the notation of Definition 9-14, we have

$$M_k(f^2) = [M_k(|f|)]^2 \qquad \text{and} \qquad m_k(f^2) = [m_k(|f|)]^2.$$

Hence we can write

$$\begin{aligned} M_k(f^2) - m_k(f^2) &= [M_k(|f|) + m_k(|f|)][M_k(|f|) - m_k(|f|)] \\ &\leq 2M[M_k(|f|) - m_k(|f|)], \end{aligned}$$

where M is an upper bound for $|f|$ on $[a, b]$. By applying Riemann's condition, the conclusion follows.

9-25 THEOREM. *Assume that* $\alpha\nearrow$ *on* $[a, b]$. *If* $f \in R(\alpha)$ *and* $g \in R(\alpha)$ *on* $[a, b]$, *then the product* $f \cdot g \in R(\alpha)$ *on* $[a, b]$.

Proof. We use Theorem 9-24 along with the identity

$$2f(x)g(x) = [f(x) + g(x)]^2 - [f(x)]^2 - [g(x)]^2.$$

9-12 Sufficient conditions for existence of Riemann-Stieltjes integrals. In most of the previous theorems we have assumed that certain integrals existed and then studied their properties. It is quite natural to ask: When does the integral exist? Two useful sufficient conditions will be obtained.

9-26 THEOREM. *If f is continuous on* $[a, b]$ *and if* α *is of bounded variation on* $[a, b]$, *then* $f \in R(\alpha)$ *on* $[a, b]$.

NOTE. By Theorem 9-6, a second sufficient condition can be obtained by interchanging f and α in the hypothesis.

Proof. It suffices to prove the theorem when α is strictly increasing. Continuity of f on $[a, b]$ implies uniform continuity, so that if $\epsilon > 0$ is given, we can find $\delta > 0$ (depending only on ϵ) such that

$$|x - y| < \delta \qquad \text{implies} \qquad |f(x) - f(y)| < \epsilon/A,$$

where $A = 2[\alpha(b) - \alpha(a)]$. If P_ϵ is a partition with norm $|P_\epsilon| < \delta$, then for P finer than P_ϵ we must have

$$M_k(f) - m_k(f) \leq \epsilon/A,$$

since $M_k(f) - m_k(f) = \sup \{f(x) - f(y) \mid x, y \text{ in } [x_{k-1}, x_k]\}$ Multiplying the inequality by $\Delta\alpha_k$ and summing, we find

$$U(P, f, \alpha) - L(P, f, \alpha) \leq \frac{\epsilon}{A} \sum_{k=1}^{n} \Delta\alpha_k = \frac{\epsilon}{2} < \epsilon$$

and we see that Riemann's condition holds Hence, $f \in R(\alpha)$ on $[a, b]$

Taking the special case in which $\alpha(x) = x$, Theorems 9-26 and 9-6 yield the following corollary

9-27 THEOREM *Each of the following conditions is sufficient for the existence of the Riemann integral* $\int_a^b f(x)\,dx$·

(a) *f is continuous on $[a, b]$*

(b) *f is of bounded variation on $[a, b]$*

9-13 Necessary conditions for existence of Riemann-Stieltjes integrals. When α is of bounded variation on $[a, b]$, continuity of f is sufficient for the existence of $\int_a^b f\,d\alpha$ Continuity of f throughout $[a, b]$ is by no means necessary, however For example, in Theorem 9-9 we found that when α is a step function, then f can be defined quite arbitrarily in $[a, b]$ provided only that f is continuous at the discontinuities of α The next theorem tells us that common discontinuities from the right or from the left must be avoided if the integral is to exist

9-28 THEOREM. *Assume that $\alpha \nearrow$ on $[a, b]$ and let $a < c < b$. Assume further that both α and f are discontinuous from the right at $x = c$, that is, assume that there exists an $\epsilon > 0$ such that for every $\delta > 0$ there are values of x and y in the interval $(c, c + \delta)$ for which*

$$|f(x) - f(c)| \geq \epsilon \quad \text{and} \quad |\alpha(y) - \alpha(c)| \geq \epsilon$$

Then the integral $\int_a^b f(x)\,d\alpha(x)$ cannot exist. The integral also fails to exist if α and f are discontinuous from the left at c

Proof Let P be a partition of $[a, b]$ containing c as a point of subdivision and form the difference

$$U(P, f, \alpha) - L(P, f, \alpha) = \sum_{k=1}^{n} [M_k(f) - m_k(f)]\,\Delta\alpha_k$$

If the ith subinterval has c as its left endpoint, then

$$U(P, f, \alpha) - L(P, f, \alpha) \geq [M_i(f) - m_i(f)][\alpha(x_i) - \alpha(c)]$$

since each term of the sum is ≥ 0. If c is a common discontinuity from the right, we can assume that the point x_i is chosen so that $\alpha(x_i) - \alpha(c) \geq \epsilon$. Furthermore, the hypothesis of the theorem implies $M_i(f) - m_i(f) \geq \epsilon$. Hence, $U(P, f, \alpha) - L(P, f, \alpha) \geq \epsilon^2$ and Riemann's condition cannot be satisfied. (If c is a common discontinuity from the left, the argument is similar.)

9-14 Mean Value Theorems for Riemann-Stieltjes integrals. Although integrals occur in a wide variety of problems, there are relatively few cases in which the explicit value of the integral can be obtained. However, it often suffices to have an *estimate* for the integral rather than its exact value. The *Mean Value Theorems* of this section are especially useful in making such estimates.

9-29 THEOREM (*First Mean Value Theorem for Riemann-Stieltjes integrals*). *Assume that* $\alpha \nearrow$ *and let* $f \in R(\alpha)$ *on* $[a, b]$. *Let* M *and* m *denote, respectively, the* sup *and* inf *of the set* $\{f(x) \mid x \in [a, b]\}$. *Then there exists a real number* c *satisfying* $m \leq c \leq M$ *such that*

$$\int_a^b f(x)\, d\alpha(x) = c \int_a^b d\alpha(x) = c[\alpha(b) - \alpha(a)].$$

In particular, if f *is continuous on* $[a, b]$, *then* $c = f(x_0)$ *for some* x_0 *in* $[a, b]$.

Proof. If $\alpha(a) = \alpha(b)$, the theorem holds trivially, both sides being 0. Hence we can assume that $\alpha(a) < \alpha(b)$. Since all upper and lower sums satisfy

$$m[\alpha(b) - \alpha(a)] \leq L(P, f, \alpha) \leq U(P, f, \alpha) \leq M[\alpha(b) - \alpha(a)],$$

the integral $\int_a^b f\, d\alpha$ must lie between the same bounds. Therefore, the quotient $c = (\int_a^b f\, d\alpha)/(\int_a^b d\alpha)$ lies between m and M. When f is continuous on $[a, b]$, the intermediate value theorem yields $c = f(x_0)$ for some x_0 in $[a, b]$.

A second theorem of this type can be obtained from the first by using integration by parts.

9-30 THEOREM (*Second Mean Value Theorem for Riemann-Stieltjes integrals*) *Assume that α is continuous and that $f \nearrow$ on $[a, b]$ Then there exists a point x_0 in $[a, b]$ such that*

$$\int_a^b f(x)\,d\alpha(x) = f(a)\int_a^{x_0} d\alpha(x) + f(b)\int_{x_0}^b d\alpha(x)$$

Proof By Theorem 9-6, we have

$$\int_a^b f(x)\,d\alpha(x) = f(b)\alpha(b) - f(a)\alpha(a) - \int_a^b \alpha(x)\,df(x)$$

Applying Theorem 9-29 to the integral on the right, we find

$$\int_a^b f(x)\,d\alpha(x) = f(a)[\alpha(x_0) - \alpha(a)] + f(b)[\alpha(b) - \alpha(x_0)],$$

where $x_0 \in [a, b]$, which is the statement we set out to prove

9-15 The integral as a function of the interval. If $f \in R(\alpha)$ on $[a, b]$ and if α is of bounded variation, then (by Theorem 9-21) the integral $\int_a^x f\,d\alpha$ exists for each x in $[a, b]$ and can be studied as a function of x Some properties of this function will now be obtained

9-31 THEOREM *Let α be of bounded variation on $[a, b]$ and assume that $f \in R(\alpha)$ on $[a, b]$ Define F by the equation*

$$F(x) = \int_a^x f\,d\alpha, \qquad \text{if } x \in [a, b]$$

Then we have

(i) *F is of bounded variation on $[a, b]$*

(ii) *Every point of continuity of α is also a point of continuity of F*

(iii) *If $\alpha \nearrow$ on $[a, b]$, the derivative $F'(x)$ exists at each point x in (a, b) where $\alpha'(x)$ exists and where f is continuous For such x, we have*

$$F'(x) = f(x)\alpha'(x)$$

Proof It suffices to assume that $\alpha \nearrow$ on $[a, b]$ If $x \neq y$, Theorem 9-29 implies

$$F(y) - F(x) = \int_x^y f\,d\alpha = c[\alpha(y) - \alpha(x)],$$

where $m \leq c \leq M$ (in the notation of Theorem 9-29). Statements (i) and (ii) follow at once from this equation. To prove (iii), we divide by $y - x$ and observe that $c \to f(x)$ as $y \to x$.

The usefulness of the derivative in evaluating Riemann integrals is shown in the next theorem.

9-32 THEOREM (*Fundamental theorem of integral calculus*). *Assume that $f \in R$ on $[a, b]$. Let g be a function defined on $[a, b]$ such that the derivative g' exists in (a, b) and has the value*

$$g'(x) = f(x) \qquad \textit{for every } x \textit{ in } (a, b).$$

At the endpoints assume that $g(a+)$ and $g(b-)$ exist and satisfy

$$g(a) - g(a+) = g(b) - g(b-).$$

Then we have

$$\int_a^b f(x)\,dx = \int_a^b g'(x)\,dx = g(b) - g(a).$$

Proof. For every partition of $[a, b]$ we can write

$$g(b) - g(a) = \sum_{k=1}^{n} [g(x_k) - g(x_{k-1})] = \sum_{k=1}^{n} g'(t_k)\,\Delta x_k = \sum_{k=1}^{n} f(t_k)\,\Delta x_k,$$

where t_k is a point in (x_{k-1}, x_k) determined by the Mean Value Theorem of differential calculus. But, for a given $\epsilon > 0$, the partition can be taken so fine that

$$\left| g(b) - g(a) - \int_a^b f(x)\,dx \right| = \left| \sum_{k=1}^{n} f(t_k)\,\Delta x_k - \int_a^b f(x)\,dx \right| < \epsilon,$$

and this proves the theorem.

9-16 Change of variable in a Riemann integral. The formula

$$\int_a^b f\,d\alpha = \int_c^d h\,d\beta$$

of Theorem 9-7 for changing the variable in an integral assumes the form

$$\int_a^b f(x)\,dx = \int_c^d f[g(t)]\,g'(t)\,dt$$

when $\alpha(x) = x$ and when g is a strictly monotonic function with a continuous derivative g'. It is valid if $f \in R$ on $[a, b]$. When f is continuous, we can use Theorem 9-31 to remove the restriction that g be monotonic. In fact, we have the following theorem.

9-33 THEOREM (*Change of variable in a Riemann integral*) *Let g have a continuous derivative g' on an interval S whose endpoints are c and d. Let f be continuous on $g(S)$ and let $a = g(c)$, $b = g(d)$ Define F by the equation*

$$F(x) = \int_a^x f(t)\,dt, \qquad \text{if } x \in g(S)$$

Then, for each x in S, the integral $\int_c^x f[g(t)]g'(t)\,dt$ exists and has the value $F[g(x)]$ In particular, we have

$$\int_a^b f(x)\,dx = \int_c^d f[g(t)]\,g'(t)\,dt$$

Proof. Since both g' and the composite function fg are continuous on S, the integral in question exists Define G on S as follows

$$G(x) = \int_c^x f[g(t)]\,g'(t)\,dt.$$

We are to show that $G(x) = F[g(x)]$ By Theorem 9-31, we have $G'(x) = f[g(x)]g'(x)$ and, by the chain rule, the derivative of $F[g(x)]$ is also $f[g(x)]g'(x)$, since $F'(x) = f(x)$. Hence, $G(x) - F[g(x)]$ is a constant. But, when $x = c$, we get $G(c) = 0$ and $F[g(c)] = F(a) = 0$, so that this constant must be 0 Hence, $G(x) = F[g(x)]$ for all x in S In particular, when $x = d$, we get $G(d) = F[g(d)] = F(b)$ and this is the last equation in the theorem

NOTE. Many texts prove the preceding theorem under the added hypothesis that g' is never zero on S, which, of course, implies monotonicity of g The above proof shows that this is not needed Furthermore, the interval joining a and b

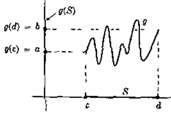

FIGURE 9-2.

need not be the image of S under g. All that is required is that $a = g(c)$ and $b = g(d)$. In fact, the result is still valid if $a = g(c) = g(d) = b$. This makes the theorem especially useful in the applications. (See Fig. 9–2 for a permissible g.)

9–17 Second Mean Value Theorem for Riemann integrals.

9–34 THEOREM. *Let g be continuous and assume that $f \nearrow$ on $[a, b]$. Let A and B be two real numbers satisfying the inequalities*

$$A \leq f(a+) \quad \textit{and} \quad B \geq f(b-).$$

Then there exists a point x_0 in $[a, b]$ such that

$$\text{(i)} \quad \int_a^b f(x)g(x)\,dx = A\int_a^{x_0} g(x)\,dx + B\int_{x_0}^b g(x)\,dx.$$

In particular, if $f(x) \geq 0$ for all x in $[a, b]$, we have

$$\text{(ii)} \quad \int_a^b f(x)g(x)\,dx = B\int_{x_0}^b g(x)\,dx, \qquad \textit{where } x_0 \in [a, b].$$

NOTE. Part (ii) is known as *Bonnet's theorem.*

Proof. If $\alpha(x) = \int_a^x g(t)\,dt$, then $\alpha' = g$, Theorem 9–30 is applicable, and we get

$$\int_a^b f(x)g(x)\,dx = f(a)\int_a^{x_0} g(x)\,dx + f(b)\int_{x_0}^b g(x)\,dx.$$

This proves (i) whenever $A = f(a)$ and $B = f(b)$. Now if A and B are any two real numbers satisfying $A \leq f(a+)$ and $B \geq f(b-)$, we can redefine f at the endpoints a and b to have the values $f(a) = A$ and $f(b) = B$. The modified f is still increasing on $[a, b]$ and, as we have remarked before, changing the value of f at a finite number of points does not affect the value of a Riemann integral. (Of course, the point x_0 in (i) will depend on the choice of A and B.) By taking $A = 0$, part (ii) follows from part (i).

9–18 Riemann-Stieltjes integrals depending on a parameter.

9–35 THEOREM. *Let f be continuous at each point (x, y) of a rectangle $R = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$. Assume that α is of bounded variation on $[a, b]$ and let F be the function defined on $[c, d]$ by the* *equation*

$$F(y) = \int_a^b f(x, y)\, d\alpha(x).$$

Then F is continuous on $[c, d]$ In other words, if $y_0 \in [c, d]$, we have

$$\lim_{y \to y_0} \int_a^b f(x, y)\, d\alpha(x) = \int_a^b \lim_{y \to y_0} f(x, y)\, d\alpha(x) = \int_a^b f(x, y_0)\, d\alpha(x)$$

Proof Assume that $\alpha \nearrow$ on $[a, b]$ Since R is a compact set, f is uniformly continuous on R Hence, given $\epsilon > 0$, there exists a $\delta > 0$ (depending only on ϵ) such that for every pair of points $z = (x, y)$ and $z' = (x', y')$ in R with $|z - z'| < \delta$, we have $|f(x, y) - f(x', y')| < \epsilon$ If $|y - y'| < \delta$, we have

$$|F(y) - F(y')| \leq \int_a^b |f(x, y) - f(x, y')|\, d\alpha(x) \leq \epsilon[\alpha(b) - \alpha(a)]$$

This establishes the continuity of F on $[c, d]$

Of course, when $\alpha(x) = x$, this becomes a continuity theorem for Riemann integrals involving a parameter However, we can derive a much more useful result for Riemann integrals than that obtained by simply setting $\alpha(x) = x$ if we employ the following theorem

9-36 THEOREM *Let α be of bounded variation on $[a, b]$, let $g \in R(\alpha)$ on $[a, b]$, and define*

$$\beta(x) = \int_a^x g(t)\, d\alpha(t), \qquad \text{if } x \in [a, b]$$

If f is continuous on $[a, b]$, then $f \in R(\beta)$ on $[a, b]$ and we have

$$\int_a^b f(x)\, d\beta(x) = \int_a^b f(x) g(x)\, d\alpha(x)$$

NOTE Theorem 9-8 is a special case

Proof. To begin with, β is of bounded variation on $[a, b]$ (by Theorem 9-31) and hence $f \in R(\beta)$ by Theorem 9-26 For every partition P of $[a, b]$, we have

$$S(P, f, \beta) = \sum_{k=1}^{n} f(t_k) \int_{x_{k-1}}^{x_k} g(t)\, d\alpha(t)$$

and

$$\int_a^b f(x) g(x)\, d\alpha(x) = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f(t) g(t)\, d\alpha(t)$$

Therefore,

$$S(P, f, \beta) - \int_a^b f(x)g(x)\, d\alpha(x) = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} [f(t_k) - f(t)]g(t)\, d\alpha(t).$$

The conclusion is now an easy consequence of the uniform continuity of f on $[a, b]$.

NOTE. When $\alpha(x) = x$, this theorem enables us to convert a Riemann integral $\int_a^b f(x)g(x)\, dx$ whose integrand is a product of a *continuous* function f with an *integrable* function g, into a Riemann-Stieltjes integral $\int_a^b f\, d\beta$ in which β is of bounded variation on $[a, b]$. In particular, we can use Theorem 9–36 to deduce the following:

COROLLARY OF THEOREM 9–35. *If f is continuous on the rectangle R of Theorem 9–35 and if $g \in R$ on $[a, b]$, then the function F defined by the equation*

$$F(y) = \int_a^b g(x)f(x, y)\, dx$$

is continuous on $[c, d]$. That is, if $y_0 \in [c, d]$, we have

$$\lim_{y \to y_0} \int_a^b g(x)f(x, y)\, dx = \int_a^b g(x)f(x, y_0)\, dx.$$

Proof. If $\beta(x) = \int_a^x g(t)\, dt$, we have $F(y) = \int_a^b f(x, y)\, d\beta(x)$. Now apply Theorem 9–35.

9–19 Differentiation under the integral sign.

9–37 THEOREM. *Let $R = \{(x, y) \mid a \le x \le b, c \le y \le d\}$. Assume that α is of bounded variation on $[a, b]$ and, for each fixed y in $[c, d]$, assume that the integral*

$$F(y) = \int_a^b f(x, y)\, d\alpha(x)$$

exists. If the partial derivative $D_2 f$ is continuous on R, the derivative $F'(y)$ exists for each y in (c, d) and is given by

$$F'(y) = \int_a^b D_2 f(x, y)\, d\alpha(x).$$

NOTE. In particular, when $g \in R$ on $[a, b]$ and $\alpha(x) = \int_a^x g(t)\, dt$, we get

$$F(y) = \int_a^b g(x)f(x, y)\, dx \qquad \text{and} \qquad F'(y) = \int_a^b g(x)\, D_2f(x, y)\, dx$$

Proof If $y_0 \in (c, d)$ and $y \neq y_0$, we have

$$\frac{F(y) - F(y_0)}{y - y_0} = \int_a^b \frac{f(x, y) - f(x, y_0)}{y - y_0}\, d\alpha(x) = \int_a^b D_2f(x, \bar{y})\, d\alpha(x),$$

where $\bar{y}$ is between y and y_0 Since D_2f is continuous on R, we obtain the conclusion by arguing as in the proof of Theorem 9–35

For Riemann integrals we have an additional theorem in which the limits of integration are allowed to depend on y

9–38 THEOREM *Let* $R = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$ *Assume that* f *and* D_2f *are continuous on* R *Let* p *and* q *be two functions having finite derivatives* p' *and* q' *on* $[c, d]$ *and assume that* $a \leq p(y) \leq b$ *and that* $a \leq q(y) \leq b$ *for each* y *in* $[c, d]$ *Define* F *by the equation*

$$F(y) = \int_{p(y)}^{q(y)} f(x, y)\, dx, \qquad \text{if } y \in [c, d]$$

Then $F'(y)$ *exists for each* y *in* (c, d) *and is given by*

$$F'(y) = \int_{p(y)}^{q(y)} D_2f(x, y)\, dx + f[q(y), y]q'(y) - f[p(y), y]p'(y)$$

Proof. Let $G(x_1, x_2, x_3) = \int_{x_1}^{x_2} f(t, x_3)\, dt$ whenever x_1 and x_2 are in $[a, b]$ and $x_3 \in [c, d]$ Then F is the composite function given by $F(y) = G[p(y), q(y), y]$ By the chain rule (Theorem 6–14), we have

$$F'(y) = D_1G[p(y), q(y), y]p'(y) + D_2G[p(y), q(y), y]q'(y) + D_3G[p(y), q(y), y]$$

Applying Theorem 9–31, we can write $D_1G(x_1, x_2, x_3) = -f(x_1, x_3)$ and $D_2G(x_1, x_2, x_3) = f(x_2, x_3)$ By Theorem 9–37, we also have

$$D_3G(x_1, x_2, x_3) = \int_{x_1}^{x_2} D_2f(t, x_3)\, dt$$

Substituting these results in the formula for $F'(y)$ yields the theorem

9–20 Interchanging the order of integration.

9–39 THEOREM. *Let $R = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$. Assume that α is of bounded variation on $[a, b]$, β is of bounded variation on $[c, d]$, and f is continuous on R. If $(x, y) \in R$, define*

$$F(y) = \int_a^b f(x, y)\, d\alpha(x), \qquad G(x) = \int_c^d f(x, y)\, d\beta(y).$$

Then $F \in R(\beta)$ on $[c, d]$, $G \in R(\alpha)$ on $[a, b]$, and we have

$$\int_c^d F(y)\, d\beta(y) = \int_a^b G(x)\, d\alpha(x).$$

In other words, we may interchange the order of integration as follows:

$$\int_a^b \left[\int_c^d f(x, y)\, d\beta(y)\right] d\alpha(x) = \int_c^d \left[\int_a^b f(x, y)\, d\alpha(x)\right] d\beta(y).$$

NOTE. If $\alpha(x) = \int_a^x g(u)\, du$ and $\beta(y) = \int_c^y h(v)\, dv$, where $g \in R$ on $[a, b]$ and $h \in R$ on $[c, d]$, Theorems 9–39 and 9–36 yield the following result for Riemann integrals:

$$\int_a^b \left[\int_c^d g(x)h(y)f(x, y)\, dy\right] dx = \int_c^d \left[\int_a^b g(x)h(y)f(x, y)\, dx\right] dy.$$

Proof. By Theorem 9–35, F is continuous on $[c, d]$ and hence $F \in R(\beta)$ on $[c, d]$. Similarly, $G \in R(\alpha)$ on $[a, b]$. To prove the equality of the two integrals, it suffices to consider the case in which $\alpha \nearrow$ on $[a, b]$ and $\beta \nearrow$ on $[c, d]$.

By uniform continuity, given $\epsilon > 0$ there is a $\delta > 0$ such that for every pair of points $z = (x, y)$ and $z' = (x', y')$ in R, with $|z - z'| < \delta$, we have $|f(x, y) - f(x', y')| < \epsilon$. Let us now subdivide R into n^2 equal rectangles by subdividing $[a, b]$ and $[c, d]$ each into n equal parts, where n is chosen so that $(b - a)/n < \delta/\sqrt{2}$ and $(d - c)/n < \delta/\sqrt{2}$. Writing

$$x_k = a + \frac{k(b - a)}{n} \quad \text{and} \quad y_k = c + \frac{k(d - c)}{n},$$

for $k = 0, 1, 2, \ldots, n$, we have

$$\int_a^b \left(\int_c^d f(x, y)\, d\beta(y) \right) d\alpha(x)$$
$$= \sum_{k=0}^{n-1} \sum_{j=0}^{n-1} \int_{x_k}^{x_{k+1}} \left(\int_{y_j}^{y_{j+1}} f(x, y)\, d\beta(y) \right) d\alpha(x)$$

Applying Theorem 9-29 twice on the right, the double sum becomes

$$\sum_{k=0}^{n-1} \sum_{j=0}^{n-1} f(x'_k, y'_j)[\beta(y_{j+1}) - \beta(y_j)][\alpha(x_{k+1}) - \alpha(x_k)],$$

where (x'_k, y'_j) is in the rectangle R_{kj} having (x_k, y_j) and (x_{k+1}, y_{j+1}) as opposite vertices Similarly, we find

$$\int_c^d \left(\int_a^b f(x, y)\, d\alpha(x) \right) d\beta(y)$$
$$= \sum_{k=0}^{n-1} \sum_{j=0}^{n-1} f(x''_k, y''_j)[\beta(y_{j+1}) - \beta(y_j)][\alpha(x_{k+1}) - \alpha(x_k)],$$

where $(x''_k, y''_j) \in R_{kj}$ But $|f(x'_k, y'_j) - f(x''_k, y''_j)| < \epsilon$ and hence

$$\left| \int_a^b G(x)\, d\alpha(x) - \int_c^d F(y)\, d\beta(y) \right|$$
$$< \epsilon \sum_{j=0}^{n-1} [\beta(y_{j+1}) - \beta(y_j)] \sum_{k=0}^{n-1} [\alpha(x_{k+1}) - \alpha(x_k)]$$
$$= \epsilon[\beta(d) - \beta(c)][\alpha(b) - \alpha(a)]$$

Since ϵ is arbitrary, this implies equality of the two integrals

9-21 Oscillation of a function. Every continuous function is Riemann-integrable. However, continuity is certainly not necessary, for we have seen that $f \in R$ when f is of bounded variation on $[a, b]$ In particular, f can be a monotonic function having a countable set of discontinuities and yet $\int_a^b f(x)\, dx$ will exist We shall see later in the Chapter (Exercise 9-32) that there are Riemann-integrable functions whose discontinuities form a noncountable set Therefore it is natural to ask "how many" discontinuities a function can have and still be Riemann-integrable A study of this interesting question will lead us to some very important necessary and sufficient conditions for the existence of the Riemann integral.

We have seen that $f \in R$ if, and only if, Riemann's condition is satisfied. If we examine the statement of this condition, we can get a rough idea of the kind of restrictions it puts on the set of discontinuities of f. The difference between the upper and lower Riemann sums is given by

$$\sum_{k=1}^{n} [M_k(f) - m_k(f)]\,\Delta x_k$$

and, roughly speaking, f will be integrable if, and only if, this sum becomes and remains "small." Let us split this sum into two parts, say $S_1 + S_2$, where S_1 comes from subintervals containing only points of continuity of f, and S_2 contains the remaining terms. In S_1, each difference $M_k(f) - m_k(f)$ is small because of continuity and hence a large number of such terms can occur and still keep S_1 small. In S_2, however, the differences $M_k(f) - m_k(f)$ need not be small; but because they are bounded (say by M), we have $|S_2| \le M \sum \Delta x_k$, so that S_2 will be small if the sum of the lengths of the subintervals corresponding to S_2 is small. Hence we may expect that the set of discontinuities of an integrable function can be covered by intervals whose total length is small. We proceed now to a precise discussion of this question.

We must first explain what is meant by the statement that

$$M_k(f) - m_k(f)$$

is "small." For this purpose we introduce the concept of oscillation.

9–40 DEFINITION. *Let f be defined and bounded on a closed interval S. Let T be a subset of S. The number*

$$\Omega_f(T) = \sup \{f(x) - f(y) \mid x \in T, y \in T\}$$

is called the oscillation of f on the set T. If $x \in S$, the oscillation of f at x is defined to be the number

$$\omega_f(x) = \lim_{h \to 0+} \Omega_f(N(x; h) \cap S).$$

NOTE. This limit always exists, since $\Omega_f(N(x; h) \cap S)$ is a monotonic function of h. In fact, $T_1 \subset T_2$ implies $\Omega_f(T_1) \le \Omega_f(T_2)$.

The notion of oscillation provides a particularly neat characterization of continuity which can be stated as follows:

9-41 THEOREM *Let f be defined and bounded on a closed interval S Then f is continuous at a point x of S if, and only if, $\omega_f(x) = 0$*

The proof of this theorem is left as an exercise for the reader (See Exercise 9-24)

The next theorem tells us that if $\omega_f(x) < \epsilon$ at each point of a closed interval $[a, b]$, then $\Omega_f(T) < \epsilon$ for all sufficiently small subintervals T

9-42 THEOREM *Let f be defined and bounded on $[a, b]$, and let $\epsilon > 0$ be given. Assume that $\omega_f(x) < \epsilon$ for every x in $[a, b]$ Then there exists a $\delta > 0$ (depending only on ϵ) such that for every closed subinterval $T \subset [a, b]$, we have $\Omega_f(T) < \epsilon$ whenever the length of T is less than δ*

Proof. For each x in $[a, b]$ there exists a neighborhood $N_x = N(x; \delta_x)$ such that

$$\Omega_f(N_x \cap [a, b]) < \omega_f(x) + [\epsilon - \omega_f(x)] = \epsilon$$

The set of all halfsize neighborhoods $N(x, \delta_x/2)$ forms an open covering of $[a, b]$ By the Heine-Borel theorem, a finite number (say k) of these cover $[a, b]$ Let their radii be $\delta_1/2, \ldots, \delta_k/2$ and let δ be the smallest of these k numbers When the interval T has length $< \delta$, then T is partly covered by at least one of these neighborhoods, say by $N(x_p; \delta_p/2)$ However, the neighborhood $N(x_p; \delta_p)$ completely covers T (since $\delta_p > 2\delta$). Moreover, in $N(x_p; \delta_p) \cap [a, b]$ the oscillation of f is less than ϵ This implies that $\Omega_f(T) < \epsilon$ and the theorem is proved

9-43 THEOREM. *Let f be defined and bounded on $[a, b]$ For each $\epsilon > 0$ define the set J_ϵ as follows:*

$$J_\epsilon = \{x \mid x \in [a, b], \omega_f(x) \geq \epsilon\}$$

Then J_ϵ is a closed set.

Proof Let x be an accumulation point of J_ϵ If $x \notin J_\epsilon$, we have $\omega_f(x) < \epsilon$. Hence there is a neighborhood $N(x)$ such that

$$\Omega_f(N(x) \cap [a, b]) < \epsilon$$

Thus no points of $N(x)$ can belong to J_ϵ, contradicting the statement that x is an accumulation point of J_ϵ Therefore, $x \in J_\epsilon$ and J_ϵ is closed

9-22 Jordan content of bounded sets in E_1. As stated earlier, it seems reasonable to expect that the set of discontinuities of an integrable function can be covered by intervals whose total length is small In an attempt to measure that portion of an interval which is occupied by the discontinuities of a function, Hankel (1882) introduced the notion of *content* of a set. Cantor and Stolz (1884) extended Hankel's ideas to subsets of E_n and in so doing laid the foundations for a branch of mathematics known as the *theory of content.* The theory, as developed by Hankel, Cantor, and Stolz, had certain defects which appeared when the sets in question were not closed. In an effort to remove these defects, Peano (1887) and Jordan (1892) introduced the notions of *inner* and *outer* content.

The theory of content was later extended by Borel (1898) and Lebesgue (1902). Their work led to a more general and more satisfactory theory, now known as the *theory of Lebesgue measure.* A detailed study of Lebesgue measure will not be attempted in this book (several excellent accounts of the subject are listed in the bibliography at the end of the chapter); we propose to discuss just enough of the theory of content and measure to show the usefulness of these ideas as they occur in the theory of Riemann integration.

Let S be a subset of an interval $[a, b]$, and let P be a partition of $[a, b]$. Among the various subintervals of P we fix our attention on two types: (1) those which contain only interior points of S, and (2) those which contain points of S and boundary points of S. (Recall Definition 8–38.) The intervals of type (1) give us an approximation to S from the "inside" and those of type (2) an approximation from the "outside."

9–44 DEFINITION. *Let $S \subset [a, b]$ and let P be a partition of $[a, b]$. Define $\underline{J}(P, S)$ to be the sum of the lengths of those subintervals of P which contain only interior points of S. Similarly, let $\bar{J}(P, S)$ denote the sum of the lengths of those subintervals of P which contain points of $S \cup \partial S$. The numbers*

$$\underline{c}(S) = \sup \{\underline{J}(P, S) \mid P \in \mathcal{P}[a, b]\},$$

$$\bar{c}(S) = \inf \{\bar{J}(P, S) \mid P \in \mathcal{P}[a, b]\}$$

are called, respectively, the inner and outer Jordan content of S. The set S is said to be Jordan-measurable if $\underline{c}(S) = \bar{c}(S)$, in which case this common value is called the Jordan content of S and is denoted by $c(S)$.

NOTE. It is clear that the numbers $\underline{c}(S)$ and $\bar{c}(S)$ depend only on the set S and not on the interval $[a, b]$ which contains S.

We always have $0 \leq \underline{c}(S) \leq \bar{c}(S)$. Moreover, when S is a finite set, then $\underline{c}(S) = \bar{c}(S) = 0$. In fact, a set S with no interior points has $\underline{c}(S) = 0$ since each sum $\underline{J}(P, S) = 0$. In particular, the set of rational numbers in any interval $[a, b]$ has inner Jordan content zero, as does the corresponding set of irrational numbers. However, the same sets each have outer Jordan content $b - a$. In fact, for each of these sets we have $\bar{J}(P, S) = b - a$

for all P, since every subinterval contains both rational and irrational points

The length of the interval $[a, b]$ is $b - a$ This is also the content of $[a, b]$ If the interval is decomposed into a union of two disjoint subintervals, the length of their union is the sum of the lengths of the component parts This "additive" property does not hold for either inner or outer Jordan content, as we see by writing $[a, b] = A \cup B$, where A is the set of rational numbers and B the set of irrational numbers in $[a, b]$ In fact, we have $\bar{c}(A) = \bar{c}(B) = \bar{c}(A \cup B) = b - a$ and $\underline{c}(A) = \underline{c}(B) = 0$, $\underline{c}(A \cup B) = b - a$ However, content *is* additive, that is, $c(A \cup B) = c(A) + c(B)$ if $A \cap B$ is empty, provided that both sets A and B are Jordan-measurable (See Exercise 9-25)

A connection can be established immediately between Jordan content and Riemann integration The connecting link is the *characteristic function* of a set

9-45 Definition *Given an arbitrary set S in E_n The function f_S defined as follows*

$$f_S(x) = 1 \quad \text{if } x \in S, \qquad f_S(x) = 0 \quad \text{if } x \in E_n - S,$$

is called the characteristic function of S

9-46 Theorem *Let $S \subset [a, b]$ and let f_S be the characteristic function of S Then we have*

$$\underline{c}(S) = \underline{\int_a^b} f_S(x)\,dx \qquad \text{and} \qquad \bar{c}(S) = \overline{\int_a^b} f_S(x)\,dx$$

A proof of this theorem may be found in Reference 9-3, pp 62-64 Theorem 9-46 will not be needed in any of the subsequent work in this book

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9-23 A necessary and sufficient condition for integrability in terms of content.

9-47 Theorem *Let f be defined and bounded on $[a, b]$ For each $\epsilon > 0$ define the set J_ϵ as follows*

$$J_\epsilon = \{x \mid x \in [a, b],\ \omega_f(x) \geq \epsilon\}$$

Then $f \in R$ on $[a, b]$ if, and only if, $c(J_\epsilon) = 0$ for every $\epsilon > 0$

Proof. Assume that $\bar{c}(J_\epsilon) \neq 0$ for some $\epsilon > 0$. We will show that Riemann's condition cannot hold. For every partition P of $[a, b]$ we have

$$U(P, f) - L(P, f) = \sum_{k=1}^{n} [M_k(f) - m_k(f)]\, \Delta x_k = S_1 + S_2 \geq S_1,$$

where S_1 contains those terms coming from subintervals containing points of J_ϵ, and S_2 contains the remaining terms. The intervals in S_1 have total length $\bar{J}(P, J_\epsilon) \geq \bar{c}(J_\epsilon)$. Moreover, in these intervals we have

$$M_k(f) - m_k(f) \geq \epsilon \qquad \text{so that} \qquad S_1 \geq \epsilon\, \bar{c}(J_\epsilon).$$

This means that

$$U(P, f) - L(P, f) \geq \epsilon\, \bar{c}(J_\epsilon)$$

and Riemann's condition cannot be satisfied. Hence, $f \in R$ implies $\bar{c}(J_\epsilon) = 0$.

Conversely, let $\epsilon > 0$ be given and assume that $\bar{c}(J_\epsilon) = 0$. Then there exists a partition P_0 of $[a, b]$ such that $\bar{J}(P_0, J_\epsilon) < \epsilon$. In each of the subintervals I of P_0 containing no points of J_ϵ, we have $\omega_f(x) < \epsilon$ if $x \in I$. Hence, by Theorem 9–42, there exists a $\delta > 0$ (depending only on ϵ) such that every such subinterval I can be further subdivided into subintervals T of length $< \delta$ in which $\Omega_f(T) < \epsilon$. Let P_ϵ denote the new partition of $[a, b]$ so obtained. If P is finer than P_ϵ, we can write

$$U(P, f) - L(P, f) = \sum_{k=1}^{n} [M_k(f) - m_k(f)]\, \Delta x_k = A_1 + A_2,$$

where A_1 contains those terms coming from subintervals containing points of J_ϵ, and A_2 contains the remaining terms. In the kth term of A_2 we have $M_k(f) - m_k(f) < \epsilon$ and hence $A_2 < \epsilon(b - a)$. For A_1 we have the estimate $A_1 \leq (M - m)\bar{J}(P_0, J_\epsilon) < \epsilon(M - m)$, where m and M are the inf and sup of f on $[a, b]$. Therefore,

$$U(P, f) - L(P, f) < \epsilon[M - m + b - a].$$

Since this inequality is valid for every $\epsilon > 0$, we see that Riemann's condition holds.

Although Theorem 9–47 gives us a necessary and sufficient condition for determining whether or not a bounded function is Riemann integrable, the condition is not too useful in practice. Its chief disadvantage is that one must compute $c(J_\epsilon)$ for infinitely many sets J_ϵ whose structure may be difficult to determine from the description of the function A much more useful necessary and sufficient condition was discovered by Lebesgue. In order to formulate Lebesgue's condition, we must introduce outer Lebesgue measure

9-24 Outer Lebesgue measure of subsets of E_1.

9-48 DEFINITION *Let $S \subset E_1$ By a Lebesgue covering of S we mean a countable collection $T = \{I_1, I_2, \ldots\}$ of open intervals which covers S. If $L(I_k)$ is the length of I_k, the length of the covering, $L(T)$, is defined to be the number*

$$L(T) = \sum_{k=1}^{\infty} L(I_k)$$

whenever the series on the right converges We set $L(T) = +\infty$ in case the series diverges The number

$$\overline{m}(S) = \inf \{L(T) \mid T \text{ is a Lebesgue covering of } S\}$$

is called the outer Lebesgue measure of S

NOTE. When S is bounded, say $S \subset [a, b]$, then $0 \leq \overline{m}(S) \leq b - a$. If $\overline{m}(S) = 0$, S is said to be a set of measure zero. (The notion of *inner* Lebesgue measure will not be developed here since we shall be principally interested in sets of measure zero, and, for such sets, the inner and outer Lebesgue measures are both zero. See Ref 9-1 for a discussion of inner Lebesgue measure) It is clear from the definition that $A \subset B$ implies $\overline{m}(A) \leq \overline{m}(B)$

The relationship between outer Lebesgue measure and outer Jordan content is illustrated in the next three theorems.

9-49 THEOREM *For every bounded set S in E_1 we have $\overline{m}(S) \leq \bar{c}(S)$.*

Proof Let $S \subset [a, b]$ and let P be a partition of $[a, b]$ Let the subintervals of P which contain points of $S \cup \partial S$ be $A_1, \ldots, A_m$, where $A_k = [a_k, b_k]$ Then

$$\bar{J}(P, S) = \sum_{k=1}^{m} L(A_k)$$

If $\epsilon > 0$ is given, the open intervals $B_k = (a_k - \epsilon/2m, b_k + \epsilon/2m)$ form a Lebesgue covering of S for which

$$\sum_{k=1}^{m} L(B_k) = \bar{J}(P, S) + \epsilon.$$

Hence, $\overline{m}(S) \leq \bar{J}(P, S) + \epsilon$, or $\overline{m}(S) - \epsilon \leq \bar{J}(P, S)$ for all P. Thus $\overline{m}(S) - \epsilon \leq \bar{c}(S)$ and, since ϵ is arbitrary, $\overline{m}(S) \leq \bar{c}(S)$.

9–50 THEOREM. *Let B be a bounded set in E_1 and let A be a compact subset of B. Then $\bar{c}(A) \leq \overline{m}(B)$.*

Proof. Assume that $B \subset [a, b]$ and let $\epsilon > 0$ be given. There exists a Lebesgue covering T of B such that $L(T) < \overline{m}(B) + \epsilon$. Now T is an open covering of A and, since A is compact, we can extract a finite collection T_1 of open intervals which covers A. The endpoints of these intervals determine a partition P of $[a, b]$ for which we have

$$\bar{c}(A) \leq \bar{J}(P, A) \leq L(T_1) \leq L(T) < \overline{m}(B) + \epsilon.$$

Since ϵ is arbitrary, this implies $\bar{c}(A) \leq \overline{m}(B)$.

Combining Theorems 9–49 and 9–50, we obtain

9–51 THEOREM. *If S is a compact set in E_1, then $\bar{c}(S) = \overline{m}(S)$.*

It is quite easy to prove that all countable sets are of measure zero. In fact, we will prove a stronger statement:

9–52 THEOREM. *Let F be a countable collection of sets in E_1, say*

$$F = \{F_1, F_2, \ldots\},$$

such that $\overline{m}(F_k) = 0$ for each k. Let

$$S = \bigcup_{k=1}^{\infty} F_k.$$

Then $\overline{m}(S) = 0$.

Proof. If $\epsilon > 0$ is given, then for each set F_k there exists a Lebesgue covering T_k of F_k such that $L(T_k) < \epsilon/2^k$. Since T_k covers F_k, the collection $T = \{T_1, T_2, \ldots\}$ covers S. Furthermore, T is a countable covering of S and we have

$$L(T) \leq \sum_{k=1}^{\infty} L(T_k) < \sum_{k=1}^{\infty} \frac{\epsilon}{2^k} = \epsilon.$$

This implies that $\overline{m}(S) = 0$ and the theorem is proved.

Since a set consisting of just one point has measure zero, it follows that every countable subset of E_1 has measure zero. In particular, the set of rational numbers has measure zero. However, there are *uncountable* sets which have measure zero. (See Exercise 9–32.)

9-25 A necessary and sufficient condition for integrability in terms of measure. The next theorem describes Lebesgue's useful criterion for Riemann-integrability

9-53 THEOREM *Let f be defined and bounded on $[a, b]$ and let D denote the set of discontinuities of f in $[a, b]$ Then $f \in R$ on $[a, b]$ if, and only if, D is a set of measure zero*

Proof Assume that $f \in R$ on $[a, b]$ and let $J_\epsilon = \{x \mid \omega_f(x) \geq \epsilon\}$ By Theorem 9-47, we have $\bar{c}(J_\epsilon) = 0$ for every $\epsilon > 0$ In particular, this holds for $\epsilon = 1/n$, $n = 1, 2, \ldots$. Now if $x \in D$, then $\omega_f(x) \neq 0$ and hence $\omega_f(x) > 1/n$ for *some* n Hence,

$$D \subset \bigcup_{n=1}^{\infty} J_{1/n}$$

Since $\bar{c}(J_{1/n}) = 0$, we also have $\overline{m}(J_{1/n}) = 0$ and therefore $\overline{m}(D) = 0$

Conversely, assume that $\overline{m}(D) = 0$ Choose $\epsilon > 0$ and form J_ϵ Then J_ϵ is a closed set, by Theorem 9-43 Also, $J_\epsilon \subset D$ Hence,

$$\bar{c}(J_\epsilon) \leq \overline{m}(D) = 0$$

and this proves that $c(J_\epsilon) = 0$ Therefore, $f \in R$ on $[a, b]$.

The following statements (some of which were proved earlier in the chapter) are immediate consequences of Lebesgue's theorem

9-54 THEOREM

(i) *If f is of bounded variation on $[a, b]$, then $f \in R$ on $[a, b]$*

(ii) *If $f \in R$ on $[a, b]$, then $f \in R$ on $[c, d]$ for every subinterval $[c, d] \subset [a, b]$, $|f| \in R$ and $f^2 \in R$ on $[a, b]$ Also, $f \cdot g \in R$ on $[a, b]$ whenever $g \in R$ on $[a, b]$*

(iii) *If $f \in R$ and $g \in R$ on $[a, b]$, then $f/g \in R$ on $[a, b]$ whenever g is bounded away from 0*

(iv) *If f and g are bounded functions having the same discontinuities on $[a, b]$, then $f \in R$ on $[a, b]$ if, and only if, $g \in R$ on $[a, b]$*

(v) *Let $g \in R$ on $[a, b]$ and assume that $m \leq g(x) \leq M$ for all x in $[a, b]$ If f is continuous on $[m, M]$, the composite function h defined by $h(x) = f[g(x)]$ is Riemann-integrable on $[a, b]$*

NOTE Statement (v) need not hold if we assume only that $f \in R$ on $[m, M]$ (See Exercise 9-29.)

9–26 Complex-valued Riemann-Stieltjes integrals. Riemann-Stieltjes integrals of the form $\int_a^b f\, d\alpha$, in which f and α are *complex-valued* functions defined and bounded on an interval $[a, b]$, are of fundamental importance in the theory of functions of a complex variable. They can be introduced by exactly the same definition we have used in the real case. In fact, every statement in Definition 9–1 still makes sense when f and α are complex-valued. The sums of the products $f(t_k)[\alpha(x_k) - \alpha(x_{k-1})]$ which are used to form Riemann-Stieltjes sums need only be interpreted as sums of products of complex numbers. Since complex numbers satisfy the commutative, associative, and distributive laws which hold for real numbers, it is not surprising that complex-valued integrals share many of the properties of real-valued integrals. In particular, Theorems 9–2, 9–3, 9–4, 9–6, and 9–7 (as well as their proofs) are all valid (word-for-word) when f and α are complex-valued functions. (In Theorems 9–2 and 9–3, the constants c_1 and c_2 may now be complex numbers.) In addition, we have the following theorem which, in effect, reduces the theory of complex Stieltjes integrals to the real case.

9–55 THEOREM. *Let $f = f_1 + if_2$ and $\alpha = \alpha_1 + i\alpha_2$ be complex-valued functions defined on an interval $[a, b]$. Then we have*

$$\int_a^b f\, d\alpha = \left(\int_a^b f_1\, d\alpha_1 - \int_a^b f_2\, d\alpha_2\right) + i\left(\int_a^b f_2\, d\alpha_1 + \int_a^b f_1\, d\alpha_2\right)$$

whenever all four integrals on the right exist.

The proof of Theorem 9–55 is immediate from the definition and is left to the reader.

The use of this theorem permits us to extend most of the important properties of real integrals to the complex case. For example, the connection between differentiation and integration established in Theorem 9–31 remains valid for complex integrals if we simply define such notions as continuity, differentiability, bounded variation, etc., "component-wise." Thus, we say that the complex-valued function $\alpha = \alpha_1 + i\alpha_2$ is of bounded variation on $[a, b]$ if each component α_1 and α_2 is of bounded variation on $[a, b]$. Similarly, the derivative $\alpha'(t)$ is defined by the equation $\alpha'(t) = \alpha_1'(t) + i\alpha_2'(t)$ whenever the derivatives $\alpha_1'(t)$ and $\alpha_2'(t)$ exist. (One-sided derivatives are defined in the same way.) It is easy to verify that all the usual rules for calculating with derivatives (for example, the chain rule) are still valid for complex-valued functions of this type. With this understanding, Theorems 9–31 and 9–32 (the fundamental theorem of integral calculus) both remain valid when f and α are complex-valued. The proofs follow from the real case by using Theorem 9–55 in a straightforward manner.

9-27 Contour integrals. In the study of functions of a complex variable, complex Riemann-Stieltjes integrals occur in connection with the theory of *contour integrals* (also known as *complex line integrals*). These can be defined as follows.

9-56 DEFINITION. *Let Γ be a curve in E_2 described by a complex-valued function z, say*

$$z(t) = x(t) + iy(t), \qquad \text{if } a \leq t \leq b,$$

where x and y are real-valued continuous functions defined on $[a, b]$. Let $\Gamma[z]$ denote the corresponding path along Γ. (See Definition 8-22.) If F is a complex-valued function defined on Γ, the contour integral of F along $\Gamma[z]$, denoted by $\int_{\Gamma[z]} F(z)\,dz$, is defined by the equation

$$\int_{\Gamma[z]} F(z)\,dz = \int_a^b F[z(t)]\,dz(t),$$

whenever the complex Riemann-Stieltjes integral on the right exists.

NOTE. If $A = z(a)$ and $B = z(b)$, we also write $\int_{\Gamma(A,B)}$ instead of $\int_{\Gamma[z]}$.

The effect of replacing the function z by an equivalent function is, at worst, a change in sign. In fact, we have

9-57 THEOREM. *Let z and w be two equivalent continuous functions describing the same curve Γ in E_2. If $\int_{\Gamma[z]} F(z)\,dz$ exists, then $\int_{\Gamma[w]} F(w)\,dw$ exists and we have*

$$\int_{\Gamma[z]} F(z)\,dz = \pm \int_{\Gamma[w]} F(w)\,dw,$$

where the plus sign holds if $\Gamma[w] = \Gamma[z]$ and the minus sign holds if $\Gamma[w] = -\Gamma[z]$.

The proof of this theorem follows easily from the formula for changing the variable in a Riemann-Stieltjes integral (Theorem 9-7). The details are left to the reader. (See Exercise 9-35.)

Contour integrals have an additive property analogous to Theorem 9-4. Before stating this property, we shall define "addition" of paths.

9-58 DEFINITION. *Let Γ be a curve in E_n described by a vector-valued function α defined on $[a, b]$, and let Γ_1 and Γ_2 be the images of the subintervals $[a, c]$ and $[c, b]$, respectively, under α, where $a < c < b$. We write*

$\Gamma = \Gamma_1 + \Gamma_2$ and refer to Γ as the sum of Γ_1 and Γ_2. Similarly, we say the path $\Gamma[\alpha]$ is the sum of the paths $\Gamma_1[\alpha]$ and $\Gamma_2[\alpha]$ and we write $\Gamma[\alpha] = \Gamma_1[\alpha] + \Gamma_2[\alpha]$.

NOTE. This addition is defined only when the endpoints are suitably matched. The sum is associative but not-commutative.

Using this concept, we see that Theorem 9–4 implies the following theorem:

9–59 THEOREM. *Let Γ be a curve in E_2 described by a continuous complex-valued function z and suppose $\Gamma[z] = \Gamma_1[z] + \Gamma_2[z]$. If two of the three contour integrals in* (3) *exist, the third also exists and we have*

$$\int_{\Gamma[z]} F(z)\,dz = \int_{\Gamma_1[z]} F(z)\,dz + \int_{\Gamma_2[z]} F(z)\,dz. \tag{3}$$

In practice, nearly all curves used as paths of integration are *rectifiable* curves. For such curves the following theorem is often quite useful for estimating the absolute value of a contour integral.

9–60 THEOREM. *If Γ is a rectifiable curve of length $\Lambda(\Gamma)$ described by a complex-valued continuous function z, and if $\int_{\Gamma[z]} F(z)\,dz$ exists, then we have the inequality*

$$\left|\int_{\Gamma[z]} F(z)\,dz\right| \le M\Lambda(\Gamma),$$

where M is such that $|F(z)| \le M$ for all z on Γ.

Proof. We simply observe that all Riemann-Stieltjes sums which occur in the definition of $\int_a^b F[z(t)]\,dz(t)$ have absolute value not exceeding $M\Lambda(\Gamma)$.

NOTE. *If* Γ is a rectifiable curve, a sufficient condition for the existence of the contour integral $\int_{\Gamma[z]} F(z)\,dz$ is that F be continuous on Γ. This follows directly from Theorems 9–26 and 9–55.

An important class of rectifiable curves are those known as *piece-wise smooth* curves. They are defined as follows:

9–61 DEFINITION. *A curve Γ in E_n is called piece-wise smooth if it can be described by a vector-valued function $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)$, defined and continuous on $[a, b]$, such that each component α_k has a bounded de-*

rivative α_k' which is continuous everywhere in $[a, b]$, except (possibly) at a finite number of points. At these exceptional points it is required that both right- and left-hand derivatives exist. In such cases the function α is also called piece-wise smooth.

NOTE. By Theorems 8-25 and 8-6, it follows that every piece-wise smooth curve is rectifiable.

If Γ is a plane curve described by a complex-valued function z, where $z(t) = x(t) + iy(t)$, then at those points t in $[a, b]$ where the derivative $z'(t)$ exists and is different from 0, the curve Γ will have a tangent line at $z(t)$ with slope $y'(t)/x'(t)$. At points where $z'(t) = 0$, we must have $x'(t) = y'(t) = 0$. (At such points, called *singular points* of Γ, there may or may not be a tangent line.) Of course, at points where $z'(t)$ does not exist, there is no tangent line. A piece-wise smooth plane curve will have a tangent line at all but a finite number of its points. (See Fig. 9-3.)

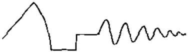

FIG 9-3 A piece-wise smooth curve in E_2

Contour integrals taken over piece-wise smooth curves can be expressed in terms of Riemann integrals. In fact, we have

9-62 THEOREM. *Let Γ be a curve in E_2 described by a piece-wise smooth function z defined on $[a, b]$ and assume that the contour integral of F along $\Gamma[z]$ exists. Then we have*

$$\int_{\Gamma[z]} F(z)\, dz = \int_a^b F[z(t)]z'(t)\, dt.$$

The proof is an easy consequence of Theorems 9-55 and 9-8. Details are left to the reader.

Polygonal curves are examples of piece-wise smooth curves. Therefore every *rectifiable* curve Γ can be approximated by piece-wise smooth curves, namely, the inscribed polygons which are used to find the arc length of Γ. (See Definition 8-24.) The line integral of a continuous function F taken along one of these polygons will serve as an approximation to the line integral of F along Γ. The precise statement of this fact is given in the next theorem.

9-63 THEOREM. *Let Γ be a rectifiable curve described by a complex-valued function z defined on an interval $[a, b]$. Suppose F is defined and con-*

tinuous on some open region S which contains Γ. Then, for every $\epsilon > 0$, there exists a partition P of $[a, b]$ and a corresponding inscribed polygon $\pi = \pi(P) \subset S$ such that

$$\left| \int_{\Gamma[z]} F(z)\, dz - \int_{\pi[w]} F(w)\, dw \right| < \epsilon.$$

Proof. Arguing as in the proof of Theorem 8–33, we can find *one* radius $r > 0$ such that for every point z on Γ we have $N(z; r) \subset S$. Let T denote the union of all closed spheres $\overline{N}(z; r/2)$, where $z \in \Gamma$. Then T is a compact subset of S and hence F is uniformly continuous on T. Therefore, given $\epsilon > 0$, there is a $\delta > 0$, $0 < \delta < r/2$, such that

$$|F(z) - F(z')| < \frac{\epsilon}{2\Lambda(\Gamma)}, \qquad \text{if } z \in T,\ z' \in T, \qquad \text{and} \qquad |z - z'| < \delta,$$

where $\Lambda(\Gamma)$ is the length of Γ. For this δ we can choose δ_1 so that

$$|z(t) - z(t')| < \delta$$

whenever t and t' are points of $[a, b]$ with $|t - t'| < \delta_1$. For this same ϵ, choose a partition P_ϵ of $[a, b]$ such that

$$\left| \int_a^b F[z(t)]\, dz(t) - S(P_\epsilon, Fz, z) \right| < \frac{\epsilon}{2}, \tag{4}$$

where $S(P_\epsilon, Fz, z)$ is a corresponding Riemann-Stieltjes sum. We can assume that the norm of P_ϵ is $< \delta_1$. If $P_\epsilon = \{t_0, t_1, \ldots, t_m\}$, the corresponding inscribed polygon $\pi = \pi(P_\epsilon)$ lies in T. The integral of F along π exists and we can write

$$\int_{\pi[w]} F(w)\, dw = \sum_{k=1}^{m} \int_{t_{k-1}}^{t_k} F[w(t)]\, dw(t),$$

where w denotes the function which describes π. We can assume that w is chosen so that $w(t_k) = z(t_k)$ for each $k = 0, 1, 2, \ldots, m$. Then we have $z(t_k) - z(t_{k-1}) = \int_{t_{k-1}}^{t_k} dw(t)$ and we can write the Riemann-Stieltjes sum $S(P_\epsilon, Fz, z)$ as follows:

$$S(P_\epsilon, Fz, z) = \sum_{k=1}^{m} F[z(v_k)][z(t_k) - z(t_{k-1})] = \sum_{k=1}^{m} F[z(v_k)] \int_{t_{k-1}}^{t_k} dw(t)$$

$$= \sum_{k=1}^{m} \int_{t_{k-1}}^{t_k} F[z(v_k)]\, dw(t),$$

where $v_k \in [t_{k-1}, t_k]$ Therefore,

$$\left| \int_{\gamma[w]} F(w)\,dw - S(P_\epsilon, Fz, z) \right|$$

$$= \left| \sum_{k=1}^{m} \int_{t_{k-1}}^{t_k} \{F[w(t)] - F[z(v_k)]\}\,dw(t) \right|$$

$$\leq \sum_{k=1}^{m} \left| \int_{t_{k-1}}^{t_k} \{F[w(t)] - F[z(v_k)]\}\,dw(t) \right|.$$

By Theorem 9-60, the last integral has absolute value not exceeding

$$\frac{\epsilon}{2\Lambda(\Gamma)} |z(t_k) - z(t_{k-1})|$$

and hence

$$\left| \int_{z[w]} F(w)\,dw - S(P_\epsilon, Fz, z) \right| \leq \sum_{k=1}^{m} \frac{\epsilon}{2\Lambda(\Gamma)} |z(t_k) - z(t_{k-1})|$$

$$= \frac{\epsilon|\pi(P_\epsilon)|}{2\Lambda(\Gamma)} \leq \frac{\epsilon}{2}.$$

If we combine this with (4), the theorem follows

9-28 The winding number. If Γ is a rectifiable closed curve and if z_0 is a point not on Γ, the contour integral

$$\int_{\Gamma[z]} \frac{1}{z - z_0}\,dz$$

always exists (since the integrand is continuous on Γ). This integral is of fundamental importance in the theory of functions of a complex variable. It can be employed to introduce an exceedingly useful concept known as the *winding number*

9-64 DEFINITION *If Γ is a rectifiable closed curve described by a complex-valued function z, and if z_0 is a point not on Γ, we define $W(\Gamma[z]; z_0)$, the winding number of the path $\Gamma[z]$ with respect to z_0, by the equation*

$$W(\Gamma[z]; z_0) = \frac{1}{2\pi i} \int_{\Gamma[z]} \frac{1}{z - z_0}\,dz.$$

NOTE The term "index" is sometimes used instead of "winding number."

The relations

$$W(-\Gamma[z]; z_0) = -W(\Gamma[z]; z_0)$$

and

$$W(\Gamma_1[z] + \Gamma_2[z]; z_0) = W(\Gamma_1[z]; z_0) + W(\Gamma_2[z]; z_0)$$

follow immediately from Theorems 9–57 and 9–59, respectively.

The term "winding number" is used because the value of this integral turns out to be an *integer* which can be used to give a mathematically precise way of counting the number of times the path $\Gamma[z]$ "winds around" the point z_0. For example, let us evaluate this integral when $\Gamma[z]$ is a positively oriented circle with center at z_0. (See Definition 8–23.)

9–65 THEOREM. *If $\Gamma[z]$ is a positively oriented circle with center at z_0, the winding number $W(\Gamma[z]; z_0) = 1$.*

Proof. In this case, Γ can be described by the equation

$$z(t) = z_0 + re^{2\pi it} = z_0 + r\cos(2\pi t) + ir\sin(2\pi t), \qquad \text{if } 0 \le t \le 1.$$

Since $z'(t) = -2\pi r\sin(2\pi t) + 2\pi ir\cos(2\pi t) = 2\pi ire^{2\pi it}$, we can evaluate the integral for the winding number by Theorem 9–62 to obtain

$$\frac{1}{2\pi i}\int_{\Gamma[z]} \frac{1}{z - z_0}\, dz = \frac{1}{2\pi ir}\int_0^1 e^{-2\pi it} z'(t)\, dt = \int_0^1 dt = 1.$$

This proves the theorem.

Note that the value $+1$ obtained in this case is in accord with the physical interpretation of $\Gamma[z]$ as the path traced out by a particle moving *once* around the circle in the positive direction. If the particle moves n times around the circle in the positive direction, the parametric interval becomes $0 \le t \le n$ and a calculation similar to the above shows that the winding number is n. On the other hand, if the particle moves n times around the circle in the *negative* direction, the winding number has the value $-n$.

The situation just illustrated for a circle is a special case of the following more general theorem:

9–66 THEOREM. *If Γ is a rectifiable closed curve, the winding number*

$$W(\Gamma[z]; z_0)$$

is always an integer.

Proof Because of Theorem 9–63, it suffices to prove the theorem for piece-wise smooth curves If Γ is described by a piece-wise smooth function z defined on $[a, b]$, the integral for the winding number can be expressed as a Riemann integral, by Theorem 9–62 We then define a complex-valued function F by the equation

$$F(x) = \int_a^x \frac{z'(t)}{z(t) - z_0}\,dt, \qquad \text{if } a \leq x \leq b,$$

where $z_0 \notin \Gamma$ To prove the theorem, we must show that $F(b) = 2n\pi i$, where n is an integer For this purpose we observe that F is continuous on $[a, b]$ and that (by Theorem 9–31) we have

$$F'(x) = \frac{z'(x)}{z(x) - z_0}$$

at each point of continuity of z' Next, introduce G, where

$$G(t) = e^{-F(t)}[z(t) - z_0] \qquad \text{if } t \in [a, b]$$

Then G is also continuous throughout $[a, b]$ Moreover, at each point t in $[a, b]$ where z' is continuous we have

$$G'(t) = e^{-F(t)}z'(t) - F'(t)e^{-F(t)}[z(t) - z_0] = 0$$

That is, $G'(t) = 0$ for each t in $[a, b]$, except (possibly) for a finite number of points In each subinterval (t_{k-1}, t_k) in which z' is continuous, we can apply Theorem 9–32 to write

$$G(x) - G(t_{k-1}) = \int_{t_{k-1}}^x G'(t)\,dt = 0$$

From this it follows that G is constant on each subinterval (t_{k-1}, t_k) and hence (by continuity) G is constant throughout $[a, b]$ That is, $G(t) = G(a)$ for all t in $[a, b]$ Therefore, the equation

$$e^{-F(t)}[z(t) - z_0] = z(a) - z_0$$

is valid for each t in $[a, b]$ If we substitute $t = b$ and use the fact that $z(b) = z(a)$, we find

$$e^{-F(b)} = 1,$$

which implies $F(b) = 2n\pi i$, where n is an integer. This completes the proof

9–67 THEOREM. *Let Γ be a rectifiable Jordan curve in E_2 described by a continuous function z defined on $[a, b]$. Then the winding number $W(\Gamma[z]; z_0) = 0$ for each point z_0 outside Γ.*

Proof. Define a function G on $E_2 - \Gamma$ by the equation

$$G(z_0) = W(\Gamma[z]; z_0) \qquad \text{if } z_0 \in E_2 - \Gamma.$$

By an argument similar to that given in the proof of Theorem 9–35, it is easy to see that G is continuous on $E_2 - \Gamma$. But, since $G(z_0)$ is always an *integer*, it follows from Theorem 8–37 (the intermediate value theorem) that G is constant on each component of $E_2 - \Gamma$ and, in particular, G is constant on the unbounded component of $E_2 - \Gamma$. It is not difficult to prove that G is identically zero on this component. In fact, let K be a constant such that $|z(t)| < K$ for all t in $[a, b]$ and let z_1 be a point in the unbounded component of $E_2 - \Gamma$ such that $|z_1| > K + \Lambda(\Gamma)$ where $\Lambda(\Gamma)$ is the length of Γ. Then we have

$$\left|\frac{1}{z(t) - z_1}\right| \leq \frac{1}{|z_1| - |z(t)|} < \frac{1}{|z_1| - K}$$

and hence, by Theorem 9–60, we can write

$$0 \leq |G(z_1)| \leq \frac{\Lambda(\Gamma)}{|z_1| - K} < 1.$$

Since $G(z_1)$ is an integer, it follows that $G(z_1) = 0$ and this proves that G has the constant value 0 everywhere on the unbounded component of $E_2 - \Gamma$.

We have already seen that if $\Gamma[z]$ is an oriented circle with center at z_0, the winding number $W(\Gamma[z]; z_0)$ is ± 1. The situation is similar for every rectifiable Jordan curve in E_2.

9–68 THEOREM. *Let Γ be a rectifiable Jordan curve in E_2 described by a continuous function z. Then the winding number $W(\Gamma[z]; z_0)$ is either $+1$ for every z_0 inside Γ or else it is -1 for every z_0 inside Γ.*

Proof. We observed in the proof of Theorem 9–67 that the winding number must be constant on the inner region of Γ, and hence we need only show that it is ± 1 for a particular point inside Γ. Let z_0 be inside Γ and let C be a circle with center at z_0 lying entirely inside Γ. Choose two points a and b on C and construct the Jordan curves G_1 and G_2 as described in Theorem 8–43. In terms of paths we can write

$$G_1(a, a) = P(a, \alpha) + \Gamma_1(\alpha, \beta) + Q(\beta, b) + C_1(b, a)$$

and

$$G_2(a, a) = P(a, \alpha) + \Gamma_2(\alpha, \beta) + Q(\beta, b) + C_2(b, a),$$

where $C_1 + C_2 = C$, $\Gamma_1 + \Gamma_2 = \Gamma$, and P and Q are polygonal curves (using the notation of Theorem 8–43). By Theorem 8–43, we can assume that z_0 is outside *both* G_1 and G_2 and hence

$$\int_{G_1(a, a)} \frac{1}{z - z_0}\, dz - \int_{G_2(a, a)} \frac{1}{z - z_0}\, dz = 0,$$

since each integral separately is 0. But this means that we have

$$(5) \qquad \int_{P(a, \alpha)} + \int_{\Gamma_1(\alpha, \beta)} + \int_{Q(\beta, b)} + \int_{C_1(b, a)} + \int_{C_2(a, b)} + \int_{Q(b, \beta)} + \int_{\Gamma_2(\beta, \alpha)} + \int_{P(\alpha, a)} = 0$$

(The integrand in each case is $(z - z_0)^{-1}$.) The integrals over the polygonal curves cancel and equation (5) becomes

$$(6) \qquad \left(\frac{1}{2\pi i}\int_{\Gamma_1(\alpha, \beta)} + \frac{1}{2\pi i}\int_{\Gamma_2(\beta, \alpha)}\right) + \left(\frac{1}{2\pi i}\int_{C_1(b, a)} + \frac{1}{2\pi i}\int_{C_2(a, b)}\right) = 0.$$

The sum of the first two integrals in (6) is $\pm W(\Gamma[z], z_0)$ and the sum of the third and fourth is ± 1. Hence, $W(\Gamma[z], z_0)$ is either $+1$ or -1 and this completes the proof.

9–29 Orientation of rectifiable Jordan curves. We have defined a positively oriented circle (Definition 8–23) by explicitly specifying the function which describes the curve. The concept of positive and negative orientations for arbitrary rectifiable Jordan curves will now be introduced by means of the winding number.

9–69 DEFINITION. *Let Γ be a rectifiable Jordan curve in E_2 described by a continuous function z. The directed path $\Gamma[z]$ is said to be positively oriented if the winding number $W(\Gamma[z], z_0)$ is $+1$ for each z_0 inside Γ, and negatively oriented if $W(\Gamma[z], z_0)$ is -1 for each z_0 inside Γ.*

Positively oriented paths are said to have a *counterclockwise direction*, whereas negatively oriented paths have a *clockwise direction*. To represent

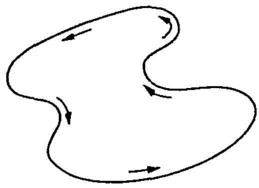

FIG. 9–4. Geometric representation of a positively oriented Jordan curve.

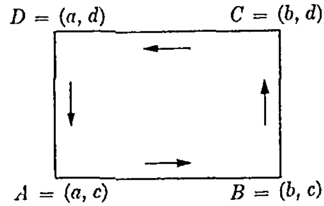

FIG. 9–5. Geometric representation of $\Gamma(R)$.

a positively oriented path geometrically we use a scheme as shown in Fig. 9–4. If one walks along the curve Γ in the direction indicated by the arrows (counterclockwise), the inner region of Γ always lies to the left.

9–70 DEFINITION. *Let $A = (a, c)$, $B = (b, c)$, $C = (b, d)$, $D = (a, d)$ be the vertices of a rectangle R in the xy-plane given by*

$$R = \{(x, y) \mid a \le x \le b,\ c \le y \le d\}.$$

The four line segments which form the boundary of R determine four directed paths $\Gamma_1(A, B)$, $\Gamma_2(B, C)$, $\Gamma_3(C, D)$, and $\Gamma_4(D, A)$ whose sum is a directed closed path which we denote by $\Gamma(R)$:

$$\Gamma(R) = \Gamma_1(A, B) + \Gamma_2(B, C) + \Gamma_3(C, D) + \Gamma_4(D, A).$$

NOTE. $\Gamma(R)$ is represented geometrically in Fig. 9–5.

9–71 THEOREM. *Let z_0 be a point interior to a rectangle R in E_2. Then the winding number of $\Gamma(R)$ with respect to z_0 is $+1$, and hence $\Gamma(R)$ is positively oriented.*

Proof. It suffices to consider the case in which z_0 is the center of the rectangle. Using Theorem 9–59 and the notation of Definition 9–70, we have

$$\int_{\Gamma(R)} = \int_{\Gamma_1(A,B)} + \int_{\Gamma_2(B,C)} + \int_{\Gamma_3(C,D)} + \int_{\Gamma_4(D,A)}$$

Now the path $\Gamma_1(A, B)$ can be described by the function z given by

$$z(t) = t + ic, \qquad \text{if } a \le t \le b,$$

and hence if we write $z_0 = x_0 + iy_0$, we obtain (by Theorem 9–62)

$$\int_{\Gamma_1(A,B)} \frac{1}{z - z_0}\,dz = \int_a^b \frac{t - x_0}{(t - x_0)^2 + (c - y_0)^2}\,dt$$

$$- i(c - y_0)\int_a^b \frac{dt}{(t - x_0)^2 + (c - y_0)^2}$$

Similarly, the path $-\Gamma_3(C, D)$ is given by $z(t) = t + id$, $a \le t \le b$, so that

$$\int_{\Gamma_3(C,D)} \frac{1}{z - z_0}\,dz = -\int_a^b \frac{t - x_0}{(t - x_0)^2 + (d - y_0)^2}\,dt$$

$$+ i(d - y_0)\int_a^b \frac{dt}{(t - x_0)^2 + (d - y_0)^2}.$$

Since z_0 is at the center of R, we have $y_0 - c = d - y_0$, and hence

$$\int_{\Gamma_1(A,B)} + \int_{\Gamma_3(C,D)} = 2i(d - y_0)\int_a^b \frac{dt}{(t - x_0)^2 + (d - y_0)^2}$$

$$= 2ih\int_{-k}^{+k} \frac{dt}{t^2 + h^2} = 4i\tan^{-1}\left(\frac{k}{h}\right),$$

where $h = d - y_0$ and $k = b - x_0 = x_0 - a$

By a similar calculation, we obtain

$$\int_{\Gamma_2(B,C)} + \int_{\Gamma_4(D,A)} = 2i(b - x_0)\int_c^d \frac{dt}{(t - y_0)^2 + (b - x_0)^2}$$

$$= 2ik\int_{-h}^{+h} \frac{dt}{t^2 + k^2} = 4i\tan^{-1}\left(\frac{h}{k}\right)$$

Hence,

$$\int_{\Gamma(R)} \frac{1}{z - z_0}\,dz = 4i\left[\tan^{-1}\left(\frac{k}{h}\right) + \tan^{-1}\left(\frac{h}{k}\right)\right] = 4i\left(\frac{\pi}{2}\right) = 2\pi i.$$

This proves, as asserted, that the winding number is $+1$

Exercises

Riemann-Stieltjes integrals.

9–1. Show that $f \in R(\alpha)$ on $[a, b]$ if, and only if, the following condition is satisfied: For every $\epsilon > 0$ there exists a partition P_ϵ such that for all partitions P' and P'' finer than P_ϵ, we have

$$|S(P', f, \alpha) - S(P'', f, \alpha)| < \epsilon.$$

[*Hint*: Let $\{\epsilon_n\}$ be a sequence of numbers such that $\epsilon_n \to 0$ as $n \to \infty$. Prove the sufficiency of the condition by constructing a corresponding sequence of partitions $\{P_{\epsilon_n}\}$ in which P_{ϵ_m} is finer than P_{ϵ_n} if $m > n$.]

9–2. Use the condition of Exercise 9–1 to prove that if $f \in R(\alpha)$ on $[a, b]$, then $f \in R(\alpha)$ on $[a, c]$ for each c in (a, b).

9–3. Prove Theorem 9–3.

9–4. The following definition of a Riemann-Stieltjes integral is often used in the literature: We say f is integrable with respect to α if there exists a real number A having the property that for every $\epsilon > 0$, there exists a $\delta > 0$ such that for every partition P of $[a, b]$ with norm $|P| < \delta$ and for every choice of t_k in $[x_{k-1}, x_k]$, we have $|S(P, f, \alpha) - A| < \epsilon$.

(a) Show that if $\int_a^b f\,d\alpha$ exists according to this definition, then it also exists according to Definition 9–1 and the two integrals are equal.

(b) Let $f(x) = \alpha(x) = 0$ for $a \le x < c$, $f(x) = \alpha(x) = 1$ for $c < x \le b$, $f(c) = 0, \alpha(c) = 1$. Show that $\int_a^b f\,d\alpha$ exists according to Definition 9–1 but does not exist by this second definition.

9–5. If $s \neq 1$, use Stieltjes integration to derive the identities (a) and (b):

(a) $$\sum_{k=1}^{n} \frac{1}{k^s} = \frac{1}{n^{s-1}} + s\int_1^n \frac{[x]}{x^{s+1}}\,dx.$$

(b) $$\sum_{k=1}^{2n} \frac{(-1)^k}{k^s} = s\int_1^{2n} \frac{2[x/2] - [x]}{x^{s+1}}\,dx.$$

(c) State and prove a theorem (analogous to Theorem 9–13) which will yield the identity (b) as a special case.

9–6. Let $\{a_n\}$ be a sequence of real numbers. For $x \ge 0$, define

$$A(x) = \sum_{n \le x} a_n = \sum_{n=1}^{[x]} a_n,$$

where $[x]$ is the greatest integer in x and empty sums are interpreted as zero Let f have a continuous derivative in the interval $1 \leq x \leq a$. Use Stieltjes integrals to derive the following formulas

(a) $$\sum_{n \leq a} a_n f(n) = -\int_1^a A(x)f'(x)\,dx + A(a)f(a)$$

(b) $$\sum_{n \leq a} a_n f(n) = \sum_{n \leq a} A(n)(f(n) - f(n+1)) + A([a])f([a]+1).$$

NOTE. When a is an integer, (b) is called *Abel's partial summation formula*

9–7. If $\alpha \nearrow$ on $[a, b]$, prove that we have

(a) $$\bar{\int_a^b} f\,d\alpha = \bar{\int_a^c} f\,d\alpha + \bar{\int_c^b} f\,d\alpha, \qquad (a < c < b),$$

(b) $$\bar{\int_a^b} (f + g)\,d\alpha \leq \bar{\int_a^b} f\,d\alpha + \bar{\int_a^b} g\,d\alpha,$$

(c) $$\underline{\int_a^b} (f + g)\,d\alpha \geq \underline{\int_a^b} f\,d\alpha + \underline{\int_a^b} g\,d\alpha$$

9–8. Complete the proof of Theorem 9–22.

9–9. Give an example of a bounded function f and an increasing function α defined on $[a, b]$ such that $|f| \in R(\alpha)$ but for which $\int_a^b f\,d\alpha$ does not exist

9–10. Let α be a continuous function of bounded variation on $[a, b]$. Assume $g \in R(\alpha)$ on $[a, b]$ and define $\beta(x) = \int_a^x g(t)\,d\alpha(t)$ if $x \in [a, b]$ Show that

(a) If $f \nearrow$ on $[a, b]$, there exists a point x_0 in $[a, b]$ such that

$$\int_a^b f\,d\beta = f(a)\int_a^{x_0} g\,d\alpha + f(b)\int_{x_0}^b g\,d\alpha$$

(b) If, in addition, f is continuous on $[a, b]$, we also have

$$\int_a^b f(x)g(x)\,d\alpha(x) = f(a)\int_a^{x_0} g\,d\alpha + f(b)\int_{x_0}^b g\,d\alpha$$

9–11. Assume $f \in R(\alpha)$ on $[a, b]$, where α is of bounded variation on $[a, b]$. Let $V(x)$ denote the total variation of α on $[a, x]$ for each x in $(a, b]$, and let $V(a) = 0$ Show that

$$\left|\int_a^b f\,d\alpha\right| \le \int_a^b |f|\,dV \le MV(b),$$

where M is an upper bound for $|f|$ on $[a, b]$. In particular, when $\alpha(x) = x$, the inequality becomes

$$\left|\int_a^b f(x)\,dx\right| \le M(b - a).$$

9–12. Let $\{\alpha_n\}$ be a sequence of functions of bounded variation on $[a, b]$. Suppose there exists a function α defined on $[a, b]$ such that the total variation of $\alpha - \alpha_n$ on $[a, b]$ tends to 0 as $n \to \infty$. Assume also that $\alpha(a) = \alpha_n(a) = 0$ for each $n = 1, 2, \ldots$. If f is continuous on $[a, b]$, prove that

$$\lim_{n\to\infty} \int_a^b f(x)\,d\alpha_n(x) = \int_a^b f(x)\,d\alpha(x).$$

9–13. If $f \in R(\alpha)$, $f^2 \in R(\alpha)$, $g \in R(\alpha)$ and $g^2 \in R(\alpha)$ on $[a, b]$, prove that

$$\frac{1}{2}\int_a^b \left[\int_a^b \begin{vmatrix} f(x) & g(x) \\ f(y) & g(y) \end{vmatrix}^2 d\alpha(y)\right] d\alpha(x)$$

$$= \left(\int_a^b f(x)^2\,d\alpha(x)\right)\left(\int_a^b g(x)^2\,d\alpha(x)\right) - \left(\int_a^b f(x)g(x)\,d\alpha(x)\right)^2.$$

When $\alpha \nearrow$ on $[a, b]$, deduce the Cauchy-Schwarz inequality

$$\left(\int_a^b f(x)g(x)\,d\alpha(x)\right)^2 \le \left(\int_a^b f(x)^2\,d\alpha(x)\right)\left(\int_a^b g(x)^2\,d\alpha(x)\right).$$

(Compare with Exercise 1–15.)

9–14. Assume that $f \in R(\alpha)$, $g \in R(\alpha)$ and $f\cdot g \in R(\alpha)$ on $[a, b]$. Show that

$$\frac{1}{2}\int_a^b \left[\int_a^b (f(y) - f(x))(g(y) - g(x))\,d\alpha(y)\right] d\alpha(x)$$

$$= (\alpha(b) - \alpha(a))\int_a^b f(x)g(x)\,d\alpha(x) - \left(\int_a^b f(x)\,d\alpha(x)\right)\left(\int_a^b g(x)\,d\alpha(x)\right).$$

If $\alpha \nearrow$ on $[a, b]$, deduce the inequality

$$\left(\int_a^b f(x)\,d\alpha(x)\right)\left(\int_a^b g(x)\,d\alpha(x)\right) \le (\alpha(b) - \alpha(a))\int_a^b f(x)g(x)\,d\alpha(x)$$

when both f and g are increasing (or both are decreasing) on $[a, b]$ Show that the reverse inequality holds if f increases and g decreases on $[a, b]$.

Riemann integrals.

9–15. Assume $f \in R$ on $[a, b]$ Prove that the limit

$$\lim_{n\to\infty} \frac{b-a}{n} \sum_{k=1}^{n} f\left(a + k\frac{b-a}{n}\right)$$

exists and has the value $\int_a^b f(x)\,dx$ Deduce that

$$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{n}{k^2+n^2} = \frac{\pi}{4}, \qquad \lim_{n\to\infty} \sum_{k=1}^{n} (n^2+k^2)^{-1/2} = \log(1+\sqrt{2}).$$

9–16. Let $\alpha = (\alpha_1, \ldots, \alpha_n)$ be a piece-wise smooth function defined on $[a, b]$. Show that the curve described by α has length

$$\Lambda_\alpha(a, b) = \int_a^b |\alpha'(t)|\,dt,$$

where $\alpha' = (\alpha_1', \ldots, \alpha_n')$

9–17. Define

$$f(x) = \left(\int_0^x e^{-t^2}\,dt\right)^2, \qquad g(x) = \int_0^1 \frac{e^{-x^2(t^2+1)}}{t^2+1}\,dt$$

(a) Show that $g'(x) + f'(x) = 0$ for all x and deduce that $g(x) + f(x) = \pi/4$.

(b) Use (a) to prove that

$$\lim_{x\to\infty} \int_0^x e^{-t^2}\,dt = \frac{1}{2}\sqrt{\pi}$$

9–18. Assume $g \in R$ on $[a, b]$ and define $f(x) = \int_a^x g(t)\,dt$ if $x \in [a, b]$ Show that the integral $\int_a^x |g(t)|\,dt$ gives the total variation of f on $[a, x]$

9–19. If $f^{(n+1)}$ is continuous on $[a, x]$, define

$$I_n(x) = \frac{1}{n!}\int_a^x (x-t)^n f^{(n+1)}(t)\,dt$$

(a) Show that

$$I_{k-1}(x) - I_k(x) = \frac{f^{(k)}(a)(x-a)^k}{k!}, \qquad k = 1, 2, \ldots, n.$$

(b) Use (a) to express the remainder in Taylor's formula (Theorem 5–14) as an integral.

9–20. Let f be continuous on $[0, a]$. If $x \in [0, a]$, define $f_0(x) = f(x)$ and let

$$f_{n+1}(x) = \frac{1}{n!}\int_0^x (x - t)^n f(t)\,dt, \qquad n = 0, 1, 2, \ldots.$$

(a) Show that the nth derivative of f_n exists and equals f.

(b) Prove the following theorem of M. Fekete: The number of changes in sign of f in $[0, a]$ is not less than the number of changes in sign in the ordered set of numbers

$$f(a), f_1(a), \ldots, f_n(a).$$

[*Hint*: Use mathematical induction.]

(c) Use (b) to prove the following theorem of L. Fejér: The number of changes in sign of f in $[0, a]$ is not less than the number of changes in sign in the ordered set

$$f(0), \qquad \int_0^a f(t)\,dt, \qquad \int_0^a tf(t)\,dt, \ldots, \qquad \int_0^a t^n f(t)\,dt.$$

9–21. Let f be a positive continuous function in $[a, b]$. Let M denote the maximum value of f on $[a, b]$. Show that

$$\lim_{n\to\infty}\left[\int_a^b f(x)^n\,dx\right]^{1/n} = M.$$

9–22. A function f of two real variables is defined for each point (x, y) in the unit square $0 \le x \le 1$, $0 \le y \le 1$ as follows:

$$f(x, y) = \begin{cases} 1, & \text{if } x \text{ is rational,} \\ 2y, & \text{if } x \text{ is irrational.} \end{cases}$$

(a) Compute $\overline{\int_0^1} f(x, y)\,dx$ and $\underline{\int_0^1} f(x, y)\,dx$ in terms of y.

(b) Show that $\int_0^1 f(x, y)\,dy$ exists for each fixed x and compute $\int_0^t f(x, y)\,dy$ in terms of x and t for $0 \le x \le 1$, $0 \le t \le 1$.

(c) Let $F(x) = \int_0^1 f(x, y)\,dy$. Show that $\int_0^1 F(x)\,dx$ exists and find its value.

9–23. Let f be defined on $[0, 1]$ as follows: $f(0) = 0$; if $2^{-n-1} < x \le 2^{-n}$, then $f(x) = 2^{-n}$, for $n = 0, 1, 2, \ldots$.

(a) Give two reasons why $\int_0^1 f(x)\,dx$ exists.

(b) Let $F(x) = \int_0^x f(t)\,dt$. Show that for $0 < x \le 1$ we have

$$F(x) = xA(x) - \tfrac{1}{3}A(x)^2, \qquad \text{where} \qquad A(x) = 2^{-[-\log x/\log 2]}$$

and where $[y]$ is the greatest integer in y.

9–24. Prove Theorem 9–41.

Jordan content

9–25. Show that if A and B are bounded sets in E_1, then

(a) $\bar{c}(A \cup B) + \bar{c}(A \cap B) \leq \bar{c}(A) + \bar{c}(B)$

(b) $\underline{c}(A \cup B) + \underline{c}(A \cap B) \geq \underline{c}(A) + \underline{c}(B)$

(c) If A and B are Jordan-measurable, show that $A \cup B$ and $A \cap B$ are Jordan-measurable and prove that

$$c(A \cup B) + c(A \cap B) = c(A) + c(B)$$

(d) If A and B are Jordan-measurable and if $A \subset B$, show that $B - A$ is Jordan-measurable and that

$$c(B - A) = c(B) - c(A)$$

9–26. Let A and B be subsets of the interval I. A partition P of I is said to *separate* A and B if there are subintervals of P whose union is S such that $A \subset S$ but $B \subset I - S$. If A and B can be so separated, show that

$$\bar{c}(A \cup B) = \bar{c}(A) + \bar{c}(B).$$

9–27. If A is a subset of the interval I, show that

$$\underline{c}(A) = c(I) - \bar{c}(I - A)$$

9–28. Given a decreasing sequence of real numbers $\{G(n)\}$ such that $G(n) \to 0$ as $n \to \infty$. Define a function f on $[0, 1]$ in terms of $\{G(n)\}$ as follows: $f(0) = 1$; if x is irrational, then $f(x) = 0$, if x is the rational m/n (in lowest terms), then $f(m/n) = G(n)$. Compute the oscillation $\omega_f(x)$ at each x in $[0, 1]$ and show that $f \in R$ on $[0, 1]$.

9–29. Let f be defined as in Exercise 9–28 with $G(n) = 1/n$. Let $g(x) = 1$ if $0 < x \leq 1$, $g(0) = 0$. Show that the composite function h defined by $h(x) = g[f(x)]$ is not Riemann-integrable on $[0, 1]$, although both $f \in R$ and $g \in R$ on $[0, 1]$.

9–30. Use Theorem 9–53 to prove Theorem 9–54.

9–31. Use Theorem 9–53 to prove that if $f \in R$ and $g \in R$ on $[a, b]$ and if $f(x) > 0$ for all x in $[a, b]$, then the function h defined by

$$h(x) = f(x)^{g(x)}$$

is Riemann-integrable on $[a, b]$.

9–32. Let $I = [0, 1]$ and let $A_1 = I - (\frac{1}{3}, \frac{2}{3})$ be that subset of I obtained by removing those points which lie in the open middle third of I; that is, $A_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$. Let A_2 be that subset of A_1 obtained by removing the open middle third of $[0, \frac{1}{3}]$ and of $[\frac{2}{3}, 1]$. Continue this process and define $A_3, A_4, \ldots$. The set $C = \bigcap_{n=1}^{\infty} A_n$ is called the *Cantor set*. Prove that:

(a) C is a compact set having Jordan content zero.

(b) $x \in C$ if, and only if, $x = \sum_{n=1}^{\infty} a_n 3^{-n}$, where each a_n is either 0 or 2.

(c) C is uncountable.

(d) The characteristic function of C is Riemann-integrable on $[0, 1]$.

Complex-valued integrals.

9–33. Prove Theorem 9–55.

9–34. Verify that Theorems 9–31 and 9–32 are valid for complex f and α.

9–35. Prove Theorem 9–57 by using Theorems 9–55 and 9–7.

9–36. Prove Theorem 9–59 by using Theorems 9–55 and 9–4.

9–37. Prove Theorem 9–62 by using Theorems 9–55 and 9–8.

9–38. Let Γ be a plane curve described by a piece-wise smooth function z defined on $[a, b]$ and assume that $\int_{\Gamma[z]} F(z)\,dz$ exists. Let S be an open region containing Γ and suppose G is a complex-valued function defined on S such that $G'(z)$ exists and has the value $G'(z) = F(z)$ for each point z on Γ. Prove that:

(a) $$\int_{\Gamma[z]} F(z)\,dz = \int_{\Gamma[z]} G'(z)\,dz = G(B) - G(A), \text{ if } A = z(a),\ B = z(b).$$

NOTE. In particular, the integral has the value 0 when Γ is a closed curve. [*Hint*: Subdivide $[a, b]$ into subintervals in which z' is continuous and apply Theorems 9–62 and 9–32 in each subinterval.]

(b) $$\int_{\Gamma[z]} z^n\,dz = \left(\frac{B^{n+1} - A^{n+1}}{n+1}\right),$$
if n is an integer, $n \neq -1$.

9–39. Let f be continuous on a rectifiable curve Γ of length $\Lambda(\Gamma)$, and suppose $|f(z)| \leq M$ if $z \in \Gamma$. Show that $|\int_\Gamma f(z)\,dz| = M\Lambda(\Gamma)$ only if $|f(z)| = M$ everywhere on Γ.

[*Hint*: If $|f(z_0)| < M$ for some point z_0 of Γ, there is a subarc Γ_1 of Γ on which $|f(z)| \leq M - \epsilon$ for some $\epsilon > 0$.]

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CHAPTER 10

MULTIPLE INTEGRALS AND LINE INTEGRALS

10–1 Introduction. Riemann's concept of the integral $\int_a^b f(x)\,dx$ can be generalized by replacing the interval $[a, b]$ by an n-dimensional region in which f is defined and bounded. The simplest regions in E_n suitable for this purpose are n-dimensional intervals. For example, in E_2 we take a rectangle R partitioned into subrectangles R_k and consider Riemann sums of the form $\sum f(x_k, y_k)A(R_k)$, where $(x_k, y_k) \in R_k$ and $A(R_k)$ denotes the area of R_k. This leads us to the concept of a double integral. Similarly, in E_3 we use rectangular parallelepipeds subdivided into smaller parallelepipeds R_k and, by considering sums of the form $\sum f(x_k, y_k, z_k)V(R_k)$, where $(x_k, y_k, z_k) \in R_k$ and $V(R_k)$ is the volume of R_k, we are led to the concept of a triple integral. It is just as easy to discuss multiple integrals in E_n, provided that we have a suitable generalization of the notions of area and volume. This "generalized volume" is called *measure* or *content* and is defined in the next section.

10–2 The measure (or content) of elementary sets in E_n. Recall that an n-dimensional open interval $(\mathbf{a}, \mathbf{b})$ is the cartesian product of n open one-dimensional intervals (a_k, b_k), where $\mathbf{a} = (a_1, \ldots, a_n)$ and $\mathbf{b} = (b_1, \ldots, b_n)$. The cartesian product of n closed intervals $[a_k, b_k]$ is called an n-dimensional closed interval and is denoted by $[\mathbf{a}, \mathbf{b}]$. A general n-dimensional interval I (not necessarily open or closed) is a set of the form

$$I = \{(x_1, \ldots, x_n) \mid a_k \le x_k \le b_k, k = 1, 2, \ldots, n\},$$

where some or all of the inequality signs $\le$ may be replaced by the sign $<$. The n-dimensional *measure* or *content* of I, denoted by $\mu(I)$, is defined to be the product

$$\mu(I) = (b_1 - a_1) \cdots (b_n - a_n).$$

Thus, when $n = 2$, $\mu(I)$ is the area of I and, when $n = 3$, it yields the volume of I. If $r < n$, an r-dimensional interval is considered to be a "degenerate" n-dimensional interval obtained by permitting $n - r$ equalities $a_k = b_k$ to hold. Accordingly, the n-dimensional measure of such a degenerate interval is zero.

A set S which is the union of a finite number of pairwise nonoverlapping intervals is called an *elementary set* Two intervals (and more generally, two sets in E_n) are said to be nonoverlapping if they have no interior points in common (They may, however, have boundary points in common) If $S = I_1 \cup \cdots \cup I_r$ is such a decomposition, we assign to S the measure

$$\mu(S) = \mu(I_1) + \cdots + \mu(I_r)$$

When S is decomposed in two ways as a union of nonoverlapping intervals, say $S = I_1 \cup \cdots \cup I_r$ and $S = J_1 \cup \cdots \cup J_p$, an easy calculation shows that we must have

$$\mu(I_1) + \cdots + \mu(I_r) = \mu(J_1) + \cdots + \mu(J_p)$$

This implies that the measure of S is a number completely determined by S It follows that for two nonoverlapping elementary sets S and T we must have $\mu(S \cup T) = \mu(S) + \mu(T)$ That is, the "set function" μ is additive with respect to elementary sets This simple property is of fundamental importance in the material that follows Of course, $\mu(S) \geq 0$

With these preliminary remarks, we are now ready to discuss Riemann integration in E_n The only essential difference between the case $n = 1$ and the case $n > 1$ is that the quantity $\Delta x_k = x_k - x_{k-1}$ which was used to measure the length of the subinterval $[x_{k-1}, x_k]$ must now be replaced by the measure $\mu(I_k)$ of an n-dimensional subinterval Since the work proceeds on exactly the same lines as the one-dimensional case, we shall omit many of the details in the discussions that follow The missing proofs can be easily supplied by anyone with a thorough grasp of the techniques developed in Chapter 9

10–3 Riemann integration of bounded functions defined on intervals in E_n.

10–1 DEFINITION *Let* $[\mathbf{a}, \mathbf{b}]$ *be an interval in* E_n, *where* $\mathbf{a} = (a_1, \ldots, a_n)$ *and* $\mathbf{b} = (b_1, \ldots, b_n)$ *If* P_k *is a partition of* $[a_k, b_k]$, *the cartesian product*

$$P = P_1 \times \cdots \times P_n$$

is said to be a partition of $[\mathbf{a}, \mathbf{b}]$ *If* P_k *divides* $[a_k, b_k]$ *into* m_k *one-dimensional subintervals, then* P *determines a decomposition of* $[\mathbf{a}, \mathbf{b}]$ *as a union of* $m_1 \cdots m_n$ *n-dimensional intervals (called subintervals of* P*). A partition* P' *of* $[\mathbf{a}, \mathbf{b}]$ *is said to be finer than* P *if* $P \subset P'$. *The set of all possible partitions of* $[\mathbf{a}, \mathbf{b}]$ *will be denoted by* $\mathcal{P}[\mathbf{a}, \mathbf{b}]$

Figure 10–1 illustrates partitions of intervals in E_2 and in E_3

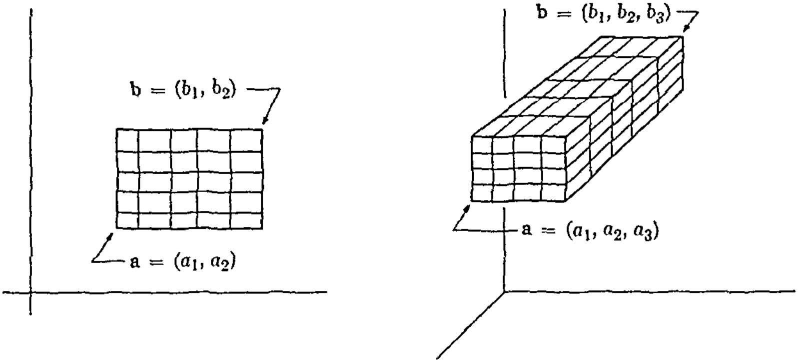


FIG. 10–1. Partitions of intervals [a, b] in E_2 and in E_3.

10–2 DEFINITION. *Let f be defined and bounded on a closed interval $I \subset E_n$. If P is a partition of I into m subintervals $I_1, \ldots, I_m$ and if $\mathbf{t}_k \in I_k$, a sum of the form*

$$S(P, f) = \sum_{k=1}^{m} f(\mathbf{t}_k)\mu(I_k)$$

is called a Riemann sum. We say f is Riemann-integrable on I and we write $f \in R$ on I, whenever there exists a real number A having the following property: For every $\epsilon > 0$ there exists a partition P_ϵ of I such that P finer than P_ϵ implies $|S(P, f) - A| < \epsilon$ for all Riemann sums $S(P, f)$. When such a number A exists, it is uniquely determined and is denoted by

$$\int_I f\,d\mathbf{x}, \qquad \int_I f(\mathbf{x})\,d\mathbf{x}, \qquad \text{or by} \int_I f(x_1, \ldots, x_n)\,d(x_1, \ldots, x_n).$$

NOTE. For $n > 1$ the integral is called a *multiple* or *n-fold* integral. When $n = 2$ and 3, the terms *double* and *triple* integral are used. As in E_1, the symbol $\mathbf{x}$ in $\int_I f(\mathbf{x})\,d\mathbf{x}$ is a "dummy variable" and may be replaced by any other convenient symbol. The notation $\int_I f(x_1, \ldots, x_n)\,dx_1 \cdots dx_n$ is also used instead of $\int_I f(x_1, \ldots, x_n)\,d(x_1, \ldots, x_n)$. Double integrals are sometimes written with two integral signs and triple integrals with three such signs, thus:

$$\iint_I f(x, y)\,dx\,dy, \qquad \iiint_I f(x, y, z)\,dx\,dy\,dz.$$

10-3 DEFINITION *Let f be defined and bounded on a closed interval $I \subset E_n$. If P is a partition of I into m subintervals $I_1, \ldots, I_m$, let*

$$m_k(f) = \inf \{f(\mathbf{x}) \mid \mathbf{x} \in I_k\}, \qquad M_k(f) = \sup \{f(\mathbf{x}) \mid \mathbf{x} \in I_k\}$$

The numbers

$$U(P, f) = \sum_{k=1}^{m} M_k(f)\mu(I_k) \qquad \textit{and} \qquad L(P, f) = \sum_{k=1}^{m} m_k(f)\mu(I_k)$$

are called upper and lower Riemann sums. The upper and lower Riemann integrals of f over I are defined as follows.

$$\overline{\int_I} f\, d\mathbf{x} = \inf \{U(P, f) \mid P \text{ is a partition of } I\},$$

$$\underline{\int_I} f\, d\mathbf{x} = \sup \{L(P, f) \mid P \text{ is a partition of } I\}.$$

The function f is said to satisfy Riemann's condition on I if, for every $\epsilon > 0$, there exists a partition P_ϵ of I such that P finer than P_ϵ implies $U(P, f) - L(P, f) < \epsilon$.

NOTE. As in the one-dimensional case, upper and lower integrals have the following properties:

(a)
$$\overline{\int_I} (f + g)\, d\mathbf{x} \le \overline{\int_I} f\, d\mathbf{x} + \overline{\int_I} g\, d\mathbf{x},$$

$$\underline{\int_I} (f + g)\, d\mathbf{x} \ge \underline{\int_I} f\, d\mathbf{x} + \underline{\int_I} g\, d\mathbf{x}$$

(b) If an interval I is decomposed into a union of two nonoverlapping intervals I_1, I_2, then we have

$$\overline{\int_I} f\, d\mathbf{x} = \overline{\int_{I_1}} f\, d\mathbf{x} + \overline{\int_{I_2}} f\, d\mathbf{x} \qquad \text{and} \qquad \underline{\int_I} f\, d\mathbf{x} = \underline{\int_{I_1}} f\, d\mathbf{x} + \underline{\int_{I_2}} f\, d\mathbf{x}$$

The proof of the following theorem is essentially the same as that of Theorem 9-19 and will be omitted.

10–4 THEOREM. *Let f be defined and bounded on a closed interval $I \subset E_n$. Then the following statements are equivalent:*

(i) *$f \in R$ on I.*

(ii) *f satisfies Riemann's condition on I.*

(iii) $\underline{\int}_I f\,d\mathbf{x} = \overline{\int}_I f\,d\mathbf{x}$.

10–4 Jordan content of bounded sets in E_n.

10–5 DEFINITION. *Let S be a subset of an interval $[\mathbf{a}, \mathbf{b}]$ in E_n. For every partition P of $[\mathbf{a}, \mathbf{b}]$ define $\underline{J}(P, S)$ to be the sum of the measures of those subintervals of P which contain only interior points of S and let $\bar{J}(P, S)$ be the sum of the measures of those subintervals of P which contain points of $S \cup \partial S$. (Recall Definitions 3–24 and 8–38.) The numbers*

$$\underline{c}(S) = \sup \{\underline{J}(P, S) \mid P \in \mathcal{P}[\mathbf{a}, \mathbf{b}]\},$$

$$\bar{c}(S) = \inf \{\bar{J}(P, S) \mid P \in \mathcal{P}[\mathbf{a}, \mathbf{b}]\}$$

are called, respectively, the (n-dimensional) inner and outer Jordan content of S. The set S is said to be Jordan-measurable if $\underline{c}(S) = \bar{c}(S)$, in which case this common value is called the Jordan content of S, denoted by $c(S)$.

NOTE. It is easy to verify that $\underline{c}(S)$ and $\bar{c}(S)$ depend only on S and not on the interval $[\mathbf{a}, \mathbf{b}]$ which contains S. Also, $0 \le \underline{c}(S) \le \bar{c}(S)$. For finite sets we have $\underline{c}(S) = \bar{c}(S) = 0$. The n-dimensional outer Jordan content of a bounded r-dimensional set is zero if $r < n$. Elementary sets S are Jordan-measurable and $c(S) = \mu(S)$. (Strictly speaking, we should write c_n rather than c to emphasize the fact that the content is n-dimensional, but no confusion arises by omitting this reference to n.) Jordan-measurable sets S in E_2 are also said to have *area* $c(S)$. In this case, the sums $\underline{J}(P, S)$ and $\bar{J}(P, S)$ represent approximations to the area from the "inside" and the "outside" of S, respectively. This is illustrated in Fig. 10–2, where the lightly shaded rectangles are counted in $\bar{J}(P, S)$, the heavily shaded rectangles in $\underline{J}(P, S)$. For sets in E_3, $c(S)$ is called the *volume* of S.

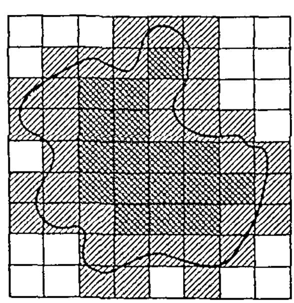


FIG. 10–2. Illustrating $\underline{J}(P, S)$ and $\bar{J}(P, S)$ in E_2.

As in the one-dimensional case, Jordan content is related to Riemann integration by the formulas

$$\underline{c}(S) = \underline{\int}_I f_S(\mathbf{x})\, d\mathbf{x}, \qquad \bar{c}(S) = \overline{\int}_I f_S(\mathbf{x})\, d\mathbf{x},$$

if S is a subset of an interval I and if f_S is the characteristic function of S.

A useful necessary and sufficient condition for bounded sets to be Jordan-measurable will now be given

10-6 THEOREM. *Let S be a bounded set in E_n and let ∂S denote its boundary. Then we have*

$$\bar{c}(\partial S) = \bar{c}(S) - \underline{c}(S).$$

Hence, S is Jordan-measurable if, and only if, ∂S has content zero.

Proof. Let $I = [\mathbf{a}, \mathbf{b}]$ be a closed interval containing S Then $\partial S \subset I$ and for every partition P of I we have

$$\bar{J}(P, \partial S) = \bar{J}(P, S) - \underline{J}(P, S)$$

Therefore, $\bar{J}(P, \partial S) \geq \bar{c}(S) - \underline{c}(S)$ and hence $\bar{c}(\partial S) \geq \bar{c}(S) - \underline{c}(S)$ To obtain the reverse inequality, let $\epsilon > 0$ be given, choose P_1 so that $\bar{J}(P_1, S) < \bar{c}(S) + \epsilon/2$ and choose P_2 so that $\underline{J}(P_2, S) > \underline{c}(S) - \epsilon/2$. Let $P = P_1 \cup P_2$ Since refinement increases the inner sums $\underline{J}$ and decreases the outer sums $\bar{J}$, we find

$$\bar{c}(\partial S) \leq \bar{J}(P, \partial S) = \bar{J}(P, S) - \underline{J}(P, S) \leq \bar{J}(P_1, S) - \underline{J}(P_2, S)$$
$$< \bar{c}(S) - \underline{c}(S) + \epsilon$$

Since ϵ is arbitrary, this means that $\bar{c}(\partial S) \leq \bar{c}(S) - \underline{c}(S)$. Therefore, $\bar{c}(\partial S) = \bar{c}(S) - \underline{c}(S)$ and the proof is complete.

At this point it is convenient to introduce the following definition.

10-7 DEFINITION. *Let S be a bounded set in E_n The diameter of S, denoted by $d(S)$, is defined to be the number*

$$d(S) = \sup \{|\mathbf{x} - \mathbf{y}| \mid \mathbf{x} \in S, \mathbf{y} \in S\}.$$

NOTE. If $S \subset T$, then $d(S) \leq d(T)$. If S is an interval, say

$$S = \{(x_1, \ldots, x_n) \mid a_k \leq x_k \leq b_k, k = 1, 2, \ldots, n\}$$

(where some or all inequality signs $\leq$ may be $<$), we have $d(S) = |\mathbf{b} - \mathbf{a}|$.

In a later chapter we shall give an example of a continuous curve in E_2 which passes through every point of a rectangle. Such curves (known as *space-filling curves*) illustrate the fact that the continuous image of a set of two-dimensional Jordan content zero can be a set of two-dimensional Jordan content greater than zero. By way of contrast, the next theorem shows that this cannot occur for functions which satisfy a uniform Lipschitz condition.

10–8 THEOREM. *Let* $\mathbf{f} = (f_1, \ldots, f_p)$ *be a vector-valued function satisfying a uniform Lipschitz condition on a set* X *in* E_n. *That is, assume that two positive numbers* $\delta > 0$ *and* $M > 0$ *exist such that for every pair of points* $\mathbf{x}$ *and* $\mathbf{y}$ *in* X,

$$|\mathbf{x} - \mathbf{y}| < \delta \qquad \textit{implies} \qquad |\mathbf{f}(\mathbf{y}) - \mathbf{f}(\mathbf{x})| \leq M|\mathbf{y} - \mathbf{x}|.$$

Let $Y = \mathbf{f}(X)$ *be the image of* X *in* E_p *and assume that* $p \geq n$. *If* X *has n-dimensional Jordan content zero, then* Y *has p-dimensional Jordan content zero.*

NOTE. It is easy to verify that $\mathbf{f}$ will satisfy a uniform Lipschitz condition on X if each component f_k satisfies such a condition on X.

Proof. Enclose X in an n-dimensional cube I, that is, in an interval I which is the cartesian product of n equal intervals $[a, b]$ (the number $b - a$ is called the *edge-length* of I). If P is a partition of I into m subintervals $I_1, \ldots, I_m$, the largest of the diameters $d(I_1), \ldots, d(I_m)$ will be called the *norm* of P, denoted by $|P|$.

Let $\epsilon > 0$ be given and assume that X has n-dimensional Jordan content zero. Let $A \doteq (2M)^p n^{n/2} \delta^{p-n}$. Then there exists a partition P_ϵ of I such that $\bar{J}(P, X) < \epsilon/A$ whenever P is finer than P_ϵ. (We can also assume that $|P_\epsilon| \leq \delta$.) Since I is a cube, there is at least one partition P_1 finer than P_ϵ whose subintervals are n-dimensional cubes with equal edge-lengths $\Delta < \delta/\sqrt{n}$. The content of each such cube is Δ^n and the diameter is $\sqrt{n}\,\Delta < \delta$. Therefore, $\bar{J}(P_1, X) = q\,\Delta^n$, where q represents the number of subintervals of P_1 which intersect X. Note that $q\,\Delta^n < \epsilon/A$, since P_1 is finer than P_ϵ. If $I_1, \ldots, I_q$ are the subintervals of P_1 which intersect X and if $\mathbf{x}_k \in I_k \cap X$, then for every $\mathbf{y}$ in $I_k \cap X$ we must have

$$|\mathbf{y} - \mathbf{x}_k| \leq \sqrt{n}\,\Delta < \delta,$$

and hence

$$|\mathbf{f}(\mathbf{y}) - \mathbf{f}(\mathbf{x}_k)| \leq M|\mathbf{y} - \mathbf{x}_k| \leq M\sqrt{n}\,\Delta.$$

This means that, in E_p, $\mathbf{f}(\mathbf{y})$ lies within a sphere of radius $M\sqrt{n}\,\Delta$ about $\mathbf{f}(\mathbf{x}_k)$. But this sphere is contained within a p-dimensional cube of edge-length $2M\sqrt{n}\,\Delta$ whose content is $(2M\sqrt{n}\,\Delta)^p$. Therefore the image $Y = \mathbf{f}(X)$ is a bounded set whose p-dimensional outer Jordan content does not exceed $q(2M\sqrt{n}\,\Delta)^p$. But we have

$$q(2M\sqrt{n}\,\Delta)^p = q\,\Delta^n n^{n/2}(2M)^p(\sqrt{n}\,\Delta)^{p-n}$$

$$< q\,\Delta^n n^{n/2}(2M)^p\,\delta^{p-n} = q\,\Delta^n A < \epsilon$$

Since ϵ is arbitrary, it follows that Y has p-dimensional Jordan content zero

---

As a consequence of the last two theorems, we have the following useful result.

10–9 THEOREM *Let $\mathbf{g} = (g_1, \ldots, g_n)$ be a continuous vector-valued function defined on an open set S in E_n. Assume that $\mathbf{g}^{-1}$ exists and is continuous on $\mathbf{g}(S)$ and that $\mathbf{g}^{-1}$ satisfies a Lipschitz condition at each point of a compact subset X of $\mathbf{g}(S)$. If X is Jordan-measurable, then the inverse image $\mathbf{g}^{-1}(X)$ is also Jordan-measurable*

Proof. By compactness it is easy to prove that $\mathbf{g}^{-1}$ satisfies a *uniform* Lipschitz condition on X. If X is Jordan-measurable, then ∂X has content zero and, by Theorem 10–8, $\mathbf{g}^{-1}(\partial X)$ also has content zero. If we can prove that $\partial[\mathbf{g}^{-1}(X)] \subset \mathbf{g}^{-1}(\partial X)$ it will follow that $\partial[\mathbf{g}^{-1}(X)]$ has content zero and hence that $\mathbf{g}^{-1}(X)$ is Jordan-measurable, as required

Now it is easy to prove that $\mathbf{g}(\partial Z) \subset \partial\mathbf{g}(Z)$ for every closed subset Z of S (This merely requires continuity of $\mathbf{g}$ and of $\mathbf{g}^{-1}$, and we leave the proof to the reader.) Therefore, $\mathbf{g}^{-1}[\mathbf{g}(\partial Z)] \subset \mathbf{g}^{-1}[\partial\mathbf{g}(Z)]$, which implies $\partial Z \subset \mathbf{g}^{-1}[\partial\mathbf{g}(Z)]$. Taking $Z = \mathbf{g}^{-1}(X)$, we obtain $\partial[\mathbf{g}^{-1}(X)] \subset \mathbf{g}^{-1}(\partial X)$, and this completes the proof.

10–5 Necessary and sufficient conditions for the existence of multiple integrals.

10–10 DEFINITION. *Let f be defined and bounded on a closed interval $I \subset E_n$. If T is a subset of I, the number*

$$\Omega_f(T) = \sup\,\{f(\mathbf{x}) - f(\mathbf{y}) \mid \mathbf{x} \in T, \mathbf{y} \in T\}$$

is called the oscillation of f *on* T. *If* $\mathbf{x} \in I$, *the oscillation of* f *at* $\mathbf{x}$ *is defined to be the number*

$$\omega_f(\mathbf{x}) = \lim_{h \to 0+} \Omega_f(N(\mathbf{x}; h) \cap I).$$

NOTE. This limit always exists since $\Omega_f(N(\mathbf{x}; h) \cap I)$ is a monotonic function of h. In fact, $T_1 \subset T_2$ implies $\Omega_f(T_1) \leq \Omega_f(T_2)$.

10-11 THEOREM. *Let* f *be defined and bounded on the closed interval* $I \subset E_n$. *Then* f *is continuous at the point* $\mathbf{x}$ *of* I *if, and only if,* $\omega_f(\mathbf{x}) = 0$.

10-12 THEOREM. *Let* f *be defined and bounded on* $[\mathbf{a}, \mathbf{b}]$ *and let* $\epsilon > 0$ *be given. If* $\omega_f(\mathbf{x}) < \epsilon$ *for every* $\mathbf{x}$ *in* $[\mathbf{a}, \mathbf{b}]$, *then there exists a* $\delta > 0$ *(depending only on* ϵ*) such that, for every closed subinterval* $T \subset [\mathbf{a}, \mathbf{b}]$, *we have* $\Omega_f(T) < \epsilon$ *whenever the diameter* $d(T) < \delta$.

The proofs of Theorems 10-11 and 10-12 and of the remaining theorems in this section are so similar to the corresponding theorems in the one-dimensional case that they need not be given in detail.

10-13 THEOREM. *Let* f *be defined and bounded on a closed interval* $I \subset E_n$. *For each* $\epsilon > 0$ *define* $J_\epsilon = \{\mathbf{x} \mid \mathbf{x} \in I, \omega_f(\mathbf{x}) \geq \epsilon\}$. *Then:*

(a) J_ϵ *is a closed set.*
(b) $f \in R$ *on* I *if, and only if,* $c(J_\epsilon) = 0$ *for every* $\epsilon > 0$.

10-14 DEFINITION. *If* $S \subset E_n$, *a countable collection* $\{I_1, I_2, \ldots\}$ *of open* n*-dimensional intervals such that* $S \subset \bigcup_{k=1}^{\infty} I_k$ *is called a Lebesgue covering of* S. *The number*

$$\overline{m}(S) = \inf \left\{ \sum_{k=1}^{\infty} c(I_k) \mid \{I_1, I_2, \ldots\} \quad \textit{is a Lebesgue covering of } S \right\}$$

is called the n*-dimensional outer Lebesgue measure of* S.

NOTE. When S is bounded, say $S \subset [\mathbf{a}, \mathbf{b}] = I$, then $0 \leq \overline{m}(S) \leq c(I)$. If $\overline{m}(S) = 0$, S is said to be of measure zero. For elementary sets we have $\overline{m}(S) = c(S)$.

10-15 THEOREM. *For every bounded set* S *in* E_n *we have* $\overline{m}(S) \leq \bar{c}(S)$.

10-16 THEOREM. *If* B *is a bounded set in* E_n *and if* A *is a compact subset of* B, *then* $\bar{c}(A) \leq \overline{m}(B)$.

10-17 THEOREM. *For compact sets* S *in* E_n *we have* $\bar{c}(S) = \overline{m}(S)$.

10-18 THEOREM. *Let* $F = \{F_1, F_2, \ldots\}$ *be a countable collection of sets in* E_n *such that* $\overline{m}(F_k) = 0$ *for each* $k = 1, 2, \ldots$. *Then* $\overline{m}(S) = 0$, *where* $S = \bigcup_{k=1}^{\infty} F_k$.

10–19 THEOREM *Let f be defined and bounded on a closed interval $I \subset E_n$. Let $D = \{\mathbf{x} \mid \mathbf{x} \in I, f$ is not continuous at $\mathbf{x}\}$ Then $f \in R$ on I if, and only if, the set D has measure zero*

10–6 Evaluation of a multiple integral by repeated integration. From elementary calculus the reader has learned to evaluate certain double and triple integrals by successive integration with respect to each variable For example, if f is a function of two variables continuous on a closed rectangle R in the xy-plane, say $R = \{(x, y) \mid a \leq x \leq b,\ c \leq y \leq d\}$, then for each fixed y in $[c, d]$ the function F defined by the equation $F(x) = f(x, y)$ is continuous (and hence integrable) on $[a, b]$ The value of the integral $\int_a^b F(x)\,dx$ depends on y and defines a new function G, where $G(y) = \int_a^b f(x, y)\,dx$ This function G is continuous (by Theorem 9–35), and hence integrable, on $[c, d]$ The integral $\int_c^d G(y)\,dy$ turns out to have the same value as the *double* integral $\int_R f(x, y)\,d(x, y)$ That is, we have the equation

$$(1) \qquad \int_R f(x, y)\,d(x, y) = \int_c^d \left[\int_a^b f(x, y)\,dx\right] dy$$

(This formula will be proved later) The question now arises as to whether a similar result holds when f is merely integrable (and not necessarily continuous) on R We can see at once that certain difficulties are inevitable. For example, the inner integral $\int_a^b f(x, y)\,dx$ may not exist for certain values of y even though the double integral exists In fact, if f is discontinuous at every point of the line segment $y = y_0$, $a \leq x \leq b$, then $\int_a^b f(x, y_0)\,dx$ will fail to exist However, this line segment is a set whose two-dimensional outer Lebesgue measure is zero and therefore does not affect the integrability of f on the whole rectangle R In a case of this kind we must make use of upper and lower integrals to obtain a suitable generalization of (1)

10–20 THEOREM *Let f be defined and bounded on the closed rectangle R in the xy-plane given by*

$$R = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$$

Then we have

$$\text{(i)}\ \underline{\int_R} f\,d(x, y) \leq \underline{\int_a^b}\left[\overline{\int_c^d} f(x, y)\,dy\right] dx$$

$$\leq \overline{\int_a^b}\left[\overline{\int_c^d} f(x, y)\,dy\right] dx \leq \overline{\int_R} f\,d(x, y)$$

(ii) *Statement* (i) *holds with* $\overline{\int_c^d}$ *replaced by* $\underline{\int_c^d}$ *throughout.*

(iii) $$\underline{\int_R} f\, d(x, y) \leq \underline{\int_c^d} \left[\overline{\int_a^b} f(x, y)\, dx \right] dy$$

$$\leq \overline{\int_c^d} \left[\overline{\int_a^b} f(x, y)\, dx \right] dy \leq \overline{\int_R} f\, d(x, y).$$

(iv) *Statement* (iii) *holds with* $\overline{\int_a^b}$ *replaced by* $\underline{\int_a^b}$ *throughout.*

(v) *When* $\int_R f(x, y)\, d(x, y)$ *exists, we have*

$$\int_R f(x, y)\, d(x, y) = \int_a^b \left[\underline{\int_c^d} f(x, y)\, dy \right] dx = \int_a^b \left[\overline{\int_c^d} f(x, y)\, dy \right] dx$$

$$= \int_c^d \left[\underline{\int_a^b} f(x, y)\, dx \right] dy = \int_c^d \left[\overline{\int_a^b} f(x, y)\, dx \right] dy.$$

Proof. To prove (i), define F by the equation

$$F(x) = \overline{\int_c^d} f(x, y)\, dy, \qquad \text{if } x \in [a, b].$$

Then $|F(x)| \leq M(d - c)$, where $M = \sup \{|f(x, y)| \mid (x, y) \in R\}$, and we can consider

$$\bar{I} = \overline{\int_a^b} F(x)\, dx = \overline{\int_a^b} \left[\overline{\int_c^d} f(x, y)\, dy \right] dx.$$

Similarly, we define

$$\underline{I} = \underline{\int_a^b} F(x)\, dx = \underline{\int_a^b} \left[\overline{\int_c^d} f(x, y)\, dy \right] dx.$$

Let $P_1 = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$ and let $P_2 = \{y_0, y_1, \ldots, y_m\}$ be a partition of $[c, d]$. Then $P = P_1 \times P_2$ is a partition of R into mn subrectangles R_{ij} and we define

$$\bar{I}_{ij} = \int_{x_{i-1}}^{\bar{x}_i} \left[\int_{y_{j-1}}^{\bar{y}_j} f(x, y)\, dy \right] dx, \qquad I_{ij} = \int_{\underline{x_{i-1}}}^{x_i} \left[\bar{\int}_{y_{j-1}}^{y_j} f(x, y)\, dy \right] dx.$$

Since we have

$$\bar{\int_c^d} f(x, y)\, dy = \sum_{j=1}^{m} \bar{\int}_{y_{j-1}}^{y_j} f(x, y)\, dy,$$

we can write

$$\bar{\int_a^b} \left[\bar{\int_c^d} f(x, y)\, dy \right] dx \leq \sum_{j=1}^{m} \bar{\int_a^b} \left[\bar{\int}_{y_{j-1}}^{y_j} f(x, y)\, dy \right] dx$$

$$= \sum_{j=1}^{m} \sum_{i=1}^{n} \bar{\int}_{x_{i-1}}^{x_i} \left[\bar{\int}_{y_{j-1}}^{y_j} f(x, y)\, dy \right] dx.$$

That is, we have the inequality

$$\bar{I} \leq \sum_{j=1}^{m} \sum_{i=1}^{n} \bar{I}_{ij}.$$

Similarly, we find

$$\underline{I} \geq \sum_{j=1}^{m} \sum_{i=1}^{n} \underline{I}_{ij}$$

If we write

$$m_{ij} = \inf \{f(x, y) \mid (x, y) \in R_{ij}\}$$

and

$$M_{ij} = \sup \{f(x, y) \mid (x, y) \in R_{ij}\},$$

then from the inequality $m_{ij} \leq f(x, y) \leq M_{ij}$, $(x, y) \in R_{ij}$, we obtain

$$m_{ij}(y_j - y_{j-1}) \leq \bar{\int}_{y_{j-1}}^{y_j} f(x, y)\, dy \leq M_{ij}(y_j - y_{j-1})$$

This, in turn, implies

$$m_{ij}\mu(R_{ij}) \leq \underline{\int}_{x_{i-1}}^{x_i} \left[\bar{\int}_{y_{j-1}}^{y_j} f(x, y)\, dy \right] dx$$

$$\leq \bar{\int}_{x_{i-1}}^{x_i} \left[\bar{\int}_{y_{j-1}}^{y_j} f(x, y)\, dy \right] dx \leq M_{ij}\mu(R_{ij})$$

Summing on i and j and using the above inequalities, we get

$$L(P,f) \leq \underline{I} \leq \bar{I} \leq U(P,f).$$

Since this holds for all partitions P of R, we must have

$$\underline{\int_R} f\,d(x, y) \leq \underline{I} \leq \bar{I} \leq \overline{\int_R} f\,d(x, y).$$

This proves statement (i).

It is clear that the preceding proof could also be carried out if the function F were originally defined by the formula

$$F(x) = \underline{\int_c^d} f(x, y)\,dy,$$

and hence (ii) follows by the same argument.

Statements (iii) and (iv) can be similarly proved by interchanging the roles of x and y. Finally, statement (v) is an immediate consequence of statements (i) through (iv).

As a corollary, we have the formula mentioned earlier:

$$\int_R f(x, y)\,d(x, y) = \int_a^b \left[\int_c^d f(x, y)\,dy\right] dx = \int_c^d \left[\int_a^b f(x, y)\,dx\right] dy,$$

which is valid when f is continuous on R.

Note. The existence of the "repeated" integrals $\int_a^b [\int_c^d f(x, y)\,dy]\,dx$ and $\int_c^d [\int_a^b f(x, y)\,dx]\,dy$ does not imply the existence of $\int_R f(x, y)\,d(x, y)$. A counter example is given in Exercise 10–9.

Before commenting on the analog of Theorem 10–20 in E_n, we first introduce some further notation and terminology. If $k \leq n$, the set of $\mathbf{x}$ in E_n for which $x_k = 0$ is called the *coordinate hyperplane* Π_k. Given a set S in E_n, the *projection* S_k of S on Π_k is defined to be the image of S under that mapping whose value at each point $(x_1, x_2, \ldots, x_n)$ in S is $(x_1, \ldots, x_{k-1}, 0, x_{k+1}, \ldots, x_n)$. It is easy to show that such a mapping is continuous on S. It follows that if S is compact, each projection S_k is compact. Also, if S is connected, each S_k is connected. Projections in E_3 are illustrated in Fig. 10–3.

A theorem entirely analogous to Theorem 10–20 holds for n-fold integrals. It will suffice to indicate how the extension goes when $n = 3$. In

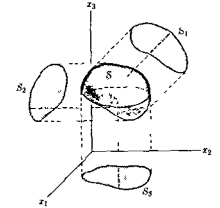


FIG 10-3 Projections in E_3

this case, f is defined and bounded on a closed interval $R = [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3]$ in E_3 and statement (i) of Theorem 10-20 is replaced by

$$(2) \qquad \underline{\int_R} f\,d\mathbf{x} \le \int_{a_1}^{b_1}\left[\bar{\int_{R_1}} f\,d(x_2, x_3)\right]dx_1 \le \bar{\int_{a_1}^{b_1}}\left[\bar{\int_{R_1}} f\,d(x_2, x_3)\right]dx_1 \le \bar{\int_R} f\,d\mathbf{x},$$

where R_1 is the projection of R on the coordinate plane Π_1 When $\int_R f(\mathbf{x})\,d\mathbf{x}$ exists, the analog of part (v) of Theorem 10-20 is the formula

$$(3) \qquad \int_R f(\mathbf{x})\,d\mathbf{x} = \int_{a_1}^{b_1}\left[\bar{\int_{R_1}} f\,d(x_2, x_3)\right]dx_1 = \int_{R_1}\left[\bar{\int_{a_1}^{b_1}} f\,dx_1\right]d(x_2, x_3).$$

As in Theorem 10-20, similar statements hold with appropriate replacements of upper integrals by lower integrals, and there are also analogous formulas for the projections R_2 and R_3

The reader should have no difficulty in stating analogous results for n-fold integrals (they can be proved by the method used in Theorem 10-20). The special case in which the n-fold integral $\int_R f(\mathbf{x})\,d\mathbf{x}$ exists is of particular importance and can be stated as follows.

10-21 THEOREM *Let f be defined and bounded on a closed interval*

$$R = [a_1, b_1] \times \cdots \times [a_n, b_n]$$

in E_n. Assume that $\int_R f(\mathbf{x})\, d\mathbf{x}$ exists. Then

$$\int_R f\, d\mathbf{x} = \int_{a_1}^{b_1} \left[\overline{\int_{R_1}} f\, d(x_2, \ldots, x_n) \right] dx_1$$

$$= \int_{R_1} \left[\overline{\int_{a_1}^{b_1}} f\, dx_1 \right] d(x_2, \ldots, x_n).$$

Similar formulas hold with upper integrals replaced by lower integrals and with R_1 replaced by R_k, the projection of R on Π_k.

We conclude this section with a two-dimensional analog of the fundamental theorem of integral calculus. This result will be of importance later in the proof of Green's theorem for a rectangle (Theorem 10–39).

10–22 THEOREM. *Assume that the double integral $\int_R f(x, y)\, d(x, y)$ exists, where R is a rectangle, $R = [a, b] \times [c, d]$. Let g be a function defined on R with the following properties:*

(i) *The partial derivative $D_1 g$ exists and is bounded in the interior of R and $D_1 g(x, y) = f(x, y)$ for each interior point (x, y).*

(ii) *For each fixed y in $[c, d]$, the limits $g(a+, y)$ and $g(b-, y)$ exist and satisfy the equation*

$$g(a, y) - g(a+, y) = g(b, y) - g(b-, y).$$

Then the Riemann integral $\int_c^d [g(b, y) - g(a, y)]\, dy$ exists and we have

$$\int_R f(x, y)\, d(x, y) = \int_c^d [g(b, y) - g(a, y)]\, dy.$$

NOTE. An analogous formula,

$$\int_R f(x, y)\, d(x, y) = \int_a^b [h(x, d) - h(x, c)]\, dx,$$

holds whenever h is a function satisfying a similar set of conditions [that is, $D_2 h = f$ in (i), and $h(x, c) - h(x, c+) = h(x, d) - h(x, d-)$ in (ii)]. The functions g and h will satisfy (ii) automatically if they are both continuous throughout R.

Proof. Let $P_1 = \{x_0, x_1, \ldots, x_n\}$ and $P_2 = \{y_0, y_1, \ldots, y_m\}$ be partitions of $[a, b]$ and $[c, d]$, respectively. Form the sum

$$S = \sum_{k=1}^{m} \sum_{i=1}^{n} [g(x_i, \bar{y}_k) - g(x_{i-1}, \bar{y}_k)](y_k - y_{k-1}),$$

where $y_{k-1} \le \bar{y}_k \le y_k$. Applying the one-dimensional Mean Value Theorem (Theorem 5-10), we can write

$$g(x_i, \bar{y}_k) - g(x_{i-1}, \bar{y}_k) = D_1 g(\bar{x}_{i,k}, \bar{y}_k)(x_i - x_{i-1}),$$

where $x_{i-1} < \bar{x}_{i,k} < x_i$. [Theorem 5-10 is applicable because of (i) and (ii).] Since $D_1 g = f$ in the interior, we can write

$$S = \sum_{k=1}^{m} \sum_{i=1}^{n} f(\bar{x}_{i,k}, \bar{y}_k)(x_i - x_{i-1})(y_k - y_{k-1}),$$

and this is a particular Riemann sum which approximates the double integral $\int_R f(x, y)\, d(x, y)$, assumed to exist.

A second expression for S can be obtained by observing that, in the formula which defines S, the sum over i telescopes to give us

$$S = \sum_{k=1}^{m} [g(b, \bar{y}_k) - g(a, \bar{y}_k)](y_k - y_{k-1}).$$

However, this is a general Riemann sum for the one-dimensional Riemann integral $\int_c^d [g(b, y) - g(a, y)]\, dy$, and hence the conclusion follows.

10-7 Multiple integration over more general sets. Up to this point, the multiple integral $\int_I f(\mathbf{x})\, d\mathbf{x}$ has been defined only for *intervals* I. This, of course, is too restrictive for the applications of integration. However, it is not difficult to extend the definition to arbitrary Jordan-measurable sets.

10-23 DEFINITION. *Let f be defined and bounded on a bounded Jordan-measurable set S in E_n. Let I be a closed interval containing S and define g on I as follows:*

$$g(\mathbf{x}) = \begin{cases} f(\mathbf{x}), & \text{if } \mathbf{x} \in S, \\ 0, & \text{if } \mathbf{x} \in I - S. \end{cases}$$

Then f is said to be Riemann-integrable on S and we write $f \in R$ on S, whenever the integral $\int_I g(\mathbf{x})\, d\mathbf{x}$ exists. We also write $\int_S f(\mathbf{x})\, d\mathbf{x} = \int_I g(\mathbf{x})\, d\mathbf{x}$. The upper and lower integrals $\overline{\int}_S f(\mathbf{x})\, d\mathbf{x}$ and $\underline{\int}_S f(\mathbf{x})\, d\mathbf{x}$ are similarly defined.

NOTE. By considering the Riemann sums which approximate $\int_I g(\mathbf{x})\, d\mathbf{x}$, it is easy to see that the integral $\int_S f(\mathbf{x})\, d\mathbf{x}$ does not depend on the choice of the interval I used to enclose S.

A necessary and sufficient condition for the existence of $\int_S f(\mathbf{x})\, d\mathbf{x}$ can now be given.

10–24 THEOREM. *Let S be a Jordan-measurable set in E_n and let f be defined and bounded on S. Then $f \in R$ on S if, and only if, the discontinuities of f in S form a set of measure zero.*

Proof. Let I be a closed interval containing S and let $g(\mathbf{x}) = f(\mathbf{x})$ when $\mathbf{x} \in S$, $g(\mathbf{x}) = 0$ when $\mathbf{x} \in I - S$. The discontinuities of f will be discontinuities of g. However, g may also have discontinuities at some or all of the boundary points of S. Since S has Jordan content, Theorem 10–6 tells us that $c(\partial S) = 0$. Therefore, $g \in R$ on I if, and only if, the discontinuities of f form a set of measure zero.

The next theorem shows that the integral is additive with respect to sets having Jordan content.

10–25 THEOREM. *Assume $f \in R$ on a Jordan-measurable set S in E_n. Suppose $S = A \cup B$, where A and B are nonoverlapping sets, each having Jordan content. Then $f \in R$ on A, $f \in R$ on B, and we have*

$$\int_S f(\mathbf{x})\, d\mathbf{x} = \int_A f(\mathbf{x})\, d\mathbf{x} + \int_B f(\mathbf{x})\, d\mathbf{x}. \tag{4}$$

Proof. Let I be a closed interval containing S and define g as follows:

$$g(x) = \begin{cases} f(\mathbf{x}), & \text{if } \mathbf{x} \in S, \\ 0, & \text{if } \mathbf{x} \in I - S. \end{cases}$$

The existence of $\int_A f(\mathbf{x})\, d\mathbf{x}$ and $\int_B f(\mathbf{x})\, d\mathbf{x}$ is an easy consequence of Theorem 10–24. To prove (4), let P be a partition of I into m subintervals $I_1, \ldots, I_m$ and form a Riemann sum

$$S(P, g) = \sum_{k=1}^{m} g(\mathbf{t}_k)\mu(I_k).$$

If S_A denotes that part of the sum arising from those subintervals containing points of A, and if S_B is similarly defined, we can write

$$S(P, g) = S_A + S_B - S_C,$$

where S_C contains those terms coming from subintervals which contain both points of A and points of B. In particular, all points common to the two boundaries ∂A and ∂B will fall in this third class. But now S_A is a Riemann sum approximating the integral $\int_A f(\mathbf{x})\,d\mathbf{x}$, and S_B is a Riemann sum approximating $\int_B f(\mathbf{x})\,d\mathbf{x}$. Since $c(\partial A \cap \partial B) = 0$, it follows that $|S_C|$ can be made arbitrarily small when P is sufficiently fine. The equation in the theorem is an easy consequence of these remarks.

NOTE. Formula (4) also holds for upper and lower integrals.

For sets S whose structure is relatively simple, Theorem 10-20 can be used to obtain formulas for evaluating double integrals by repeated integration. These formulas are given in the next theorem.

10-26 THEOREM. *Let ϕ_1 and ϕ_2 be two continuous functions defined on $[a, b]$ such that $\phi_1(x) \le \phi_2(x)$ for each x in $[a, b]$. Let S be the closed region in E_2 given by*

$$S = \{(x, y) \mid a \le x \le b, \phi_1(x) \le y \le \phi_2(x)\}.$$

If $f \in R$ on S, we have

$$\int_S f(x, y)\,d(x, y) = \int_a^b \left[\int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\,dy\right] dx.$$

Analogous statements hold for n-fold integrals. The extensions are too obvious to require further comment.

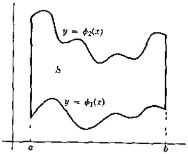


FIG. 10-4. A region in E_2 of the type described in Theorem 10-26.

Figure 10–4 illustrates the type of region described in the theorem. For sets which can be decomposed into a finite number of nonoverlapping regions of this type, we can apply repeated integration to each separate part and add the results in accordance with Theorem 10–25. Of course, all the sets under consideration must be Jordan-measurable.

10–8 Mean Value Theorem for multiple integrals. As in the one-dimensional case, multiple integrals satisfy a mean value property. This can be obtained as an easy consequence of the following theorem, the proof of which is left as an exercise.

10–27 THEOREM. *Assume $f \in R$ and $g \in R$ on a bounded set S in E_n. If $f(\mathbf{x}) \leq g(\mathbf{x})$ for each $\mathbf{x}$ in S, then we have*

$$\int_S f(\mathbf{x})\, d\mathbf{x} \leq \int_S g(\mathbf{x})\, d\mathbf{x}.$$

10–28 THEOREM (*Mean Value Theorem for multiple integrals*). *Assume that $g \in R$ and $f \in R$ on a bounded set S in E_n and suppose that $g(\mathbf{x}) \geq 0$ for each $\mathbf{x}$ in S. Let $m = \inf f(S)$, $M = \sup f(S)$. Then there exists a real number λ in the interval $m \leq \lambda \leq M$ such that*

$$\int_S f(\mathbf{x})g(\mathbf{x})\, d\mathbf{x} = \lambda \int_S g(\mathbf{x})\, d\mathbf{x}. \tag{5}$$

In particular, we have

$$mc(S) \leq \int_S f(\mathbf{x})\, d\mathbf{x} \leq Mc(S). \tag{6}$$

NOTE. If, in addition, S is connected and f is continuous on S, then $\lambda = f(\mathbf{x}_0)$ for some $\mathbf{x}_0$ in S (by Theorem 8–37) and (5) becomes

$$\int_S f(\mathbf{x})g(\mathbf{x})\, d\mathbf{x} = f(\mathbf{x}_0) \int_S g(\mathbf{x})\, d\mathbf{x}. \tag{7}$$

In particular, (7) implies $\int_S f(\mathbf{x})\, d\mathbf{x} = f(\mathbf{x}_0)c(S)$, where $\mathbf{x}_0 \in S$.

Proof. Since $g(\mathbf{x}) \geq 0$, we have $mg(\mathbf{x}) \leq f(\mathbf{x})\, g(\mathbf{x}) \leq Mg(\mathbf{x})$ for each $\mathbf{x}$ in S. By Theorem 10–27, we can write

$$m \int_S g(\mathbf{x})\, d\mathbf{x} \leq \int_S f(\mathbf{x})g(\mathbf{x})\, d\mathbf{x} \leq M \int_S g(\mathbf{x})\, d\mathbf{x}.$$

If $\int_S g(\mathbf{x})\,d\mathbf{x} = 0$, (5) holds for every λ. If $\int_S g(\mathbf{x})\,d\mathbf{x} > 0$, (5) holds with $\lambda = \int_S f(\mathbf{x})g(\mathbf{x})\,d\mathbf{x}/\int_S g(\mathbf{x})\,d\mathbf{x}$. Taking $g(\mathbf{x}) \equiv 1$, we obtain (6)

We can use (6) to prove that the integrand f can be "disturbed" on a set of content zero without affecting the value of the integral. In fact, we have the following theorem:

10–29 THEOREM. *Assume that $f \in R$ on a bounded set S in E_n. Let T be a subset of S having n-dimensional Jordan content zero. Let g be a function, defined and bounded on S, such that $g(\mathbf{x}) = f(\mathbf{x})$ when $\mathbf{x} \in S - T$. Then $g \in R$ on S and*

$$\int_S f(\mathbf{x})\,d\mathbf{x} = \int_S g(\mathbf{x})\,d\mathbf{x}.$$

Proof. Let $h = f - g$. Then $\int_S h(\mathbf{x})\,d\mathbf{x} = \int_T h(\mathbf{x})\,d\mathbf{x} + \int_{S-T} h(\mathbf{x})\,d\mathbf{x}$. However, $\int_T h(\mathbf{x})\,d\mathbf{x} = 0$ because of (6), and $\int_{S-T} h(\mathbf{x})\,d\mathbf{x} = 0$ since $h(\mathbf{x}) = 0$ for each $\mathbf{x}$ in $S - T$.

NOTE. This theorem suggests a way of extending the definition of the Riemann integral $\int_S f(\mathbf{x})\,d\mathbf{x}$ for functions which may not be defined and bounded on the *whole* of S. In fact, let S be a bounded set in E_n having Jordan content and let T be a subset of S having content zero. If f is defined and bounded on $S - T$ and if $\int_{S-T} f(\mathbf{x})\,d\mathbf{x}$ exists, we agree to write

$$\int_S f(\mathbf{x})\,d\mathbf{x} = \int_{S-T} f(\mathbf{x})\,d\mathbf{x}$$

and to say that f is Riemann-integrable on S. In view of the theorem just proved, this is essentially the same as extending the domain of definition of f to the whole of S by defining f on T in such a way that it remains bounded.

10–9 Change of variable in a multiple integral.

One of the most important results in the theory of multiple integration is the formula for making a "change of variable." This is an extension of the equation

$$\int_a^b f(x)\,dx = \int_c^d f[g(t)]g'(t)\,dt, \qquad a = g(c),\, b = g(d),$$

which was proved (in Theorem 9–33) under the assumptions that g has a continuous derivative g' on an interval S joining c and d and that f is continuous on the image $X = g(S)$. An important special case occurs when the derivative g' is never zero (and hence of constant sign) on S. For definiteness, suppose $c < d$ so that $S = [c, d]$. If g' is positive on S, we have $a < b$ and $X = g(S) = [a, b]$. In this case, the above formula can be written as follows:

$$\int_X f(x)\,dx = \int_{g^{-1}(X)} f[g(t)]g'(t)\,dt.$$

On the other hand, if g' is negative on S, then $X = g(S) = [b, a]$, and the above formula becomes

$$\int_X f(x)\,dx = -\int_{g^{-1}(X)} f[g(t)]g'(t)\,dt.$$

Both cases are included, therefore, in the *single* formula

$$\int_X f(x)\,dx = \int_{g^{-1}(X)} f[g(t)]\,|g'(t)|\,dt.$$

This last equation is also valid when $c > d$, and it is in this form that the result will be generalized to multiple integrals. The function g which "transforms the variable" must now be replaced by a vector-valued function whose Jacobian plays the role of the derivative g'.

10–30 THEOREM. *Let $\mathbf{g} = (g_1, \ldots, g_n)$ be a vector-valued function defined on an open region S in E_n. Assume that $\mathbf{g} \in C'$ on S, that $\mathbf{g}$ is one-to-one on S, and that the Jacobian $J_{\mathbf{g}}$ is never zero on S. Let f be a real-valued function defined and continuous on the image $\mathbf{g}(S)$. Then, for every Jordan-measurable compact subregion X of $\mathbf{g}(S)$, we have the formula*

$$\int_X f(\mathbf{x})\,d\mathbf{x} = \int_{\mathbf{g}^{-1}(X)} f[\mathbf{g}(\mathbf{t})]\,|J_{\mathbf{g}}(\mathbf{t})|\,d\mathbf{t}. \tag{8}$$

NOTE. Formula (8) is still valid if the Jacobian vanishes on a subset of S having content zero. (See Exercise 10–16.)

Proof. The proof is by induction on the dimension n. When $n = 1$, the result follows as a special case of Theorem 9–33. Therefore, we assume the formula is correct for $n - 1$ and show that this implies the result for n.

Since $J_{\mathbf{g}}(\mathbf{t}) \neq 0$ for each $\mathbf{t}$ in S, not all derivatives $D_k g_k(\mathbf{t})$ can be zero at any point of S. For definiteness, let us suppose that $D_n g_n(\mathbf{t}_0) \neq 0$ at some particular point $\mathbf{t}_0$. By continuity, there will be a neighborhood

$N(\mathbf{t}_0) \subset S$ such that $D_n g_n(\mathbf{t}) \neq 0$ for all $\mathbf{t}$ in $N(\mathbf{t}_0)$. We will now show that there is an open set $A \subset N(\mathbf{t}_0)$ in which $\mathbf{g}$ can be expressed as the composition of two vector-valued functions $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$ and $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_n)$ having rather special properties.

First, define $\boldsymbol{\phi}$ as follows:

$$\boldsymbol{\phi}(\mathbf{t}) = (t_1, \ldots, t_{n-1}, g_n(\mathbf{t})), \qquad \text{if } \mathbf{t} \in S.$$

That is, $\boldsymbol{\phi}(\mathbf{t})$ leaves all components of $\mathbf{t}$ fixed except for t_n, which is replaced by $g_n(\mathbf{t})$. Clearly, $\boldsymbol{\phi}$ is one-to-one on S and $\boldsymbol{\phi} \in C'$ on S. Moreover, the Jacobian of $\boldsymbol{\phi}$ is $D_n g_n$ and, since this is not zero in $N(\mathbf{t}_0)$, we can apply the inverse function theorem (Theorem 7–5) to obtain a "local inverse" function $\boldsymbol{\psi}$ and two open sets $A \subset N(\mathbf{t}_0)$ and $B \subset \boldsymbol{\phi}(S)$ such that $\mathbf{t}_0 \in A$, $\boldsymbol{\phi}(\mathbf{t}_0) \in B$, $B = \boldsymbol{\phi}(A)$, $\boldsymbol{\psi} \in C'$ on B, and

$$\boldsymbol{\psi}[\boldsymbol{\phi}(\mathbf{t})] = \mathbf{t} \qquad \text{for each } \mathbf{t} \text{ in } A.$$

Now define $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$ as follows. If $\mathbf{t} \in B$, let

$$\theta_k(\mathbf{t}) = g_k(t_1, \ldots, t_{n-1}, \psi_n(\mathbf{t})) \qquad \text{for } 1 \leq k \leq n-1,$$

and let $\theta_n(\mathbf{t}) = t_n$. Then, for each $\mathbf{t}$ in A, we have

$$\theta_k[\boldsymbol{\phi}(\mathbf{t})] = g_k(t_1, \ldots, t_{n-1}, \psi_n[\boldsymbol{\phi}(\mathbf{t})]) = g_k(t_1, \ldots, t_n) = g_k(\mathbf{t}), \qquad \text{if } 1 \leq k \leq n-1,$$

and $\theta_n[\boldsymbol{\phi}(\mathbf{t})] = \phi_n(\mathbf{t}) = g_n(\mathbf{t})$. Hence we have

$$\boldsymbol{\theta}[\boldsymbol{\phi}(\mathbf{t})] = \mathbf{g}(\mathbf{t}) \qquad \text{for every } \mathbf{t} \text{ in } A.$$

That is to say, in the open set A containing $\mathbf{t}_0$, $\mathbf{g}$ has been expressed as a composite function $\mathbf{g} = \boldsymbol{\theta}\boldsymbol{\phi}$, where $\boldsymbol{\phi}(\mathbf{t})$ leaves all components of $\mathbf{t}$ fixed except for t_n, and $\boldsymbol{\theta}(\mathbf{t})$ leaves t_n fixed. Furthermore, $\boldsymbol{\theta}$ is one-to-one and $\boldsymbol{\theta} \in C'$ on B.

Of course, all this was based on the assumption that $D_n g_n(\mathbf{t}_0) \neq 0$. If, instead, we had assumed that $D_k g_k(\mathbf{t}_0) \neq 0$, there would again be an open set A in which $\mathbf{g}$ was the composition $\mathbf{g} = \boldsymbol{\theta}\boldsymbol{\phi}$, except that now $\boldsymbol{\phi}(\mathbf{t})$ would leave all components of $\mathbf{t}$ fixed except t_k, and $\boldsymbol{\theta}(\mathbf{t})$ would leave t_k fixed.

Now let X be a Jordan-measurable compact subregion of $\mathbf{g}(S)$. The inverse image $\mathbf{g}^{-1}(X)$ is then a compact subset of S. For each $\mathbf{t}_0$ in $\mathbf{g}^{-1}(X)$, there corresponds an open set A containing $\mathbf{t}_0$ in which $\mathbf{g}$ is a composition of the form $\mathbf{g} = \boldsymbol{\theta}\boldsymbol{\phi}$, where $\boldsymbol{\phi}(\mathbf{t})$ leaves all components of $\mathbf{t}$ fixed except one, and $\boldsymbol{\theta}(\mathbf{t})$ leaves this one component fixed. (There is no loss in generality in assuming that each such open set A is an open *interval.*) As we vary $\mathbf{t}_0$ over the set $\mathbf{g}^{-1}(X)$, these open intervals A form an open covering of $\mathbf{g}^{-1}(X)$ and (by compactness) a finite collection of these intervals, say $A^{(1)}, \ldots, A^{(p)}$, covers $\mathbf{g}^{-1}(X)$. (We can also assume that these intervals are nonoverlapping.) A corresponding finite collection of vector-valued functions $\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(p)}$ and $\boldsymbol{\phi}^{(1)}, \ldots, \boldsymbol{\phi}^{(p)}$ exists such that

$$\mathbf{g}(\mathbf{t}) = \boldsymbol{\theta}^{(k)}[\boldsymbol{\phi}^{(k)}(\mathbf{t})]$$

for each $\mathbf{t}$ in $A^{(k)}$. For each k, $\boldsymbol{\phi}^{(k)}(\mathbf{t})$ leaves all components of $\mathbf{t}$ fixed except one, and $\boldsymbol{\theta}^{(k)}$ leaves this one component fixed. (This preferred component need not be the same for all k.) By Theorem 7–2, we have the product formula

$$J_{\mathbf{g}}(\mathbf{t}) = J_{\boldsymbol{\theta}^{(k)}}[\boldsymbol{\phi}^{(k)}(\mathbf{t})]J_{\boldsymbol{\phi}^{(k)}}(\mathbf{t}), \qquad (k = 1, 2, \ldots, p)$$

relating the Jacobians. It is important to observe that this product is *independent* of k.

Let $T^{(k)} = \mathbf{g}(A^{(k)})$ denote the image of the interval $A^{(k)}$ under $\mathbf{g}$. (See Fig. 10–5.) The hypothesis that $\mathbf{g}$ is one-to-one on S implies that the sets $T^{(1)}, \ldots, T^{(p)}$ are nonoverlapping. Moreover, we have

$$X \subset T^{(1)} \cup \cdots \cup T^{(p)} \subset \mathbf{g}(S).$$

Therefore, if X has Jordan content, we can write

$$\int_X f(\mathbf{x})\, d\mathbf{x} = \sum_{k=1}^{p} \int_{T^{(k)}} F(\mathbf{x})\, d\mathbf{x},$$

where $F(\mathbf{x}) = f(\mathbf{x})$ if $\mathbf{x} \in X$, and $F(\mathbf{x}) = 0$ if $\mathbf{x} \in \mathbf{g}(S) - X$. The proof of the theorem will be complete if we can show that the transformation formula

$$\int_{T^{(k)}} F(\mathbf{x})\, d\mathbf{x} = \int_{\mathbf{g}^{-1}(T^{(k)})} F[\mathbf{g}(\mathbf{t})]\, |J_{\mathbf{g}}(\mathbf{t})|\, dt \tag{9}$$

holds for each $k = 1, 2, \ldots, p$.

Dropping superscripts, suppose that $\mathbf{g} = \boldsymbol{\theta}\boldsymbol{\phi}$ in the interval A and, for definiteness, assume that $\boldsymbol{\phi}(\mathbf{t})$ leaves $t_1, \ldots, t_{n-1}$ fixed, whereas $\boldsymbol{\theta}(\mathbf{t})$ leaves t_n fixed. By Theorem 10–21, we can write

$$\int_T F(\mathbf{x})\, d\mathbf{x} = \int_I \left[\int_{T_n} F^*(x_1, \ldots, x_n)\, d(x_1, \ldots, x_{n-1})\right] dx_n, \tag{10}$$

where $T = \mathbf{g}(A)$, T_n is the projection of T on the coordinate plane Π_n, I is a one-dimensional interval such that $T \subset I \times T_n$, and $F^*(\mathbf{x}) = f_T(\mathbf{x})F(\mathbf{x})$, where f_T is the characteristic function of T. The inner integral in (10) is an $(n-1)$-fold integral to which we apply the induction hypothesis. If we make the $(n-1)$-dimensional change of variable

$$x_1 = \theta_1(t_1, \ldots, t_{n-1}, x_n), \quad \ldots, x_{n-1} = \theta_{n-1}(t_1, \ldots, t_{n-1}, x_n),$$

the inner integral becomes

$$(11) \quad \int_{\theta^{-1}(T_n)} G(t_1, \ldots, t_{n-1}, x_n)|J_\theta(t_1, \ldots, t_{n-1}, x_n)|\, d(t_1, \ldots, t_{n-1}),$$

where

$$\begin{aligned} G(t_1, &\ldots, t_{n-1}, x_n) \\ &= F^*[\theta_1(t_1, \ldots, t_{n-1}, x_n), \ldots, \theta_{n-1}(t_1, \ldots, t_{n-1}, x_n), x_n]. \end{aligned}$$

In the outer integral of (10) we make the one-dimensional change of variable $x_n = \phi_n(\mathbf{t})$ and use (11) to obtain

$$\begin{aligned} \int_T F(\mathbf{x})\, d\mathbf{x} \\ &= \int_{\phi^{-1}(I)} \left[\int_{\theta^{-1}(T_n)} G[\phi(\mathbf{t})]\, |J_\theta[\phi(\mathbf{t})]|\, |J_\phi(\mathbf{t})|\, d(t_1, \ldots, t_{n-1})\right] dt_n \\ &= \int_{\mathbf{g}^{-1}(T)} F[\mathbf{g}(\mathbf{t})]\, |J_{\mathbf{g}}(\mathbf{t})|\, d\mathbf{t}, \end{aligned}$$

since $\phi^{-1}(I)$ is an interval such that $\mathbf{g}^{-1}(T) \subset \phi^{-1}(I) \times \theta^{-1}(T_n)$. A similar argument can be applied to each of the integrals in (9) and the theorem follows.

Theorem 10-30 can be used to give the Jacobian a geometric interpretation.

10-31 THEOREM *Under the hypotheses of Theorem 10-30, we have*

$$c(X) = |J_{\mathbf{g}}(\mathbf{y}_0)|c(Y),$$

where Y is any compact subregion of S having Jordan content, $X = \mathbf{g}(Y)$, and $\mathbf{y}_0 \in Y$.

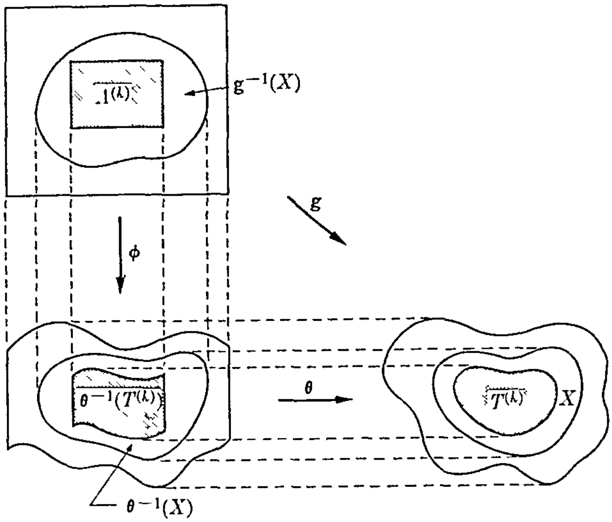


FIG. 10–5. Proof of Theorem 10–30 when $n = 2$.

Proof. If we take $f(\mathbf{x}) = 1$ for all $\mathbf{x}$ in X, Theorem 10–30 yields

$$c(X) = \int_Y |J_{\mathbf{g}}(\mathbf{t})|\,d\mathbf{t}.$$

Since Y is connected, the result follows at once from Theorem 10–28.

NOTE. The formula in Theorem 10–31 shows that the absolute value of the Jacobian is, in a sense, a "local magnification factor" for Jordan content.

10–10 Line integrals. The multiple integral has been introduced as a generalization of the one-dimensional Riemann integral $\int_a^b f(x)\,dx$. There is another way of extending the notion of integral in which the interval $[a, b]$ is replaced by a curve in E_n described by a vector-valued function $\alpha = (\alpha_1, \ldots, \alpha_n)$. In this generalization the integrand is a vector-valued function $\mathbf{f} = (f_1, \ldots, f_n)$ defined and bounded on this curve and the resulting integral is called a *line integral* or a *curvilinear integral*, denoted by $\int \mathbf{f} \cdot d\alpha$ or by some similar symbol. The dot in this symbol is purposely used to suggest an inner product of two vectors. In fact, line integrals can be thought of as generalizations of Riemann-Stieltjes integrals in which both the integrand $\mathbf{f}$ and the integrator α are vector-valued functions.

Line integrals arise quite naturally in certain types of physical problems involving vector fields For example, the *work* done by a force-field $\mathbf{f}$ in moving a particle along the curve described by $\boldsymbol{\alpha}$ turns out to be the numerical value of the line integral $\int \mathbf{f} \cdot d\boldsymbol{\alpha}$

Line integrals are defined as follows

10–32 DEFINITION *Let Γ be a curve in E_n described by a continuous vector-valued function $\boldsymbol{\alpha} = (\alpha_1, \quad, \alpha_n)$ defined on $[a, b]$ Let*

$$P = \{t_0, t_1, \quad, t_m\}$$

be a partition of $[a, b]$ and assume that $v_k \in [t_{k-1}, t_k]$ If

$$\mathbf{f} = (f_1, \quad, f_n)$$

is a vector-valued function defined on Γ, we form the sum of inner products

$$S(P, \mathbf{f}, \boldsymbol{\alpha}) = \sum_{k=1}^{m} \mathbf{f}[\boldsymbol{\alpha}(v_k)] \quad [\boldsymbol{\alpha}(t_k) - \boldsymbol{\alpha}(t_{k-1})] \tag{12}$$

We say that the line integral of $\mathbf{f}$ with respect to $\boldsymbol{\alpha}$ along the path $\Gamma[\boldsymbol{\alpha}]$ exists if there is a real number A having the following property: For every $\epsilon > 0$, there is a partition P_ϵ of $[a, b]$ such that for every partition P finer than P_ϵ and for every choice of the points v_k in $[t_{k-1}, t_k]$, we have

$$|A - S(P, \mathbf{f}, \boldsymbol{\alpha})| < \epsilon$$

When such a number A exists, it is uniquely determined and is denoted by the symbol $\int_{\Gamma[\boldsymbol{\alpha}]} \mathbf{f} \cdot d\boldsymbol{\alpha}$ We also define

$$\int_{-\Gamma[\boldsymbol{\alpha}]} \mathbf{f} \cdot d\boldsymbol{\alpha} = -\int_{\Gamma[\boldsymbol{\alpha}]} \mathbf{f} \quad d\boldsymbol{\alpha}$$

NOTE If $\mathbf{x} = \boldsymbol{\alpha}(a)$ and $\mathbf{y} = \boldsymbol{\alpha}(b)$, we also write $\int_{\Gamma(\mathbf{x}, \mathbf{y})}$ instead of $\int_{\Gamma[\boldsymbol{\alpha}]}$.

Observe that in the special case $n = 1$, Definition 10–32 is the same as the definition of the Riemann-Stieltjes integral given in Chapter 9 For general n, line integrals are related to Riemann-Stieltjes integrals by the following theorem·

10–33 THEOREM *Let Γ be a curve in E_n described by a continuous vector-valued function $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)$ defined on $[a, b]$, and let $\mathbf{f} = (f_1, \ldots, f_n)$ be defined and bounded on Γ. Then we have*

$$\int_{\Gamma[\alpha]} \mathbf{f} \cdot d\alpha = \sum_{r=1}^{n} \int_a^b f_r[\alpha(t)]\, d\alpha_r(t) \tag{13}$$

whenever each Riemann-Stieltjes integral on the right exists.

Proof. The sum (12) defining $S(P, \mathbf{f}, \alpha)$ can be written in the form

$$S(P, \mathbf{f}, \alpha) = \sum_{r=1}^{n} \sum_{k=1}^{m} f_r[\alpha(v_k)][\alpha_r(t_k) - \alpha_r(t_{k-1})].$$

The proof follows by observing that, for each r, the inner sum is a Riemann-Stieltjes sum approximating $\int_a^b f_r[\alpha(t)]\, d\alpha_r(t)$.

NOTE. For most line integrals which occur in practice, Γ is *rectifiable* and f is *continuous* on Γ. In this case, Theorem 9–26 guarantees the existence of each integral on the right of equation (13).

Because of Theorem 10–33, the notation $f_1\, d\alpha_1 + \cdots + f_n\, d\alpha_n$ is also used in place of the symbol $\mathbf{f} \cdot d\alpha$ which appears under the integral sign. For curves in E_2, it is a common practice to write the parametric equations of Γ in the form

$$x = x(t), \qquad y = y(t), \qquad a \le t \le b.$$

Consequently, if α is the vector-valued function defined by

$$\alpha(t) = (x(t), y(t)),$$

the line integral $\int_{\Gamma[\alpha]} \mathbf{f} \cdot d\alpha$ is written in the form $\int_{\Gamma[\alpha]} f_1\, dx + f_2\, dy$. In case Γ is a rectifiable *Jordan* curve (in E_2), the symbol $\oint_\Gamma$ is used instead of $\int_{\Gamma[\alpha]}$ if $\Gamma[\alpha]$ is a positively oriented path. (See Definition 9–69.)

NOTE. Despite similarity in appearance, the complex-valued contour integral $\int_{\Gamma[z]} F(z)\, dz$ of Definition 9–56 is conceptually distinct from the real-valued line integral obtained by putting $n = 2$ in Definition 10–32. However, there is a connection between the two types of integrals. (See Exercise 10–21.)

Since the line integral is defined in essentially the same way as a Riemann-Stieltjes integral, it is not surprising that many properties of Stieltjes integrals also hold for line integrals. In particular, the linearity property

$$\int_{\Gamma[\alpha]} (c_1\mathbf{f} + c_2\mathbf{g}) \cdot d\alpha = c_1\int_{\Gamma[\alpha]} \mathbf{f} \cdot d\alpha + c_2\int_{\Gamma[\alpha]} \mathbf{g} \cdot d\alpha$$

follows as in the proof of Theorem 9-2 We also have the following additive property which is analogous to Theorem 9-4

10-34 THEOREM. *Let Γ denote a curve described by a vector-valued function α and assume that $\Gamma[\alpha] = \Gamma_1[\alpha] + \Gamma_2[\alpha]$ Whenever two of the three line integrals in (14) exist, the third also exists and we have*

$$\int_{\Gamma[\alpha]} \mathbf{f} \cdot d\alpha = \int_{\Gamma_1[\alpha]} \mathbf{f} \cdot d\alpha + \int_{\Gamma_2[\alpha]} \mathbf{f} \cdot d\alpha \tag{14}$$

The proof is similar to that of Theorem 9-4

The effect of replacing the integrator α by an equivalent integrator is, at worst, a change in sign In fact, we have the following theorem:

10-35 THEOREM *Let α and β be two equivalent functions describing the same curve Γ If $\int_{\Gamma[\alpha]} \mathbf{f} \cdot d\alpha$ exists, then $\int_{\Gamma[\beta]} \mathbf{f} \cdot d\beta$ also exists and we have*

$$\int_{\Gamma[\alpha]} \mathbf{f} \cdot d\alpha = \pm \int_{\Gamma[\beta]} \mathbf{f} \cdot d\beta,$$

where the plus sign holds if $\Gamma[\alpha] = \Gamma[\beta]$ and the minus sign holds if $\Gamma[\alpha] = -\Gamma[\beta]$

The proof follows by an argument similar to that given in the proof of Theorem 9-7 Details are left to the reader (See Exercise 10-22)

For *rectifiable* curves the following theorem is very useful for estimating the absolute value of a line integral

10-36 THEOREM *Let Γ be a rectifiable curve of length $\Lambda(\Gamma)$ described by a continuous function α defined on $[a, b]$. If $\mathbf{f}$ is defined and bounded on Γ, say $|\mathbf{f}(\mathbf{x})| \le M$ for each $\mathbf{x}$ on Γ, and if $\int_{\Gamma[\alpha]} \mathbf{f} \cdot d\alpha$ exists, then we have*

$$\left| \int_{\Gamma[\alpha]} \mathbf{f} \cdot d\alpha \right| \le M\Lambda(\Gamma).$$

Proof Because of the inequality $|\mathbf{a} \cdot \mathbf{b}| \le |\mathbf{a}||\mathbf{b}|$, each sum $S(P, \mathbf{f}, \alpha)$ in (12) satisfies the relation

$$|S(P, \mathbf{f}, \boldsymbol{\alpha})| \leq M \sum_{k=1}^{m} |\boldsymbol{\alpha}(t_k) - \boldsymbol{\alpha}(t_{k-1})| \leq M\Lambda(\Gamma).$$

Hence, $|\int_{\Gamma[\boldsymbol{\alpha}]} \mathbf{f} \cdot d\boldsymbol{\alpha}|$ cannot exceed $M\Lambda(\Gamma)$.

10–11 Line integrals with respect to arc length. Let Γ be a rectifiable *simple* curve, of length $\Lambda(\Gamma)$, described by a function $\boldsymbol{\alpha}$ defined on $[a, b]$. There exists a function $\boldsymbol{\beta}$, defined on $[0, \Lambda(\Gamma)]$, such that $\boldsymbol{\alpha}(t) = \boldsymbol{\beta}[s(t)]$, where s is the arc length, defined by $s(x) = \Lambda_{\boldsymbol{\alpha}}(a, x)$ if $a < x \leq b$, $s(a) = 0$. (See Exercise 8–11.) Since $\boldsymbol{\beta}$ is properly equivalent to $\boldsymbol{\alpha}$, we can write $\int_{\Gamma[\boldsymbol{\alpha}]} \mathbf{f} \cdot d\boldsymbol{\alpha} = \int_{\Gamma[\boldsymbol{\beta}]} \mathbf{f} \cdot d\boldsymbol{\beta}$ if either integral exists. On the other hand, we also have

$$\int_{\Gamma[\boldsymbol{\beta}]} \mathbf{f} \cdot d\boldsymbol{\beta} = \sum_{r=1}^{n} \int_{0}^{\Lambda(\Gamma)} f_r[\boldsymbol{\beta}(s)]\, d\beta_r(s),$$

provided, of course, that each Riemann-Stieltjes integral on the right exists. If the resulting function $\boldsymbol{\beta}$ is piece-wise smooth, the integral can also be written as a single Riemann integral, since we have

$$\sum_{r=1}^{n} \int_{0}^{\Lambda(\Gamma)} f_r[\boldsymbol{\beta}(s)]\beta_r'(s)\, ds = \int_{0}^{\Lambda(\Gamma)} \mathbf{f}[\boldsymbol{\beta}(s)] \cdot \boldsymbol{\beta}'(s)\, ds, \tag{15}$$

where $\boldsymbol{\beta}' = (\beta_1', \ldots, \beta_n')$. The Riemann integral on the right of (15) is often denoted by the symbol $\int_\Gamma \mathbf{f} \cdot \boldsymbol{\beta}'\, ds$ and it is referred to as a "line integral with respect to arc length." Thus the equation

$$\int_{\Gamma} \mathbf{f} \cdot \boldsymbol{\beta}'\, ds = \int_{0}^{\Lambda(\Gamma)} \mathbf{f}[\boldsymbol{\beta}(s)] \cdot \boldsymbol{\beta}'(s)\, ds$$

serves to define the symbol on the left, whenever the integral on the right exists.

10–12 The line integral of a gradient. The fundamental theorem of integral calculus involves the formula

$$\int_a^b f'(x)\, dx = f(b) - f(a).$$

The importance of this formula for theoretical purposes is that it establishes a connection between differentiation and Riemann integration. It is also useful for computational purposes because it gives us an elegant method for evaluating a Riemann integral in terms of an expression involving only the *endpoints* of the interval $[a, b]$, whenever the integrand is a derivative In the next theorem we derive the analog of this result for line integrals

10-37 THEOREM *Let* $\mathbf{f}$ *be continuous on an open region* $S \subset E_n$, *and assume that there exists a real-valued function* ϕ *such that* $\phi \in C'$ *on* S *and such that* $\mathbf{f}(\mathbf{x}) = \nabla\phi(\mathbf{x})$ *for each* $\mathbf{x}$ *in* S *Then, for every pair of points* $\mathbf{x}$ *and* $\mathbf{y}$ *in* S *and for every piece-wise smooth curve* Γ *in* S *joining* $\mathbf{x}$ *and* $\mathbf{y}$, *we have*

$$\int_{\Gamma(\mathbf{x},\mathbf{y})} \mathbf{f} \quad d\boldsymbol{\alpha} = \int_{\Gamma(\mathbf{x},\mathbf{y})} \nabla\phi \cdot d\boldsymbol{\alpha} = \phi(\mathbf{y}) - \phi(\mathbf{x}),$$

where $\boldsymbol{\alpha} \in \Gamma(\mathbf{x}, \mathbf{y})$

NOTE A real-valued function ϕ such that $\mathbf{f} = \nabla\phi$ is called a *potential function* for the vector-valued function $\mathbf{f}$.

Proof. Let $\boldsymbol{\alpha} = (\alpha_1, \quad, \alpha_n)$ be a piece-wise smooth function defined on $[a, b]$ such that $\boldsymbol{\alpha}$ describes Γ and $\boldsymbol{\alpha}(a) = \mathbf{x}$, $\boldsymbol{\alpha}(b) = \mathbf{y}$ Write $\boldsymbol{\alpha}' = (\alpha_1', \ldots, \alpha_n')$ and subdivide $[a, b]$ into a finite number of subintervals $[t_{k-1}, t_k]$ such that $\boldsymbol{\alpha}'$ is continuous in each open interval (t_{k-1}, t_k), $a = t_0 < t_1 < \cdots < t_m = b$ Define a real-valued function F on $[a, b]$ by the equation

$$F(t) = \phi[\boldsymbol{\alpha}(t)], \qquad \text{if } t \in [a, b]$$

By the chain rule, the equation

$$F'(t) = \nabla\phi[\boldsymbol{\alpha}(t)] \cdot \boldsymbol{\alpha}'(t) = \mathbf{f}[\boldsymbol{\alpha}(t)] \cdot \boldsymbol{\alpha}'(t) \tag{16}$$

is valid in each open interval (t_{k-1}, t_k) By Theorem 10-33, we have

$$\int_{\Gamma(\mathbf{x},\mathbf{y})} \mathbf{f} \quad d\boldsymbol{\alpha} = \sum_{r=1}^{n} \int_a^b f_r[\boldsymbol{\alpha}(t)]\, d\alpha_r(t), \tag{17}$$

and, by Theorem 9-8, we have

$$\int_a^b f_r[\boldsymbol{\alpha}(t)]\, d\alpha_r(t) = \sum_{k=1}^{m} \int_{t_{k-1}}^{t_k} f_r[\boldsymbol{\alpha}(t)]\, \alpha_r'(t)\, dt. \tag{18}$$

Combining (16), (17), and (18), we get

$$(19)\quad \int_{\Gamma(\mathbf{x},\mathbf{y})} \mathbf{f}\cdot d\alpha = \sum_{k=1}^{m}\int_{t_{k-1}}^{t_k} F'(t)\,dt = \sum_{k=1}^{m}[F(t_k) - F(t_{k-1})]$$

$$= F(b) - F(a) = \phi(\mathbf{y}) - \phi(\mathbf{x}).$$

NOTE. Observe that the value of the integral in Theorem 10-37 is *independent* of the particular curve Γ used to join $\mathbf{x}$ and $\mathbf{y}$ (provided that Γ is piece-wise smooth). Remarkably enough, this property suffices to establish the existence of a potential function ϕ. In fact, we have the following theorem:

10-38 THEOREM. *Let* $\mathbf{f} = (f_1, \ldots, f_n)$ *be continuous on an open region* $S \subset E_n$. *Let* $\mathbf{x}_0$ *be a fixed point in* S. *For each* $\mathbf{x}$ *in* S *and for every pair of polygonal curves* Γ_1 *and* Γ_2 *in* S *joining* $\mathbf{x}_0$ *to* $\mathbf{x}$, *assume that we have*

$$\int_{\Gamma_1(\mathbf{x}_0,\mathbf{x})} \mathbf{f}\cdot d\alpha = \int_{\Gamma_2(\mathbf{x}_0,\mathbf{x})} \mathbf{f}\cdot d\beta$$

whenever $\alpha \in \Gamma_1(\mathbf{x}_0, \mathbf{x})$ *and* $\beta \in \Gamma_2(\mathbf{x}_0, \mathbf{x})$. *Then there exists a function* ϕ *such that* $\phi \in C'$ *on* S *and such that* $\mathbf{f}(\mathbf{x}) = \nabla\phi(\mathbf{x})$ *for each* $\mathbf{x}$ *in* S.

Proof. For each $\mathbf{x}$ in S, let Γ be a polygonal curve in S joining $\mathbf{x}_0$ to $\mathbf{x}$ (such a curve exists, since S is polygonally connected) and define

$$\phi(\mathbf{x}) = \int_{\Gamma(\mathbf{x}_0,\mathbf{x})} \mathbf{f}\cdot d\alpha,$$

where $\alpha \in \Gamma(\mathbf{x}_0, \mathbf{x})$. By hypothesis, the number $\phi(\mathbf{x})$ is completely determined by $\mathbf{x}$ and does not depend on the particular polygonal curve used to join $\mathbf{x}_0$ and $\mathbf{x}$. We shall complete the proof by showing that each partial derivative $D_k\phi(\mathbf{x})$ exists and equals $f_k(\mathbf{x})$, $k = 1, 2, \ldots, n$.

For this purpose take a neighborhood $N(\mathbf{x}; \delta) \subset S$, let λ be a real number satisfying $0 < |\lambda| \le \delta$, and consider the difference quotient

$$\frac{\phi(\mathbf{x} + \lambda\mathbf{u}_k) - \phi(\mathbf{x})}{\lambda}.$$

(As usual, $\mathbf{u}_k$ is the kth unit coordinate vector.) By additivity of the integral, we can write

$$\phi(\mathbf{x} + \lambda \mathbf{u}_k) - \phi(\mathbf{x}) = \int_{\Gamma(\mathbf{x}, \mathbf{x}+\lambda\mathbf{u}_k)} \mathbf{f} \cdot d\alpha,$$

where the curve Γ which joins $\mathbf{x}$ and $\mathbf{x} + \lambda \mathbf{u}_k$ is immaterial so long as it is a polygonal curve lying in S. One such path can be given as follows:

$$\alpha(t) = (1 - t)\mathbf{x} + t(\mathbf{x} + \lambda \mathbf{u}_k), \qquad \text{where} \qquad 0 \le t \le 1.$$

The curve so described is a line segment joining $\mathbf{x}$ to $\mathbf{x} + \lambda \mathbf{u}_k$, and hence lies in $N(\mathbf{x}, \delta)$. Each component of α has a continuous derivative on $(0, 1)$ given by

$$\alpha_i'(t) = 0, \qquad \text{if } i \neq k, \qquad \alpha_k'(t) = \lambda, \qquad \text{if } t \in (0, 1).$$

Therefore we can write

$$\phi(\mathbf{x} + \lambda \mathbf{u}_k) - \phi(\mathbf{x}) = \sum_{i=1}^{n} \int_0^1 f_i[\alpha(t)]\alpha_i'(t)\, dt$$

$$= \lambda \int_0^1 f_k[\alpha(t)]\, dt = \lambda \int_0^1 g(t, \lambda)\, dt,$$

where $g(t, \lambda) = f_k[(1 - t)\mathbf{x} + t(\mathbf{x} + \lambda \mathbf{u}_k)]$. Since g is continuous on the closed rectangle

$$\{(t, \lambda) \mid 0 \le t \le 1, -\delta \le \lambda \le \delta\},$$

we can apply Theorem 9–35 to obtain

$$\lim_{\lambda \to 0} \int_0^1 g(t, \lambda)\, dt = \int_0^1 g(t, 0)\, dt = \int_0^1 f_k(\mathbf{x})\, dt = f_k(\mathbf{x}).$$

This proves that

$$\lim_{\lambda \to 0} \frac{\phi(\mathbf{x} + \lambda \mathbf{u}_k) - \phi(\mathbf{x})}{\lambda} = f_k(\mathbf{x}), \qquad (k = 1, 2, \ldots, n)$$

and hence $\nabla\phi(\mathbf{x}) = \mathbf{f}(\mathbf{x})$ for each $\mathbf{x}$ in S.

Of course, the potential function ϕ so constructed depends on the choice of the point $\mathbf{x}_0$. If ψ is another function such that $\nabla\psi = \mathbf{f}$, then $\nabla(\phi - \psi) = \mathbf{0}$ throughout S and this means that ϕ and ψ can differ at most by an additive constant. [Why?]

10–13 Green's theorem for rectangles. There is a two-dimensional analog of the fundamental theorem of integral calculus which expresses a double integral $\int_R f(x, y)\, d(x, y)$ as a line integral taken along a closed path determined by the boundary of R, whenever the integrand f is a partial derivative. This result is usually attributed to G. Green (1828) and is known as *Green's theorem.* [Actually, it appeared much earlier in the works of Gauss (1813) and Lagrange (1760).]

The proof of Green's theorem presents no particular difficulties when the region R is of a rather simple type such as a rectangle or a disk. Ultimately we shall prove Green's theorem for every region R in E_2 bounded by a rectifiable Jordan curve. For the present, however, we shall confine ourselves to the case in which R is a rectangle.

10–39 THEOREM. *(Green's theorem for rectangles). Let P and Q be two real-valued functions defined and continuous on a closed rectangle*

$$R = [a, b] \times [c, d].$$

Assume that the partial derivatives D_1Q and D_2P exist and are bounded in the interior of R and that both double integrals $\int_R D_1Q(x, y)\, d(x, y)$ and $\int_R D_2P(x, y)\, d(x, y)$ exist. If $\alpha = (\alpha_1, \alpha_2) \in \Gamma(R)$, where $\Gamma(R)$ is the positively oriented boundary of R, the line integral

$$\int_{\Gamma(R)} P\, d\alpha_1 + Q\, d\alpha_2$$

also exists and we have

$$\int_R [D_1Q(x, y) - D_2P(x, y)]\, d(x, y) = \int_{\Gamma(R)} P\, d\alpha_1 + Q\, d\alpha_2. \tag{20}$$

NOTE. This formula is usually written as follows:

$$\iint\limits_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy = \int_{\Gamma(R)} P\, dx + Q\, dy.$$

Proof. The hypotheses of Theorem 10–22 are satisfied when $g = Q$ and $h = P$. The conclusion of Theorem 10–22 then yields

$$\int_R (D_1Q - D_2P)\, d(x, y) = \int_c^d [Q(b, y) - Q(a, y)]\, dy - \int_a^b [P(x, d) - P(x, c)]\, dx. \tag{21}$$

On the other hand (using the notation of Definition 9–70), we have $\Gamma(R) = \Gamma_1(A, B) + \Gamma_2(B, C) + \Gamma_3(C, D) + \Gamma_4(D, A)$ and hence

$$(22) \qquad \int_{\Gamma(R)} = \int_{\Gamma_1(A,B)} + \int_{\Gamma_2(B\ C)} + \int_{\Gamma_3(C\ D)} + \int_{\Gamma_4(D\ A)}$$

The path $\Gamma_1(A, B)$ is described by the function $\alpha = (\alpha_1, \alpha_2)$, where

$$\alpha_1(t) = t, \quad \alpha_2(t) = c, \quad a \leq t \leq b$$

(See Fig 10–6) Thus, Theorem 10–33 yields

$$\int_{\Gamma_1(A,B)} P\,d\alpha_1 + Q\,d\alpha_2 = \int_a^b P(t, c)\,dt$$

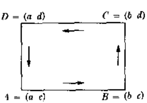


FIG 10–6 Geometric representation of the path $\Gamma(R)$

Similarly, we find

$$\int_{\Gamma_2(B\ C)} = \int_c^d Q(b, t)\,dt, \qquad \int_{\Gamma_3(C\ D)} = -\int_a^b P(t, d)\,dt,$$

$$\int_{\Gamma_4(D\ A)} = -\int_c^d Q(a, t)\,dt$$

Combining these integrals as indicated in (22) and comparing with (21), we find that the theorem follows at once

10–14 Green's theorem for regions bounded by rectifiable Jordan curves. By a *Jordan region* we shall mean the union of a rectifiable Jordan curve (in E_2) with its interior. In this section we will prove that formula (20) of Green's theorem is valid whenever R is a Jordan region The proof of Green's theorem for *rectangles* was particularly simple because we were able to express both the double integral and the line integral in (20) in terms of Riemann integrals. It is clear that this method will work for many other regions (for example, circles or polygons) However, we have no theorems which allow us to evaluate either the double integral or the line integral in (20) when R is an *arbitrary* region bounded by a rectifiable Jordan curve. Therefore, a proof of Green's theorem in this degree of generality requires a different approach. The next three theorems prepare the ground for such a proof.

10–40 THEOREM. *Let Γ be a rectifiable Jordan curve in E_2 described by a vector-valued function $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)$. If $\Lambda(\Gamma)$ is the length of Γ and if P is a real-valued function defined and continuous on Γ, we have*

$$\left|\int_{\Gamma[\boldsymbol{\alpha}]} P\, d\alpha_1\right| \leq \tfrac{1}{2}(M - m)\Lambda(\Gamma),$$

where M and m denote, respectively, the maximum and minimum of P on Γ.

NOTE. A similar result holds for the integral $\int_{\Gamma[\boldsymbol{\alpha}]} P\, d\alpha_2$.

Proof. If c is a constant, we have $\int_{\Gamma[\boldsymbol{\alpha}]} c\, d\alpha_1 = 0$. Hence

$$\int_{\Gamma[\boldsymbol{\alpha}]} P\, d\alpha_1 = \int_{\Gamma[\boldsymbol{\alpha}]} (P - c)\, d\alpha_1.$$

Taking $c = \frac{1}{2}(M + m)$, we have $|P(x, y) - c| \leq \frac{1}{2}(M - m)$ for each (x, y) on Γ, and the result follows by applying Theorem 10–36.

10–41 THEOREM. *Let Γ be a rectifiable simple curve in E_2 of length $\Lambda(\Gamma)$. For each $h > 0$, let $\Gamma(h)$ denote the set of points in the plane at a distance not exceeding h from Γ. That is,*

$$\Gamma(h) = \{\mathbf{x} \mid \mathbf{x} \in E_2, |\mathbf{x} - \mathbf{y}| \leq h \text{ for some } \mathbf{y} \text{ on } \Gamma\}.$$

Then $\Gamma(h)$ has outer Jordan content not exceeding $\pi h^2 + 2h\Lambda(\Gamma)$.

Proof. If Γ is a line segment, the region $\Gamma(h)$ is the union of a rectangle and two semicircular disks (as shown in Fig. 10–7), and in this case the area of $\Gamma(h)$ is exactly equal to $\pi h^2 + 2h\Lambda(\Gamma)$. If Γ is a polygonal curve consisting of two segments, the set $\Gamma(h)$ is the union of two overlapping

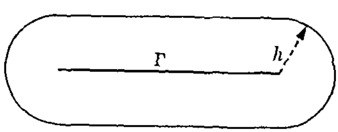


FIG. 10–7. $\Gamma(h)$ when Γ is a segment.

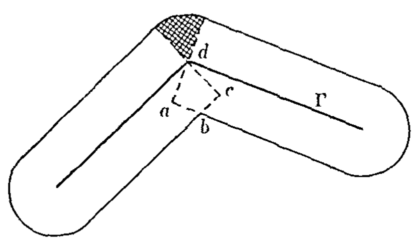


FIG. 10–8. $\Gamma(h)$ when Γ is a polygon with two sides.

rectangles, two semicircular disks, and a circular sector (as shown in Fig 10–8, where the sector is shaded).

Since the area of the circular sector does not exceed the area of the quadrilateral formed by the overlapping rectangles (*abcd* in Fig 10–8), it follows that in this case the area of $\Gamma(h)$ does not exceed $\pi h^2 + 2h\Lambda(\Gamma)$ A similar argument can be used to prove the inequality

$$c[\Gamma(h)] \leq \pi h^2 + 2h\Lambda(\Gamma)$$

for every polygon Γ by using induction on the number of sides (as usual, $c(S)$ denotes the Jordan content of S).

To prove the theorem for an arbitrary rectifiable curve Γ, let δ be a given positive number and construct an inscribed polygon P with vertices $\mathbf{p}_0, \mathbf{p}_1, \ldots, \mathbf{p}_n$ such that every point of the subarc of Γ joining $\mathbf{p}_{i-1}$ and $\mathbf{p}_i$ is at a distance $< \delta$ from the line segment $L[\mathbf{p}_{i-1}, \mathbf{p}_i]$ (This can be done by using an argument similar to that given in the proof of Theorem 8–33) If a point is at a distance $\leq h$ from Γ, then it is at a distance $< h + \delta$ from P Hence, $\Gamma(h) \subset P(h + \delta)$ and we have

$$\bar{c}[\Gamma(h)] \leq c[P(h + \delta)] \leq \pi(h + \delta)^2 + 2(h + \delta)\Lambda(P)$$
$$\leq \pi(h + \delta)^2 + 2(h + \delta)\Lambda(\Gamma).$$

But, since δ is arbitrary, it follows that we must have

$$\bar{c}[\Gamma(h)] \leq \pi h^2 + 2h\Lambda(\Gamma),$$

and this completes the proof

A horizontal line may intersect a rectifiable Jordan curve infinitely often (See Fig 10–9, where the "upper" part of the boundary, having the equation $y = x^2 \sin(1/x)$, is intersected infinitely often by the x-axis) Such a line decomposes the inner region of the curve into an infinite collection of subregions (For example, there are infinitely many shaded regions in

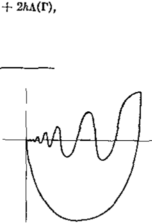


FIGURE 10–9

Fig. 10-9.) In view of this fact, it may seem surprising to discover that by using horizontal and vertical line segments, the inner region of a rectifiable Jordan curve can always be decomposed into a *finite* collection of subregions of arbitrarily small diameter. This, in effect, is what we shall prove in the next theorem—a crucial result which makes possible a simple proof of Green's theorem in its general form.

10-42 THEOREM. *Let $\delta > 0$ be given and let $S(\delta)$ denote the collection of squares in the xy-plane determined by the lines $x = m\delta$, $y = m\delta$ ($m = 0, \pm 1, \pm 2, \ldots$). Let R be a Jordan region bounded by a rectifiable Jordan curve Γ, and let $\Gamma(R)$ denote the positively oriented boundary of R. Then R can be expressed as the union of a finite collection of nonoverlapping regions $R_1, \ldots, R_n$ having the following properties:*

(a) *Those subregions inside Γ are squares of the collection $S(\delta)$.*

(b) *The remaining subregions (called border regions) have boundaries composed of arcs of Γ and segments of the lines $x = m\delta$, $y = m\delta$ (where m is an integer).*

(c) *Each border region can be enclosed in a square of edge-length 2δ.*

(d) *If $\Gamma(R_1), \ldots, \Gamma(R_n)$ are the positively oriented boundaries of $R_1, \ldots, R_n$, we have $\Gamma(R) = \Gamma(R_1) + \cdots + \Gamma(R_n)$.*

(e) *The number of squares in $S(\delta)$ having points in common with Γ does not exceed $4 + 4\Lambda(\Gamma)/\delta$, where $\Lambda(\Gamma)$ is the length of Γ.*

Proof. The idea of the proof is this: First, we show that by drawing a finite number of horizontal segments taken from the lines $y = m\delta$ (where m is an integer), we can divide R into a finite number of subregions, each of which has an "altitude" not exceeding 2δ. (The *altitude* of a set A in E_2 is the sup of all differences $y - y'$, where $(x, y) \in A$, $(x', y') \in A$. The *width* of A is similarly defined.) Then, by drawing a finite number of vertical segments, taken from the lines $x = m\delta$ (where m is an integer), each of the subregions so formed can be further subdivided into a finite number of regions whose width is at most 2δ. At each step we observe that the boundary curves satisfy (d).

To carry out this plan, let $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)$ be a function describing Γ, such that $\boldsymbol{\alpha} \in \Gamma(R)$. Then α_1 and α_2 are continuous functions of bounded variation defined on an interval $[a, b]$. Let y_1 and y_2 denote the minimum and maximum, respectively, of α_2 on $[a, b]$, and suppose $y_1 = \alpha_2(t_1)$, $y_2 = \alpha_2(t_2)$. Let $\mathbf{p}_1 = \boldsymbol{\alpha}(t_1)$, $\mathbf{p}_2 = \boldsymbol{\alpha}(t_2)$ be the corresponding points on Γ. If $y_2 - y_1 \le 2\delta$, we need not draw any horizontal lines. Therefore, assume that $y_2 - y_1 > 2\delta$. Then there exists an integer m between

$y_1/\delta + \frac{1}{2}$ and $y_2/\delta - \frac{1}{2}$ such that

$$(23) \qquad m\delta - y_1 > \delta/2 \quad \text{and} \quad y_2 - m\delta > \delta/2$$

Since $y_1 < m\delta < y_2$, the line $y = m\delta$ intersects Γ (at least twice) We will now show that there is a segment $L[\mathbf{u}, \mathbf{v}]$ of the line $y = m\delta$ which forms, with Γ, two rectifiable Jordan curves Γ_1 and Γ_2, with $\mathbf{p}_1$ on Γ_1 and $\mathbf{p}_2$ on Γ_2, such that $\Gamma_1 \cap \Gamma_2 = L[\mathbf{u}, \mathbf{v}]$

Let C_1 be an arc joining $\mathbf{p}_1$ to a point $\mathbf{q}_1$ inside Γ, where $\mathbf{q}_1$ is below the line $y = m\delta$ and each point of C_1 (except $\mathbf{p}_1$) is inside Γ (This is possible by Theorem 8-41.) We can also assume that C_1 does not intersect the line $y = m\delta$ Similarly, let C_2 be an arc joining $\mathbf{p}_2$ to a point $\mathbf{q}_2$ inside Γ and above the line $y = m\delta$ Since $\mathbf{q}_1$ and $\mathbf{q}_2$ are both inside Γ, they can be joined by a polygonal curve C inside Γ and we can assume that no side of C coincides with part of the line $y = m\delta$ (See Fig 10-10) The polygon C must intersect the line $y = m\delta$ at least once (since $\mathbf{q}_1$ is below and $\mathbf{q}_2$ is above the line) If an intersection occurs at a point $\mathbf{p}$, there will be a "maximal" segment $L[\mathbf{u}, \mathbf{v}]$ of the line $y = m\delta$ which contains $\mathbf{p}$ such that $\mathbf{u} \in \Gamma$, $\mathbf{v} \in \Gamma$, but such that the open line segment $L(\mathbf{u}, \mathbf{v})$ is inside Γ There can be only a finite number of such segments, since C has only a finite number of points in common with the line $y = m\delta$ But there must be at least one such segment on which C has an odd number of intersections with the line $y = m\delta$ [Why?] Using such a segment, it is clear that $L[\mathbf{u}, \mathbf{v}]$ forms, with Γ, two rectifiable Jordan curves Γ_1 and Γ_2, one of which contains $\mathbf{p}_1$ (call this one Γ_1) and the other contains $\mathbf{p}_2$ Moreover, these curves form the boundaries of two regions R_1 and R_2 whose union is R The positively

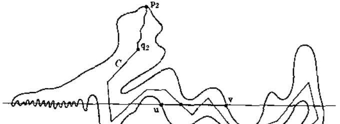


FIG 10-10 Illustrating the first step in the proof of Theorem 10-42

oriented boundaries $\Gamma(R_1)$ and $\Gamma(R_2)$ of R_1 and R_2 are such that $\Gamma(R) = \Gamma(R_1) + \Gamma(R_2)$.

Now let us write $v(\Gamma)$ for the total variation of α_2 on the parametric interval of Γ, and define the numbers $v(\Gamma_1)$ and $v(\Gamma_2)$ similarly. Because of the manner in which Γ_1 and Γ_2 were constructed, we have

$$v(\Gamma_1) + v(\Gamma_2) = v(\Gamma).$$

Furthermore, using inequalities (23), we obtain $v(\Gamma_1) > \delta$ and $v(\Gamma_2) > \delta$. Therefore,

$$\delta < v(\Gamma_1) \leq v(\Gamma) - \delta \qquad \text{and} \qquad \delta < v(\Gamma_2) \leq v(\Gamma) - \delta.$$

If the altitude of either curve, say Γ_1, is $>2\delta$, we can repeat the above construction with Γ replaced by Γ_1 and arrive at a new curve Γ_3 for which we have

$$\delta < v(\Gamma_3) \leq v(\Gamma_1) - \delta \leq v(\Gamma) - 2\delta.$$

After k such steps, we would arrive at an inequality of the form

$$\delta < v(\Gamma) - k\delta, \qquad \text{or} \qquad (k + 1)\delta < v(\Gamma).$$

Therefore the number k is restricted and the process cannot continue indefinitely. That is, after a finite number of steps, all curves obtained by the above construction will have altitude $\leq 2\delta$. The corresponding regions bounded by these curves (finite in number) will also have altitude $\leq 2\delta$. In a similar manner, we can divide each of these regions by vertical segments taken from the lines $x = m\delta$ (where m is an integer) to complete the proof of parts (a), (b), (c), and (d) of the theorem.

To prove part (e), we simply observe that a subarc of Γ of length $<\delta$ can have points in common with at most four squares of $S(\delta)$. The number of such arcs does not exceed $1 + \Lambda(\Gamma)/\delta$.

It is now quite easy to prove Green's theorem in the following general form:

10–43 THEOREM (*Green's theorem for regions bounded by rectifiable Jordan curves*). *Let P and Q be two real-valued functions defined and continuous on a Jordan region R bounded by a rectifiable Jordan curve Γ. Assume that the partial derivatives D_2P and D_1Q exist and are bounded in the interior of R and that both double integrals $\int_R D_1Q(x, y)\, d(x, y)$ and $\int_R D_2P(x, y)\, d(x, y)$ exist. If $\alpha = (\alpha_1, \alpha_2) \in \Gamma(R)$, where $\Gamma(R)$ is* *the positively oriented boundary of R (see Definition 9–69), then the line integral $\int_{\Gamma(R)} P\,d\alpha_1 + Q\,d\alpha_2$ exists and we have*

$$\int_R [D_1Q(x, y) - D_2P(x, y)]\,d(x, y) = \int_{\Gamma(R)} P\,d\alpha_1 + Q\,d\alpha_2$$

Proof The formula in Green's theorem is equivalent to the *two* formulas

$$-\int_R D_2P(x, y)\,d(x, y) = \int_{\Gamma(R)} P\,d\alpha_1$$

and

$$\int_R D_1Q(x, y)\,d(x, y) = \int_{\Gamma(R)} Q\,d\alpha_2$$

We will prove the first of these formulas (The proof of the second is similar)

If δ is given, we consider the collection $S(\delta)$ of squares in the xy-plane determined by the lines $x = m\delta$, $y = m\delta$ ($m = 0, \pm 1, \pm 2,$), and decompose R into a finite collection of nonoverlapping subregions as described in Theorem 10–42 Let $R_1,$, R_k denote those subregions which are squares of $S(\delta)$ lying completely inside the boundary Γ of R, and let $R_{k+1}, \ldots, R_n$ be the border regions Let T_δ denote the union of the border regions

Since no point of a border region is more than a distance $\delta\sqrt{2}$ from Γ, we must have $T_\delta \subset \Gamma(\delta\sqrt{2})$ (in the notation of Theorem 10–41), and hence we can write

$$c(T_\delta) \le \bar{c}[\Gamma(\delta\sqrt{2})] \le \pi(\delta\sqrt{2})^2 + 2\delta\sqrt{2}\,\Lambda(\Gamma),$$

where $\Lambda(\Gamma)$ is the length of Γ

Let A be a constant such that $|D_2P(x, y)| \le A$ for all (x, y) in R and let $B = 16 + 17\Lambda(\Gamma)$ If $\epsilon > 0$ is given, we can choose $\delta > 0$ (depending on ϵ) so that $\delta < 1$ and so that

$$\pi(\delta\sqrt{2})^2 + 2\delta\sqrt{2}\,\Lambda(\Gamma) < \frac{\epsilon}{2A}$$

Let M_p and m_p denote the maximum and minimum of P on the subregion R_p By continuity we can also assume that δ is chosen so that

$$M_p - m_p < \frac{\epsilon}{2B} \qquad \text{for} \qquad p = 1, 2, \ldots, n.$$

For this δ, we have

$$(24)\quad \left|\int_R D_2P(x,y)\,d(x,y) - \sum_{p=1}^{k}\int_{R_p} D_2P(x,y)\,d(x,y)\right|$$

$$= \left|\int_{T_\delta} D_2P(x,y)\,d(x,y)\right| \le Ac(T_\delta) < \frac{\epsilon}{2}.$$

Also, by Theorem 10–40, we have

(25)

$$\left|\sum_{p=k+1}^{n}\int_{\Gamma(R_p)} P\,d\alpha_1\right| \le \frac{1}{2}\sum_{p=k+1}^{n}(M_p - m_p)\Lambda(\Gamma_p) < \frac{\epsilon}{2B}\sum_{p=k+1}^{n}\Lambda(\Gamma_p),$$

where Γ_p is the boundary of R_p. Each arc Γ_p is composed of portions of Γ and portions of the edges of squares of the system $S(\delta)$. By Theorem 10–42, at most $4 + 4\Lambda(\Gamma)/\delta$ of these squares have points in common with Γ, and hence we can write

$$\sum_{p=k+1}^{n}\Lambda(\Gamma_p) \le \Lambda(\Gamma) + 4\delta\left(4 + \frac{4\Lambda(\Gamma)}{\delta}\right)$$

$$= 16\delta + 17\Lambda(\Gamma) < 16 + 17\Lambda(\Gamma) = B.$$

Therefore, (25) implies

$$\left|\sum_{p=k+1}^{n}\int_{\Gamma(R_p)} P\,d\alpha_1\right| < \frac{\epsilon}{2}.$$

Since

$$\int_{\Gamma(R)} P\,d\alpha_1 = \sum_{p=1}^{k}\int_{\Gamma(R_p)} P\,d\alpha_1 + \sum_{p=k+1}^{n}\int_{\Gamma(R_p)} P\,d\alpha_1,$$

the last inequality implies

$$(26)\quad \left|\int_{\Gamma(R)} P\,d\alpha_1 - \sum_{p=1}^{k}\int_{\Gamma(R_p)} P\,d\alpha_1\right| < \frac{\epsilon}{2}.$$

Applying Green's theorem to each square R_p $(p = 1, 2, \ldots, k)$, we find

$$\sum_{p=1}^{k} \int_{\Gamma(R_p)} P\,d\alpha_1 = -\sum_{p=1}^{k} \int_{R_p} D_2P(x, y)\,d(x, y).$$

Combining this with (24) and (26), we obtain

$$\left|\iint_R D_2P(x, y)\,d(x, y) + \int_{\Gamma(R)} P\,d\alpha_1\right| < \epsilon$$

Since ϵ is arbitrary, this proves that $-\int_R D_2P(x, y)\,d(x, y) = \int_{\Gamma(R)} P\,d\alpha_1$.

The above proof of Green's theorem was published in 1951 by D H Potts (*Journal of the London Mathematical Society*, Vol 26) and is based in part on ideas due to T. Estermann, K Hu, J Ridder, and S Verblunsky.

10-15 Independence of the path.

10-44 DEFINITION. *Let* $\mathbf{f}$ *be continuous on an open region* S *in* E_n *and let* $\mathbf{p}$ *and* $\mathbf{q}$ *be two points in* S. *The line integral* $\int_{\Gamma(\mathbf{p},\mathbf{q})} \mathbf{f} \cdot d\boldsymbol{\alpha}$ *is said to be independent of the path in* S *if*

$$\int_{\Gamma(\mathbf{p}, \mathbf{q})} \mathbf{f} \cdot d\boldsymbol{\alpha} = \int_{\Gamma_1(\mathbf{p}, \mathbf{q})} \mathbf{f} \cdot d\boldsymbol{\beta}$$

for all paths $\boldsymbol{\alpha}$ *and* $\boldsymbol{\beta}$ *describing curves* Γ *and* Γ_1 *such that* $\Gamma \subset S$, $\Gamma_1 \subset S$, $\boldsymbol{\alpha} \in \Gamma(\mathbf{p}, \mathbf{q})$, $\boldsymbol{\beta} \in \Gamma_1(\mathbf{p}, \mathbf{q})$ (See Fig. 10-11)

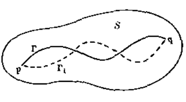


FIGURE 10-11.

An important part of the theory of line integrals deals with the following question: For which continuous functions $\mathbf{f}$ is the integral $\int_{\Gamma(\mathbf{p},\mathbf{q})} \mathbf{f} \cdot d\boldsymbol{\alpha}$ independent of the path for *all* choices of $\mathbf{p}$ and $\mathbf{q}$ in S? This is the same as asking for those continuous functions $\mathbf{f}$ whose line integral is zero along every *closed* path in S and it is a question too difficult to discuss here in its fullest generality. However, it can be completely answered if we insist that the curves used in the integration be *piece-wise smooth* We shall adopt this restriction in the remainder of this section It then follows, from Theorem 10-37, that a *sufficient* condition for independence is the existence of a potential function ϕ such that $\mathbf{f} = \nabla\phi$ On the other hand, Theorem 10-38 shows that this condition is also *necessary*.

In practice it may be difficult to determine independence by establishing the existence of a potential function. A much more useful test for independence (which does not involve any integration or construction of potential functions) will be discussed in this section. However, for this test we require that the integrand $\mathbf{f}$ be continuously differentiable on S.

First, we derive a simple necessary condition for the existence of a potential function.

10–45 THEOREM. *Assume that* $\mathbf{f} = (f_1, \ldots, f_n) \in C'$ *on an open region* S *in* E_n. *If there exists a real-valued function* ϕ *such that* $\mathbf{f}(\mathbf{x}) = \nabla\phi(\mathbf{x})$ *for each* $\mathbf{x}$ *in* S, *then the partial derivatives of the components of* $\mathbf{f}$ *satisfy the following equations:*

$$D_i f_j(\mathbf{x}) = D_j f_i(\mathbf{x}) \qquad (\mathbf{x} \in S,\ i \neq j,\ i, j = 1, 2, \ldots, n). \tag{27}$$

Proof. Since $\mathbf{f} = \nabla\phi$, we have $f_j = D_j\phi$. The hypothesis $\mathbf{f} \in C'$ implies $D_{i,j}\phi = D_{j,i}\phi$ for each pair $i, j = 1, 2, \ldots, n$, $i \neq j$. Since $D_i f_j = D_{i,j}\phi$, we obtain the conclusion at once.

Of course, because of Theorem 10–38, the $n(n-1)/2$ equations in (27) provide a simple necessary condition for independence of the path. However, what we really need in the applications is a simple *sufficient* condition for independence. It turns out that the equations in (27) furnish such a condition, provided that we put a restriction on the "structure" of the region S. To see that some kind of restriction is needed, consider the following example in E_2.

EXAMPLE. Let $S = \{(x, y) \mid \frac{1}{2} < x^2 + y^2 < 2\}$ be an open "annulus" as shown in Fig. 10–12. Define two real-valued functions f_1 and f_2 on S as follows:

$$f_1(x, y) = \frac{-y}{x^2 + y^2}, \qquad f_2(x, y) = \frac{x}{x^2 + y^2}.$$

An easy calculation shows that we have

$$D_2 f_1(x, y) = \frac{y^2 - x^2}{(x^2 + y^2)^2} = D_1 f_2(x, y)$$

for each (x, y) in S.

Consider the two points $A = (1, 0)$ and $B = (0, 1)$. One curve Γ_1 in S joining A and B is the circular arc described by

$$x(t) = \cos t, \qquad y(t) = \sin t, \qquad 0 \leq t \leq \pi/2.$$

Since we have

$$f_1(x(t), y(t)) = -\sin t, \qquad f_2(x(t), y(t)) = \cos t,$$
$$x'(t) = -\sin t, \qquad y'(t) = \cos t,$$

we find

$$\int_{\Gamma_1(A,B)} f_1\,dx + f_2\,dy = \int_0^{\pi/2} (\sin^2 t + \cos^2 t)\,dt = \frac{\pi}{2}$$

The same functions x and y defined on the interval $[\pi/2, 2\pi]$ describe another circular arc Γ_2 in S joining B and A. The integral along Γ_2 from A to B is

$$\int_{\Gamma_2(A,B)} f_1\,dx + f_2\,dy = -\int_{\pi/2}^{2\pi} dt = -\frac{3\pi}{2}$$

Hence two paths joining the same points A and B yield different values for the line integral although the equations (27) are satisfied everywhere in S.

This example shows that the converse of Theorem 10–45 cannot be true in general. The converse holds only for a restricted class of regions S. In E_2, the regions of this class are known as *simply connected* regions. They can be defined as follows.

10–46 DEFINITION *A region S in E_2 is called simply connected if, for every Jordan curve Γ in S, the inner region of Γ is also a subset of S.*

The annulus S in Fig. 10–12 is not simply connected because the inner region of the unit circle $x^2 + y^2 = 1$ is not a subset of S. Intuitively speaking, simple connectedness is equivalent to the absence of "holes."

The converse of Theorem 10–45 will now be proved for simply connected regions in E_2.

10–47 THEOREM *Assume that S is an open simply connected region in E_2. Let $\mathbf{f} = (f_1, f_2) \in C'$ on S and assume that*

$$D_1 f_2(\mathbf{x}) = D_2 f_1(\mathbf{x})$$

for each $\mathbf{x}$ in S. Then there exists a real-valued function ϕ such that $\mathbf{f}(\mathbf{x}) = \nabla\phi(\mathbf{x})$ for all $\mathbf{x}$ in S.

Proof. Let $\mathbf{x}_0$ be a fixed point in S. If $\mathbf{x}$ is an arbitrary point of S, let Γ_1 and Γ_2 be two simple step polygons in S joining $\mathbf{x}_0$ and $\mathbf{x}$. (Such curves always exist because of the remarks following Theorem 8–33.) Various

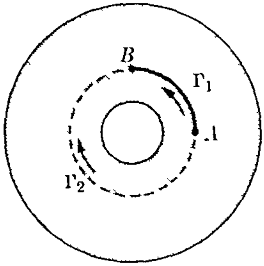


FIGURE 10–12.

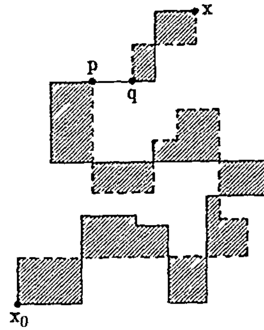


FIGURE 10–13.

portions of these two polygons may coincide along certain line segments. The remaining portions will intersect at most a finite number of times, and they will form the boundaries of a finite number of regions, say $R_1, \ldots, R_m$. Since S is assumed to be simply connected, each of these regions R_k is a subset of S. An example is illustrated in Fig. 10–13. The solid line represents Γ_1, the dotted line represents Γ_2, and the shaded regions represent $R_1, \ldots, R_m$. The two curves coincide along the segment **pq**.

We observe next that the line integral of **f** along the *closed* path $\Gamma_1(\mathbf{x}_0, \mathbf{x}) + \Gamma_2(\mathbf{x}, \mathbf{x}_0)$ is zero by writing it as a sum of integrals taken over the line segments common to Γ_1 and Γ_2 plus a sum of integrals taken around the boundaries of the regions R_k. The integrals over the common segments cancel in pairs (since each common segment is traversed twice in opposite directions) and their sum is zero. The integral over the boundary of each region R_k is also zero because of Green's theorem and the hypothesis $D_1 f_2 = D_2 f_1$. It follows that we have

$$\int_{\Gamma_1(\mathbf{x}_0, \mathbf{x})} \mathbf{f} \cdot d\boldsymbol{\alpha} = \int_{\Gamma_2(\mathbf{x}_0, \mathbf{x})} \mathbf{f} \cdot d\boldsymbol{\beta}, \tag{28}$$

where $\boldsymbol{\alpha} \in \Gamma_1(\mathbf{x}_0, \mathbf{x})$ and $\boldsymbol{\beta} \in \Gamma_2(\mathbf{x}_0, \mathbf{x})$. Once we know this, we can define

$$\phi(\mathbf{x}) = \int_{\Gamma(\mathbf{x}_0, \mathbf{x})} \mathbf{f} \cdot d\boldsymbol{\alpha},$$

where Γ is any simple step polygon in S and $\boldsymbol{\alpha} \in \Gamma(\mathbf{x}_0, \mathbf{x})$. Because of (28), the number $\phi(\mathbf{x})$ so defined is completely determined by **x** and does not depend on the particular step polygon used to join $\mathbf{x}_0$ and **x**. The argument used in the proof of Theorem 10–38 can be used again here to prove that $\nabla\phi(\mathbf{x}) = \mathbf{f}(\mathbf{x})$.

A general result similar to that in Theorem 10–47 is not easy to prove in E_n $(n > 2)$ The crucial step in the above proof was in the application of Green's theorem In order to attempt the same kind of argument in E_n, we need an appropriate generalization of Green's theorem from E_2 to E_n. Such a generalization exists and, for the case $n = 3$, it is known as *Stokes' theorem* This will be discussed in the next chapter in connection with surface integrals

For the present, however, we can obtain an extension of Theorem 10–47 to E_n by restricting S to be an n-dimensional *interval*

10–48 THEOREM *Let* $\mathbf{f} = (f_1, \ldots, f_n) \in C'$ *on an open interval* S *in* E_n, *say*

$$S = (a_1, b_1) \times \cdots \times (a_n, b_n)$$

Assume that

$$D_i f_j(\mathbf{x}) = D_j f_i(\mathbf{x}), \qquad (i \neq j,\ i, j = 1, 2, \ldots, n)$$

for each $\mathbf{x}$ *in* S *Then there exists a real-valued function* ϕ *such that* $\mathbf{f}(\mathbf{x}) = \nabla\phi(\mathbf{x})$ *for all* $\mathbf{x}$ *in* S

Proof Let $\mathbf{y} = (y_1, \ldots, y_n)$ be a fixed point in S For each $\mathbf{x} = (x_1, \ldots, x_n)$ in S, define $\phi(\mathbf{x})$ to be the sum of the n one-dimensional Riemann integrals

$$\phi(\mathbf{x}) = \int_{y_1}^{x_1} f_1(t_1, x_2, \ldots, x_n)\, dt_1 + \int_{y_2}^{x_2} f_2(y_1, t_2, x_3, \ldots, x_n)\, dt_2$$
$$+ \int_{y_3}^{x_3} f_3(y_1, y_2, t_3, x_4, \ldots, x_n)\, dt_3 + \cdots$$
$$+ \int_{y_n}^{x_n} f_n(y_1, \ldots, y_{n-1}, t_n)\, dt_n$$

Applying Theorem 9–37, we find $D_1\phi(\mathbf{x})$ exists and has the value

$$D_1\phi(\mathbf{x}) = [f_1(x_1, \ldots, x_n) - f_1(y_1, x_2, \ldots, x_n)]$$
$$+ \int_{y_2}^{x_2} D_1 f_2(y_1, t_2, x_3, \ldots, x_n)\, dt_2$$
$$+ \int_{y_3}^{x_3} D_1 f_3(y_1, y_2, t_3, x_4, \ldots, x_n)\, dt_3 + \cdots$$
$$+ \int_{y_n}^{x_n} D_1 f_n(y_1, \ldots, y_{n-1}, t_n)\, dt_n$$

But, since $D_1 f_j = D_j f_1$ for $j > 1$, we can evaluate each integral on the right by the fundamental theorem of calculus and we obtain

$$\begin{aligned} D_1\phi(\mathbf{x}) &= [f_1(x_1, \ldots, x_n) - f_1(y_1, x_2, \ldots, x_n)] \\ &\quad + [f_1(y_1, x_2, x_3, \ldots, x_n) - f_1(y_1, y_2, x_3, \ldots, x_n)] \\ &\quad + [f_1(y_1, y_2, x_3, \ldots, x_n) - f_1(y_1, y_2, y_3, x_4, \ldots, x_n)] + \cdots \\ &\quad + [f_1(y_1, \ldots, y_{n-1}, x_n) - f(y_1, \ldots, y_n)] \\ &= f_1(x_1, \ldots, x_n) = f_1(\mathbf{x}). \end{aligned}$$

Similarly, we find $D_j\phi(\mathbf{x}) = f_j(\mathbf{x})$ for $j = 2, 3, \ldots, n$.

Exercises

Jordan content.

10–1. Let S be an elementary set in E_n. Assume that S can be decomposed in two ways as a union of nonoverlapping intervals, say $S = I_1 \cup \cdots \cup I_r$ and $S = J_1 \cup \cdots \cup J_p$. Show that $\mu(I_1) + \cdots + \mu(I_r) = \mu(J_1) + \cdots + \mu(J_p)$. From this, deduce that $\mu(S \cup T) = \mu(S) + \mu(T)$ whenever S and T are nonoverlapping elementary sets.

10–2. (a) Let Γ be a rectifiable curve in E_n. Show that Γ has n-dimensional Jordan content zero.

(b) Let R be a region in E_2 formed by taking the union of a rectifiable Jordan curve and its interior. Show that R is Jordan-measurable.

10–3. Extend the results of Exercises 9–25 and 9–26 to E_n.

10–4. Let S be a bounded set in E_n having at most a finite number of accumulation points. Show that $c(S) = 0$.

10–5. Let f be a continuous real-valued function defined on $[a, b]$. Let S denote the graph of f, that is, $S = \{(x, y) \mid y = f(x), a \leq x \leq b\}$. Show that S has two-dimensional Jordan content zero.

10–6. Show that Theorem 10–8 is not true in general without the hypothesis $p \geq n$.

Multiple integration

10–7. Let f be a real-valued function defined on a set S in E_n and suppose $f(\mathbf{x}) \geq 0$ for each $\mathbf{x}$ in S. The *ordinate set* of f over S is defined to be the following subset of E_{n+1}:

$$\{(x_1, \ldots, x_n, x_{n+1}) \mid (x_1, \ldots, x_n) \in S,\ 0 \leq x_{n+1} \leq f(x_1, \ldots, x_n)\}.$$

If S is a Jordan-measurable region in E_n and if f is continuous on S, show that the ordinate set of f over S has $(n+1)$-dimensional Jordan content whose value is $\int_S f(x_1, \ldots, x_n)\, d(x_1, \ldots, x_n)$. Interpret this problem geometrically when $n = 1$ and $n = 2$.

10–8. Let f be defined on the unit square $R = [0, 1] \times [0, 1]$ as follows:

$$f(x, y) = \begin{cases} 1, & \text{if } x \text{ is rational,} \\ 2y, & \text{if } x \text{ is irrational.} \end{cases}$$

(a) Compute the oscillation of f at each interior point of R.

(b) If $0 < \epsilon \leq 1$, compute $\bar{c}(J_\epsilon)$, where J_ϵ is defined in Theorem 10–13.

(c) Show that $\int_0^1 (\bar{\int}_0^1 f(x, y)\, dx)\, dy$ exists and find its value.

(d) Show that $\int_0^1 (\bar{\int}_0^1 f(x, y)\, dy)\, dx$ exists and find its value.

(e) Show that $\int_R f(x, y)\, d(x, y)$ does not exist.

10–9. For each rational number $x \neq 0$ in $[0, 1]$, say $x = p/q$ (in lowest terms), let $S(x)$ be the following subset of E_2:

$$S(x) = \left\{\left(\frac{n}{q}, \frac{m}{q}\right) \middle| n = 0, 1, 2, \ldots, p, \quad m = 0, 1, 2, \ldots, p\right\}.$$

Let $S(0) = \{(0, 0)\}$. If $\{x_1, x_2, \ldots\}$ denotes the collection of rational numbers in $[0, 1]$, let

$$S = \bigcup_{k=1}^{\infty} S(x_k).$$

Let f be defined on the unit square $R = [0, 1] \times [0, 1]$ as follows:

$$f(x, y) = 0 \quad \text{if } (x, y) \in S, \qquad f(x, y) = 1 \quad \text{if } (x, y) \in R - S.$$

Show that $\int_0^1 [\int_0^1 f(x, y)\, dy]\, dx = \int_0^1 [\int_0^1 f(x, y)\, dx]\, dy = 1$, but that the double integral $\int_R f(x, y)\, d(x, y)$ does not exist.

10–10. If $f_1 \in R$ on $[a_1, b_1], \ldots, f_n \in R$ on $[a_n, b_n]$, show that

$$\int_S f_1(x_1) \cdots f_n(x_n)\, d(x_1, \ldots, x_n) = \left(\int_{a_1}^{b_1} f_1(x_1)\, dx_1\right) \cdots \left(\int_{a_n}^{b_n} f_n(x_n)\, dx_n\right),$$

where $S = [a_1, b_1] \times \cdots \times [a_n, b_n]$.

10–11. Assume that $f \in R$ on S and suppose $\int_S f(\mathbf{x})\, d\mathbf{x} = 0$. ($S$ is a subset of E_n.) Let $A = \{\mathbf{x} \mid \mathbf{x} \in S, f(\mathbf{x}) < 0\}$ and assume that $c(A) = 0$. Show that there exists a set B of content zero such that $f(\mathbf{x}) = 0$ for each $\mathbf{x}$ in $S - B$.

10–12. Assume that $f \in R$ on S, where S is a region in E_n and f is continuous on S. Show that there exists an *interior* point $\mathbf{x}_0$ of S such that

$$\int_S f(\mathbf{x})\, d\mathbf{x} = f(\mathbf{x}_0)c(S).$$

10–13. Let S be an open region in E_n. Let S_J denote the collection of Jordan-measurable subsets of S. Let F be a real-valued function defined on S_J. We say that F has a *space derivative* $F'(\mathbf{x})$ at a point $\mathbf{x}$ of S whenever the following statement holds: For every $\epsilon > 0$ there exists a $\delta > 0$ such that for every T in S_J

containing $\mathbf{x}$, having positive Jordan content $c(T)$ and diameter $d(T) < \delta$, we have

$$\left|\frac{F(T)}{c(T)} - F'(\mathbf{x})\right| < \epsilon$$

This is written more briefly as follows

$$F'(\mathbf{x}) = \lim_{d(T)\to 0} \frac{F(T)}{c(T)}$$

If f is continuous on S, define $F(T) = \int_T f(\mathbf{x})\,d\mathbf{x}$, if $T \in S_J$. Prove that F has a space derivative at each point of S and that

$$F'(\mathbf{x}) = f(\mathbf{x})$$

(This can be considered as an extension of Theorem 9-31 (iii) to multiple integrals.)

10-14. Let f be continuous on a rectangle $R = [a, b] \times [c, d]$. For each interior point (x_1, x_2) in R, define

$$F(x_1, x_2) = \int_a^{x_1} \left(\int_c^{x_2} f(x, y)\,dy\right) dx.$$

Show that $D_{1,2}F(x_1, x_2) = D_{2,1}F(x_1, x_2) = f(x_1, x_2)$.

10-15. Let T denote the following triangular region in the plane

$$T = \left\{(x, y) \mid 0 \le \frac{x}{a} + \frac{y}{b} \le 1\right\}, \qquad \text{where } a > 0, b > 0$$

Assume that f has a continuous second-order partial derivative $D_{1,2}f$ on T. Show that there exists a point (x_0, y_0) on the segment joining $(a, 0)$ and $(0, b)$ such that

$$\int_T D_{1,2}f(x, y)\,d(x, y) = f(0, 0) - f(a, 0) + D_1f(x_0, y_0).$$

Change of variable in a multiple integral

10-16. Prove that the transformation formula (8) of Theorem 10-30 is still valid if the Jacobian vanishes on a subset of S having content zero.

[*Hint.* Using the notation of Theorem 10-30, show that $\mathbf{g}$ satisfies a uniform Lipschitz condition on $\mathbf{g}^{-1}(X)$ and hence, for every subset T of $\mathbf{g}^{-1}(X)$ having content zero, the set $\mathbf{g}(T)$ also has content zero.]

10-17. The following formulas for transforming double and triple integrals occur in elementary calculus. Obtain these formulas as consequences of Theorem

10–30. In each case, describe what restrictions must be placed on R and R' for the validity of these formulas.

(a) $$\iint_R f(x, y)\, dx\, dy = \iint_{R'} f(r\cos\theta, r\sin\theta) r\, dr\, d\theta.$$

(b) $$\iiint_R f(x, y, z)\, dx\, dy\, dz = \iiint_{R'} f(r\cos\theta, r\sin\theta, z) r\, dr\, d\theta\, dz.$$

(c) $$\iiint_R f(x, y, z)\, dx\, dy\, dz$$
$$= \iiint_{R'} f(r\cos\theta\sin\phi, r\sin\theta\sin\phi, r\cos\phi) r^2 \sin\phi\, dr\, d\theta\, d\phi.$$

Line integrals.

10–18. Let Γ be a curve in E_2 described by α, where $\alpha(t) = (x(t), y(t))$ if $t \in [0, 1]$. Evaluate the line integral $\int_{\Gamma[\alpha]} y\, dx + (y^2 - x)\, dy$ if

(a) $x(t) = y(t) = t$, (b) $x(t) = t^2, y(t) = t^3$,

(c) $x(t) = t, y(t) = t^n, n > 0$, (d) $x(t) = t, y(t) = \sin(\pi t/2)$.

10–19. Evaluate the following line integrals if Γ is the positively oriented unit circle with center at the origin:

(a) $\int_\Gamma (2x^3 - y^3)\, dx + (x^3 + y^3)\, dy$.

(b) $\int_\Gamma (2x - y)\, dx + (x + 3y)\, dy$.

(c) $\int_\Gamma (x^2 - y^2)\, ds$ (s denotes arc length).

10–20. Let S be an open simply connected region in E_2. If u and v satisfy the Cauchy-Riemann equations on S (see Theorem 6–24), we say that v is conjugate to u.

(a) If v is conjugate to u, show that $-u$ is conjugate to v.

(b) Given that u is harmonic on S, show that a function v conjugate to u can be found by means of a line integral of the form $\int_\Gamma -\partial u/\partial y\, dx + \partial u/\partial x\, dy$ taken along a suitable curve Γ in S.

Illustrate part (b) if $S = E_2$, and

(c) $u(x, y) = e^y \cos x$,

(d) $u(x, y) = x^3 - 3xy^2$.

10–21. Let Γ be a curve in E_2 described by a complex-valued function z, where $z(t) = x(t) + iy(t)$ if $t \in [a, b]$. Let u and v be real-valued functions defined on Γ and let $f = u + iv$. Assuming that the contour integral $\int_{\Gamma[z]} f(z)\, dz$

exists, show that it can be expressed in terms of real-valued line integrals as follows

$$\int_{\Gamma[z]} f(z)\,dz = \int_{\Gamma[z]} u\,dx - v\,dy + i\int_{\Gamma[z]} v\,dx + u\,dy.$$

10–22. Prove Theorem 10–35

Green's theorem

10–23. Let S be an open simply connected region in E_2 Assume that u and v have continuous partial derivatives D_1u, D_2u, D_1v, D_2v on S satisfying the Cauchy-Riemann equations (See Theorem 6–24) If Γ is a rectifiable Jordan curve in S, use Exercise 10–21 in conjunction with Green's theorem to prove that

$$\int_{\Gamma[z]} f(z)\,dz = 0, \quad \text{where } f = u + iv$$

(This is a very important result in complex-variable theory and is known as *Cauchy's integral theorem* Some of its consequences will be discussed in Chapter 16)

In Exercises 10–24 and 10–25, R denotes a Jordan region in E_2 bounded by a rectifiable Jordan curve Γ, and $\Gamma(R)$ denotes the positively oriented boundary of R, described by a function α where $\alpha(t) = (x(t), y(t))$ if $t \in [a, b]$ The functions u and v and their partial derivatives (as they occur) are assumed to be continuous on R

10–24. Derive the following identities

(a) $$\int_{\Gamma(R)} uv\,dx + uv\,dy = \int_R \left\{v\left(\frac{\partial u}{\partial x} - \frac{\partial u}{\partial y}\right) + u\left(\frac{\partial v}{\partial x} - \frac{\partial v}{\partial y}\right)\right\} d(x, y)$$

(b) $$\frac{1}{2}\int_{\Gamma(R)} \left(v\frac{\partial u}{\partial x} - u\frac{\partial v}{\partial x}\right)dx + \left(u\frac{\partial v}{\partial y} - v\frac{\partial u}{\partial y}\right)dy = \int_R \left(u\frac{\partial^2 v}{\partial x\,\partial y} - v\frac{\partial^2 u}{\partial x\,\partial y}\right)d(x, y)$$

10–25. Write $\alpha'(t) = (x'(t), y'(t))$ when both derivatives $x'(t)$ and $y'(t)$ exist At points where $\alpha'(t) \neq 0$, let $\mathbf{n}(t)$ denote the unit *outer normal* vector, defined by the equation

$$\mathbf{n}(t) = \frac{1}{|\alpha'(t)|}(y'(t), -x'(t)).$$

The *normal derivative* of u, denoted by $\partial u/\partial n$, is defined on Γ by the equation $\partial u/\partial n = \nabla u \cdot \mathbf{n}$ If s denotes the arc length of Γ, derive *Green's identities*

(a) $$\int_{\Gamma(R)} v\frac{\partial u}{\partial n}\,ds = \int_R (v\nabla^2 u + \nabla u \cdot \nabla v)\,d(x, y).$$

(b) $$\int_{\Gamma(R)} \left(v\frac{\partial u}{\partial n} - u\frac{\partial v}{\partial n}\right) ds = \int_R (v\nabla^2 u - u\nabla^2 v)\,d(x, y).$$

NOTE. Part (b) shows that

$$\int_{\Gamma(R)} v\frac{\partial u}{\partial n}\,ds = \int_{\Gamma(R)} u\frac{\partial v}{\partial n}\,ds$$

when u and v are both harmonic on R.

CHAPTER 11

VECTOR ANALYSIS

11-1 Introduction. In the preceding chapters we saw many instances in which complicated mathematical formulas assumed elegant and simple forms when expressed vectorially. Up to this point we have used vectors chiefly for the purpose of accomplishing such simplification. Historically, however, the widespread use of vector methods came about (near the end of the nineteenth century) when it was found that vectors were ideal tools for the exposition of certain ideas in geometry and physics. The study of these methods is now called *vector analysis* and the names chiefly associated with the early development of the subject are those of Hamilton, Grassmann, Gibbs, Maxwell, and Heaviside. In this chapter we propose to discuss the elements of this theory.

11-2 Linear independence and bases in E_n. Every vector $\mathbf{x}$ in E_n can be expressed uniquely as a linear combination of n particular vectors, namely, the unit coordinate vectors

$$\mathbf{u}_1 = (1, 0, \ldots, 0), \quad \mathbf{u}_2 = (0, 1, 0, \ldots, 0), \quad \ldots, \mathbf{u}_n = (0, 0, \ldots, 1).$$

In fact, if $\mathbf{x} = (x_1, \ldots, x_n)$, we have $\mathbf{x} = x_1\mathbf{u}_1 + \cdots + x_n\mathbf{u}_n$. The uniqueness of the representation follows from the fact that

$$\lambda_1\mathbf{u}_1 + \cdots + \lambda_n\mathbf{u}_n = \mathbf{0}$$

implies $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$. In this section we wish to investigate the possibility of expressing each $\mathbf{x}$ in E_n as a linear combination of other special sets of vectors.

11-1 DEFINITION. *Let $B = \{\mathbf{b}_1, \ldots, \mathbf{b}_m\}$ be a finite collection of vectors in E_n. The set B is said to span E_n if every vector in E_n can be expressed as a linear combination of $\mathbf{b}_1, \ldots, \mathbf{b}_m$. The set B is called linearly independent if, for every choice of scalars $\lambda_1, \ldots, \lambda_m$,*

$$\lambda_1\mathbf{b}_1 + \cdots + \lambda_m\mathbf{b}_m = \mathbf{0} \quad \textit{implies} \quad \lambda_1 = \cdots = \lambda_m = 0.$$

A linearly independent set which spans E_n is called a basis for E_n.

Thus, the set $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is a basis for E_n. Many other bases exist, however. For example, the set $\{\mathbf{u}_1, 2\mathbf{u}_2, 3\mathbf{u}_3, \ldots, n\mathbf{u}_n\}$ is also a basis for E_n. It is clear that the vectors of a linearly independent set must be distinct and that none of them can be the zero vector. Some further facts concerning linear independence and bases in E_n will be briefly noted here. (For proofs, see § 2–4, 2–5 of Ref. 11–5.)

(a) If A is a linearly independent set of vectors in E_n, there exists a basis B for E_n such that $A \subset B$.

(b) Every basis for E_n contains exactly n elements.

(c) A set B of n vectors in E_n forms a basis for E_n if, and only if, B is a linearly independent set.

NOTE. Statement (b) provides the justification for referring to E_n as an n-dimensional space.

If $B = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ is a basis for E_n and if $\mathbf{x} \in E_n$, we have

$$\mathbf{x} = \xi_1\mathbf{b}_1 + \cdots + \xi_n\mathbf{b}_n, \tag{1}$$

where $\xi_1, \ldots, \xi_n$ are scalars (uniquely determined, since B is linearly independent). The numbers $\xi_1, \ldots, \xi_n$ are called the *components of* $\mathbf{x}$ *relative to* B. Multiplying $\mathbf{x}$ in (1) by a scalar λ yields

$$\lambda\mathbf{x} = (\lambda\xi_1)\mathbf{b}_1 + \cdots + (\lambda\xi_n)\mathbf{b}_n$$

and hence this operation can be performed componentwise, just as in the case when $B = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$. Similarly, if

$$\mathbf{y} = \eta_1\mathbf{b}_1 + \cdots + \eta_n\mathbf{b}_n, \tag{2}$$

we have $\mathbf{x} + \mathbf{y} = (\xi_1 + \eta_1)\mathbf{b}_1 + \cdots + (\xi_n + \eta_n)\mathbf{b}_n$. This means that no matter what basis B we choose for E_n, the sum $\mathbf{x} + \mathbf{y}$ may be formed by adding corresponding components.

The situation is slightly different when we consider the dot product. If $\mathbf{x}$ and $\mathbf{y}$ are given by (1) and (2), we have

$$\mathbf{x}\cdot\mathbf{y} = \left(\sum_{i=1}^{n} \xi_i\mathbf{b}_i\right)\cdot\left(\sum_{j=1}^{n} \eta_j\mathbf{b}_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{n} \xi_i\eta_j(\mathbf{b}_i\cdot\mathbf{b}_j) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}\xi_i\eta_j, \tag{3}$$

where $a_{ij} = (\mathbf{b}_i \cdot \mathbf{b}_j)$. Thus $\mathbf{x} \cdot \mathbf{y}$ is a "bilinear form" in the components ξ_i and η_j with coefficients a_{ij} determined by the basis vectors in B. There is an important class of bases for which the coefficients a_{ij} are *independent* of B. These are known as *orthonormal bases*.

11-2 DEFINITION. *A collection of vectors* $B = \{\mathbf{b}_1, \ldots, \mathbf{b}_m\}$ *is called orthogonal if* $\mathbf{b}_i \cdot \mathbf{b}_j = 0$ *whenever* $i \neq j$. *If, in addition, each* $\mathbf{b}_i$ *is of unit length* (*that is*, $\mathbf{b}_i \cdot \mathbf{b}_i = 1$), *the collection is said to be orthonormal.*

For orthonormal bases, the bilinear form in (3) becomes

$$\mathbf{x} \cdot \mathbf{y} = \xi_1 \eta_1 + \cdots + \xi_n \eta_n$$

Thus, if we restrict ourselves to orthonormal bases, we can compute $\mathbf{x} \cdot \mathbf{y}$ componentwise by the same type of formula used in Definition 6-12. In particular, the number $\mathbf{x} \cdot \mathbf{x} = |\mathbf{x}|^2$ will be the sum of the squares of the components of $\mathbf{x}$, regardless of which orthonormal basis we use.

From any given basis, say $\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$, it is possible to construct an orthogonal basis $\{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ by the following procedure (known as the *Gram-Schmidt process*). Define $\mathbf{b}_1 = \mathbf{a}_1$ and for $m = 2, 3, \ldots, n$, let

$$\mathbf{b}_m = \mathbf{a}_m - \sum_{k=1}^{m-1} \frac{(\mathbf{a}_m \cdot \mathbf{a}_k)}{|\mathbf{b}_k|^2} \mathbf{b}_k$$

It is easy to prove (by induction) that the set $\{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ so defined is indeed an orthogonal basis (see Exercise 11-1). To obtain an *orthonormal* basis, we need only replace each $\mathbf{b}_m$ by $\mathbf{b}_m/|\mathbf{b}_m|$.

11-3 Geometric representation of vectors in E_3. Vectors were defined in Chapter 3 as n-tuples of real numbers. When $n = 2$ or $n = 3$, vectors can be represented geometrically by the use of directed line segments. The line segment joining two distinct points P and Q is said to be directed if one of the two points, say P, is called the *initial point*, and the other, Q, the *terminal point*. We then denote the segment by $\overrightarrow{PQ}$ and represent it graphically by an arrow as in Fig. 11-1.

To represent vectors in E_3 geometrically, a rectangular cartesian coordinate system with origin O is introduced. Let A and B be two points in space with coordinates (a_1, a_2, a_3) and (b_1, b_2, b_3), respectively. The directed line segment $\overrightarrow{AB}$ is said to be a geometric representation of the vector $\mathbf{x} = (x_1, x_2, x_3)$ if we have

$$x_1 = b_1 - a_1, \qquad x_2 = b_2 - a_2, \qquad x_3 = b_1 - a_3,$$

in which case we write $\mathbf{x} = \overrightarrow{AB}$ to denote this relationship. (See Fig. 11-2.) We know from analytic geometry that the segment $\overrightarrow{AB}$ has length equal to

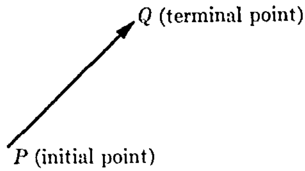


FIG. 11-1. Directed line segment.

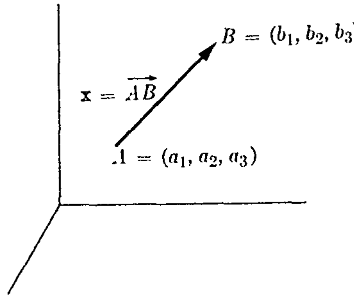


FIG. 11-2. Geometric representation of a vector.

$|\mathbf{x}|$ and direction cosines equal to $x_1/|\mathbf{x}|$, $x_2/|\mathbf{x}|$, $x_3/|\mathbf{x}|$. Therefore, every directed segment having the same length and the same direction cosines as $\overrightarrow{AB}$ will be a geometric representation of the same vector $\mathbf{x}$. Conversely, given a vector $\mathbf{x} = (x_1, x_2, x_3) \neq \mathbf{0}$ in E_3, the directed segment $\overrightarrow{OP}$ from the origin O to the point P with coordinates (x_1, x_2, x_3) will be a geometric representation of $\mathbf{x}$. The zero vector $\mathbf{0} = (0, 0, 0)$ is said to correspond to a "degenerate" segment in which the initial and terminal points coincide. Such a "segment" is assigned length 0 but does not have direction cosines. For convenience, we shall use the terms "vector" and "directed line segment" interchangeably, with the understanding that we make no distinction between segments having the same length and the same direction cosines.

The algebraic operations on vectors have simple geometric interpretations when $n = 2$ or $n = 3$. The *sum* $\mathbf{x} + \mathbf{y}$ of two vectors is formed in accordance with the *parallelogram law* as illustrated in Fig. 11-3. The difference $\mathbf{x} - \mathbf{y}$ is similarly interpreted. Multiplying $\mathbf{x}$ by a scalar λ has the effect of multiplying the length of $\mathbf{x}$ by $|\lambda|$ and the direction cosines of $\mathbf{x}$ by $\lambda/|\lambda|$ (if $\lambda \neq 0$). The discussion in the preceding section shows that the operations of addition, multiplication by scalars, and the dot product are "geometrically invariant." That is, they are completely independent of the particular rectangular coordinate system we happen to introduce.

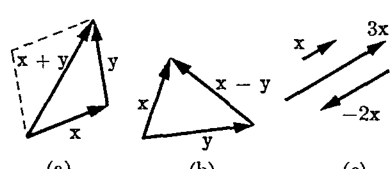


FIG. 11-3. Geometric interpretation of algebraic operations on vectors: (a) sum, (b) difference, (c) multiplication by scalars.

The reader is undoubtedly familiar with the fact that various quantities in physics (for example, forces, velocities, accelerations) can also be represented by directed line segments. If, for example, in Fig 11–3, $\mathbf{x}$ and $\mathbf{y}$ represent forces acting on a particle, the combined effect of these two forces is represented by $\mathbf{x} + \mathbf{y}$ It is for the reason that so many physical quantities combine by the parallelogram law that vector analysis has become such an important tool in physics

11–4 Geometric interpretation of the dot product in E_3. The dot product $\mathbf{x} \cdot \mathbf{y}$ of two vectors $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$ in E_3 is the real number $\mathbf{x} \cdot \mathbf{y} = x_1y_1 + x_2y_2 + x_3y_3$ In particular, $\mathbf{x} \cdot \mathbf{x} = |\mathbf{x}|^2$ We can describe the dot product in geometric terms by referring to Fig 11–4. The law of cosines of trigonometry yields

$$|\mathbf{x} - \mathbf{y}|^2 = |\mathbf{x}|^2 + |\mathbf{y}|^2 - 2|\mathbf{x}||\mathbf{y}| \cos \theta,$$

where θ is the angle ($0 \leq \theta \leq 180°$) between $\mathbf{x}$ and $\mathbf{y}$. On the other hand, we have

$$|\mathbf{x} - \mathbf{y}|^2 = (\mathbf{x} - \mathbf{y}) \cdot (\mathbf{x} - \mathbf{y}) = \mathbf{x} \cdot \mathbf{x} - 2\mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{y},$$

and, by comparing the last two equations, we find

$$\mathbf{x} \cdot \mathbf{y} = |\mathbf{x}||\mathbf{y}| \cos \theta$$

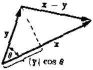


FIGURE 11–4

This formula describes $\mathbf{x} \cdot \mathbf{y}$ geometrically. It shows also that $\mathbf{x} \cdot \mathbf{y}$ remains unchanged under a rotation of axes (that is, when we change from one orthonormal basis to another)

11–5 The cross product of vectors in E_3 Besides the dot product $\mathbf{a} \cdot \mathbf{b}$ of two vectors (in which the result is a scalar), there is another way of multiplying vectors in E_3 which is particularly useful in the applications of vector analysis This is the *cross product* (also called the *vector product* or *outer product*), denoted by $\mathbf{a} \times \mathbf{b}$, in which the result is a new vector in E_3.

11–3 DEFINITION *Let* $\mathbf{a} = (a_1, a_2, a_3)$ *and* $\mathbf{b} = (b_1, b_2, b_3)$ *be two vectors in* E_3 *The cross product* $\mathbf{a} \times \mathbf{b}$ *is defined to be the vector*

$$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2, \quad a_3b_1 - a_1b_3, \quad a_1b_2 - a_2b_1).$$

NOTE The definition is easily remembered by writing $\mathbf{a} \times \mathbf{b}$ as a determinant:

$$\text{(4)} \qquad \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix},$$

where $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$ are the usual coordinate vectors.

Unlike the dot product, the cross product is *not commutative.* In fact, we have $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$. (This follows at once from the definition.) In particular, this implies $\mathbf{a} \times \mathbf{a} = \mathbf{0}$. On the other hand, an easy calculation shows that the cross product does satisfy the distributive law:

$$\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c}).$$

Furthermore, we clearly have $(\lambda\mathbf{a}) \times (\mu\mathbf{b}) = (\lambda\mu)(\mathbf{a} \times \mathbf{b})$ for all scalars λ and μ.

The length of $\mathbf{a} \times \mathbf{b}$ is easily determined by the following theorem:

11–4 THEOREM. $|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2|\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2$.

Proof.

$$|\mathbf{a} \times \mathbf{b}|^2 = (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2.$$

$$\begin{aligned} |\mathbf{a}|^2|\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2 &= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) \\ &\quad - (a_1b_1 + a_2b_2 + a_3b_3)^2. \end{aligned}$$

The result follows by comparing these equations. (See also Exercise 1–15.)

Since $(\mathbf{a} \cdot \mathbf{b})^2 = |\mathbf{a}|^2|\mathbf{b}|^2 \cos^2 \theta$, where θ is the angle ($0 \leq \theta \leq 180°$) between $\mathbf{a}$ and $\mathbf{b}$, the result of Theorem 11–4 implies

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin \theta.$$

Geometrically, the product $|\mathbf{a}||\mathbf{b}| \sin \theta$ represents the area of the parallelogram determined by $\mathbf{a}$ and $\mathbf{b}$. (See Fig. 11–5.)

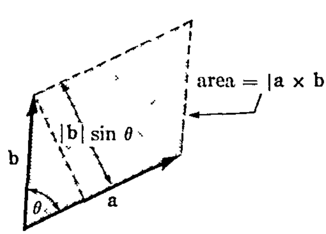


FIGURE 11–5.

11–6 The scalar triple product. Before deriving further properties of the cross product, it is convenient to study the *scalar triple product* $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$ [which can only mean $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$].

11-5 THEOREM *Let* $\mathbf{a} = (a_1, a_2, a_3)$, $\mathbf{b} = (b_1, b_2, b_3)$, $\mathbf{c} = (c_1, c_2, c_3)$. *Then*

$$\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}. \tag{5}$$

Proof If we write $\mathbf{b} \times \mathbf{c} = \lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \lambda_3 \mathbf{u}_3$, the numbers $\lambda_1, \lambda_2, \lambda_3$ are the cofactors of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ obtained by expanding the determinant

$$\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

along the first row. On the other hand, the dot product of $\mathbf{a}$ with $\mathbf{b} \times \mathbf{c}$ is $\lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3$. Since $\lambda_1, \lambda_2, \lambda_3$ are also the cofactors of a_1, a_2, a_3 in the determinant appearing in (5), the theorem follows at once.

The next theorem tells us that we can interchange the dot and cross in a scalar triple product without affecting the value of the product.

11-6 THEOREM $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} = \mathbf{a} \times \mathbf{b} \cdot \mathbf{c}$

Proof

$$\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \mathbf{c} \cdot \mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{b} \cdot \mathbf{c}.$$

We can use Theorem 11-6 to prove that the cross product $\mathbf{a} \times \mathbf{b}$ is a vector orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

11-7 THEOREM *Let* $\mathbf{c} = \mathbf{a} \times \mathbf{b}$. *Then* $\mathbf{a} \cdot \mathbf{c} = \mathbf{b} \cdot \mathbf{c} = 0$.

Proof $\mathbf{a} \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{a} \cdot \mathbf{b} = \mathbf{0} \cdot \mathbf{b} = 0$. Similarly, $\mathbf{b} \cdot \mathbf{c} = 0$.

Geometrically, then, $\mathbf{a} \times \mathbf{b}$ will be perpendicular to the plane determined by $\mathbf{a}$ and $\mathbf{b}$.

11-8 THEOREM *Let* $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ *be an orthonormal set of vectors in* E_3. *Then*

$$\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3 = \pm 1.$$

Proof. Write $\mathbf{a}_i = \alpha_{i1}\mathbf{u}_1 + \alpha_{i2}\mathbf{u}_2 + \alpha_{i3}\mathbf{u}_3$ $(i = 1, 2, 3)$. The determinant for $\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3$ is unchanged when its rows and columns are interchanged. The product of the determinant with itself is therefore equal to

$$\begin{aligned}(\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3)^2 &= \det[\alpha_{ij}]\det[\alpha_{ji}] = \det\left[\sum_{k=1}^{3}\alpha_{ik}\alpha_{jk}\right] \\ &= \det[\mathbf{a}_i \cdot \mathbf{a}_j] = 1.\end{aligned}$$

11-9 THEOREM. *Let* $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ *be an orthonormal basis for* E_3. *If* $\mathbf{v}_1 \cdot \mathbf{v}_2 \times \mathbf{v}_3 = 1$, *we have*

$$\mathbf{v}_1 \times \mathbf{v}_2 = \mathbf{v}_3, \qquad \mathbf{v}_2 \times \mathbf{v}_3 = \mathbf{v}_1, \qquad \mathbf{v}_3 \times \mathbf{v}_1 = \mathbf{v}_2.$$

On the other hand, if $\mathbf{v}_1 \cdot \mathbf{v}_2 \times \mathbf{v}_3 = -1$, *we have*

$$\mathbf{v}_1 \times \mathbf{v}_2 = -\mathbf{v}_3, \qquad \mathbf{v}_2 \times \mathbf{v}_3 = -\mathbf{v}_1, \qquad \mathbf{v}_3 \times \mathbf{v}_1 = -\mathbf{v}_2.$$

Proof. Write $\mathbf{v}_1 \times \mathbf{v}_2 = \lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \lambda_3\mathbf{v}_3$. Then, if $\mathbf{v}_1 \cdot \mathbf{v}_2 \times \mathbf{v}_3 = 1$, we have

$$\lambda_1 = \mathbf{v}_1 \cdot \mathbf{v}_1 \times \mathbf{v}_2 = 0, \qquad \lambda_2 = \mathbf{v}_2 \cdot \mathbf{v}_1 \times \mathbf{v}_2 = 0,$$

$$\lambda_3 = \mathbf{v}_3 \cdot \mathbf{v}_1 \times \mathbf{v}_2 = \mathbf{v}_1 \cdot \mathbf{v}_2 \times \mathbf{v}_3 = 1.$$

Hence, $\mathbf{v}_1 \times \mathbf{v}_2 = \mathbf{v}_3$. The other relations are proved in the same way.

Now we are ready to study the behavior of the cross product under a change of basis.

11-10 THEOREM. *Let* $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ *be an orthonormal basis for* E_3. *Let* $\mathbf{a}$ *and* $\mathbf{b}$ *be two vectors in* E_3 *and write*

$$\begin{aligned}\mathbf{a} &= \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3, \\ \mathbf{b} &= \beta_1\mathbf{v}_1 + \beta_2\mathbf{v}_2 + \beta_3\mathbf{v}_3.\end{aligned}$$

Then we have

$$\mathbf{a} \times \mathbf{b} = (\mathbf{v}_1 \cdot \mathbf{v}_2 \times \mathbf{v}_3)\begin{vmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \\ \alpha_1 & \alpha_2 & \alpha_3 \\ \beta_1 & \beta_2 & \beta_3 \end{vmatrix}.$$

Proof. By the distributive law, we have

$$\mathbf{a} \times \mathbf{b} = \left(\sum_{i=1}^{3} \alpha_i \mathbf{v}_i\right) \times \left(\sum_{j=1}^{3} \beta_j \mathbf{v}_j\right) = \sum_{i=1}^{3} \sum_{j=1}^{3} \alpha_i \beta_j (\mathbf{v}_i \times \mathbf{v}_j)$$

If $\mathbf{v}_1 \cdot \mathbf{v}_2 \times \mathbf{v}_3 = +1$, Theorem 11-9 yields

$$\mathbf{a} \times \mathbf{b} = (\alpha_1\beta_2 - \alpha_2\beta_1)\mathbf{v}_3 + (\alpha_2\beta_3 - \alpha_3\beta_2)\mathbf{v}_1 + (\alpha_3\beta_1 - \alpha_1\beta_3)\mathbf{v}_2$$

$$= \begin{vmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \\ \alpha_1 & \alpha_2 & \alpha_3 \\ \beta_1 & \beta_2 & \beta_3 \end{vmatrix}.$$

If $\mathbf{v}_1 \cdot \mathbf{v}_2 \times \mathbf{v}_3 = -1$, the cross product $\mathbf{a} \times \mathbf{b}$ is the negative of this determinant.

11-11 THEOREM *A set $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ of vectors in E_3 is linearly independent if, and only if,*

$$\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} \neq 0$$

Proof Assume that there exist scalars $\lambda_1, \lambda_2, \lambda_3$ (not all zero), such that

$$\lambda_1\mathbf{a} + \lambda_2\mathbf{b} + \lambda_3\mathbf{c} = \mathbf{0}$$

For definiteness, say $\lambda_1 \neq 0$ Then we can write $\mathbf{a} = \mu_1\mathbf{b} + \mu_2\mathbf{c}$, where μ_1, μ_2 are scalars, and we have

$$\mathbf{a} \times \mathbf{b} = (\mu_1\mathbf{b} + \mu_2\mathbf{c}) \times \mathbf{b} = \mu_2(\mathbf{c} \times \mathbf{b}).$$

Therefore,

$$\mathbf{a} \times \mathbf{b} \cdot \mathbf{c} = \mu_2(\mathbf{c} \times \mathbf{b} \cdot \mathbf{c}) = -\mu_2(\mathbf{b} \times \mathbf{c} \cdot \mathbf{c}) = -\mu_2(\mathbf{b} \cdot \mathbf{c} \times \mathbf{c}) = 0.$$

Hence, if $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} \neq 0$, the set $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ must be linearly independent

Conversely, assume that $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ is linearly independent. We will prove that $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} \neq 0$ Because $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ is linearly independent, we can use the Gram-Schmidt process to construct an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ In fact, the process yields

$$\mathbf{v}_1 = \mathbf{a}, \qquad \mathbf{v}_2 = \mathbf{b} - \frac{\mathbf{b} \cdot \mathbf{a}}{|\mathbf{a}|^2}\mathbf{a}, \qquad \mathbf{v}_3 = \mathbf{c} - \frac{\mathbf{c} \cdot \mathbf{a}}{|\mathbf{a}|^2}\mathbf{a} - \frac{\mathbf{c} \cdot \mathbf{b}}{|\mathbf{v}_2|^2}\mathbf{v}_2$$

Hence, $\mathbf{v}_1 \times \mathbf{v}_2 = \mathbf{a} \times \mathbf{b} - (\mathbf{b} \cdot \mathbf{a})\,(\mathbf{a} \times \mathbf{a})/|\mathbf{a}|^2 = \mathbf{a} \times \mathbf{b}$. Similarly, we find

$$\mathbf{v}_1 \times \mathbf{v}_2 \cdot \mathbf{v}_3 = \mathbf{a} \times \mathbf{b} \cdot \mathbf{c},$$

since $\mathbf{a} \times \mathbf{b} \cdot \mathbf{a} = \mathbf{a} \times \mathbf{b} \cdot \mathbf{b} = 0$. But, by Theorem 11–8, we have

$$\mathbf{a} \times \mathbf{b} \cdot \mathbf{c} = \left(\frac{\mathbf{v}_1}{|\mathbf{v}_1|} \times \frac{\mathbf{v}_2}{|\mathbf{v}_2|} \cdot \frac{\mathbf{v}_3}{|\mathbf{v}_3|}\right) |\mathbf{v}_1||\mathbf{v}_2||\mathbf{v}_3| = \pm|\mathbf{v}_1||\mathbf{v}_2||\mathbf{v}_3| \neq 0.$$

11–12 DEFINITION. *Let* $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ *be a linearly independent set of vectors in* E_3. *The ordered triple of vectors* $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ *is said to be a positive triple if* $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} > 0$ *and a negative triple if* $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} < 0$.

NOTE. If $\mathbf{a}$ and $\mathbf{b}$ are not collinear, the triple $(\mathbf{a}, \mathbf{b}, \mathbf{a} \times \mathbf{b})$ is always a positive triple, since $\mathbf{a} \times \mathbf{b} \cdot \mathbf{a} \times \mathbf{b} = |\mathbf{a} \times \mathbf{b}|^2 > 0$.

When representing vectors geometrically, it is customary to draw the coordinate axes so that the vectors representing $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$ form a "right-handed system" as illustrated in Fig. 11–6. When this is done, the "direction" of the cross product $\mathbf{a} \times \mathbf{b}$ can be determined by the "right-hand rule." That is to say, when $\mathbf{a}$ is rotated into $\mathbf{b}$ in such a way that the fingers of the right hand point in the direction of rotation, then the thumb will point in the direction of $\mathbf{a} \times \mathbf{b}$. (An example is illustrated in Fig. 11–6.)

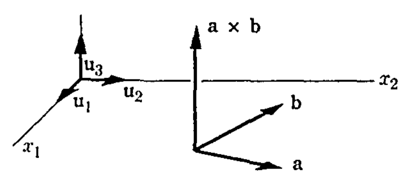


FIG. 11–6. The relative positions of a, b, and a × b in a right-handed coordinate system.

In geometric terms, Theorem 11–11 tells us that $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} = 0$ if, and only if, $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are coplanar vectors. When they are not coplanar, the absolute value of $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$ yields the volume of the parallelepiped determined by $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$. This is easily verified by using the formula

$$\mathbf{a} \times \mathbf{b} \cdot \mathbf{c} = |\mathbf{a} \times \mathbf{b}||\mathbf{c}| \cos \phi = |\mathbf{a}||\mathbf{b}| \sin \theta \, |\mathbf{c}| \cos \phi$$

in conjunction with Fig. 11–7. The area of the base is $|\mathbf{a}||\mathbf{b}| \sin \theta$ and the altitude is $|\mathbf{c}| \cos \phi$. (In Fig. 11–7, $\cos \phi > 0$ since $\phi < 90°$.)

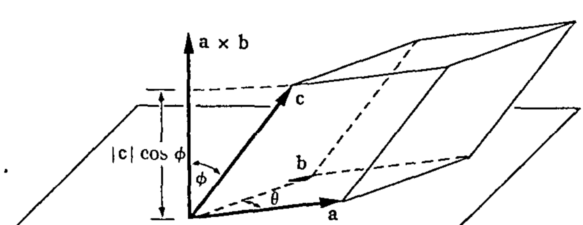


FIG. 11–7. Geometric interpretation of the scalar triple product.

11-7 Derivatives of vector-valued functions.

11-13 DEFINITION *Let* $\mathbf{f}$ *be a vector-valued function defined on an interval* $[a, b]$ *in* E_1. *The vector-valued function* $\mathbf{f}'$, *defined by the equation*

$$\mathbf{f}'(t) = \lim_{h \to 0} \frac{1}{h}[\mathbf{f}(t+h) - \mathbf{f}(t)], \qquad \text{if } t \in [a, b],$$

is called the derivative of $\mathbf{f}$. *(One-sided limits are understood at the endpoints* a *and* b.)

NOTE. If $\mathbf{f} = (f_1, f_2, f_3)$, it is clear that $\mathbf{f}'$ exists if, and only if, all three derivatives f_1', f_2', f_3' exist, in which case we have $\mathbf{f}' = (f_1', f_2', f_3')$.

In the work that follows we shall often have occasion to compute derivatives of vector-valued functions and it is convenient to note the following rules of differentiation.

11-14 THEOREM *Let* λ *be a real-valued function and let* $\mathbf{f}$ *and* $\mathbf{g}$ *be vector-valued functions defined on an interval* $[a, b]$. *Assuming the existence of the derivatives* λ', $\mathbf{f}'$, *and* $\mathbf{g}'$, *we have*

$$(\mathbf{f} + \mathbf{g})' = \mathbf{f}' + \mathbf{g}', \qquad (\lambda\mathbf{f})' = \lambda\mathbf{f}' + \lambda'\mathbf{f},$$

$$(\mathbf{f} \cdot \mathbf{g})' = \mathbf{f} \cdot \mathbf{g}' + \mathbf{f}' \cdot \mathbf{g}, \qquad (\mathbf{f} \times \mathbf{g})' = \mathbf{f} \times \mathbf{g}' + \mathbf{f}' \times \mathbf{g}.$$

These formulas are proved exactly as in the real case. (See Exercise 11-9.) The next theorem, however, has no analog in the case of real-valued functions.

11-15 THEOREM *If* $\mathbf{f}'$ *exists and if* $|\mathbf{f}|$ *is constant on* $[a, b]$, *then*

$$\mathbf{f}(t) \cdot \mathbf{f}'(t) = 0 \qquad \text{for each } t \text{ in } [a, b].$$

Proof. Since $\mathbf{f} \cdot \mathbf{f}$ is constant, we have $(\mathbf{f} \cdot \mathbf{f})' = 0$. But $(\mathbf{f} \cdot \mathbf{f})' = 2\mathbf{f} \cdot \mathbf{f}'$.

Geometrically, this states that the derivative $\mathbf{f}'(t)$ of a vector-valued function of constant length is perpendicular to $\mathbf{f}(t)$, provided that $\mathbf{f}'(t) \neq 0$.

11-8 Elementary differential geometry of space curves. Let $\mathbf{x} = (x_1, x_2, x_3)$ be a continuous vector-valued function defined on an interval $[a, b]$. The image of $[a, b]$ under $\mathbf{x}$ is a curve Γ in E_3. In Chapter 8 we observed that such a function $\mathbf{x}$ can be thought of as describing a *motion* of a particle along the curve Γ. The parametric interval $[a, b]$ is then interpreted as a "time interval," and the position of the particle is specified at time t by the vector $\mathbf{x}(t)$. If we represent each vector $\mathbf{x}(t)$ geometrically as in Fig. 11-8, the terminal point of $\mathbf{x}(t)$ traces out the curve Γ as t varies from a to b.

In Chapter 8 we found that a curve Γ is rectifiable if, and only if, the components of $\mathbf{x}$ are of bounded variation on $[a, b]$. In this chapter we shall discuss certain aspects of the theory of curves that make use of differentiability properties of $\mathbf{x}$. The corresponding theorems provide an introduction to what is called the *differential geometry of space curves.*

11-9 The tangent vector of a curve. Throughout this section and the next, Γ will denote a curve described by a continuous vector-valued function $\mathbf{x}$ defined on an interval $[a, b]$.

11-16 DEFINITION. *A point* $\mathbf{x}(t)$ *is called a regular point of* Γ *if the derivative* $\mathbf{x}'(t)$ *exists and is not zero, in which case* $\mathbf{x}'(t)$ *is called the tangent vector at this point. Points where* $\mathbf{x}'(t)$ *fails to exist or where* $\mathbf{x}'(t) = \mathbf{0}$ *are called singular points of* Γ.

By referring to Fig. 11-9, it is easy to see that this definition is in accord with our intuitive notion of tangency. It is convenient also to use physical terminology when referring to concepts associated with curves in E_3. For example, the vector $\mathbf{x}(t)$ is referred to as the *position* vector and $\mathbf{x}'(t)$ is called the *velocity* vector, also denoted by $\mathbf{v}(t)$. The scalar $|\mathbf{v}(t)|$ is called the *speed* and is denoted by $v(t)$.

11-17 DEFINITION. *At the regular points of* Γ *we define the unit tangent vector* $\mathbf{T}$ *as follows:*

$$\mathbf{T}(t) = \frac{\mathbf{x}'(t)}{|\mathbf{x}'(t)|}. \tag{6}$$

In terms of velocity, equation (6) becomes

$$\mathbf{v}(t) = v(t)\mathbf{T}(t). \tag{7}$$

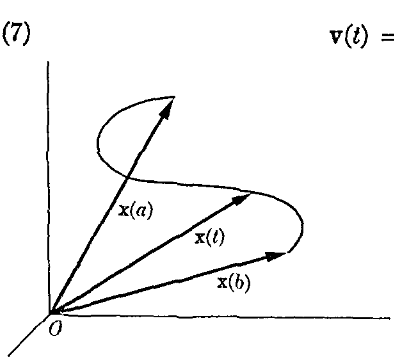


FIG. 11-8. Vector representation of a curve.

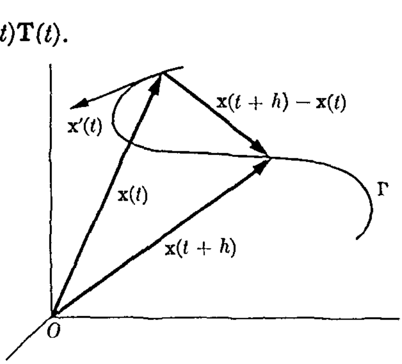


FIG. 11-9. Tangent vector.

Geometrically, $\mathbf{T}(t)$ is a vector of unit length which indicates the "direction" of the motion at time t. The scalar $v(t)$ tells us the speed at that instant. Clearly, Γ is a straight line if, and only if, $\mathbf{T}$ is constant.

11-10 Normal vectors, curvature, torsion. In this section we assume that the curve has no singular points and that both the second and third derivatives $\mathbf{x}''(t)$ and $\mathbf{x}'''(t)$ exist (when needed). Existence of $\mathbf{x}''(t)$ implies existence of $\mathbf{T}'(t)$ [because of (6)]. Since the length of $\mathbf{T}(t)$ is constant, the derivative $\mathbf{T}'(t)$ gives us a way of measuring the tendency of $\mathbf{T}(t)$ to change its "direction."

11-18 DEFINITION. *The non-negative function κ defined by the equation*

$$\kappa(t) = \frac{|\mathbf{T}'(t)|}{|\mathbf{x}'(t)|} \tag{8}$$

is called the curvature of the curve.

If Γ is a straight line, $\kappa(t)$ is zero for all t, since $\mathbf{T}' = \mathbf{0}$. At points where $\mathbf{T}'(t) \neq \mathbf{0}$, Theorem 11-15 tells us that $\mathbf{T}'(t)$ is perpendicular to $\mathbf{T}(t)$. This justifies the terminology used in the following definition.

11-19 DEFINITION. *At those points of Γ for which $\mathbf{T}'(t) \neq \mathbf{0}$, we define the unit principal normal vector $\mathbf{N}$ by the equation*

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|}. \tag{9}$$

Introducing the arc length $s(t) = \int_a^t |\mathbf{x}'(u)|\,du$ and using equation (8), (9) becomes

$$\mathbf{T}'(t) = \kappa(t)s'(t)\mathbf{N}(t). \tag{10}$$

For each t the vectors $\mathbf{T}(t)$ and $\mathbf{N}(t)$ determine a plane known as the *osculating plane* of the curve. If the curve itself is a *plane curve*, the osculating plane coincides with the plane of the curve. In general, however, the osculating plane changes with t and, to study this change, we introduce the binormal vector.

11-20 DEFINITION. *The unit vector $\mathbf{B}$ defined by the equation*

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t) \tag{11}$$

is called the binormal vector.

The binormal is perpendicular to the osculating plane since it is perpendicular to both **T** and **N**. Moreover, **B**, **T**, and **N** form an orthonormal set. (See Fig. 11–10.)

11–21 THEOREM. *$\mathbf{B}'(t)$ is a scalar multiple of $\mathbf{N}(t)$.*

Proof. We can write $\mathbf{B}' = \lambda_1\mathbf{N} + \lambda_2\mathbf{T} + \lambda_3\mathbf{B}$, where $\lambda_1 = \mathbf{B}'\cdot\mathbf{N}$, $\lambda_2 = \mathbf{B}'\cdot\mathbf{T}$, $\lambda_3 = \mathbf{B}'\cdot\mathbf{B}$. Now $\lambda_3 = 0$, since $|\mathbf{B}| = 1$. We prove next that $\lambda_2 = 0$. Differentiation of (11) gives us

$$\mathbf{B}' = \mathbf{T}' \times \mathbf{N} + \mathbf{T} \times \mathbf{N}' = \mathbf{T} \times \mathbf{N}',$$

since $\mathbf{T}' \times \mathbf{N} = 0$ [because of (10)]. Therefore,

$$\lambda_2 = \mathbf{B}'\cdot\mathbf{T} = \mathbf{T} \times \mathbf{N}'\cdot\mathbf{T} = 0.$$

Hence, $\mathbf{B}' = \lambda_1\mathbf{N}$, as asserted.

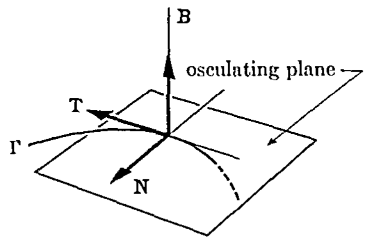


FIGURE 11–10.

11–22 DEFINITION. *The real-valued function τ defined by the equation*

$$\mathbf{B}'(t) = -\tau(t)s'(t)\mathbf{N}(t) \tag{12}$$

is called the torsion of the curve.

NOTE. Comparing formula (12) with (10), we see that the torsion plays a role similar to that of the curvature κ. In fact, the torsion is sometimes referred to as the "second curvature" of the curve. It gives us a way of measuring the tendency of the curve to "twist" out of the osculating plane.

Although the curvature $\kappa(t)$ (as we have defined it) is never negative, the torsion may be positive or negative. For a plane curve, **B** is constant and $\tau = 0$.

Formulas (10) and (12) express $\mathbf{T}'$ and $\mathbf{B}'$ in terms of the orthonormal set $\{\mathbf{N}, \mathbf{T}, \mathbf{B}\}$. There is a similar formula for $\mathbf{N}'$.

11–23 THEOREM. $\mathbf{N}'(t) = -\kappa(t)s'(t)\mathbf{T}(t) + \tau(t)s'(t)\mathbf{B}(t)$.

Proof. Differentiation of the equation $\mathbf{N} = \mathbf{B} \times \mathbf{T}$ gives us

$$\mathbf{N}' = \mathbf{B}' \times \mathbf{T} + \mathbf{B} \times \mathbf{T}' = -\tau s'\mathbf{N} \times \mathbf{T} + \kappa s'\mathbf{B} \times \mathbf{N}.$$

Since $\mathbf{N} \times \mathbf{T} = -\mathbf{B}$ and $\mathbf{B} \times \mathbf{N} = -\mathbf{T}$, this proves the theorem.

All the formulas of the last two sections simplify considerably when the speed $s'(t) = 1$ for all t. In this case, we get $\mathbf{v} = \mathbf{T}$, $\kappa = |\mathbf{T}'|$, and

$$\text{(13)} \qquad \mathbf{T}' = \kappa\mathbf{N}, \qquad \mathbf{B}' = -\tau\mathbf{N}, \qquad \mathbf{N}' = -\kappa\mathbf{T} + \tau\mathbf{B}$$

The three formulas in (13) are known as the *Frenet formulas*. They play a fundamental role in the differential geometry of space curves. (See Ref. 11-6.)

NOTE. The assumption that $|\mathbf{x}'(t)|$ is never zero implies that s is a strictly increasing function. Hence we can always obtain a representation of Γ with arc length as parameter. (See Exercise 8-11.) When this is done, we obtain the simplification just mentioned.

11-11 Vector fields. In the applications of mathematics to physics and to certain branches of engineering (for example, in fluid mechanics), we often deal with the concept of a *vector field*. Mathematically, a vector field is simply a vector-valued function defined on some set. For example, if to each point $\mathbf{x}$ in the atmosphere we assign a vector $\mathbf{v}(\mathbf{x})$ representing the wind velocity, this defines a vector field. If $\mathbf{v}(\mathbf{x})$ is expressed in terms of its components relative to some basis, say $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$, we can write

$$\mathbf{v}(\mathbf{x}) = v_1(\mathbf{x})\mathbf{u}_1 + v_2(\mathbf{x})\mathbf{u}_2 + v_3(\mathbf{x})\mathbf{u}_3$$

The components v_1, v_2, v_3 are three real-valued functions and a study of vector fields is, in a sense, equivalent to the study of "triples" of real-valued functions. To distinguish between vector fields and real-valued functions, the latter are sometimes called *scalar fields*. For example, the temperature at each point of the atmosphere defines a scalar field.

One of the most useful physical models of a vector field occurs when we consider the motion of a fluid. At each point $\mathbf{x}$ (or "particle") of the fluid we attach a vector $\mathbf{v}(\mathbf{x})$ which represents the velocity of that particular particle. Of course, the field may or may not change with time. In a *stationary flow*, $\mathbf{v}(\mathbf{x})$ is completely determined by $\mathbf{x}$ and does not depend on time. We shall restrict our considerations to stationary vector fields.

In physical problems involving vector fields it is important to know not only the vector $\mathbf{v}(\mathbf{x})$ at each point $\mathbf{x}$, but also how this vector changes as we move from one point to another. To study this change, we have at our disposal the machinery of partial differentiation which can be applied to the components of $\mathbf{v}$. In general, the partial derivatives of these components will depend on the choice of the basis relative to which the components have been determined. Hence, partial derivatives are somewhat unsatisfactory for describing certain physical quantities, particularly if these quantities have a meaning quite independent of the basis. Instead, special combinations of the partial derivatives, known as the *divergence* and *curl*, are used to describe the behavior of vector fields. The divergence and curl turn out to be *independent* of the basis (if the basis is orthonormal) and they have definite physical significance. The divergence of a vector field is a *scalar field* which, in the case of fluid flow, measures the rate at which fluid is flowing away from the immediate vicinity of each point. In an incompressible fluid with no "sources" or "sinks," the divergence is zero at each point. The curl of a vector field is, on the other hand, another vector field which in a certain sense measures the tendency of the fluid to rotate at each point. These concepts, divergence and curl, will be precisely defined later and their properties will be studied in detail.

11-12 The gradient field in E_n. We have already encountered one example of a vector field in connection with the study of partial differentiation. If ϕ is a real-valued function (a scalar field) defined on an open set S in E_n, the gradient of ϕ, denoted by $\nabla\phi$ or by grad ϕ, is a vector-valued function defined by

$$\text{grad}\,\phi(\mathbf{x}) = \nabla\phi(\mathbf{x}) = (D_1\phi(\mathbf{x}), \ldots, D_n\phi(\mathbf{x})) \tag{14}$$

at each point $\mathbf{x}$ in S where the partial derivatives exist. The following properties of the gradient are immediate consequences of the definition:

11-24 THEOREM. *Let ϕ and ψ be real-valued functions such that $\nabla\phi$ and $\nabla\psi$ both exist on an open set S in E_n. Then we have*

(a) $\nabla(\phi + \psi) = \nabla\phi + \nabla\psi$,

(b) $\nabla(\phi \cdot \psi) = \phi\nabla\psi + \psi\nabla\phi$,

(c) $\nabla(\phi/\psi) = (\psi\nabla\phi - \phi\nabla\psi)/\psi^2$ (*at points* $\mathbf{x}$ *where* $\psi(\mathbf{x}) \neq 0$).

In the case $n = 3$, the gradient has a useful geometric interpretation. Let c be a constant and consider the set S_c of points $\mathbf{x}$ in S where $\phi(\mathbf{x}) = c$. In many cases, S_c is a surface.† If S_c has a tangent plane at a point $\mathbf{a} = (a_1, a_2, a_3)$, we know from elementary calculus that the equation of this plane is

$$D_1\phi(\mathbf{a})(x_1 - a_1) + D_2\phi(\mathbf{a})(x_2 - a_2) + D_3\phi(\mathbf{a})(x_3 - a_3) = 0.$$

This means that the vector $\nabla\phi(\mathbf{a})$ is *normal* to the plane (and hence normal to S_c) at the point $\mathbf{a}$. (The tangent plane exists whenever $\nabla\phi(\mathbf{a}) \neq \mathbf{0}$.)

† A precise definition of a surface is given in Definition 11-32.

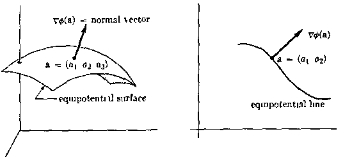


FIG 11-11. Geometric interpretation of the gradient vector

The scalar field ϕ whose gradient is $\nabla\phi$ is called the *potential function* of the vector field $\nabla\phi$. The corresponding surfaces S_c are called *equipotential surfaces* (or *level surfaces*). In the case of two-dimensional fields, each set S_c is a plane curve known as an *equipotential line* (or *level line*). The equipotential surfaces (lines) are orthogonal to the gradient vector at each point $\mathbf{a}$ where $\nabla\phi(\mathbf{a}) \neq \mathbf{0}$. (See Fig. 11-11.)

In Theorem 10-45 we proved that if $\mathbf{f} = (f_1, \ldots, f_n) \in C'$ on an open set S in E_n and if $\mathbf{f}$ has a potential function ϕ, then for each $\mathbf{x}$ in S we must have

$$(15) \qquad D_i f_j(\mathbf{x}) = D_j f_i(\mathbf{x}), \qquad (i \neq j,\ i, j = 1, 2, \ldots, n)$$

We also observed that the condition (15) is *not sufficient* to establish the existence of the potential unless S is suitably restricted. For example, Theorem 10-48 tells us that (15) suffices if S is an open interval in E_n. In the special case $n = 2$, we can get by with a less stringent requirement on S, namely, simple connectedness. (See Definition 10-46 and Theorem 10-47.)

11-13 The curl of a vector field in E_3.

11-25 DEFINITION *Let $\mathbf{f} = (f_1, f_2, f_3)$ be defined on an open set S in E_3. We denote by* curl $\mathbf{f}$ *the vector-valued function defined by*

$$(16) \qquad \operatorname{curl} \mathbf{f} = (D_2 f_3 - D_3 f_2,\quad D_3 f_1 - D_1 f_3,\quad D_1 f_2 - D_2 f_1)$$

whenever the partial derivatives on the right exist.

NOTE The expression for curl $\mathbf{f}$ is easily remembered by introducing the following device. Let us treat the operator ∇ as though it were a symbolic "vector," namely, $\nabla = (D_1, D_2, D_3)$, and form the cross product $\nabla \times \mathbf{f}$ in a purely formal way (using Definition 11–3). We then find that $\nabla \times \mathbf{f}$ has the same "components" as curl $\mathbf{f}$ and we agree to write curl $\mathbf{f} = \nabla \times \mathbf{f}$. This is a convenient notational device which makes certain formulas easy to remember. Expressing $\nabla \times \mathbf{f}$ as a determinant, we have

$$\operatorname{curl} \mathbf{f} = \nabla \times \mathbf{f} = \begin{vmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \\ D_1 & D_2 & D_3 \\ f_1 & f_2 & f_3 \end{vmatrix}$$

with the understanding that this is to be expanded in the obvious way.

The following theorem is an easy consequence of Definition 11–25:

11–26 THEOREM. *Let* $\mathbf{f}$ *and* $\mathbf{g}$ *be vector fields such that* curl $\mathbf{f}$ *and* curl $\mathbf{g}$ *exist on an open set* S *in* E_3, *and let* ϕ *be a scalar field such that* $\nabla\phi$ *exists on* S. *Then we have*

(a) $\operatorname{curl}(\mathbf{f} + \mathbf{g}) = \operatorname{curl} \mathbf{f} + \operatorname{curl} \mathbf{g}$,

(b) $\operatorname{curl}(\phi\mathbf{f}) = \phi \operatorname{curl} \mathbf{f} + \nabla\phi \times \mathbf{f}$.

The curl can be given a physical interpretation as follows: Suppose a rigid body is rotating about a fixed axis with constant angular speed ω. The basis $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$ can be chosen so that the velocity vector $\mathbf{x}'$ of an arbitrary point P of the body is given by

$$\mathbf{x}' = (\omega\mathbf{u}_3) \times \mathbf{x} = -\omega x_2\mathbf{u}_1 + \omega x_1\mathbf{u}_2,$$

where $\mathbf{x} = x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + x_3\mathbf{u}_3$ is the position vector $\overrightarrow{OP}$. (See Exercise 11–14.) The vector $\boldsymbol{\omega} = \omega\mathbf{u}_3$ is called the *angular velocity* vector of the body. The curl of $\mathbf{x}'$ is

$$\operatorname{curl} \mathbf{x}' = \begin{vmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \\ D_1 & D_2 & D_3 \\ -\omega x_2 & \omega x_1 & 0 \end{vmatrix} = 2\omega\mathbf{u}_3 = 2\boldsymbol{\omega}.$$

That is, the curl of the velocity vector of a rigid body rotating with angular velocity $\boldsymbol{\omega}$ is $2\boldsymbol{\omega}$.

NOTE. Vector fields $\mathbf{f}$ for which curl $\mathbf{f} = \mathbf{0}$ are called *irrotational.*

Using (15) with $n = 3$, we obtain the important formula

$$\operatorname{curl}(\operatorname{grad} \phi) = \mathbf{0}, \tag{17}$$

provided that grad $\phi \in C'$ on an open set in E_3. Theorem 10–48, which provides a partial converse of this statement, can now be combined with (17) to yield.

11–27 THEOREM *Let* $\mathbf{f} = (f_1, f_2, f_3)$ *be a vector field such that* $\mathbf{f} \in C'$ *on an open interval* S *in* E_3 *Then* $\mathbf{f}$ *is the gradient of a potential if, and only if,* curl $\mathbf{f} = \mathbf{0}$

11–14 The divergence of a vector field in E_n. We have discussed the possibility of solving the equation $\nabla\phi = \mathbf{f}$ if $\mathbf{f}$ is a given vector field Now we ask a similar question in connection with the equation $\nabla \times \mathbf{g} = \mathbf{f}$ That is, when is a given vector field $\mathbf{f}$ the curl of another vector field $\mathbf{g}$? A necessary and sufficient condition for solving such an equation can be stated very simply in terms of a scalar field known as the *divergence*, whose properties will be discussed next

11–28 DEFINITION *Let* $\mathbf{f} = (f_1, \ldots, f_n)$ *be a vector field defined on an open set* S *in* E_n *We denote by* div $\mathbf{f}$ *the scalar field defined by*

$$\operatorname{div} \mathbf{f} = D_1 f_1 + D_2 f_2 + \cdots + D_n f_n,$$

whenever the partial derivatives on the right exist

NOTE We also write $\operatorname{div} \mathbf{f} = \nabla \cdot \mathbf{f}$

The divergence has the following elementary properties, analogous to those of the curl

11–29 THEOREM *If* $\mathbf{f}$ *and* $\mathbf{g}$ *are vector fields such that* div $\mathbf{f}$ *and* div $\mathbf{g}$ *exist on an open set* S *in* E_n *and if* ϕ *is a scalar field such that* $\nabla\phi$ *exists on* S, *then we have*

(a) $\operatorname{div}(\mathbf{f} + \mathbf{g}) = \operatorname{div} \mathbf{f} + \operatorname{div} \mathbf{g}$,

(b) $\operatorname{div}(\phi\mathbf{f}) = \phi \operatorname{div} \mathbf{f} + \nabla\phi \cdot \mathbf{f}$

The proof is left as an exercise For three-dimensional vector fields we have the following additional property

11–30 THEOREM *Let* $\mathbf{f} = (f_1, f_2, f_3)$ *be a vector field whose components have continuous mixed partials of second order on an open set* S *in* E_3 *Then*

$$\operatorname{div}(\operatorname{curl} \mathbf{f}) = 0$$

Proof

$$\begin{aligned}\operatorname{div}(\operatorname{curl} \mathbf{f}) = D_1(D_2 f_3 - D_3 f_2) &+ D_2(D_3 f_1 - D_1 f_3) \\ &+ D_3(D_1 f_2 - D_2 f_1) = 0\end{aligned}$$

The above theorem provides a clue for answering the question raised at the beginning of this section, namely: Which vector fields are curls?

11–31 THEOREM. *Let* $\mathbf{f} = (f_1, f_2, f_3)$ *be a given vector field whose components have continuous first order partial derivatives on an open interval* S *in* E_3. *If* $\operatorname{div} \mathbf{f}(\mathbf{x}) = 0$ *for each* $\mathbf{x}$ *in* S, *then there exists a vector field* $\mathbf{g} = (g_1, g_2, g_3)$ *such that* $\operatorname{curl} \mathbf{g} = \mathbf{f}$.

Proof. We will construct $\mathbf{g}$ explicitly as follows: Let $\mathbf{y} = (y_1, y_2, y_3)$ be a fixed point in S. For each point $\mathbf{x} = (x_1, x_2, x_3)$ in S, define

$$g_1(x_1, x_2, x_3) = \int_{y_3}^{x_3} f_2(x_1, y_2, t_3)\, dt_3 - \int_{y_2}^{x_2} f_3(x_1, t_2, y_3)\, dt_2.$$

Then, by differentiation, we find

(18)

$$D_3g_1(x_1, x_2, x_3) = f_2(x_1, y_2, x_3), \qquad D_2g_1(x_1, x_2, x_3) = -f_3(x_1, x_2, y_3).$$

Now put

$$g_2(x_1, x_2, x_3) = \int_{y_3}^{x_3} \left[\int_{y_1}^{x_1} D_3f_3(t_1, x_2, t_3)\, dt_1\right] dt_3.$$

Differentiation then yields

$$D_3g_2(x_1, x_2, x_3) = \int_{y_1}^{x_1} D_3f_3(t_1, x_2, x_3)\, dt_1, \tag{19}$$

$$\begin{aligned} D_1g_2(x_1, x_2, x_3) &= \int_{y_3}^{x_3} D_3f_3(x_1, x_2, t_3)\, dt_3 \\ &= f_3(x_1, x_2, x_3) - f_3(x_1, x_2, y_3). \end{aligned} \tag{20}$$

Next, define

$$g_3(x_1, x_2, x_3) = -\int_{y_2}^{x_2} \left[\int_{y_1}^{x_1} D_2f_2(t_1, t_2, x_3)\, dt_1\right] dt_2$$

$$+\int_{y_2}^{x_2} \left\{f_1(x_1, t_2, x_3) + \int_{y_1}^{x_1} [D_2f_2(t_1, t_2, x_3) + D_3f_3(t_1, t_2, x_3)]\, dt_1\right\} dt_2.$$

Differentiation gives us

(21)

$$D_2g_3(x_1, x_2, x_3) = -\int_{y_1}^{x_1} D_2f_2(t_1, x_2, x_3)\, dt_1 + f_1(\mathbf{x})$$

$$+\int_{y_1}^{x_1} [D_2f_2(t_1, x_2, x_3) + D_3f_3(t_1, x_2, x_3)]\, dt_1$$

$$= f_1(\mathbf{x}) + \int_{y_1}^{x_1} D_3f_3(t_1, x_2, x_3)\, dt_1,$$

$$D_1g_3(x_1, x_2, x_3) = -\int_{y_2}^{x_2} D_2f_2(x_1, t_2, x_3)\, dt_2 + \int_{x_2}^{y_2} \operatorname{div} \mathbf{f}(x_1, t_2, x_3)\, dt_2$$

(22)
$$= -\int_{y_2}^{x_2} D_2f_2(x_1, t_2, x_3)\, dt_2 = -f_2(\mathbf{x}) + f_2(x_1, y_2, x_3).$$

From (21) and (19) we obtain $D_2g_3(\mathbf{x}) - D_3g_2(\mathbf{x}) = f_1(\mathbf{x})$, whereas (18) and (22) yield $D_3g_1(\mathbf{x}) - D_1g_3(\mathbf{x}) = f_2(\mathbf{x})$ Similarly, (20) and (18) give us $D_1g_2(\mathbf{x}) - D_2g_1(\mathbf{x}) = f_3(\mathbf{x})$ Therefore curl $\mathbf{g}(\mathbf{x}) = \mathbf{f}(\mathbf{x})$ and the proof is complete.

NOTE Vector fields $\mathbf{f}$ for which div $\mathbf{f} = 0$ are called *solenoidal*

Theorem 11–30 states that a necessary condition for a vector field $\mathbf{f}$ to be the curl of another vector field is that div $\mathbf{f} = 0$ It is important to realize that this condition is *not always sufficient* If div $\mathbf{f} = 0$ on some set S, then a vector field $\mathbf{g}$ may or may not exist such that curl $\mathbf{g} = \mathbf{f}$ A counter example is given in Exercise 11–34 We have shown in Theorem 11–31 that such a $\mathbf{g}$ *does* exist if S is a three-dimensional *interval* The result is also true for certain other open sets in E_3 (for example, spheres and, more generally, convex open sets), but the counter example shows that it does not hold without some restriction on the "structure" of S Intuitively speaking, S must have the property that every *closed surface* in S is the boundary of a solid lying entirely in S This is analogous to the notion of simple connectivity in E_2 A precise formulation of this property requires a discussion of certain concepts in combinatorial topology which we will not enter into here

11–15 The Laplacian operator. Let ϕ be a scalar field defined on an open set S in E_n Then the definition of divergence yields the formula

$$\text{(23)} \qquad \operatorname{div}(\nabla\phi) = D_{1,1}\phi + D_{2,2}\phi + \cdots + D_{n,n}\phi,$$

whenever the partial derivatives on the right exist.

The divergence of a gradient is expressed symbolically as $\nabla \cdot (\nabla \mathbf{f})$ and is also written more briefly as $\nabla^2\phi$. The "operator" ∇^2 is called the *Laplacian operator* and when it is applied to scalar fields the result is given by formula (23). The partial differential equation $\nabla^2\phi = 0$ is known as *Laplace's equation.* A function ϕ is said to be *harmonic* on S if it satisfies Laplace's equation on S.

The operator ∇^2 can also be applied to a vector field $\mathbf{f} = (f_1, \ldots, f_n)$ by defining

$$\nabla^2\mathbf{f} = (\nabla^2 f_1, \ldots, \nabla^2 f_n).$$

The four operators—gradient, curl, divergence, Laplacian—are related by the following identity (whose proof is left as an exercise):

$$\text{(24)} \qquad \operatorname{curl}(\operatorname{curl}\mathbf{f}) = \operatorname{grad}(\operatorname{div}\mathbf{f}) - \nabla^2\mathbf{f}.$$

Further properties of the curl and divergence are given in the exercises at the end of the chapter.

11–16 Surfaces. A deeper study of vector fields in E_3 requires the use of *surface integrals.* A surface integral can be thought of as a two-dimensional analog of a line integral in which the "path" of integration is a *surface* rather than a curve. Before we can discuss surface integrals intelligently, we must come to some understanding as to what we shall mean by a surface.

Roughly speaking, a surface is the locus of a point moving in space with two "degrees of freedom." There are several ways of describing such a locus by mathematical formulas. If we set up the usual xyz cartesian coordinate system of analytic geometry, we obtain a surface by imposing one restriction on a variable point (x, y, z). This restriction can be written as an equation of the form

$$F(x, y, z) = 0.$$

An equation of this sort is called an *implicit representation* of the surface, and examples are familiar to us from solid analytic geometry. If we are fortunate enough to be able to solve this equation explicitly for one of the variables in terms of the other two, say for z in terms of x and y, we then get an equation of the form

$$z = f(x, y),$$

which is known as an *explicit representation* of the surface. Each explicit equation, of course, gives rise to an implicit equation by writing

$$f(x, y) - z = 0$$

Although these two ways of representing a surface are very useful and quite common, it turns out that for theoretical purposes a different way of describing surfaces is more useful This is the *parametric representation* or *vector representation* of a surface Instead of having *one* equation involving x, y, and z, we have *three* equations in which x, y, and z are expressed as functions of two parameters, say u and v

$$\text{(25)} \qquad x = x(u, v), \qquad y = y(u, v), \qquad z = z(u, v)$$

Here, the point (u, v) is allowed to vary over some two-dimensional region R in the uv-plane, and the corresponding points (x, y, z) trace out a portion of a surface in xyz space This is analogous to representing a space curve by three parametric equations involving just *one* parameter The presence of two parameters makes it possible to transmit two degrees of freedom to the variable point (x, y, z), as suggested by Fig 11–12

So far, we have not said anything about what sort of restrictions must be placed on the functions defined by (25) and on the plane region R. Any attempt at great generality in this respect introduces difficulties which cannot be treated properly in a text of this sort Therefore, to avoid complications in the theory, we shall place a large number of restrictions on the types of surfaces to be considered here. Fortunately, however, most of the familiar surfaces of analytic geometry come under the scope of the definitions which follow

NOTE. To make effective use of vector notation, we shall write (x_1, x_2, x_3) instead of (x, y, z), and (t_1, t_2) instead of (u, v)

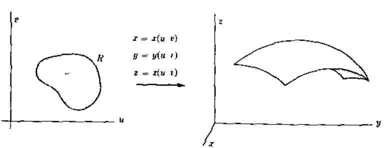


FIGURE 11–12

11–32 DEFINITION. *Let Γ be a rectifiable Jordan curve in E_2 and let R denote the union of Γ with its interior. Suppose there exists an open set R' containing R and a vector-valued function $\mathbf{x} = (x_1, x_2, x_3)$ such that $\mathbf{x} \in C'$ on R'. Then the image of R under $\mathbf{x}$, say $S = \mathbf{x}(R)$, is said to be a parametric surface described by $\mathbf{x}$. If, in addition, $\mathbf{x}$ is one-to-one on R, then S is said to be a simple parametric surface. In this case the image of Γ will be a rectifiable Jordan curve called the edge of S.*

NOTE. This definition is still too general for our purposes and we shall impose the following further restrictions on the function $\mathbf{x}$. Define vectors $D_1\mathbf{x}$ and $D_2\mathbf{x}$ as follows:

$$(26)\qquad \begin{aligned} D_1\mathbf{x}(\mathbf{t}) &= D_1x_1(\mathbf{t})\mathbf{u}_1 + D_1x_2(\mathbf{t})\mathbf{u}_2 + D_1x_3(\mathbf{t})\mathbf{u}_3, \\ D_2\mathbf{x}(\mathbf{t}) &= D_2x_1(\mathbf{t})\mathbf{u}_1 + D_2x_2(\mathbf{t})\mathbf{u}_2 + D_2x_3(\mathbf{t})\mathbf{u}_3, \end{aligned}$$

if $\mathbf{t} = (t_1, t_2) \in R$. Points $\mathbf{x}(\mathbf{t})$ on S where the cross product

$$D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t}) \neq \mathbf{0}$$

are called *regular points* of $\mathbf{x}$, and points where $D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t}) = \mathbf{0}$ are called *singular points* of $\mathbf{x}$. We shall assume in the sequel that all except possibly a finite number of points of S are regular points of $\mathbf{x}$. The geometric meaning of this restriction will be explained in the next paragraph.

Consider a horizontal line segment in R. Its image under $\mathbf{x}$ is a curve (called a t_1-curve) lying on the surface S. The vector $D_1\mathbf{x}$ represents the velocity vector of this curve. Likewise, $D_2\mathbf{x}$ is the velocity vector of a t_2-curve, obtained by setting t_1 = constant. There is a t_1-curve and a t_2-curve passing through each point of the surface. The restriction $D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t}) \neq \mathbf{0}$ means that the velocity vectors $D_1\mathbf{x}(\mathbf{t})$ and $D_2\mathbf{x}(\mathbf{t})$ are not collinear at this point. Hence, for each regular point, $D_1\mathbf{x}(\mathbf{t})$ and $D_2\mathbf{x}(\mathbf{t})$ determine a plane called the *tangent plane* to the surface at the point $\mathbf{x}(\mathbf{t})$. The vector $D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})$ is *normal* to this plane.

Figure 11–13 illustrates these ideas for a hemisphere. One function $\mathbf{x}$ which describes this hemisphere is defined on a rectangular region $R = [0, \pi/2] \times [0, 2\pi]$ by the equation

$$(27)\qquad \mathbf{x}(\mathbf{t}) = \sin t_1 \cos t_2\mathbf{u}_1 + \sin t_1 \sin t_2\mathbf{u}_2 + \cos t_1\mathbf{u}_3.$$

In this case, the parameters t_1 and t_2 are the familiar spherical coordinates (illustrated in Fig. 11–13). An easy computation yields

$$\begin{aligned} D_1\mathbf{x}(\mathbf{t}) &= \cos t_1 \cos t_2\mathbf{u}_1 + \cos t_1 \sin t_2\mathbf{u}_2 - \sin t_1\mathbf{u}_3, \\ D_2\mathbf{x}(\mathbf{t}) &= -\sin t_1 \sin t_2\mathbf{u}_1 + \sin t_1 \cos t_2\mathbf{u}_2, \\ D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t}) &= \sin t_1\mathbf{x}(\mathbf{t}). \end{aligned}$$

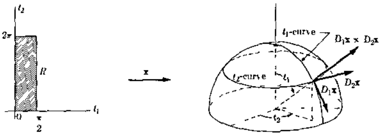


FIGURE 11-13.

In this case the image of R is not a simple parametric surface because the mapping $\mathbf{x}$ is not one-to-one on R. In fact, every point of the segment $t_1 = 0, 0 \leq t_2 \leq 2\pi$ is mapped onto the same point $(0, 0, 1)$ (the North Pole). The North Pole is a singular point of $\mathbf{x}$, since $\sin t_1 = 0$. However, the image $\mathbf{x}(R_0)$ of any closed rectangle R_0 lying in the interior of R will be a simple parametric surface.

The cross product $D_1\mathbf{x} \times D_2\mathbf{x}$ plays an important role in the theory of surfaces. Its components can be expressed as Jacobians by means of the following theorem:

11-33 THEOREM. *If $D_1\mathbf{x}$ and $D_2\mathbf{x}$ are defined by* (26), *we have*

$$(28) \qquad D_1\mathbf{x} \times D_2\mathbf{x} = \frac{\partial(x_2, x_3)}{\partial(t_1, t_2)}\mathbf{u}_1 + \frac{\partial(x_3, x_1)}{\partial(t_1, t_2)}\mathbf{u}_2 + \frac{\partial(x_1, x_2)}{\partial(t_1, t_2)}\mathbf{u}_3$$

Proof. We have

$$D_1\mathbf{x} \times D_2\mathbf{x} = \begin{vmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \\ D_1x_1 & D_1x_2 & D_1x_3 \\ D_2x_1 & D_2x_2 & D_2x_3 \end{vmatrix}$$

$$= \begin{vmatrix} D_1x_2 & D_1x_3 \\ D_2x_2 & D_2x_3 \end{vmatrix}\mathbf{u}_1 + \begin{vmatrix} D_1x_3 & D_1x_1 \\ D_2x_3 & D_2x_1 \end{vmatrix}\mathbf{u}_2$$

$$+ \begin{vmatrix} D_1x_1 & D_1x_2 \\ D_2x_1 & D_2x_2 \end{vmatrix}\mathbf{u}_3$$

11–17 Explicit representation of a parametric surface. Let us write (28) in the form

$$D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t}) = J_1(\mathbf{t})\mathbf{u}_1 + J_2(\mathbf{t})\mathbf{u}_2 + J_3(\mathbf{t})\mathbf{u}_3, \quad \text{if } \mathbf{t} = (t_1, t_2) \in R,$$

where J_1, J_2, J_3 denote the corresponding Jacobians. At a regular point not all three of these Jacobians can vanish. For definiteness, suppose $J_3(\mathbf{t}_0) \neq 0$ at an interior point of R. Write the vector equation of S as three scalar equations, say

(29)

$$x_1 - x_1(t_1, t_2) = 0, \qquad x_2 - x_2(t_1, t_2) = 0, \qquad x_3 - x_3(t_1, t_2) = 0.$$

Since $J_3(\mathbf{t}_0) \neq 0$, we can apply the implicit function theorem to solve the first two equations in (29) for t_1 and t_2 in terms of x_1 and x_2. That is to say, if $y_1 = x_1(\mathbf{t}_0)$ and $y_2 = x_2(\mathbf{t}_0)$, there is a two-dimensional neighborhood $N(y_1, y_2)$ and a vector-valued function $\mathbf{g} = (g_1, g_2)$ such that the equations

$$t_1 = g_1(x_1, x_2), \qquad t_2 = g_2(x_1, x_2) \tag{30}$$

hold whenever $(x_1, x_2) \in N(y_1, y_2)$ and, when we substitute (30) into (29), the first two equations in (29) are identically satisfied. The third equation in (29) becomes

$$x_3 = x_3[g_1(x_1, x_2), g_2(x_1, x_2)] = \phi(x_1, x_2), \tag{31}$$

let us say. This means that we always have an *explicit representation* of S, at least *locally*, in a neighborhood of each regular point.

Of course, it may happen that the equation (31) describes *all* of S. In this case, we can "identify" the t_1t_2-plane and the x_1x_2-plane and the vector equation of S can be written as follows:

(32)

$$\mathbf{x}(\mathbf{t}) = t_1\mathbf{u}_1 + t_2\mathbf{u}_2 + \phi(t_1, t_2)\mathbf{u}_3, \quad \text{if } \mathbf{t} \in R.$$

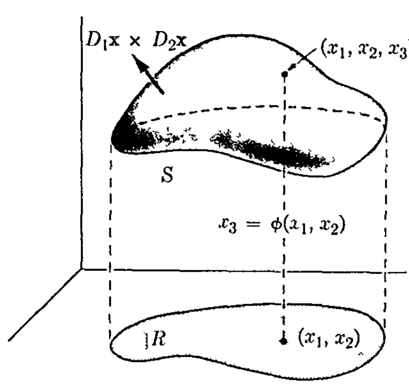


FIG. 11–14. Explicit representation of a parametric surface.

When a parametric surface S is described by an equation of the form (32), the set R is called the *projection* of S on the x_1x_2-plane. (See Fig 11-14) When (32) holds, it is easy to see that the fundamental vector product becomes

$$D_1\mathbf{x} \times D_2\mathbf{x} = -D_1\phi\mathbf{u}_1 - D_2\phi\mathbf{u}_2 + \mathbf{u}_3.$$

Thus the vector $D_1\mathbf{x} \times D_2\mathbf{x}$ always has a positive component in the $\mathbf{u}_3$ direction Of course, similar statements hold if we interchange the roles of x_2 and x_3 or those of x_1 and x_3

11-18 Area of a parametric surface. Consider a parametric surface S described by a vector-valued function $\mathbf{x}$ defined on a region R in E_2 Let us write $\mathbf{V}_1 = D_1\mathbf{x}(\mathbf{t})$ and $\mathbf{V}_2 = D_2\mathbf{x}(\mathbf{t})$ if $\mathbf{t} = (t_1, t_2) \in R$ If we think of t_1 and t_2 as representing "time," then, when t_1 increases by an amount Δt_1, a point originally at $\mathbf{x}(\mathbf{t})$ moves along a t_1-curve a distance approximately equal to $|\mathbf{V}_1|\,\Delta t_1$ (since $|\mathbf{V}_1|$ represents the "speed" along the t_1-curve) Similarly, in time Δt_2 a point of a t_2-curve moves a distance nearly equal to $|\mathbf{V}_2|\,\Delta t_2$ Hence a rectangle in R having area $\Delta t_1\,\Delta t_2$ is traced onto a portion of S which is "nearly" a parallelogram whose sides are the vectors $\mathbf{V}_1\,\Delta t_1$ and $\mathbf{V}_2\,\Delta t_2$ (See Fig 11-15) The area of the parallelogram determined by the vectors $\mathbf{V}_1\,\Delta t_1$ and $\mathbf{V}_2\,\Delta t_2$ is the length of their cross product, which is

$$|(\mathbf{V}_1\,\Delta t_1) \times (\mathbf{V}_2\,\Delta t_2)| = |\mathbf{V}_1 \times \mathbf{V}_2|\,\Delta t_1\,\Delta t_2 = |D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})|\,\Delta t_1\,\Delta t_2$$

Therefore, the number $|D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})|$ represents a local "magnification" factor for area This observation suggests the following definition for surface area:

11-34 DEFINITION *Let S be a parametric surface described by a vector-valued function $\mathbf{x}$ defined on a region R in E_2 The area of S is defined to be the value of the following double integral*

$$\int_R |D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})|\, d(t_1, t_2) \tag{33}$$

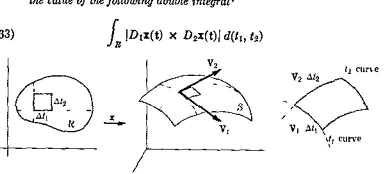


FIGURE 11-15.

The reader should observe the similarity between (33) and the integral $\int_a^b |\mathbf{x}'(t)|\,dt$ for finding the arc length of a curve described by $\mathbf{x}$. The integrand in (33) can be expressed as the square root of the sum of the squares of the Jacobians which occur in (28). If we write $\mathbf{V}_1 = D_1\mathbf{x}(\mathbf{t})$ and $\mathbf{V}_2 = D_2\mathbf{x}(\mathbf{t})$, Theorem 11–4 gives us an alternative formula for computing $|\mathbf{V}_1 \times \mathbf{V}_2|$, namely,

$$|\mathbf{V}_1 \times \mathbf{V}_2|^2 = |\mathbf{V}_1|^2|\mathbf{V}_2|^2 - (\mathbf{V}_1 \cdot \mathbf{V}_2)^2. \tag{34}$$

In differential geometry it is customary to introduce the notation

$$E = |\mathbf{V}_1|^2, \qquad F = \mathbf{V}_1 \cdot \mathbf{V}_2, \qquad G = |\mathbf{V}_2|^2,$$

in terms of which (34) yields $|\mathbf{V}_1 \times \mathbf{V}_2| = \sqrt{EG - F^2}$.

11–19 The sum of parametric surfaces. Let R_1 and R_2 denote two *closed* regions in E_2 whose boundaries Γ_1 and Γ_2 are rectifiable Jordan curves. Assume that the inner region of Γ_2 is outside Γ_1 and that $\Gamma_1 \cap \Gamma_2$ is an arc joining two distinct points, as illustrated in Fig. 11–16(a). Let S_1 and S_2 be two parametric surfaces described by vector-valued functions $\mathbf{x}$ and $\mathbf{y}$ defined on R_1 and R_2, respectively. Assume that $\mathbf{x}$ and $\mathbf{y}$ map $\Gamma_1 \cap \Gamma_2$ onto the same arc, that is, assume that $\mathbf{x}(\Gamma_1 \cap \Gamma_2) = \mathbf{y}(\Gamma_1 \cap \Gamma_2)$. Let $C_1 = \mathbf{x}(\Gamma_1)$ and $C_2 = \mathbf{y}(\Gamma_2)$ denote the edges of S_1 and S_2 and assume that $S_1 \cap S_2 = C_1 \cap C_2$. In other words, S_1 and S_2 must intersect at least along part of an edge but at no points other than points of $C_1 \cap C_2$. The union $S_1 \cup S_2$ is then called the *sum* of the surfaces S_1 and S_2 and is denoted by $S_1 + S_2$. A few of the possibilities that can occur are shown in Fig. 11–16, where R_1 and R_2 are taken to be rectangles.

If $C_1 \cap C_2 = C_1 = C_2$, the sum $S_1 + S_2$ is said to be a *closed surface.* Otherwise, the nonempty set $(C_1 \cup C_2) - (C_1 \cap C_2)$ is called the *edge* of $S_1 + S_2$. We shall restrict ourselves to those surfaces $S_1 + S_2$ whose edges are the union of at most a finite number of simple closed curves. For example, in Fig. 11–16(b) and (d), the edge of $S_1 + S_2$ consists of one simple closed curve, but in (c) the edge consists of two curves. In (e), it consists of three curves, whereas the surface in (f) has no edge and is therefore a closed surface.

If the sum $S_1 + S_2$ is not a closed surface, it has an edge (call it C) and we can proceed to define $(S_1 + S_2) + S_3$, if S_3 is an appropriate parametric surface. We must assume that the regions R_2 and R_3 associated with S_2 and S_3 have exactly one arc in common. The functions which describe S_2 and S_3 must map $R_2 \cap R_3$ onto the same set and we must have $(S_1 + S_2) \cap S_3 = C \cap C_3$. When these conditions are satisfied, the union $(S_1 + S_2) \cup S_3$ is called the *sum* $(S_1 + S_2) + S_3$. An easy exercise in

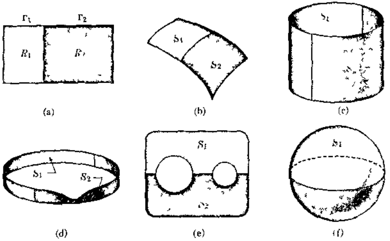


FIG 11-16. The sum of two parametric surfaces

set-algebra shows that whenever we can form $(S_1 + S_2) + S_3$ and $S_2 + S_3$, then we can also form $S_1 + (S_2 + S_3)$ and we have $(S_1 + S_2) + S_3 = S_1 + (S_2 + S_3)$, so that this addition is associative The process is extended to a finite number of summands in the same way, with the agreement that the addition is not defined if one of the summands is a closed surface In the work that follows we shall restrict ourselves to surfaces that are formed in this way by adding a finite number of parametric surfaces Also, we always assume that the edge (if any) is the union of a finite number of simple closed curves The area of a sum of parametric surfaces is defined to be the sum of the areas of the individual parts

11-20 Surface integrals. Let S be a parametric surface described by a vector-valued function $\mathbf{x} = (x_1, x_2, x_3)$ defined on a region R in E_2. At the regular points we can define two vector-valued functions $\mathbf{n}_1$ and $\mathbf{n}_2$ as follows:

$$(35)\quad \mathbf{n}_1(\mathbf{t}) = \frac{D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})}{|D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})|}, \qquad \mathbf{n}_2(\mathbf{t}) = -\mathbf{n}_1(\mathbf{t}), \qquad \text{if } \mathbf{t} \in R.$$

For each $\mathbf{t}$, both vectors $\mathbf{n}_1(\mathbf{t})$ and $\mathbf{n}_2(\mathbf{t})$ are unit vectors *normal* to the surface.

11-35 DEFINITION. *Let $\mathbf{f} = (f_1, f_2, f_3)$ be a vector-valued function defined on the parametric surface S described above. Define*

$$\mathbf{F}(\mathbf{t}) = \mathbf{f}[\mathbf{x}(\mathbf{t})], \qquad \text{if } \mathbf{t} \in R,$$

and let $\mathbf{n}$ *denote either of the two normals* $\mathbf{n}_1$ *or* $\mathbf{n}_2$ *described by* (35). *The surface integral of* $\mathbf{f} \cdot \mathbf{n}$ *over* S, *denoted by the symbol* $\iint_S \mathbf{f} \cdot \mathbf{n}\, d\sigma$, *is defined by the following equation:*

$$\iint\limits_S \mathbf{f} \cdot \mathbf{n}\, d\sigma = \int_R \mathbf{F}(\mathbf{t}) \cdot \mathbf{n}(\mathbf{t}) |D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})|\, d(t_1, t_2), \tag{36}$$

whenever the double integral on the right exists.

NOTE. Because of Theorem 11–33, the double integral in (36) can be expressed as a sum of three double integrals, namely,

$$\pm \left\{ \int_R F_1 \frac{\partial(x_2, x_3)}{\partial(t_1, t_2)}\, d(t_1, t_2) + \int_R F_2 \frac{\partial(x_3, x_1)}{\partial(t_1, t_2)}\, d(t_1, t_2) \right.$$
$$\left. + \int_R F_3 \frac{\partial(x_1, x_2)}{\partial(t_1, t_2)}\, d(t_1, t_2) \right\},$$

where the $+$ or $-$ sign is used according as $\mathbf{n} = \mathbf{n}_1$ or $\mathbf{n} = \mathbf{n}_2$. These double integrals are also written more briefly as follows:

$$\pm \left\{ \iint\limits_S f_1\, dx_2\, dx_3 + \iint\limits_S f_2\, dx_3\, dx_1 + \iint\limits_S f_3\, dx_1\, dx_2 \right\} \tag{37}$$

or

$$\pm \iint\limits_S f_1\, dx_2\, dx_3 + f_2\, dx_3\, dx_1 + f_3\, dx_1\, dx_2. \tag{38}$$

The integrals in (37) and (38) are also referred to as *surface integrals*. Thus, for example, the surface integral $\iint_S f_3\, dx_1\, dx_2$ is *defined* by the equation

$$\iint\limits_S f_3\, dx_1\, dx_2 = \int_R f_3[\mathbf{x}(\mathbf{t})] \frac{\partial(x_1, x_2)}{\partial(t_1, t_2)}\, d(t_1, t_2). \tag{39}$$

This notation is suggested by the formula for changing variables in a double integral. Note that the order in which the symbols dx_1 and dx_2 appear in (39) is important since

$$\frac{\partial(x_1, x_2)}{\partial(t_1, t_2)} = -\frac{\partial(x_2, x_1)}{\partial(t_1, t_2)}.$$

and hence we have

$$\iint_S f_3\,dx_1\,dx_2 = -\iint_S f_3\,dx_2\,dx_1$$

The unit normal $\mathbf{n}$ can be expressed in the form

$$\mathbf{n} = \cos\alpha_1\mathbf{u}_1 + \cos\alpha_2\mathbf{u}_2 + \cos\alpha_3\mathbf{u}_3$$

and hence $\mathbf{f}\cdot\mathbf{n} = f_1\cos\alpha_1 + f_2\cos\alpha_2 + f_3\cos\alpha_3$ For this reason, formula (36) is sometimes written as follows.

$$\iint_S (f_1\cos\alpha_1 + f_2\cos\alpha_2 + f_3\cos\alpha_3)\,d\sigma$$
$$= \pm\iint_S f_1\,dx_2\,dx_3 + f_2\,dx_3\,dx_1 + f_3\,dx_1\,dx_2,$$

the choice of the normal determining the + or − sign

Note that if S is described *explicitly* by an equation of the form

$$\mathbf{x}(t) = t_1\mathbf{u}_1 + t_2\mathbf{u}_2 + \phi(t_1, t_2)\mathbf{u}_3,$$

we have

$$\iint_S \mathbf{f}\cdot\mathbf{n}\,d\sigma = \pm\int_R (-F_1\,D_1\phi - F_2\,D_2\phi + F_3)\,d(t_1, t_2).$$

11-21 The theorem of Stokes. We recall that Green's theorem expresses a relation between a certain double integral extended over a plane region and a line integral taken around its boundary. There are two ways to generalize this in E_3. One of these extensions, known as Stokes' theorem, relates a surface integral taken over a parametric surface S to a line integral taken around its edge C. A second generalization of Green's theorem (known as the *divergence theorem*) will be discussed later.

For the purposes of proving Stokes' theorem we will assume that S is a simple parametric surface described by a vector-valued function $\mathbf{x}$ defined on a region R in E_2, where the boundary of R is a *piece-wise smooth* Jordan curve Γ. Also, we will assume that the components of $\mathbf{x}$ have continuous mixed partials of second order on R. We denote by $\Gamma(R)$ the positively oriented boundary of R and assume that it is described by a piece-wise smooth function $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)$ defined on $[a, b]$ Then the edge of S will

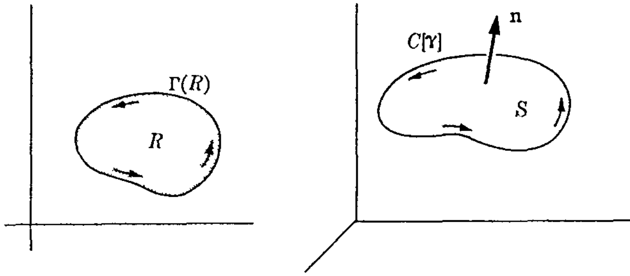


FIG. 11-17. Stokes' theorem.

be a simple closed curve C (see Fig. 11-17) described by the function γ defined by

$$\gamma(t) = \mathbf{x}[\alpha(t)], \qquad \text{if } t \in [a, b].$$

11-36 THEOREM (*Stokes*). *Under the assumptions of the preceding paragraph, let* $\mathbf{f} = (f_1, f_2, f_3)$ *be a vector-valued function defined on* S *such that* curl $\mathbf{f}$ *exists on* S. *Let*

$$\mathbf{n}(\mathbf{t}) = \frac{D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})}{|D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})|}.$$

Then we have

$$\iint\limits_S \operatorname{curl} \mathbf{f} \cdot \mathbf{n}\, d\sigma = \int_{C[\gamma]} \mathbf{f} \cdot d\gamma, \tag{40}$$

or, what is the same thing,

$$\begin{aligned}\iint\limits_S (D_2f_3 - D_3f_2)\, dx_2\, dx_3 + (D_3f_1 - D_1f_3)\, dx_3\, dx_1 \\ + (D_1f_2 - D_2f_1)\, dx_1\, dx_2 \\ = \int_{C[\gamma]} f_1\, d\gamma_1 + f_2\, d\gamma_2 + f_3\, d\gamma_3.\end{aligned}$$

Proof. The plan of the proof is quite simple. First, we express the line integral in (40) as a line integral taken around the boundary $\Gamma(R)$. Then we apply Green's theorem to express this line integral as a double integral over R, which turns out to have the same value as the surface integral in (40).

To carry out this plan, we write the line integral in (40) as a Riemann integral (using Theorem 10-33) and we find

$$\int_{C(\Upsilon)} \mathbf{f} \cdot d\boldsymbol{\gamma} = \sum_{k=1}^{3} \int_a^b f_k[\gamma(t)]\gamma_k'(t)\, dt$$

Computing $\gamma_k'(t)$ by the chain rule, we obtain

$$\int_{C(\Upsilon)} \mathbf{f} \cdot d\boldsymbol{\gamma} = \sum_{k=1}^{3} \int_a^b f_k(\mathbf{x}[\alpha(t)])\, D_1x_k[\alpha(t)]\alpha_1'(t)\, dt$$

$$+ \sum_{k=1}^{3} \int_a^b f_k(\mathbf{x}[\alpha(t)])\, D_2x_k[\alpha(t)]\alpha_2'(t)\, dt$$

$$= \int_a^b \mathbf{f}(\mathbf{x}[\alpha(t)]) \cdot D_1\mathbf{x}[\alpha(t)]\, d\alpha_1(t)$$

$$+ \int_a^b \mathbf{f}(\mathbf{x}[\alpha(t)]) \cdot D_2\mathbf{x}[\alpha(t)]\, d\alpha_2(t) = \int_{\Gamma(R)} g_1\, d\alpha_1 + g_2\, d\alpha_2,$$

where $\mathbf{g} = (g_1, g_2)$ is defined on R as follows:

(41) $\quad g_1(\mathbf{t}) = \mathbf{f}[\mathbf{x}(\mathbf{t})] \cdot D_1\mathbf{x}(\mathbf{t}) \quad$ and $\quad g_2(\mathbf{t}) = \mathbf{f}[\mathbf{x}(\mathbf{t})] \cdot D_2\mathbf{x}(\mathbf{t}),$

if $\quad \mathbf{t} = (t_1, t_2) \in R.$

Applying Green's theorem, we get

$$\int_{C(\Upsilon)} \mathbf{f} \cdot d\boldsymbol{\gamma} = \int_R (D_1g_2 - D_2g_1)\, d(t_1, t_2)$$

On the other hand, the surface integral in (40) can also be expressed as a double integral, namely,

(42)
$$\iint_S \operatorname{curl} \mathbf{f} \cdot \mathbf{n}\, d\sigma$$

$$= \int_R \left\{ (D_2F_3 - D_3F_2) \frac{\partial(x_2, x_3)}{\partial(t_1, t_2)} + (D_3F_1 - D_1F_3) \frac{\partial(x_3, x_1)}{\partial(t_1, t_2)} \right.$$

$$\left. + (D_1F_2 - D_2F_1) \frac{\partial(x_1, x_2)}{\partial(t_1, t_2)} \right\} d(t_1, t_2),$$

where $\mathbf{F} = (F_1, F_2, F_3)$ is the function defined on R by the equation

$\mathbf{F}(\mathbf{t}) = \mathbf{f}[\mathbf{x}(\mathbf{t})]$. Therefore the proof will be complete if we establish that the integrand on the right of (42) is equal to $D_1g_2 - D_2g_1$. From (41), we see that $\mathbf{g}$ and $\mathbf{F}$ are related as follows:

$$g_1(\mathbf{t}) = \mathbf{F}(\mathbf{t}) \cdot D_1\mathbf{x}(\mathbf{t}) \qquad \text{and} \qquad g_2(\mathbf{t}) = \mathbf{F}(\mathbf{t}) \cdot D_2\mathbf{x}(\mathbf{t}).$$

Using the chain rule, we find

$$D_1g_2 = (D_1\mathbf{F}\, D_1x_1 + D_2\mathbf{F}\, D_1x_2 + D_3\mathbf{F}\, D_1x_3) \cdot D_2\mathbf{x} + \mathbf{F} \cdot D_{1,2}\mathbf{x},$$

where

$$D_k\mathbf{F} = (D_kF_1, D_kF_2, D_kF_3)$$

and

$$D_{1,2}\mathbf{x} = (D_{1,2}x_1, D_{1,2}x_2, D_{1,2}x_3).$$

Similarly,

$$D_2g_1 = (D_1\mathbf{F}\, D_2x_1 + D_2\mathbf{F}\, D_2x_2 + D_3\mathbf{F}\, D_2x_3) \cdot D_1\mathbf{x} + \mathbf{F} \cdot D_{2,1}\mathbf{x}.$$

When we form $D_1g_2 - D_2g_1$, the term involving $\mathbf{F}$ drops out and we obtain

$$(43) \quad D_1g_2 - D_2g_1 = D_1\mathbf{F} \cdot \begin{vmatrix} D_1x_1 & D_2x_1 \\ D_1\mathbf{x} & D_2\mathbf{x} \end{vmatrix} + D_2\mathbf{F} \cdot \begin{vmatrix} D_1x_2 & D_2x_2 \\ D_1\mathbf{x} & D_2\mathbf{x} \end{vmatrix} + D_3\mathbf{F} \cdot \begin{vmatrix} D_1x_3 & D_2x_3 \\ D_1\mathbf{x} & D_2\mathbf{x} \end{vmatrix}.$$

But we have

$$D_1\mathbf{F} \cdot \begin{vmatrix} D_1x_1 & D_2x_1 \\ D_1\mathbf{x} & D_2\mathbf{x} \end{vmatrix} = D_1F_2 \frac{\partial(x_1, x_2)}{\partial(t_1, t_2)} + D_1F_3 \frac{\partial(x_1, x_3)}{\partial(t_1, t_2)},$$

$$D_2\mathbf{F} \cdot \begin{vmatrix} D_1x_2 & D_2x_2 \\ D_1\mathbf{x} & D_2\mathbf{x} \end{vmatrix} = D_2F_1 \frac{\partial(x_2, x_1)}{\partial(t_1, t_2)} + D_2F_3 \frac{\partial(x_2, x_3)}{\partial(t_1, t_2)},$$

$$D_3\mathbf{F} \cdot \begin{vmatrix} D_1x_3 & D_2x_3 \\ D_1\mathbf{x} & D_2\mathbf{x} \end{vmatrix} = D_3F_1 \frac{\partial(x_3, x_1)}{\partial(t_1, t_2)} + D_3F_2 \frac{\partial(x_3, x_2)}{\partial(t_1, t_2)},$$

and hence the right member of (43) is the same as the integrand in (42). This completes the proof.

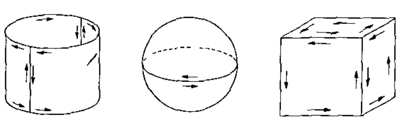
11-22 Orientation of surfaces. Let S_1 be a simple parametric surface described by a vector-valued function $\mathbf{x}$ defined on a closed region R_1. Let $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)$ be a vector-valued function defined on an interval $[a, b]$ in E_1 such that $\boldsymbol{\alpha}$ describes the positively oriented boundary of R_1. (See Definition 9-69.) Then the simple closed curve C_1 which forms the edge of S_1 can be described by the composite function $\boldsymbol{\gamma}$ given by

$$\boldsymbol{\gamma}(t) = \mathbf{x}[\boldsymbol{\alpha}(t)], \qquad \text{if } t \in [a, b] \tag{44}$$

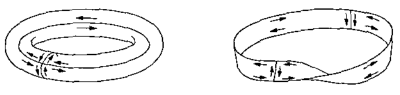
Similarly, let S_2 be another parametric surface described by a function $\mathbf{y}$ defined on a closed region R_2 such that we can form the sum $S_1 + S_2$. If $\boldsymbol{\beta}$ is a function which describes the positively oriented boundary of R_2, then the function $\boldsymbol{\delta}$ defined by the equation

$$\boldsymbol{\delta}(t) = \mathbf{y}[\boldsymbol{\beta}(t)] \tag{45}$$

describes the simple closed curve C_2 which forms the edge of S_2. The surface $S_1 + S_2$ is said to be *orientable* if $\mathbf{x}$ and $\mathbf{y}$ can be chosen so that the directed paths determined by $\boldsymbol{\delta}$ and $\boldsymbol{\gamma}$ are *opposite* on each of the boundary curves which comprise $C_1 \cap C_2$. That is to say, if Γ is an arc of $C_1 \cap C_2$ joining two points $\mathbf{p}$ and $\mathbf{q}$ and if $\boldsymbol{\delta}$ describes the directed path from $\mathbf{p}$ to $\mathbf{q}$ along Γ, then $\boldsymbol{\gamma}$ must describe the directed path from $\mathbf{q}$ to $\mathbf{p}$ along Γ. Of course, it may not be possible to choose $\mathbf{x}$ and $\mathbf{y}$ in such a way that this can be accomplished, in which case we say that $S_1 + S_2$ is *nonorientable*. For



(a) (b) (c)



(d) (e)

FIG. 11-18. Orientable surfaces and a nonorientable surface.

example, the surface illustrated in Fig. 11–18(e) is a nonorientable surface known as a *Möbius band*. On the other hand, in Fig. 11–18(a), (b), and (d), we have examples of orientable surfaces of the form $S_1 + S_2$.

Similar terminology applies, of course, to the sum of any finite number of parametric surfaces. Fig. 11–18(c) illustrates the oriented surface of a cube, the sum of six parametric surfaces (its faces).

Stokes' theorem can be extended at once to orientable surfaces of the form $S_1 + S_2$. Having chosen $\mathbf{x}$ and $\mathbf{y}$ so that the functions $\boldsymbol{\delta}$ and $\boldsymbol{\gamma}$ defined by (44) and (45) are opposite on each of the curves which comprise $C_1 \cap C_2$, let us write

$$\mathbf{n}_1(\mathbf{t}) = \frac{D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})}{|D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})|}, \qquad \mathbf{n}_2(\mathbf{t}) = \frac{D_1\mathbf{y}(\mathbf{t}) \times D_2\mathbf{y}(\mathbf{t})}{|D_1\mathbf{y}(\mathbf{t}) \times D_2\mathbf{y}(\mathbf{t})|}$$

and let us define $\iint\limits_{S_1+S_2} \mathbf{f}\cdot\mathbf{n}\,d\sigma$ to mean $\iint\limits_{S_1} \mathbf{f}\cdot\mathbf{n}_1\,d\sigma + \iint\limits_{S_2} \mathbf{f}\cdot\mathbf{n}_2\,d\sigma$. Then, if the hypotheses of Stokes' theorem are satisfied on $S_1 + S_2$, we can write

$$(46) \qquad \iint\limits_{S_1+S_2} \operatorname{curl}\mathbf{f}\cdot\mathbf{n}\,d\sigma = \int_{C_1\,[\boldsymbol{\gamma}]} \mathbf{f}\cdot d\boldsymbol{\gamma} + \int_{C_2\,[\boldsymbol{\delta}]} \mathbf{f}\cdot d\boldsymbol{\delta}.$$

The two integrals on the right of (46) can be replaced by a sum of line integrals taken around those curves which form the *edge* of $S_1 + S_2$, because the integrals along those arcs of $C_1 \cap C_2$ cancel out since they are traversed in opposite directions. (This is where the orientability of $S_1 + S_2$ enters in.) If $S_1 + S_2$ is a *closed* surface, there is no edge and all the line integrals on the right cancel, and we are left, in this case, with the formula

$$\iint\limits_{S_1+S_2} \operatorname{curl}\mathbf{f}\cdot\mathbf{n}\,d\sigma = 0.$$

11–23 Gauss' theorem (the divergence theorem). There is another way to extend Green's theorem from E_2 to E_3. In this extension we consider a "solid" V whose boundary is an orientable closed surface S and the corresponding theorem, known as Gauss' theorem, states a relationship which holds between a certain triple integral extended over the solid V and a surface integral over the boundary S. We shall treat only a very special case in which V and S can be described as follows:

Let R denote a region in E_2 whose boundary is a rectifiable Jordan curve Γ and let

$$V = \{(t_1, t_2, t_3) \mid \psi(t_1, t_2) \le t_3 \le \phi(t_1, t_2), \quad (t_1, t_2) \in R\},$$

where $\phi \in C'$ and $\psi \in C'$ on R. Assume that $\psi(t_1, t_2) < \phi(t_1, t_2)$ if (t_1, t_2) is interior to R, and that $\psi(t_1, t_2) = \phi(t_1, t_2)$ if $(t_1, t_2) \in \Gamma$. Let S_1 and S_2 be the two parametric surfaces described, respectively, by the functions $\mathbf{x}$ and $\mathbf{y}$ given by

$$\mathbf{x}(\mathbf{t}) = t_1\mathbf{u}_1 + t_2\mathbf{u}_2 + \phi(t_1, t_2)\mathbf{u}_3,$$
$$\mathbf{y}(\mathbf{t}) = t_1\mathbf{u}_1 + t_2\mathbf{u}_2 + \psi(t_1, t_2)\mathbf{u}_3,$$

if $\mathbf{t} = (t_1, t_2) \in R$, and let $S = S_1 \cup S_2$. Figure 11-19 illustrates an example of such a solid. Gauss' theorem for such a region can be stated as follows:

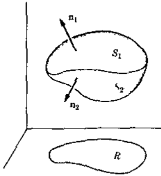


FIG. 11-19. The divergence theorem.

11-37 THEOREM (*Gauss*). *Under the assumptions of the preceding paragraph, let f_3 be a real-valued function such that f_3 is continuous on S and $f_3 \in C'$ on the interior of V. Then we have*

$$(47) \qquad \int_V D_3 f_3 \, d(t_1, t_2, t_3) = \iint_{S_1} f_3 \, dx_1 \, dx_2 - \iint_{S_2} f_3 \, dy_1 \, dy_2.$$

Proof. Applying the analog of Theorem 10-26 for triple integrals, we obtain

$$\begin{aligned}
\int_V D_3 f_3 \, d(t_1, t_2, t_3) &= \int_R \left[\int_{\psi(t_1, t_2)}^{\phi(t_1, t_2)} D_3 f_3 \, dt_3\right] d(t_1, t_2) \\
&= \int_R \{f_3[t_1, t_2, \phi(t_1, t_2)] - f_3[t_1, t_2, \psi(t_1, t_2)]\} \, d(t_1, t_2) \\
&= \int_R f_3[\mathbf{x}(\mathbf{t})] \frac{\partial(x_1, x_2)}{\partial(t_1, t_2)} \, d(t_1, t_2) \\
&\quad - \int_R f_3[\mathbf{y}(\mathbf{t})] \frac{\partial(y_1, y_2)}{\partial(t_1, t_2)} \, d(t_1, t_2) \\
&= \iint_{S_1} f_3 \, dx_1 \, dx_2 - \iint_{S_2} f_3 \, dy_1 \, dy_2.
\end{aligned}$$

NOTE. If we introduce the vector-valued function $\mathbf{f} = (0, 0, f_3)$ and define $\iint_S \mathbf{f} \cdot \mathbf{n}\, d\sigma$ by the equation

$$\iint\limits_S \mathbf{f} \cdot \mathbf{n}\, d\sigma = \iint\limits_{S_1} \mathbf{f} \cdot \mathbf{n}_1\, d\sigma + \iint\limits_{S_2} \mathbf{f} \cdot \mathbf{n}_2\, d\sigma,$$

where

$$\mathbf{n}_1(\mathbf{t}) = \frac{D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})}{|D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})|} \quad \text{and} \quad \mathbf{n}_2(\mathbf{t}) = -\frac{D_1\mathbf{y}(\mathbf{t}) \times D_2\mathbf{y}(\mathbf{t})}{|D_1\mathbf{y}(\mathbf{t}) \times D_2\mathbf{y}(\mathbf{t})|},$$

then formula (47) can be expressed as follows:

$$\int_V \operatorname{div} \mathbf{f}\, d(t_1, t_2, t_3) = \iint\limits_S \mathbf{f} \cdot \mathbf{n}\, d\sigma.$$

Because of the appearance of div **f**, Gauss' theorem is sometimes called the *divergence theorem.*

11-24 Coordinate transformations.

11-38 DEFINITION. *Let* $\mathbf{x} = (x_1, x_2, x_3)$ *be a vector-valued function defined on a region* T *in* E_3. *The function* $\mathbf{x}$ *is called a coordinate transformation if the following conditions are satisfied:*

(i) $\mathbf{x} \in C'$ *on an open set containing* T.

(ii) *The Jacobian* $J_{\mathbf{x}}(\mathbf{t}) \neq 0$ *for each point* $\mathbf{t} = (t_1, t_2, t_3)$ *in* T.

(iii) $\mathbf{x}$ *is one-to-one on* T.

In the remainder of this section we assume that $\mathbf{x}$ is a function satisfying these conditions, and we adopt the following notations:

$$D_i\mathbf{x}(\mathbf{t}) = (D_ix_1(\mathbf{t}), D_ix_2(\mathbf{t}), D_ix_3(\mathbf{t})) \qquad (i = 1, 2, 3). \tag{48}$$

$$h_1(\mathbf{t}) = |D_1\mathbf{x}(\mathbf{t})|, \qquad h_2(\mathbf{t}) = |D_2\mathbf{x}(\mathbf{t})|, \qquad h_3(\mathbf{t}) = |D_3\mathbf{x}(\mathbf{t})|. \tag{49}$$

Because of (ii), each vector $D_i\mathbf{x}(\mathbf{t}) \neq \mathbf{0}$ for each $\mathbf{t}$ in T, and hence we can define the unit vectors:

$$\mathbf{v}_1(\mathbf{t}) = \frac{D_1\mathbf{x}(\mathbf{t})}{h_1(\mathbf{t})}, \qquad \mathbf{v}_2(\mathbf{t}) = \frac{D_2\mathbf{x}(\mathbf{t})}{h_2(\mathbf{t})}, \qquad \mathbf{v}_3(\mathbf{t}) = \frac{D_3\mathbf{x}(\mathbf{t})}{h_3(\mathbf{t})}. \tag{50}$$

11-39 THEOREM. *The vectors in* (50) *form a linearly independent set. Moreover, we have*

$$J_{\mathbf{x}}(\mathbf{t}) = h_1(\mathbf{t})h_2(\mathbf{t})h_3(\mathbf{t})\mathbf{v}_1(\mathbf{t}) \cdot \mathbf{v}_2(\mathbf{t}) \times \mathbf{v}_3(\mathbf{t}).$$

Proof. $J_{\mathbf{x}}(\mathbf{t}) = D_1\mathbf{x}(\mathbf{t}) \cdot D_2\mathbf{x}(\mathbf{t}) \times D_3\mathbf{x}(\mathbf{t})$.

NOTE The triple $(\mathbf{v}_1(\mathbf{t}), \mathbf{v}_2(\mathbf{t}), \mathbf{v}_3(\mathbf{t}))$ is a positive or negative triple according as $J_{\mathbf{x}}(\mathbf{t})$ is positive or negative The set $\{\mathbf{v}_1(\mathbf{t}), \mathbf{v}_2(\mathbf{t}), \mathbf{v}_3(\mathbf{t})\}$ is usually referred to as a *curvilinear basis* associated with the point $\mathbf{x}(\mathbf{t})$ Because of Theorem 11–39, every vector $\mathbf{a}$ in E_3 can be expressed as a linear combination of these vectors, thus

$$\mathbf{a} = \alpha_1\mathbf{v}_1(\mathbf{t}) + \alpha_2\mathbf{v}_2(\mathbf{t}) + \alpha_3\mathbf{v}_3(\mathbf{t})$$

The scalars α_1, α_2, α_3 (which depend on $\mathbf{t}$) are called the *curvilinear coordinates* of $\mathbf{a}$. Two familiar special cases are described in the following examples:

EXAMPLE 1 (*Cylindrical coordinates*). In this case, $\mathbf{x}$ is given by

$$x_1(\mathbf{t}) = t_1 \cos t_2, \qquad x_2(\mathbf{t}) = t_1 \sin t_2, \qquad x_3(\mathbf{t}) = t_3$$

An easy calculation yields

$$h_1(\mathbf{t}) = 1, \qquad h_2(\mathbf{t}) = t_1, \qquad h_3(\mathbf{t}) = 1,$$

$$\mathbf{v}_1(\mathbf{t}) = \cos t_2\mathbf{u}_1 + \sin t_2\mathbf{u}_2, \quad \mathbf{v}_2(\mathbf{t}) = -\sin t_2\mathbf{u}_1 + \cos t_2\mathbf{u}_2, \quad \mathbf{v}_3(\mathbf{t}) = \mathbf{u}_3,$$

$$J_{\mathbf{x}}(\mathbf{t}) = t_1, \qquad \mathbf{v}_1(\mathbf{t}) \cdot \mathbf{v}_2(\mathbf{t}) \times \mathbf{v}_3(\mathbf{t}) = 1.$$

Hence, $\mathbf{x}$ is a coordinate transformation defined on the following set

$$T = \{(t_1, t_2, t_3) \mid t_1 > 0, \quad 0 \leq t_2 < 2\pi, \quad -\infty < t_3 < +\infty\}.$$

The image $\mathbf{x}(T)$ is $E_3 - \{(x_1, x_2, x_3) \mid x_1 = x_2 = 0\}$.

EXAMPLE 2 (*Spherical coordinates*). In this case, we have

$$x_1(t) = t_1 \sin t_2 \cos t_3, \qquad x_2(t) = t_1 \sin t_2 \sin t_3, \qquad x_3(t) = t_1 \cos t_2,$$

$$\mathbf{v}_1(\mathbf{t}) = \sin t_2 \cos t_3\mathbf{u}_1 + \sin t_2 \sin t_3\mathbf{u}_2 + \cos t_2\mathbf{u}_3, \qquad h_1(t) = 1,$$

$$\mathbf{v}_2(\mathbf{t}) = \cos t_2 \cos t_3\mathbf{u}_1 + \cos t_2 \sin t_3\mathbf{u}_2 - \sin t_2\mathbf{u}_3, \qquad h_2(t) = t_1,$$

$$\mathbf{v}_3(\mathbf{t}) = -\sin t_3\mathbf{u}_1 + \cos t_3\mathbf{u}_2, \qquad h_3(t) = t_1 \sin t_2,$$

$$J_{\mathbf{x}}(\mathbf{t}) = t_1^2 \sin t_2, \qquad \mathbf{v}_1(t) \cdot \mathbf{v}_2(t) \times \mathbf{v}_3(t) = 1$$

The coordinate transformation $\mathbf{x}$ is defined on the following set

$$T = \{(t_1, t_2, t_3) \mid t_1 > 0, 0 < t_2 < \pi, 0 < t_3 \leq 2\pi\},$$

whose image $\mathbf{x}(T)$ is $E_3 - \{(x_1, x_2, x_3) \mid x_1 = x_2 = 0\}$

If t_3 is kept fixed and t_1 and t_2 are allowed to vary over a rectangle, the point $\mathbf{x}(\mathbf{t})$ traces out a parametric surface called a t_1t_2*-surface.* We define t_1t_3-surfaces and t_2t_3-surfaces similarly. Each t_1t_2-surface intersects a t_1t_3-surface along a t_1-curve, that is, a curve traced out by $\mathbf{x}(\mathbf{t})$ when both t_2 and t_3 are held fixed and t_1 varies over an interval. The vector $\mathbf{v}_1(\mathbf{t})$ is a unit tangent to the t_1-curve and the scalar $h_1(\mathbf{t})$ represents the "speed" with which the t_1-curve is traced out. Similar remarks apply, of course, to t_2-curves and t_3-curves. For small changes in "time" Δt_1, Δt_2, Δt_3, a rectangular box of volume $\Delta t_1\, \Delta t_2\, \Delta t_3$ is mapped by $\mathbf{x}$ onto a solid which is nearly a parallelepiped having edges determined by the vectors $D_1\mathbf{x}(\mathbf{t})\, \Delta t_1$, $D_2\mathbf{x}(\mathbf{t})\, \Delta t_2$, $D_3\mathbf{x}(\mathbf{t})\, \Delta t_3$. (See Fig. 11–20.) The volume of such a parallelepiped is given by the scalar triple product

$$|D_1\mathbf{x}(\mathbf{t}) \cdot D_2\mathbf{x}(\mathbf{t}) \times D_3\mathbf{x}(\mathbf{t})|\, \Delta t_1\, \Delta t_2\, \Delta t_3 = |J_{\mathbf{x}}(\mathbf{t})|\, \Delta t_1\, \Delta t_2\, \Delta t_3.$$

This means that $|J_{\mathbf{x}}(\mathbf{t})|$ can be interpreted as a local "magnification factor" for volumes. (Compare with Theorem 10–31.)

Let $\mathbf{x}$ be a coordinate transformation as described in Definition 11–38 and let $S = \mathbf{x}(T)$. Let $\mathbf{f}$ be a vector field defined on S. If we express $\mathbf{f}$ in terms of the basis $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$, we can write

$$\mathbf{f} = f_1\mathbf{u}_1 + f_2\mathbf{u}_2 + f_3\mathbf{u}_3, \tag{51}$$

let us say. Now form the composite function $\mathbf{F}$ defined on T by the equation $\mathbf{F}(\mathbf{t}) = \mathbf{f}[\mathbf{x}(\mathbf{t})]$ and let us express $\mathbf{F}(\mathbf{t})$ in terms of the curvilinear basis $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$, say,

$$\mathbf{F}(\mathbf{t}) = F_1(\mathbf{t})\mathbf{v}_1(\mathbf{t}) + F_2(\mathbf{t})\mathbf{v}_2(\mathbf{t}) + F_3(\mathbf{t})\mathbf{v}_3(\mathbf{t}). \tag{52}$$

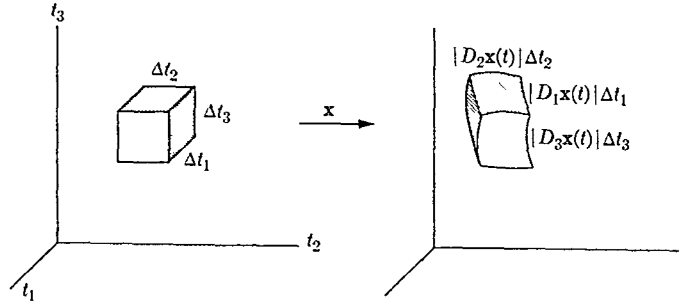


FIGURE 11–20.

The next theorem tells us how the curvilinear coordinates F_1, F_2, F_3 are related to the "rectangular" coordinates f_1, f_2, f_3, under the assumption that $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ is an orthonormal set. (This is always the case for the more familiar coordinate transformations.)

11–40 THEOREM. *Assume that $\mathbf{v}_1 \cdot \mathbf{v}_2 \times \mathbf{v}_3 = 1$. If $\mathbf{f}$ is given by (51) and $\mathbf{F}$ by (52), then, for each $i = 1, 2, 3$, we have*

$$f_i[\mathbf{x}(\mathbf{t})] = \frac{F_1(\mathbf{t})}{h_1(\mathbf{t})} D_1 x_i(\mathbf{t}) + \frac{F_2(\mathbf{t})}{h_2(\mathbf{t})} D_2 x_i(\mathbf{t}) + \frac{F_3(\mathbf{t})}{h_3(\mathbf{t})} D_3 x_i(\mathbf{t}) \tag{53}$$

and

$$F_i(\mathbf{t}) = \frac{1}{h_i(\mathbf{t})} (D_i x_1(\mathbf{t}) f_1[\mathbf{x}(\mathbf{t})] + D_i x_2(\mathbf{t}) f_2[\mathbf{x}(\mathbf{t})] + D_i x_3(\mathbf{t}) f_3[\mathbf{x}(\mathbf{t})]). \tag{54}$$

Proof. Writing $\mathbf{p} = \mathbf{x}(\mathbf{t})$, we have

$$\begin{aligned} f_i(\mathbf{p}) &= \mathbf{f}(\mathbf{p}) \cdot \mathbf{u}_i \\ &= F_1(\mathbf{t})(\mathbf{u}_i \cdot \mathbf{v}_1(\mathbf{t})) + F_2(\mathbf{t})(\mathbf{u}_i \cdot \mathbf{v}_2(\mathbf{t})) + F_3(\mathbf{t})(\mathbf{u}_i \cdot \mathbf{v}_3(\mathbf{t})). \end{aligned}$$

Since $\mathbf{u}_i \cdot \mathbf{v}_j(\mathbf{t}) = D_j x_i(\mathbf{t})/h_j(\mathbf{t})$, we get (53) at once. Formula (54) is similarly proved.

To illustrate the use of Theorem 11–40, consider the special case in which $\mathbf{f}$ is the gradient of a potential, say $\mathbf{f} = \nabla\phi$. Then $f_i = D_i\phi$ and (54) becomes

$$F_i(\mathbf{t}) = \frac{1}{h_i(\mathbf{t})} \sum_{k=1}^{3} D_k\phi[\mathbf{x}(\mathbf{t})]\, D_i x_k(\mathbf{t}) \tag{55}$$

If we write $\Phi(\mathbf{t}) = \phi[\mathbf{x}(\mathbf{t})]$, the chain rule (Theorem 6–14) can be applied in (55) to yield $F_i(\mathbf{t}) = D_i\Phi(\mathbf{t})/h_i(\mathbf{t})$. Hence we have the following expression for the gradient $\nabla\phi[\mathbf{x}(\mathbf{t})]$ in orthogonal curvilinear coordinates:

(56)

$$\nabla\phi[\mathbf{x}(\mathbf{t})] = \frac{1}{h_1(\mathbf{t})} D_1\Phi(\mathbf{t})\mathbf{v}_1(\mathbf{t}) + \frac{1}{h_2(\mathbf{t})} D_2\Phi(\mathbf{t})\mathbf{v}_2(\mathbf{t}) + \frac{1}{h_3(\mathbf{t})} D_3\Phi(\mathbf{t})\mathbf{v}_3(\mathbf{t}),$$

where $\Phi(\mathbf{t}) = \phi[\mathbf{x}(\mathbf{t})]$.

Formulas for the curl, divergence, and Laplacian are developed in Exercises 11–38 through 11–40.

EXERCISES

Vector algebra.

11–1. Let $\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$ be a basis for E_n. Let $\mathbf{b}_1 = \mathbf{a}_1$ and define

$$\mathbf{b}_m = \mathbf{a}_m - \sum_{k=1}^{m-1} \frac{(\mathbf{a}_m \cdot \mathbf{a}_k)}{|\mathbf{b}_k|^2} \mathbf{b}_k, \qquad \text{if } m = 2, 3, \ldots, n.$$

Show that $\{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ is an orthogonal basis for E_n.

11–2. Prove that $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$.

11–3. Let $A = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$ and $B = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ be two ordered triples of vectors in E_3 satisfying the nine equations:

$$\mathbf{a}_i \cdot \mathbf{b}_j = \delta_{i,j} \qquad (i, j = 1, 2, 3).$$

(A and B are called *reciprocal* sets of vectors.)

(a) Let $\alpha = \mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3$. Show that $\alpha \neq 0$ and derive the formulas

$$\alpha\mathbf{b}_1 = \mathbf{a}_2 \times \mathbf{a}_3, \qquad \alpha\mathbf{b}_2 = \mathbf{a}_3 \times \mathbf{a}_1, \qquad \alpha\mathbf{b}_3 = \mathbf{a}_1 \times \mathbf{a}_2,$$

$$\mathbf{a}_1 = \alpha(\mathbf{b}_2 \times \mathbf{b}_3), \qquad \mathbf{a}_2 = \alpha(\mathbf{b}_3 \times \mathbf{b}_1), \qquad \mathbf{a}_3 = \alpha(\mathbf{b}_1 \times \mathbf{b}_2).$$

(b) Show that $(\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3)(\mathbf{b}_1 \cdot \mathbf{b}_2 \times \mathbf{b}_3) = 1$.

(c) Prove that both A and B are bases for E_3.

(d) Prove that A is an orthogonal set if, and only if, B is an orthogonal set.

(e) Show that $A = B$ if, and only if, A is an orthonormal set.

11–4. Let $\mathbf{a}$ and $\mathbf{c}$ be given vectors in E_3 with $\mathbf{a} \neq 0$, $\mathbf{a} \cdot \mathbf{c} = 0$, and let λ be a given scalar. Find a vector $\mathbf{b}$ which simultaneously satisfies both equations $\mathbf{a} \times \mathbf{b} = \mathbf{c}$ and $\mathbf{a} \cdot \mathbf{b} = \lambda$. Show that no such $\mathbf{b}$ exists if $\mathbf{a} \cdot \mathbf{c} \neq 0$.

11–5. Prove that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{c} \cdot \mathbf{a})\mathbf{b} - (\mathbf{b} \cdot \mathbf{a})\mathbf{c}$. Use this to deduce

(a) $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \times \mathbf{b} \cdot \mathbf{d})\mathbf{c} - (\mathbf{a} \times \mathbf{b} \cdot \mathbf{c})\mathbf{d}$.

(b) $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{b} \cdot \mathbf{d})(\mathbf{a} \cdot \mathbf{c}) - (\mathbf{b} \cdot \mathbf{c})(\mathbf{a} \cdot \mathbf{d})$.

(c) $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = 0$.

(d) $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$ if, and only if, $\mathbf{b} \times (\mathbf{c} \times \mathbf{a}) = 0$.

11–6. Prove that $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a}) = (\mathbf{a} \cdot \mathbf{b} \times \mathbf{c})^2$.

11–7. If $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} \neq 0$, show that for an arbitrary vector $\mathbf{v}$ in E_3 we have

$$(\mathbf{a} \cdot \mathbf{b} \times \mathbf{c})\mathbf{v} = (\mathbf{v} \cdot \mathbf{b} \times \mathbf{c})\mathbf{a} + (\mathbf{v} \cdot \mathbf{c} \times \mathbf{a})\mathbf{b} + (\mathbf{v} \cdot \mathbf{a} \times \mathbf{b})\mathbf{c}$$

11–8. Prove or disprove the formula $\mathbf{a} \times [\mathbf{a} \times (\mathbf{a} \times \mathbf{b})] \cdot \mathbf{c} = -|\mathbf{a}|^2\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$

11–9. Prove Theorem 11–14

Space curves (In Exercises 11–10 through 11–15, the notation is that of Section 11–10)

11–10. A particle moves on a space curve with velocity $\mathbf{x}'$, constant speed 1, and constant curvature 1. Let $\theta(t)$ denote the angle $(0 \leq \theta(t) \leq 180°)$ between the vectors $\mathbf{x}'(t)$ and $\mathbf{x}'''(t)$ at time t Show that $|\tau(t)| = |\mathbf{x}'''(t)| \sin \theta(t)$

11–11. Let $\mathbf{R} = s'\tau\mathbf{T} + s'\kappa\mathbf{B}$ Show that the Frenet formulas become

$$\mathbf{T}' = \mathbf{R} \times \mathbf{T}, \qquad \mathbf{N}' = \mathbf{R} \times \mathbf{N}, \qquad \mathbf{B}' = \mathbf{R} \times \mathbf{B}$$

11–12. Derive the following formulas

(a) $\tau s' = \mathbf{T} \cdot \mathbf{N} \times \mathbf{N}'$, (b) $\tau\kappa(s')^2 = \mathbf{T} \cdot \mathbf{N} \times \mathbf{T}''$,
(c) $\tau\kappa^2(s')^3 = \mathbf{T} \cdot \mathbf{T}' \times \mathbf{T}''$, (d) $\kappa(s')^3 = |\mathbf{x}' \times \mathbf{x}''|$,
(e) $\tau|\mathbf{x}' \times \mathbf{x}''|^2 = \mathbf{x}' \cdot \mathbf{x}'' \times \mathbf{x}'''$.

11–13. A portion of a circular helix is described by the function $\mathbf{x}$ given by

$$\mathbf{x}(t) = a \cos t\mathbf{u}_1 + a \sin t\mathbf{u}_2 + bt\mathbf{u}_3, \qquad 0 \leq t \leq t_1$$

Compute the vectors $\mathbf{x}'$, $\mathbf{x}''$, $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$. Verify that $\kappa^2 = a^2/c^4$ and $\tau = b/c^2$, where $c^2 = a^2 + b^2$.

11–14. Suppose a particle is rotating about a fixed axis through the origin in E_3. Let $\mathbf{x}(t)$ denote the position vector at time t Show that there exists a vector $\boldsymbol{\omega}(t)$ parallel to this axis such that

$$\mathbf{x}'(t) = \boldsymbol{\omega}(t) \times \mathbf{x}(t)$$

NOTE. The vector $\boldsymbol{\omega}(t)$ is called the *angular velocity* and $\omega(t) = |\boldsymbol{\omega}(t)|$ is called the *angular speed*

11–15. A body is said to undergo a *rigid motion* if, for every pair of particles $\mathbf{p}$ and $\mathbf{q}$ in the body, the distance $|\mathbf{x}_p(t) - \mathbf{x}_q(t)|$ is independent of t, where $\mathbf{x}_p(t)$ and $\mathbf{x}_q(t)$ denotes the position vectors of $\mathbf{p}$ and $\mathbf{q}$ at time t Prove that for a rigid motion in which each particle $\mathbf{p}$ rotates about the same axis through the origin we have

$$\mathbf{x}_p'(t) = \boldsymbol{\omega}(t) \times \mathbf{x}_p(t),$$

where $\boldsymbol{\omega}(t)$ is the same for each particle.

Gradient, curl, divergence.

11–16. If $\mathbf{x} \in E_3$, define $\phi(\mathbf{x}) = |\mathbf{x}|$.

(a) Show that $\nabla\phi(\mathbf{x}) = \mathbf{x}/|\mathbf{x}|$ if $\mathbf{x} \neq \mathbf{0}$.

(b) Compute $\nabla\psi(\mathbf{x})$, where $\psi(\mathbf{x}) = \phi(\mathbf{x})^n$.

(c) Find a scalar field λ such that $\nabla\lambda(\mathbf{x}) = \mathbf{x}$.

11–17. Find (when possible) a scalar field ϕ such that $\nabla\phi = \mathbf{f}$, given that

(a) $\mathbf{f}(x_1, x_2) = (\sin x_2 - x_2 \sin x_1 + x_1)\mathbf{u}_1 + (\cos x_1 + x_1 \cos x_2 + x_2)\mathbf{u}_2$.

(b) $\mathbf{f}(x_1, x_2) = (2x_1e^{x_2} + x_2)\mathbf{u}_1 + (x_1^2e^{x_2} + x_1 - 2x_2)\mathbf{u}_2$.

(c) $\mathbf{f}(x_1, x_2, x_3) = (x_1 + x_3)\mathbf{u}_1 - (x_2 + x_3)\mathbf{u}_2 + (x_1 - x_2)\mathbf{u}_3$.

11–18. Let $\mathbf{a}$ and $\mathbf{b}$ be given vectors in E_3 and define $\phi(\mathbf{x}) = (\mathbf{x} \times \mathbf{a}) \cdot (\mathbf{x} \times \mathbf{b})$. Show that $\nabla\phi(\mathbf{x}) = \mathbf{b} \times (\mathbf{x} \times \mathbf{a}) + \mathbf{a} \times (\mathbf{x} \times \mathbf{b})$.

11–19. Let F be a real-valued function such that $F \in C'$ on E_1. Define $\phi(\mathbf{x}) = F(|\mathbf{x}|)$ if $\mathbf{x} \in E_3$. Show that there exists a scalar field λ such that $\nabla\phi(\mathbf{x}) = \lambda(\mathbf{x})\mathbf{x}$ if $\mathbf{x} \neq \mathbf{0}$.

11–20. Given two scalar fields λ and ϕ such that $\nabla\phi(\mathbf{x}) = \lambda(\mathbf{x})\mathbf{x}$ for each $\mathbf{x} \neq \mathbf{0}$ in E_3, where $\phi \in C'$ on E_3. Show that there exists a real-valued function F defined on E_1 such that $\phi(\mathbf{x}) = F(|\mathbf{x}|)$.

11–21. Let $\mathbf{f}$ and $\mathbf{g}$ be three-dimensional vector fields such that $\operatorname{div} \mathbf{f}$, $\operatorname{curl} \mathbf{f}$, and $\operatorname{curl} \mathbf{g}$ exist on an open set S in E_3. Derive the following formulas where $D_i\mathbf{f} = (D_if_1, D_if_2, D_if_3)$, $i = 1, 2, 3$:

(a) $\operatorname{div} \mathbf{f} = \mathbf{u}_1 \cdot D_1\mathbf{f} + \mathbf{u}_2 \cdot D_2\mathbf{f} + \mathbf{u}_3 \cdot D_3\mathbf{f}$.

(b) $\operatorname{curl} \mathbf{f} = \mathbf{u}_1 \times D_1\mathbf{f} + \mathbf{u}_2 \times D_2\mathbf{f} + \mathbf{u}_3 \times D_3\mathbf{f}$.

(c) Use (a) and (b) and properties of the scalar triple product to derive the formula: $\operatorname{div} (\mathbf{f} \times \mathbf{g}) = \mathbf{g} \cdot \operatorname{curl} \mathbf{f} - \mathbf{f} \cdot \operatorname{curl} \mathbf{g}$.

11–22. Let F be a real-valued function such that F' exists and is continuous everywhere on E_1. Given a fixed vector $\mathbf{a} \neq \mathbf{0}$ in E_3, define

$$\mathbf{g}(\mathbf{x}) = F(\mathbf{x} \cdot \mathbf{a})\mathbf{f}(\mathbf{x}),$$

where $\mathbf{f}(\mathbf{x}) = 2x_1x_2x_3\mathbf{u}_1 + x_1^2x_3\mathbf{u}_2 + x_1^2x_2\mathbf{u}_3$ if $\mathbf{x} \in E_3$. Show that $\operatorname{curl} \mathbf{g}(\mathbf{x})$ is orthogonal to both $\mathbf{a}$ and $\mathbf{f}(\mathbf{x})$.

11–23. Let $\mathbf{a}$ be a fixed vector in E_3. Define $f(\mathbf{x}) = |\mathbf{x}|^n\, \mathbf{a} \times \mathbf{x}$ if $\mathbf{x} \in E_3$.

(a) Find scalar fields λ and μ such that $\operatorname{curl} \mathbf{f}(\mathbf{x}) = \lambda(\mathbf{x})\mathbf{a} + \mu(\mathbf{x})\mathbf{x}$.

(b) Prove or disprove the formula

$$\nabla\left(\frac{\mathbf{a} \cdot \mathbf{x}}{|\mathbf{x}|^3}\right) = -\operatorname{curl}\left(\frac{\mathbf{a} \times \mathbf{x}}{|\mathbf{x}|^3}\right).$$

11-24. Find a three-dimensional vector field $\mathbf{f}$ such that

$$\operatorname{div} \mathbf{f}(\mathbf{x}) = 2x_1 + x_2 - 1 \quad \text{and} \quad \operatorname{curl} \mathbf{f}(\mathbf{x}) = \mathbf{u}_3.$$

11-25. Let $\mathbf{f} = (f_1, f_2, f_3)$ be a vector field and let $\mathbf{g} = (g_1, g_2, g_3) = \operatorname{curl} \mathbf{f}$, $\phi = \operatorname{div} \mathbf{f}$. If each f_i has continuous second-order partials, show that

$$\nabla^2 f_1 = D_1\phi - D_2 g_3 + D_3 g_2$$

and find similar formulas for $\nabla^2 f_2$ and $\nabla^2 f_3$.

11-26. Let f_1, f_2, f_3 be three real-valued functions having second order derivatives on E_1. Define $g(\mathbf{x}) = f_1(x_1)f_2(x_2)f_3(x_3)$ if $\mathbf{x} = (x_1, x_2, x_3) \in E_3$. Show that whenever $g(\mathbf{x}) \neq 0$ we have

$$\frac{\nabla^2 g(\mathbf{x})}{g(\mathbf{x})} = \frac{f_1''(x_1)}{f_1(x_1)} + \frac{f_2''(x_2)}{f_2(x_2)} + \frac{f_3''(x_3)}{f_3(x_3)}$$

11-27. Let F be a real-valued function having a continuous second derivative F'' on E_1. If $\mathbf{x} \in E_3$, define $\phi(\mathbf{x}) = F(|\mathbf{x}|)$. Show that for $\mathbf{x} \neq \mathbf{0}$ we have

$$\nabla^2\phi(\mathbf{x}) = F''(|\mathbf{x}|) + \frac{2F'(|\mathbf{x}|)}{|\mathbf{x}|}$$

If $\nabla^2\phi = 0$, what can you conclude about F?

Surfaces and surface integrals.

11-28. In each case find a rectangular region R whose image $\mathbf{x}(R)$ will be a parametric surface:

(a) Elliptic paraboloid

$$\mathbf{x}(\mathbf{t}) = at_1 \cos t_2\mathbf{u}_1 + bt_1 \sin t_2\mathbf{u}_2 + t_1^2\mathbf{u}_3.$$

(b) Cone:

$$\mathbf{x}(\mathbf{t}) = (\cos\alpha)t_1 \cos t_2\mathbf{u}_1 + (\cos\alpha)t_1 \sin t_2\mathbf{u}_2 + (\sin\alpha)t_1\mathbf{u}_3.$$

What does α represent on the cone?

11-29. In each case find two rectangular regions R_1 and R_2 having one side in common such that the sum $S_1 + S_2$ will be a closed surface, where $S_1 = \mathbf{x}(R_1)$ and $S_2 = \mathbf{x}(R_2)$.

(a) Ellipsoid:

$$\mathbf{x}(\mathbf{t}) = a \sin t_1 \cos t_2\mathbf{u}_1 + b \sin t_1 \sin t_2\mathbf{u}_2 + c \cos t_1\mathbf{u}_3.$$

(b) Torus:

$$\mathbf{x}(\mathbf{t}) = (a + b \cos t_1) \sin t_2\mathbf{u}_1 + (a + b \cos t_1) \cos t_2\mathbf{u}_2 + b \sin t_1\mathbf{u}_3,$$

where $0 < b < a$. What are the geometric meanings of a and b on the torus?

11–30. If a parametric surface S can be described explicitly by an equation of the form $z = f(x, y)$ and if R is the projection of S on the xy-plane, show that the area of S is given by the integral

$$\int_R \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, d(x, y).$$

11–31. Let S denote the parametric surface (a hemisphere) described by

$$\mathbf{x}(\mathbf{t}) = \sin t_1 \cos t_2\mathbf{u}_1 + \sin t_1 \sin t_2\mathbf{u}_2 + \cos t_1\mathbf{u}_3,$$

where $(t_1, t_2) \in [0, \pi/2] \times [0, 2\pi]$. Let

$$\mathbf{n}(\mathbf{t}) = D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})/|D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})|.$$

(a) If $\mathbf{f}(x_1, x_2, x_3) = x_1\mathbf{u}_1 + x_2\mathbf{u}_2$, evaluate the surface integral

$$\iint_S \mathbf{f} \cdot \mathbf{n}\, d\sigma.$$

(b) Evaluate $\iint_S x_1x_3\, dx_2\, dx_3 + x_2x_3\, dx_3\, dx_1 + x_1^2\, dx_1\, dx_2$.

(c) Let $\mathbf{f}(x_1, x_2, x_3) = (x_1^2 + x_1x_2 - x_3^2)\mathbf{u}_3$ and evaluate $\iint_S \operatorname{curl} \mathbf{f} \cdot \mathbf{n}\, d\sigma$.

11–32. Let S be a parametric surface represented by an equation of the form $\mathbf{x}(\mathbf{t}) = t_1\mathbf{u}_1 + t_2\mathbf{u}_2 + \phi(t_1, t_2)\mathbf{u}_3$, if $(t_1, t_2) \in R$. Let f be a real-valued function defined on S and define $F(\mathbf{t}) = f[\mathbf{x}(\mathbf{t})]$ if $\mathbf{t} \in R$. Show that

$$\iint_S f\, dx_1\, dx_2 = \int_R F\, d(t_1, t_2),$$

$$\iint_S f\, dx_2\, dx_3 = -\int_R F\, D_1\phi\, d(t_1, t_2),$$

$$\iint_S f\, dx_3\, dx_1 = -\int_R F\, D_2\phi\, d(t_1, t_2).$$

11–33. Let $R = \{(t_1, t_2) \mid t_1^2 + t_2^2 \le r\}$, where $r > 0$. Define

$$\mathbf{x}(\mathbf{t}) = t_1\mathbf{u}_1 + t_2\mathbf{u}_2 + \sqrt{r^2 - t_1^2 - t_2^2}\,\mathbf{u}_3,$$

$$\mathbf{y}(\mathbf{t}) = t_1\mathbf{u}_1 - t_2\mathbf{u}_2 - \sqrt{r^2 - t_1^2 - t_2^2}\,\mathbf{u}_3,$$

if $\mathbf{t} = (t_1, t_2) \in R$. Let $\mathbf{n}_1(\mathbf{t}) = D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})/|D_1\mathbf{x}(\mathbf{t}) \times D_2\mathbf{x}(\mathbf{t})|$ and $\mathbf{n}_2(\mathbf{t}) = D_1\mathbf{y}(\mathbf{t}) \times D_2\mathbf{y}(\mathbf{t})/|D_1\mathbf{y}(\mathbf{t}) \times D_2\mathbf{y}(\mathbf{t})|$. Let $\mathbf{f}(\mathbf{t}) = \mathbf{t}/|\mathbf{t}|^3$ if $\mathbf{t} \neq 0$, $\mathbf{f}(0) = 0$.

(a) If $S_1 = \mathbf{x}(R)$, $S_2 = \mathbf{y}(R)$, show that

$$\iint_{S_1} \mathbf{f} \cdot \mathbf{n}_1\, d\sigma = \iint_{S_2} \mathbf{f} \cdot \mathbf{n}_2\, d\sigma = 2\pi$$

(b) The function $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)$, defined by $\alpha_1(t) = r\cos t$, $\alpha_2(t) = r\sin t$, $0 \leq t \leq 2\pi$, describes the positively oriented boundary of R. Let $\boldsymbol{\gamma}(t) = \mathbf{x}[\boldsymbol{\alpha}(t)]$, $\boldsymbol{\delta}(t) = \mathbf{y}[\boldsymbol{\alpha}(t)]$ if $t \in [0, 2\pi]$. Let $C = S_1 \cap S_2$ and show that

$$\int_{C(\gamma)} \mathbf{f} \cdot d\boldsymbol{\gamma} = -\int_{C(\delta)} \mathbf{f} \cdot d\boldsymbol{\delta}$$

11–34. If $0 < a < b$, let $V = \{\mathbf{x} \mid \mathbf{x} \in E_3, 0 < a < |\mathbf{x}| < b\}$ and let $\mathbf{f}(\mathbf{x}) = \mathbf{x}/|\mathbf{x}|^3$ if $\mathbf{x} \neq 0$.

(a) Show that $\operatorname{div} \mathbf{f}(\mathbf{x}) = 0$ for each $\mathbf{x}$ in V.

(b) Use Exercise 11–33 in conjunction with Stokes' theorem to prove that there does not exist a vector field $\mathbf{g}$ such that $\mathbf{f}(\mathbf{x}) = \operatorname{curl} \mathbf{g}(\mathbf{x})$ for each $\mathbf{x}$ in V.

11–35. Let $\mathbf{a}$ be a fixed vector in E_3, define $\mathbf{f}(\mathbf{x}) = \mathbf{a} \times \mathbf{x}$ if $\mathbf{x} \in E_3$. Using the notation of Theorem 11–36, show that

$$2\iint_{S} \mathbf{a} \cdot \mathbf{n}\, d\sigma = \int_{C(\gamma)} \mathbf{f} \cdot d\boldsymbol{\gamma}$$

11–36. Using the notation of Exercise 11–33, evaluate

$$\iint_{S_1} \mathbf{f} \cdot \mathbf{n}_1\, d\sigma + \iint_{S_2} \mathbf{f} \cdot \mathbf{n}_2\, d\sigma$$

when $\mathbf{f}(\mathbf{x}) = x_1^2\mathbf{u}_1 + x_2^2\mathbf{u}_2 + x_3^2\mathbf{u}_3$.

Coordinate transformations.

11–37. Let $\mathbf{x} = (x_1, x_2, x_3)$ be a coordinate transformation defined on a region T in E_3 and let $\mathbf{y} = (y_1, y_2, y_3) = \mathbf{x}^{-1}$ be the inverse function, defined on the image $S = \mathbf{x}(T)$. Assuming that $\mathbf{y} \in C'$ on an open set containing S, prove the following statements.

(a) The Jacobian $J_{\mathbf{y}}(\mathbf{s}) = \nabla y_1(\mathbf{s}) \cdot \nabla y_2(\mathbf{s}) \times \nabla y_3(\mathbf{s})$ if $\mathbf{s} \in S$.

(b) If $\mathbf{t} \in T$ and $\mathbf{s} = \mathbf{x}(\mathbf{t})$, the two triples $(D_1\mathbf{x}(\mathbf{t}), D_2\mathbf{x}(\mathbf{t}), D_3\mathbf{x}(\mathbf{t}))$ and $(\nabla y_1(\mathbf{s}), \nabla y_2(\mathbf{s}), \nabla y_3(\mathbf{s}))$ are reciprocal sets of vectors (in the sense of Exercise 11–3).

(c) Derive the identity

$$D_1\mathbf{x}(\mathbf{t}) = J_{\mathbf{x}}(\mathbf{t})(\nabla y_2(\mathbf{s}) \times \nabla y_3(\mathbf{s}))$$

and obtain similar formulas for $D_2\mathbf{x}(\mathbf{t})$ and $D_3\mathbf{x}(\mathbf{t})$.

(d) Show that

$$\nabla y_1(\mathbf{s}) = J_{\mathbf{y}}(\mathbf{s})(D_2\mathbf{x}(\mathbf{t}) \times D_3\mathbf{x}(\mathbf{t}))$$

and obtain similar formulas for $\nabla y_2(\mathbf{s})$ and $\nabla y_3(\mathbf{s})$.

(e) From (c), deduce that

$$\frac{D_1\mathbf{x}(\mathbf{t})}{J_{\mathbf{x}}(\mathbf{t})} = \operatorname{curl}\,(y_2(\mathbf{s})\,\nabla y_3(\mathbf{s}))$$

and obtain similar formulas for $D_2\mathbf{x}(\mathbf{t})/J_{\mathbf{x}}(\mathbf{t})$ and $D_3\mathbf{x}(\mathbf{t})/J_{\mathbf{x}}(\mathbf{t})$.

(f) Use (e) to prove that $\operatorname{div}(D_i\mathbf{x}(\mathbf{t})/J_{\mathbf{x}}(\mathbf{t})) = 0$ for $i = 1, 2, 3$.

In Exercises 11–38 and 11–39 the curvilinear basis $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ is assumed to be orthonormal.

11–38. (a) Show that formula (52) can be written as follows:

$$\mathbf{F} = (F_1h_2h_3)\,\frac{D_1\mathbf{x}}{J_{\mathbf{x}}} + (F_2h_3h_1)\,\frac{D_2\mathbf{x}}{J_{\mathbf{x}}} + (F_3h_1h_2)\,\frac{D_3\mathbf{x}}{J_{\mathbf{x}}}.$$

(b) Use Theorem 11–29(b) in conjunction with Exercise 11–37(f) to prove

$$\operatorname{div}\mathbf{F} = \frac{1}{J_{\mathbf{x}}}\,[\nabla(F_1h_2h_3)\cdot D_1\mathbf{x} + \nabla(F_2h_3h_1)\cdot D_2\mathbf{x} + \nabla(F_3h_1h_2)\cdot D_3\mathbf{x}].$$

(c) Apply the chain rule in (b) to get

$$\operatorname{div}\mathbf{F} = \frac{1}{h_1h_2h_3}\,[D_1(F_1h_2h_3) + D_2(F_2h_3h_1) + D_3(F_3h_1h_2)].$$

11–39. (a) Use Theorem 11–26(b) and formula (52) to obtain

$$\operatorname{curl}\mathbf{F} = \nabla(h_1F_1) \times \left(\frac{\mathbf{v}_1}{h_1}\right) + \nabla(h_2F_2) \times \left(\frac{\mathbf{v}_2}{h_2}\right) + \nabla(h_3F_3) \times \left(\frac{\mathbf{v}_3}{h_3}\right).$$

(b) From (a), deduce

$$\operatorname{curl}\mathbf{F} = \frac{1}{h_2h_3}\,[D_2(h_3F_3) - D_3(h_2F_2)]\mathbf{v}_1 + \frac{1}{h_3h_1}\,[D_3(h_1F_1) - D_1(h_3F_3)]\mathbf{v}_2$$

$$+ \frac{1}{h_1h_2}\,[D_1(h_2F_2) - D_2(h_1F_1)]\mathbf{v}_3.$$

11–40. Use formula (56) along with Exercise 11–38(c) to obtain the following formula for the Laplacian in orthogonal curvilinear coordinates

$$\nabla^2\phi[\mathbf{x}(t)] = \frac{1}{h_1h_2h_3}\left[D_1\left(\frac{h_2h_3}{h_1}\,D_1\Phi\right) + D_2\left(\frac{h_3h_1}{h_2}\,D_2\Phi\right) + D_3\left(\frac{h_1h_2}{h_3}\,D_3\Phi\right)\right],$$

where $\Phi(\mathbf{t}) = \phi[\mathbf{x}(\mathbf{t})]$

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CHAPTER 12

INFINITE SERIES AND INFINITE PRODUCTS

12–1 Introduction. In this chapter we intend to give a brief development of the theory of infinite series and infinite products. These are merely special infinite sequences whose terms are complex numbers, and hence we begin by developing some properties of such sequences. The reader should recall the definitions of *infinite sequence* (Definition 2–13), *limit of a sequence* (Definition 4–2), and *Cauchy condition for sequences* (Theorem 4–6). Sequences which satisfy the Cauchy condition will be called *Cauchy sequences.* Unless otherwise specified, all sequences in this chapter will be assumed to have complex terms.

12–2 Convergent and divergent sequences.

12–1 DEFINITION. *A sequence $\{a_n\}$ is called convergent if there exists a point a in E_2 such that $\lim_{n\to\infty} a_n = a$, in which case we also say that the sequence converges to a. A sequence is called divergent if it is not convergent.*

Theorem 4–6 tells us that a sequence is convergent if, and only if, it is a Cauchy sequence. The Cauchy condition is particularly useful in establishing convergence when we do not know the actual value to which the sequence converges. Every Cauchy sequence is bounded (see the proof of Theorem 4–6), and therefore, an unbounded sequence necessarily diverges.

Using the Cauchy condition, we can easily see that if a sequence $\{a_n\}$ converges to a, then every subsequence $\{a_{k_n}\}$ also converges to a. (See Exercise 12–1.)

A sequence $\{a_n\}$ whose terms are real numbers is said to diverge to $+\infty$ if $\lim_{n\to\infty} a_n = +\infty$, and it is said to diverge to $-\infty$ if $\lim_{n\to\infty} a_n = -\infty$. Of course, there are divergent real-valued sequences which do not diverge to $+\infty$ or to $-\infty$. For example, the sequence $\{(-1)^n(1 + 1/n)\}$ diverges but does not diverge to $+\infty$ or to $-\infty$.

12–3 Limit superior and limit inferior of a real-valued sequence.

12–2 DEFINITION. *Let $\{a_n\}$ be a sequence of real numbers. Suppose there is a real number U satisfying the following two conditions:*

(i) *For every* $\epsilon > 0$ *there exists an integer* N *such that* $n > N$ *implies*

$$a_n < U + \epsilon$$

(ii) *Given* $\epsilon > 0$ *and given* $m > 0$, *there exists an integer* $n > m$ *such that*

$$a_n > U - \epsilon$$

Then U *is called the limit superior (or upper limit) of* $\{a_n\}$ *and we write*

$$U = \limsup_{n\to\infty} a_n$$

Statement (i) *implies that the set* $\{a_1, a_2, \ldots\}$ *is bounded above. If this set is not bounded above, we define* $\limsup_{n\to\infty} a_n = +\infty$ *The limit inferior (or lower limit) of* $\{a_n\}$ *is defined as follows:*

$$\liminf_{n\to\infty} a_n = -\limsup_{n\to\infty} b_n, \qquad \textit{where } b_n = -a_n \textit{ for } n = 1, 2, \ldots$$

NOTE Statement (i) means that ultimately *all* terms of the sequence lie to the left of $U + \epsilon$ Statement (ii) means that *infinitely many* terms lie to the right of $U - \epsilon$ It is clear that there cannot be more than one U which satisfies both (i) and (ii).

The reader should supply the proofs of the following theorems:

12-3 THEOREM *Let* $\{a_n\}$ *be a sequence of real numbers Then we have:*

(a) $\liminf_{n\to\infty} a_n \le \limsup_{n\to\infty} a_n$

(b) *The sequence converges if, and only if,* $\limsup_{n\to\infty} a_n$ *and* $\liminf_{n\to\infty} a_n$ *are both finite and equal, in which case* $\lim_{n\to\infty} a_n = \liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n$

(c) *The sequence diverges to* $+\infty$ *if, and only if,* $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = +\infty$

(d) *The sequence diverges to* $-\infty$ *if, and only if,* $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = -\infty$.

NOTE A sequence for which $\liminf_{n\to\infty} a_n \neq \limsup_{n\to\infty} a_n$ is said to oscillate

12-4 THEOREM *Assume that* $a_n \le b_n$ *for each* $n = 1, 2, \ldots$ *Then we have:*

$$\liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} b_n \qquad \textit{and} \qquad \limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} b_n.$$

EXAMPLES.

1. $a_n = (-1)^n(1 + 1/n)$, $\liminf_{n\to\infty} a_n = -1$, $\limsup a_n = +1$.
2. $a_n = (-1)^n$, $\liminf_{n\to\infty} a_n = -1$, $\limsup_{n\to\infty} a_n = +1$.
3. $a_n = (-1)^n n$, $\liminf_{n\to\infty} a_n = -\infty$, $\limsup_{n\to\infty} a_n = +\infty$.
4. $a_n = n^2 \sin^2(\frac{1}{2}n\pi)$, $\liminf_{n\to\infty} a_n = 0$, $\limsup a_n = +\infty$.

12-4 Monotonic sequences of real numbers.

12-5 DEFINITION. *Let $\{a_n\}$ be a sequence of real numbers. We say the sequence is increasing and we write $a_n \nearrow$ if $a_n \le a_{n+1}$ for $n = 1, 2, \ldots$. If $a_n \ge a_{n+1}$ for all n, we say the sequence is decreasing and we write $a_n \searrow$. A sequence is called monotonic if it is increasing or if it is decreasing.*

The convergence or divergence of a monotonic sequence is particularly easy to determine. In fact, we have

12-6 THEOREM. *A monotonic sequence converges if, and only if, it is bounded.*

Proof. If $a_n \nearrow$, $\lim_{n\to\infty} a_n = \sup\{a_n \mid n = 1, 2, \ldots\}$. If $a_n \searrow$, $\lim_{n\to\infty} a_n = \inf\{a_n \mid n = 1, 2, \ldots\}$.

12-5 Infinite series.

12-7 DEFINITION. *Let $\{a_n\}$ be a given sequence of real or complex numbers. Form a new sequence $\{s_n\}$ as follows:*

$$s_n = a_1 + \cdots + a_n = \sum_{k=1}^{n} a_k \qquad (n = 1, 2, \ldots). \tag{1}$$

A sequence $\{s_n\}$ formed in this way is called a series (or an infinite series). The number s_n is called the nth *partial sum of the series and a_n is called the* nth *term of the series. The series is said to converge or to diverge according as $\{s_n\}$ is convergent or divergent. The following symbols are used to denote the series defined by* (1):

$$a_1 + a_2 + \cdots + a_n + \cdots, \qquad a_1 + a_2 + a_3 + \cdots, \qquad \sum_{k=1}^{\infty} a_k.$$

NOTE. The letter k used in $\sum_{k=1}^{\infty} a_k$ is a "dummy variable" and may be replaced by any other convenient symbol. If p is an integer ≥ 0, a symbol of the form $\sum_{n=p}^{\infty} b_n$ is interpreted to mean $\sum_{n=1}^{\infty} a_n$ where $b_n = a_{n-p+1}$.

When there is no danger of misunderstanding, we write $\sum b_n$ instead of $\sum_{n=p}^{\infty} b_n$

If the series $\{s_n\}$ defined by (1) converges to s, the number s is called the *sum* of the series and we write

$$s = \sum_{k=1}^{\infty} a_k$$

Thus, for convergent series the same symbol is used to denote both the series and its sum

12-8 THEOREM. *Let $a = \sum a_n$ and $b = \sum b_n$ be convergent series Then, for every pair of constants α and β, the series $\sum(\alpha a_n + \beta b_n)$ converges to the sum $\alpha a + \beta b$ That is,*

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n) = \alpha \sum_{n=1}^{\infty} a_n + \beta \sum_{n=1}^{\infty} b_n$$

Proof $\sum_{k=1}^{n} (\alpha a_k + \beta b_k) = \alpha \sum_{k=1}^{n} a_k + \beta \sum_{k=1}^{n} b_k$

12-9 THEOREM. *Assume that $a_n \geq 0$ for each $n = 1, 2,$. Then $\sum a_n$ converges if, and only if, the sequence of partial sums is bounded above*

Proof. Let $s_n = a_1 + \cdots + a_n$ Then $s_n \nearrow$ and we can apply Theorem 12-6.

12-10 THEOREM (*Telescoping series*). *Let $\{a_n\}$ and $\{b_n\}$ be two sequences such that $a_n = b_{n+1} - b_n$ for $n = 1, 2,$. Then $\sum a_n$ converges if, and only if, $\lim_{n\to\infty} b_n$ exists, in which case we have*

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} b_n - b_1.$$

Proof. $\sum_{k=1}^{n} a_k = \sum_{k=1}^{n} (b_{k+1} - b_k) = b_{n+1} - b_1$

12-11 THEOREM (*Cauchy condition for series*) *The series $\sum a_n$ converges if, and only if, for every $\epsilon > 0$ there exists an integer N such that $n > N$ implies*

(2) $\quad |a_{n+1} + \cdots + a_{n+p}| < \epsilon \qquad$ *for each* $p = 1, 2,$.

Proof Let $s_n = \sum_{k=1}^{n} a_k$, write $s_{n+p} - s_n = a_{n+1} + \cdots + a_{n+p}$, and apply Theorem 4-6

Taking $p = 1$ in (2), we find that $\lim_{n\to\infty} a_n = 0$ is a *necessary* condition for convergence of $\sum a_n$ That this condition is *not sufficient* is seen by considering the example in which $a_n = 1/n$. When $n = 2^m$ and $p = 2^m$ in (2), we find

$$a_{n+1} + \cdots + a_{n+p} = \frac{1}{2^m + 1} + \cdots + \frac{1}{2^m + 2^m} \geq \frac{2^m}{2^m + 2^m} = \frac{1}{2}$$

and hence the Cauchy condition cannot be satisfied when $\epsilon \leq \frac{1}{2}$. Therefore the series $\sum_{n=1}^{\infty} 1/n$ diverges. This series is called the *harmonic series.*

12–6 Inserting and removing parentheses.

12–12 DEFINITION. *Let p be a function whose domain is the set of positive integers and whose range is a subset of the positive integers such that*

$$\text{(i)} \qquad p(n) < p(m), \qquad \text{if } n < m.$$

Let $\sum a_n$ and $\sum b_n$ be two series related as follows:

$$\text{(ii)} \qquad \begin{aligned} b_1 &= a_1 + a_2 + \cdots + a_{p(1)}, \\ b_{n+1} &= a_{p(n)+1} + a_{p(n)+2} + \cdots + a_{p(n+1)}, \qquad \text{if } n = 1, 2, \ldots. \end{aligned}$$

Then we say $\sum b_n$ is obtained from $\sum a_n$ by inserting parentheses, and that $\sum a_n$ is obtained from $\sum b_n$ by removing parentheses.

12–13 THEOREM. *If $\sum a_n$ converges to s, every series $\sum b_n$ obtained from $\sum a_n$ by inserting parentheses also converges to s.*

Proof. Let $\sum a_n$ and $\sum b_n$ be related by (ii) and write $s_n = \sum_{k=1}^{n} a_k$, $t_n = \sum_{k=1}^{n} b_k$. Then $\{t_n\}$ is a subsequence of $\{s_n\}$. In fact, $t_n = s_{p(n)}$. Therefore, convergence of $\{s_n\}$ to s implies convergence of $\{t_n\}$ to s.

Removing parentheses may destroy convergence. To see this, consider the series $\sum b_n$ in which each term is 0 (obviously convergent). Let $p(n) = 2n$ and let $a_n = (-1)^n$. Then (i) and (ii) hold but $\sum a_n$ diverges.

Parentheses *can* be removed if we put further restrictions on $\sum a_n$ and on p.

12–14 THEOREM. *Let $\sum a_n$, $\sum b_n$ be related as in Definition 12–12. Assume that there exists a constant $M > 0$ such that $p(n+1) - p(n) < M$ for all n, and assume that $\lim_{n\to\infty} a_n = 0$. Then $\sum a_n$ converges if, and only if, $\sum b_n$ converges, in which case they have the same sum.*

Proof. If $\sum a_n$ converges, the result follows from Theorem 12–13. The whole difficulty lies in the converse deduction. Let $s_n = a_1 + \cdots + a_n$,

$t_n = b_1 + \cdots + b_n$, $t = \lim_{n\to\infty} t_n$. Let $\epsilon > 0$ be given and choose N so that $n > N$ implies

$$|t_n - t| < \frac{\epsilon}{2} \quad \text{and} \quad |a_n| < \frac{\epsilon}{2M}.$$

If $n > p(N)$, we can find $m \geq N$ so that $N \leq p(m) \leq n < p(m + 1)$. [Why?] For such n, we have

$$\begin{aligned} s_n &= a_1 + \cdots + a_{p(m+1)} - (a_{n+1} + a_{n+2} + \cdots + a_{p(m+1)}) \\ &= t_{m+1} - (a_{n+1} + a_{n+2} + \cdots + a_{p(m+1)}), \end{aligned}$$

and hence

$$\begin{aligned} |s_n - t| &\leq |t_{m+1} - t| + |a_{n+1} + a_{n+2} + \cdots + a_{p(m+1)}| \\ &\leq |t_{m+1} - t| + |a_{p(m)+1}| + |a_{p(m)+2}| + \cdots + |a_{p(m+1)}| \\ &< \frac{\epsilon}{2} + (p(m + 1) - p(m)) \frac{\epsilon}{2M} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

This proves that $\lim_{n\to\infty} s_n = t$.

12-7 Alternating series.

12-15 DEFINITION. *If $a_n > 0$ for each n, the series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ is called an alternating series.*

12-16 THEOREM. *If $\{a_n\}$ is a decreasing sequence converging to 0, the alternating series $\sum(-1)^{n+1} a_n$ converges. If s denotes its sum and s_n its nth partial sum, we have the inequality*

$$(3) \qquad 0 < (-1)^n(s - s_n) < a_{n+1}, \qquad \text{for } n = 1, 2, \ldots$$

NOTE. Inequality (3) tells us that when we "approximate" s by s_n, the error made has the same sign as the first neglected term and is less than the absolute value of this term.

Proof. $s_{2n} = a_1 - (a_2 - a_3) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n} < a_1$. Also, $s_{2n+2} - s_{2n} = a_{2n+1} - a_{2n+2} > 0$. Therefore, $\{s_{2n}\}$ is a bounded increasing sequence and must converge (by Theorem 12-6). Similarly, $\{s_{2n-1}\}$, being a bounded decreasing sequence, converges. These two sequences converge to the same limit since $s_{2n+1} - s_{2n} = a_{2n+1}$ and $a_n \to 0$ as $n \to \infty$. The convergence of $\{s_n\}$ now follows easily by observing that for each n there is an m such that $s_{2m} \leq s_n \leq s_{2m+1}$.

Inequality (3) is a consequence of the following relations:

$$(-1)^n(s - s_n) = \sum_{k=1}^{\infty} (-1)^{k+1} a_{n+k} = \sum_{k=1}^{\infty} (a_{n+2k-1} - a_{n+2k}) > 0,$$

and

$$(-1)^n(s - s_n) = a_{n+1} - \sum_{k=1}^{\infty} (a_{n+2k} - a_{n+2k+1}) < a_{n+1}.$$

12–8 Absolute and conditional convergence.

12–17 DEFINITION. *A series $\sum a_n$ is called absolutely convergent if $\sum |a_n|$ converges. It is called conditionally convergent if $\sum a_n$ converges but $\sum |a_n|$ diverges.*

12–18 THEOREM. *Absolute convergence of $\sum a_n$ implies convergence.*

Proof. Apply the Cauchy condition to the inequality

$$|a_{n+1} + \cdots + a_{n+p}| \leq |a_{n+1}| + \cdots + |a_{n+p}|.$$

To see that the converse is not true, consider the example

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}.$$

This alternating series converges, by Theorem 12–16, but it does not converge absolutely.

12–19 THEOREM. *Let $\sum a_n$ be a given series with real-valued terms and define*

$$p_n = \frac{|a_n| + a_n}{2}, \qquad q_n = \frac{|a_n| - a_n}{2} \qquad (n = 1, 2, \ldots). \tag{4}$$

Then:

(i) *If $\sum a_n$ is conditionally convergent, both $\sum p_n$ and $\sum q_n$ diverge.*

(ii) *If $\sum |a_n|$ converges, both $\sum p_n$ and $\sum q_n$ converge and we have*

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} p_n - \sum_{n=1}^{\infty} q_n.$$

NOTE. $p_n = a_n$ and $q_n = 0$ if $a_n \geq 0$, whereas $q_n = -a_n$ and $p_n = 0$ if $a_n \leq 0$.

Proof. We have $a_n = p_n - q_n$, $|a_n| = p_n + q_n$. To prove (i), assume that $\sum a_n$ converges and $\sum |a_n|$ diverges. If $\sum q_n$ converges, then $\sum p_n$ also converges (by Theorem 12-8), since $p_n = a_n + q_n$ Similarly, if $\sum p_n$ converges, then $\sum q_n$ also converges Hence, if either $\sum p_n$ or $\sum q_n$ converges, *both* must converge and we deduce that $\sum|a_n|$ converges, since $|a_n| = p_n + q_n$ This contradiction proves (i)

To prove (ii), we simply use (4) in conjunction with Theorem 12-8

12-9 Real and imaginary parts of a complex series. Let $\sum c_n$ be a series with complex terms and write $c_n = a_n + ib_n$, where a_n and b_n are real The series $\sum a_n$ and $\sum b_n$ are called, respectively, the real and imaginary parts of $\sum c_n$ In situations involving complex series, it is often convenient to treat the real and imaginary parts separately Of course, convergence of both $\sum a_n$ and $\sum b_n$ implies convergence of $\sum c_n$ Conversely, convergence of $\sum c_n$ implies convergence of both $\sum a_n$ and $\sum b_n$ The same remarks hold for *absolute* convergence However, when $\sum c_n$ is *conditionally* convergent, one (but not both) of $\sum a_n$ and $\sum b_n$ might be absolutely convergent (See Exercise 12-19.)

If $\sum c_n$ converges absolutely, we can apply part (ii) of Theorem 12-19 to the real and imaginary parts separately to obtain the decomposition

$$\sum c_n = \sum(p_n + iu_n) - \sum(q_n + iv_n),$$

where $\sum p_n$, $\sum q_n$, $\sum u_n$, $\sum v_n$ are convergent series of non-negative terms.

12-10 Tests for convergence of series with positive terms.

12-20 THEOREM (*Comparison test*). *If $a_n > 0$ and $b_n > 0$ for $n = 1, 2, \ldots$, and if there exist positive constants c and N such that*

$$a_n < cb_n \qquad \text{for } n \geq N,$$

then convergence of $\sum b_n$ implies convergence of $\sum a_n$

Proof. The partial sums of $\sum a_n$ are bounded if the partial sums of $\sum b_n$ are bounded By Theorem 12-9, this completes the proof

12-21 THEOREM (*Limit comparison test*) *Assume that $a_n > 0$ and $b_n > 0$ for $n = 1, 2, \ldots$, and suppose that*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 1$$

Then $\sum a_n$ converges if, and only if, $\sum b_n$ converges

Proof There exists N such that $n \geq N$ implies $\frac{1}{2} < a_n/b_n < \frac{3}{2}$ The theorem follows by applying Theorem 12-20 twice

NOTE. Theorem 12-21 also holds if $\lim_{n\to\infty} a_n/b_n = c$, provided that $c \neq 0$. If $\lim_{n\to\infty} a_n/b_n = 0$, we can only conclude that convergence of $\sum b_n$ implies convergence of $\sum a_n$.

To use comparison tests effectively, we must have at our disposal some examples of series of known behavior. One of the most important series for comparison purposes is the *geometric series.*

12-22 THEOREM. *If $|x| < 1$, the series $1 + x + x^2 + \cdots$ converges and has sum $1/(1 - x)$. If $|x| \geq 1$, the series diverges.*

Proof. $(1 - x) \sum_{k=0}^{n} x^k = \sum_{k=0}^{n} (x^k - x^{k+1}) = 1 - x^{n+1}$. When $|x| < 1$, we find $\lim_{n\to\infty} x^{n+1} = 0$. If $|x| \geq 1$, the general term does not tend to zero and the series cannot converge.

Further examples of series of known behavior can be obtained very simply by applying the *integral test.*

12-23 THEOREM (*Integral test*). *Let f be a positive decreasing function defined on $[1, +\infty)$ such that $\lim_{x\to+\infty} f(x) = 0$. For $n = 1, 2, \ldots,$ define*

$$s_n = \sum_{k=1}^{n} f(k), \qquad t_n = \int_1^n f(x)\,dx, \qquad d_n = s_n - t_n.$$

Then we have:

(i) $0 < f(n + 1) \leq d_{n+1} \leq d_n \leq f(1)$, *for* $n = 1, 2, \ldots.$
(ii) $\lim_{n\to\infty} d_n$ *exists.*
(iii) $\sum_{n=1}^{\infty} f(n)$ *converges if, and only if, the sequence $\{t_n\}$ converges.*
(iv) $0 \leq d_k - \lim_{n\to\infty} d_n \leq f(k)$, *for* $k = 1, 2, \ldots.$

Proof. To prove (i), write

$$t_{n+1} = \int_1^{n+1} f(x)\,dx = \sum_{k=1}^{n} \int_k^{k+1} f(x)\,dx \leq \sum_{k=1}^{n} \int_k^{k+1} f(k)\,dx$$
$$= \sum_{k=1}^{n} f(k) = s_n.$$

This implies $f(n + 1) = s_{n+1} - s_n \leq s_{n+1} - t_{n+1} = d_{n+1}$, and we obtain

$$0 < f(n + 1) \leq d_{n+1}.$$

But we also have

(5)
$$d_n - d_{n+1} = t_{n+1} - t_n - (s_{n+1} - s_n) = \int_n^{n+1} f(x)\,dx - f(n+1)$$
$$\geq \int_n^{n+1} f(n+1)\,dx - f(n+1) = 0,$$

and hence $d_{n+1} \leq d_n \leq d_1 = f(1)$ This proves (i) But now it is clear that (i) implies (ii) and that (ii) implies (iii)

To prove part (iv), we use (5) again to write

$$0 \leq d_n - d_{n+1} \leq \int_n^{n+1} f(n)\,dx - f(n+1) = f(n) - f(n+1)$$

Summing on n, we get

$$0 \leq \sum_{n=k}^{\infty} (d_n - d_{n+1}) \leq \sum_{n=k}^{\infty} (f(n) - f(n+1)), \qquad \text{if } k \geq 1.$$

When we evaluate the sums of these telescoping series, we get (iv).

NOTE. Let $D = \lim_{n\to\infty} d_n$ Then (i) implies $0 \leq D \leq f(1)$, whereas (iv) gives us

$$(6) \qquad 0 \leq \sum_{k=1}^{n} f(k) - \int_1^n f(x)\,dx - D \leq f(n).$$

This inequality is extremely useful for approximating certain finite sums by integrals

12-24 DEFINITION. *Given two sequences $\{a_n\}$ and $\{b_n\}$ such that $b_n \geq 0$ for all n. We write*

$$a_n = O(b_n)$$

if there exists a constant $M > 0$ such that $|a_n| \leq Mb_n$ for all n We write

$$a_n = o(b_n) \qquad \text{as } n \to \infty$$

if $\lim_{n\to\infty} a_n/b_n = 0$

NOTE. An equation of the form $a_n = c_n + O(b_n)$ means $a_n - c_n = O(b_n)$. Similarly, $a_n = c_n + o(b_n)$ means $a_n - c_n = o(b_n)$.

The advantage of this notation is that it allows us to replace certain inequalities by equations. For example, (6) implies

$$\sum_{k=1}^{n} f(k) = \int_1^n f(x)\,dx + D + O(f(n)). \tag{7}$$

EXAMPLES.

1. Let $f(x) = 1/x$ in Theorem 12–23. We find $t_n = \log n$ and hence $\sum 1/n$ diverges. However, (ii) establishes the existence of the limit

$$\lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k} - \log n\right),$$

a famous number known as *Euler's constant*, usually denoted by C (or by γ). Equation (7) becomes

$$\sum_{k=1}^{n}\frac{1}{k} = \log n + C + O\left(\frac{1}{n}\right). \tag{8}$$

2. Let $f(x) = x^{-s}$, $s \neq 1$, in Theorem 12–23. We find that $\sum n^{-s}$ converges if $s > 1$ and diverges if $s < 1$. For $s > 1$, this series defines a very important function in analysis known as the *Riemann zeta function*:

$$\zeta(s) = \sum_{n=1}^{\infty}\frac{1}{n^s} \qquad (s > 1).$$

For $s > 0$, $s \neq 1$, we can apply (7) to write

$$\sum_{k=1}^{n}\frac{1}{k^s} = \frac{n^{1-s} - 1}{1 - s} + C(s) + O\left(\frac{1}{n^s}\right),$$

where $C(s) = \lim_{n\to\infty}\left(\sum_{k=1}^{n} k^{-s} - (n^{1-s}-1)/(1-s)\right)$.

12–11 The ratio test and the root test.

12–25 THEOREM (*Ratio test*). *Given a series $\sum a_n$ of nonzero complex terms, let*

$$r = \liminf_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|, \qquad R = \limsup_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|.$$

(a) *The series $\sum a_n$ converges absolutely if $R < 1$.*

(b) *The series $\sum a_n$ diverges if $r > 1$.*

(c) *The test is inconclusive if $r \leq 1 \leq R$.*

Proof Assume that $R < 1$ and choose x so that $R < x < 1$ The definition of R implies the existence of N such that $|a_{n+1}/a_n| < x$ if $n \geq N$ Since $x = x^{n+1}/x^n$, this means that

$$\frac{|a_{n+1}|}{x^{n+1}} < \frac{|a_n|}{x^n} \leq \frac{|a_N|}{x^N}, \qquad \text{if } n \geq N,$$

and hence $|a_n| \leq cx^n$ if $n \geq N$, where $c = |a_N|x^{-N}$ Statement (a) now follows by applying the comparison test

To prove (b), we simply observe that $r > 1$ implies $|a_{n+1}| > |a_n|$ for all $n \geq N$ for some N and hence we cannot have $\lim_{n\to\infty} a_n = 0$

To prove (c), consider the two examples $\sum n^{-1}$ and $\sum n^{-2}$ In both cases, $r = R = 1$ but $\sum n^{-1}$ diverges, whereas $\sum n^{-2}$ converges

12-26 THEOREM (*Root test*) *Given a series $\sum a_n$ of complex terms, let*

$$\rho = \limsup_{n\to\infty} \sqrt[n]{|a_n|}$$

(a) *The series $\sum a_n$ converges absolutely if $\rho < 1$*

(b) *The series $\sum a_n$ diverges if $\rho > 1$*

(c) *The test is inconclusive if $\rho = 1$*

Proof Assume that $\rho < 1$ and choose x so that $\rho < x < 1$ The definition of ρ implies the existence of N such that $|a_n| < x^n$ for $n \geq N$ Hence, $\sum|a_n|$ converges by the comparison test. This proves (a)

To prove (b), we observe that $\rho > 1$ implies $|a_n| > 1$ infinitely often and hence we cannot have $\lim_{n\to\infty} a_n = 0$

Finally, (c) is proved by using the same examples as in Theorem 12-25

NOTE The root test is more "powerful" than the ratio test That is, whenever the root test is inconclusive, so is the ratio test But there are examples where the ratio test fails and the root test *is* conclusive (See Exercise 12-4)

12-12 Dirichlet's test and Abel's test. All the tests in the previous section help us to determine *absolute* convergence of a series with complex terms It is also important to have tests for determining convergence when the series might not converge absolutely The tests in this section are particularly useful for this purpose They all depend on the *partial summation formula* of Abel (equation (9) in the next theorem)

12-27 THEOREM *If $\{a_n\}$ and $\{b_n\}$ are two sequences of complex numbers, define*

$$A_n = a_1 + \cdots + a_n$$

Then we have the identity

$$\sum_{k=1}^{n} a_k b_k = A_n b_{n+1} - \sum_{k=1}^{n} A_k(b_{k+1} - b_k). \tag{9}$$

Therefore, $\sum_{k=1}^{\infty} a_k b_k$ *converges if both the series* $\sum_{k=1}^{\infty} A_k(b_{k+1} - b_k)$ *and the sequence* $\{A_n b_{n+1}\}$ *converge.*

Proof. Writing $A_0 = 0$, we have

$$\sum_{k=1}^{n} a_k b_k = \sum_{k=1}^{n} (A_k - A_{k-1})b_k = \sum_{k=1}^{n} A_k b_k - \sum_{k=1}^{n} A_k b_{k+1} + A_n b_{n+1}.$$

The second assertion follows at once from this identity.

NOTE. Formula (9) is analogous to the formula for integration by parts in a Riemann-Stieltjes integral. In fact, it can be viewed as a special case of Theorem 9–6. (See Exercise 9–6.)

12–28 THEOREM (*Dirichlet's test*). *Let* $\sum a_n$ *be a series of complex terms whose partial sums form a bounded sequence. Let* $\{b_n\}$ *be a decreasing sequence which converges to* 0. *Then* $\sum a_n b_n$ *converges.*

Proof. Let $A_n = a_1 + \cdots + a_n$ and assume that $|A_n| \leq M$ for all n. Then

$$\lim_{n\to\infty} A_n b_{n+1} = 0.$$

Therefore, to establish convergence of $\sum a_n b_n$ we need only show that $\sum A_k(b_{k+1} - b_k)$ is convergent. Since $b_n \searrow$, we have

$$|A_k(b_{k+1} - b_k)| \leq M(b_k - b_{k+1}).$$

But the series $\sum(b_{k+1} - b_k)$ is a convergent telescoping series. Hence the comparison test implies *absolute* convergence of $\sum A_k(b_{k+1} - b_k)$.

12–29 THEOREM (*Abel's test*). *The series* $\sum a_n b_n$ *converges if* $\sum a_n$ *converges and if* $\{b_n\}$ *is a monotonic convergent sequence.*

Proof. Convergence of $\sum a_n$ and of $\{b_n\}$ establishes the existence of the limit $\lim_{n\to\infty} A_n b_{n+1}$, where $A_n = a_1 + \cdots + a_n$. Also, $\{A_n\}$ is a bounded sequence. The remainder of the proof is similar to that of Theorem 12–28. (Two further tests, similar to the above, are given in Exercise 12–27.)

To use Dirichlet's test effectively, we must be acquainted with a few series having bounded partial sums Of course, all *convergent* series have this property. In the next theorem we give an example of a divergent series whose partial sums are bounded The formula for the partial sums of the series in this particular example is of fundamental importance in the theory of Fourier series

12-30 THEOREM *For every real $x \neq 2m\pi$ (m is an integer), we have*

$$\sum_{k=1}^{n} e^{ikx} = e^{ix}\frac{1 - e^{inx}}{1 - e^{ix}} = \frac{\sin(nx/2)}{\sin(x/2)}\, e^{i(n+1)x/2} \tag{10}$$

NOTE This identity yields the following estimate

$$\left|\sum_{k=1}^{n} e^{ikx}\right| \le \frac{1}{|\sin(x/2)|} \tag{11}$$

Proof $(1 - e^{ix})\sum_{k=1}^{n} e^{ikx} = \sum_{k=1}^{n}(e^{ikx} - e^{i(k+1)x}) = e^{ix} - e^{i(n+1)x}$. This establishes the first equality in (10) The second follows from the identity

$$e^{ix}\frac{1 - e^{inx}}{1 - e^{ix}} = \frac{e^{inx/2} - e^{-inx/2}}{e^{ix/2} - e^{-ix/2}}\, e^{i(n+1)x/2}$$

NOTE. By considering the real and imaginary parts of (10), we obtain

$$\sum_{k=1}^{n} \cos kx = \sin\frac{nx}{2}\cos(n+1)\frac{x}{2}\Big/\sin\frac{x}{2} \tag{12}$$

$$= -\frac{1}{2} + \frac{1}{2}\sin(2n+1)\frac{x}{2}\Big/\sin\frac{x}{2},$$

$$\sum_{k=1}^{n} \sin kx = \sin\frac{nx}{2}\sin(n+1)\frac{x}{2}\Big/\sin\frac{x}{2} \tag{13}$$

Using (10), we can also write

$$\sum_{k=1}^{n} e^{i(2k-1)x} = e^{-ix}\sum_{k=1}^{n} e^{ik(2x)} = \frac{\sin nx}{\sin x}\, e^{inx}, \tag{14}$$

an identity valid for every $x \neq m\pi$ (m is an integer) Taking real and imaginary parts of (14) yields

$$\sum_{k=1}^{n} \cos(2k-1)x = \frac{\sin 2nx}{2\sin x}, \tag{15}$$

$$\sum_{k=1}^{n} \sin (2k-1)x = \frac{\sin^2 nx}{\sin x}. \tag{16}$$

Formulas (12) and (16) occur in the theory of Fourier series.

12–13 Rearrangements of series. In this section, P denotes the set of positive integers, $P = \{1, 2, 3, \dots\}$.

12–31 DEFINITION. *Let f be a function whose domain is P and whose range is P, and assume that f is one-to-one on P. Let $\sum a_n$ and $\sum b_n$ be two series such that*

$$a_n = b_{f(n)} \qquad \textit{for } n = 1, 2, \dots. \tag{17}$$

Then $\sum b_n$ is said to be a rearrangement of $\sum a_n$.

NOTE. Equation (17) implies $b_n = a_{f^{-1}(n)}$ and hence $\sum a_n$ is also a rearrangement of $\sum b_n$.

12–32 THEOREM. *Let $\sum a_n$ be an absolutely convergent series having sum s. Then every rearrangement of $\sum a_n$ also converges absolutely and has sum s.*

Proof. Let $\{b_n\}$ be defined by (17). Given $\epsilon > 0$, choose N so that $n \geq N$ implies

$$|a_{n+1}| + \cdots + |a_{n+p}| < \frac{\epsilon}{2} \qquad \text{for } p = 1, 2, \dots.$$

It follows that $\sum_{k=1}^{\infty} |a_{n+k}| \leq \epsilon/2$. Let $g = f^{-1}$ and choose M so that $\{1, 2, \dots, N\} \subset \{g(1), g(2), \dots, g(M)\}$. Then $n > M$ implies $g(n) > N$, and hence for such n we have

$$\begin{aligned} &|b_{n+1}| + \cdots + |b_{n+p}| \\ &\qquad = |a_{g(n+1)}| + \cdots + |a_{g(n+p)}| \leq \sum_{k=1}^{\infty} |a_{N+k}| \leq \frac{\epsilon}{2} < \epsilon. \end{aligned}$$

This proves that $\sum b_n$ converges absolutely.

To show that $\sum b_n = s$, let $t_n = b_1 + \cdots + b_n$, $s_n = a_1 + \cdots + a_n$. Given $\epsilon > 0$, choose N so that $|s_N - s| < \epsilon/2$ and so that $\sum_{k=1}^{\infty} |a_{N+k}| \leq \epsilon/2$. Then

$$|t_n - s| \leq |t_n - s_N| + |s_N - s| < |t_n - s_N| + \frac{\epsilon}{2}.$$

If, now, we choose M as in the first part of the proof, $n > M$ implies

$$\begin{aligned}|t_n - s_N| &= |b_1 + \cdots + b_n - (a_1 + \cdots + a_N)| \\ &= |a_{g(1)} + \cdots + a_{g(n)} - (a_1 + \cdots + a_N)| \\ &\le |a_{N+1}| + |a_{N+2}| + \cdots \le \frac{\epsilon}{2},\end{aligned}$$

since all the terms $a_1, \ldots, a_N$ cancel out in the subtraction. Hence, $n > M$ implies $|t_n - s| < \epsilon$ and this means $\sum b_n = s$.

The hypothesis of absolute convergence is essential in Theorem 12-32. Riemann discovered that any *conditionally* convergent series of real terms can be rearranged to yield a series which converges to any prescribed sum. This remarkable fact is a consequence of the following theorem:

12-33 THEOREM *Let $\sum a_n$ be a conditionally convergent series with real-valued terms. Let x and y be given numbers in the closed interval $[-\infty, +\infty]$, with $x \le y$. Then there exists a rearrangement $\sum b_n$ of $\sum a_n$ such that*

$$\liminf_{n\to\infty} t_n = x \qquad \textit{and} \qquad \limsup_{n\to\infty} t_n = y,$$

where $t_n = b_1 + \cdots + b_n$.

Proof. Discarding those terms of a series which are zero does not affect its convergence or divergence. Hence we might as well assume that no terms of $\sum a_n$ are zero. Let p_n denote the nth positive term of $\sum a_n$ and let $-q_n$ denote its nth negative term. Then $\sum p_n$ and $\sum q_n$ are both divergent series of positive terms. [Why?] Next, construct two sequences of real numbers, say $\{x_n\}$ and $\{y_n\}$, such that $\lim_{n\to\infty} x_n = x$, $\lim_{n\to\infty} y_n = y$, $x_n < y_n$, $y_1 > 0$. The idea of the proof is now quite simple. We take just enough (say k_1) positive terms so that

$$p_1 + \cdots + p_{k_1} > y_1,$$

followed by just enough (say r_1) negative terms so that

$$p_1 + \cdots + p_{k_1} - q_1 - \cdots - q_{r_1} < x_1.$$

Next, we take just enough *further* positive terms so that

$$p_1 + \cdots + p_{k_1} - q_1 - \cdots - q_{r_1} + p_{k_1+1} + \cdots + p_{k_2} > y_2,$$

followed by just enough further negative terms to satisfy the inequality

$$p_1 + \cdots + p_{k_1} - q_1 - \cdots - q_{r_1} + p_{k_1+1} + \cdots$$
$$+ p_{k_2} - q_{r_1+1} - \cdots - q_{r_2} < x_2.$$

These steps are possible since $\sum p_n$ and $\sum q_n$ are both divergent series of positive terms. If the process is continued in this way, we obviously obtain a rearrangement of $\sum a_n$. We leave it to the reader to show that the partial sums of this rearrangement have limit superior y and limit inferior x.

12–34 DEFINITION. *Let f be a function whose domain is P and whose range is an infinite subset Q of P, and assume that f is one-to-one on P. Let $\sum a_n$ and $\sum b_n$ be two given series such that*

$$b_n = a_{f(n)}, \qquad \textit{if } n \in P.$$

Then $\sum b_n$ is said to be a subseries of $\sum a_n$.

12–35 THEOREM. *If $\sum a_n$ converges absolutely, every subseries $\sum b_n$ also converges absolutely. Moreover, we have*

$$\left|\sum_{n=1}^{\infty} b_n\right| \le \sum_{n=1}^{\infty} |b_n| \le \sum_{n=1}^{\infty} |a_n|.$$

Proof. Given n, let N be the largest integer in the set $\{f(1), \ldots, f(n)\}$. Then

$$\left|\sum_{k=1}^{n} b_k\right| \le \sum_{k=1}^{n} |b_k| \le \sum_{k=1}^{N} |a_k| \le \sum_{k=1}^{\infty} |a_k|.$$

The inequality $\sum_{k=1}^{n} |b_k| \le \sum_{k=1}^{\infty} |a_k|$ implies absolute convergence of $\sum b_n$.

12–36 THEOREM. *Let $\{f_1, f_2, \ldots\}$ be a countable collection of functions, each defined on P, having the following properties:*

(a) *Each f_n is one-to-one on P.*

(b) *The range $f_n(P)$ is a subset Q_n of P.*

(c) *$\{Q_1, Q_2, \ldots\}$ is a collection of disjoint sets whose union is P.*

Let $\sum a_n$ be an absolutely convergent series and define

$$b_k(n) = a_{f_k(n)}, \qquad \textit{if } n \in P,\ k \in P.$$

Then:

(i) *For each k, $\sum_{n=1}^{\infty} b_k(n)$ is an absolutely convergent subseries of $\sum a_n$*

(ii) *If $s_k = \sum_{n=1}^{\infty} b_k(n)$, the series $\sum_{k=1}^{\infty} s_k$ converges absolutely and has the same sum as $\sum_{k=1}^{\infty} a_k$*

Proof Theorem 12-35 implies (i) To prove (ii), let $t_k = |s_1| + \cdots + |s_k|$. Then

$$t_k \le \sum_{n=1}^{\infty} |b_1(n)| + \cdots + \sum_{n=1}^{\infty} |b_k(n)| = \sum_{n=1}^{\infty} (|b_1(n)| + \cdots + |b_k(n)|)$$

$$= \sum_{n=1}^{\infty} (|a_{f_1(n)}| + \cdots + |a_{f_k(n)}|).$$

But $\sum_{n=1}^{\infty} (|a_{f_1(n)}| + \cdots + |a_{f_k(n)}|) \le \sum_{n=1}^{\infty} |a_n|$ This proves that $\sum |s_k|$ has bounded partial sums and hence $\sum s_k$ converges absolutely

To find the sum of $\sum s_k$, we proceed as follows. Let $\epsilon > 0$ be given and choose N so that $n \ge N$ implies

$$\sum_{k=1}^{\infty} |a_k| - \sum_{k=1}^{n} |a_k| < \frac{\epsilon}{2}. \tag{18}$$

Choose enough functions $f_1, \ldots, f_r$ so that each term $a_1, a_2, \ldots, a_N$ will appear somewhere in the sum

$$\sum_{n=1}^{\infty} a_{f_1(n)} + \cdots + \sum_{n=1}^{\infty} a_{f_r(n)}$$

The number r depends on N and hence on ϵ If $n > r$ and $n > N$, we have

$$\left| s_1 + s_2 + \cdots + s_n - \sum_{k=1}^{n} a_k \right| \le |a_{N+1}| + |a_{N+2}| + \cdots < \frac{\epsilon}{2} \tag{19}$$

because the terms $a_1, a_2, \ldots, a_N$ cancel in the subtraction Now (18) implies

$$\left| \sum_{k=1}^{\infty} a_k - \sum_{k=1}^{n} a_k \right| < \frac{\epsilon}{2}.$$

When this is combined with (19) we find

$$\left| s_1 + \cdots + s_n - \sum_{k=1}^{\infty} a_k \right| < \epsilon$$

if $n > r$, $n > N$. This completes the proof of (ii).

12–14 Double sequences. As in the last section, P denotes the set of positive integers. Also, we write $P \times P$ for the cartesian product of P with itself.

12–37 DEFINITION. *A function f whose domain is $P \times P$ is called a double sequence.*

NOTE. We shall be interested only in real- or complex-valued double sequences.

12–38 DEFINITION. *If $a \in E_2$, we write $\lim_{p,q\to\infty} f(p, q) = a$ and we say that the double sequence f converges to a, provided that the following condition is satisfied: For every $\epsilon > 0$, there exists an N such that $|f(p, q) - a| < \epsilon$ whenever both $p > N$ and $q > N$.*

12–39 THEOREM. *Assume that $\lim_{p,q\to\infty} f(p, q) = a$. For each fixed p, assume that the limit $\lim_{q\to\infty} f(p, q)$ exists. Then the limit $\lim_{p\to\infty} (\lim_{q\to\infty} f(p, q))$ also exists and has the value a.*

NOTE. To distinguish $\lim_{p,q\to\infty} f(p, q)$ from $\lim_{p\to\infty} (\lim_{q\to\infty} f(p, q))$, the first is called a *double limit*, the second an *iterated limit.*

Proof. Let $F(p) = \lim_{q\to\infty} f(p, q)$. Given $\epsilon > 0$, choose N_1 so that

$$(20) \qquad |f(p, q) - a| < \frac{\epsilon}{2}, \qquad \text{if } p > N_1 \text{ and } q > N_1.$$

For each p we can choose N_2 so that

$$(21) \qquad |F(p) - f(p, q)| < \frac{\epsilon}{2}, \qquad \text{if } q > N_2.$$

(Note that N_2 depends on p as well as on ϵ.) For each $p > N_1$ choose N_2, and then choose a fixed q greater than both N_1 and N_2. Then both (20) and (21) hold and hence

$$|F(p) - a| < \epsilon, \qquad \text{if } p > N_1.$$

Therefore, $\lim_{p\to\infty} F(p) = a$.

NOTE A similar result holds if we interchange the roles of p and q

Thus the existence of the double limit $\lim_{p,q\to\infty} f(p, q)$ and of $\lim_{q\to\infty} f(p, q)$ implies the existence of the iterated limit

$$\lim_{p\to\infty} (\lim_{q\to\infty} f(p, q))$$

The following example shows that the converse is not true. Let

$$f(p, q) = \frac{pq}{p^2 + q^2}, \qquad (p = 1, 2, \ldots, q = 1, 2, \ldots)$$

Then $\lim_{q\to\infty} f(p, q) = 0$ and hence $\lim_{p\to\infty} (\lim_{q\to\infty} f(p, q)) = 0$ But $f(p, q) = \frac{1}{2}$ when $p = q$ and $f(p, q) = \frac{2}{5}$ when $p = 2q$, and hence it is clear that the double limit cannot exist in this case

A suitable converse of Theorem 12-39 can be established by introducing the notion of *uniform convergence* (This will be done in the next chapter. See Theorem 13-2)

Further examples illustrating the behavior of double sequences are given in Exercise 12-28.

12-15 Double series.

12-40 DEFINITION *Let f be a double sequence The double sequence s defined by the equation*

$$s(p, q) = \sum_{m=1}^{p} \sum_{n=1}^{q} f(m, n)$$

is called a double series and is denoted by the symbol $\sum_{m,n} f(m, n)$ or, more briefly, by $\sum f(m, n)$ The double series is said to converge to the sum a if

$$\lim_{p,q\to\infty} s(p, q) = a$$

Each number $f(m, n)$ is called a *term* of the double series and each $s(p, q)$ is a partial sum If $\sum f(m, n)$ has only positive terms, it is easy to show that it converges if, and only if, the set of partial sums is bounded (See Exercise 12-29) We say $\sum f(m, n)$ converges *absolutely* if $\sum |f(m, n)|$ converges Theorem 12-18 is valid for double series (See Exercise 12-29.)

12-41 DEFINITION *Let f be a double sequence and let g be a one-to-one function defined on P with range $P \times P$ Let G be the sequence defined by $G(n) = f[g(n)]$ if $n \in P$. Then g is said to be an arrangement of the double sequence f into the sequence G*

12–42 THEOREM. *Let $\sum f(m, n)$ be a given double series and let g be an arrangement of the double sequence f into a sequence G. Then*

(a) *$\sum G(n)$ converges absolutely if, and only if, $\sum f(m, n)$ converges absolutely. Assuming that $\sum f(m, n)$ does converge absolutely, with sum S, we have further:*

(b) *$\sum_{n=1}^{\infty} G(n) = S$.*

(c) *$\sum_{n=1}^{\infty} f(m, n)$ and $\sum_{m=1}^{\infty} f(m, n)$ both converge absolutely.*

(d) *If $A_m = \sum_{n=1}^{\infty} f(m, n)$ and $B_n = \sum_{m=1}^{\infty} f(m, n)$, both series $\sum A_m$ and $\sum B_n$ converge absolutely and both have sum S. That is,*

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} f(m, n) = S.$$

Proof. Let $T_k = |G(1)| + \cdots + |G(k)|$ and let

$$S(p, q) = \sum_{m=1}^{p} \sum_{n=1}^{q} |f(m, n)|.$$

Then, for each k, there exists a pair (p, q) such that $T_k \le S(p, q)$ and, conversely, for each pair (p, q) there exists an integer r such that $S(p, q) \le T_r$. These inequalities tell us that $\sum |G(n)|$ has bounded partial sums if, and only if, $\sum |f(m, n)|$ has bounded partial sums. This proves (a).

Now assume that $\sum |f(m, n)|$ converges. Before we prove (b), we will show that the sum of the series $\sum G(n)$ is independent of the function g used to construct G from f. To see this, let h be another arrangement of the double sequence f into a sequence H. Then we have

$$G(n) = f[g(n)] \qquad \text{and} \qquad H(n) = f[h(n)].$$

But this means that $G(n) = H[k(n)]$, where $k(n) = h^{-1}[g(n)]$. Since k is a one-to-one mapping of P onto P, the series $\sum H(n)$ is a rearrangement of $\sum G(n)$, and hence has the same sum. Let us denote this common sum by S'. We will show later that $S' = S$.

Now observe that each series in (c) is a subseries of $\sum G(n)$. Hence (c) follows from (a). Applying Theorem 12–36, we conclude that $\sum A_m$ converges absolutely and has sum S'. The same thing is true of $\sum B_n$. It remains to prove that $S' = S$.

For this purpose let $T = \lim_{p,q\to\infty} S(p, q)$. Given $\epsilon > 0$, choose N so that $0 \le T - S(p, q) < \epsilon/2$ whenever $p > N$ and $q > N$. Now write

$$t_k = \sum_{n=1}^{k} G(n), \qquad s(p, q) = \sum_{m=1}^{p} \sum_{n=1}^{q} f(m, n).$$

Choose M so that t_M includes all terms $f(m, n)$ with $1 \leq m \leq N + 1$, $1 \leq n \leq N + 1$. Then $t_M - s(N + 1, N + 1)$ is a sum of terms $f(m, n)$ with either $m > N$ or $n > N$. Therefore, if $n \geq M$, we have

$$|t_n - s(N + 1, N + 1)| \leq T - S(N + 1, N + 1) < \frac{\epsilon}{2}$$

Similarly,

$$|S - s(N + 1, N + 1)| \leq T - S(N + 1, N + 1) < \frac{\epsilon}{2}$$

Thus, given $\epsilon > 0$, we can always find M so that $|t_n - S| < \epsilon$ whenever $n \geq M$. Since $\lim_{n\to\infty} t_n = S'$, it follows that $S' = S$.

NOTE. The series $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n)$ and $\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} f(m, n)$ are called "repeated series." Convergence of both repeated series does not imply their equality. For example, suppose

$$f(m, n) = \begin{cases} 1, & \text{if } m = n + 1, n = 1, 2, \ldots, \\ -1, & \text{if } m = n - 1, n = 1, 2, \ldots, \\ 0, & \text{otherwise} \end{cases}$$

Then

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n) = -1, \qquad \text{but} \qquad \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} f(m, n) = 1$$

An important case in which the two repeated series are equal is described in the next theorem.

12-43 THEOREM. *Suppose $f(m, n) \geq 0$ for all m, n. Assume that $\sum_{n=1}^{\infty} f(m, n)$ converges for each fixed $m = 1, 2, \ldots$, and that $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n)$ converges. Then*

(a) *$\sum_{m=1}^{\infty} f(m, n)$ converges for each $n = 1, 2, \ldots$,*

(b) *$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} f(m, n)$ converges and equals $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n)$.*

Proof. For each fixed n, the series in (a) is a subseries of

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n)$$

and hence converges. This proves (a). To prove (b), let $A_n = \sum_{m=1}^{\infty} f(m, n)$. Then

$$\sum_{n=1}^{q} A_n = \sum_{n=1}^{q} \sum_{m=1}^{\infty} f(m, n) = \sum_{m=1}^{\infty} \sum_{n=1}^{q} f(m, n) \leq \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n) = S,$$

let us say. Hence $\sum A_n$ converges and has a sum $S' \le S$. But a similar argument proves $S \le S'$. Therefore, $S = S'$ and (b) holds.

NOTE. In the proof we made essential use of the fact that the terms were non-negative. A more general theorem for arbitrary complex terms will be proved later. (See Theorem 13–9.)

12–44 THEOREM. *Let $\sum a_m$ and $\sum b_n$ be two absolutely convergent series with sums A and B, respectively. Let f be the double sequence defined by the equation*

$$f(m, n) = a_m b_n, \qquad \text{if } (m, n) \in P \times P.$$

Then $\sum_{m,n} f(m, n)$ converges absolutely and has the sum AB.

Proof. Let g be any arrangement of the double sequence f into a sequence G, and let $T_k = |G(1)| + \cdots + |G(k)|$. Each term in this partial sum is of the form $|a_m|\,|b_n|$. If p denotes the largest index m, and q the largest n, such that $|a_m|\,|b_n|$ appears in the partial sum T_k, then we certainly have

$$T_k \le \sum_{m=1}^{p} |a_m| \sum_{n=1}^{q} |b_n| \le \sum_{m=1}^{\infty} |a_m| \sum_{n=1}^{\infty} |b_n|.$$

Hence $\sum |G(n)|$ converges, since its partial sums are bounded. From part (a) of Theorem 12–42, we conclude that $\sum f(m, n)$ also converges absolutely.

To show that $\sum f(m, n)$ has sum AB, let us consider the particular arrangement g defined as follows: $g(1) = (1, 1)$; having defined $g(1)$, $g(2), \ldots, g(n^2)$, let

$$g(n^2 + k) = (n + 1, k) \qquad \text{if } 1 \le k \le n + 1,$$
$$g(n^2 + k) = (2n + 2 - k, n + 1) \qquad \text{if } n + 2 \le k \le 2n + 1.$$

The function g so defined maps P in a one-to-one fashion onto $P \times P$. Let $G(n) = f[g(n)]$ and let $t_n = G(1) + \cdots + G(n)$. Now g has been constructed so that t_{n^2} is the sum of exactly those n^2 terms $a_p b_q$ for which $p = 1, 2, \ldots, n$ and $q = 1, 2, \ldots, n$. Hence

$$t_{n^2} = \sum_{p=1}^{n} a_p \sum_{q=1}^{n} b_q.$$

This proves that $\lim_{n\to\infty} t_{n^2} = AB$ and hence $\lim_{n\to\infty} t_n = AB$. That is, for this particular arrangement we have $\sum G(n) = AB$. It follows, by Theorem 12–42, that $\sum f(m, n)$ also has sum AB.

12-16 Multiplication of series. Given two series $\sum a_n$ and $\sum b_n$, we can always form the double series $\sum f(m, n)$, where $f(m, n) = a_m b_n$. For every arrangement g of f into a sequence G, we are led to a further series $\sum G(n)$. By analogy with finite sums, it seems natural to refer to $\sum f(m, n)$ or to $\sum G(n)$ as the "product" of $\sum a_n$ and $\sum b_n$, and Theorem 12-44 justifies this terminology when the two given series $\sum a_n$ and $\sum b_n$ are absolutely convergent. However, if either $\sum a_n$ or $\sum b_n$ is *conditionally* convergent, we have no guarantee that either $\sum f(m, n)$ or $\sum G(n)$ will converge. Moreover, if one of them does converge, its sum need not be AB. The convergence and the sum will depend on the arrangement g. Different choices of g may yield different values of the product. There is one very important case in which the terms $f(m, n)$ are arranged "diagonally" to produce $\sum G(n)$, and then parentheses are inserted by grouping together those terms $a_m b_n$ for which $m + n$ has a fixed value. This product is called the *Cauchy product* and is defined as follows.

12-45 DEFINITION. *Given two series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$, define*

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}, \qquad \text{if } n = 0, 1, 2, \ldots. \tag{22}$$

The series $\sum_{n=0}^{\infty} c_n$ is called the Cauchy product of $\sum a_n$ and $\sum b_n$.

NOTE. The Cauchy product arises in a natural way when we multiply two power series. (See Exercise 12-33.)

Because of Theorems 12-44 and 12-13, absolute convergence of *both* $\sum a_n$ and $\sum b_n$ implies convergence of the Cauchy product to the value

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right). \tag{23}$$

This equation may fail to hold if both $\sum a_n$ and $\sum b_n$ are *conditionally* convergent. (See Exercise 12-32.) However, we can prove that (23) is valid if at least *one* of $\sum a_n$, $\sum b_n$ is absolutely convergent.

12-46 THEOREM (*Mertens*). *Assume that $\sum_{n=0}^{\infty} a_n$ converges absolutely and has sum A, and suppose $\sum_{n=0}^{\infty} b_n$ converges with sum B. Then the Cauchy product of these two series converges and has sum AB.*

Proof. Define $A_n = \sum_{k=0}^{n} a_k$, $B_n = \sum_{k=0}^{n} b_k$, $C_n = \sum_{k=0}^{n} c_k$, where c_k is given by (22). Let $d_n = B - B_n$ and $e_n = \sum_{k=0}^{n} a_k d_{n-k}$. Then

$$C_p = \sum_{n=0}^{p} \sum_{k=0}^{n} a_k b_{n-k} = \sum_{n=0}^{p} \sum_{k=0}^{p} f_n(k), \tag{24}$$

where

$$f_n(k) = \begin{cases} a_k b_{n-k}, & \text{if } n \geq k, \\ 0, & \text{if } n < k. \end{cases}$$

Then (24) becomes

$$C_p = \sum_{k=0}^{p} \sum_{n=0}^{p} f_n(k) = \sum_{k=0}^{p} \sum_{n=k}^{p} a_k b_{n-k} = \sum_{k=0}^{p} a_k \sum_{m=0}^{p-k} b_m = \sum_{k=0}^{p} a_k B_{p-k}$$

$$= \sum_{k=0}^{p} a_k(B - d_{p-k}) = A_p B - e_p.$$

To complete the proof, it suffices to show that $e_p \to 0$ as $p \to \infty$. The sequence $\{d_n\}$ converges to 0, since $B = \sum b_n$. Choose $M > 0$ so that $|d_n| \leq M$ for all n, and let $K = \sum_{n=0}^{\infty} |a_n|$. Given $\epsilon > 0$, choose N so that $n > N$ implies $|d_n| < \epsilon/(2K)$ and also so that

$$\sum_{n=N+1}^{\infty} |a_n| < \frac{\epsilon}{2M} \cdot$$

Then, for $p > 2N$, we can write

$$|e_p| \leq \sum_{k=0}^{N} |a_k d_{p-k}| + \sum_{k=N+1}^{p} |a_k d_{p-k}|$$

$$\leq \frac{\epsilon}{2K} \sum_{k=0}^{N} |a_k| + M \sum_{k=N+1}^{p} |a_k|$$

$$\leq \frac{\epsilon}{2K} \sum_{k=0}^{\infty} |a_k| + M \sum_{k=N+1}^{\infty} |a_k| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

This proves that $e_p \to 0$ as $p \to \infty$, and hence $C_p \to AB$ as $p \to \infty$.

A related theorem (due to Abel), in which no absolute convergence is assumed, will be proved in the next chapter. (See Theorem 13–34.)

Another product, known as the *Dirichlet product*, is of particular importance in the Theory of Numbers. We take $a_0 = b_0 = 0$ and, instead of defining c_n by (22), we use the formula

$$c_n = \sum_{d|n} a_d b_{n/d}, \qquad (n = 1, 2, \ldots), \tag{25}$$

where $\sum_{d|n}$ means a sum extended over all the *divisors* of n (including 1 and n) For example, $c_6 = a_1b_6 + a_2b_3 + a_3b_2 + a_6b_1$, and $c_7 = a_1b_7 + a_7b_1$ The analog of Mertens' theorem holds also for this product The Dirichlet product arises in a natural way when we multiply Dirichlet series (See Exercise 12-34)

12-17 Cesàro summability.

12-47 Definition. *Let s_n denote the* nth *partial sum of the series* $\sum a_n$, *and let* $\{\sigma_n\}$ *be the sequence of arithmetic means defined by*

$$\sigma_n = \frac{s_1 + \cdots + s_n}{n}, \qquad \text{if } n = 1, 2, . \tag{26}$$

The series $\sum a_n$ *is said to be Cesàro summable (or* (C, 1) *summable) if* $\{\sigma_n\}$ *converges If* $\lim_{n\to\infty} \sigma_n = S$, *then* S *is called the Cesàro sum (or* (C, 1) *sum) of* $\sum a_n$, *and we write*

$$\sum a_n = S \qquad (C, 1)$$

Examples

1 Let $a_n = z^n$, $|z| = 1$, $z \neq 1$ Then

$$s_n = \frac{1}{1-z} - \frac{z^n}{1-z}$$

and

$$\sigma_n = \frac{1}{1-z} - \frac{1}{n}\frac{z(1-z^n)}{(1-z)^2}$$

Therefore,

$$\sum_{n=1}^{\infty} z^{n-1} = \frac{1}{1-z} \qquad (C, 1)$$

In particular,

$$\sum_{n=1}^{\infty} (-1)^{n-1} = \tfrac{1}{2} \qquad (C, 1)$$

2. Let $a_n = (-1)^{n+1}n$. In this case,

$$\limsup_{n\to\infty} \sigma_n = \tfrac{1}{2}, \qquad \liminf_{n\to\infty} \sigma_n = 0,$$

and hence $\sum(-1)^{n+1} n$ is not (C, 1) summable

12-48 Theorem *If a series is convergent with sum* S, *then it is also* (C, 1) *summable with Cesàro sum* S.

Proof. Let s_n denote the nth partial sum of the series, define σ_n by (26), and introduce $t_n = s_n - S$, $\tau_n = \sigma_n - S$. Then we have

$$\tau_n = \frac{t_1 + \cdots + t_n}{n} \tag{27}$$

and we must prove that $\tau_n \to 0$ as $n \to \infty$. Choose $A > 0$ so that each $|t_n| \le A$. Given $\epsilon > 0$, choose N so that $n > N$ implies $|t_n| < \epsilon$. Taking $n > N$ in (27), we obtain

$$|\tau_n| \le \frac{|t_1| + \cdots + |t_N|}{n} + \frac{|t_{N+1}| + \cdots + |t_n|}{n} < \frac{NA}{n} + \epsilon.$$

Hence, $\limsup_{n\to\infty} |\tau_n| \le \epsilon$. Since ϵ is arbitrary, it follows that $\lim_{n\to\infty} |\tau_n| = 0$.

NOTE. We have really proved that if a sequence $\{s_n\}$ converges, then the sequence $\{\sigma_n\}$ of arithmetic means also converges and, in fact, to the same limit.

Cesàro summability is just one of a large class of "summability methods" which can be used to assign a "sum" to an infinite series. Theorem 12–48 and Example 1 (following Definition 12–47) show us that Cesàro's method has a wider scope than ordinary convergence. The theory of summability methods is an important and fascinating subject, but one which we cannot enter into here. For an excellent account of the subject the reader is referred to Hardy's *Divergent Series* (Ref. 12–1). We shall see later that (C, 1) summability plays an important role in the theory of Fourier series. (See Theorem 15–21.)

12–18 Infinite products. In this section we give a brief introduction to the theory of infinite products.

12–49 DEFINITION. *Given a sequence $\{u_n\}$ of real or complex numbers, let*

$$p_1 = u_1, \qquad p_2 = u_1u_2, \qquad p_n = u_1u_2\cdots u_n = \prod_{k=1}^{n} u_k. \tag{28}$$

A sequence $\{p_n\}$ formed in this way is called an infinite product (or simply, a product). The number p_n is called the nth *partial product and u_n is called the* nth *factor of the product. The following symbols are used to denote the product defined by* (28):

$$u_1u_2\cdots u_n\cdots, \qquad \prod_{n=1}^{\infty} u_n. \tag{29}$$

NOTE. The symbol $\prod_{n=N+1}^{\infty} u_n$ means $\prod_{n=1}^{\infty} u_{N+n}$. We also write $\prod u_n$ when there is no danger of misunderstanding.

By analogy with infinite series, it would seem natural to call the product (29) convergent if $\{p_n\}$ converges However, this definition would be inconvenient since every product having one factor equal to zero would converge, regardless of the behavior of the remaining factors The following definition turns out to be more useful.

12-50 DEFINITION. *Given an infinite product* $\prod_{n=1}^{\infty} u_n$, *let* $p_n = \prod_{k=1}^{n} u_k$

(a) *If infinitely many factors* u_n *are zero, we say the product diverges to zero.*

(b) *If no factor* u_n *is zero, we say the product converges if there exists a number* $p \neq 0$ *such that* $\{p_n\}$ *converges to* p *In this case,* p *is called the value of the product and we write* $p = \prod_{n=1}^{\infty} u_n$ *If* $\{p_n\}$ *converges to zero, we say the product diverges to zero*

(c) *If there exists an* N *such that* $n > N$ *implies* $u_n \neq 0$, *we say* $\prod_{n=1}^{\infty} u_n$ *converges, provided that* $\prod_{n=N+1}^{\infty} u_n$ *converges as described in* (b) *In this case, the value of the product* $\prod_{n=1}^{\infty} u_n$ *is*

$$u_1u_2 \cdots u_N \prod_{n=N+1}^{\infty} u_n$$

(d) $\prod_{n=1}^{\infty} u_n$ *is called divergent if it does not converge as described in* (b) *or* (c).

Note that the value of a convergent infinite product can be zero But this happens if, and only if, a finite number of factors are zero The convergence of an infinite product is not affected by inserting or removing a finite number of factors, zero or not It is this fact which makes Definition 12-50 very convenient.

EXAMPLE $\prod_{n=1}^{\infty} (1 + 1/n)$ and $\prod_{n=2}^{\infty} (1 - 1/n)$ are both divergent In the first case, $p_n = n + 1$, and in the second case, $p_n = 1/n$

12-51 THEOREM (*Cauchy condition for products*) *The infinite product* $\prod u_n$ *converges if, and only if, for every* $\epsilon > 0$ *there exists an* N *such that* $n > N$ *implies*

$$(30) \qquad |u_{n+1}u_{n+2} \cdots u_{n+k} - 1| < \epsilon, \qquad \textit{for } k = 1, 2, 3, \ldots$$

Proof. Assume that the product $\prod u_n$ converges We can assume that no u_n is zero (discarding a few terms if necessary). Let $p_n = u_1 \cdots u_n$ and $p = \lim_{n\to\infty} p_n$. Then $p \neq 0$ and hence there exists an $M > 0$ such that $|p_n| > M$. Now $\{p_n\}$ satisfies the Cauchy condition for sequences. Hence, given $\epsilon > 0$, there is an N such that $n > N$ implies $|p_{n+k} - p_n| < \epsilon M$ for $k = 1, 2, \ldots$. Dividing through by $|p_n|$, we obtain (30).

Now assume that condition (30) holds. Then $n > N$ implies $u_n \neq 0$.

[Why?] Take $\epsilon = \frac{1}{2}$ in (30), let N_0 be the corresponding N, and let $q_n = u_{N_0+1}u_{N_0+2}\cdots u_n$ if $n > N_0$. Then (30) implies $\frac{1}{2} < |q_n| < \frac{3}{2}$. Therefore, if $\{q_n\}$ converges, it cannot converge to zero. To show that $\{q_n\}$ *does* converge, let $\epsilon > 0$ be arbitrary and write (30) as follows:

$$\left|\frac{q_{n+k}}{q_n} - 1\right| < \epsilon.$$

This gives us $|q_{n+k} - q_n| < \epsilon|q_n| < \frac{3}{2}\epsilon$. Therefore, $\{q_n\}$ satisfies the Cauchy condition for sequences and hence is convergent. This means that the product $\prod u_n$ converges.

NOTE. Taking $k = 1$ in (30), we find that convergence of $\prod u_n$ implies $\lim_{n\to\infty} u_n = 1$. For this reason, the factors of a product are written as $u_n = 1 + a_n$. Thus convergence of $\prod(1 + a_n)$ implies $\lim_{n\to\infty} a_n = 0$.

12–52 THEOREM. *Assume that each $a_n > 0$. Then the product $\prod(1 + a_n)$ converges if, and only if, the series $\sum a_n$ converges.*

Proof. Part of the proof is based on the following inequality:

$$1 + x \le e^x. \tag{31}$$

Although (31) holds for all real x, we need it only for $x \ge 0$. When $x > 0$, (31) is a simple consequence of the Mean Value Theorem, which gives us

$$e^x - 1 = xe^{x_0}, \qquad \text{where } 0 < x_0 < x.$$

Since $e^{x_0} \ge 1$, (31) follows at once from this equation.

Now let $s_n = a_1 + a_2 + \cdots + a_n$, $p_n = (1 + a_1)(1 + a_2)\cdots(1 + a_n)$. Both sequences $\{s_n\}$ and $\{p_n\}$ are increasing, and hence to prove the theorem we need only show that $\{s_n\}$ is bounded if, and only if, $\{p_n\}$ is bounded.

First, the inequality $p_n > s_n$ is obvious. Next, taking $x = a_k$ in (31), $k = 1, 2, \ldots, n$, and multiplying, we find $p_n < e^{s_n}$. Hence, $\{s_n\}$ is bounded if, and only if, $\{p_n\}$ is bounded. Note that $\{p_n\}$ cannot converge to zero since each $p_n \ge 1$. Note also that $p_n \to +\infty$ if $s_n \to +\infty$.

12–53 DEFINITION. *The product $\prod(1 + a_n)$ is said to converge absolutely if $\prod(1 + |a_n|)$ converges.*

12–54 THEOREM. *Absolute convergence of $\prod(1 + a_n)$ implies convergence.*

Proof. Use the Cauchy condition along with the inequality

$$\begin{aligned}&|(1 + a_{n+1})(1 + a_{n+2})\cdots(1 + a_{n+k}) - 1| \\ &\qquad\le (1 + |a_{n+1}|)(1 + |a_{n+2}|)\cdots(1 + |a_{n+k}|) - 1.\end{aligned}$$

NOTE Theorem 12-52 tells us that $\prod(1 + a_n)$ converges absolutely if, and only if, $\sum a_n$ converges absolutely In Exercise 12-43 we give an example in which $\prod(1 + a_n)$ converges but $\sum a_n$ diverges

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A result analogous to Theorem 12-52 is the following

12-55 THEOREM *Assume that each $a_n \geq 0$ Then the product $\prod(1 - a_n)$ converges if, and only if, the series $\sum a_n$ converges*

Proof. Convergence of $\sum a_n$ implies absolute convergence (and hence convergence) of $\prod(1 - a_n)$

To prove the converse, assume that $\sum a_n$ diverges If $\{a_n\}$ does not converge to zero, then $\prod(1 - a_n)$ also diverges Therefore we can assume that $a_n \to 0$ as $n \to \infty$. Discarding a few terms if necessary, we can assume that each $a_n \leq \frac{1}{2}$ Then each factor $1 - a_n \geq \frac{1}{2}$ (and hence $\neq 0$) Let $p_n = (1 - a_1)(1 - a_2) \cdots (1 - a_n)$, $q_n = (1 + a_1)(1 + a_2) \cdots (1 + a_n)$ Since we have

$$(1 - a_k)(1 + a_k) = 1 - a_k^2 \leq 1,$$

we can write $p_n \leq 1/q_n$. But in the proof of Theorem 12-52 we observed that $q_n \to +\infty$ if $\sum a_n$ diverges Therefore, $p_n \to 0$ as $n \to \infty$ and, by part (b) of Definition 12-50, it follows that $\prod(1 - a_n)$ diverges to 0

Exercises

Sequences.

12–1. Given a real-valued sequence $\{a_n\}$, let $U = \limsup_{n\to\infty} a_n$, $V = \liminf_{n\to\infty} a_n$. If U and V are finite, show that:

(a) There exists a subsequence of $\{a_n\}$ which converges to U and a subsequence which converges to V.

(b) If $U = V$, every subsequence of $\{a_n\}$ converges to U.

12–2. Given two real-valued sequences $\{a_n\}$ and $\{b_n\}$. Prove that

(a) $\limsup_{n\to\infty} (a_n + b_n) \le \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n$.

(b) $\limsup_{n\to\infty} (a_n b_n) \le (\limsup_{n\to\infty} a_n)(\limsup_{n\to\infty} b_n)$ if $a_n > 0$, $b_n > 0$ for all n.

12–3. Prove Theorems 12–3 and 12–4.

12–4. If each $a_n > 0$, prove that

$$\liminf_{n\to\infty} \frac{a_{n+1}}{a_n} \le \liminf_{n\to\infty} \sqrt[n]{a_n} \le \limsup_{n\to\infty} \sqrt[n]{a_n} \le \limsup_{n\to\infty} \frac{a_{n+1}}{a_n}.$$

12–5. Let $\{a_n\}$ be a real-valued sequence and let $\sigma_n = (a_1 + \cdots + a_n)/n$. Show that

$$\liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} \sigma_n \le \limsup_{n\to\infty} \sigma_n \le \limsup_{n\to\infty} a_n.$$

12–6. Find $\limsup_{n\to\infty} a_n$ and $\liminf_{n\to\infty} a_n$ if a_n is given by

(a) $\cos n$, (b) $\left(1 + \frac{1}{n}\right)\cos n\pi$, (c) $n \sin \frac{n\pi}{3}$,

(d) $\sin \frac{n\pi}{2} \cos \frac{n\pi}{2}$, (e) $(-1)^n n/(1+n)^n$, (f) $\frac{n}{3} - \left[\frac{n}{3}\right]$.

Note. In (f), $[x]$ means the greatest integer $\le x$.

12–7. If $a_1 > 0$ and $a_{n+1} = a_1^{a_n}$ for $n = 1, 2, 3, \ldots$, show that $\{a_n\}$ converges if, and only if, $e^{-e} \le a_1 \le e^{1/e}$.

12–8. Let $a_n = n^n/n!$. Show that $\lim_{n\to\infty} a_{n+1}/a_n = e$ and use Exercise 12–4 to deduce that

$$\lim_{n\to\infty} \frac{n}{(n!)^{1/n}} = e.$$

In Exercises 12–9 through 12–14, show that the real-valued sequence $\{a_n\}$ is convergent. The given conditions are assumed to hold for all $n \ge 1$. In Exercises 12–10 through 12–14, show that $\{a_n\}$ has the limit L indicated.

12-9. $|a_n| \le 2, |a_{n+2} - a_{n+1}| \le \frac{1}{8}|a_{n+1}^2 - a_n^2|$

12-10. $a_1 \ge 0, a_2 \ge 0, a_{n+2} = (a_n a_{n+1})^{1/2}, L = (a_1 a_2^2)^{1/3}.$

12-11. $a_1 = 2, a_2 = 8, a_{2n+1} = \frac{1}{2}(a_{2n} + a_{2n-1}), a_{2n+2} = \dfrac{a_{2n}a_{2n-1}}{a_{2n+1}},$

$L = 4$

12-12. $a_1 = -\frac{3}{2},\quad 3a_{n+1} = 2 + a_n^3,\quad L = 1$ Modify a_1 to make $L = -2$

12-13. $a_1 = 3,\quad a_{n+1} = \dfrac{3(1 + a_n)}{3 + a_n},\quad L = \sqrt{3}.$

12-14. $a_n = \dfrac{b_{n+1}}{b_n}$, where $b_1 = b_2 = 1, b_{n+2} = b_n + b_{n+1}, L = \dfrac{1 + \sqrt{5}}{2}$

Hint Show that $b_{n+2}b_n - b_{n+1}^2 = (-1)^{n+1}$ and deduce that

$$|a_n - a_{n+1}| < n^{-2}, \qquad \text{if } n > 4$$

Series

12-15. Test for convergence (p and q denote fixed real numbers)

(a) $\displaystyle\sum_{n=1}^{\infty} n^3 e^{-n}$, (b) $\displaystyle\sum_{n=2}^{\infty} (\log n)^p$,

(c) $\displaystyle\sum_{n=1}^{\infty} p^n n^p \quad (p > 0)$, (d) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n^p - n^q} \quad (0 < q < p)$,

(e) $\displaystyle\sum_{n=1}^{\infty} n^{-1-1/n}$, (f) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{p^n - q^n} \quad (0 < q < p)$,

(g) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n \log (1 + 1/n)}$, (h) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{(\log n)^{\log n}}$,

(i) $\displaystyle\sum_{n=3}^{\infty} \frac{1}{n \log n (\log \log n)^p}$, (j) $\displaystyle\sum_{n=3}^{\infty} \left(\frac{1}{\log \log n}\right)^{\log \log n}$,

(k) $\displaystyle\sum_{n=1}^{\infty} (\sqrt{1 + n^2} - n)$, (l) $\displaystyle\sum_{n=2}^{\infty} n^p \left(\frac{1}{\sqrt{n-1}} - \frac{1}{\sqrt{n}}\right)$,

(m) $\displaystyle\sum_{n=1}^{\infty} (\sqrt[n]{n} - 1)^n$, (n) $\displaystyle\sum_{n=1}^{\infty} n^p(\sqrt{n+1} - 2\sqrt{n} + \sqrt{n-1})$.

12-16. Let $S = \{n_1, n_2, \ldots\}$ denote the collection of those positive integers that do not involve the digit 0 in their decimal representation (For example, $7 \in S$ but $101 \notin S$) Show that $\sum_{k=1}^{\infty} 1/n_k$ converges and has a sum less than 90.

12–17. Given integers $a_1, a_2, \dots$ such that $1 \le a_n \le n - 1$, $n = 2, 3, \dots$. Show that the sum of the series $\sum_{n=1}^{\infty} a_n/n!$ is rational if, and only if, there exists an integer N such that $a_n = n - 1$ for all $n \ge N$.

[*Hint*: For the sufficiency, show that $\sum_{n=2}^{\infty} (n-1)/n!$ is a telescoping series with sum 1.]

12–18. Let p and q be fixed integers, $p \ge q \ge 1$, and let

$$x_n = \sum_{k=qn+1}^{pn} \frac{1}{k}, \qquad s_n = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k}.$$

(a) Use formula (8) to prove that $\lim_{n\to\infty} x_n = \log (p/q)$.
(b) When $q = 1$, $p = 2$, show that $s_{2n} = x_n$ and deduce that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \log 2.$$

(c) Rearrange the series in (b), writing alternately p positive terms followed by q negative terms and use (a) to show that this rearrangement has sum $\log 2 + \frac{1}{2}\log (p/q)$.
(d) Find the sum of $\sum_{n=1}^{\infty} (-1)^{n+1}(1/(3n-2) - 1/(3n-1))$.

12–19. Let $c_n = a_n + ib_n$, where $a_n = (-1)^n/\sqrt{n}$, $b_n = 1/n^2$. Show that $\sum c_n$ is conditionally convergent.

12–20. Use Theorem 12–23 to derive the following formulas:

(a) $$\sum_{k=1}^{n} \frac{\log k}{k} = \frac{1}{2}\log^2 n + A + O\left(\frac{\log n}{n}\right) \qquad (A \text{ is constant}).$$

(b) $$\sum_{k=2}^{n} \frac{1}{k \log k} = \log(\log n) + B + O\left(\frac{1}{n \log n}\right) \qquad (B \text{ is constant}).$$

12–21. If $0 < a \le 1$, $s > 1$, define $\zeta(s, a) = \sum_{n=0}^{\infty} (n + a)^{-s}$.

(a) Show that this series converges absolutely for $s > 1$ and prove that

$$\sum_{h=1}^{k} \zeta\left(s, \frac{h}{k}\right) = k^s\zeta(s) \qquad \text{if } k = 1, 2, \dots,$$

where $\zeta(s) = \zeta(s, 1)$ is the Riemann zeta function.
(b) Prove that $\sum_{n=1}^{\infty} (-1)^{n-1}/n^s = (1 - 2^{1-s})\zeta(s)$ if $s > 1$.

12–22. Given a convergent series $\sum a_n$, where each $a_n \ge 0$. Prove that $\sum \sqrt{a_n}\, n^{-p}$ converges if $p > \frac{1}{2}$. Give a counter example for $p = \frac{1}{2}$.

12-23. Given that $\sum a_n$ diverges Prove that $\sum na_n$ also diverges

12-24. Given that $\sum a_n$ converges, where each $a_n > 0$. Prove that

$$\sum (a_n a_{n+1})^{1/2}$$

also converges Show that the converse is also true if $\{a_n\}$ is monotonic

12-25. Given that $\sum a_n$ converges absolutely Show that each of the following series also converges absolutely

(a) $\sum a_n^2$,

(b) $\sum \dfrac{a_n}{1 + a_n}$ (if no $a_n = -1$),

(c) $\sum \dfrac{a_n^2}{1 + a_n^2}$

12-26. Determine all real values of x for which the following series converges

$$\sum_{n=1}^{\infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right) \frac{\sin nx}{n}$$

12-27. Prove the following statements

(a) $\sum a_n b_n$ converges if $\sum a_n$ converges and if $\sum (b_n - b_{n+1})$ converges absolutely

(b) $\sum a_n b_n$ converges if $\sum a_n$ has bounded partial sums and if $\sum (b_n - b_{n+1})$ converges absolutely, provided that $b_n \to 0$ as $n \to \infty$.

Double sequences and double series.

12-28. Investigate the existence of the two iterated limits and the double limit of the double sequence f defined by

(a) $f(p, q) = \dfrac{1}{p + q}$,

(b) $f(p, q) = \dfrac{p}{p + q}$,

(c) $f(p, q) = \dfrac{(-1)^p p}{p + q}$,

(d) $f(p, q) = (-1)^{p+q}\left(\dfrac{1}{p} + \dfrac{1}{q}\right)$,

(e) $f(p, q) = \dfrac{(-1)^p}{q}$,

(f) $f(p, q) = (-1)^{p+q}$,

(g) $f(p, q) = \dfrac{\cos p}{q}$,

(h) $f(p, q) = \dfrac{p}{q^2} \displaystyle\sum_{n=1}^{q} \sin \frac{n}{p}$

Answer Double limit exists in (a), (d), (e), (g). Both iterated limits exist in (a), (b), (h) Only one iterated limit exists in (c), (e). Neither iterated limit exists in (d), (f).

12–29. Prove the following statements:

(a) A double series of positive terms converges if, and only if, the set of partial sums is bounded.

(b) A double series converges if it converges absolutely.

(c) $\sum_{m,n} e^{-(m^2+n^2)}$ converges.

12–30. Assume that the double series $\sum_{m,n} a(n)x^{mn}$ converges absolutely for $|x| < 1$. Call its sum $S(x)$. Show that each of the following series also converges absolutely for $|x| < 1$ and has sum $S(x)$:

$$\sum_{n=1}^{\infty} a(n) \frac{x^n}{1 - x^n}, \qquad \sum_{n=1}^{\infty} A(n)x^n, \qquad \text{where } A(n) = \sum_{d|n} a(d).$$

12–31. If α is real, show that the double series $\sum_{m,n} (m + in)^{-\alpha}$ converges absolutely if, and only if, $\alpha > 2$.

Hint: Let $s(p, q) = \sum_{m=1}^{p} \sum_{n=1}^{q} |m + in|^{-\alpha}$. The set

$$\{m + in \mid m = 1, 2, \ldots, p,\ n = 1, 2, \ldots, p\}$$

consists of p^2 complex numbers of which one has absolute value $\sqrt{2}$, three satisfy $|1 + 2i| \le |m + in| \le 2\sqrt{2}$, five satisfy $|1 + 3i| \le |m + in| \le 3\sqrt{2}$, etc. Verify this geometrically and deduce the inequality

$$2^{-\alpha/2} \sum_{n=1}^{p} \frac{2n - 1}{n^\alpha} \le s(p, p) \le \sum_{n=1}^{p} \frac{2n - 1}{(n^2 + 1)^{\alpha/2}}.$$

12–32. (a) Show that the Cauchy product of $\sum_{n=0}^{\infty} (-1)^{n+1}/\sqrt{n + 1}$ with itself is a divergent series.

(b) Show that the Cauchy product of $\sum_{n=0}^{\infty} (-1)^{n+1}/(n + 1)$ with itself is the series

$$2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n+1} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right).$$

Does this converge? Why?

12–33. Given two absolutely convergent power series, say $\sum_{n=0}^{\infty} a_n x^n$ and $\sum_{n=0}^{\infty} b_n x^n$, having sums $A(x)$ and $B(x)$, respectively, show that $\sum_{n=0}^{\infty} c_n x^n = A(x)B(x)$ where $c_n = \sum_{k=0}^{n} a_k b_{n-k}$.

12–34. A series of the form $\sum_{n=1}^{\infty} a_n/n^s$ is called a *Dirichlet series*. Given two absolutely convergent Dirichlet series, say $\sum_{n=1}^{\infty} a_n/n^s$ and $\sum_{n=1}^{\infty} b_n/n^s$, having sums $A(s)$ and $B(s)$, respectively, show that $\sum_{n=1}^{\infty} c_n/n^s = A(s)B(s)$ where $c_n = \sum_{d|n} a_d b_{n/d}$.

12-35. If $\zeta(s) = \sum_{n=1}^{\infty} 1/n^s$, $s > 1$, show that $\zeta^2(s) = \sum_{n=1}^{\infty} d(n)/n^s$, where $d(n)$ is the number of divisors of n (including 1 and n).

Cesàro summability

12-36. Show that each of the following series has (C, 1) sum 0

(a) $1 - 1 - 1 + 1 + 1 - 1 - 1 + 1 + 1 - - + + \cdots$.

(b) $\frac{1}{2} - 1 + \frac{1}{2} + \frac{1}{2} - 1 + \frac{1}{2} + \frac{1}{2} - 1 + + - \cdots$.

(c) $\cos x + \cos 3x + \cos 5x + \cdots$ (x real, $x \neq m\pi$)

12-37. Given a series $\sum a_n$, let

$$s_n = \sum_{k=1}^{n} a_k, \qquad t_n = \sum_{k=1}^{n} k a_k, \qquad \sigma_n = \frac{1}{n} \sum_{k=1}^{n} s_k$$

Prove that.

(a) $t_n = (n + 1)s_n - n\sigma_n$.

(b) If $\sum a_n$ is (C, 1) summable, then $\sum a_n$ converges if, and only if, $t_n = o(n)$ as $n \to \infty$.

(c) $\sum a_n$ is (C, 1) summable if, and only if, $\sum_{n=1}^{\infty} t_n/n(n + 1)$ converges

12-38. Given a monotonic sequence $\{a_n\}$ of positive terms, such that $\lim_{n\to\infty} a_n = 0$. Let

$$s_n = \sum_{k=1}^{n} a_k, \qquad u_n = \sum_{k=1}^{n} (-1)^k a_k, \qquad v_n = \sum_{k=1}^{n} (-1)^k s_k$$

Prove that·

(a) $v_n = \frac{1}{2}u_n + (-1)^n s_n/2$.

(b) $\sum_{n=1}^{\infty} (-1)^n s_n$ is (C, 1) summable and has Cesàro sum $\frac{1}{2}\sum_{n=1}^{\infty} (-1)^n a_n$.

(c) $\sum_{n=1}^{\infty} (-1)^n(1 + \frac{1}{2} + \cdots + 1/n) = -\log \sqrt{2}$ (C, 1)

Infinite products.

12-39. Determine whether or not the following infinite products converge Find the value of each convergent product

(a) $\displaystyle\prod_{n=2}^{\infty} \left(1 - \frac{2}{n(n+1)}\right)$, (b) $\displaystyle\prod_{n=2}^{\infty} (1 - n^{-2})$,

(c) $\displaystyle\prod_{n=2}^{\infty} \frac{n^3 - 1}{n^3 + 1}$, (d) $\displaystyle\prod_{n=0}^{\infty} (1 + z^{2^n})$ if $|z| < 1$

12-40. If each partial sum s_n of the convergent series $\sum a_n$ is not zero and if the sum itself is not zero, show that the infinite product $a_1 \prod_{n=2}^{\infty} (1 + a_n/s_{n-1})$ converges and has the value $\sum_{n=1}^{\infty} a_n$.

12–41. Find the values of the following products by establishing the following identities and summing the series:

(a) $$\prod_{n=2}^{\infty}\left(1+\frac{1}{2^n-2}\right) = 2\sum_{n=1}^{\infty} 2^{-n}.$$

(b) $$\prod_{n=2}^{\infty}\left(1+\frac{1}{n^2-1}\right) = 2\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

12–42. Determine all real x for which the product $\prod_{n=1}^{\infty} \cos(x/2^n)$ converges and find the value of the product when it does converge.

12–43. (a) Let $a_n = (-1)^n/\sqrt{n}$ for $n = 1, 2, \ldots$. Show that $\prod(1 + a_n)$ diverges but that $\sum a_n$ converges.

(b) Let $a_{2n-1} = -1/\sqrt{n}$, $a_{2n} = 1/\sqrt{n} + 1/n$ for $n = 1, 2, \ldots$. Show that $\prod(1 + a_n)$ converges but that $\sum a_n$ diverges.

12–44. Assume that $a_n \geq 0$ for each $n = 1, 2, \ldots$. Assume further that

$$\frac{a_{2n+2}}{1 + a_{2n+2}} < a_{2n+1} < \frac{a_{2n}}{1 + a_{2n}} \qquad \text{for } n = 1, 2, \ldots.$$

Show that $\prod_{k=1}^{\infty} (1 + (-1)^k a_k)$ converges if, and only if, $\sum_{k=1}^{\infty} (-1)^k a_k$ converges.

12–45. Let p_n denote the nth prime integer. (Thus, $p_1 = 2$, $p_2 = 3$, $p_3 = 5, \ldots$.) If $s > 1$, prove that

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{k=1}^{\infty} \frac{1}{1 - p_k^{-s}}.$$

Hint: Let $P_m = \prod_{k=1}^{m} 1/(1 - p_k^{-s})$. Express each factor $(1 - p_k^{-s})^{-1}$ as a geometric series and show that $0 \leq \sum_{n=1}^{\infty} 1/n^s - P_m \leq \sum_{n=m}^{\infty} 1/n^s$.

Suggested References

12–1. Hardy, G. H., *Divergent Series*. Oxford: University Press, 1949.

12–2. Knopp, K., *Theory and Application of Infinite Series*. R. C. Young, translator. New York: Hafner, 1951.

CHAPTER 13

SEQUENCES OF FUNCTIONS

13–1 Introduction. In this chapter we shall be dealing with complex-valued functions defined on certain subsets of E_2. Some of the results hold only for real-valued functions defined on subsets of E_1, and in such cases this restriction will be explicitly stated.

Given a sequence $\{f_n\}$, each term of which is a function defined on a set S, for each x in S we can form another sequence $\{f_n(x)\}$ whose terms are the corresponding function values. Let T denote the set of those points x in S for which this second sequence converges. The function f defined by the equation

$$f(x) = \lim_{n\to\infty} f_n(x), \qquad \text{if } x \in T,$$

will be referred to as the *limit function* of the sequence $\{f_n\}$ and we will say that $\{f_n\}$ *converges pointwise* on the set T.

Our chief interest in this chapter will be the following type of problem: Given that each term of the sequence $\{f_n\}$ has a certain property (for example, continuity, differentiability, integrability), to what extent does the limit function f also possess this property? As an example, suppose each f_n is continuous at a point x_0 in T. Is the limit function f also continuous at x_0? We shall see that, in general, it is not. In fact, we shall find that pointwise convergence is usually not "strong enough" to transfer any of the properties mentioned above from the individual terms f_n to the limit function f. Therefore we are led to a study of "stronger" methods of convergence that *do* preserve these properties. The most important of these is the notion of *uniform* convergence.

Before we begin a discussion of uniform convergence, let us formulate one of our basic questions in another way. When we ask whether continuity of each f_n at x_0 implies continuity of the limit function f at x_0, we are really asking whether the equation

$$\lim_{x\to x_0} f_n(x) = f_n(x_0)$$

implies the equation

(1) $$\lim_{x\to x_0} f(x) = f(x_0).$$

But (1) can also be written as follows:

(2) $$\lim_{x \to x_0} \lim_{n \to \infty} f_n(x) = \lim_{n \to \infty} \lim_{x \to x_0} f_n(x).$$

Therefore our question about continuity amounts to this: Can we interchange the limit symbols in (2)? We shall see that, in general, we cannot. First of all, the limit in (1) may not exist. Secondly, even if it does exist, it need not be equal to $f(x_0)$. We encountered a similar situation in Chapter 12 in connection with repeated series when we found that $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n)$ is not necessarily equal to $\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} f(m, n)$. (See Note preceding Theorem 12–43.)

The general question of whether we can reverse the order of two limit processes arises again and again in mathematical analysis. We shall find that uniform convergence is a far-reaching *sufficient* condition for the validity of interchanging certain limits, but it does not provide the complete answer to the question. We shall encounter examples in which the order of two limits can be interchanged although the sequence is not uniformly convergent.

13–2 Examples of sequences of real-valued functions. The following examples illustrate some of the possibilities that might arise when we form the limit function of a sequence of real-valued functions.

EXAMPLE 1. Let $f_n(x) = nx(1 - x)^n$ if $x \in E_1$, $n = 1, 2, \ldots$. Then $\lim_{n\to\infty} f_n(x)$ exists if $0 \le x \le 1$. The limit function f has the value 0 for each point in $[0, 1]$. Note that f_n has a local maximum at $x = 1/(n + 1)$ and that $f_n(1/(n + 1)) = (n/(n + 1))^{1+n} \to e^{-1}$ as $n \to \infty$. (See Fig. 13–1.)

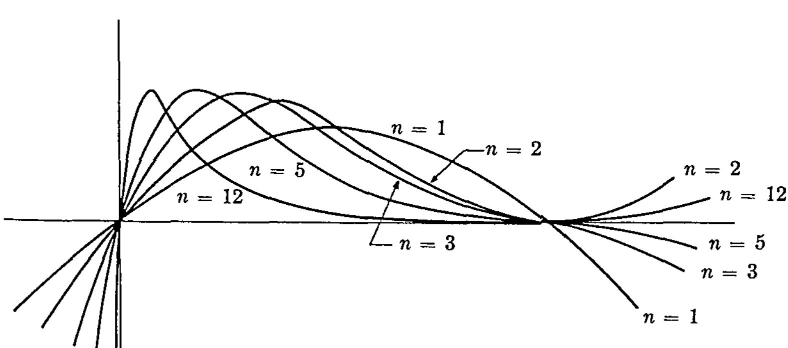


FIG. 13–1. $f_n(x) = nx(1 - x)^n, n = 1, 2, 3, 5, 12.$

EXAMPLE 2. Let $f_n(x) = n^2x(1-x)^n$ if $x \in E_1$, $n = 1, 2, \ldots$ Again, $\lim_{n\to\infty} f_n(x)$ exists if $0 \leq x \leq 1$ and the limit function has the value 0 at each point in $[0, 1]$ As in Example 1, f_n has a local maximum at $x = 1/(n+1)$, but $f_n(1/(n+1)) \to +\infty$ as $n \to \infty$. (See Fig 13-2)

EXAMPLE 3 Let $f_n(x) = x^{2n}/(1+x^{2n})$ ($x \in E_1$, $n = 1, 2, \ldots$) In this case, $\lim_{n\to\infty} f_n(x)$ exists for every x and the limit function f is given by

$$f(x) = \begin{cases} 0, & \text{if } |x| < 1, \\ \frac{1}{2}, & \text{if } |x| = 1, \\ 1, & \text{if } |x| > 1. \end{cases}$$

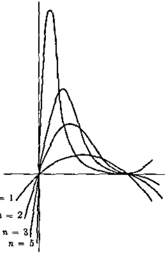
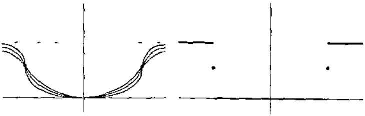


FIG. 13-2. $f_n(x) = n^2x(1-x)^n$, $n = 1, 2, 3, 5$

Each f_n is continuous on E_1, but f is discontinuous at $x = 1$ and $x = -1$. (See Fig. 13-3)

EXAMPLE 4 Let $f_n(x) = \sin nx/\sqrt{n}$ ($x \in E_1$, $n = 1, 2, \ldots$) Then $\lim_{n\to\infty} f_n(x) = 0$ for every x Note that $f_n'(x) = \sqrt{n}\cos nx$. Hence, $\lim_{n\to\infty} f_n'(x)$ does not exist for any x. (See Fig 13-4)



$f_n(x) = \dfrac{x^{2n}}{1+x^{2n}}$ $n = 1, 2, 3$ $\qquad$ $f(x) = \lim_{n\to\infty} f_n(x)$

FIGURE 13-3.

13-3 Definition of uniform convergence. Let $\{f_n\}$ be a sequence of functions which converges pointwise on a set T to a limit function f. Going back to the basic definition of limit, this means that for each point x

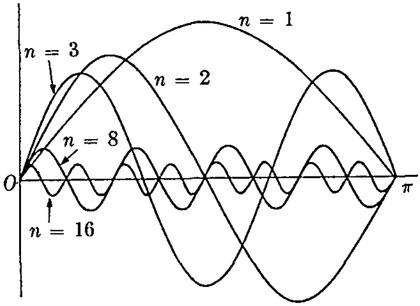


FIG. 13-4. $f_n(x) = \sin nx/\sqrt{n}$, $n = 1, 2, 3, 8, 16$.

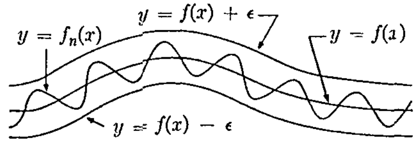


FIG. 13-5. Geometric meaning of uniform convergence.

in T and for each $\epsilon > 0$ there exists an N (depending on both x and ϵ) such that $n > N$ implies $|f_n(x) - f(x)| < \epsilon$. If the same N works equally well for *every* point in T, the convergence is said to be *uniform* on T. That is, we have

13-1 DEFINITION. *A sequence of functions $\{f_n\}$ is said to converge uniformly to f on a set T if, for every $\epsilon > 0$, there exists an N (depending only on ϵ) such that $n > N$ implies*

$$|f_n(x) - f(x)| < \epsilon, \qquad \textit{for every } x \textit{ in } T.$$

We denote this symbolically by writing

$$f_n \to f \textit{ uniformly on } T.$$

When each term of the sequence $\{f_n\}$ is real-valued, there is a useful geometric interpretation of uniform convergence. The inequality $|f_n(x) - f(x)| < \epsilon$ is then equivalent to the *two* inequalities

$$f(x) - \epsilon < f_n(x) < f(x) + \epsilon. \tag{3}$$

If (3) is to hold for all $n > N$ and for all x in T, this means that the entire graph of f_n (that is, the set $\{(x, y) \mid y = f_n(x), x \in T\}$) lies within a "band" of height 2ϵ situated symmetrically about the graph of f. (See Fig. 13-5.)

A sequence $\{f_n\}$ is said to be *uniformly bounded* on T if there exists a constant $M > 0$ such that $|f_n(x)| \leq M$ for all x in T and all $n = 1, 2, \ldots$. The number M is called a *uniform bound* for $\{f_n\}$. If each individual function is bounded and if $f_n \to f$ uniformly on T, then it is easy to prove that

$\{f_n\}$ is uniformly bounded on T (See Exercise 13-1.) This observation often enables us to conclude that a sequence is *not* uniformly convergent. For instance, a glance at Fig. 13-2 tells us at once that the sequence of Example 2 cannot converge uniformly on any subset containing a neighborhood of the origin. However, the convergence in this example *is* uniform on every closed subinterval not containing the origin.

13-4 An application to double sequences. As an application of the concept of uniform convergence, we deduce the following theorem, which can be viewed as a converse of Theorem 12-39.

13-2 THEOREM. *Let f be a double sequence and let P denote the set of positive integers. For each $n = 1, 2, \ldots$, define a function g_n on P as follows:*

$$g_n(m) = f(m, n), \qquad \text{if } m \in P.$$

Assume that $g_n \to g$ uniformly on P, where $g(m) = \lim_{n\to\infty} f(m, n)$. If the iterated limit $\lim_{m\to\infty} (\lim_{n\to\infty} f(m, n))$ exists, then the double limit $\lim_{m,n\to\infty} f(m, n)$ also exists and has the same value.

Proof. Given $\epsilon > 0$, choose N_1 so that $n > N_1$ implies

$$|f(m, n) - g(m)| < \frac{\epsilon}{2}, \qquad \text{for every } m \text{ in } P.$$

Let $a = \lim_{m\to\infty} (\lim_{n\to\infty} f(m, n)) = \lim_{m\to\infty} g(m)$. For the same ϵ, choose N_2 so that $m > N_2$ implies $|g(m) - a| < \epsilon/2$. Then, if N is the larger of N_1 and N_2, we have $|f(m, n) - a| < \epsilon$ whenever both $m > N$ and $n > N$. In other words, $\lim_{m,n\to\infty} f(m, n) = a$.

13-5 Uniform convergence and continuity.

13-3 THEOREM. *Assume that $f_n \to f$ uniformly on T. If each f_n is continuous at a point x_0 of T, then the limit function f is also continuous at x_0.*

NOTE. If x_0 is an accumulation point of T, the conclusion implies that

$$\lim_{x\to x_0} \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \lim_{x\to x_0} f_n(x).$$

Proof. If x_0 is an isolated point of T, then f is automatically continuous at x_0. Suppose, then, that x_0 is an accumulation point of T. By hypothesis, for every $\epsilon > 0$ there is an M such that $n \geq M$ implies

$$|f_n(x) - f(x)| < \frac{\epsilon}{3} \qquad \text{for every } x \text{ in } T.$$

Since f_M is continuous at x_0, there is a neighborhood $N(x_0)$ such that

$x \in N(x_0) \cap T$ implies

$$|f_M(x) - f_M(x_0)| < \frac{\epsilon}{3}.$$

But

$$|f(x) - f(x_0)| \le |f(x) - f_M(x)| + |f_M(x) - f_M(x_0)| + |f_M(x_0) - f(x_0)|.$$

If $x \in N(x_0) \cap T$, each term on the right is less than $\epsilon/3$ and hence $|f(x) - f(x_0)| < \epsilon$. This proves the theorem.

NOTE. Uniform convergence of $\{f_n\}$ is sufficient but not necessary to transmit continuity from the individual terms to the limit function. In Example 2 (Section 13-2), we have a nonuniformly convergent sequence of continuous functions with a continuous limit function.

13-6 The Cauchy condition for uniform convergence.

13-4 THEOREM. *Let $\{f_n\}$ be a sequence of functions defined on a set T. There exists a function f such that $f_n \to f$ uniformly on T if, and only if, the following condition (called the Cauchy condition) is satisfied: For every $\epsilon > 0$ there exists an N such that $m > N$ and $n > N$ implies*

$$|f_m(x) - f_n(x)| < \epsilon, \qquad \textit{for every } x \textit{ in } T.$$

Proof. Assume that $f_n \to f$ uniformly on T. Then, given $\epsilon > 0$, we can find N so that $n > N$ implies $|f_n(x) - f(x)| < \epsilon/2$ for all x in T. Taking $m > N$, we also have $|f_m(x) - f(x)| < \epsilon/2$, and hence $|f_m(x) - f_n(x)| < \epsilon$ for every x in T.

Conversely, suppose the Cauchy condition is satisfied. Then, for each x in T, the sequence $\{f_n(x)\}$ converges. Let $f(x) = \lim_{n\to\infty} f_n(x)$ if $x \in T$. We must show that $f_n \to f$ uniformly on T. If $\epsilon > 0$ is given, we can choose N so that $n > N$ implies $|f_n(x) - f_{n+k}(x)| < \epsilon/2$ for every $k = 1, 2, \ldots$, and every x in T. Therefore, $\lim_{k\to\infty} |f_n(x) - f_{n+k}(x)| = |f_n(x) - f(x)| \le \epsilon/2$. Hence, $n > N$ implies $|f_n(x) - f(x)| < \epsilon$ for every x in T. This proves that $f_n \to f$ uniformly on T.

13-7 Uniform convergence of infinite series.

13-5 DEFINITION. *Given a sequence $\{f_n\}$ of functions defined on a set T. For each x in T, let*

$$s_n(x) = \sum_{k=1}^{n} f_k(x) \qquad (n = 1, 2, \ldots). \tag{4}$$

If there exists a function f such that $s_n \to f$ uniformly on T, we say the series $\sum f_n(x)$ converges uniformly on T and we write

$$\sum_{n=1}^{\infty} f_n(x) = f(x) \qquad (\textit{uniformly on } T)$$

13-6 THEOREM *(Cauchy condition for uniform convergence of series) The series $\sum f_n(x)$ converges uniformly on T if, and only if, for every $\epsilon > 0$ there is an N such that $n > N$ implies*

$$\left| \sum_{k=n+1}^{n+p} f_k(x) \right| < \epsilon, \qquad \textit{for each } p = 1, 2, \ldots, \textit{ and every } x \textit{ in } T$$

Proof Define s_n by (4) and apply Theorem 13-4

13-7 THEOREM *(Weierstrass' M-test) Let $\{M_n\}$ be a sequence of nonnegative numbers such that*

$$0 \leq |f_n(x)| \leq M_n, \qquad \textit{for } n = 1, 2, \ldots, \textit{ and for every } x \textit{ in } T.$$

Then $\sum f_n(x)$ converges uniformly on T if $\sum M_n$ converges

Proof Apply Theorems 12-11 and 13-6 in conjunction with the inequality

$$\left| \sum_{k=n+1}^{n+p} f_k(x) \right| \leq \sum_{k=n+1}^{n+p} M_k$$

13-8 THEOREM *Assume that $\sum f_n(x) = f(x)$ (uniformly on T) If each f_n is continuous at a point x_0 of T, then f is also continuous at x_0.*

Proof Define s_n by (4) Continuity of each f_n at x_0 implies continuity of s_n at x_0, and the conclusion follows at once from Theorem 13-3

NOTE If x_0 is an accumulation point of T, this theorem permits us to write

$$\lim_{x \to x_0} \sum_{n=1}^{\infty} f_n(x) = \sum_{n=1}^{\infty} \lim_{x \to x_0} f_n(x).$$

13-8 A space-filling curve. We can apply Theorem 13-8 to construct an example of what is known as a *space-filling curve* This is a continuous curve in E_2 that passes through every point of the unit square $[0, 1] \times [0, 1]$ Peano (1890) was the first to give an example of such a curve The example to be presented here is due to I. J. Schoenberg (*Bulletin of the American Mathematical Society*, 1938) and can be described as follows:

Let ϕ be defined on the interval $[0, 2]$ by the following formulas:

$$\phi(t) = \begin{cases} 0, & \text{if } 0 \le t \le \frac{1}{3}, \text{ or if } \frac{5}{3} \le t \le 2, \\ 3t - 1, & \text{if } \frac{1}{3} \le t \le \frac{2}{3}, \\ 1, & \text{if } \frac{2}{3} \le t \le \frac{4}{3}, \\ -3t + 5, & \text{if } \frac{4}{3} \le t \le \frac{5}{3}. \end{cases}$$

Extend the definition of ϕ to all of E_1 by the equation

$$\phi(t + 2) = \phi(t).$$

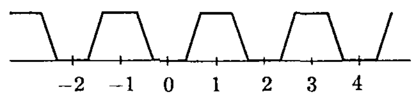


FIGURE 13–6.

This makes ϕ periodic with period 2. (The graph of ϕ is shown in Fig. 13–6.)

Now define two functions α_1 and α_2 by the following equations:

$$\alpha_1(t) = \sum_{n=1}^{\infty} \frac{\phi(3^{2n-2}t)}{2^n}, \qquad \alpha_2(t) = \sum_{n=1}^{\infty} \frac{\phi(3^{2n-1}t)}{2^n}.$$

Both series converge absolutely for each real t and they converge *uniformly* on E_1. In fact, since $|\phi(t)| \le 1$ for all t, the Weierstrass M-test is applicable with $M_n = 2^{-n}$. Since ϕ is continuous on E_1, Theorem 13–8 tells us that α_1 and α_2 are also continuous on E_1. Let $\alpha = (\alpha_1, \alpha_2)$ and let Γ denote the image of the unit interval $[0, 1]$ under α. We will show that Γ "fills" the unit square, i.e., that $\Gamma = [0, 1] \times [0, 1]$.

First, it is clear that $0 \le \alpha_1(t) \le 1$ and $0 \le \alpha_2(t) \le 1$ for each t, since $\sum_{n=1}^{\infty} 2^{-n} = 1$. Hence, Γ is a subset of the unit square. Next, we must show that $(a, b) \in \Gamma$ whenever $(a, b) \in [0, 1] \times [0, 1]$. For this purpose we write a and b in the binary system. That is, we write

$$a = \sum_{n=1}^{\infty} \frac{a_n}{2^n}, \qquad b = \sum_{n=1}^{\infty} \frac{b_n}{2^n},$$

where each a_n and each b_n is either 0 or 1. (See Exercise 1–14.) Now let

$$c = 2 \sum_{n=1}^{\infty} \frac{c_n}{3^n}, \qquad \text{where } c_{2n-1} = a_n \text{ and } c_{2n} = b_n,\ n = 1, 2, \ldots.$$

Clearly, $0 \le c \le 1$ since $2\sum_{n=1}^{\infty} 3^{-n} = 1$. We will show that $\alpha_1(c) = a$ and that $\alpha_2(c) = b$.

If we can prove that

$$\phi(3^k c) = c_{k+1}, \qquad \text{for each } k = 0, 1, 2, \ldots, \tag{5}$$

then we will have $\phi(3^{2n-2}c) = c_{2n-1} = a_n$ and $\phi(3^{2n-1}c) = c_{2n} = b_n$, and this will give us $\alpha_1(c) = a$, $\alpha_2(c) = b$. To prove (5), we write

$$3^k c = 2\sum_{n=1}^{k} \frac{c_n}{3^{n-k}} + 2\sum_{n=k+1}^{\infty} \frac{c_n}{3^{n-k}} = (\text{an even integer}) + d_k,$$

where $d_k = 2\sum_{n=1}^{\infty} c_{n+k}/3^n$. Since ϕ has period 2, it follows that

$$\phi(3^k c) = \phi(d_k).$$

If $c_{k+1} = 0$, then we have $0 \le d_k \le 2\sum_{n=2}^{\infty} 3^{-n} = \frac{1}{3}$, and hence $\phi(d_k) = 0$. Therefore, $\phi(3^k c) = c_{k+1}$ in this case. The only other case to consider is $c_{k+1} = 1$. But then we get $\frac{2}{3} \le d_k \le 1$ and hence $\phi(d_k) = 1$. Therefore, $\phi(3^k c) = c_{k+1}$ in all cases and this proves that $\alpha_1(c) = a$, $\alpha_2(c) = b$. Hence, Γ fills the unit square.

13-9 An application to repeated series. As a second application of Theorem 13-8, we derive the following generalization of Theorem 12-43:

13-9 THEOREM. *Let f be a complex-valued double sequence. Assume that $\sum_{n=1}^{\infty} f(m, n)$ converges absolutely for each fixed m and that*

$$\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} |f(m, n)|$$

converges. Then:

(a) *$\sum_{m=1}^{\infty} f(m, n)$ converges absolutely for each n.*

(b) *$\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} f(m, n)$ converges absolutely and equals*

$$\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} f(m, n).$$

Proof. Absolute convergence of the series in (a) and (b) is a consequence of Theorem 12-43. The problem is to show equality of the repeated series. For this purpose, let $T = \{0\} \cup \{1/q \mid q = 1, 2, \ldots\}$. For each fixed

$m = 1, 2, \ldots$, define a function g_m on T as follows:

$$g_m(0) = \sum_{n=1}^{\infty} f(m, n), \qquad g_m\left(\frac{1}{q}\right) = \sum_{n=1}^{q} f(m, n), \qquad \text{if } q = 1, 2, \ldots.$$

Now define g on T by the equation

$$g(x) = \sum_{m=1}^{\infty} g_m(x), \qquad \text{if } x \in T. \tag{6}$$

Since $|g_m(x)| \le \sum_{n=1}^{\infty} |f(m, n)|$, the Weierstrass M-test tells us that the series in (6) converges uniformly on T. Also, convergence of $\sum_{n=1}^{\infty} f(m, n)$ implies continuity of g_m at 0. Hence, g is also continuous at 0. This means that we have

$$\sum_{m=1}^{\infty} g_m(0) = \lim_{q\to\infty} \sum_{m=1}^{\infty} g_m\left(\frac{1}{q}\right), \tag{7}$$

the existence of the limit being part of the conclusion. But

$$\sum_{m=1}^{\infty} g_m\left(\frac{1}{q}\right) = \sum_{m=1}^{\infty} \sum_{n=1}^{q} f(m, n) = \sum_{n=1}^{q} \sum_{m=1}^{\infty} f(m, n).$$

Substituting in (7), we obtain

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n) = \lim_{q\to\infty} \sum_{n=1}^{q} \sum_{m=1}^{\infty} f(m, n) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} f(m, n).$$

13–10 Uniform convergence and Riemann-Stieltjes integration.

13–10 THEOREM. *Let α be of bounded variation on $[a, b]$. Assume that each term of the sequence $\{f_n\}$ is a real-valued function such that $f_n \in R(\alpha)$ on $[a, b]$ for each $n = 1, 2, \ldots$. Assume that $f_n \to f$ uniformly on $[a, b]$ and define $g_n(x) = \int_a^x f_n(t)\, d\alpha(t)$ if $x \in [a, b]$, $n = 1, 2, \ldots$. Then we have:*

(a) *$f \in R(\alpha)$ on $[a, b]$.*

(b) *$g_n \to g$ uniformly on $[a, b]$, where $g(x) = \int_a^x f(t)\, d\alpha(t)$.*

NOTE. The conclusion implies that, for each x in $[a, b]$, we can write

$$\lim_{n\to\infty} \int_a^x f_n(t)\, d\alpha(t) = \int_a^x \lim_{n\to\infty} f_n(t)\, d\alpha(t).$$

Proof We can assume that α is strictly increasing To prove (a), we will show that f satisfies Riemann's condition with respect to α on $[a, b]$. (See Theorem 9–19)

Given $\epsilon > 0$, choose N so that

$$|f(x) - f_N(x)| < \frac{\epsilon}{3[\alpha(b) - \alpha(a)]}, \qquad \text{for all } x \text{ in } [a, b]$$

Then, for every partition P of $[a, b]$, we have

$$|U(P, f - f_N, \alpha)| \leq \frac{\epsilon}{3} \qquad \text{and} \qquad |L(P, f - f_N, \alpha)| \leq \frac{\epsilon}{3}$$

(using the notation of Definition 9–14) For this N, choose P_ϵ so that P finer than P_ϵ implies $U(P, f_N, \alpha) - L(P, f_N, \alpha) < \epsilon/3$ Then for such P we have

$$\begin{aligned} U(P, f, \alpha) - L(P, f, \alpha) &\leq U(P, f - f_N, \alpha) - L(P, f - f_N, \alpha) \\ &\quad + U(P, f_N, \alpha) - L(P, f_N, \alpha) \\ &< |U(P, f - f_N, \alpha)| + |L(P, f - f_N, \alpha)| \\ &\quad + \frac{\epsilon}{3} \leq \epsilon \end{aligned}$$

This proves (a) To prove (b), let $\epsilon > 0$ be given and choose N so that

$$|f_n(t) - f(t)| < \frac{\epsilon}{2[\alpha(b) - \alpha(a)]}$$

for all $n > N$ and every t in $[a, b]$ If $x \in [a, b]$, we have

$$|g_n(x) - g(x)| \leq \int_a^x |f_n(t) - f(t)|\, d\alpha(t) \leq \frac{\alpha(x) - \alpha(a)}{\alpha(b) - \alpha(a)} \frac{\epsilon}{2} \leq \frac{\epsilon}{2} < \epsilon$$

This proves that $g_n \to g$ uniformly on $[a, b]$

13–11 THEOREM *Let α be of bounded variation on $[a, b]$ and assume that $\sum f_n(x) = f(x)$ (uniformly on $[a, b]$), where each f_n is a real-valued function such that $f_n \in R(\alpha)$ on $[a, b]$ Then we have.*

(a) $f \in R(\alpha)$ *on* $[a, b]$

(b) $\int_a^x \sum_{n=1}^\infty f_n(t)\, d\alpha(t) = \sum_{n=1}^\infty \int_a^x f_n(t)\, d\alpha(t)$ *(uniformly on* $[a, b]$*)*

Proof. Apply Theorem 13–10 to the sequence of partial sums

Part (b) of the above theorem tells us that a uniformly convergent series may be integrated term-by-term. Uniform convergence is a sufficient but not a necessary condition for termwise integration. For example, the series $\sum f_n(x)$, in which $f_n(x) = nx(1-x)^n - (n-1)x(1-x)^{n-1}$, has partial sums $s_n(x) = nx(1-x)^n$. (See Fig. 13–1.) The convergence of this series is not uniform on the interval $[0, 1]$. Nevertheless, it is easily seen that term-by-term integration leads to a correct result in this case. (See Exercise 13–8.)

On the other hand, the example in which $s_n(x) = n^2x(1-x)^n$ shows that some kind of condition is needed. In this example, $\{s_n\}$ converges pointwise to $f = 0$ on $[0, 1]$ (see Fig. 13–2), and hence $\int_0^1 f(t)\,dt = 0$. However, we have

$$\int_0^1 s_n(t)\,dt = \frac{n^2}{(n+1)(n+2)}$$

and this approaches 1 as $n \to \infty$. Therefore term-by-term integration leads to an incorrect result in this case.

13–12 THEOREM. *Let Γ be a rectifiable curve in E_2 described by a complex-valued function z, say*

$$z(t) = x(t) + iy(t), \qquad a \le t \le b,$$

where x and y are real-valued continuous functions of bounded variation defined on $[a, b]$. Let $\{f_n\}$ be a sequence of complex-valued functions defined and continuous on Γ. Assume that $f_n \to f$ uniformly on Γ. Then we have

$$\lim_{n\to\infty} \int_{\Gamma[z]} f_n(z)\,dz = \int_{\Gamma[z]} f(z)\,dz.$$

The proof can be obtained as a consequence of Theorem 13–11 by treating real and imaginary parts separately. Of course, a similar result holds for infinite series. That is, we have

$$\sum_{n=1}^{\infty} \int_{\Gamma[z]} f_n(z)\,dz = \int_{\Gamma[z]} f(z)\,dz$$

if $\sum f_n(z) = f(z)$ (uniformly on Γ) and if each f_n is continuous on Γ.

13–11 Uniform convergence and differentiation. By analogy with Theorems 13–3 and 13–10, one might expect the following result to hold: If $f_n \to f$ uniformly on $[a, b]$ and if f_n' exists for each n, then f' exists and

$f_n' \to f'$ uniformly on $[a, b]$ However, Example 4 (of Section 13-2) shows that this cannot be true Although the sequence $\{f_n\}$ of Example 4 converges uniformly on E_1, the sequence $\{f_n'\}$ does not even converge pointwise on E_1 For example, $\{f_n'(0)\}$ diverges since $f_n'(0) = \sqrt{n}$ Therefore the analog of Theorems 13-3 and 13-10 for differentiation must take a different form

13-13 THEOREM *Assume that each term of $\{f_n\}$ is a real-valued function having a finite derivative at each point of an open interval (a, b) Assume that for at least one point x_0 in (a, b) the sequence $\{f_n(x_0)\}$ converges Assume further that there exists a function g such that $f_n' \to g$ uniformly on (a, b) Then*

(a) *There exists a function f such that $f_n \to f$ uniformly on (a, b)*

(b) *For each x in (a, b) the derivative $f'(x)$ exists and equals $g(x)$*

Proof Assume that $c \in (a, b)$ and define a new sequence $\{g_n\}$ as follows

$$g_n(x) = \begin{cases} \dfrac{f_n(x) - f_n(c)}{x - c}, & \text{if } x \neq c, \\ f_n'(c), & \text{if } x = c \end{cases} \tag{8}$$

The sequence $\{g_n\}$ so formed depends on the choice of c Convergence of $\{g_n(c)\}$ follows from the hypothesis, since $g_n(c) = f_n'(c)$ We will prove next that $\{g_n\}$ converges uniformly on (a, b) If $x \neq c$, we have

$$g_n(x) - g_m(x) = \frac{h(x) - h(c)}{x - c}, \tag{9}$$

where $h(x) = f_n(x) - f_m(x)$ Now $h'(x)$ exists for each x in (a, b) and has the value $f_n'(x) - f_m'(x)$ Applying the Mean Value Theorem in (9), we get

$$g_n(x) - g_m(x) = f_n'(x_1) - f_m'(x_1), \tag{10}$$

where x_1 lies between x and c Since $\{f_n'\}$ converges uniformly on (a, b) (by hypothesis), we can use (10), together with the Cauchy condition (Theorem 13-4), to deduce that $\{g_n\}$ converges uniformly on (a, b)

Now we can show that $\{f_n\}$ converges uniformly on (a, b) Let us form the particular sequence $\{g_n\}$ corresponding to the special point $c = x_0$ for which $\{f_n(x_0)\}$ is assumed to converge From (8) we can write

$$f_n(x) = f_n(x_0) + (x - x_0)g_n(x),$$

an equation which holds for every x in (a, b). Hence we have

$$f_n(x) - f_m(x) = f_n(x_0) - f_m(x_0) + (x - x_0)[g_n(x) - g_m(x_0)].$$

This equation, with the help of the Cauchy condition, establishes the uniform convergence of $\{f_n\}$ on (a, b). This proves (a).

To prove (b), return to the sequence $\{g_n\}$ defined by (8) for an *arbitrary* point c in (a, b) and let $G(x) = \lim_{n\to\infty} g_n(x)$. The hypothesis that f'_n exists means that $\lim_{x\to c} g_n(x) = g_n(c)$. In other words, each g_n is continuous at c. Since $g_n \to G$ uniformly on (a, b), the limit function G is also continuous at c. This means that

$$G(c) = \lim_{x\to c} G(x), \tag{11}$$

the existence of the limit being part of the conclusion. But, for $x \neq c$, we have

$$G(x) = \lim_{n\to\infty} g_n(x) = \lim_{n\to\infty} \frac{f_n(x) - f_n(c)}{x - c} = \frac{f(x) - f(c)}{x - c}.$$

Hence, (11) states that the derivative $f'(c)$ exists and equals $G(c)$. But $G(c) = \lim_{n\to\infty} g_n(c) = \lim_{n\to\infty} f'_n(c) = g(c)$, and hence $f'(c) = g(c)$. Since c is an arbitrary point of (a, b), this proves (b).

When we reformulate Theorem 13–13 in terms of series, we obtain

13–14 THEOREM. *Assume that each f_n is a real-valued function defined on (a, b) such that the derivative $f'_n(x)$ exists for each x in (a, b). Assume that, for at least one point x_0 in (a, b), the series $\sum f_n(x_0)$ converges. Assume further that there exists a function g such that $\sum f'_n(x) = g(x)$ (uniformly on (a, b)). Then:*

(a) *There exists a function f such that $\sum f_n(x) = f(x)$ (uniformly on (a, b)).*

(b) *If $x \in (a, b)$, the derivative $f'(x)$ exists and equals $\sum f'_n(x)$.*

13–12 Sufficient conditions for uniform convergence of a series. The importance of uniformly convergent series has been amply illustrated in some of the preceding theorems. Therefore it seems natural to seek some simple ways of testing a series for uniform convergence without resorting to the definition in each case. One such test, the *Weierstrass M-test*, was described in Theorem 13–7. There are other tests that may be useful when the M-test is not applicable. One of these is the analog of Theorem 12–28.

13–15 THEOREM (*Dirichlet's test for uniform convergence*) *Let $F_n(x)$ denote the nth partial sum of the series $\sum f_n(x)$, where each f_n is a complex-valued function defined on a set T. Assume that $\{F_n\}$ is uniformly bounded on T. Let $\{g_n\}$ be a sequence of real-valued functions such that $g_{n+1}(x) \le g_n(x)$ for each x in T and for every $n = 1, 2, \ldots$, and assume that $g_n \to 0$ uniformly on T. Then the series $\sum f_n(x)g_n(x)$ converges uniformly on T.*

Proof. Let $s_n(x) = \sum_{k=1}^{n} f_k(x)g_k(x)$. By partial summation we have

$$s_n(x) = \sum_{k=1}^{n} F_k(x)(g_k(x) - g_{k+1}(x)) + g_{n+1}F_n(x),$$

and hence if $n > m$, we can write

$$s_n(x) - s_m(x) = \sum_{k=m+1}^{n} F_k(x)(g_k(x) - g_{k+1}(x)) + g_{n+1}(x)F_n(x) - g_{m+1}(x)F_m(x).$$

Therefore, if M is a uniform bound for $\{F_n\}$, we have

$$\begin{aligned} |s_n(x) - s_m(x)| &\le M \sum_{k=m+1}^{n} (g_k(x) - g_{k+1}(x)) + Mg_{n+1}(x) + Mg_{m+1}(x) \\ &= M(g_{m+1}(x) - g_{n+1}(x)) + Mg_{n+1}(x) + Mg_{m+1}(x) \\ &= 2Mg_{m+1}(x). \end{aligned}$$

Since $g_n \to 0$ uniformly on T, this inequality (together with the Cauchy condition) implies that $\sum f_n(x)g_n(x)$ converges uniformly on T.

The reader should have no difficulty in extending Theorem 12–29 (Abel's test) in a similar way so that it yields a test for uniform convergence. (See Exercise 13–11.)

EXAMPLE. Let $F_n(x) = \sum_{k=1}^{n} e^{ikx}$. In the last chapter (see Theorem 12–30), we derived the inequality $|F_n(x)| \le 1/|\sin(x/2)|$, valid for every real $x \ne 2m\pi$ (m is an integer). Therefore, if $0 < \delta < \pi$, we have the estimate $|F_n(x)| \le 1/\sin(\delta/2)$ if $\delta \le x \le 2\pi - \delta$. Hence, $\{F_n\}$ is uniformly bounded on the interval $[\delta, 2\pi - \delta]$. If $\{g_n\}$ satisfies the conditions of Theorem 13–15, we can conclude that the series $\sum g_n(x)e^{inx}$ converges uniformly on $[\delta, 2\pi - \delta]$. In particular, if we take $g_n(x) = 1/n$, this establishes the uniform convergence of the series

$$\sum_{n=1}^{\infty} \frac{e^{inx}}{n}$$

on $[\delta, 2\pi - \delta]$ if $0 < \delta < \pi$. Note that the Weierstrass M-test cannot be used to establish uniform convergence in this case, since $|e^{inx}| = 1$.

13–13 Bounded convergence. Arzelà's theorem. In Example 1 of Section 13–2 we encountered a nonuniformly convergent sequence in which term-by-term integration led to a correct result. This example has the property of *bounded convergence,* which is defined as follows:

13–16 DEFINITION. *A sequence of functions $\{f_n\}$ is said to be boundedly convergent on T if $\{f_n\}$ is pointwise convergent and uniformly bounded on T.*

The following theorem explains why term-by-term integration was permissible in Example 1:

13–17 THEOREM (*Arzelà*). *Assume that $\{f_n\}$ is boundedly convergent on $[a, b]$ and suppose each f_n is Riemann-integrable on $[a, b]$. Assume also that the limit function f is Riemann-integrable on $[a, b]$. Then*

$$\lim_{n\to\infty} \int_a^b f_n(x)\, dx = \int_a^b \lim_{n\to\infty} f_n(x)\, dx = \int_a^b f(x)\, dx.$$

Proof. Let $g_n(x) = |f_n(x) - f(x)|$. We will prove that

$$\lim_{n\to\infty} \int_a^b g_n(x)\, dx = 0.$$

For this purpose we define a new sequence of functions $\{h_n\}$ as follows:

$$h_n(x) = \sup\,\{g_n(x), g_{n+1}(x), \ldots\}, \qquad \text{if } x \in [a, b] \text{ and } n = 1, 2, \ldots.$$

Note that, for each fixed x, we have

$$0 \le g_n(x) \le h_n(x), \qquad h_{n+1}(x) \le h_n(x), \qquad \lim_{n\to\infty} h_n(x) = 0.$$

Hence, $0 \le \int_a^b g_n(x)\, dx = \underline{\int}_a^b g_n(x)\, dx \le \underline{\int}_a^b h_n(x)\, dx$. (The lower integral $\underline{\int}_a^b h_n(x)\, dx$ exists because each h_n is bounded on $[a, b]$. However, h_n need not be Riemann-integrable. On the other hand, g_n *is* Riemann-integrable, since both f_n and f are integrable) Let $I_n = \int_a^b h_n(x)\,dx$ The proof will be complete if we can show that $I_n \to 0$ as $n \to \infty$ Observe that the sequence $\{I_n\}$ converges to *some* non-negative limit, since $0 \le I_{n+1} \le I_n$ Let $I = \lim_{n\to\infty} I_n$ We will show that the inequality $I > 0$ leads to a contradiction

Assume that $I > 0$ Since $I_n \ge I > I/2$ for each n, there is a partition P_n of $[a, b]$ and a lower Riemann sum $L(P_n, h_n)$ such that $L(P_n, h_n) > I/2$ Let $\epsilon = \frac{1}{2}I/(M + b - a)$, where M is a uniform bound for $\{h_n\}$ on $[a, b]$ Then the lower sum $L(P_n, h_n)$ can be split into two parts as follows

$$L(P_n, h_n) = \sum_{i\in A(n)} m_i(h_n)\,\Delta x_i + \sum_{i\in B(n)} m_i(h_n)\,\Delta x_i,$$

where $m_i(h_n)$ denotes the infimum of h_n in the ith subinterval of P_n and

$$A(n) = \{i \mid m_i(h_n) > \epsilon\}, \qquad B(n) = \{i \mid m_i(h_n) \le \epsilon\}$$

The inequality $L(P_n, h_n) > I/2$ implies

$$\tfrac{1}{2} I < \sum_{i\in A(n)} M\,\Delta x_i + \epsilon \sum_{i\in B(n)} \Delta x_i \le M \sum_{i\in A(n)} \Delta x_i + \epsilon(b - a),$$

from which we obtain $\sum_{i\in A(n)} \Delta x_i > \epsilon$ Since refinement of partitions increases lower sums, there is no loss in generality in assuming that the P_n are chosen so that $P_n \subset P_{n+1}$

Now let S_n denote the union of those subintervals $[x_{i-1}, x_i]$ of P_n for which $i \in A(n)$ Then S_n is a closed set and its Jordan content is

$$c(S_n) = \sum_{i\in A(n)} \Delta x_i > \epsilon$$

This implies that there is at least one x belonging to infinitely many of the sets S_n Hence for this x we have $h_n(x) > \epsilon$ for infinitely many n, contradicting the fact that $h_n(x) \to 0$ as $n \to \infty$ This contradiction implies that $I = 0$.

NOTE It is easy to give an example of a boundedly convergent sequence $\{f_n\}$ of Riemann-integrable functions whose limit f is not Riemann-integrable If $\{r_1, r_2, \ldots\}$ denotes the set of rational numbers in $[0, 1]$, define $f_n(x)$ to have the value 1 if $x = r_k$ for some $k = 1, 2, \ldots, n$, and put $f_n(x) = 0$ otherwise Then the integral $\int_0^1 f_n(x)\,dx = 0$ for each n, but the limit function f is not Riemann-integrable on $[0, 1]$

13–14 Mean convergence. The functions in this section may be real- or complex-valued.

13–18 DEFINITION. *Let $\{f_n\}$ be a sequence of Riemann-integrable functions defined on $[a, b]$. Assume that $f \in R$ on $[a, b]$. The sequence $\{f_n\}$ is said to converge in the mean to f on $[a, b]$, and we write* $\text{l.i.m.}_{n\to\infty} f_n = f$ *on $[a, b]$, if*

$$\lim_{n\to\infty} \int_a^b |f_n(x) - f(x)|^2\, dx = 0.$$

If the inequality $|f(x) - f_n(x)| < \epsilon$ holds for every x in $[a, b]$, then we have $\int_a^b |f(x) - f_n(x)|^2\, dx \le \epsilon^2(b - a)$. Therefore, uniform convergence of $\{f_n\}$ to f on $[a, b]$ implies mean convergence, provided that each f_n is Riemann-integrable on $[a, b]$. A rather surprising fact is that convergence in the mean need not imply pointwise convergence at any point of the interval. This can be seen as follows: For each integer $n \ge 0$, subdivide the interval $[0, 1]$ into 2^n equal subintervals and let I_{2^n+k} denote that subinterval whose right endpoint is $(k + 1)/2^n$, where $k = 0, 1, 2, \ldots, 2^n - 1$. This yields a collection $\{I_1, I_2, \ldots\}$ of subintervals of $[0, 1]$, of which the first few are:

$$I_1 = [0, 1], \quad I_2 = [0, \tfrac{1}{2}], \quad I_3 = [\tfrac{1}{2}, 1],$$
$$I_4 = [0, \tfrac{1}{4}], \quad I_5 = [\tfrac{1}{4}, \tfrac{1}{2}], \quad I_6 = [\tfrac{1}{2}, \tfrac{3}{4}],$$

and so forth. Define f_n on $[0, 1]$ as follows:

$$f_n(x) = \begin{cases} 1, & \text{if } x \in I_n, \\ 0, & \text{if } x \in [0, 1] - I_n. \end{cases}$$

Then $\{f_n\}$ converges in the mean to 0, since $\int_0^1 |f_n(x)|^2\, dx$ is the length of I_n, and this approaches 0 as $n \to \infty$. On the other hand, for each x in $[0, 1]$ we have

$$\limsup_{n\to\infty} f_n(x) = 1 \qquad \text{and} \qquad \liminf_{n\to\infty} f_n(x) = 0.$$

[Why?] Hence, $\{f_n(x)\}$ does not converge for any x in $[0, 1]$.

The next theorem illustrates the importance of mean convergence.

13–19 THEOREM. *Assume that* $\text{l.i.m.}_{n\to\infty} f_n = f$ *on $[a, b]$. If $g \in R$ on $[a, b]$, define*

$$h(x) = \int_a^x f(t)g(t)\,dt, \qquad h_n(x) = \int_a^x f_n(t)g(t)\,dt,$$

if $x \in [a, b]$ Then $h_n \to h$ uniformly on $[a, b]$.

Proof. The proof is based on the inequality

$$(12) \qquad 0 \le \left(\int_a^x |f(t) - f_n(t)||g(t)|\,dt\right)^2 \le \left(\int_a^x |f(t) - f_n(t)|^2\,dt\right)\left(\int_a^x |g(t)|^2\,dt\right),$$

which is a direct application of the Cauchy-Schwarz inequality for integrals. (See Exercise 9-13 for the statement of the Cauchy-Schwarz inequality and a sketch of its proof) Given $\epsilon > 0$, we can choose N so that $n > N$ implies

$$(13) \qquad \int_a^b |f(t) - f_n(t)|^2\,dt < \frac{\epsilon^2}{A},$$

where $A = 1 + \int_a^b |g(t)|^2\,dt$. Substituting (13) in (12), we find that $n > N$ implies $0 \le |h(x) - h_n(x)| < \epsilon$ for every x in $[a, b]$

This theorem is particularly useful in the theory of Fourier series. (See Theorem 15-22) The following generalization is also of interest.

13-20 THEOREM *Assume that $\lim_{n\to\infty} f_n = f$ and $\lim_{n\to\infty} g_n = g$ on $[a, b]$. Define*

$$h(x) = \int_a^x f(t)g(t)\,dt, \qquad h_n(x) = \int_a^x f_n(t)g_n(t)\,dt,$$

if $x \in [a, b]$ Then $h_n \to h$ uniformly on $[a, b]$

Proof. We have

$$h_n(x) - h(x) = \int_a^x (f - f_n)(g - g_n)\,dt + \left(\int_a^x f_n g\,dt - \int_a^x fg\,dt\right) + \left(\int_a^x f g_n\,dt - \int_a^x fg\,dt\right).$$

Applying the Cauchy-Schwarz inequality, we can write

$$0 \leq \left(\int_a^x |f - f_n||g - g_n|\,dt\right)^2 \leq \left(\int_a^b |f - f_n|^2\,dt\right)\left(\int_a^b |g - g_n|^2\,dt\right).$$

The proof is now an easy consequence of Theorem 13–19.

13–15. Power series. An infinite series of the form

$$a_0 + \sum_{n=1}^{\infty} a_n(z - z_0)^n,$$

written more briefly as

$$\sum_{n=0}^{\infty} a_n(z - z_0)^n, \tag{14}$$

is called a power series in $z - z_0$. Here z, z_0, and $a_n (n = 0, 1, 2, \ldots)$ are complex numbers. With every power series (14) there is associated a circle, called the *circle of convergence,* such that the series converges absolutely for every z interior to this circle and diverges for every z outside this circle. The center of the circle is at z_0 and its radius is called the *radius of convergence* of the power series. (The radius may be 0 or $+\infty$ in extreme cases.) The next theorem establishes the existence of the circle of convergence and provides us with a way of calculating its radius.

13–21 THEOREM. *Given a power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$, let*

$$\lambda = \limsup_{n\to\infty} \sqrt[n]{|a_n|}, \qquad r = \frac{1}{\lambda},$$

(where $r = 0$ if $\lambda = +\infty$ and $r = +\infty$ if $\lambda = 0$). Then the series converges absolutely if $|z - z_0| < r$ and diverges if $|z - z_0| > r$. Furthermore, the series converges uniformly on every compact subset interior to the circle of convergence.

Proof. Applying the root test (Theorem 12–26), we have

$$\limsup_{n\to\infty} \sqrt[n]{|a_n(z - z_0)^n|} = \frac{|z - z_0|}{r},$$

and hence $\sum a_n(z - z_0)^n$ converges absolutely if $|z - z_0| < r$ and diverges if $|z - z_0| > r$.

To prove the second assertion, we simply observe that if T is a compact subset of the circle of convergence, there is a point p in T such that $z \in T$ implies $|z - z_0| \le |p - z_0| < r$. Hence, $|a_n(z - z_0)^n| \le |a_n(p - z_0)^n|$ for each z in T, and the Weierstrass M-test is applicable.

NOTE. If the limit $\lim_{n\to\infty} |a_n/a_{n+1}|$ exists (or if this limit is $+\infty$), its value is also equal to the radius of convergence of (14). (See Exercise 13-27.)

EXAMPLES

1. The two series $\sum_{n=0}^{\infty} z^n$ and $\sum_{n=1}^{\infty} z^n/n^2$ have the same radius of convergence, namely, $r = 1$. On the boundary of the circle of convergence, the first converges nowhere, the second converges everywhere.

2. The series $\sum_{n=1}^{\infty} z^n/n$ has radius of convergence $r = 1$, but it does not converge at $z = 1$. However, it does converge everywhere else on the boundary because of Dirichlet's test (Theorem 12-28).

These examples illustrate why Theorem 13-21 makes no assertion about the behavior of a power series on the *boundary* of the circle of convergence.

13-22 THEOREM. *Assume that the power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ converges for each z in $N(z_0, r)$. Then the function f defined by the equation*

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n, \qquad \text{if } z \in N(z_0; r), \tag{15}$$

is continuous on $N(z_0, r)$.

Proof. Since each point in $N(z_0; r)$ belongs to some compact subset of $N(z_0, r)$, the conclusion follows at once from Theorem 13-8.

NOTE. The series in (15) is said to *represent* f in $N(z_0, r)$. It is also called a *power series expansion* of f about z_0. Functions having power series expansions are continuous inside the circle of convergence. Much more than this is true, however. We will later prove that such functions have derivatives of every order inside the circle of convergence. The proof will make use of the following theorem.

13-23 THEOREM. *Assume that $\sum a_n(z - z_0)^n$ converges if $z \in N(z_0, r)$. Suppose that the equation*

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$$

is known to be valid for each z in some open subset S of $N(z_0; r)$. Then, for each point z_1 in S, there exists a neighborhood $N(z_1, R) \subset S$ in which f has a power series expansion of the form

(16) $$f(z) = \sum_{k=0}^{\infty} b_k(z - z_1)^k,$$

where

(17) $$b_k = \sum_{n=k}^{\infty} \binom{n}{k} a_n(z_1 - z_0)^{n-k} \qquad (k = 0, 1, 2, \ldots).$$

Proof. If $z \in S$, we have

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n = \sum_{n=0}^{\infty} a_n(z - z_1 + z_1 - z_0)^n$$

$$= \sum_{n=0}^{\infty} a_n \sum_{k=0}^{n} \binom{n}{k} (z - z_1)^k(z_1 - z_0)^{n-k}$$

$$= \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} c_n(k),$$

where

$$c_n(k) = \begin{cases} \binom{n}{k} a_n(z - z_1)^k(z_1 - z_0)^{n-k}, & \text{if } k \le n, \\ 0, & \text{if } k > n. \end{cases}$$

Now choose R so that $N(z_1; R) \subset S$ and assume that $z \in N(z_1; R)$. Then the repeated series $\sum_{n=0}^{\infty} \sum_{k=0}^{\infty} c_n(k)$ converges absolutely, since

(18)
$$\sum_{n=0}^{\infty} \sum_{k=0}^{\infty} |c_n(k)| = \sum_{n=0}^{\infty} |a_n|(|z - z_1| + |z_1 - z_0|)^n = \sum_{n=0}^{\infty} |a_n|(z_2 - z_0)^n,$$

where

$$z_2 = z_0 + |z - z_1| + |z_1 - z_0|.$$

But

$$|z_2 - z_0| < R + |z_1 - z_0| \le r,$$

and hence the series in (18) converges. Therefore, by Theorem 13-9, we can interchange the order of summation to obtain

$$f(z) = \sum_{k=0}^{\infty} \sum_{n=0}^{\infty} c_n(k) = \sum_{k=0}^{\infty} \sum_{n=k}^{\infty} \binom{n}{k} a_n(z - z_1)^k (z_1 - z_0)^{n-k}$$

$$= \sum_{k=0}^{\infty} b_k(z - z_1)^k,$$

where b_k is given by (17) This completes the proof

NOTE In the course of the proof we have shown that we may use any $R > 0$ that satisfies the condition

$$N(z_1, R) \subset S \tag{19}$$

13-24 THEOREM *Assume that* $\sum a_n(z - z_0)^n$ *converges for each* z *in* $N(z_0; r)$ *Then the function f defined by the equation*

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n, \qquad \text{if } z \in N(z_0, r), \tag{20}$$

has a derivative $f'(z)$ *for each* z *in* $N(z_0, r)$, *given by*

$$f'(z) = \sum_{n=1}^{\infty} na_n(z - z_0)^{n-1} \tag{21}$$

NOTE The series in (20) and (21) have the same radius of convergence

Proof Assume that $z_1 \in N(z_0, r)$ and expand f in a power series about z_1, as indicated in (16) Then, if $z \in N'(z_1, R)$, we have

$$\frac{f(z) - f(z_1)}{z - z_1} = b_1 + \sum_{k=1}^{\infty} b_{k+1}(z - z_1)^k. \tag{22}$$

By continuity, the right member of (22) tends to b_1 as $z \to z_1$. Hence, $f'(z_1)$ exists and equals b_1 Using (17) to compute b_1, we find

$$b_1 = \sum_{n=1}^{\infty} na_n(z_1 - z_0)^{n-1}$$

Since z_1 is an arbitrary point of $N(z_0, r)$, this proves (21) The two series have the same radius of convergence because $\sqrt[n]{n} \to 1$ as $n \to \infty$

NOTE By repeated application of (21), we find that for each $k = 1, 2, \ldots$, the derivative $f^{(k)}(z)$ exists in $N(z_0; r)$ and is given by the series

$$f^{(k)}(z) = \sum_{n=k}^{\infty} \frac{n!}{(n-k)!} a_n(z - z_0)^{n-k}. \tag{23}$$

If we put $z = z_0$ in (23), we obtain the important formula

$$f^{(k)}(z_0) = k!a_k \qquad (k = 1, 2, \ldots). \tag{24}$$

This equation tells us that if two power series $\sum a_n(z - z_0)^n$ and $\sum b_n(z - z_0)^n$ both represent the same function in a neighborhood $N(z_0; r)$, then $a_n = b_n$ for every n. That is, the power series expansion of a function f about a given point z_0 is uniquely determined (if it exists at all), and it is given by the formula

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n,$$

valid for each z in the circle of convergence.

13–25 THEOREM. *Assume that $\sum a_n(z - z_0)^n$ converges for each z in $N(z_0; r)$. Let Γ be a piece-wise smooth curve in $N(z_0; r)$ joining z_0 to z_1. Then*

$$\int_{\Gamma(z_0, z_1)} \sum_{n=0}^{\infty} a_n(z - z_0)^n \, dz = \sum_{n=0}^{\infty} \frac{a_n}{n+1} (z_1 - z_0)^{n+1}.$$

Proof. Termwise integration is permissible by Theorem 13–12 and the integral of each term can be evaluated by means of Exercise 9–38.

13–16. Multiplication of power series.

13–26 THEOREM. *Given two power series expansions about the origin, say*

$$f(z) = \sum_{n=0}^{\infty} a_n z^n, \qquad \text{if } z \in N(0; r),$$

and

$$g(z) = \sum_{n=0}^{\infty} b_n z^n, \qquad \text{if } z \in N(0; R).$$

Then the product $f(z)g(z)$ is given by the power series

$$f(z)g(z) = \sum_{n=0}^{\infty} c_n z^n, \qquad \text{if } z \in N(0; r) \cap N(0; R),$$

where

$$c_n = \sum_{k=0}^{n} a_k b_{n-k} \qquad (n = 0, 1, 2, \ldots)$$

Proof The Cauchy product of the two given series is

$$\sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_k z^k b_{n-k} z^{n-k} \right) = \sum_{n=0}^{\infty} c_n z^n,$$

and the conclusion follows from Theorem 12–46 (Mertens' Theorem).

NOTE. If the two series are identical, we get

$$f(z)^2 = \sum_{n=0}^{\infty} c_n z^n,$$

where $c_n = \sum_{k=0}^{n} a_k a_{n-k} = \sum_{m_1+m_2=n} a_{m_1} a_{m_2}$. The symbol $\sum_{m_1+m_2=n}$ indicates that the summation is to be extended over all non-negative integers m_1 and m_2 whose sum is n. Similarly, for any integer $p > 0$, we have

$$f(z)^p = \sum_{n=0}^{\infty} c_n(p) z^n,$$

where

$$c_n(p) = \sum_{m_1+\cdots+m_p=n} a_{m_1} \cdots a_{m_p}.$$

13–17 The substitution theorem.

13–27 THEOREM. *Given two power series expansions about the origin, say*

$$f(z) = \sum_{n=0}^{\infty} a_n z^n, \qquad \text{if } z \in N(0; r),$$

and

$$g(z) = \sum_{n=0}^{\infty} b_n z^n, \qquad \text{if } z \in N(0; R).$$

If, for a fixed z in $N(0; R)$, we have $\sum_{n=0}^{\infty} |b_n z^n| < r$, then for this z we can write

$$f[g(z)] = \sum_{k=0}^{\infty} c_k z^k,$$

where the coefficients c_k are obtained as follows: Define the numbers $b_k(n)$ by the equation

$$g(z)^n = \left(\sum_{k=0}^{\infty} b_k z^k\right)^n = \sum_{k=0}^{\infty} b_k(n) z^k.$$

Then $c_k = \sum_{n=0}^{\infty} a_n b_k(n)$ *for* $k = 0, 1, 2, \ldots$.

NOTE. The series $\sum_{k=0}^{\infty} c_k z^k$ is the power series which arises *formally* by substituting the series for $g(z)$ in place of z in the expansion of f and then rearranging terms in increasing powers of z.

Proof. By hypothesis, we can choose z so that $\sum_{n=0}^{\infty} |b_n z^n| < r$. For this z we have $|g(z)| < r$ and hence we can write

$$f[g(z)] = \sum_{n=0}^{\infty} a_n g(z)^n = \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} a_n b_k(n) z^k.$$

If we are allowed to interchange the order of summation, we obtain

$$f[g(z)] = \sum_{k=0}^{\infty} \left(\sum_{n=0}^{\infty} a_n b_k(n)\right) z^k = \sum_{k=0}^{\infty} c_k z^k,$$

which is the statement we set out to prove. To justify the interchange, we will establish the convergence of the series

$$\sum_{n=0}^{\infty} \sum_{k=0}^{\infty} |a_n b_k(n) z^k| = \sum_{n=0}^{\infty} |a_n| \sum_{k=0}^{\infty} |b_k(n) z^k|. \tag{25}$$

Now each number $b_k(n)$ is a finite sum of the form

$$b_k(n) = \sum_{m_1 + \cdots + m_n = k} b_{m_1} \cdots b_{m_n},$$

and hence $|b_k(n)| \le \sum_{m_1 + \cdots + m_n = k} |b_{m_1}| \cdots |b_{m_n}|$. On the other hand, we have

$$\left(\sum_{k=0}^{\infty} |b_k| z^k\right)^n = \sum_{k=0}^{\infty} B_k(n) z^k,$$

where $B_k(n) = \sum_{m_1 + \cdots + m_n = k} |b_{m_1}| \cdots |b_{m_n}|$. Returning to (25), we have

$$\sum_{n=0}^{\infty} |a_n| \sum_{k=0}^{\infty} |b_k(n)z^k| \le \sum_{n=0}^{\infty} |a_n| \sum_{k=0}^{\infty} B_k(n)|z^k|$$

$$= \sum_{n=0}^{\infty} |a_n| \left(\sum_{k=0}^{\infty} |b_k z^k| \right)^n,$$

and this establishes the convergence of (25)

As an application of the substitution theorem, we will show that the reciprocal of a power series in z is again a power series in z, provided that the constant term is not 0.

13-28 THEOREM *Assume that we have*

$$p(z) = \sum_{n=0}^{\infty} p_n z^n, \qquad \text{if } z \in N(0; h),$$

where $p(0) \neq 0$ *Then there exists a neighborhood* $N(0, \delta)$ *in which the reciprocal of* p *has a power series expansion of the form*

$$\frac{1}{p(z)} = \sum_{n=0}^{\infty} q_n z^n.$$

Furthermore, $q_0 = 1/p_0$.

Proof Without loss in generality we can assume that $p_0 = 1$ [Why?] Then $p(0) = 1$. Let $P(z) = 1 + \sum_{n=1}^{\infty} |p_n z^n|$ if $z \in N(0, h)$. By continuity, there exists a neighborhood $N(0, \delta)$ such that $|P(z) - 1| < 1$ if $z \in N(0; \delta)$. The conclusion follows by applying Theorem 13-27 with

$$f(z) = \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

and

$$g(z) = 1 - p(z) = \sum_{n=1}^{\infty} p_n z^n.$$

13-18 Real power series. If x, x_0, and a_n ($n = 0, 1, 2, \ldots$) are real numbers, the series $\sum a_n(x - x_0)^n$ is called a *real power series* Its circle of convergence intersects the real axis in an interval called the *interval of convergence.* If neighborhoods are taken to mean one-dimensional neighborhoods, all the results of the last three sections can be interpreted in an obvious way as theorems on the real line.

We shall restrict our considerations in the remainder of this chapter to real power series. In this section we shall deal with the following type of problem: Suppose we are given a real-valued function f defined in some neighborhood of a point x_0 in E_1, and suppose f has derivatives of every order in this neighborhood. Then we can certainly *form* the power series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n.$$

Does this series converge for any x besides $x = x_0$? If so, is its sum equal to $f(x)$? In general, the answer to both questions is "No." (See Exercise 13–30 for a counter example.) A necessary and sufficient condition for answering both questions in the affirmative can be given by using Taylor's formula (Theorem 5–14), but first we introduce some further terminology and notation.

13–29 DEFINITION. *Let f be a real-valued function defined on an interval I in E_1. If f has derivatives of every order at each point of I, we write $f \in C^\infty$ on I.*

If $f \in C^\infty$ on some neighborhood of a point x_0, the power series $\sum_{n=0}^{\infty} (f^{(n)}(x_0)/n!)(x - x_0)^n$ is called the *Taylor's series about x_0 generated by f*. To indicate that f generates this series, we write

$$f(x) \sim \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n.$$

The question we are interested in is this: When can we replace the symbol $\sim$ by the symbol $=$? Taylor's formula states that if $f \in C^\infty$ on the closed interval $[a, b]$ and if $x_0 \in [a, b]$, then, for every x in $[a, b]$ and for every n, we have

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \frac{f^{(n)}(x_1)}{n!}(x - x_0)^n, \tag{26}$$

where x_1 is some point between x and x_0. The point x_1 depends on x, x_0, and on n. Hence a necessary and sufficient condition for the Taylor's series to converge to $f(x)$ is that

$$\lim_{n\to\infty} \frac{f^{(n)}(x_1)}{n!}(x - x_0)^n = 0. \tag{27}$$

In practice it may be quite difficult to deal with this limit because of the unknown position of x_1. In some cases, however, a suitable upper bound can be obtained for $f^{(n)}(x_1)$ and the limit can be shown to be zero. Since $(x - x_0)^n/n! \to 0$ as $n \to \infty$, equation (27) will certainly hold if the sequence $\{f^{(n)}\}$ is *uniformly bounded* on $[a, b]$. Thus we can state the following sufficient condition for representing a function by a Taylor's series.

13-30 THEOREM. *Assume that $f \in C^\infty$ on $[a, b]$ and let $x_0 \in [a, b]$. Assume that there is a neighborhood $N(x_0)$ and a constant M (which might depend on x_0) such that $|f^{(n)}(x)| \le M$ for every x in $N(x_0) \cap [a, b]$ and every $n = 1, 2, \ldots$. Then, for each x in $N(x_0) \cap [a, b]$, we have*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n.$$

13-19 Bernstein's theorem. Another useful sufficient condition, due to S. Bernstein, will now be derived.

13-31 THEOREM (*Bernstein*). *Assume that $f \in C^\infty$ on an open interval of the form $(a - \delta, b)$, where $\delta > 0$, and suppose that f and all its derivatives are non-negative in the half-open interval $[a, b)$. Then, for every x_0 in $[a, b)$, we have*

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k, \qquad \text{if } x_0 \le x < b. \tag{28}$$

Proof. Let $x_0 \in [a, b)$ and assume that $x \in [x_0, b)$. Then the series in (28) has positive terms and, because of (26), its partial sums are all bounded above by $f(x)$. Hence the series in (28) converges and has a sum $\le f(x)$. Applying the same argument to $f^{(n)}$, we find that the nth derivative of the series in (28) converges and has a sum $\le f^{(n)}(x)$. Therefore, if we write

$$g(x) = f(x) - \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k, \tag{29}$$

then $g(x)$ and all the derivatives $g^{(n)}(x)$ are non-negative if $x_0 \le x < b$.

Next, we will prove that $g(x) = 0$ if $x_0 \le x < (x_0 + b)/2$. This restriction on x implies $x_0 \le x \le 2x - x_0 < b$. Hence we can choose a number y satisfying $2x - x_0 < y < b$ and use Taylor's formula to write

$$f(y) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x)}{k!} (y - x)^k + \frac{f^{(n)}(y_1)}{n!} (y - x)^n, \tag{30}$$

where $x < y_1 < y$. Now $f^{(n)}$ is an increasing function [Why?] and hence we have $f^{(n)}(y_1) \geq f^{(n)}(x)$. Therefore, (30) gives us

$$f(y) \geq \frac{f^{(n)}(x)}{n!}(y - x)^n,$$

or

$$\frac{f^{(n)}(x)}{n!} \leq \frac{f(y)}{(y - x)^n}.$$

Returning to the remainder term in (26), we can write

$$0 \leq \frac{f^{(n)}(x_1)}{n!}(x - x_0)^n \leq \frac{f^{(n)}(x)}{n!}(x - x_0)^n \leq f(y)\left(\frac{x - x_0}{y - x}\right)^n.$$

But $0 \leq (x - x_0)/(y - x) < 1$ because of the way y was chosen. Hence,

$$\lim_{n\to\infty} \frac{f^{(n)}(x_1)}{n!}(x - x_0)^n = 0$$

and this means that we have

$$(31) \qquad f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k, \qquad \text{if } x_0 \leq x < \frac{x_0 + b}{2}.$$

Now let us make the following observation: Since f is non-negative and increasing on $[a, b)$, if $f(x_0) = 0$ it follows that $f(x) = 0$ for every x in $[a, x_0]$. This, in turn, implies that all derivatives $f^{(n)}(x) = 0$ if $x \in [a, x_0]$ (provided that $x_0 > a$), and hence [by (31)] $f(x)$ must be 0 for every x in $[a, (x_0 + b)/2)$. In other words, if there exists a point x_0 interior to (a, b) such that $f(x_0) = 0$, then $f(x) = 0$ for every x in $[a, (x_0 + b)/2)$. But this implies that f must be identically zero in the whole interval $[a, b)$, since the supremum of the set $\{x \mid x \in (a, b), f(x) = 0\}$ cannot be a point in (a, b). Now the function g defined by (29) satisfies the same hypotheses as f on the interval $[x_0, b)$ and g *does* vanish at interior points of the interval [by (31)]. Hence, g is identically zero on $[x_0, b)$. In other words, (28) holds. (This proof is due to Dr. Basil Gordon.)

Note that Bernstein's theorem gives us a power series expansion about x_0 valid in a half-open interval of the form $[x_0, b)$. By Theorem 13–21, this series also converges in a neighborhood of radius $b - x_0$ about x_0, but we have no guarantee that it represents $f(x)$ if $x < x_0$. However, if x_0 is an *interior* point of $[a, b)$, we can always increase the range of validity of (28) so that it extends to the left of x_0.

13-32 THEOREM *Assume that f satisfies the hypotheses of Theorem 13-31 and suppose x_0 is an interior point of $[a, b)$. Let R be such that $N(x_0; R) \subset (a, b)$. Then formula (28) is valid for x in $N(x_0, R)$.*

Proof. Applying (28) with $x_0 = a$, we obtain the expansion

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n,$$

valid for each x in $[a, b)$. By Theorem 13-21, this series still converges if x is in the interval $2a - b < x < b$ (although it need not represent $f(x)$ if $x < a$). Hence we can apply Theorem 13-23 with $S = (a, b)$ to obtain an expansion about x_0 of the form

$$f(x) = \sum_{k=0}^{\infty} b_k(x - x_0)^k, \tag{32}$$

valid for each x in $N(x_0, R)$, where R is any positive number satisfying the condition in (19). In this case, (19) requires

$$N(x_0, R) \subset (a, b).$$

By (24), the coefficient b_k must be $f^{(k)}(x_0)/k!$. This completes the proof.

13-20 The binomial series. As an example illustrating the use of Bernstein's theorem, we will obtain the following expansion, known as the *binomial series:*

$$(1 + x)^a = \sum_{n=0}^{\infty} \binom{a}{n} x^n, \qquad \text{if } -1 < x < 1, \tag{33}$$

where a is an arbitrary real number and

$$\binom{a}{n} = \frac{a(a-1)\cdots(a-n+1)}{n!}.$$

Bernstein's theorem is not directly applicable in this case. However, we can argue as follows. Let $f(x) = (1 - x)^{-c}$, where $c > 0$ and $x < 1$. Then

$$f^{(n)}(x) = c(c+1)\cdots(c+n-1)(1-x)^{-c-n}$$

and hence $f^{(n)}(x) \geq 0$ for each n, provided that $x < 1$. Therefore we may apply Theorem 13-32, with $x_0 = 0$, $a = -1$ and $b = 1$. We find

$$\frac{1}{(1-x)^c} = \sum_{k=0}^{\infty} \binom{-c}{k} (-1)^k x^k, \qquad \text{if } -1 < x < 1. \tag{34}$$

Replacing c by $-a$ and x by $-x$ in (34), we find that (33) is valid for each $a < 0$. But now the validity of (33) can be extended to all real a by successive integration.

Of course, if a is a positive integer, say $a = m$, then $\binom{m}{n} = 0$ for $n > m$, and (33) reduces to a finite sum (the Binomial Theorem).

13-21 Abel's limit theorem. If $-1 < x < 1$, integration of the geometric series

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

gives us the series expansion

$$\log(1-x) = -\sum_{n=1}^{\infty} \frac{x^n}{n}, \tag{35}$$

also valid for $-1 < x < 1$. If we put $x = -1$ in the right-hand side of (35), we obtain a convergent alternating series, namely, $\sum(-1)^{n+1}/n$. Can we also put $x = -1$ in the left-hand side of (35)? The next theorem answers this question in the affirmative.

13-33 THEOREM (*Abel's limit theorem*). *Assume that we have*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \qquad \text{if } -r < x < r. \tag{36}$$

If the series also converges at $x = r$, then the limit $\lim_{x\to r-} f(x)$ *exists and we have*

$$\lim_{x\to r-} f(x) = \sum_{n=0}^{\infty} a_n r^n.$$

Proof. For simplicity, assume that $r = 1$ (this amounts to a change in scale). Then we are given that $f(x) = \sum a_n x^n$ for $-1 < x < 1$ and that $\sum a_n$ converges. Let us write $f(1) = \sum_{n=0}^{\infty} a_n$. We are to prove that $\lim_{x\to 1-} f(x) = f(1)$, or, in other words, that f is continuous from the left at $x = 1$.

If we multiply the series for $f(x)$ by the geometric series and use Theorem 13-26, we find

$$\frac{1}{1-x} f(x) = \sum_{n=0}^{\infty} c_n x^n, \qquad \text{where } c_n = \sum_{k=0}^{n} a_k.$$

Hence we have

$$(37) \quad f(x) - f(1) = (1 - x) \sum_{n=0}^{\infty} [c_n - f(1)]x^n, \qquad \text{if } -1 < x < 1$$

By hypothesis, $\lim_{n\to\infty} c_n = f(1)$ Therefore, given $\epsilon > 0$, we can find N such that $n \geq N$ implies $|c_n - f(1)| < \epsilon/2$ If we split the sum (37) into two parts, we get

(38)

$$f(x) - f(1) = (1 - x) \sum_{n=0}^{N-1} [c_n - f(1)]x^n + (1 - x) \sum_{n=N}^{\infty} [c_n - f(1)]x^n.$$

Let M denote the largest of the N numbers $|c_n - f(1)|$, $n = 0, 1, 2, \ldots, N - 1$ If $0 < x < 1$, (38) gives us

$$|f(x) - f(1)| \leq (1 - x)NM + (1 - x)\frac{\epsilon}{2} \sum_{n=N}^{\infty} x^n$$

$$= (1 - x)NM + (1 - x)\frac{\epsilon}{2}\frac{x^N}{1 - x} < (1 - x)NM + \frac{\epsilon}{2}$$

Now let $\delta = \epsilon/2NM$ Then $0 < 1 - x < \delta$ implies $|f(x) - f(1)| < \epsilon$, which means $\lim_{x\to 1-} f(x) = f(1)$ This completes the proof

Example We may put $x = -1$ in (35) to obtain

$$\log 2 = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}.$$

(See Exercise 12-18 for another derivation of this formula)

As an application of Abel's theorem we can derive the following result on multiplication of series

13-34 Theorem *Let $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be two convergent series and let $\sum_{n=0}^{\infty} c_n$ denote their Cauchy product If $\sum_{n=0}^{\infty} c_n$ converges, we have*

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right)$$

Note. This result is similar to Theorem 12-46 except that we do not assume absolute convergence of either of the two given series However, we *do* assume convergence of their Cauchy product.

Proof. The two power series $\sum a_n x^n$ and $\sum b_n x^n$ both converge for $x = 1$, and hence they converge in the neighborhood $N(0; 1)$. Keep $|x| < 1$ and write

$$\sum_{n=0}^{\infty} c_n x^n = \left(\sum_{n=0}^{\infty} a_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right),$$

using Theorem 13–26. Now let $x \to 1-$ and apply Abel's theorem.

13–22 Tauber's theorem. The converse of Abel's limit theorem is false in general. That is, if f is given by (36), the limit $f(r-)$ may exist but yet the series $\sum a_n r^n$ may fail to converge. For example, take $a_n = (-1)^n$. Then $f(x) = 1/(1 + x)$ if $-1 < x < 1$ and $f(x) \to \frac{1}{2}$ as $x \to 1-$. However, $\sum(-1)^n$ diverges. A. Tauber (1897) discovered that by placing further restrictions on the coefficients a_n, one can obtain a converse to Abel's theorem. A large number of such results are now known and they are referred to as *Tauberian theorems.* The simplest of these, sometimes called *Tauber's first theorem,* is the following:

13–35 THEOREM (*Tauber*). *Let* $f(x) = \sum_{n=0}^{\infty} a_n x^n$ *for* $-1 < x < 1$, *and assume that* $\lim_{n\to\infty} n a_n = 0$. *If* $f(x) \to S$ *as* $x \to 1-$, *then* $\sum_{n=0}^{\infty} a_n$ *converges and has sum* S.

Proof. Let $n\sigma_n = \sum_{k=0}^{n} k|a_k|$. Then $\sigma_n \to 0$ as $n \to \infty$. (See Note following Theorem 12–48.) Also, $\lim_{n\to\infty} f(x_n) = S$ if $x_n = 1 - 1/n$. Hence, given $\epsilon > 0$, we can choose N so that $n \geq N$ implies

$$|f(x_n) - S| < \frac{\epsilon}{3}, \qquad \sigma_n < \frac{\epsilon}{3}, \qquad n|a_n| < \frac{\epsilon}{3}.$$

Now let $s_n = \sum_{k=0}^{n} a_k$. Then, for $-1 < x < 1$, we can write

$$s_n - S = f(x) - S + \sum_{k=0}^{n} a_k(1 - x^k) - \sum_{k=n+1}^{\infty} a_k x^k.$$

Now keep x in $(0, 1)$. Then

$$(1 - x^k) = (1 - x)(1 + x + \cdots + x^{k-1}) \leq k(1 - x)$$

for each k. Therefore, if $n \geq N$ and $0 < x < 1$, we have

$$|s_n - S| \leq |f(x) - S| + (1 - x)\sum_{k=0}^{n} k|a_k| + \frac{\epsilon}{3n(1 - x)}.$$

Taking $x = x_n = 1 - 1/n$, we find $|s_n - S| < \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon$. This completes the proof.

NOTE. See Exercise 13–34 for another Tauberian theorem.

Uniform convergence

13–1. Assume that $f_n \to f$ uniformly on S and that each f_n is bounded on S Show that $\{f_n\}$ is uniformly bounded on S

13–2. Define two sequences $\{f_n\}$ and $\{g_n\}$ as follows

$$f_n(x) = x\left(1 + \frac{1}{n}\right), \qquad \text{if } x \in E_1, \quad n = 1, 2, \ldots,$$

$$g_n(x) = \begin{cases} \dfrac{1}{n}, & \text{if } x = 0 \text{ or if } x \text{ is irrational,} \\ b + \dfrac{1}{n}, & \text{if } x \text{ is rational, say } x = \dfrac{a}{b}, \quad b > 0. \end{cases}$$

Let $h_n(x) = f_n(x)g_n(x)$.

(a) Show that both $\{f_n\}$ and $\{g_n\}$ converge uniformly on every finite interval.

(b) Show that $\{h_n\}$ does not converge uniformly on any finite interval.

13–3. Assume that $f_n \to f$ uniformly on S, $g_n \to g$ uniformly on S.

(a) Show that $f_n + g_n \to f + g$ uniformly on S.

(b) Let $h_n(x) = f_n(x)g_n(x)$, $h(x) = f(x)g(x)$, if $x \in S$ Exercise 13–2 shows that the assertion $h_n \to h$ uniformly on S is, in general, incorrect. Prove that it is correct if each f_n and each g_n is bounded on S. Also, point out why this hypothesis is not satisfied in Exercise 13–2

13–4. Assume that $f_n \to f$ uniformly on S and suppose there is a constant $M > 0$ such that $|f_n(x)| \leq M$ for all x in S and all n Let g be continuous on the closed disk $\bar{N}(0, M)$ and define $h_n(x) = g[f_n(x)]$, $h(x) = g[f(x)]$, if $x \in S$. Show that $h_n \to h$ uniformly on S.

13–5. (a) Let $f_n(x) = 1/(nx + 1)$ if $0 < x < 1$, $n = 1, 2, \ldots$ Show that $\{f_n\}$ converges pointwise but not uniformly on $(0, 1)$.

(b) Let $g_n(x) = x/(nx + 1)$ if $0 < x < 1$, $n = 1, 2, \ldots$ Show that $g_n \to 0$ uniformly on $(0, 1)$.

13–6. Let $\{f_n\}$ be a sequence of continuous functions defined on a compact set S and assume that $\{f_n\}$ converges pointwise on S to a limit function f. Show that $f_n \to f$ uniformly on S if, and only if, the following two conditions hold.

(i) The limit function f is continuous on S

(ii) For every $\epsilon > 0$, there exists an $m > 0$ and a $\delta > 0$ such that $n > m$ and $|f_k(x) - f(x)| < \delta$ implies $|f_{k+n}(x) - f(x)| < \epsilon$ for all x in S and all $k = 1, 2, \ldots$

Hint: To prove the sufficiency of (i) and (ii), show that for each x_0 in S there is neighborhood $N(x_0)$ and an integer k (depending on x_0) such that

$$|f_k(x) - f(x)| < \delta \qquad \text{if } x \in N(x_0).$$

y compactness, a finite set of integers, say $A = \{k_1, \ldots, k_r\}$, has the property at, for each x in S, some k in A satisfies $|f_k(x) - f(x)| < \delta$. Uniform conergence is an easy consequence of this fact.

13–7. (a) Use Exercise 13–6 to prove the following theorem of Dini: *If $\{f_n\}$ is a sequence of real-valued continuous functions converging pointwise to a continuous limit function f on a compact set S, and if $f_n(x) \geq f_{n+1}(x)$ for each x in S and every $n = 1, 2, \ldots$, then $f_n \to f$ uniformly on S.*

(b) Use the sequence in Exercise 13–5(a) to show that compactness of S is essential in Dini's theorem.

13–8. Let $f_n(x) = nx(1-x)^n$. Show that $\{f_n\}$ converges pointwise but not niformly on the interval [0, 1]. (See Fig. 13–1.) Verify that term-by-term tegration leads to a correct result in this case.

13–9. Show that $\sum x^n(1-x)$ converges pointwise but not uniformly on [0, 1], hereas $\sum(-1)^n x^n(1-x)$ converges uniformly on [0, 1]. This illustrates that niform convergence of $\sum f_n(x)$ along with pointwise convergence of $\sum|f_n(x)|$ oes not necessarily imply uniform convergence of $\sum|f_n(x)|$.

13–10. Assume that $g_{n+1}(x) \leq g_n(x)$ for each x in T and each $n = 1, 2, \ldots$, nd suppose that $g_n \to 0$ uniformly on T. Show that $\sum(-1)^{n+1}g_n(x)$ converges niformly on T.

13–11. Prove Abel's test for uniform convergence: Let $\{g_n\}$ be a sequence of eal-valued functions such that $g_{n+1}(x) \leq g_n(x)$ for each x in T and for every $= 1, 2, \ldots$. If $\{g_n\}$ is uniformly bounded on T and if $\sum f_n(x)$ converges unirmly on T, then $\sum f_n(x)g_n(x)$ also converges uniformly on T.

13–12. Let $f_n(x) = x/(1 + nx^2)$ if $x \in E_1$, $n = 1, 2, \ldots$. Find the limit nction f of the sequence $\{f_n\}$ and the limit function g of the sequence $\{f'_n\}$.

(a) Show that $f'(x)$ exists for every x but that $f'(0) \neq g(0)$. For what values f x is $f'(x) = g(x)$?

(b) In what subintervals of E_1 does $f_n \to f$ uniformly?

(c) In what subintervals of E_1 does $f'_n \to g$ uniformly?

13–13. Let $f_n(x) = (1/n)e^{-n^2x^2}$ if $x \in E_1$, $n = 1, 2, \ldots$. Show that $f_n \to 0$ niformly on E_1, that $f'_n \to 0$ pointwise on E_1, but that the convergence of $\{f'_n\}$ not uniform on any interval containing the origin.

13–14. Let $\{f_n\}$ be a sequence of real-valued continuous functions defined on), 1] and assume that $f_n \to f$ uniformly on [0, 1]. Prove or disprove

$$\lim_{n\to\infty} \int_0^{1-1/n} f_n(x)\,dx = \int_0^1 f(x)\,dx.$$

13–15. Let $f_n(x) = 1/(1 + n^2x^2)$ if $0 \le x \le 1$, $n = 1, 2, \ldots$. Show that $\{f_n\}$ converges pointwise but not uniformly on $[0, 1]$. Is term-by-term integration permissible?

13–16. Prove that $\sum_{n=1}^{\infty} x/n^{\alpha}(1 + nx^2)$ converges uniformly on every finite interval in E_1 if $\alpha > \frac{1}{2}$. Is the convergence uniform on E_1?

13–17. Show that the series $\sum_{n=1}^{\infty} ((-1)^n/\sqrt{n}) \sin (1 + (x/n))$ converges uniformly on E_1.

13–18. Show that the series $\sum_{n=0}^{\infty} (x^{2n+1}/(2n + 1) - x^{n+1}/(2n + 2))$ converges pointwise but not uniformly on $[0, 1]$.

13–19. Show that $\sum_{n=1}^{\infty} a_n \sin nx$ and $\sum_{n=1}^{\infty} a_n \cos nx$ are uniformly convergent on E_1 if $\sum_{n=1}^{\infty} |a_n|$ converges.

13–20. Let $\{a_n\}$ be a decreasing sequence of positive terms. Show that the series $\sum a_n \sin nx$ converges uniformly on E_1 if, and only if, $na_n \to 0$ as $n \to \infty$.

13–21. Given a convergent series $\sum_{n=1}^{\infty} a_n$. Show that the Dirichlet series $\sum_{n=1}^{\infty} a_n n^{-s}$ converges uniformly on the half-infinite interval $0 \le s < +\infty$. Use this to prove that $\lim_{s\to 0+} \sum_{n=1}^{\infty} a_n n^{-s} = \sum_{n=1}^{\infty} a_n$.

13–22. Prove that the series $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ converges uniformly on every half-infinite interval $1 + h \le s < +\infty$, where $h > 0$. Show that the equation

$$\zeta'(s) = -\sum_{n=1}^{\infty} \frac{\log n}{n^s}$$

is valid for each $s > 1$ and obtain a similar formula for the kth derivative $\zeta^{(k)}(s)$.

Mean convergence

13–23. Let $f_n(x) = n^{3/2}xe^{-n^2x^2}$. Show that $\{f_n\}$ converges pointwise to 0 on $[-1, 1]$ but that $\text{l.i.m.}_{n\to\infty} f_n \ne 0$ on $[-1, 1]$.

13–24. Assume that $\{f_n\}$ converges pointwise to f on $[a, b]$ and that $\text{l.i.m.}_{n\to\infty} f_n = g$ on $[a, b]$. Show that $f = g$ if both f and g are continuous on $[a, b]$.

13–25. Let $f_n(x) = \cos^n x$ if $0 \le x \le \pi$.

(a) Show that $\text{l.i.m.}_{n\to\infty} f_n = 0$ on $[0, \pi]$ but that $\{f_n(\pi)\}$ does not converge.

(b) Show that $\{f_n\}$ converges pointwise but not uniformly on $[0, \pi/2]$.

13–26. Let $f_n(x) = 0$ if $0 \le x \le 1/n$ or if $2/n \le x \le 1$, and let $f_n(x) = n$ if $1/n < x < 2/n$. Show that $\{f_n\}$ converges pointwise to 0 on $[0, 1]$ but that $\text{l.i.m.}_{n\to\infty} f_n \ne 0$ on $[0, 1]$.

Power series.

13–27. If r is the radius of convergence of $\sum a_n(z - z_0)^n$, show that

$$\liminf_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right| \le r \le \limsup_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right|.$$

13–28. Given that the power series $\sum_{n=0}^{\infty} a_n z^n$ has radius of convergence 2. Find the radius of convergence of each of the following series:

$$\text{(a)}\ \sum_{n=0}^{\infty} a_n^k z^n, \qquad \text{(b)}\ \sum_{n=0}^{\infty} a_n z^{kn}, \qquad \text{(c)}\ \sum_{n=0}^{\infty} a_n z^{n^2}.$$

In (a) and (b), k is a fixed positive integer.

13–29. Given a power series $\sum_{n=0}^{\infty} a_n x^n$ whose coefficients are related by an equation of the form $a_n + Aa_{n-1} + Ba_{n-2} = 0$ ($n = 2, 3, \ldots$). Show that for any x for which the series converges, its sum is

$$\frac{a_0 + (a_1 + Aa_0)x}{1 + Ax + Bx^2}.$$

13–30. Let $f(x) = e^{-1/x^2}$ if $x \neq 0$, $f(0) = 0$.

(a) Show that $f^{(n)}(0)$ exists for all $n \ge 1$.

(b) Show that the Taylor's series about $x_0 = 0$ generated by f converges everywhere on E_1 but that it represents f only at the origin.

13–31. Show that the binomial series $(1 + x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$ exhibits the following behavior at the points $x = \pm 1$.

(a) If $x = -1$, the series converges for $\alpha \ge 0$ and diverges for $\alpha < 0$.

(b) If $x = 1$, the series diverges for $\alpha \le -1$, converges conditionally for $-1 < \alpha < 0$, and converges absolutely for $\alpha \ge 0$.

13–32. Show that $\sum a_n x^n$ converges uniformly on $[0, 1]$ if $\sum a_n$ converges. Use this fact to give another proof of Abel's limit theorem.

13–33. If each $a_n \ge 0$ and if $\sum a_n$ diverges, show that $\sum a_n x^n \to +\infty$ as $x \to 1-$. (Assume $\sum a_n x^n$ converges for $|x| < 1$.)

13–34. If each $a_n \ge 0$ and if $\lim_{x\to 1-} \sum a_n x^n$ exists and equals A, then $\sum a_n$ converges and has sum A. (Compare with Theorem 13–35.)

13–35. For each real t, define $f_t(x) = xe^{xt}/(e^x - 1)$ if $x \in E_1$, $x \neq 0$, $f_t(0) = 1$.

(a) Show that there exists a neighborhood $N(0; \delta)$ of the origin in which f_t is represented by a power series in x.

(b) Define $P_0(t), P_1(t), P_2(t), \ldots$, by the equation

$$f_t(x) = \sum_{n=0}^{\infty} P_n(t) \frac{x^n}{n!}, \qquad \text{if } x \in N(0, \delta),$$

and use the identity

$$\sum_{n=0}^{\infty} P_n(t) \frac{x^n}{n!} = e^{tx} \sum_{n=0}^{\infty} P_n(0) \frac{x^n}{n!}$$

to prove that $P_n(t) = \sum_{k=0}^{n} \binom{n}{k} P_k(0) t^{n-k}$. This shows that each function P_n is a polynomial. These are the *Bernoulli polynomials*. The numbers $B_n = P_n(0) (n = 0, 1, 2, \ldots)$ are called the *Bernoulli numbers*. Derive the following further properties:

(c) $B_0 = 1, \quad B_1 = -\frac{1}{2}, \quad \sum_{k=0}^{n-1} \binom{n}{k} B_k = 0, \qquad$ if $n = 2, 3, \ldots$

(d) $P_n'(t) = nP_{n-1}(t), \qquad$ if $n = 1, 2, \ldots$

(e) $P_n(t + 1) - P_n(t) = nt^{n-1} \qquad$ if $n = 1, 2, \ldots$

(f) $P_n(1 - t) = (-1)^n P_n(t) \qquad$ (g) $B_{2n+1} = 0 \qquad$ if $n = 1, 2, \ldots$

(h) $1^n + 2^n + \cdots + (k - 1)^n = \dfrac{P_{n+1}(k) - P_{n+1}(0)}{n + 1} \quad (n = 2, 3, \ldots)$

References

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CHAPTER 14

IMPROPER RIEMANN-STIELTJES INTEGRALS

14–1 Introduction. Riemann-Stieltjes integrals of the form $\int_a^b f\,d\alpha$ were discussed in detail in Chapter 9 under the restrictions that both f and α are defined and bounded on a finite interval $[a, b]$. In this chapter we shall extend the scope of integration by relaxing these restrictions. To begin with, we shall study the behavior of $\int_a^b f\,d\alpha$ as $b \to +\infty$. This leads to the notion of an *infinite integral* (also called an *improper integral of the first kind*), denoted by $\int_a^\infty f\,d\alpha$. Such integrals are, in many respects, analogous to infinite series. As a matter of fact, every convergent series can be expressed as an infinite Riemann-Stieltjes integral. Many important functions in analysis appear as infinite integrals and a study of some of the general properties of such functions will be undertaken later in this chapter.

We shall also be concerned with integrals in which the interval $[a, b]$ remains finite but the integrand becomes unbounded at one or more points. These integrals, called *improper integrals of the second kind*, are also defined by a limit process. To distinguish between improper integrals and those discussed in Chapter 9, the latter are usually referred to as *proper* integrals.

NOTE. All the functions in this chapter will be assumed to be *real-valued* unless a statement is made to the contrary. Improper Riemann-Stieltjes integrals in which both the integrand f and the integrator α are complex-valued can be introduced by exactly the same definitions we shall use in the real case. However, for our purposes it will suffice to deal with complex improper integrals by treating the real and imaginary parts separately.

14–2 Infinite Riemann-Stieltjes integrals. We shall adopt the following notations in this chapter: The set of all functions f such that $f \in R(\alpha)$ on $[a, b]$ will be denoted by $R(\alpha; a, b)$. When $\alpha(x) = x$, we simply write $R(a, b)$ for this set. It must be understood that when this notation is used, $[a, b]$ is a *finite interval* and both f and α are real-valued functions *defined and bounded* on $[a, b]$. The notation $\alpha \nearrow$ on $[a, +\infty)$ will mean that α is monotonically increasing on $[a, +\infty)$.

14–1 DEFINITION. *Assume that $f \in R(\alpha; a, b)$ for every $b \geq a$. Keep a, α, and f fixed and define a function I on $[a, +\infty)$ as follows:*

$$(1) \qquad I(b) = \int_a^b f(x)\,d\alpha(x), \qquad \text{if } b \geq a.$$

The function I so defined is called an infinite integral (or an improper integral of the first kind) and is denoted by the symbol $\int_a^\infty f(x)\,d\alpha(x)$ or by $\int_a^\infty f\,d\alpha$. The integral $\int_a^\infty f\,d\alpha$ is said to converge if the limit

$$\lim_{b \to +\infty} I(b) \tag{2}$$

exists (finite). Otherwise, $\int_a^\infty f\,d\alpha$ is said to diverge. If the limit in (2) exists and equals A, the number A is called the value of the integral and we write $\int_a^\infty f\,d\alpha = A$.

NOTE. This definition is entirely analogous to that of an infinite series. (See Definition 12–7.) The function values $I(b)$ play the role of partial sums and may be referred to as "partial integrals."

EXAMPLE 1. Since $\int_1^b x^{-p}\,dx = (1 - b^{1-p})/(p - 1)$ if $p \neq 1$, the integral $\int_1^\infty x^{-p}\,dx$ diverges if $p < 1$. When $p > 1$, it converges and has the value $1/(p - 1)$. If $p = 1$, we get $\int_1^b x^{-1}\,dx = \log b$, so that $\int_1^\infty x^{-1}\,dx$ also diverges.

EXAMPLE 2. Since $\int_0^b \sin 2\pi x\,dx = (1 - \cos 2\pi b)/2\pi$, the integral $\int_0^\infty \sin 2\pi x\,dx$ diverges.

Infinite integrals of the form $\int_{-\infty}^b f\,d\alpha$ are similarly defined. If $\int_{-\infty}^a f\,d\alpha$ and $\int_a^\infty f\,d\alpha$ are both convergent for some value of a, we say that the integral $\int_{-\infty}^\infty f\,d\alpha$ is convergent and its value is defined to be the sum

$$\int_{-\infty}^{\infty} f\,d\alpha = \int_{-\infty}^{a} f\,d\alpha + \int_{a}^{\infty} f\,d\alpha.$$

The choice of the point a is clearly immaterial.

If the integral $\int_{-\infty}^\infty f\,d\alpha$ converges, its value is equal to the limit

$$\lim_{b \to +\infty} \int_{-b}^{b} f\,d\alpha. \tag{3}$$

However, it is important to realize that the limit in (3) might still exist even when the integral $\int_{-\infty}^\infty f\,d\alpha$ is divergent (for example, take $f(x) = \alpha(x) = x$ for every x). In this case, the limit in (3) is called the *Cauchy principal value* of $\int_{-\infty}^\infty f\,d\alpha$.

Many of the theorems in Chapter 9 can be converted into theorems on infinite integrals. To illustrate the straightforward manner in which some of these extensions can be made, let us consider the formula for integration by parts (Theorem 9–6):

$$\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a).$$

Since b appears in three terms of this equation, there are three limits to take into consideration as $b \to +\infty$. If two of these limits exist, the third also exists and we get the formula

$$\int_a^\infty f\,d\alpha + \int_a^\infty \alpha\,df = \lim_{b\to+\infty} f(b)\alpha(b) - f(a)\alpha(a).$$

We can extend Theorems 9–2 through 9–13 to infinite integrals in much the same way. However, it is not necessary to develop the details of these extensions any further, since, in any particular situation, it suffices to apply the required theorem to a finite interval $[a, b]$ and then let $b \to +\infty$.

14–3 Tests for convergence of infinite integrals.

14–2 THEOREM. *Assume that $\alpha \nearrow$ on $[a, +\infty)$ and suppose that $f \in R(\alpha; a, b)$ for every $b \geq a$. Assume that $f(x) \geq 0$ for each $x \geq a$. Then $\int_a^\infty f\,d\alpha$ converges if, and only if, there exists a constant $M > 0$ such that $\int_a^b f\,d\alpha \leq M$ for every $b \geq a$.*

Proof. Define $I(b)$ by equation (1). Then $I \nearrow$ on $[a, +\infty)$ and hence $\lim_{b\to+\infty} I(b) = \sup\{I(b) \mid b \geq a\}$ and the theorem follows easily from this fact. Note also that $\int_a^\infty f\,d\alpha \leq M$ whenever the integral converges.

14–3 THEOREM (*Comparison test*). *Assume that $\alpha \nearrow$ on $[a, +\infty)$. If $f \in R(\alpha; a, b)$ for every $b \geq a$, if $0 \leq f(x) \leq g(x)$ for every $x \geq a$, and if $\int_a^\infty g\,d\alpha$ converges, then $\int_a^\infty f\,d\alpha$ also converges and we have*

$$\int_a^\infty f\,d\alpha \leq \int_a^\infty g\,d\alpha.$$

Proof. $\int_a^b f\,d\alpha \leq \int_a^b g\,d\alpha \leq \int_a^\infty g\,d\alpha$.

14–4 THEOREM (*Limit comparison test*). *Assume that $\alpha \nearrow$ on $[a, +\infty)$. Suppose that $f \in R(\alpha; a, b)$ and that $g \in R(\alpha; a, b)$ for every $b \geq a$, where $f(x) \geq 0$ and $g(x) \geq 0$ if $x \geq a$. If*

$$\lim_{x\to+\infty} \frac{f(x)}{g(x)} = 1,$$

then $\int_a^\infty f\,d\alpha$ and $\int_a^\infty g\,d\alpha$ both converge or both diverge.

Proof. There exists an N such that $\frac{1}{2} \le f(x)/g(x) \le \frac{3}{2}$ if $x \ge N$. The conclusion follows by applying Theorem 14-3 twice

NOTE Theorem 14-4 also holds if $\lim_{x\to+\infty} f(x)/g(x) = c$, provided that $c \ne 0$ If $c = 0$, we can only conclude that convergence of $\int_a^\infty g\,d\alpha$ implies convergence of $\int_a^\infty f\,d\alpha$ (Compare with Theorem 12-21)

EXAMPLE 3 For every real p, the integral $\int_1^\infty e^{-x}x^p\,dx$ converges This is seen by comparison with $\int_1^\infty x^{-2}\,dx$, since $\lim_{x\to+\infty} e^{-x}x^p/x^{-2} = 0$

14-5 THEOREM *Assume that $\alpha \nearrow$ on $[a, +\infty)$ If $f \in R(\alpha, a, b)$ for every $b \ge a$ and if $\int_a^\infty |f|\,d\alpha$ converges, then $\int_a^\infty f\,d\alpha$ also converges*

Proof. If $x \ge a$, we have $0 \le |f(x)| - f(x) \le 2|f(x)|$ Hence, $\int_a^\infty (|f| - f)\,d\alpha$ converges (by Theorem 14-3) Subtracting from $\int_a^\infty |f|\,d\alpha$, we find that $\int_a^\infty f\,d\alpha$ converges (the last step requires the extension of Theorem 9-2 to infinite integrals)

14-6 DEFINITION *Assume that $\alpha \nearrow$ on $[a, +\infty)$ and suppose $f \in R(\alpha, a, b)$ for every $b \ge a$ The integral $\int_a^\infty f\,d\alpha$ is said to converge absolutely if $\int_a^\infty |f|\,d\alpha$ converges It is said to converge conditionally if $\int_a^\infty f\,d\alpha$ converges but $\int_a^\infty |f|\,d\alpha$ diverges*

The comparison tests given above can be used to determine absolute convergence of an infinite integral, but they are of no use in determining conditional convergence The following theorem, somewhat analogous to Dirichlet's test for series (Theorem 12-28), is particularly useful for establishing conditional convergence in certain cases

14-7 THEOREM *Let f be a positive decreasing function defined on $[a, +\infty)$ such that $f(x) \to 0$ as $x \to +\infty$ Let α be bounded on $[a, +\infty)$ and assume that $f \in R(\alpha, a, b)$ for every $b \ge a$ Then the integral $\int_a^\infty f\,d\alpha$ is convergent*

Proof. Integration by parts gives us

$$\int_a^b f\,d\alpha = f(b)\alpha(b) - f(a)\alpha(a) + \int_a^b \alpha\,d(-f)$$

Since α is bounded, $f(b)\alpha(b) \to 0$ as $b \to +\infty$. Therefore, to complete the proof it suffices to establish the convergence of $\int_a^\infty \alpha\,d(-f)$. Actually, this integral converges *absolutely* To see this, suppose $|\alpha(x)| \le M$ for all $x \ge a$ Then $\int_a^\infty M\,d(-f)$ converges and

$$\int_a^b M\, d(-f) = Mf(a) - Mf(b) \to Mf(a) \qquad \text{as } b \to +\infty.$$

Since $-f$ is an increasing function, we can apply Theorem 14-3 to conclude that $\int_a^\infty |\alpha|\, d(-f)$ is convergent. This completes the proof.

EXAMPLE 4. Taking $\alpha(x) = \cos(Ax + B)$, $A \neq 0$, we conclude that

$$\int_a^\infty f(x) \sin(Ax + B)\, dx$$

is convergent whenever f tends monotonically to 0 on $[a, +\infty)$. In particular, the integral

$$\int_1^\infty x^{-p} \sin x\, dx$$

converges if $p > 0$. This integral is absolutely convergent if $p > 1$. However, if $0 < p \leq 1$, the convergence is *conditional*. To see this, write

$$\int_\pi^{n\pi} x^{-p} |\sin x|\, dx = \sum_{k=2}^{n} \int_{(k-1)\pi}^{k\pi} x^{-p} |\sin x|\, dx \geq \frac{2}{\pi} \sum_{k=2}^{n} \frac{1}{k}$$

and recall that $\sum_{k=1}^\infty 1/k$ diverges.

14-8 THEOREM (*Cauchy condition for infinite integrals*). *Assume that*

$$f \in R(\alpha; a, b)$$

for every $b \geq a$. Then the integral $\int_a^\infty f\, d\alpha$ converges if, and only if, for every $\epsilon > 0$ there exists a $B > 0$ such that $c > b > B$ implies

$$\left| \int_b^c f(x)\, d\alpha(x) \right| < \epsilon. \tag{4}$$

Proof. Assume that $\int_a^\infty f\, d\alpha$ converges. Then, for every $\epsilon > 0$, there is a $B > 0$ such that $|\int_a^b f\, d\alpha - \int_a^\infty f\, d\alpha| < \epsilon/2$ whenever $b > B$. Taking $c > b > B$, we get (4).

Conversely, assume that the Cauchy condition holds and define

$$a_n = \int_a^{a+n} f\, d\alpha, \qquad \text{if } n = 1, 2, \ldots.$$

The sequence $\{a_n\}$ is a Cauchy sequence and, therefore, converges. Let $A = \lim_{n\to\infty} a_n$. We will prove that $\int_a^\infty f\,d\alpha = A$. Given $\epsilon > 0$, choose B so that $|\int_b^c f\,d\alpha| < \epsilon/2$ if $c > b > B$ and also so that $|a_n - A| < \epsilon/2$ whenever $a + n \geq B$. Choose an integer $N > B - a$. Then, if $b > a + N$, we have

$$\left|\int_a^b f\,d\alpha - A\right| = \left|\int_a^{a+N} f\,d\alpha - A + \int_{a+N}^b f\,d\alpha\right|$$

$$\leq |a_N - A| + \left|\int_{a+N}^b f\,d\alpha\right| < \epsilon$$

This completes the proof.

NOTE. It follows from Theorem 14-8 that convergence of $\int_a^\infty f\,d\alpha$ implies

$$\lim_{b\to+\infty} \int_b^{b+\epsilon} f\,d\alpha = 0, \qquad \text{for every fixed } \epsilon > 0.$$

However, this does not imply that $f(x) \to 0$ as $x \to +\infty$. A counterexample is the integral $\int_1^\infty \sin x^2\,dx$. Since $\int_1^b \sin x^2\,dx = \frac{1}{2}\int_1^{b^2} t^{-1/2}\sin t\,dt$, Example 4 shows that $\int_1^\infty \sin x^2\,dx$ converges. Nevertheless, the integrand does not tend to 0 as $x \to +\infty$.

14-4 Infinite series and infinite integrals.

14-9 THEOREM. *Every convergent infinite series of real terms, say $\sum_{k=1}^\infty a_k$, can be expressed as a convergent infinite integral. In fact, we have*

$$\sum_{k=1}^\infty a_k = \sum_{k=1}^\infty f(k) = \int_0^\infty f(x)\,d[x],$$

where f is defined on $[0, +\infty)$ as follows:

$$f(x) = a_k \quad \text{if } k - 1 < x \leq k \; (k = 1, 2, \ldots), \; f(0) = 0.$$

As usual, $[x]$ denotes the greatest integer $\leq x$.

Proof. By Theorem 9-12, we have $\int_0^b f(x)\,d[x] = \sum_{k=1}^{[b]} f(k) = \sum_{k=1}^{[b]} a_k$. Now let $b \to +\infty$.

NOTE. If $\sum_{k=1}^\infty a_k$ diverges, then $\int_0^\infty f(x)\,d[x]$ also diverges.

14-10 THEOREM. *Every convergent infinite integral $\int_a^\infty f(x)\,d\alpha(x)$ can be written as a convergent infinite series. In fact, we have*

(5) $$\int_a^\infty f(x)\,d\alpha(x) = \sum_{k=1}^{\infty} a_k, \qquad \textit{where } a_k = \int_{a+k-1}^{a+k} f(x)\,d\alpha(x).$$

Proof. Since $\int_a^\infty f\,d\alpha$ converges, the sequence $\{\int_a^{a+n} f\,d\alpha\}$ also converges. But $\int_a^{a+n} f\,d\alpha = \sum_{k=1}^{n} a_k$. Hence the series $\sum_{k=1}^{\infty} a_k$ converges and equals $\int_a^\infty f\,d\alpha$.

NOTE. Convergence of the series in (5) does not always imply convergence of the integral. For example, suppose $a_k = \int_{k-1}^{k} \sin 2\pi x\,dx$. Then each $a_k = 0$ and $\sum a_k$ converges. However, we found in Example 2 that $\int_0^\infty \sin 2\pi x\,dx$ diverges. On the other hand, if f is *non-negative* and if $f \in R(a, b)$ for every $b \geq a$, convergence of $\sum_{k=1}^{\infty} \int_{a+k-1}^{a+k} f(x)\,dx$ *does* imply convergence of the integral $\int_a^\infty f(x)\,dx$. In this case, the partial integrals $\int_a^b f\,dx$ increase monotonically with b and are all bounded above by the sum of the corresponding series.

14–5 Improper integrals of the second kind.

14–11 DEFINITION. *Let f be defined on the half-open interval $(a, b]$ and assume that $f \in R(\alpha; x, b)$ for every x in $(a, b]$. Define a function I on $(a, b]$ as follows:*

(6) $$I(x) = \int_x^b f\,d\alpha, \qquad \textit{if } x \in (a, b].$$

The function I so defined is called an improper integral of the second kind and is denoted by the symbol $\int_{a+}^b f(t)\,d\alpha(t)$ or by $\int_{a+}^b f\,d\alpha$. The integral $\int_{a+}^b f\,d\alpha$ is said to converge if the limit

(7) $$\lim_{t\to a+} I(t)$$

exists (finite). Otherwise, $\int_{a+}^b f\,d\alpha$ is said to diverge. If the limit in (7) exists and equals A, the number A is called the value of the integral and we write $\int_{a+}^b f\,d\alpha = A$.

EXAMPLE 5. If $b > 0$ and $\epsilon > 0$, we have

$$\int_\epsilon^b x^{-p}\,dx = \frac{b^{1-p} - \epsilon^{1-p}}{1-p}, \qquad \text{if } p \neq 1.$$

When $\epsilon \to 0+$, we find that $\int_{0+}^b x^{-p}\,dx$ converges if $p < 1$ and diverges if $p > 1$. When $p = 1$, we get $\int_\epsilon^b x^{-1}\,dx = \log b - \log \epsilon$, and hence $\int_{0+}^b x^{-1}\,dx$ also diverges. This example can also be treated another way by making a change of variable which converts the integral into an infinite integral In fact, we have

$$\int_{\epsilon}^{b} x^{-p}\,dx = \int_{1/b}^{1/\epsilon} t^{p-2}\,dt,$$

and, when $\epsilon \to 0+$, we find $\int_{0+}^{b} x^{-p}\,dx = \int_{1/b}^{\infty} t^{p-2}\,dt$ Therefore we have convergence if, and only if, $p < 1$

NOTE. Improper integrals of the form $\int_{a}^{b-} f\,d\alpha$ are defined in a similar fashion If the two integrals $\int_{a+}^{c} f\,d\alpha$ and $\int_{c}^{b-} f\,d\alpha$ both converge, we write

$$\int_{a+}^{b-} f\,d\alpha = \int_{a+}^{c} f\,d\alpha + \int_{c}^{b-} f\,d\alpha$$

The definition can be extended in an obvious way to cover the case of any finite number of summands Furthermore, we can also consider "mixed" combinations such as $\int_{a+}^{b} f\,d\alpha + \int_{b}^{\infty} f\,d\alpha$, which we write as $\int_{a+}^{\infty} f\,d\alpha$

EXAMPLE 6 If $p > 0$, the integral $\int_{0+}^{\infty} e^{-x}x^{p-1}\,dx$ converges This integral must be interpreted as a sum, say

$$\int_{0+}^{1} e^{-x}x^{p-1}\,dx + \int_{1}^{\infty} e^{-x}x^{p-1}\,dx. \tag{8}$$

The second integral converges for every real p (by Example 3) To test the first integral, let $t = 1/x$ and write

$$\int_{\epsilon}^{1} e^{-x}x^{p-1}\,dx = \int_{1}^{1/\epsilon} e^{-1/t}t^{-p-1}\,dt.$$

But $\int_{1}^{\infty} e^{-1/t}t^{-p-1}\,dt$ converges for $p > 0$ by comparison with $\int_{1}^{\infty} t^{-p-1}\,dt$ Therefore, $\int_{0+}^{1} e^{-x}x^{p-1}\,dx$ converges for $p > 0$ When $p > 0$, the value of the sum in (8) is denoted by $\Gamma(p)$. The function Γ so defined is called the *Gamma function* Some of its properties are developed in the exercises at the end of the chapter.

NOTE Sometimes the notation $\int_{a}^{b} f(x)\,dx$ is used rather than $\int_{a+}^{b} f(x)\,dx$ for an improper *Riemann* integral of the second kind This is justified by Theorem 9–31 (ii), which tells us that the limit $\lim_{\epsilon\to 0+} \int_{a+\epsilon}^{b} f(x)\,dx$ always exists and equals $\int_{a}^{b} f(x)\,dx$ if $f \in R$ on $[a, b]$. However, for a general integrator α, the limit $\lim_{\epsilon\to 0+} \int_{a+\epsilon}^{b} f(x)\,d\alpha(x)$ and the integral $\int_{a}^{b} f(x)\,d\alpha(x)$ need not be equal if α has a discontinuity at a. (See Exercise 14–9.)

The next four theorems (and their proofs) are similar to Theorems 14–2 through 14–5.

14–12 THEOREM. *Assume that $\alpha \nearrow$ on $(a, b]$ and suppose that $f \in R(\alpha; x, b)$ for every x in $(a, b]$. Assume that $f(x) \geq 0$ for each x in $(a, b]$. Then $\int_{a+}^{b} f\,d\alpha$ converges if, and only if, there exists a constant $M > 0$ such that $\int_{x}^{b} f\,d\alpha \leq M$ for every x in $(a, b]$.*

14–13 THEOREM (*Comparison test*). *Assume that $\alpha \nearrow$ on $(a, b]$. If $f \in R(\alpha; x, b)$ for every x in $(a, b]$, if $0 \leq f(x) \leq g(x)$ for every x in $(a, b]$, and if $\int_{a+}^{b} g\,d\alpha$ converges, then $\int_{a+}^{b} f\,d\alpha$ also converges and we have*

$$\int_{a+}^{b} f\,d\alpha \leq \int_{a+}^{b} g\,d\alpha.$$

14–14 THEOREM (*Limit comparison test*). *Assume that $\alpha \nearrow$ on $(a, b]$. Suppose that $f \in R(\alpha; x, b)$ and that $g \in R(\alpha; x, b)$ for every x in $(a, b]$, where $f(x) \geq 0$ and $g(x) \geq 0$ if $a < x \leq b$. If*

$$\lim_{x \to a+} \frac{f(x)}{g(x)} = c \neq 0,$$

then $\int_{a+}^{b} f\,d\alpha$ and $\int_{a+}^{b} g\,d\alpha$ both converge or both diverge. If $c = 0$, convergence of $\int_{a+}^{b} g\,d\alpha$ implies convergence of $\int_{a+}^{b} f\,d\alpha$.

14–15 THEOREM. *Assume that $\alpha \nearrow$ on $(a, b]$. If $f \in R(\alpha; x, b)$ for every x in $(a, b]$ and if $\int_{a+}^{b} |f|\,d\alpha$ converges, then $\int_{a+}^{b} f\,d\alpha$ also converges.*

14–16 DEFINITION. *Assume that $\alpha \nearrow$ on $(a, b]$ and suppose that $f \in R(\alpha; x, b)$ for every x in $(a, b]$. The integral $\int_{a+}^{b} f\,d\alpha$ is said to converge absolutely if $\int_{a+}^{b} |f|\,d\alpha$ converges. It is said to converge conditionally if $\int_{a+}^{b} f\,d\alpha$ converges but $\int_{a+}^{b} |f|\,d\alpha$ diverges.*

14–6 Uniform convergence of improper integrals. Suppose f is a real-valued function of two variables defined on a subset of E_2 of the form $[a, +\infty) \times S$, where $S \subset E_1$. Suppose further that, for each y in S, the integral $\int_{a}^{\infty} f(x, y)\,d\alpha(x)$ is convergent. If F denotes the function defined by the equation

$$(9) \qquad F(y) = \int_{a}^{\infty} f(x, y)\,d\alpha(x), \qquad \text{if } y \in S,$$

the integral is said to converge *pointwise* to F on S. We wish to investigate some of the properties of functions F defined in this way.

In Chapter 13 we studied properties of functions defined as limits of sequences The situation here is quite similar except that now there are really *two* limit processes involved in the definition of the limit function F First, we must form the proper integral $\int_a^b f(x, y)\, d\alpha(x)$ (this is one limit process), and then we must let $b \to +\infty$ In spite of this added degree of complexity, there is a strong parallel between the theory for infinite integrals and the theory for infinite sequences As in the case of sequences, the concept of *uniform convergence* plays a fundamental role

14-17 DEFINITION *Assume that the integral $\int_a^\infty f(x, y)\, d\alpha(x)$ converges pointwise to F on S The integral is said to converge uniformly on S if, for every $\epsilon > 0$, there exists a $B > 0$ (depending only on ϵ) such that $b > B$ implies*

$$\left| F(y) - \int_a^b f(x, y)\, d\alpha(x) \right| < \epsilon, \qquad \textit{for every } y \textit{ in } S$$

We denote this symbolically by writing

$$\int_a^\infty f(x, y)\, d\alpha(x) = F(y) \qquad \textit{uniformly on } S$$

NOTE A similar definition of uniform convergence of improper integrals of the second kind should be formulated by the reader The reader should also state the analogs of the remaining theorems in this chapter as they apply to improper integrals of the second kind

14-18 THEOREM (*Cauchy condition for uniform convergence*) *The integral $\int_a^\infty f(x, y)\, d\alpha(x)$ converges uniformly on S if, and only if, for every $\epsilon > 0$ there exists a $B > 0$ (depending only on ϵ) such that $c > b > B$ implies*

$$\left| \int_b^c f(x, y)\, d\alpha(x) \right| < \epsilon, \qquad \textit{for every } y \textit{ in } S$$

Proof This is an easy consequence of Theorem 14-8.

14-19 THEOREM (*Weierstrass' M-test*) *Assume that $\alpha \nearrow$ on $[a, +\infty)$ and suppose the integral $\int_a^b f(x, y)\, d\alpha(x)$ exists for every $b \geq a$ and for every y in S If there is a positive function M defined on $[a, +\infty)$ such that the integral $\int_a^\infty M(x)\, d\alpha(x)$ converges and such that*

$$|f(x, y)| \leq M(x), \qquad \textit{for each } x \geq a \textit{ and every } y \textit{ in } S,$$

then the integral $\int_a^\infty f(x, y)\, d\alpha(x)$ converges uniformly on S

Proof. Apply the Cauchy condition in conjunction with the inequality

$$\left| \int_b^c f(x, y)\, d\alpha(x) \right| \leq \int_b^c M(x)\, d\alpha(x).$$

EXAMPLE 7. The integral $\int_0^\infty e^{-xy} \sin x\, dx$ converges uniformly on $[c, +\infty)$ for every $c > 0$. To see this, note that

$$|e^{-xy} \sin x| \leq e^{-cx}, \qquad \text{for every } y \geq c.$$

Now apply the M-test with $M(x) = e^{-cx}$. It is of interest to note here that the value of the proper integral $\int_0^b e^{-xy} \sin x\, dx$ can be computed by the methods of elementary calculus (using integration by parts twice). When $b \to +\infty$, this result leads to the formula

$$\int_0^\infty e^{-xy} \sin x\, dx = \frac{1}{1 + y^2}, \qquad \text{if } y > 0.$$

14–20 THEOREM (*Dirichlet's test for uniform convergence*). *Assume that α is bounded on $[a, +\infty)$ and suppose the integral $\int_a^b f(x, y)\, d\alpha(x)$ exists for every $b \geq a$ and for every y in S. For each fixed y in S, assume that $f(x, y) \leq f(x', y)$ if $a \leq x' < x < +\infty$. Furthermore, suppose there exists a positive function g, defined on $[a, +\infty)$, such that $g(x) \to 0$ as $x \to +\infty$ and such that $x \geq a$ implies*

$$|f(x, y)| \leq g(x), \qquad \textit{for every } y \textit{ in } S.$$

Then the integral $\int_a^\infty f(x, y)\, d\alpha(x)$ converges uniformly on S.

Proof. Let $M > 0$ be an upper bound for $|\alpha|$ on $[a, +\infty)$. Given $\epsilon > 0$, choose $B > a$ such that $x \geq B$ implies $g(x) < \epsilon/(4M)$. If $c > b$, integration by parts yields

$$\int_b^c f\, d\alpha = f(c, y)\alpha(c) - f(b, y)\alpha(b) + \int_b^c \alpha\, d(-f). \tag{10}$$

But, since $-f$ is increasing (for each fixed y), we have

$$\left| \int_b^c \alpha\, d(-f) \right| \leq M \int_b^c d(-f) = Mf(b, y) - Mf(c, y). \tag{11}$$

Now if $c > b > B$, we can combine (10) and (11) to write

$$\left| \int_b^c f\, d\alpha \right| \leq 2Mg(b) + 2Mg(c) < \epsilon$$

for every y in S Therefore the Cauchy condition is satisfied and

$$\int_a^\infty f(x, y)\, d\alpha(x)$$

converges uniformly on S

EXAMPLE 8. Take $\alpha(x) = \cos x$ and $f(x, y) = e^{-xy}/x$ if $x > 0, y \geq 0$ If $S = [0, +\infty)$, the hypotheses of Theorem 14–20 are satisfied with $g(x) = 1/x$ on $[\epsilon, +\infty)$ for every $\epsilon > 0$ Therefore the integral

$$\int_\epsilon^\infty e^{-xy} \frac{\sin x}{x}\, dx$$

converges uniformly on $[0, +\infty)$ if $\epsilon > 0$ If we agree that $\sin x/x$ is to be replaced by 1 when $x = 0$, we also have

$$\int_0^\epsilon e^{-xy} \frac{\sin x}{x}\, dx \leq \epsilon,$$

and hence the integral $\int_0^\infty e^{-xy} \sin x/x\, dx$ also converges uniformly on $[0, +\infty)$.

NOTE. In much of the material that follows we shall have occasion to deal with integrals involving the quotient $\sin x/x$ It will be understood that this quotient is to be replaced by 1 when $x = 0$ Similarly, a quotient of the form $\sin xy/x$ is to be replaced by y, its limit as $x \to 0$. More generally, if we are dealing with an integrand which has removable discontinuities at certain isolated points within the interval of integration, we will agree that these discontinuities are to be "removed" by redefining the integrand suitably at these exceptional points

14–7 Properties of functions defined by improper integrals. In this section we shall be dealing with functions F defined pointwise on an interval $[c, d]$ by an equation of the type

$$F(y) = \int_a^\infty f(x, y)\, d\alpha(x),$$

where the integrand f is defined on a rectangular strip of the form $[a, +\infty) \times [c, d]$. (See Fig. 14–1.) Our aim will be to find conditions on f and α that will endow F with such properties as continuity, differentiability, and integrability.

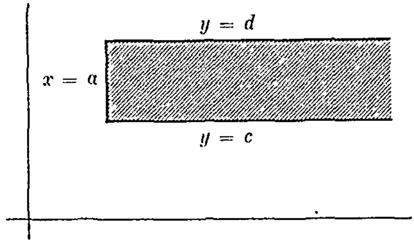

FIG. 14–1. A rectangular strip $[a, +\infty) \times [c, d]$.

NOTE. *All the theorems of this section and the next section have straightforward analogs for improper integrals of the second kind. The reader should formulate these results for himself.*

14–21 THEOREM. *Assume that α is of bounded variation on $[a, b]$ for each $b \geq a$. Let f be continuous on the rectangular strip $[a, +\infty) \times [c, d]$ and assume that*

$$\int_a^\infty f(x, y)\, d\alpha(x) = F(y) \qquad \textit{uniformly on } [c, d].$$

Then F is continuous on $[c, d]$. In other words, if $y_0 \in [c, d]$, we have

$$\lim_{y \to y_0} \int_a^\infty f(x, y)\, d\alpha(x) = \int_a^\infty \lim_{y \to y_0} f(x, y)\, d\alpha(x) = \int_a^\infty f(x, y_0)\, d\alpha(x).$$

Proof. For each $n = 1, 2, \ldots$, define

$$F_n(y) = \int_a^{a+n} f(x, y)\, d\alpha(x), \qquad \text{if } y \in [c, d].$$

Then $F_n \to F$ uniformly on $[c, d]$. Furthermore, each F_n is continuous on $[c, d]$ because of Theorem 9–35. Hence, by Theorem 13–3, the limit function F is also continuous on $[c, d]$.

An important consequence is the following theorem for improper Riemann integrals:

14–22 THEOREM. *Let f be continuous on the rectangular strip $[a, +\infty) \times [c, d]$ and suppose $g \in R$ on $[a, b]$ for every $b \geq a$. If*

$$\int_a^\infty g(x) f(x, y)\, dx = F(y) \qquad \textit{uniformly on } [c, d],$$

then F is continuous on $[c, d]$.

Proof If $\alpha(x) = \int_a^x g(t)\,dt$, then α is of bounded variation on every finite interval $[a, b]$ and $\int_a^b g(x)f(x, y)\,dx = \int_a^b f(x, y)\,d\alpha(x)$, by Theorem 9-36

EXAMPLE 9 Let $F(y) = \int_0^\infty e^{-xy} \sin x/x\,dx$ The integrand is continuous on $[0, +\infty) \times [c, d]$ for every finite interval $[c, d]$ By Example 8, the integral converges uniformly on $[0, +\infty)$ Hence, F is continuous on every finite interval of the form $[0, d]$ In particular, we have

$$\lim_{y \to 0+} F(y) = F(0) = \int_0^\infty \frac{\sin x}{x}\,dx \tag{12}$$

14-23 THEOREM *Suppose that $\int_a^\infty f(x, y)\,d\alpha(x)$ converges pointwise to F on (c, d) and that $\int_a^\infty D_2 f(x, y)\,d\alpha(x)$ converges uniformly on (c, d) Let*

$$F_b(y) = \int_a^{a+b} f(x, y)\,d\alpha(x), \qquad \text{if } b \geq 0 \quad \text{and } y \in (c, d)$$

Assume that for each y in (c, d) the derivative $F_b'(y)$ exists and is given by the formula

$$F_b'(y) = \int_a^{a+b} D_2 f(x, y)\,d\alpha(x), \qquad \text{whenever } b \geq 0. \tag{13}$$

Then, for each y in (c, d), the derivative $F'(y)$ exists and we have

$$F'(y) = \int_a^\infty D_2 f(x, y)\,d\alpha(x) \tag{14}$$

Proof. For each $n = 1, 2, \ldots$, we have

$$F_n(y) = \int_a^{a+n} f(x, y)\,d\alpha(x), \qquad \text{if } y \in (c, d)$$

For each y in (c, d), the sequence $\{F_n(y)\}$ converges to $F(y)$ and the sequence of derivatives $\{F_n'\}$ converges uniformly to G on (c, d), where

$$G(y) = \int_a^\infty D_2 f(x, y)\,d\alpha(x), \qquad \text{if } y \in (c, d)$$

Hence, by Theorem 13–13, we can conclude that $F'(y)$ exists and equals $G(y)$. In other words, (14) holds.

NOTE. In applying Theorem 14–23, the whole difficulty lies in verifying that (13) holds for every $b \geq 0$. The next theorem gives a sufficient condition for the validity of (13).

14–24 THEOREM. *Assume that α is of bounded variation on $[a, b]$ for every $b \geq a$ and suppose the integral $\int_a^\infty f(x, y)\, d\alpha(x)$ converges pointwise to F on $[c, d]$. If the partial derivative $D_2 f$ is continuous on the rectangular strip $[a, +\infty) \times [c, d]$ and if the integral $\int_a^\infty D_2 f(x, y)\, d\alpha(x)$ converges uniformly on (c, d), then $F'(y)$ exists for each y in (c, d) and we have*

$$F'(y) = \int_a^\infty D_2 f(x, y)\, d\alpha(x).$$

Proof. The hypotheses of the present theorem, in conjunction with Theorem 9–37, imply that (13) is valid. Hence (14) is also valid.

NOTE. In particular, when $g \in R(a, b)$ for every $b \geq a$ and $\alpha(x) = \int_a^x g(t)\, dt$, we get

$$F(y) = \int_a^\infty g(x) f(x, y)\, dx \qquad \text{and} \qquad F'(y) = \int_a^\infty g(x)\, D_2 f(x, y)\, dx,$$

provided that the first integral converges pointwise on $[c, d]$ and the second converges uniformly on (c, d).

EXAMPLE 10. Let $f(x, y) = e^{-xy} \sin x / x$ if $x \neq 0$, $f(0, y) = 1$. Then $D_2 f(x, y) = -e^{xy} \sin x$ and hence $D_2 f$ is continuous everywhere on E_2. In Example 7 we found that the integral $\int_0^\infty e^{-xy} \sin x\, dx$ converges uniformly on $[c, +\infty)$ if $c > 0$. Therefore, if we write

$$F(y) = \int_0^\infty e^{-xy} \frac{\sin x}{x}\, dx, \qquad \text{whenever } y > 0,$$

the derivative $F'(y)$ exists for each $y > 0$ and is given by

$$F'(y) = -\int_0^\infty e^{-xy} \sin x\, dx = -\frac{1}{(1 + y^2)}$$

(by Example 7). Hence, $F(y) = C - \tan^{-1} y$ if $y > 0$, where C is a constant. To evaluate C, we will show that $F(y) \to 0$ as $y \to +\infty$. For this purpose we apply Bonnet's theorem (part (ii) of Theorem 9-34) to write

$$\left|\int_{\epsilon}^{b} e^{-xy}\frac{\sin x}{x}\,dx\right| = \frac{e^{-\epsilon y}}{\epsilon}\left|\int_{x_0}^{b}\sin x\,dx\right| \le \frac{2e^{-\epsilon y}}{\epsilon},$$

if $\epsilon > 0$. Hence, $\int_{\epsilon}^{b} e^{-xy}\sin x/x\,dx \to 0$ as $y \to +\infty$. Since

$$\int_{0}^{\epsilon} e^{-xy}\frac{\sin x}{x}\,dx \le \epsilon,$$

it follows that $F(y) \to 0$ as $y \to +\infty$. Therefore, $C = \pi/2$. In other words,

$$\int_{0}^{\infty} e^{-xy}\frac{\sin x}{x}\,dx = \frac{\pi}{2} - \tan^{-1} y, \qquad \text{if } y > 0.$$

But, by formula (12) of Example 9, we can let $y \to 0+$ to obtain

$$\int_{0}^{\infty}\frac{\sin x}{x}\,dx = \frac{\pi}{2}. \tag{15}$$

This formula will be required in Chapter 15.

14-25 THEOREM. *Assume that $\int_a^\infty f(x, y)\,d\alpha(x) = F(y)$ uniformly on $[c, d]$. Let β be of bounded variation on $[c, d]$ and suppose that $F \in R(\beta)$ on $[c, d]$. If the equation*

$$\int_{c}^{d}\left[\int_{a}^{b} f(x, y)\,d\alpha(x)\right]d\beta(y) = \int_{a}^{b}\left[\int_{c}^{d} f(x, y)\,d\beta(y)\right]d\alpha(x) \tag{16}$$

is known to be valid for each $b \ge a$, then we also have

$$\int_{c}^{d}\left[\int_{a}^{\infty} f(x, y)\,d\alpha(x)\right]d\beta(y) = \int_{a}^{\infty}\left[\int_{c}^{d} f(x, y)\,d\beta(y)\right]d\alpha(x). \tag{17}$$

Proof. Assume that β is strictly increasing on $[c, d]$. If $\epsilon > 0$ is given, there exists a $B > 0$ (depending only on ϵ) such that $b > B$ implies

$$\left| \int_a^b f(x, y)\, d\alpha(x) - F(y) \right| < \frac{\epsilon}{2[\beta(d) - \beta(c)]}, \qquad \text{for every } y \text{ in } [c, d].$$

Integration yields

$$\left| \int_c^d \left[\int_a^b f(x, y)\, d\alpha(x) \right] d\beta(y) - \int_c^d F(y)\, d\beta(y) \right| \le \frac{\epsilon}{2} < \epsilon.$$

By (16), we can rewrite this as follows:

$$\left| \int_a^b \left[\int_c^d f(x, y)\, d\beta(y) \right] d\alpha(x) - \int_c^d F(y)\, d\beta(y) \right| < \epsilon.$$

Since this holds for every $b > B$, we obtain (17) as a consequence.

NOTE. In applying Theorem 14–25, the whole difficulty lies in verifying that (16) holds for each $b \ge a$. Sufficient conditions for the validity of (16) are given in the next two theorems.

14–26 THEOREM. *Assume that α is of bounded variation on $[a, b]$ for every $b \ge a$ and let β be of bounded variation on $[c, d]$. If f is continuous on the rectangular strip $[a, +\infty) \times [c, d]$ and if $\int_a^\infty f(x, y)\, d\alpha(x) = F(y)$ uniformly on $[c, d]$, then F is continuous on $[c, d]$ and we have*

$$\int_c^d \left[\int_a^\infty f(x, y)\, d\alpha(x) \right] d\beta(y) = \int_a^\infty \left[\int_c^d f(x, y)\, d\beta(y) \right] d\alpha(x). \tag{18}$$

Proof. By Theorem 14–21, F is continuous on $[c, d]$ and hence $F \in R(\beta)$ on $[c, d]$, by Theorem 9–26. Also, equation (16) is satisfied for each $b \ge a$ because of Theorem 9–39. Hence (17) also holds.

NOTE. If $g \in R$ on $[a, b]$ for every $b \ge a$ and if $h \in R$ on $[c, d]$, we can take $\alpha(x) = \int_a^x g(u)\, du$ and $\beta(y) = \int_c^y h(v)\, dv$. Then (18) becomes

$$\int_c^d \left[\int_a^\infty g(x)h(y)f(x, y)\, dx \right] dy = \int_a^\infty \left[\int_c^d g(x)h(y)f(x, y)\, dy \right] dx, \tag{19}$$

provided that the integral $\int_a^\infty g(x)f(x, y)\, dx$ converges uniformly on $[c, d]$. In this case, the theorem tells us that if f is continuous on $[a, +\infty) \times [c, d]$ and if we have

$$F(y) = \int_a^\infty g(x)f(x, y)\, dx \qquad \text{uniformly on } [c, d], \tag{20}$$

then we may multiply both sides of (20) by $h(y)$ and integrate with respect to y under the integral sign The important thing to note here is that the factors g and h are only assumed to be Riemann-integrable This form of the theorem is especially useful in many applications In particular, it is needed in the proof of the Fourier integral theorem (Theorem 15–24)

14–27 THEOREM *Let f be defined on the rectangular strip $[a, +\infty) \times [c, d]$ If $R = [a, b] \times [c, d]$, assume that the double integral*

$$\int_R f(x, y)\, d(x, y)$$

exists for each $b > a$ If the integral $\int_a^\infty f(x, y)\, dx = F(y)$ uniformly on $[c, d]$ and if $\int_c^d F(y)dy$ exists, then we have

$$\int_c^d \left[\int_a^\infty f(x, y)\, dx\right] dy = \int_a^\infty \left[\int_c^d f(x, y)\, dy\right] dx.$$

Proof By part (v) of Theorem 10–20, equation (16) holds for every $b \geq a$, since both sides are equal to the double integral $\int_R f(x, y)\, d(x, y)$ Hence (17) also holds

14–8 Repeated improper integrals. The last three theorems provide sufficient conditions for the validity of the equation

$$\text{(21)} \qquad \int_c^d \left[\int_a^\infty f(x, y)\, d\alpha(x)\right] d\beta(y) = \int_a^\infty \left[\int_c^d f(x, y)\, d\beta(y)\right] d\alpha(x)$$

We face a much more difficult problem when we seek conditions under which we may write

$$\text{(22)} \qquad \int_c^\infty \left[\int_a^\infty f(x, y)\, d\alpha(x)\right] d\beta(y) = \int_a^\infty \left[\int_c^\infty f(x, y)\, d\beta(y)\right] d\alpha(x)$$

Such "repeated improper integrals" are analogous to "repeated series" Even if both sides of (22) are convergent, they need not have the same value (See Exercise 14–14) In this section we shall give some conditions which allow us to conclude equality of the two repeated integrals in (22)

To see what sort of difficulties arise when we try to prove (22), let us operate in a purely formal way (disregarding, for the moment, questions of convergence) and write

$$\int_c^\infty \left[\int_a^\infty f\,d\alpha\right] d\beta = \lim_{d\to+\infty} \int_c^d \left[\int_a^\infty f\,d\alpha\right] d\beta \tag{23}$$

$$= \lim_{d\to+\infty} \int_a^\infty \left[\int_c^d f\,d\beta\right] d\alpha \tag{24}$$

$$= \int_a^\infty \left[\lim_{d\to+\infty} \int_c^d f\,d\beta\right] d\alpha \tag{25}$$

$$= \int_a^\infty \left[\int_c^\infty f\,d\beta\right] d\alpha.$$

This procedure will lead to a legitimate proof of (22) if we can justify the steps leading from (23) to (24) and from (24) to (25). To get (24) from (23), we need to know that (21) holds for all $d \geq c$. Conditions for the validity of (21) can be obtained from the theorems of the previous section. However, a new difficulty arises in trying to justify the step leading from (24) to (25). The next theorem deals with this problem.

14–28 THEOREM. *Assume that α is of bounded variation on $[a, b]$ for every $b \geq a$. Let f be defined on the set $[a, +\infty) \times [c, +\infty)$ and let*

$$g(x, t) = \int_c^t f(x, y)\,d\beta(y), \qquad \textit{if } x \geq a \quad \textit{and} \quad t \geq c.$$

Suppose that

(a) $\int_a^\infty g(x, t)\,d\alpha(x)$ *converges uniformly on* $[c, +\infty)$,

(b) $\int_c^\infty f(x, y)\,d\beta(y)$ *converges uniformly on* $[a, +\infty)$,

(c) $\int_a^\infty \left[\int_c^\infty f(x, y)\,d\beta(y)\right] d\alpha(x)$ *converges.*

Then we have

$$\lim_{t\to+\infty} \int_a^\infty g(x, t)\,d\alpha(x) = \int_a^\infty \left[\lim_{t\to+\infty} g(x, t)\right] d\alpha(x) \tag{26}$$

or, what is the same thing,

$$(27) \qquad \lim_{t \to +\infty} \int_a^\infty \left[\int_c^t f(x, y)\, d\beta(y)\right] d\alpha(x) = \int_a^\infty \left[\int_c^\infty f(x, y)\, d\beta(y)\right] d\alpha(x)$$

Proof Given $\epsilon > 0$, we can choose B (depending only on ϵ) so that

$$(28) \qquad \left|\int_a^B g(x, t)\, d\alpha(x) - \int_a^\infty g(x, t)\, d\alpha(x)\right| < \frac{\epsilon}{4},$$

for every t in $[c, +\infty)$ [by (a)], and also so that

$$(29) \qquad \left|\int_a^B \left[\int_c^\infty f(x, y)\, d\beta(y)\right] d\alpha(x) - \int_a^\infty \left[\int_c^\infty f(x, y)\, d\beta(y)\right] d\alpha(x)\right| < \frac{\epsilon}{4}.$$

Inequality (29) follows from (c) Furthermore, we can assume that α is strictly increasing on the interval $[a, B]$ Then, for this same ϵ, we can choose T (depending only on ϵ) so that $t \geq T$ implies

$$(30) \qquad \left|g(x, t) - \int_c^\infty f(x, y)\, d\beta(y)\right| < \frac{\epsilon}{4[\alpha(B) - \alpha(a)]},$$

for every x in $[a, +\infty)$ [by (b)]. Integration in (30) yields

$$(31) \qquad \left|\int_a^B g(x, t)\, d\alpha(x) - \int_a^B \left[\int_c^\infty f(x, y)\, d\beta(y)\right] d\alpha(x)\right| \leq \frac{\epsilon}{4} < \frac{\epsilon}{2}.$$

Combining (31), (29), and (28), we find that $t \geq T$ implies

$$\left|\int_a^\infty g(x, t)\, d\alpha(x) - \int_a^\infty \left[\int_c^\infty f(x, y)\, d\beta(y)\right] d\alpha(x)\right| < \epsilon.$$

In other words, (26) holds.

By combining the results of Theorem 14-28 with those of Theorems 14-25, 14-26, and 14-27, we can obtain various sufficient conditions for the validity of (22). For example, combining Theorems 14-28 and 14-26, we get

14-29 THEOREM *Assume that α is of bounded variation on $[a, b]$ for every $b \geq a$ and that β is of bounded variation on $[c, d]$ for every $d \geq c$.*

Let f be continuous on $[a, +\infty) \times [c, +\infty)$ and define

$$g(x, t) = \int_c^t f(x, y)\, d\beta(y), \qquad \text{if } x \geq a \quad \text{and} \quad t \geq c.$$

Suppose that

(a) $\displaystyle\int_a^\infty f(x, y)\, d\alpha(x)$ *converges uniformly on* $[c, +\infty)$,

(b) $\displaystyle\int_a^\infty g(x, t)\, d\alpha(x)$ *converges uniformly on* $[c, +\infty)$,

(c) $\displaystyle\int_c^\infty f(x, y)\, d\beta(y)$ *converges uniformly on* $[a, +\infty)$,

(d) $\displaystyle\int_a^\infty \left[\int_c^\infty f(x, y)\, d\beta(y)\right] d\alpha(x)$ *converges.*

Then we have

$$\int_c^\infty \left[\int_a^\infty f(x, y)\, d\alpha(x)\right] d\beta(y) = \int_a^\infty \left[\int_c^\infty f(x, y)\, d\beta(y)\right] d\alpha(x). \tag{32}$$

Proof. Using (b), (c), (d), along with Theorem 14–28, we have

(33)

$$\int_a^\infty \left[\int_c^\infty f\, d\beta\right] d\alpha = \int_a^\infty \left[\lim_{t\to+\infty} g(x, t)\right] d\alpha(x) = \lim_{t\to+\infty} \int_a^\infty \left[\int_c^t f\, d\beta\right] d\alpha.$$

By (a) and Theorem 14–26, we have $\int_a^\infty [\int_c^t f\, d\beta]\, d\alpha = \int_c^t [\int_a^\infty f\, d\alpha]\, d\beta$. Hence (32) follows from (33).

Another theorem of this type can be obtained by an entirely different kind of argument.

14–30 THEOREM. *Assume that $\alpha \nearrow$ on $[a, +\infty)$, that $\beta \nearrow$ on $[c, +\infty)$, and let f be a non-negative function defined on $[a, +\infty) \times [c, +\infty)$. Suppose that the following two equations are known to be valid:*

$$(34)\quad \int_c^d \left[\int_a^\infty f(x, y)\, d\alpha(x)\right] d\beta(y) = \int_a^\infty \left[\int_c^d f(x, y)\, d\beta(y)\right] d\alpha(x),$$

for all $d \geq c$,

$$(35)\quad \int_a^b \left[\int_c^\infty f(x, y)\, d\beta(y)\right] d\alpha(x) = \int_c^\infty \left[\int_a^b f(x, y)\, d\alpha(x)\right] d\beta(y),$$

for all $b \geq a$

Then, whenever one of the repeated integrals in (36) *converges, the other also converges and we have*

$$(36)\quad \int_c^\infty \left[\int_a^\infty f(x, y)\, d\alpha(x)\right] d\beta(y) = \int_a^\infty \left[\int_c^\infty f(x, y)\, d\beta(y)\right] d\alpha(x).$$

Proof. Suppose the integral $I = \int_c^\infty [\int_a^\infty f\, d\alpha]\, d\beta$ converges Since f is non-negative and α is increasing, we have

$$(37)\quad \int_a^b f(x, y)\, d\alpha(x) \leq \int_a^\infty f(x, y)\, d\alpha(x), \qquad \text{for every } b \geq a$$

Since β is increasing, (37) implies $\int_c^\infty [\int_a^b f\, d\alpha]\, d\beta \leq I$, which, by (35), states that $\int_a^b [\int_c^\infty f\, d\beta]\, d\alpha \leq I$ By Theorem 14–2, this means that $\int_a^\infty [\int_c^\infty f\, d\beta]\, d\alpha$ converges and has a value $\leq I$ A similar argument, using (34) instead of (35), yields the opposite inequality Hence (36) holds

NOTE. We can omit the hypothesis that f be non-negative if, instead, we assume that at least one side of (36) is convergent when f is replaced by $|f|$. When stated in this form, Theorem 14–30 also holds for *complex-valued* functions The proof (for both real and complex f) can be reduced to the case already treated When f is real-valued, we consider the functions $f_1 = |f| + f$ and $f_2 = |f| - f$, both of which are non-negative If one of the integrals in (36) converges when f is replaced by $|f|$, then it also converges when f is replaced by f_1 or by f_2 Furthermore, the validity of (36) for f_1 and f_2 implies its validity for f, since $2f = f_1 - f_2$ When f is complex-valued, say $f = u + iv$, we introduce

$$u_1 = |u| + u, \quad u_2 = |u| - u, \quad v_1 = |v| + v, \quad v_2 = |v| - v$$

(each of which is non-negative) and argue as before on the real and imaginary parts separately.

14–9 Integration of infinite series when improper integrals are involved. In Theorem 13–11 we established the validity of the equation

$$\int_a^b \sum_{n=1}^{\infty} f_n(x)\, d\alpha(x) = \sum_{n=1}^{\infty} \int_a^b f_n(x)\, d\alpha(x) \tag{38}$$

under the following hypotheses:

(a) The series $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on $[a, b]$.

(b) Each $f_n \varepsilon R(\alpha)$ on $[a, b]$, where α is of bounded variation on $[a, b]$.

When we attempt to extend equation (38) to improper integrals, we face essentially the same difficulties that were discussed in the previous section. In fact, by expressing the series $\sum f_n(x)$ as an infinite Riemann-Stieltjes integral, we can reduce the problem to that considered in the previous section. In practice, however, the following simple theorem usually suffices to justify term-by-term integration of an infinite series when improper integrals are involved.

14–31 THEOREM. *Assume that $\alpha \nearrow$ on $[a, +\infty)$ and let $\{f_n\}$ be a sequence of functions defined on $[a, +\infty)$ such that $f_n(x) \geq 0$ for each $x \geq a$ and each $n = 1, 2, \ldots$. If equation* (38) *is known to be valid for each $b \geq a$, then we also have*

$$\int_a^{\infty} \sum_{n=1}^{\infty} f_n(x)\, d\alpha(x) = \sum_{n=1}^{\infty} \int_a^{\infty} f_n(x)\, d\alpha(x), \tag{39}$$

provided that either side of (39) *is convergent.*

Proof. Argue as in the proof of Theorem 14–30.

NOTE. We can omit the hypothesis $f_n(x) \geq 0$, if, instead, we assume that at least one side of (39) is convergent when $f_n(x)$ is replaced by $|f_n(x)|$. When stated in this form, Theorem 14–31 also holds for complex-valued functions. (See the Note following Theorem 14–30.) Of course, a similar result holds for improper integrals of the second kind.

EXAMPLE 11. For each fixed $s > 1$ and each fixed $a > 0$, we have

$$x^{s-1} \sum_{n=1}^{\infty} e^{-nx} = \frac{x^{s-1}}{e^x - 1} \qquad \text{(uniformly on } [a, b]\text{)},$$

for each $b > a$. Hence, if $0 < a < b$, we can apply Theorem 13–11 to write

$$\int_a^b \frac{x^{s-1}}{e^x - 1}\,dx = \int_a^b x^{s-1} \sum_{n=1}^{\infty} e^{-nx}\,dx = \sum_{n=1}^{\infty} \int_a^b x^{s-1} e^{-nx}\,dx.$$

But the series $\sum_{n=1}^{\infty} \int_0^{\infty} x^{s-1} e^{-nx}\,dx$ converges, since its general term is

$$\int_0^{\infty} x^{s-1} e^{-nx}\,dx = n^{-s} \int_0^{\infty} t^{s-1} e^{-t}\,dt = n^{-s}\Gamma(s),$$

where Γ denotes the Gamma function (see Example 6) Hence all the hypotheses of Theorem 14-31 (and its analog for improper integrals of the second kind) are fulfilled and we obtain

$$\int_0^{\infty} \frac{x^{s-1}}{e^x - 1}\,dx = \sum_{n=1}^{\infty} \int_0^{\infty} x^{s-1} e^{-nx}\,dx = \zeta(s)\Gamma(s), \tag{40}$$

where ζ denotes the Riemann zeta function. Formula (40) is of fundamental importance in the theory of the zeta function.

Exercises

14–1. Show that the following improper integrals are convergent:

(a) $\displaystyle\int_2^{\infty} \frac{1}{x\,(\log x)^p}\,dx \quad (p > 1)$, (b) $\displaystyle\int_0^{\infty} x^p e^{-x^q}\,dx \quad (p > 0, q > 0)$,

(c) $\displaystyle\int_1^{\infty} \sin^2 \frac{1}{x}\,dx$, (d) $\displaystyle\int_1^{\infty} \frac{\sin^2 x}{x^2}\,dx$.

14–2. If f is positive and decreasing on $[a, +\infty)$ and if the integral $\int_a^{\infty} f(x)\,dx$ converges, show that $xf(x) \to 0$ as $x \to +\infty$ and that $\int_a^{\infty} x\,df(x)$ converges.

14–3. Show that the following improper integrals are convergent:

(a) $\displaystyle\int_0^{1} \frac{x \log x}{(1 + x)^2}\,dx$, (b) $\displaystyle\int_0^{1} \frac{x^p - 1}{\log x}\,dx \quad (p > -1)$,

(c) $\displaystyle\int_0^{1} \log x \log (1 + x)\,dx$, (d) $\displaystyle\int_0^{1} \frac{\log (1 - x)}{(1 - x)^{1/2}}\,dx$.

14–4. Assume that f is continuous on $[0, 1]$, $f(0) = 0$, $f'(0)$ exists. Show that the integral $\int_0^1 f(x)x^{-3/2}\,dx$ converges.

14–5. Test the following improper integrals for convergence:

(a) $\displaystyle\int_0^{\infty} e^{-(t^2 + t^{-2})}\,dt$, (b) $\displaystyle\int_0^{\infty} \frac{\cos x}{\sqrt{x}}\,dx$,

(c) $\displaystyle\int_1^{\infty} \frac{\log x}{x(x^2 - 1)^{1/2}}\,dx$, (d) $\displaystyle\int_0^{\infty} e^{-x} \sin \frac{1}{x}\,dx$,

(e) $\displaystyle\int_0^{1} \log x \sin \frac{1}{x}\,dx$, (f) $\displaystyle\int_0^{\infty} e^{-x} \log (\cos^2 x)\,dx$,

(g) $\displaystyle\int_0^{\infty} \frac{\sin^2 x}{x}\,dx$, (h) $\displaystyle\int_0^{\infty} \left\{\frac{5e^{-x}}{x} + \frac{x + 2}{x^2}\,(e^{-3x} - e^{-x/2})\right\} dx$.

14-6. Determine those values of p and q for which the following integrals converge

(a) $\displaystyle\int_0^1 x^p(1-x^2)^q\,dx$, (b) $\displaystyle\int_0^\infty x^x e^{-x^p}\,dx$,

(c) $\displaystyle\int_0^\infty \frac{x^{p-1}-x^{q-1}}{1-x}\,dx$, (d) $\displaystyle\int_0^\infty \frac{\sin(x^p)}{x^q}\,dx$,

(e) $\displaystyle\int_0^\infty \frac{x^{p-1}}{1+x^q}\,dx$, (f) $\displaystyle\int_\pi^\infty (\log x)^p (\sin x)^{-1/3}\,dx$

14-7. Derive the following equations (m and n denote positive integers)

(a) $\displaystyle\int_0^\infty \frac{\sin^{2n+1} x}{x}\,dx = \frac{\pi(2n)!}{2^{2n+1}(n!)^2}$, (b) $\displaystyle\int_1^\infty \frac{\log x}{x^{n+1}}\,dx = n^{-2}$,

(c) $\displaystyle\int_0^\infty x^n(1+x)^{-n-m-1}\,dx = \frac{n!(m-1)!}{(m+n)!}$

14-8. Show that the integrals in (a) and (c) are convergent but that those in (b) and (d) are divergent.

(a) $\displaystyle\int_0^\infty x^2 e^{-x^6\sin^2 x}\,dx$, (b) $\displaystyle\int_0^\infty x^3 e^{-x^6\sin^2 x}\,dx$,

(c) $\displaystyle\int_1^\infty \frac{dx}{1+x^4\sin^2 x}$, (d) $\displaystyle\int_1^\infty \frac{dx}{1+x^2\sin^2 x}$

[*Hint* Obtain upper and lower bounds for the integrals over suitably chosen neighborhoods of the points $n\pi$ ($n = 1, 2, 3, \ldots$).]

14-9. Construct an example of a proper Riemann-Stieltjes integral

$$\int_a^b f(x)\,d\alpha(x)$$

in which α is of bounded variation on $[a, b]$ but such that

$$\lim_{\epsilon\to 0+}\int_{a+\epsilon}^b f(x)\,d\alpha(x) \neq \int_a^b f(x)\,d\alpha(x)$$

14-10. Given that $f \in R$ on $[0, 1]$, that f is periodic with period 1 and that $\int_0^1 f(x)\,dx = 0$. Prove that the improper integral $\int_1^\infty f(x)x^{-s}\,dx$ converges if $s > 0$. [*Hint* Introduce $g(x) = \int_1^x f(t)\,dt$ and write $\int_1^b f(x)x^{-s}\,dx = \int_1^b x^{-s}\,dg(x)$.]

14–11. Determine the set S of those real values of y on which each of the following integrals is uniformly convergent:

(a) $\displaystyle\int_0^\infty \frac{\cos xy}{1+x^2}\,dx,$ (b) $\displaystyle\int_0^\infty (x^2+y^2)^{-1}\,dx,$

(c) $\displaystyle\int_0^\infty \frac{\sin^2 xy}{x^2}\,dx,$ (d) $\displaystyle\int_0^\infty e^{-x^2}\cos 2xy\,dx.$

14–12. Let $F(y) = \int_0^\infty e^{-x^2}\cos 2xy\,dx$ if $y \in E_1$. Show that F satisfies the differential equation $F'(y) + 2y\,F(y) = 0$ and deduce that $F(y) = \frac{1}{2}\sqrt{\pi}\,e^{-y^2}$. (Use the result $\int_0^\infty e^{-x^2}\,dx = \frac{1}{2}\sqrt{\pi}$, derived in Exercise 9–17.)

14–13. Let $F(y) = \int_0^\infty \sin xy/x(x^2+1)\,dx$ if $y > 0$. Show that F satisfies the differential equation $F''(y) - F(y) + \pi/2 = 0$ and deduce that $F(y) = \frac{1}{2}\pi(1-e^{-y})$. Use this result to deduce the following equations, valid for $y > 0$ and $a > 0$:

$$\int_0^\infty \frac{\sin xy}{x(x^2+a^2)}\,dx = \frac{\pi}{2a^2}(1-e^{-ay}), \qquad \int_0^\infty \frac{\cos xy}{x^2+a^2}\,dx = \frac{\pi e^{-ay}}{2a},$$

$$\int_0^\infty \frac{x\sin xy}{x^2+a^2}\,dx = \frac{\pi}{2}e^{-ay}, \qquad \int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}.$$

14–14. Show that $\int_1^\infty [\int_1^\infty f(x,y)\,dx]\,dy \neq \int_1^\infty [\int_1^\infty f(x,y)\,dy]\,dx$ if

(a) $f(x,y) = \dfrac{x-y}{(x+y)^3},$ (b) $f(x,y) = \dfrac{x^2-y^2}{(x^2+y^2)^2}.$

14–15. Show that the order of integration cannot be interchanged in the following integrals:

(a) $\displaystyle\int_0^1\left[\int_0^1 \frac{x-y}{(x+y)^3}\,dx\right]dy,$ (b) $\displaystyle\int_0^1\left[\int_1^\infty (e^{-xy}-2e^{-2xy})\,dy\right]dx.$

14–16. Let $f(x,y) = \int_0^\infty dt/[(1+x^2t^2)(1+y^2t^2)]$ if $(x,y) \neq (0,0)$. Show (by methods of elementary calculus) that $f(x,y) = \frac{1}{2}\pi(x+y)^{-1}$. Evaluate the repeated integral $\int_0^1 [\int_0^1 f(x,y)\,dx]\,dy$ to derive the formula:

$$\int_0^\infty \frac{(\tan^{-1} x)^2}{x^2}\,dx = \pi\log 2.$$

14–17. Let $f(y) = \int_0^\infty \sin x\cos xy/x\,dx$ if $y \geq 0$. Show (by methods of elementary calculus) that $f(y) = \pi/2$ if $0 \leq y < 1$ and that $f(y) = 0$ if $y > 1$.

Evaluate the integral $\int_0^a f(y)\,dy$ to derive the formula

$$\int_0^\infty \frac{\sin ax \sin x}{x^2}\,dx = \begin{cases} \dfrac{\pi a}{2}, & \text{if } 0 \le a \le 1, \\ \dfrac{\pi}{2}, & \text{if } a \ge 1 \end{cases}$$

14-18. (a) Use the trigonometric identity $\sin 2x = 2 \sin x \cos x$, along with the formula $\int_0^\infty \sin x/x\,dx = \pi/2$, to evaluate the integral

$$\int_0^\infty \frac{\sin x \cos x}{x}\,dx = \frac{\pi}{4}$$

(b) Use integration by parts in (a) to derive the formula

$$\int_0^\infty \frac{\sin^2 x}{x^2}\,dx = \frac{\pi}{2}$$

(c) Use the identity $\sin^2 x + \cos^2 x = 1$, along with (b), to obtain

$$\int_0^\infty \frac{\sin^4 x}{x^2}\,dx = \frac{\pi}{4}$$

(d) Use the result of (c) to obtain

$$\int_0^\infty \frac{\sin^4 x}{x^4}\,dx = \frac{\pi}{3}$$

14-19. Assume that $f \in R$ on $[a, b]$ for every $b > a > 0$. Define g by the equation $xg(x) = \int_1^x f(t)\,dt$ if $x > 0$, assume that the limit $\lim_{x\to+\infty} g(x)$ exists, and denote this limit by B. If a and b are fixed positive numbers, prove that

(a) $\displaystyle\int_a^b \frac{f(x)}{x}\,dx = g(b) - g(a) + \int_a^b \frac{g(x)}{x}\,dx$

(b) $\displaystyle\lim_{T\to+\infty} \int_{aT}^{bT} \frac{f(x)}{x}\,dx = B \log \frac{b}{a}.$

(c) $\displaystyle\int_1^\infty \frac{f(ax) - f(bx)}{x}\,dx = B \log \frac{a}{b} + \int_a^b \frac{f(t)}{t}\,dt.$

(d) Assume that the limit $\lim_{x\to0+} x\int_x^1 f(t)t^{-2}\,dt$ exists, denote this limit by A, and prove that

$$\int_0^1 \frac{f(ax) - f(bx)}{x}\,dx = A \log \frac{b}{a} - \int_a^b \frac{f(t)}{t}\,dt$$

(e) Combine (c) and (d) to deduce

$$\int_0^\infty \frac{f(ax) - f(bx)}{x}\, dx = (B - A) \log \frac{a}{b}$$

and use this result to evaluate the following integrals:

$$\int_0^\infty \frac{\cos ax - \cos bx}{x}\, dx, \qquad \int_0^\infty \frac{e^{-ax} - e^{-bx}}{x}\, dx.$$

14–20. If $f_n(x) = e^{-nx} - 2e^{-2nx}$, show that

$$\sum_{n=1}^\infty \int_0^\infty f_n(x)\, dx \neq \int_0^\infty \sum_{n=1}^\infty f_n(x)\, dx.$$

14–21. State and prove a theorem similar to Theorem 14–31 for improper integrals of the second kind.

14–22. Justify the following equations:

(a) $$\int_0^1 \log \frac{1}{1-x}\, dx = \int_0^1 \sum_{n=1}^\infty \frac{x^n}{n}\, dx = \sum_{n=1}^\infty \frac{1}{n} \int_0^1 x^n\, dx = 1.$$

(b) $$\int_0^1 \frac{x^{p-1}}{1-x} \log\left(\frac{1}{x}\right) dx = \sum_{n=0}^\infty \frac{1}{(n+p)^2} \quad (p > 0).$$

14–23. (a) If $s > 0$ and $a > 0$, show that the series

$$\sum_{n=1}^\infty \frac{1}{n} \int_a^\infty \frac{\sin 2n\pi x}{x^s}\, dx$$

converges and prove that

$$\lim_{a \to +\infty} \sum_{n=1}^\infty \frac{1}{n} \int_a^\infty \frac{\sin 2n\pi x}{x^s}\, dx = 0.$$

(b) Let $f(x) = \sum_{n=1}^\infty \sin(2n\pi x)/n$. Show that

$$\int_0^\infty \frac{f(x)}{x^s}\, dx = (2\pi)^{s-1} \zeta(2-s) \int_0^\infty \frac{\sin t}{t^s}\, dt, \qquad \text{if } 0 < s < 1,$$

where ζ denotes the Riemann zeta function.

14–24. (a) Derive the following formula for the nth derivative of the Gamma function

$$\Gamma^{(n)}(x) = \int_0^\infty e^{-t}t^{x-1} (\log t)^n \, dt \qquad (x > 0)$$

(b) When $x = 1$, show that this can be written as follows

$$\Gamma^{(n)}(1) = \int_0^1 (t^2 + (-1)^n e^{t-1/t})e^{-1/t}t^{-2} (\log t)^n \, dt$$

(c) Use (b) to show that $\Gamma^{(n)}(1)$ has the same sign as $(-1)^n$

NOTE In Exercises 14–25 and 14–26, the limit $\lim_{x\to+\infty} f(x)$ is assumed to exist (finite) Its value is denoted by $f(\infty)$

14–25. Assume that f is of bounded variation on $[0, b]$ for every $b > 0$ Prove that

$$\lim_{y\to 0+} y\int_0^\infty e^{-xy}f(x)\, dx = f(\infty)$$

[*Hint* Use integration by parts]

14–26. Deduce the result of Exercise 14–25 under the weaker hypothesis that $f \in R$ on $[0, b]$ for every $b > 0$ *Hint* Let $F(y) = y\int_0^\infty e^{-xy}f(x)\, dx$ and write

$$|F(y) - f(\infty)| \le y\int_0^B e^{-xy}|f(x) - f(\infty)|\, dx + y\int_B^\infty e^{-xy}|f(x) - f(\infty)|\, dx.$$

14–27. Assume that f is of bounded variation on $[0, 1]$ Prove that

$$\lim_{y\to 0+} y\int_0^1 x^{y-1}f(x)\, dx = f(0+)$$

14–28. Deduce the result of Exercise 14–27 under the weaker hypothesis that $f \in R$ on $[0, 1]$ and that the limit $f(0+)$ exists

14–29. Prove the following theorem (*Tannery's theorem*) *Given a sequence of functions* $\{f_n\}$ *and a sequence of real numbers* $\{p_n\}$ *such that*

(a) $f_n \to f$ *uniformly on* $[a, b]$ *for every* $b \ge a$,

(b) $f_n \in R$ *on* $[a, b]$ *for every* $b \ge a$,

(c) $p_n \to +\infty$ *as* $n \to \infty$,

(d) $|f_n(x)| \le g(x)$ *for every* $n = 1, 2, \ldots$ *and every* $x \ge a$, *where* g *is such that the integral* $\int_a^\infty g(x)\, dx$ *converges*

Then the integral $\int_a^\infty f(x)\, dx$ *converges absolutely and we have*

$$\lim_{n\to\infty} \int_a^{p_n} f_n(x)\,dx = \int_a^{\infty} f(x)\,dx.$$

14–30. Use the result of Exercise 14–29 to prove that

$$\lim_{n\to\infty} \int_0^n \left(1 - \frac{x}{n}\right)^n x^p\,dx = \int_0^{\infty} e^{-x}x^p\,dx, \qquad \text{if } p > -1.$$

In Exercises 14–31 and 14–32, Γ denotes the Gamma function.

14–31. Show that $\Gamma(x + 1) = x\Gamma(x)$ if $x > 0$ and that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Deduce that $\Gamma(n + 1) = n!$ and that $\Gamma(n + \frac{1}{2}) = (2n)!\sqrt{\pi}/4^n n!$ if $n = 0, 1, 2, \ldots$.

14–32. (a) Show that for $x > 0$ we have the series representation

$$\Gamma(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{1}{n + x} + \sum_{n=0}^{\infty} c_n x^n,$$

where $c_n = (1/n!) \int_1^{\infty} t^{-1}e^{-t} (\log t)^n\,dt$. [*Hint:* Write $\int_0^{\infty} = \int_0^1 + \int_1^{\infty}$ and use an appropriate power series expansion in each integral.]

(b) Show that the power series $\sum_{n=0}^{\infty} c_n z^n$ converges for every complex z and that the series $\sum_{n=0}^{\infty} [(-1)^n/n!]/(n + z)$ converges for every complex $z \neq 0, -1, -2, \ldots$.

14–33. Prove *Lebesgue's dominated convergence theorem for improper integrals: Let $\{f_n\}$ be a sequence of functions such that each improper integral $\int_a^{\infty} f_n(x)\,dx$ converges. Assume that $\{f_n\}$ converges pointwise on $[a, +\infty)$ to a function f for which the integral $\int_a^{\infty} f(x)\,dx$ converges. Suppose there is a non-negative function g such that $\int_a^{\infty} g(x)\,dx$ converges and such that $|f_n(x)| \leq g(x)$ for each $x \geq a$ and each $n = 1, 2, \ldots$. Then*

$$\lim_{n\to\infty} \int_a^{\infty} f_n(x)\,dx = \int_a^{\infty} \lim_{n\to\infty} f_n(x)\,dx = \int_a^{\infty} f(x)\,dx.$$

[*Hint:* Use Theorem 13–17.]

CHAPTER 15

FOURIER SERIES AND FOURIER INTEGRALS

15–1 Introduction. In 1807, Fourier astounded some of his contemporaries by asserting that an "arbitrary" function could be expressed as a linear combination of sines and cosines. These linear combinations, now called *Fourier series*, have become an indispensable tool in the analysis of certain periodic phenomena (such as vibrations, planetary and wave motion) which are studied in physics and engineering. The many important questions of a purely mathematical nature which have also arisen in connection with the theory of Fourier series have been the object of intensive study by a large number of outstanding mathematicians, and it is a remarkable historical fact that much of the development of modern mathematical analysis has been profoundly influenced by the search for the answers to these questions. For a brief but excellent account of the history of the subject of Fourier series and its impact on the development of mathematics see Ref. 15–1.

15–2 Orthogonal systems of functions. Fourier series are now just one part of a much more general discipline known as the *theory of orthogonal expansions*. Certain features of this theory are remarkably similar to some of the results in vector analysis. For example, the problem of expressing an arbitrary vector in n-dimensional space as a linear combination of basis vectors has its counterpart in the theory of orthogonal expansions, where it is required to express a more-or-less "arbitrary" function as a linear combination of members of some particular collection of functions. Many of the tools employed in the study of vectors have analogs in the theory of orthogonal expansions. The most important of these is the *inner product* of two functions (analogous to the dot product of two vectors).

15–1 DEFINITION. *If f and g are real-valued functions, both Riemann-integrable on $[a, b]$, the integral*

$$\int_a^b f(x)g(x)\,dx \tag{1}$$

is called the inner product of f and g, denoted by (f, g). The non-negative number $(f, f)^{1/2}$, denoted by $\|f\|$, is called the norm of f.

NOTE. The integral in (1) resembles the sum $\sum_{k=1}^{n} x_k y_k$ which defines the dot product of two vectors $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$. The function values $f(x)$ and $g(x)$ in (1) play the role of the components x_k and y_k, and integration takes the place of summation. The norm of f is analogous to the length of a vector.

It follows immediately from (1) that the inner product has the following elementary properties:

$$(2)\ (f, g) = (g, f), \qquad (cf, g) = (f, cg) = c(f, g) \qquad (c = \text{constant}),$$

$$(3)\ (f_1 + f_2, g) = (f_1, g) + (f_2, g), \qquad (f, g_1 + g_2) = (f, g_1) + (f, g_2).$$

The Cauchy-Schwarz inequality (see Exercise 9–13) assumes the form

$$(4) \qquad |(f, g)| \le ||f||\ ||g||.$$

Using (4), it is an easy matter to derive the *Minkowski inequality:*

$$(5) \qquad ||f + g|| \le ||f|| + ||g||.$$

(See Exercise 15–1.) This, of course, is analogous to the "triangle inequality" for vectors. We also have the relation

$$||f+g||^2 = (f+g, f+g) = (f, f) + 2(f, g) + (g, g) = ||f||^2 + ||g||^2 + 2(f, g).$$

Therefore the equation $||f + g||^2 = ||f||^2 + ||g||^2$ is equivalent to the equation $(f, g) = 0$. Functions f and g satisfying $(f, g) = 0$ are said to be *orthogonal* (by analogy with the theorem of Pythagoras for right triangles). The concept of orthogonality is very basic in this theory and is introduced for infinite collections of functions as follows:

15–2 DEFINITION. *Let $S = \{\phi_0, \phi_1, \phi_2, \ldots\}$ be a collection of functions, each of which is Riemann-integrable on $[a, b]$. If*

$$(\phi_n, \phi_m) = 0, \qquad \textit{whenever } m \ne n,$$

the collection S is said to be an orthogonal system on $[a, b]$. If, in addition, each ϕ_n has norm 1, then S is said to be orthonormal on $[a, b]$.

NOTE. Every orthogonal system for which each $||\phi_n|| \ne 0$ can be converted into an orthonormal system by dividing each ϕ_n by its norm.

We shall be particularly interested in the special trigonometric system $S = \{\phi_0, \phi_1, \phi_2, \ldots\}$ for which

(6)

$$\phi_0(x) = \frac{1}{\sqrt{2\pi}}, \quad \phi_{2n-1}(x) = \frac{\cos nx}{\sqrt{\pi}}, \quad \phi_{2n}(x) = \frac{\sin nx}{\sqrt{\pi}} \quad (n = 1, 2, \ldots).$$

It is a simple matter to verify that S is orthonormal on the interval $[0, 2\pi]$ (or on any other interval of length 2π) (See Exercise 15-2)

NOTE The notion of an inner product can also be extended to *complex-valued* functions defined and Riemann-integrable on an interval $[a, b]$ In this case, (f, g) is defined by the equation

$$(f, g) = \int_a^b f(x)\overline{g(x)}\, dx, \tag{7}$$

where $\overline{g(x)}$ denotes the complex conjugate of $g(x)$ The complex conjugate is introduced so that the inner product of f with itself will be a non-negative quantity, namely, $(f, f) = \int_a^b |f(x)|^2\, dx$ The norm of f is, as before, $\|f\| = (f, f)^{1/2}$ The equations in (3) are still valid but those in (2) must be modified by writing

$$(f, g) = \overline{(g, f)}, \qquad (cf, g) = c(f, g), \qquad (f, cg) = \bar{c}(f, g), \tag{8}$$

where now, of course, c can be a complex constant The definition of orthogonal and orthonormal systems is unaltered Probably the most important example of an orthonormal system of complex-valued functions is the collection $\{\phi_0, \phi_1, \phi_2, \ldots\}$ given by

$$\phi_n(x) = \frac{e^{inx}}{\sqrt{2\pi}} = \frac{\cos nx + i \sin nx}{\sqrt{2\pi}} \qquad (n = 0, 1, 2, \ldots), [a, b] = [0, 2\pi]$$

Linear independence for functions can be introduced as in the case of vectors A finite collection of functions, say $\{\phi_0, \phi_1, \ldots, \phi_m\}$, is said to be *linearly dependent* on $[a, b]$ whenever there exist constants $c_0, c_1, \ldots, c_m$ (not all zero) such that

$$c_0\phi_0(x) + c_1\phi_1(x) + \cdots + c_m\phi_m(x) = 0, \qquad \text{for every } x \text{ in } [a, b] \tag{9}$$

Otherwise, the set $\{\phi_0, \phi_1, \ldots, \phi_m\}$ is called *linearly independent* on $[a, b]$ An infinite collection $\{\phi_0, \phi_1, \ldots\}$ is called linearly independent on $[a, b]$ if every finite subset is linearly independent on $[a, b]$ It is easy to see that every orthonormal system is linearly independent In fact, if an equation such as (9) holds, we can multiply by $\overline{\phi_k(x)}$ and integrate from a to b to get $c_k = 0$ for each $k = 0, 1, 2, \ldots, m$

This analogy with vectors suggests that there may be some hope of expressing an "arbitrary" function f as a linear combination of elements of an orthonormal set $\{\phi_0, \phi_1, \ldots\}$ Such an "expansion" would be an equation of the form

$$f(x) = \sum_{n=0}^{\infty} c_n \phi_n(x), \tag{10}$$

valid for each x in $[a, b]$. There are really two types of problems involved here. First, we must seek conditions on f and on the set $\{\phi_0, \phi_1, \ldots\}$ which will guarantee that an expansion such as (10) actually exists. Second, assuming that the expansion *does* exist, we must find a way to determine the coefficients $c_0, c_1, c_2, \ldots$. The second problem will be treated first.

Let us operate with (10) in a purely formal way, disregarding for the moment questions of convergence. If we multiply both sides of (10) by $\overline{\phi_k(x)}$ and integrate the resulting series term-by-term from a to b, we get

$$(f, \phi_k) = \sum_{n=0}^{\infty} c_n(\phi_n, \phi_k) = c_k. \tag{11}$$

Thus, if we can justify these formal operations, we find that the kth coefficient in (10) is merely the inner product (f, ϕ_k). One sufficient condition for the validity of these steps is given in the next theorem.

15–3 THEOREM. *Let $\{\phi_0, \phi_1, \phi_2, \ldots\}$ be an orthonormal system on $[a, b]$. Assume that the series*

$$\sum_{n=0}^{\infty} c_n \phi_n(x) \tag{12}$$

converges uniformly on $[a, b]$ and let $f(x)$ denote the sum of this series. Then the function f so defined is Riemann-integrable on $[a, b]$ and the nth coefficient c_n is given by

$$c_n = (f, \phi_n) = \int_a^b f(x)\overline{\phi_n(x)}\, dx \qquad (n = 0, 1, 2, \ldots). \tag{13}$$

Proof. Uniform convergence implies Riemann-integrability of f and justifies the formal steps leading to (11). (See Theorem 13–12.)

Suppose now that we adopt the following point of view. Assume that we are given a Riemann-integrable function f defined on $[a, b]$ and a set $\{\phi_0, \phi_1, \phi_2, \ldots\}$ which is orthonormal on $[a, b]$. Then we can always define numbers $c_0, c_1, c_2, \ldots$ by means of (13) and it makes sense to form the series in (12). Two questions now arise: Does this series converge pointwise on $[a, b]$? If (12) does converge for a fixed x, is its sum $f(x)$? In general, the answer to both questions is "No." To answer these questions in the affirmative, it is necessary to put further restrictions on both the function f

and the orthonormal set $\{\phi_0, \phi_1, \phi_2, \ldots\}$ We shall give a detailed discussion of these questions later for the special trigonometric system described in (6) At present we shall be interested in properties of the numbers defined by (13).

15-3 The Fourier series of a function relative to an orthonormal system.

15-4 DEFINITION *Let $S = \{\phi_0, \phi_1, \phi_2, \ldots\}$ be orthonormal on $[a, b]$ and assume that $f \in R$ on $[a, b]$. The notation*

$$f(x) \sim \sum_{n=0}^{\infty} c_n \phi_n(x) \tag{14}$$

will mean that the numbers $c_0, c_1, c_2, \ldots$ are given by the following formulas:

$$c_n = (f, \phi_n) = \int_a^b f(x)\overline{\phi_n(x)}\,dx \qquad (n = 0, 1, 2, \ldots). \tag{15}$$

The series in (14) is called the Fourier series of f relative to the system S and the numbers $c_0, c_1, c_2, \ldots$ are called the Fourier coefficients of f relative to S

NOTE When S is the particular orthonormal set of trigonometric functions described in (6), the series is called simply the *Fourier series generated by f* In this case, we write (14) in the form

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx),$$

the coefficients being given by the following formulas.

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(t) \cos nt\,dt, \qquad b_n = \frac{1}{\pi}\int_0^{2\pi} f(t) \sin nt\,dt. \tag{16}$$

15-4 Mean-square approximation. Theorem 15-3 provides some justification for considering the series whose coefficients are given by (15) Further justification is provided by the following theorem

15-5 THEOREM *Let $\{\phi_0, \phi_1, \phi_2, \ldots\}$ be orthonormal on $[a, b]$, assume that $f \in R$ on $[a, b]$, and suppose that*

$$f(x) \sim \sum_{n=0}^{\infty} c_n \phi_n(x).$$

Let $s_n(x)$ denote the partial sum

$$(17) \qquad s_n(x) = \sum_{k=0}^{n} c_k\phi_k(x) \qquad (n = 0, 1, 2, \ldots).$$

If $b_0, b_1, b_2, \ldots$, are arbitrary complex numbers, let

$$(18) \qquad t_n(x) = \sum_{k=0}^{n} b_k\phi_k(x) \qquad (n = 0, 1, 2, \ldots).$$

Then we have the following identity:

$$(19) \quad \int_a^b |f(x) - t_n(x)|^2\, dx = \int_a^b |f(x)|^2\, dx - \sum_{k=0}^{n} |c_k|^2 + \sum_{k=0}^{n} |c_k - b_k|^2,$$

from which we obtain the inequality

$$(20) \qquad \int_a^b |f(x) - s_n(x)|^2\, dx \le \int_a^b |f(x) - t_n(x)|^2\, dx.$$

NOTE. Each linear combination t_n can be thought of as an "approximation" to f. The integral $\int_a^b |f(x) - t_n(x)|^2\, dx = \|f - t_n\|^2$ measures the "mean-square error" of this approximation. Inequality (20) states that, among all possible linear combinations t_n, the partial sums s_n of the Fourier series of f yield the "best" approximation to f (in the sense that the mean-square error is least).

Proof. It is clear that (19) implies (20) because the right member of (19) has its smallest value when $b_k = c_k$ for each $k = 0, 1, 2, \ldots, n$. To prove (19), write

$$\begin{aligned}\int_a^b |f - t_n|^2\, dx &= (f - t_n, f - t_n)\\ &= (f, f) - (f, t_n) - (t_n, f) + (t_n, t_n).\end{aligned}$$

Using (3) and (8), we find

$$(t_n, t_n) = \left(\sum_{k=0}^{n} b_k\phi_k, \sum_{m=0}^{n} b_m\phi_m\right) = \sum_{k=0}^{n}\sum_{m=0}^{n} b_k\bar{b}_m(\phi_k, \phi_m) = \sum_{k=0}^{n} |b_k|^2$$

and

$$(f, t_n) = \left(f, \sum_{k=0}^{n} b_k \phi_k\right) = \sum_{k=0}^{n} \bar{b}_k (f, \phi_k) = \sum_{k=0}^{n} \bar{b}_k c_k$$

Also, $(t_n, f) = \overline{(f, t_n)} = \sum_{k=0}^{n} b_k \bar{c}_k$, and hence

$$\int_a^b |f - t_n|^2 \, dx = \|f\|^2 + \sum_{k=0}^{n} |b_k|^2 - \sum_{k=0}^{n} \bar{b}_k c_k - \sum_{k=0}^{n} b_k \bar{c}_k$$

$$= \|f\|^2 - \sum_{k=0}^{n} |c_k|^2 + \sum_{k=0}^{n} (b_k - c_k)(\bar{b}_k - \bar{c}_k)$$

$$= \|f\|^2 - \sum_{k=0}^{n} |c_k|^2 + \sum_{k=0}^{n} |b_k - c_k|^2$$

15-6 THEOREM *Let $\{\phi_0, \phi_1, \phi_2, \ldots\}$ be orthonormal on $[a, b]$, assume that $f \in R$ on $[a, b]$, and suppose that*

$$f(x) \sim \sum_{n=0}^{\infty} c_n \phi_n(x)$$

Then

(a) *The series $\sum |c_n|^2$ converges and satisfies the inequality*

(21) $$\sum_{n=0}^{\infty} |c_n|^2 \le \int_a^b |f(x)|^2 \, dx \qquad \textit{(Bessel's inequality)}$$

(b) *The equation*

$$\sum_{n=0}^{\infty} |c_n|^2 = \int_a^b |f(x)|^2 \, dx \qquad \textit{(Parseval's formula)}$$

holds if, and only if, we also have

$$\lim_{n \to \infty} \|f - s_n\| = 0,$$

where $\{s_n\}$ is the sequence of partial sums, defined by (17)

Proof Taking $b_k = c_k$ in (19) and observing that the left member is non-negative, we obtain

$$\sum_{k=0}^{n} |c_k|^2 \le \int_a^b |f(x)|^2\, dx.$$

This establishes (a). To prove (b), we again put $b_k = c_k$ in (19) to obtain

$$||f - s_n||^2 = \int_a^b |f(x) - s_n(x)|^2\, dx = \int_a^b |f(x)|^2\, dx - \sum_{k=0}^{n} |c_k|^2.$$

Part (b) follows at once from this equation.

NOTE. The equation $\lim_{n\to\infty} ||f - s_n|| = 0$ is the same as the statement

$$\underset{n\to\infty}{\text{l.i.m.}}\, s_n = f \text{ on } [a, b].$$

(See Definition 13–18.) Part (b) of Theorem 15–6 tells us that Parseval's formula holds if, and only if, the sequence $\{s_n\}$ *converges in the mean* to f on $[a, b]$. As a matter of interest, the Parseval formula can be written in the form

$$||f||^2 = c_0^2 + c_1^2 + c_2^2 + \cdots,$$

which is analogous to the vector formula $|\mathbf{x}|^2 = x_1^2 + \cdots + x_n^2$.

15–7 DEFINITION. *Let S be a collection of Riemann-integrable functions defined on $[a, b]$. Let $\{\phi_0, \phi_1, \phi_2, \ldots\}$ be orthonormal on $[a, b]$, assume that*

$$f(x) \sim \sum_{n=0}^{\infty} c_n\phi_n(x),$$

and let

$$s_n(x) = \sum_{k=0}^{n} c_k\phi_k(x) \qquad (n = 0, 1, 2, \ldots).$$

The orthonormal set $\{\phi_0, \phi_1, \phi_2, \ldots\}$ is said to be complete for S if, for every f in S, we have $||s_n - f|| \to 0$ as $n \to \infty$.

Thus, for a *complete* orthonormal system the Parseval equation holds for every f in S. We will show later (in Theorem 15–22) that the trigonometric system described in (6) is complete for the set S of all functions which are continuous on $[0, 2\pi]$.

As a further consequence of part (a) of Theorem 15–6, we observe that the Fourier coefficients c_n of f relative to any orthonormal system

$\{\phi_0, \phi_1, \phi_2, \ldots\}$ must tend to 0 as $n \to \infty$ (since $\sum |c_n|^2$ converges) In particular, when $\phi_n(x) = e^{inx}$, we find

$$\lim_{n\to\infty} \int_0^{2\pi} f(x)e^{-inx}\,dx = 0,$$

from which we obtain the important formulas:

$$\lim_{n\to\infty} \int_0^{2\pi} f(x)\cos nx\,dx = \lim_{n\to\infty} \int_0^{2\pi} f(x)\sin nx\,dx = 0. \tag{22}$$

15–5 Trigonometric Fourier series. At this point we abandon the study of general orthonormal systems and fix our attention on the special trigonometric system described in (6) Ever since Fourier's time, an enormous number of papers have been published in the mathematical literature dealing with trigonometric Fourier series The object of much of the research has been to find sufficient conditions to be satisfied by a given integrable function f in order that its Fourier series may converge, either throughout the interval $[0, 2\pi]$ or at particular points of the interval It turns out that the convergence or divergence of the series at a particular point really depends on the behavior of the function in an arbitrarily small neighborhood of the point. (See Theorem 15–17.)

The efforts of Fourier and Dirichlet in the early nineteenth century, followed by the contributions of Riemann, Lipschitz, Heine, Cantor, du Bois-Reymond, Dini, Jordan, and de la Vallée Poussin in the latter part of the century, have resulted in the discovery of sufficient conditions of a wide scope for establishing convergence of the series, either at particular points, or generally, throughout the interval Useful *necessary and sufficient* conditions for the convergence of the series at any point of the interval have not, as yet, been obtained In view of the results which are now known, it seems unlikely that such conditions exist

After the discovery by Lebesgue, in 1902, of his general theory of measure and integration, the field of investigation was considerably widened and the names chiefly associated with the subject since then are those of Fejér, Hobson, W. H. Young, Hardy, and Littlewood. Fejér showed, in 1903, that *divergent* Fourier series may be utilized by considering, instead of the sequence of partial sums $\{s_n\}$, the sequence of arithmetic means $\{\sigma_n\}$, where

$$\sigma_n(x) = \frac{s_0(x) + s_1(x) + \cdots + s_{n-1}(x)}{n}.$$

He established the remarkable theorem that the sequence $\{\sigma_n(x)\}$ is convergent and its limit is $\frac{1}{2}(f(x+) + f(x-))$ at every point in $[0, 2\pi]$ where $f(x+)$ and $f(x-)$ exist, the only restriction on f being that it is Riemann-integrable on $[0, 2\pi]$ or that the integral $\int_0^{2\pi} f(x)\,dx$ exists as an absolutely convergent improper integral.

A thorough and complete study of the theory of Fourier series cannot be undertaken without the use of the Lebesgue integral. However, the Riemann integral is quite adequate for the introduction to the theory that will be presented here. Furthermore, the results contained in this chapter suffice for most applications of Fourier series to problems in physics and engineering.

We turn now to the convergence problem for Fourier series. The whole theory rests on two fundamental limit formulas which will be discussed in the next few sections. These limit formulas, which also form the basis for the theory of Fourier integrals, deal with integrals depending on a real parameter α, and we are interested in the behavior of these integrals as $\alpha \to +\infty$. The first of these is a generalization of (22) and is known as the Riemann-Lebesgue lemma.

15-6 The Riemann-Lebesgue lemma.

15-8 THEOREM. *Assume that $f \in R$ on $[a, b]$. Then, for each real β, we have*

$$\lim_{\alpha\to+\infty} \int_a^b f(t)\sin(\alpha t+\beta)\,dt = 0. \tag{23}$$

Proof. If f is constant on $[a, b]$, the result is obvious since we have

$$\left|\int_a^b \sin(\alpha t+\beta)\,dt\right| = \left|\frac{\cos(a\alpha+\beta) - \cos(b\alpha+\beta)}{\alpha}\right| \le \frac{2}{\alpha}, \quad \text{if } \alpha > 0.$$

The result also holds if f is constant on the *open* interval (a, b), regardless of how we define $f(a)$ and $f(b)$. Hence, (23) is valid if f is a step function. But now it is easy to prove (23) for every Riemann-integrable f. If $\epsilon > 0$ is given, choose a partition P of $[a, b]$ such that the corresponding upper and lower Riemann sums satisfy the inequality $U(P, f) - L(P, f) < \epsilon/2$. Now write

$$L(P, f) = \int_a^b m(t)\,dt \qquad \text{and} \qquad U(P, f) = \int_a^b M(t)\,dt,$$

where m and M are step functions (determined in an obvious way by the partition P) such that the inequalities $m(t) \le f(t) \le M(t)$ hold throughout $[a, b]$. Then we have

$$\left|\int_a^b [f(t) - m(t)]\sin(\alpha t+\beta)\,dt\right| \le \int_a^b |M(t) - m(t)|\,dt \le \frac{\epsilon}{2}. \tag{24}$$

But, for this same ϵ, we can choose A so that $\alpha \geq A$ implies

$$(25) \qquad \left| \int_a^b m(t) \sin(\alpha t + \beta)\, dt \right| < \frac{\epsilon}{2}.$$

Combining (24) with (25), we obtain $|\int_a^b f(t) \sin(\alpha t + \beta)\, dt| < \epsilon$ whenever $\alpha \geq A$. This proves (23)

NOTE. Taking $\beta = 0$ and $\beta = \pi/2$, we find

$$\lim_{\alpha\to+\infty} \int_a^b f(t) \sin \alpha t\, dt = \lim_{\alpha\to+\infty} \int_a^b f(t) \cos \alpha t\, dt = 0$$

The result of Theorem 15-8 is known as the *Riemann-Lebesgue lemma for proper Riemann integrals* It can also be extended to absolutely convergent improper integrals

15-9 THEOREM *Assume that $f \in R(a, b)$ for every $b \geq a$ and suppose that the improper integral $\int_a^\infty f(t)\, dt$ converges absolutely Then, for each real β, we have*

$$\lim_{\alpha\to+\infty} \int_a^\infty f(t) \sin(\alpha t + \beta)\, dt = 0$$

Proof. Given $\epsilon > 0$, choose b so that $\int_b^\infty |f(t)|\, dt < \epsilon/2$ Having chosen b, choose A so that $\alpha \geq A$ implies $|\int_a^b f(t) \sin(\alpha t + \beta)\, dt| < \epsilon/2$. Then

$$\left| \int_a^\infty f(t) \sin(\alpha t + \beta)\, dt \right| \leq \left| \int_a^b f(t) \sin(\alpha t + \beta)\, dt \right| + \int_b^\infty |f(t)|\, dt < \epsilon$$

In a similar fashion we can prove the Riemann-Lebesgue lemma for absolutely convergent improper integrals of the second kind.

15-10 THEOREM. *Assume that $f \in R(a + \epsilon, b)$ for every $\epsilon > 0$ and suppose that the integral $\int_{a+}^b |f(t)|\, dt$ converges Then, for each real β, we have*

$$\lim_{\alpha\to+\infty} \int_{a+}^b f(t) \sin(\alpha t + \beta)\, dt = 0$$

15-7 Absolutely integrable functions. At this point it is convenient to introduce a class of functions that we shall be working with rather extensively in the latter part of this chapter. The functions in this class are called *absolutely integrable* and they are described as follows:

15-11 DEFINITION. *A function f is said to be absolutely integrable on an interval $[a, b]$ (where a may be $-\infty$ and b may be $+\infty$) if there exist a finite number of points in $[a, b,]$, say $a = c_0 < c_1 < c_2 < \cdots < c_n = b$, such that:*

(a) *$f \in R$ on every finite subinterval of $[a, b]$ containing none of the points $c_0, c_1, \ldots, c_n$.*

(b) *For each $k = 1, 2, \ldots, n$, the integral $\int_{c_{k-1}}^{c_k} f(t)\,dt$ either exists as a proper Riemann integral or else is an absolutely convergent improper Riemann integral.*

When these conditions are satisfied, we write $f \in R^*(a, b)$ and we define

$$\int_a^b f(t)\,dt = \sum_{k=1}^{n} \int_{c_{k-1}}^{c_k} f(t)\,dt.$$

NOTE. It is important to observe that an absolutely integrable function is defined everywhere on $[a, b]$, except possibly for a finite number of points. The function need not be bounded on $[a, b]$.

The following is an example of a function that is absolutely integrable on $(-\infty, +\infty)$:

$$f(t) = \frac{e^t}{|t|^{1/2}} \text{ if } t < 0, \qquad f(t) = \frac{1}{8\sqrt{t(1-t)}} \text{ if } 0 < t < 1,$$

$$f(t) = \frac{\log (t-1)}{e^t} \text{ if } t > 1.$$

Figure 15-1 represents an approximation of the graph of this function.

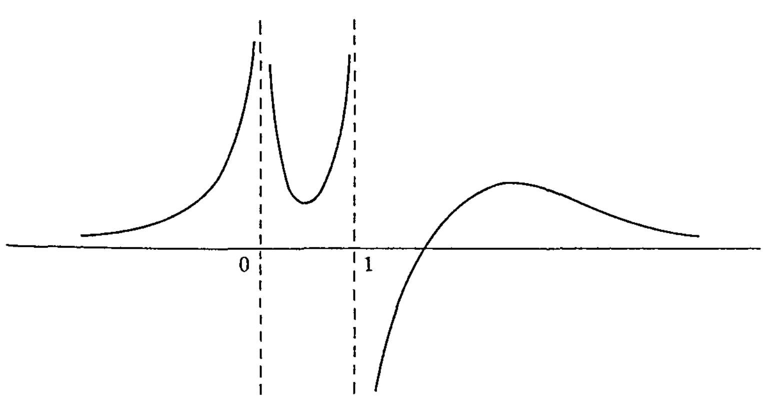

FIG. 15-1. An example in which $f \in R^*(-\infty, +\infty)$.

The results of the previous section can be collected into a single theorem which we can refer to as the *Riemann-Lebesgue lemma for absolutely integrable functions*

15-12 THEOREM *If $f \in R^*(a, b)$ (where a may be $-\infty$ and b may be $+\infty$), then for each real β we have*

$$\lim_{\alpha\to+\infty} \int_a^b f(t) \sin(\alpha t + \beta)\, dt = 0.$$

As an application of this theorem, we derive a result that will later be needed in the theory of Fourier integrals

15-13 THEOREM *If $f \in R^*(-\infty, +\infty)$, we have*

$$\lim_{\alpha\to+\infty} \int_{-\infty}^{\infty} f(t) \frac{1 - \cos \alpha t}{t}\, dt = \int_0^{\infty} \frac{f(t) - f(-t)}{t}\, dt \tag{26}$$

whenever the integral on the right is absolutely convergent

Proof. For each fixed α the integral on the left of (26) converges absolutely [Why?] Hence we can write

$$\begin{aligned}
\int_{-\infty}^{\infty} f(t) \frac{1 - \cos \alpha t}{t}\, dt &= \int_0^{\infty} f(t) \frac{1 - \cos \alpha t}{t}\, dt + \int_{-\infty}^{0} f(t) \frac{1 - \cos \alpha t}{t}\, dt \\
&= \int_0^{\infty} [f(t) - f(-t)] \frac{1 - \cos \alpha t}{t}\, dt \\
&= \int_0^{\infty} \frac{f(t) - f(-t)}{t}\, dt - \int_0^{\infty} \frac{f(t) - f(-t)}{t} \cos \alpha t\, dt
\end{aligned}$$

When $\alpha \to +\infty$, the last integral tends to 0, by the Riemann-Lebesgue lemma.

15-8 The Dirichlet integrals. Integrals of the form $\int_0^\delta g(t)\,(\sin \alpha t)/t\, dt$ (called *Dirichlet integrals*) play a very important role in the theory of Fourier series and also in the theory of Fourier integrals The function g in the integrand is assumed to have a finite right-hand limit $g(0+) = \lim_{t\to0+} g(t)$ and we are interested in formulating further conditions on g which will guarantee the validity of the following equation

$$\lim_{\alpha\to+\infty} \frac{2}{\pi} \int_0^{\delta} g(t) \frac{\sin \alpha t}{t}\, dt = g(0+) \tag{27}$$

To get an idea why we might expect a formula like (27) to hold, let us first consider the case when g is constant ($g(t) = g(0+)$) on $[0, \delta]$. Then (27) is a trivial consequence of the equation $\int_0^\infty (\sin t)/t\, dt = \pi/2$ (see formula 14–15), since

$$\int_0^\delta \frac{\sin \alpha t}{t}\, dt = \int_0^{\alpha\delta} \frac{\sin t}{t}\, dt \to \frac{\pi}{2} \qquad \text{as } \alpha \to +\infty.$$

More generally, if $g \in R$ on $[0, \delta]$ and if $0 < \epsilon < \delta$, we have

$$\lim_{\alpha \to +\infty} \frac{2}{\pi} \int_\epsilon^\delta g(t) \frac{\sin \alpha t}{t}\, dt = 0,$$

by the Riemann-Lebesgue lemma. Hence the validity of (27) is governed entirely by the *local behavior* of g near 0. Since $g(t)$ is "nearly" $g(0+)$ when t is "nearly" 0, there is some hope of proving (27) without placing too many additional restrictions on g. At first sight it would seem that continuity of g at 0 should certainly be enough to insure the existence of the limit in (27). Dirichlet showed that continuity of g on $[0, \delta]$ is sufficient to prove (27), if, in addition, g has only a finite number of maxima or minima on $[0, \delta]$. Jordan later proved (27) under the less restrictive condition that g be of bounded variation on $[0, \delta]$. However, all attempts to prove (27) under the sole hypothesis that g is continuous on $[0, \delta]$ have resulted in failure. In fact, du Bois-Reymond discovered an example of a continuous function g for which the limit in (27) fails to exist. Jordan's result, and a related theorem due to Dini, will be discussed here.

15–14 THEOREM (*Jordan*). *If g is of bounded variation on* $[0, \delta]$, *then*

$$\lim_{\alpha \to +\infty} \frac{2}{\pi} \int_0^\delta g(t) \frac{\sin \alpha t}{t}\, dt = g(0+). \tag{28}$$

Proof. It suffices to consider the case in which g is increasing on $[0, \delta]$. If $\alpha > 0$ and if $0 < h < \delta$, we have

$$\begin{aligned} \int_0^\delta g(t) \frac{\sin \alpha t}{t}\, dt &= \int_0^h [g(t) - g(0+)] \frac{\sin \alpha t}{t}\, dt \\ &\quad + g(0+) \int_0^h \frac{\sin \alpha t}{t}\, dt + \int_h^\delta g(t) \frac{\sin \alpha t}{t}\, dt \qquad (29) \\ &= I_1(\alpha, h) + I_2(\alpha, h) + I_3(\alpha, h), \end{aligned}$$

let us say We can apply the Riemann-Lebesgue lemma to $I_3(\alpha, h)$ (since the integral $\int_h^\delta g(t)/t\,dt$ exists) and we find $I_3(\alpha, h) \to 0$ as $\alpha \to +\infty$ Also,

$$I_2(\alpha, h) = g(0+)\int_0^h \frac{\sin \alpha t}{t}\,dt$$

$$= g(0+)\int_0^{h\alpha} \frac{\sin t}{t}\,dt \to \frac{\pi}{2}g(0+) \qquad \text{as } \alpha \to +\infty.$$

Next, choose $M > 0$ so that $|\int_a^b (\sin t)/t\,dt| < M$ for every $b \geq a \geq 0$ It follows that $|\int_a^b (\sin \alpha t)/t\,dt| < M$ for every $b \geq a \geq 0$ if $\alpha > 0$ Now let $\epsilon > 0$ be given and choose h in $(0, \delta)$ so that $|g(h) - g(0+)| < \epsilon/(3M)$ Since $g(t) - g(0+) \geq 0$ if $0 \leq t \leq h$, we can apply Bonnet's theorem (Theorem 9-34) in $I_1(\alpha, h)$ to write

$$I_1(\alpha, h) = \int_0^h [g(t) - g(0+)]\frac{\sin \alpha t}{t}\,dt = [g(h) - g(0+)]\int_c^h \frac{\sin \alpha t}{t}\,dt,$$

where $c \in [0, h]$ The definition of h gives us

$$(30) \qquad |I_1(\alpha, h)| = |g(h) - g(0+)|\left|\int_c^h \frac{\sin \alpha t}{t}\,dt\right| < \frac{\epsilon}{3M}M = \frac{\epsilon}{3}$$

For the same h we can choose A so that $\alpha \geq A$ implies

$$(31) \qquad |I_3(\alpha, h)| < \frac{\epsilon}{3} \qquad \text{and} \qquad \left|I_2(\alpha, h) - \frac{\pi}{2}g(0+)\right| < \frac{\epsilon}{3}.$$

Then, for $\alpha \geq A$, we can combine (29), (30), and (31) to get

$$\left|\int_0^\delta g(t)\frac{\sin \alpha t}{t}\,dt - \frac{\pi}{2}g(0+)\right| < \epsilon$$

This proves (28)

A different kind of condition for the validity of (28) was found by Dini and can be described as follows

15-15 THEOREM (*Dini*) *Assume that $g(0+)$ exists and suppose that for some $\delta > 0$ the improper integral*

$$\int_{0+}^\delta \frac{g(t) - g(0+)}{t}\,dt$$

converges absolutely Then we have

$$\lim_{\alpha\to+\infty} \frac{2}{\pi}\int_0^\delta g(t)\,\frac{\sin \alpha t}{t}\,dt = g(0+).$$

Proof. Write

$$\int_0^\delta g(t)\,\frac{\sin \alpha t}{t}\,dt = \int_0^\delta \frac{g(t) - g(0+)}{t}\sin \alpha t\,dt + g(0+)\int_0^{\alpha\delta} \frac{\sin t}{t}\,dt.$$

When $\alpha \to +\infty$, the first term on the right tends to 0 (by the Riemann-Lebesgue lemma) and the second term tends to $\frac{1}{2}\pi g(0+)$.

NOTE. If $g \in R$ on $[a, \delta]$ for every positive $a < \delta$, it is easy to show that Dini's condition is satisfied whenever g satisfies a "right-handed" Lipschitz condition at 0; that is, whenever there exist two positive constants M and p such that

$$|g(t) - g(0+)| < Mt^p, \qquad \text{for every } t \text{ in } (0, \delta].$$

(See Exercise 15–18.) In particular, the Lipschitz condition holds with $p = 1$ whenever g has a right-hand derivative at 0. It is of interest to note that there exist functions which satisfy Dini's condition but which do not satisfy Jordan's condition. Similarly, there are functions which satisfy Jordan's condition but not Dini's. (See Ref. 15–12.)

15–9 An integral representation for the partial sums of a Fourier series.

15–16 THEOREM. *Assume that $f \in R$ on $[0, 2\pi]$ and suppose that f is periodic with period 2π. Let $\{s_n\}$ denote the sequence of partial sums of the Fourier series generated by f, say*

$$(32)\qquad s_n(x) = \frac{a_0}{2} + \sum_{k=1}^{n} (a_k \cos kx + b_k \sin kx) \qquad (n = 1, 2, \ldots).$$

Then we have the integral representation

$$(33)\qquad s_n(x) = \frac{2}{\pi}\int_0^{\pi/2} \frac{f(x + 2t) + f(x - 2t)}{2}\,\frac{\sin (2n + 1)t}{\sin t}\,dt.$$

Proof. The Fourier coefficients of f are given by the integrals in (16). Substituting these integrals in (32) yields

$$s_n(x) = \frac{1}{\pi}\int_0^{2\pi} f(t)\left\{\frac{1}{2} + \sum_{k=1}^{n} (\cos kt \cos kx + \sin kt \sin kx)\right\} dt$$

$$= \frac{1}{\pi}\int_0^{2\pi} f(t)\left\{\frac{1}{2} + \sum_{k=1}^{n} \cos k(t - x)\right\} dt = \frac{1}{\pi}\int_0^{2\pi} f(t)\,D_n(t - x)\,dt,$$

where $D_n(t)$ denotes the sum

$$D_n(t) = \frac{1}{2} + \sum_{k=1}^{n} \cos kt$$

Since both f and D_n are periodic with period 2π, we can replace the interval of integration by $[x - \pi, x + \pi]$ and make a change of variable to get

$$s_n(x) = \frac{1}{\pi} \int_{x-\pi}^{x+\pi} f(t)\, D_n(t - x)\, dt = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x + t)\, D_n(t)\, dt$$

Using the equation $D_n(-t) = D_n(t)$, we find

$$\begin{aligned} s_n(x) &= \frac{2}{\pi} \int_0^{\pi} \frac{f(x + t) + f(x - t)}{2}\, D_n(t)\, dt \\ &= \frac{4}{\pi} \int_0^{\pi/2} \frac{f(x + 2t) + f(x - 2t)}{2}\, D_n(2t)\, dt. \end{aligned}$$

But, by formula (12–12), we have

$$D_n(2t) = \begin{cases} \dfrac{\sin (2n + 1)t}{2 \sin t}, & \text{if } t \neq m\pi \ (m \text{ is an integer}), \\[2ex] n + \dfrac{1}{2}, & \text{if } t = m\pi \ (m \text{ is an integer}). \end{cases}$$

This completes the proof of (33)

15–10 Riemann's localization theorem. Formula (33) tells us that the Fourier series generated by f will converge at a point x if, and only if, the following limit exists

(34) $$\lim_{n\to\infty} \frac{2}{\pi} \int_0^{\pi/2} \frac{f(x + 2t) + f(x - 2t)}{2}\, \frac{\sin (2n + 1)t}{\sin t}\, dt,$$

in which case the value of this limit will be the sum of the series This integral is essentially a Dirichlet integral of the type discussed in the previous section, except that $\sin t$ appears in the denominator rather than t. However, the Riemann-Lebesgue lemma allows us to replace $\sin t$ by t in (34) without affecting either the existence or the value of the limit More precisely, the Riemann-Lebesgue lemma implies

$$\lim_{n\to\infty} \frac{2}{\pi}\int_0^{\pi/2} \left[\frac{1}{t} - \frac{1}{\sin t}\right] \frac{f(x+2t) + f(x-2t)}{2} \sin (2n+1)t\, dt = 0,$$

because the function F defined by the equation

$$F(t) = \begin{cases} \dfrac{1}{t} - \dfrac{1}{\sin t}, & \text{if } 0 < t \le \dfrac{\pi}{2}, \\ 0, & \text{if } t = 0, \end{cases}$$

is continuous on $[0, \pi/2]$. (The reader should verify continuity at $t = 0$.) Therefore the convergence problem for Fourier series amounts to finding conditions on f which will guarantee the existence of the following limit:

$$(35) \qquad \lim_{n\to\infty} \frac{2}{\pi}\int_0^{\pi/2} \frac{f(x+2t) + f(x-2t)}{2} \frac{\sin (2n+1)t}{t}\, dt.$$

Using the Riemann-Lebesgue lemma once more, we need only consider the limit in (35) when the integral $\int_0^{\pi/2}$ is replaced by $\int_0^{\delta}$, where δ is any positive number $< \pi/2$, because the integral $\int_\delta^{\pi/2}$ tends to 0 as $n \to \infty$. Therefore we can sum up the results of the previous section in the following theorem:

15–17 THEOREM. *Assume that $f \in R$ on $[0, 2\pi]$ and suppose f has period 2π. Then the Fourier series generated by f will converge for a given value of x if, and only if, for some positive $\delta < \pi/2$ the following limit exists:*

$$(36) \qquad \lim_{n\to\infty} \frac{2}{\pi}\int_0^{\delta} \frac{f(x+2t) + f(x-2t)}{2} \frac{\sin (2n+1)t}{t}\, dt,$$

in which case the value of this limit is the sum of the Fourier series.

This theorem is known as *Riemann's localization theorem.* It tells us that the convergence or divergence of a Fourier series at a particular point is governed entirely by the behavior of f in an arbitrarily small neighborhood of the point. This is rather surprising in view of the fact that the coefficients of the Fourier series depend on the values which the function assumes throughout the entire interval $[0, 2\pi]$.

NOTE. In Theorem 15–17 the hypothesis $f \in R$ on $[0, 2\pi]$ can be weakened to read $f \in R^*(0, 2\pi)$. The integrals in (16) which define the Fourier coefficients still exist as absolutely convergent improper integrals under this hypothesis. All the above analysis goes through as before if we apply the Riemann-Lebesgue lemma in the form given in Theorem 15–12.

15–11 Sufficient conditions for convergence of a Fourier series.

15–18 THEOREM (*Jordan*) *Assume that $f \in R^*(0, 2\pi)$ and suppose that f has period 2π Suppose there is a point x and an interval $[x - \delta, x + \delta]$ about x on which f is of bounded variation Then the Fourier series generated by f converges for this value of x to the sum $[f(x+) + f(x-)]/2$. If, in addition, f is continuous at x, the series converges to $f(x)$*

Proof Keep x fixed and define g as follows.

$$g(t) = \frac{f(x + 2t) + f(x - 2t)}{2}, \qquad \text{if } t \in [0, \delta/2]$$

Then g satisfies the hypotheses of Theorem 15–14 and hence the limit in (36) exists and equals $g(0+)$. This proves the theorem

15–19 THEOREM (*Dini*) *Assume that $f \in R^*(0, 2\pi)$ and suppose that f has period 2π Suppose there is a point x and a positive $\delta < \pi/2$ such that both limits $f(x+)$ and $f(x-)$ exist and such that both improper integrals*

$$\int_{0+}^{\delta} \frac{f(x + t) - f(x+)}{t}\,dt \qquad \text{and} \qquad \int_{0+}^{\delta} \frac{f(x - t) - f(x-)}{t}\,dt$$

are absolutely convergent Then the Fourier series generated by f converges for this x to the value $[f(x+) + f(x-)]/2$.

Proof Keep x fixed and define g on $[0, \delta]$ as follows·

$$g(t) = \frac{f(x + t) + f(x - t)}{2}.$$

Then g satisfies the hypotheses of Theorem 15–15 and hence the limit in (36) exists and equals $g(0+)$ This proves the theorem

15–12 Cesàro summability of Fourier series. Continuity of a function f is not a very fruitful hypothesis when it comes to studying convergence of the Fourier series generated by f In 1873, du Bois-Reymond gave an example of a function, continuous throughout the interval $[0, 2\pi]$, whose Fourier series fails to converge on an *uncountable subset* of $[0, 2\pi]$ To this day it is not known whether or not continuity of a function on $[0, 2\pi]$ always implies convergence of its Fourier series for *at least one point* of the interval On the other hand, continuity *does* suffice to establish Cesàro summability of the series This result (due to Fejér) and some of its consequences will be discussed next

Our first task is to obtain an integral representation for the arithmetic means of the partial sums of a Fourier series

15–20 THEOREM. *Assume that $f \in R^*(0, 2\pi)$ and suppose that f is periodic with period 2π. Let s_n denote the nth partial sum of the Fourier series generated by f and let*

$$\sigma_n(x) = \frac{s_0(x) + s_1(x) + \cdots + s_{n-1}(x)}{n} \qquad (n = 1, 2, \ldots). \tag{37}$$

Then we have the integral representation

$$\sigma_n(x) = \frac{2}{n\pi} \int_0^{\pi/2} \frac{f(x + 2t) + f(x - 2t)}{2} \left(\frac{\sin nt}{\sin t}\right)^2 dt. \tag{38}$$

Proof. If we use the integral representation for $s_n(x)$ given in (33) and form the sum defining $\sigma_n(x)$, we immediately obtain the required result because of formula (12–16).

NOTE. If we apply Theorem 15–20 to the constant function whose value is 1 at each point we find $\sigma_n(x) = s_n(x) = 1$ for each n and hence (38) becomes

$$\frac{2}{n\pi} \int_0^{\pi/2} \left(\frac{\sin nt}{\sin t}\right)^2 dt = 1. \tag{39}$$

Therefore, given any s, we can combine (39) with (38) to write

$$\sigma_n(x) - s = \frac{2}{n\pi} \int_0^{\pi/2} \left\{\frac{f(x + 2t) + f(x - 2t)}{2} - s\right\} \left(\frac{\sin nt}{\sin t}\right)^2 dt. \tag{40}$$

If we can choose a value of s such that the integral on the right of (40) tends to 0 as $n \to \infty$, it will follow that $\sigma_n(x) \to s$ as $n \to \infty$. The next theorem shows that it suffices to take $s = [f(x+) + f(x-)]/2$.

15–21 THEOREM (*Fejér*). *Assume that $f \in R^*(0, 2\pi)$ and suppose that f is periodic with period 2π. Define a function s by the following equation:*

$$s(x) = \lim_{t \to 0+} \frac{f(x + t) + f(x - t)}{2},$$

whenever the limit exists. Then, for each x for which $s(x)$ is defined, the Fourier series generated by f is Cesàro summable and has $(C, 1)$ sum $s(x)$. That is, we have

$$\lim_{n \to \infty} \sigma_n(x) = s(x), \tag{41}$$

where $\{\sigma_n\}$ is the sequence of arithmetic means defined by (37). If, in addition, f is continuous on a closed interval $[a, b]$, the sequence $\{\sigma_n\}$ converges uniformly to f on $[a, b]$.

Proof Let $g_x(t) = [f(x + 2t) + f(x - 2t)]/2 - s(x)$ whenever $s(x)$ is defined Then $g_x(t) \to 0$ as $t \to 0+$. Therefore, given $\epsilon > 0$, there exists a $\delta < \pi/2$ such that $|g_x(t)| < \epsilon/2$ whenever $0 < t \leq \delta$ (Note that δ depends on x as well as on ϵ However, if f is continuous on a closed interval $[a, b]$, then f is uniformly continuous on $[a, b]$ and there exists a δ which serves equally well for *every* x in $[a, b]$) Using (39), we have

$$\left|\frac{2}{n\pi}\int_0^\delta g_x(t)\left(\frac{\sin nt}{\sin t}\right)^2 dt\right| \leq \frac{\epsilon}{2}\,\frac{2}{n\pi}\int_0^{\pi/2}\left(\frac{\sin nt}{\sin t}\right)^2 dt = \frac{\epsilon}{2}$$

Next, we have

$$\left|\frac{2}{n\pi}\int_\delta^{\pi/2} g_x(t)\left(\frac{\sin nt}{\sin t}\right)^2 dt\right| \leq \frac{2}{n\pi}\,\frac{1}{\sin^2\delta}\int_\delta^{\pi/2}|g_x(t)|\,dt \leq \frac{2}{n\pi}\,\frac{I}{\sin^2\delta},$$

where $I = \int_0^{\pi/2}|g_x(t)|\,dt$. (Note that the hypothesis $f \in R^*(0, 2\pi)$ implies convergence of $\int_0^{\pi/2}|g_x(t)|\,dt$ in case this is an improper integral) Next, choose N so that $2I/(N\pi\sin^2\delta) < \frac{1}{2}\epsilon$ Then $n \geq N$ implies

$$|\sigma_n(x) - s(x)| = \left|\frac{2}{n\pi}\int_0^{\pi/2} g_x(t)\left(\frac{\sin nt}{\sin t}\right)^2 dt\right| < \epsilon.$$

In other words, (41) holds Observe that when f is continuous on $[a, b]$, the function g_x is bounded on $[a, b]$ (by M, say, where M is independent of x) and we may replace I by $\pi M/2$ in the above argument The resulting N is then independent of x and hence $\sigma_n \to s$ uniformly on $[a, b]$ In this case, of course, $s(x) = f(x)$ at each point of $[a, b]$ This completes the proof.

15–13 Consequences of Fejér's theorem.

15–22 THEOREM *Let f be continuous on $[0, 2\pi]$ and periodic with period 2π. Let $\{s_n\}$ denote the sequence of partial sums of the Fourier series generated by f, say*

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx) \tag{42}$$

Then we have

(a) $\lim_{n\to\infty} s_n = f$ *on every interval* $[a, b]$

(b) $\displaystyle\frac{1}{\pi}\int_0^{2\pi}|f(x)|^2\,dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)$ (*Parseval's formula*)

$$\text{(c)} \quad \int_0^x f(t)\,dt = \frac{a_0 x}{2} + \sum_{n=1}^{\infty} \int_0^x (a_n \cos nt + b_n \sin nt)\,dt.$$

NOTE. The series in (c) converges for every x, even when the series in (42) diverges. The convergence is uniform on every interval $[a, b]$.

Proof. Applying formula (20) of Theorem 15–5, with $t_n(x) = \sigma_n(x) = (1/n) \sum_{k=0}^{n-1} s_k(x)$, we obtain the inequality

$$\int_0^{2\pi} |f(x) - s_n(x)|^2\,dx \leq \int_0^{2\pi} |f(x) - \sigma_n(x)|^2\,dx. \tag{43}$$

But, since $\sigma_n \to f$ uniformly on $[0, 2\pi]$, it follows that $\operatorname{l.i.m.}_{n\to\infty} \sigma_n = f$ on $[0, 2\pi]$ and (43) implies (a). (The result holds for every interval $[a, b]$ because of periodicity.) Part (b) follows from (a) because of Theorem 15–6. Finally, part (c) also follows from (a), by Theorem 13–19.

15–23 THEOREM (*Weierstrass' approximation theorem*). *Let f be real-valued and continuous on a closed interval $[a, b]$. Then, given any $\epsilon > 0$, there exists a polynomial p (which may depend on ϵ) such that*

$$|f(x) - p(x)| < \epsilon, \qquad \textit{for every } x \textit{ in } [a, b]. \tag{44}$$

NOTE. This theorem is described by saying that every continuous function can be "uniformly approximated" by a polynomial.

Proof. If $t \in [0, 2\pi]$, let $g(t) = f[a + \frac{1}{2}t(b - a)/\pi]$ and define g outside $[0, 2\pi]$ so that g has period 2π. For the ϵ given in the theorem, we can apply Fejér's theorem to find a function σ defined by an equation of the form

$$\sigma(t) = A_0 + \sum_{k=1}^{N} (A_k \cos kt + B_k \sin kt)$$

such that $|g(t) - \sigma(t)| < \epsilon/2$ for every t in $[0, 2\pi]$. (Note that N, and hence σ, depends on ϵ.) Since σ is a finite sum of trigonometric functions, it generates a power series expansion about the origin which converges uniformly on every finite interval. The partial sums of this power series expansion constitute a sequence of polynomials, say $\{p_n\}$, such that $p_n \to \sigma$ uniformly on $[0, 2\pi]$. Hence, for the same ϵ, there exists an m such that

$$|p_m(t) - \sigma(t)| < \frac{\epsilon}{2}, \qquad \text{for every } t \text{ in } [0, 2\pi].$$

Therefore we have

$$|p_m(t) - g(t)| < \epsilon, \qquad \text{for every } t \text{ in } [0, 2\pi] \tag{45}$$

Now define the polynomial p by the formula $p(x) = p_m[2\pi(x - a)/(b - a)]$ Then inequality (45) becomes (44) when we put $t = 2\pi(x - a)/(b - a)$

15-14 Other forms of Fourier series. Using the formulas $2 \cos nx = e^{inx} + e^{-inx}$ and $2i \sin nx = e^{inx} - e^{-inx}$, the Fourier series generated by f can be expressed in terms of complex exponentials as follows.

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (\alpha_n e^{inx} + \beta_n e^{-inx}),$$

where $\alpha_n = (a_n - ib_n)/2$ and $\beta_n = (a_n + ib_n)/2$ By defining $\alpha_0 = a_0$ and $\alpha_{-n} = \beta_n$, we can write the exponential form more briefly as follows

$$f(x) \sim \sum_{n=-\infty}^{\infty} \alpha_n e^{inx}$$

The formulas (16) for the coefficients now become

$$\alpha_n = \frac{1}{2\pi} \int_0^{2\pi} f(t) e^{-int}\, dt \qquad (n = 0, \pm 1, \pm 2, \quad)$$

If f has period 2π, the interval of integration can be replaced by any other interval of length 2π.

More generally, if $f \in R$ on $[0, p]$ and if f has period p, we write

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{2\pi n x}{p} + b_n \sin \frac{2\pi n x}{p} \right)$$

to mean that the coefficients are given by the formulas

$$a_n = \frac{2}{p} \int_0^p f(t) \cos \frac{2\pi n t}{p}\, dt,$$

$$b_n = \frac{2}{p} \int_0^p f(t) \sin \frac{2\pi n t}{p}\, dt \qquad (n = 0, 1, 2, . \quad)$$

In exponential form we can write

$$f(x) \sim \sum_{n=-\infty}^{\infty} \alpha_n e^{2\pi i n x/p},$$

where

$$\alpha_n = \frac{1}{p}\int_0^p f(t)e^{-2\pi i n t/p}\,dt, \qquad \text{if } n = 0, \pm 1, \pm 2, \ldots.$$

All the convergence theorems for Fourier series of period 2π can also be applied to the case of a general period p by making a suitable change of scale.

15–15 The Fourier integral theorem. The hypothesis of periodicity, which appears in all the convergence theorems dealing with Fourier series, is not as serious a restriction as it may appear to be at first sight. If a function is initially defined on a finite interval, say $[a, b]$, we can always extend the definition of f outside $[a, b]$ by imposing some sort of periodicity condition. For example, if $f(a) = f(b)$, we can define f everywhere on $(-\infty, +\infty)$ by requiring the equation $f(x + p) = f(x)$ to hold for every x, where $p = b - a$. (The condition $f(a) = f(b)$ can always be brought about by changing the value of f at one of the endpoints if necessary. This does not affect the existence or the values of the integrals which are used to compute the Fourier coefficients of f.) However, if the given function is already defined everywhere on $(-\infty, +\infty)$ and is *not* periodic, then there is no hope of obtaining a Fourier series which represents the function everywhere on $(-\infty, +\infty)$. Nevertheless, in such a case the function can sometimes be represented by an *infinite integral* rather than by an infinite series. These integrals, which are in many ways analogous to Fourier series, are known as *Fourier integrals,* and the theorem which gives sufficient conditions for representing a function by such an integral is known as the *Fourier integral theorem.* The basic tools used in the theory are, as in the case of Fourier series, the Dirichlet integrals and the Riemann-Lebesgue lemma.

15–24 THEOREM (*Fourier integral theorem*). *Assume that* $f \in R^*(-\infty, +\infty)$. *Suppose there is a point* x *in* E_1 *and an interval* $[x - \delta, x + \delta]$ *about* x *such that either*

(a) f *is of bounded variation on* $[x - \delta, x + \delta]$,

or else

(b) *both limits* $f(x+)$ *and* $f(x-)$ *exist and both improper integrals*

$$\int_{0+}^{\delta} \frac{f(x+t) - f(x+)}{t}\,dt \quad \text{and} \quad \int_{0+}^{\delta} \frac{f(x-t) - f(x-)}{t}\,dt$$

are absolutely convergent.

Then we have the formula

$$\frac{f(x+) + f(x-)}{2} = \frac{1}{\pi} \int_0^\infty \left[\int_{-\infty}^\infty f(u) \cos v(u - x) \, du \right] dv \tag{46}$$

Proof The first step in the proof is to establish the following formula.

$$\lim_{\alpha \to +\infty} \frac{1}{\pi} \int_{-\infty}^\infty f(x + t) \frac{\sin \alpha t}{t} \, dt = \frac{f(x+) + f(x-)}{2} \tag{47}$$

For this purpose we write

$$\int_{-\infty}^\infty f(x + t) \frac{\sin \alpha t}{\pi t} \, dt = \int_{-\infty}^{-\delta} + \int_{-\delta}^0 + \int_0^\delta + \int_\delta^\infty$$

When $\alpha \to +\infty$, the first and fourth integrals on the right tend to 0, because of the Riemann-Lebesgue lemma In the third integral, we can apply either Theorem 15–14 or Theorem 15–15 (depending on whether (a) or (b) is satisfied) to get

$$\lim_{\alpha \to +\infty} \int_0^\delta f(x + t) \frac{\sin \alpha t}{\pi t} \, dt = \frac{f(x+)}{2}.$$

Similarly, we have

$$\int_{-\delta}^0 f(x + t) \frac{\sin \alpha t}{\pi t} \, dt = \int_0^\delta f(x - t) \frac{\sin \alpha t}{\pi t} \, dt \to \frac{f(x-)}{2} \qquad \text{as } \alpha \to +\infty$$

Thus we have established (47) If we make a change of variable, we get

$$\int_{-\infty}^\infty f(x + t) \frac{\sin \alpha t}{t} \, dt = \int_{-\infty}^\infty f(u) \frac{\sin \alpha(u - x)}{u - x} \, du,$$

and if we use the elementary formula

$$\frac{\sin \alpha(u - x)}{u - x} = \int_0^\alpha \cos v(u - x) \, dv,$$

the limit relation in (47) becomes

$$\lim_{\alpha \to +\infty} \frac{1}{\pi} \int_{-\infty}^\infty f(u) \left[\int_0^\alpha \cos v(u - x) \, dv \right] du = \frac{f(x+) + f(x-)}{2}. \tag{48}$$

But the formula we seek to prove is nothing else than (48) with the order of integration reversed. Because of the rather weak hypothesis we have imposed on f, some care is required to verify that this interchange is permissible. The problem we face is to justify the equation

(49)
$$\int_0^{\alpha}\left[\int_{-\infty}^{\infty} f(u)\cos v(u - x)\,du\right]dv = \int_{-\infty}^{\infty}\left[\int_0^{\alpha} f(u)\cos v(u - x)\,dv\right]du$$

for every $\alpha > 0$. The hypothesis $f \in R^*(-\infty, +\infty)$ means that, for each $\alpha > 0$, the integral $\int_0^\alpha f(u)\,du$ exists, either as a proper Riemann integral or as a finite sum of absolutely convergent improper Riemann integrals. If $\int_0^\alpha f(u)\,du$ is a proper Riemann integral, then formula (49) follows at once from Theorem 14–26 [in the form given in formula (14–19)]. It remains to prove that (49) still holds when the integral $\int_0^\alpha f(u)\,du$ is a finite sum of improper integrals. For simplicity, let us assume that

$$\int_0^{\alpha} f(u)\,du = \lim_{\beta\to\alpha-}\int_0^{\beta} f(u)\,du,$$

where $\int_0^\beta f(u)\,du$ is a proper Riemann integral for each $\beta < \alpha$. Then (49) holds with α replaced by β, and we need only show that we have

(50)
$$\lim_{\beta\to\alpha-}\int_{-\infty}^{\infty}\left[\int_0^{\beta} f(u)\cos v(u - x)\,dv\right]du = \int_{-\infty}^{\infty}\left[\int_0^{\alpha} f(u)\cos v(u - x)\,dv\right]du.$$

Let $g(\beta) = \int_{-\infty}^{\infty}\left[\int_0^\beta f(u)\cos v(u - x)\,dv\right]du$ if $0 \le \beta \le \alpha$. Then, for each $\beta < \alpha$, we have

$$|g(\alpha) - g(\beta)| \le \int_{-\infty}^{\infty} |f(u)|\left|\int_\beta^{\alpha}\cos v(u - x)\,dv\right|du$$
$$\le (\alpha - \beta)\int_{-\infty}^{\infty} |f(u)|\,du.$$

Since the integral $\int_{-\infty}^{\infty}|f(u)|\,du$ is finite (by hypothesis), we see that $g(\beta) \to g(\alpha)$ as $\beta \to \alpha-$. In other words, (50) holds. The remaining cases are similarly treated and this completes the proof.

15-16 The exponential form of the Fourier integral theorem.

15-25 THEOREM *If f satisfies the hypotheses of the Fourier integral theorem and if, in addition, the integral $\int_{-\infty}^{\infty} |f(x+t) - f(x-t)|/t\,dt$ is convergent, then we have*

$$\frac{f(x+) + f(x-)}{2} = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} f(u)e^{iv(u-x)}\,du\right]dv \tag{51}$$

Proof Let $F(v) = \int_{-\infty}^{\infty} f(u)\cos v(u-x)\,du$ Then $F(v) = F(-v)$ and hence $\int_{-\infty}^{0} F(v)\,dv = \int_0^{\infty} F(-v)\,dv = \int_0^{\infty} F(v)\,dv$ Therefore, (46) becomes

$$\begin{aligned}\frac{f(x+) + f(x-)}{2} &= \frac{1}{2\pi}\int_{-\infty}^{\infty} F(v)\,dv \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} f(u)\cos v(u-x)\,du\right]dv.\end{aligned} \tag{52}$$

Now define G on $(-\infty, +\infty)$ by the equation

$$G(v) = \int_{-\infty}^{\infty} f(u)\sin v(u-x)\,du.$$

We will prove that both integrals $\int_0^{\infty} G(v)\,dv$ and $\int_{-\infty}^{0} G(v)\,dv$ converge and that their sum is zero The same argument we used to prove (49) also proves that we have

$$\int_0^{\alpha} G(v)\,dv = \int_{-\infty}^{\infty}\left[\int_0^{\alpha} f(u)\sin v(u-x)\,dv\right]du \tag{53}$$

for every $\alpha > 0$ Because of the elementary formula

$$\int_0^{\alpha} \sin v(u-x)\,dv = \frac{1-\cos\alpha(u-x)}{u-x},$$

equation (53) becomes

$$\int_0^{\alpha} G(v)\,dv = \int_{-\infty}^{\infty} f(u)\,\frac{1-\cos\alpha(u-x)}{u-x}\,du = \int_{-\infty}^{\infty} f(t+x)\,\frac{1-\cos\alpha t}{t}\,dt$$

By Theorem 15-13, we find

$$\lim_{\alpha\to+\infty}\int_{-\infty}^{\infty} f(t+x)\,\frac{1-\cos\alpha t}{t}\,dt = \int_0^{\infty}\frac{f(x+t) - f(x-t)}{t}\,dt,$$

since the integral on the right is absolutely convergent (by hypothesis). The existence of the limit in this equation shows that the integral $\int_0^\infty G(v)\,dv$ is convergent. Since $\int_{-\alpha}^0 G(v)\,dv = \int_0^\alpha G(-v)\,dv = -\int_0^\alpha G(v)\,dv$, the integral $\int_{-\infty}^0 G(v)\,dv$ is also convergent and its value is $-\int_0^\infty G(v)\,dv$. This proves that $\int_{-\infty}^\infty G(v)\,dv = 0$. Combining this with (52), we find

$$\frac{f(x+) + f(x-)}{2} = \frac{1}{2\pi}\int_{-\infty}^{\infty} (F(v) + iG(v))\,dv.$$

This is formula (51).

NOTE. The hypothesis that $\int_{-\infty}^\infty |f(x+t) - f(x-t)|/t\,dt$ converges was used only to prove that $\int_0^\infty G(v)\,dv$ converges. It is important to realize that the Cauchy principal value of $\int_{-\infty}^\infty G(v)\,dv$ exists without this hypothesis. In fact, we have $\int_{-\alpha}^\alpha G(v)\,dv = 0$ for every α, since G is an odd function and hence $\lim_{\alpha\to+\infty}\int_{-\alpha}^\alpha G(v)\,dv = 0$. Therefore, equation (51) is still valid under the hypotheses of the Fourier integral theorem provided that we interpret the outer integral in (51) as a Cauchy principal value. In other words, under the assumptions of the Fourier integral theorem we have

$$\text{(54)} \qquad \frac{f(x+) + f(x-)}{2} = \frac{1}{2\pi}\lim_{\alpha\to+\infty}\int_{-\alpha}^{\alpha}\left[\int_{-\infty}^{\infty} f(u)e^{iv(u-x)}\,du\right]dv.$$

It should be further noted that the value of the integral in (54) is not changed when v is replaced by $-v$ in the integrand.

15–17 Integral transforms. A large number of important functions in analysis can be expressed as improper integrals of the form

$$\text{(55)} \qquad g(y) = \int_{-\infty}^{\infty} K(x, y)f(x)\,dx.$$

A function g defined by an equation of this sort (in which y may be either real or complex) is called an *integral transform* of f. The function K which appears in the integrand is referred to as the *kernel* of the transform.

Integral transforms are employed very extensively in both pure and applied mathematics. They are especially useful in solving certain boundary value problems and certain types of integral equations. Some of the more commonly used transforms are listed below:

Exponential Fourier transform: $\displaystyle\int_{-\infty}^{\infty} e^{-ixy}f(x)\,dx.$

Fourier cosine transform: $\displaystyle\int_0^{\infty} \cos xy\, f(x)\,dx.$

Fourier sine transform: $\int_0^\infty \sin xy\, f(x)\, dx$

Laplace transform: $\int_0^\infty e^{-xy} f(x)\, dx$

Mellin transform: $\int_0^\infty x^{y-1} f(x)\, dx$

Since $e^{-ixy} = \cos xy - i \sin xy$, the sine and cosine transforms are merely special cases of the exponential Fourier transform in which the function f vanishes on the negative real axis The Laplace transform is also related to the exponential Fourier transform If we consider a complex value of y, say $y = u + iv$, where u and v are real, we can write

$$\int_0^\infty e^{-xy} f(x)\, dx = \int_0^\infty e^{-ixv} e^{-xu} f(x)\, dx = \int_0^\infty e^{-ixv} \phi_u(x)\, dx,$$

where $\phi_u(x) = e^{-xu} f(x)$ Therefore the Laplace transform can also be regarded as a special case of the exponential Fourier transform.

NOTE. An equation such as (55) is sometimes written more briefly in the form $g = K(f)$ or $g = Kf$, where K denotes the "operator" which converts f into g Since integration is involved in this equation, the operator K is referred to as an *integral operator* It is clear that K is also a linear operator That is, $K(a_1 f_1 + a_2 f_2) = a_1 K f_1 + a_2 K f_2$, if a_1 and a_2 are constants The operator defined by the Fourier transform is often denoted by F and that defined by the Laplace transform is denoted by L.

The exponential form of the Fourier integral theorem can be expressed in terms of Fourier transforms as follows Let g denote the Fourier transform of f, so that

$$g(u) = \int_{-\infty}^{\infty} f(t) e^{-itu}\, dt \tag{56}$$

Then, at points of continuity of f, formula (51) becomes

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} g(u) e^{ixu}\, du \tag{57}$$

and this is called the *inversion formula* for Fourier transforms It tells us that a continuous function f satisfying the conditions of Theorem 15–25 is uniquely determined by its Fourier transform g

NOTE If F denotes the operator defined by (56), it is customary to denote by F^{-1} the operator defined by (57). Equations (56) and (57) can be expressed symbolically by writing $g = Ff$ and $f = F^{-1}g$. Roughly speaking, the inversion formula tells us how to "solve" the equation $g = Ff$ for f in terms of g.

Before we pursue the study of Fourier transforms any further, we must introduce a new notion, the *convolution* of two functions. This can be interpreted as a special kind of integral transform in which the kernel $K(x, y)$ depends only on the *difference* $x - y$.

15–18 Convolutions.

15–26 DEFINITION. *Given two functions f and g, both absolutely integrable on $(-\infty, +\infty)$, let S denote the set of x for which the improper integral*

$$h(x) = \int_{-\infty}^{\infty} f(t)g(x - t)\, dt \tag{58}$$

converges. This integral defines a function h on S called the convolution of f and g. We also write $h = f*g$ to denote this function.

NOTE. It is easy to see (by a change of variable) that $f*g = g*f$ whenever the integral converges.

An important special case occurs when both f and g vanish on the negative real axis. In this case, $g(x - t) = 0$ if $t > x$, and (58) becomes

$$h(x) = \int_{0}^{x} f(t)g(x - t)\, dt. \tag{59}$$

It is clear that, in this case, the convolution will be defined at each point of an interval $[a, b]$ if both f and g are properly Riemann-integrable on $[a, b]$. However, this need not be so if we assume only that f and g are absolutely integrable on $[a, b]$. For example, let

$$f(t) = \frac{1}{\sqrt{t}} \quad \text{and} \quad g(t) = \frac{1}{\sqrt{1 - t}}, \quad \text{if } 0 < t < 1,$$

and let $f(t) = g(t) = 0$ if $t \le 0$ or if $t \ge 1$. Then f has an infinite discontinuity at $t = 0$. Nevertheless, the improper integral $\int_{-\infty}^{\infty} f(t)\, dt = \int_0^1 t^{-1/2}\, dt$ converges. Similarly, the improper integral $\int_{-\infty}^{\infty} g(t)\, dt = \int_0^1 (1 - t)^{-1/2}\, dt$ converges, although g has an infinite discontinuity at $t = 1$. However, when we form the convolution integral in (58) corresponding to $x = 1$, we find

$$\int_{-\infty}^{\infty} f(t)g(1 - t)\, dt = \int_0^1 t^{-1}\, dt.$$

Observe that the two discontinuities of f and g have "coalesced" into *one* discontinuity of such nature that the convolution integral diverges

This example shows that there may be certain points on the real axis at which the integral in (58) fails to exist, even though both f and g are absolutely integrable on $(-\infty, +\infty)$ Let us refer to such points as "singularities" of h It is easy to show that such singularities cannot occur unless *both* f and g have infinite discontinuities More precisely, we have the following theorem

15–27 THEOREM *If $f \in R^*(-\infty, +\infty)$ and $g \in R^*(-\infty, +\infty)$, and if either f or g is bounded on $(-\infty, +\infty)$, then the convolution integral $\int_{-\infty}^{\infty} f(t)g(x-t)\,dt$ converges absolutely for every x in E_1*

Proof Since $f*g = g*f$, it suffices to consider the case in which f is bounded Suppose $|f(t)| \le M$ for every t in E_1 Then, for every finite interval $[a, b]$ in which both f and g are Riemann-integrable, we have

$$\int_a^b |f(t)g(x-t)|\,dt \le M\int_a^b |g(x-t)|\,dt$$

$$= M\int_{x-b}^{x-a} |g(t)|\,dt \le M\int_{-\infty}^{\infty} |g(t)|\,dt$$

The conclusion is now an easy consequence of Theorems 14–2 and 14–12.

Another sufficient condition for the existence of the convolution can be described as follows

15–28 THEOREM. *If $f \in R^*(-\infty, +\infty)$ and $g \in R^*(-\infty, +\infty)$, and if both integrals $\int_{-\infty}^{\infty} |f(t)|^2\,dt$ and $\int_{-\infty}^{\infty} |g(t)|^2\,dt$ converge, then the convolution integral $\int_{-\infty}^{\infty} f(t)g(x-t)\,dt$ converges absolutely for every x in E_1.*

Proof If $f \in R$ and $g \in R$ on $[a, b]$, we can apply the Cauchy-Schwarz inequality to get

$$\left(\int_a^b |f(t)g(x-t)|\,dt\right)^2 \le \int_a^b |f(t)|^2\,dt \int_a^b |g(x-t)|^2\,dt$$

$$\le \int_{-\infty}^{\infty} |f(t)|^2\,dt \int_{-\infty}^{\infty} |g(t)|^2\,dt.$$

The conclusion is an easy consequence of these inequalities.

In spite of the existence of singularities, we can prove that the convolution is always absolutely integrable on $(-\infty, +\infty)$.

15–29 THEOREM. *If* $f \in R^*(-\infty, +\infty)$ *and* $g \in R^*(-\infty, +\infty)$, *and if* h *denotes the function defined by the equation*

$$h(x) = \int_{-\infty}^{\infty} f(t)g(x - t)\, dt$$

whenever the integral converges, then $h \in R^*(-\infty, +\infty)$.

Proof. Since only *absolute* convergence is involved here, we might as well assume that both f and g are non-negative. Then h will also be non-negative. By Theorem 15–27, singularities of h can arise only from infinite discontinuities of f and g, of which there are at most a finite number. For every finite closed interval $[a, b]$ which contains no singularities of h, the integral $\int_{-\infty}^{\infty} f(t)\, g(x - t)\, dt$ converges *uniformly* on $[a, b]$. (The reader should verify this statement.) Therefore, by the discussion which follows Theorem 14–26, the integral $\int_a^b h(x)\, dx$ exists and is given by

$$\begin{aligned} \int_a^b h(x)\, dx &= \int_{-\infty}^{\infty} f(t) \left[\int_a^b g(x - t)\, dx\right] dt \\ &= \int_{-\infty}^{\infty} f(t) \left[\int_{a-t}^{b-t} g(y)\, dy\right] dt \le \int_{-\infty}^{\infty} f(t)\, dt \int_{-\infty}^{\infty} g(y)\, dy. \end{aligned} \tag{60}$$

If we let b increase to the nearest singularity of h greater than b (or to $+\infty$ if there is no singularity to the right of b), we find that $\int_a^b h(x)\, dx$ increases to a finite limit. A similar statement holds when a decreases to the nearest singularity less than a (or to $-\infty$). Thus, $h \in R^*(c_1, c_2)$ for every pair of consecutive singularities c_1 and c_2 of h. Since the whole real axis is the union of a finite sum of such intervals, it follows that $h \in R^*(-\infty, +\infty)$, as asserted.

15–19 The convolution theorem for Fourier transforms.

15–30 THEOREM. *Let* f *and* g *be absolutely integrable on* $(-\infty, +\infty)$ *and let* h *denote their convolution,* $h = f*g$. *Then, for every real* u, *we have*

$$\int_{-\infty}^{\infty} h(x)e^{-ixu}\, dx = \left(\int_{-\infty}^{\infty} f(t)e^{-itu}\, dt\right)\left(\int_{-\infty}^{\infty} g(y)e^{-iyu}\, dy\right). \tag{61}$$

NOTE. In operational notation, (61) can be expressed as follows:

$$F(f*g) = F(f) \cdot F(g).$$

Proof. The integral which defines h exists everywhere on $(-\infty, +\infty)$, except (possibly) for a finite number of singularities, say $c_1 < c_2 < \cdots < c_n$. Let us write

$$\int_{-\infty}^{\infty} h(x)e^{-ixu}\,dx = \int_{-\infty}^{c_1} + \sum_{k=2}^{n-1}\int_{c_k}^{c_{k+1}} + \int_{c_n}^{\infty}.$$

If we can establish the formulas

$$\int_{-\infty}^{c_1} h(x)e^{-ixu}\,dx = \int_{-\infty}^{\infty} f(t)e^{-itu}\left[\int_{-\infty}^{c_1-t} e^{-iyu}g(y)\,dy\right]dt, \tag{62}$$

$$\int_{c_k}^{c_{k+1}} h(x)e^{-ixu}\,dx = \int_{-\infty}^{\infty} f(t)e^{-itu}\left[\int_{c_k-t}^{c_{k+1}-t} e^{-iyu}g(y)\,dy\right]dt, \tag{63}$$

$$\int_{c_n}^{\infty} h(x)e^{-ixu}\,dx = \int_{-\infty}^{\infty} f(t)e^{-itu}\left[\int_{c_n-t}^{\infty} e^{-iyu}g(y)\,dy\right]dt, \tag{64}$$

then, by addition, we get

$$\int_{-\infty}^{\infty} h(x)e^{-ixu}\,dx = \int_{-\infty}^{\infty} f(t)e^{-itu}\left[\int_{-\infty}^{\infty} e^{-iyu}g(y)\,dy\right]dt,$$

which is the same as (61). Therefore we need only prove (62), (63), and (64).

We treat (63) first. Let $[a, b]$ be a closed interval interior to $[c_k, c_{k+1}]$. Then, arguing as in the proof of (60), we have

$$\int_a^b h(x)e^{-ixu}\,dx = \int_{-\infty}^{\infty} f(t)\left[\int_a^b e^{-ixu}g(x-t)\,dx\right]dt$$

$$= \int_{-\infty}^{\infty} f(t)e^{-itu}\left[\int_{a-t}^{b-t} e^{-iyu}g(y)\,dy\right]dt.$$

If we introduce

$$\phi(a, b, t) = \int_{a-t}^{b-t} e^{-iyu}g(y)\,dy,$$

we can write the last equation as follows:

$$\int_a^b h(x)e^{-ixu}\,dx = \int_{-\infty}^{\infty} f(t)e^{-itu}\phi(a, b, t)\,dt. \tag{65}$$

If $c \in (a, b)$, we have $\phi(a, b, t) = \phi(a, c, t) + \phi(c, b, t)$, and hence to prove (63) it suffices to verify the following two equations:

$$\lim_{b \to c_{k+1}} \int_{-\infty}^{\infty} f(t)e^{-itu}\phi(c, b, t)\, dt = \int_{-\infty}^{\infty} f(t)e^{-itu} \lim_{b \to c_{k+1}} \phi(c, b, t)\, dt$$

and

$$\lim_{a \to c_k} \int_{-\infty}^{\infty} f(t)e^{-itu}\phi(a, c, t)\, dt = \int_{-\infty}^{\infty} f(t)e^{-itu} \lim_{a \to c_k} \phi(a, c, t)\, dt.$$

But these equations follow at once from Theorem 14–22 and hence (63) is established. (The reader should verify that the hypotheses of Theorem 14–22 are satisfied.)

To prove (64), we first note that (65) is still valid if $b > a > c_n$. Choosing c in (a, b) as before and applying Theorem 14–22 again, we find

$$\int_{c_n}^{c} h(x)e^{-ixu}\, dx = \int_{-\infty}^{\infty} f(t)e^{-itu} \left[\int_{c_n - t}^{c-t} e^{-iyu}g(y)\, dy\right] dt. \tag{66}$$

Next, we must verify that

$$\lim_{b \to +\infty} \int_{-\infty}^{\infty} f(t)e^{-itu}\phi(c, b, t)\, dt = \int_{-\infty}^{\infty} f(t)e^{-itu} \lim_{b \to +\infty} \phi(c, b, t)\, dt. \tag{67}$$

But this follows from Theorem 14–28. (The reader should verify that the hypotheses of Theorem 14–28 are satisfied.) Combining (66) and (67), we get (64). A similar argument yields (62) and this completes the proof.

15–20 The Laplace transform. If f is a real-valued function defined on $(-\infty, +\infty)$ such that $f(t) = 0$ for $t < 0$, the function F defined by the equation

$$F(z) = \int_{0}^{\infty} e^{-zt} f(t)\, dt \tag{68}$$

is called the *Laplace transform* of f, also denoted by $L(f)$ or Lf. Here z is a complex number, say $z = x + iy$, and $F(z)$ is defined for those values of z in the complex plane for which the integral converges. We shall see presently that the set of such z is a *half-plane*, that is, a set of the form $\{x + iy \mid x > a\}$. Since

$$\int_{0}^{\infty} e^{-zt} f(t)\, dt = \int_{-\infty}^{\infty} e^{-iyt} \phi_x(t)\, dt,$$

where $\phi_x(t) = e^{-xt}f(t)$, we see that for each fixed x the Laplace transform is the same as the Fourier transform of ϕ_x. This fact is often helpful in developing some of the properties of Laplace transforms. For example, the reader can easily verify that the convolution theorem for Fourier transforms, when stated in terms of Laplace transforms, yields the formula

$$L(f*g) = L(f) \cdot L(g) \tag{69}$$

(see Exercise 15-29), where now the convolution $f*g$ is given by the integral

$$(f*g)(x) = \int_0^x f(t)g(x - t)\, dt. \tag{70}$$

Whenever we think of the Laplace transform in (68) as a Fourier transform, we are actually restricting the complex number z to lie on a fixed vertical line. It is more natural to study the Laplace transform as a function of an "unrestricted" complex variable, and this is the viewpoint we shall adopt in the remainder of this chapter.

NOTE. For simplicity, we shall confine our study of Laplace transforms to functions f which satisfy the following restrictions:

(a) $f \in R$ on every finite interval.

(b) There exist positive constants M, a, t_0, such that

$$|f(t)| \le Me^{at}, \qquad \text{for every } t \ge t_0. \tag{71}$$

Under these restrictions, the integral in (68) converges *absolutely* for each $z = x + iy$ having real part $x > a$. In fact, when $x > a$, we have

$$|e^{-zt}f(t)| = e^{-xt}|f(t)| \le Me^{-(x-a)t}$$

whenever $t \ge t_0$. Since $\int_{t_0}^{\infty} e^{-(x-a)t}\, dt$ converges, it follows that

$$\int_0^{\infty} e^{-zt}f(t)\, dt$$

converges absolutely when $x > a$.

15-31 THEOREM. *Assume that the integral $\int_0^{\infty} e^{-zt}f(t)\, dt$ is convergent for $z = z_0 = x_0 + iy_0$. For each $K > 0$ define the sector $S(K)$ with vertex z_0 to be the set*

$$S(K) = \left\{x + iy \mid x > x_0, \left|\frac{y - y_0}{x - x_0}\right| < K\right\}.$$

Then the integral $\int_0^{\infty} e^{-zt}f(t)\, dt$ converges uniformly on $S(K)$.

Proof. Define β by the equation

$$\beta(x) = \int_0^x e^{-z_0 t} f(t)\, dt - \int_0^\infty e^{-z_0 t} f(t)\, dt, \qquad \text{if } x \geq 0. \tag{72}$$

Then, by Theorem 9–36, we can write

$$\int_0^x e^{-zt} f(t)\, dt = \int_0^x e^{-(z-z_0)t}\, d\beta(t). \tag{73}$$

Equation (73) is valid for every complex z and for every $x \geq 0$. Note that $\beta(x) \to 0$ as $x \to +\infty$. Therefore, if $\epsilon > 0$ is given, we can choose t_0 so that

$$|\beta(t)| < \epsilon', \qquad \text{for every } t \geq t_0, \tag{74}$$

where $\epsilon' = \frac{1}{2}\epsilon/(1 + \sqrt{1 + K^2})$. In view of (73), the theorem will be proved if we can show that $t_2 \geq t_1 \geq t_0$ implies

$$\left| \int_{t_1}^{t_2} e^{-(z-z_0)t}\, d\beta(t) \right| < \epsilon, \qquad \text{for every } z \text{ in } S(K). \tag{75}$$

To do this we use integration by parts to obtain

$$\int_{t_1}^{t_2} e^{-(z-z_0)t}\, d\beta(t) = e^{-(z-z_0)t_2}\beta(t_2) - e^{-(z-z_0)t_1}\beta(t_1) + (z - z_0)\int_{t_1}^{t_2} e^{-(z-z_0)t}\beta(t)\, dt.$$

When $z \in S(K)$, we have $x > x_0$ and hence $|e^{-(z-z_0)t}| = e^{-(x-x_0)t} \leq 1$. Using this inequality along with (74), we obtain

$$\begin{aligned} \left| \int_{t_1}^{t_2} e^{-(z-z_0)t}\, d\beta(t) \right| &\leq \epsilon' + \epsilon' + \epsilon'|z - z_0| \int_{t_1}^{t_2} e^{-(x-x_0)t}\, dt \\ &= 2\epsilon' + \epsilon' \frac{|z - z_0|}{x - x_0} (e^{-(x-x_0)t_1} - e^{-(x-x_0)t_2}) \\ &< 2\epsilon' \left(1 + \frac{|z - z_0|}{x - x_0}\right) = 2\epsilon'(1 + \sec\theta), \end{aligned}$$

where $\theta = \arg(z - z_0)$. (See Fig. 15–2.) Since $\sec\theta = \sqrt{1 + \tan^2\theta} < \sqrt{1 + K^2}$, we have $2\epsilon'(1 + \sec\theta) < 2\epsilon'(1 + \sqrt{1 + K^2}) < \epsilon$. This proves (75).

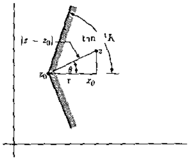

FIG. 15-2. The sector $S(K)$ with vertex at z_0.

Since every point in the half-plane $x > x_0$ lies within *some* sector $S(K)$ with vertex at x_0, we have the following corollary of Theorem 15-31

15-32 THEOREM *If the integral $\int_0^\infty e^{-zt}f(t)\,dt$ converges for $z = z_0 = x_0 + iy_0$, then it also converges for each $z = x + iy$ in the half-plane $x > x_0$*

NOTE The convergence may or may not be uniform in the half-plane $x > x_0$ Also, the integral may or may not converge at points of the line $x = x_0$ other than z_0

The infimum of the set of all x_0 such that $\int_0^\infty e^{-zt}f(t)\,dt$ converges for $z = x_0 + iy_0$ is called the *abscissa of convergence* of the integral and is denoted by $\sigma(f)$ [The abscissa may be $-\infty$, but it cannot be $+\infty$ because of the restriction (71)] The half-plane $x > \sigma(f)$ is called the *half-plane of convergence* of the integral.

15-33 THEOREM *Let $\sigma(f)$ denote the abscissa of convergence of the integral*

$$F(z) = \int_0^\infty e^{-zt} f(t)\,dt$$

Then, for each $z = x + iy$ with $x > \sigma(f)$, the derivative $F'(z)$ exists and is given by the integral

$$F'(z) = -\int_0^\infty e^{-zt} t f(t)\,dt. \tag{76}$$

The integral in (76) converges uniformly on every sector $S(K)$ interior to the half-plane $x > \sigma(f)$

NOTE. Repeated application of this theorem tells us that F has derivatives of every order and that they are given by the formula

$$F^{(n)}(z) = (-1)^n \int_0^\infty e^{-zt} t^n f(t)\, dt, \tag{77}$$

if z is in the half-plane of convergence.

Proof. First, we establish the convergence of the integral in (76) in the half-plane $x > \sigma(f)$. To do this, let $z_0 = x_0 + iy_0$, where $x_0 > \sigma(f)$, define β by (72), and apply Theorem 9–36 to write

$$\begin{aligned} \int_0^T e^{-zt} t f(t)\, dt &= \int_0^T t e^{-(z-z_0)t}\, d\beta(t) \\ &= Te^{-(z-z_0)T}\beta(T) - \int_0^T \beta(t)\, d(te^{-(z-z_0)t}) \\ &= Te^{-(z-z_0)T}\beta(T) - \int_0^T e^{-(z-z_0)t}\beta(t)\, dt \\ &\qquad + (z - z_0)\int_0^T t\beta(t)e^{-(z-z_0)t}\, dt. \end{aligned} \tag{78}$$

Now β is bounded, say $|\beta(t)| \le M$ if $t \ge 0$. If $z = x + iy$, where $x > x_0$, we can choose $h > 0$ so that $x_0 < x_0 + h < x$. Then, for $t \ge 0$, we have

$$|e^{-(z-z_0)t}\, t\beta(t)| \le Mte^{-(x-x_0)t} \le Mte^{-ht},$$

and the Weierstrass M-test tells us that the integral $\int_0^\infty e^{-(z-z_0)t}t\beta(t)\, dt$ converges (in fact, converges *uniformly* on the half-plane $x \ge x_0 + h$). When $T \to +\infty$ in (78), each term on the right tends to a limit and this establishes the convergence of the integral in (76) whenever $x > x_0$. But, since x_0 is subject only to the restriction $x_0 > \sigma(f)$, it follows that this integral converges for each z in the half-plane $x > \sigma(f)$. Applying Theorem 15–31 to this integral, we find that it converges *uniformly* on every sector $S(K)$ interior to this half-plane.

To prove the existence of the derivative $F'(z)$ and the validity of (76), we write $F(z) = u(x, y) + iv(x, y)$, where

$$u(x, y) = \int_0^\infty e^{-tx} \cos ty\, f(t)\, dt, \qquad v(x, y) = -\int_0^\infty e^{-tx} \sin ty\, f(t)\, dt.$$

We can also write $-\int_0^\infty e^{-zt} t f(t)\,dt = U(x, y) + iV(x, y)$, where

$$U(x, y) = -\int_0^\infty te^{-tx} \cos ty\, f(t)\, dt, \qquad V(x, y) = \int_0^\infty te^{-tx} \sin ty\, f(t)\, dt$$

By what we have just proved, the integrals for U and V converge uniformly on every sector $S(K)$ interior to the half-plane $x > \sigma(f)$ On the other hand, these integrals are also the result of differentiation under the integral sign (with respect to x) of the integrals for u and v Hence, by Theorem 14-24, we must have $U = D_1u$ and $V = D_1v$ Similarly, $V = -D_2u$ and $U = D_2v$ It follows that we have $D_1u = D_2v$ and $D_2u = -D_1v$ In other words, u and v satisfy the Cauchy-Riemann equations everywhere in the half-plane $x > \sigma(f)$ Since the partial derivatives (being equal to U and V) are continuous (by Theorem 14-22), it follows from Theorem 6-25 that the derivative $F'(z)$ exists for each z in this half-plane Furthermore, by Theorem 6-24, we have

$$F'(z) = D_1u(x, y) + i\, D_1v(x, y) = U(x, y) + iV(x, y)$$

$$= -\int_0^\infty te^{-zt} f(t)\, dt$$

This completes the proof

15-21 The inversion formula for Laplace transforms.

15-34 THEOREM *Let c be a positive number such that the integral*

$$F(z) = \int_0^\infty e^{-zt} f(t)\, dt$$

converges absolutely for each $z = x + iy$ with $x > c$ Let t be a point at which f satisfies one of the "local" conditions (a) *or* (b) *of the Fourier integral theorem* (Theorem 15-24) *Then, for each $a > c$, we have*

$$\frac{f(t+) + f(t-)}{2} = \frac{1}{2\pi} \lim_{T\to+\infty} \int_{-T}^{T} e^{(a+iv)t} F(a + iv)\, dv \tag{79}$$

NOTE. The integral on the right can be expressed as a *contour integral* taken along a vertical line segment joining $a - iT$ to $a + iT$, in which case (79) is written as follows:

$$\frac{f(t+) + f(t-)}{2} = \frac{1}{2\pi i} \lim_{T\to+\infty} \int_{a-iT}^{a+iT} e^{zt} F(z)\, dz. \tag{80}$$

Sometimes the symbol $\int_{a-\infty i}^{a+\infty i}$ is used as an abbreviation for $\lim_{T\to+\infty}\int_{a-iT}^{a+iT}$.

Proof. Let $g(t) = e^{-at}f(t)$ if $t \geq 0$, $g(t) = 0$ if $t < 0$. Then we may apply formula (54) to get

$$\frac{g(t+) + g(t-)}{2} = \frac{1}{2\pi} \lim_{T\to+\infty} \int_{-T}^{T} \left[\int_{-\infty}^{\infty} g(u)e^{iv(t-u)}\, du\right] dv.$$

When this formula is written in terms of f, it becomes

$$\frac{f(t+) + f(t-)}{2} = \frac{1}{2\pi} \lim_{T\to+\infty} \int_{-T}^{T} e^{(a+iv)t}\left[\int_{0}^{\infty} e^{-(a+iv)u} f(u)\, du\right] dv,$$

which is the same as (79).

Further discussion of this formula is given in Chapter 16. (See Theorem 16–21.)

Exercises

Orthonormal systems

15-1. Derive the Minkowski inequality $\|f+g\| \leq \|f\| + \|g\|$ from the Cauchy-Schwarz inequality

15-2. Verify that the trigonometric system in (6) is orthonormal on $[0, 2\pi]$

15-3. Show that the Gram-Schmidt process for vectors (see Exercise 11-1) can be applied to a linearly independent system of functions $\{\phi_0, \phi_1, \phi_2, \ldots\}$ to yield an orthogonal system

15-4. If $x \in E_1$ and $n = 1, 2, \ldots$, let $f_n(x) = (x^2 - 1)^n$ and define

$$\phi_0(x) = 1, \qquad \phi_n(x) = \frac{1}{2^n n!} f_n^{(n)}(x).$$

It is clear that ϕ_n is a polynomial This is called the *Legendre polynomial* of order n. The first few are

$$\phi_1(x) = x, \qquad \phi_2(x) = \tfrac{3}{2}x^2 - \tfrac{1}{2},$$

$$\phi_3(x) = \tfrac{5}{2}x^3 - \tfrac{3}{2}x, \qquad \phi_4(x) = \tfrac{35}{8}x^4 - \tfrac{15}{4}x^2 + \tfrac{3}{8}$$

Derive the following properties of Legendre polynomials:

(a) $\phi_n'(x) = x\phi_{n-1}'(x) + n\phi_{n-1}(x)$

(b) $\phi_n(x) = x\phi_{n-1}(x) + \dfrac{x^2 - 1}{n}\phi_{n-1}'(x)$

(c) $(n+1)\phi_{n+1}(x) = (2n+1)x\phi_n(x) - n\phi_{n-1}(x)$

(d) ϕ_n satisfies the differential equation $[(1 - x^2)y']' + n(n+1)y = 0$

(e) $[(1 - x^2)\Delta(x)]' + [m(m+1) - n(n+1)]\phi_m(x)\phi_n(x) = 0$, where $\Delta = \phi_n\phi_m' - \phi_m\phi_n'$

(f) The set $\{\phi_0, \phi_1, \phi_2, \ldots\}$ is orthogonal on $[-1, 1]$

(g) $\displaystyle\int_{-1}^{1} \phi_n^2\,dx = \frac{2n-1}{2n+1}\int_{-1}^{1} \phi_{n-1}^2\,dx$

(h) $\displaystyle\int_{-1}^{1} \phi_n^2\,dx = \frac{2}{2n+1}$

15–5. Let $\{\phi_0, \phi_1, \phi_2, \ldots\}$ be orthonormal on $[a, b]$ and assume this set is complete for a collection S. If $f(x) \sim \sum_{n=0}^{\infty} a_n\phi_n(x)$ and $g(x) \sim \sum_{n=0}^{\infty} b_n\phi_n(x)$, where f and g are in S, show that the series $\sum_{n=0}^{\infty} a_n\bar{b}_n$ converges and has a sum equal to (f, g). [*Hint:* $4f\bar{g} = (f + \bar{g})^2 - (f - \bar{g})^2$.]

Trigonometric Fourier series.

15–6. Assume that $f \in R$ on $[-\pi, \pi]$ and that f has period 2π. Show that the Fourier series generated by f assumes the following special forms under the conditions stated:

(a) If $f(-x) = f(x)$ when $0 < x < \pi$, then

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx, \quad \text{where } a_n = \frac{2}{\pi}\int_0^{\pi} f(t) \cos nt\, dt.$$

(b) If $f(-x) = -f(x)$ when $0 < x < \pi$, then

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin nx, \quad \text{where } b_n = \frac{2}{\pi}\int_0^{\pi} f(t) \sin nt\, dt.$$

In Exercises 15–7 through 15–13, show that each of the expansions is valid in the range indicated. *Suggestion:* Use Exercise 15–6 and Theorem 15–22(c) when possible.

15–7. (a) $x = \pi - 2\sum_{n=1}^{\infty} \frac{\sin nx}{n}$, if $0 < x < 2\pi$.

(b) $\frac{x^2}{2} = \pi x - \frac{\pi^2}{3} + 2\sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$, if $0 \le x \le 2\pi$.

15–8. (a) $\frac{\pi}{4} = \sum_{n=1}^{\infty} \frac{\sin (2n-1)x}{2n-1}$, if $0 < x < \pi$.

(b) $x = \frac{\pi}{2} - \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{\cos (2n-1)x}{(2n-1)^2}$, if $0 \le x \le \pi$.

15–9. (a) $x = 2\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \sin nx}{n}$, if $-\pi < x < \pi$.

(b) $x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n \cos nx}{n^2}$, if $-\pi \le x \le \pi$.

15–10. $x^2 = \frac{4}{3}\pi^2 + 4\sum_{n=1}^{\infty}\left(\frac{\cos nx}{n^2} - \frac{\pi \sin nx}{n}\right), \qquad \text{if } 0 < x < 2\pi$

15–11. (a) $\cos x = \frac{8}{\pi}\sum_{n=1}^{\infty}\frac{n \sin 2nx}{4n^2 - 1}, \qquad \text{if } 0 < x < \pi$

(b) $\sin x = \frac{2}{\pi} - \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\cos 2nx}{4n^2 - 1}, \qquad \text{if } 0 < x < \pi$

15–12. (a) $x \cos x = -\frac{1}{2}\sin x + 2\sum_{n=2}^{\infty}\frac{(-1)^n n \sin nx}{n^2 - 1}, \qquad \text{if } -\pi < x < \pi.$

(b) $x \sin x = 1 - \frac{1}{2}\cos x - 2\sum_{n=2}^{\infty}\frac{(-1)^n \cos nx}{n^2 - 1}, \quad \text{if } -\pi \le x \le \pi.$

15–13. (a) $\log\left|\sin\frac{x}{2}\right| = -\log 2 - \sum_{n=1}^{\infty}\frac{\cos nx}{n}, \quad \text{if } x \ne 2k\pi \ (k \text{ an integer}).$

(b) $\log\left|\cos\frac{x}{2}\right| = -\log 2 - \sum_{n=1}^{\infty}\frac{(-1)^n \cos nx}{n}, \qquad \text{if } x \ne (2k+1)\pi.$

(c) $\log\left|\tan\frac{x}{2}\right| = -2\sum_{n=1}^{\infty}\frac{\cos (2n-1)x}{2n-1}, \qquad \text{if } x \ne k\pi$

15–14. A sequence $\{\overline{B}_n\}$ of periodic functions (of period 1) is defined on E_1 as follows:

$$\overline{B}_{2n}(x) = (-1)^{n+1}\frac{2(2n)!}{(2\pi)^{2n}}\sum_{k=1}^{\infty}\frac{\cos 2\pi kx}{k^{2n}} \qquad (n = 1, 2, \ldots),$$

$$\overline{B}_{2n+1}(x) = (-1)^{n+1}\frac{2(2n+1)!}{(2\pi)^{2n+1}}\sum_{k=1}^{\infty}\frac{\sin 2\pi kx}{k^{2n+1}} \qquad (n = 0, 1, 2, \ldots).$$

($\overline{B}_n$ is called the *Bernoulli function* of order n.) Show that

(a) $\overline{B}_1(x) = x - [x] - \frac{1}{2}$ if x is not an integer ($[x]$ is the greatest integer $\le x$.)

(b) $\int_0^1 \overline{B}_n(x)\,dx = 0$ if $n \ge 1$ and $\overline{B}_n'(x) = n\overline{B}_{n-1}(x)$ if $n \ge 2$.

(c) $\overline{B}_n(x) = P_n(x)$ if $0 < x < 1$, where P_n is the nth Bernoulli polynomial. (See Exercise 13–35 for the definition of P_n.)

(d) $$\overline{B}_n(x) = -\frac{n!}{(2\pi i)^n} \sum_{\substack{k=-\infty \\ k\neq 0}}^{\infty} \frac{e^{2\pi i k x}}{k^n} \qquad (n = 1, 2, \ldots).$$

15–15. (a) Show that Parseval's formula for Fourier series becomes

$$\frac{1}{\pi}\int_0^{2\pi} |f(x)|^2\, dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

(b) Use an appropriate Fourier series in conjunction with Parseval's formula to show that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}.$$

15–16. Let f be the function of period 2π whose values on $[-\pi, \pi]$ are

$$f(x) = 1 \quad \text{if } 0 < x < \pi, \qquad f(x) = -1 \quad \text{if } -\pi < x < 0,$$
$$f(x) = 0 \quad \text{if } x = 0 \text{ or } x = \pi.$$

(a) Show that $f(x) = \dfrac{4}{\pi}\displaystyle\sum_{n=1}^{\infty} \frac{\sin (2n-1)x}{2n-1}$, for every x.

This is one example of a class of Fourier series which have a curious property known as *Gibbs' phenomenon*. This exercise is designed to illustrate this phenomenon. In that which follows, $s_n(x)$ denotes the nth partial sum of the series in part (a).

(b) Show that $s_n(x) = \dfrac{2}{\pi}\displaystyle\int_0^x \frac{\sin 2nt}{\sin t}\, dt.$

(c) Show that, in $(0, \pi)$, s_n has local maxima at $x_1, x_3, \ldots, x_{2n-1}$ and local minima at $x_2, x_4, \ldots, x_{2n-2}$, where $x_m = \frac{1}{2}m\pi/n$ $(m = 1, 2, \ldots, 2n-1)$.

(d) Show that $s_n(\frac{1}{2}\pi/n)$ is the largest of the numbers

$$s_n(x_m) \qquad (m = 1, 2, \ldots, 2n-1).$$

(e) Interpret $s_n(\frac{1}{2}\pi/n)$ as a Riemann sum and prove that

$$\lim_{n\to\infty} s_n\left(\frac{\pi}{2n}\right) = \frac{2}{\pi}\int_0^{\pi} \frac{\sin t}{t}\, dt.$$

The value of the limit in (e) is about 1.179. Thus, although f has a jump equal to 2 at the origin, the graphs of the approximating curves s_n tend to approximate a vertical segment of length 2.358 in the vicinity of the origin. This is the Gibbs phenomenon.

15–17. If $f(x) \sim a_0/2 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$ and if f is of bounded variation on $[0, 2\pi]$, show that $a_n = O(1/n)$ and $b_n = O(1/n)$. *Hint.* Write $f = g - h$, where g and h are increasing on $[0, 2\pi]$. Then

$$a_n = \frac{1}{n\pi} \int_0^{2\pi} g(x)\, d(\sin nx) - \frac{1}{n\pi} \int_0^{2\pi} h(x)\, d(\sin nx).$$

Now apply Theorem 9–30.

15–18. Suppose $g \in R$ on $[a, \delta]$ for every a in $(0, \delta)$ and assume that g satisfies a "right-handed" Lipschitz condition at 0. (See the Note following Theorem 15–15.) Show that the improper integral $\int_{0+}^{\delta} |g(t) - g(0+)|/t\, dt$ converges.

15–19. Use Exercise 15–18 to prove that differentiability of f at a point implies convergence of its Fourier series at the point.

15–20. Assume that $f(x) \sim a_0/2 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$ and let $\{\sigma_n\}$ be the sequence of arithmetic means of the partial sums of this series, as defined in (37). Show that:

(a) $$\sigma_n(x) = \frac{a_0}{2} + \sum_{k=1}^{n-1} \left(1 - \frac{k}{n}\right)(a_k \cos kx + b_k \sin kx).$$

(b) $$\int_0^{2\pi} |f(x) - \sigma_n(x)|^2\, dx = \int_0^{2\pi} |f(x)|^2\, dx - \frac{\pi}{2} a_0^2 - \pi \sum_{k=1}^{n-1} (a_k^2 + b_k^2) + \frac{\pi}{n^2} \sum_{k=1}^{n-1} k^2 (a_k^2 + b_k^2).$$

(c) If f is continuous on $[0, 2\pi]$ and has period 2π, then

$$\lim_{n\to\infty} \frac{\pi}{n^2} \sum_{k=1}^{n} k^2 (a_k^2 + b_k^2) = 0.$$

15–21. Fejér's theorem can be used to prove that a function of bounded variation on $[0, 2\pi]$ has a convergent Fourier series by using Exercise 15–17, along with the following:

> *Theorem of Hardy and Landau.* *Let $\sum_{n=1}^{\infty} f_n(x)$ be a series whose terms satisfy an inequality of the form*
>
> $$|f_n(x)| < \frac{M}{n}, \qquad \textit{for every } x \textit{ in } [a, b]$$

Let $s_n(x) = \sum_{k=1}^n f_k(x)$ *and* $\sigma_n(x) = (1/n)\sum_{k=1}^n s_k(x)$. *If, for some* x *in* $[a, b]$, *the limit* $\lim_{n\to\infty} \sigma_n(x)$ *exists and has the value* $s(x)$, *then the series* $\sum f_n(x)$ *converges and has sum* $s(x)$. *If, in addition, the sequence* $\{\sigma_n\}$ *converges uniformly on* $[a, b]$, *then the series* $\sum f_n(x)$ *also converges uniformly on* $[a, b]$.

Sketch of proof: If $m > n$, we have the identity

$$(m - n)s_m(x) - (m - n)s(x) = m(\sigma_m(x) - s(x)) - n(\sigma_n(x) - s(x)) + \sum_{r=n+2}^{m} (r - 1 - n)f_r(x).$$

Given $0 < \epsilon < 1$, choose N so that $n > N$ implies $|\sigma_n(x) - s(x)| < \epsilon$. When $m > n > N$, the identity yields

$$\begin{aligned}
|s_m(x) - s(x)| &< \frac{m+n}{m-n}\epsilon + \frac{M}{n(m-n)}\sum_{r=n+2}^{m}(r-1-n) \\
&= \frac{m+n}{m-n}\epsilon + \frac{M}{n(m-n)}\sum_{k=1}^{m-n-1} k \\
&< \frac{m+n}{m-n}\epsilon + \frac{M}{n}\frac{(m-n)^2}{m-n} \\
&= \epsilon + \frac{2\epsilon}{(m/n - 1)} + M\left(\frac{m}{n} - 1\right).
\end{aligned}$$

Take $m > 6/\sqrt{\epsilon}$, $m > (1 + 2\sqrt{\epsilon})N$, choose n to satisfy $m/(1 + 2\sqrt{\epsilon}) < n < m/(1 + \sqrt{\epsilon})$. Then $|s_m(x) - s(x)| < \epsilon + (2 + 2M)\sqrt{\epsilon}$. (Note that N is independent of x if $\{\sigma_n\}$ converges uniformly on $[a, b]$.)

Fourier integrals and Laplace transforms.

15–22. If f satisfies the hypotheses of the Fourier integral theorem, show that:

(a) If f is even, that is, if $f(-t) = f(t)$ for every t, then

$$\frac{f(x+) + f(x-)}{2} = \frac{2}{\pi}\int_0^\infty \cos vx\left[\int_0^\infty f(u)\cos vu\, du\right]dv.$$

(b) If f is odd, that is, if $f(-t) = -f(t)$ for every t, then

$$\frac{f(x+) + f(x-)}{2} = \frac{2}{\pi}\int_0^\infty \sin vx \left[\int_0^\infty f(u)\sin vu\,du\right] dv$$

Use the Fourier integral theorem to evaluate the improper integrals in Exercises 15–23 through 15–25 *Suggestion* Use Exercise 15–22 when possible

15–23. $$\frac{2}{\pi}\int_0^\infty \frac{\sin v\cos vx}{v}\,dv = \begin{cases} 1, & \text{if } -1 < x < 1, \\ 0, & \text{if } |x| > 1, \\ \frac{1}{2}, & \text{if } |x| = 1 \end{cases}$$

15–24. $$\int_0^\infty \frac{\cos ax}{b^2 + x^2}\,dx = \frac{\pi}{2b}e^{-|a|b}, \qquad \text{if } b > 0$$

Hint Apply Exercise 15–22 with $f(u) = e^{-b|u|}$.

15–25. $$\int_0^\infty \frac{x\sin ax}{1 + x^2}\,dx = \frac{a}{|a|}\,\frac{\pi}{2}e^{-|a|}, \qquad \text{if } a \neq 0.$$

15–26. Derive the following properties of the convolution, assuming that all functions which come into consideration are absolutely integrable on $(-\infty, +\infty)$.

(a) $f*g = g*f$.

(b) $f*(g*h) = (f*g)*h = g*(f*h)$

(c) $\delta(f*g) = f*\delta g + g*\delta f$, where $\delta f(t) = tf(t)$, $\delta g(t) = tg(t)$, etc

15–27. Let $F = Lf$ and $G = Lg$ denote the Laplace transforms of f and g, respectively. Let c denote a positive constant Show that for each z in a suitable half-plane we have

(a) $G(z) = \dfrac{1}{c}F\left(\dfrac{z}{c}\right)$, if $g(t) = f(ct)$

(b) $G(z) = e^{-cz}F(z)$, if $g(t) = f(t - c)$ when $t \geq c$, $g(t) = 0$ when $t < c$.

(c) $G(z) = (-1)^n F^{(n)}(z)$, if $g(t) = t^n f(t)$, $n = 1, 2, \ldots$

(d) $G(z) = z^n F(z) - \sum_{k=1}^{n} z^{n-k} f^{(k-1)}(0+)$, if $g(t) = f^{(n)}(t)$, $n = 1, 2, \ldots$

NOTE In part (d) it is assumed that $e^{-zt}f^{(k)}(t) \to 0$ as $t \to +\infty$ for $k = 0, 1, \ldots, n - 1$

15–28. Verify the entries in the following table of Laplace transforms.

$f(t)$	$F(z) = \int_0^\infty e^{-zt}f(t)\,dt,$	$z = x + iy.$
$e^{\alpha t}$	$(z - \alpha)^{-1}$	$(x > \alpha)$
$\cos \alpha t$	$z/(z^2 + \alpha^2)$	$(x > 0)$
$\sin \alpha t$	$\alpha/(z^2 + \alpha^2)$	$(x > 0)$
$t^p e^{\alpha t}$	$\Gamma(p + 1)/(z - \alpha)^{p+1}$	$(x > \alpha, p > 0)$

15–29. Show that the convolution $h = f*g$ assumes the form

$$h(t) = \int_0^t f(x)g(t - x)\,dx$$

when f and g both vanish on the negative real axis. Use the convolution theorem for Fourier transforms to prove that $L(f*g) = L(f) \cdot L(g)$.

15–30. Let g be continuous on $[0, 1]$ and assume that $\int_0^1 t^n g(t)\,dt \doteq 0$ for $n = 0, 1, 2, \ldots.$ Show that:

(a) $\int_0^1 g(t)^2\,dt = \int_0^1 g(t)(g(t) - P(t))\,dt$ for every polynomial P.

(b) $\int_0^1 g(t)^2\,dt = 0.$ [*Hint:* Use Theorem 15–23.]

(c) $g(t) = 0$ for every t in $[0, 1]$.

15–31. Let f be continuous on $(0, +\infty)$ and assume that $F(s) = \int_0^\infty e^{-st}f(t)\,dt$ converges for $s > \sigma(f)$. If $s_0 > \sigma(f)$ and $a > 0$, show that:

(a) $F(s_0 + a) = a\int_0^\infty g(t)e^{-at}\,dt$, where $g(x) = \int_0^x e^{-s_0 t}f(t)\,dt$.

(b) If $F(s_0 + na) = 0$ for $n = 0, 1, 2, \ldots,$ then $f(t) = 0$ for $t > 0$.

(c) If h is continuous on $(0, +\infty)$ and if f and h have the same Laplace transform, then $f(t) = h(t)$ for every $t > 0$.

15–32. Given $F(s) = \int_0^\infty e^{-st}f(t)\,dt$ converges absolutely for $s = s_0$, where $s_0 \geq 0$. Show that:

(a) F cannot be a constant $\neq 0$.

(b) F cannot be a polynomial of positive degree.

15–33. Let p denote a positive constant. Define $f(t) = t^{p-1}$ if $t > 0$, $f(t) = 0$ if $t \leq 0$. Show that $\int_0^\infty e^{-zt}f(t)\,dt = \Gamma(p)z^{-p}$ if $z = x + iy$, $x > 0$.

15–34. Let p and q denote positive constants. If $t > 0$, define $f(t) = t^{p-1}$, $g(t) = t^{q-1}$, and put $f(t) = g(t) = 0$ if $t \leq 0$.

(a) Let $h = f*g$ and use the convolution theorem to prove that

$$h(t) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p + q)}\,t^{p+q-1}, \qquad \text{if } t > 0.$$

(b) Use part (a) to derive the formula

$$\int_0^1 x^{p-1}(1-x)^{q-1}\,dx = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

15–35. (a) Use Exercise 15–34 (b) to derive the formula

$$\frac{\Gamma(p)\Gamma(p)}{\Gamma(2p)} = 2\int_0^{1/2} x^{p-1}(1-x)^{p-1}\,dx$$

(b) Make a suitable change of variable in (a) and derive the *duplication formula* for the Gamma function

$$\Gamma(2p)\Gamma(\tfrac{1}{2}) = 2^{2p-1}\Gamma(p)\Gamma(p+\tfrac{1}{2})$$

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CHAPTER 16

CAUCHY'S THEOREM AND THE RESIDUE CALCULUS

16–1 Analytic functions. The concept of derivative for functions of a complex variable was introduced in the latter part of Chapter 6 (Definition 6–23). The functions which turn out to be most important in complex variable theory are those which possess a continuous derivative at each point of an open set. These are called *analytic* functions.

16–1 DEFINITION. *Let $f = u + iv$ be a complex-valued function defined on an open set S in E_2. The function f is said to be analytic on S if the derivative f' exists and is continuous* at every point of S.*

NOTE. The terminology "f is analytic at a point z_0" is also used. This means that there is a neighborhood $N(z_0)$ on which f is analytic. It is possible for a function to have a derivative at a point without being analytic at the point. For example, the function f defined by $f(z) = |z|^2$ has a derivative at 0 but at no other point of E_2.

Some examples of analytic functions have already been encountered in Chapter 6. The function f defined by the equation $f(z) = z^n$ (where n is a positive integer) is analytic everywhere in E_2 and its derivative is $f'(z) = nz^{n-1}$. When n is a negative integer, the equation $f(z) = z^n$ if $z \neq 0$ defines a function analytic everywhere except at 0. Polynomials are analytic everywhere in E_2, and rational functions are analytic everywhere except at points where the denominator vanishes. The exponential function, defined by the formula $e^z = e^x(\cos y + i \sin y)$ if $z = x + iy$, is analytic everywhere in E_2 and is equal to its derivative. The complex sine and cosine functions (being linear combinations of exponentials) are also analytic everywhere in E_2.

The function f defined by the equation $f(z) = \operatorname{Log} z$ if $z \neq 0$, where $\operatorname{Log} z$ denotes the principal logarithm of z (see Definition 1–36), is analytic everywhere in E_2 except for those points $z = x + iy$ for which $x \leq 0$ and $y = 0$. At these points, the principal logarithm fails to be continuous.

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* It can be shown that the existence of f' on S automatically implies continuity of f' on S (a fact discovered by Goursat in 1900). Hence an analytic function can be defined as one which merely possesses a derivative everywhere on S. However, we shall take continuity of f' as being part of the definition of analyticity, since this allows some of the proofs to run more smoothly.

We shall see later that analyticity at a point z_0 puts a very severe restriction on a function. It implies the existence of *all higher derivatives* in a neighborhood of z_0 and also guarantees the existence of a convergent power series which represents the function in a neighborhood of z_0. This is in marked contrast to the behavior of real-valued functions, where it is possible to have existence and continuity of the first derivative without existence of the second derivative.

16-2 The Cauchy integral theorem. A peculiar aspect of complex derivative theory is the way which complex integration enters in. Early in the nineteenth century, Cauchy developed the theory of complex integration and used it to prove that analyticity of f at z_0 implies the existence of all higher derivatives in a neighborhood of z_0. To this day no one has been able to deduce this result without the use of contour integration. The principal tool in this sort of investigation is the *Cauchy integral theorem*, which states that the contour integral $\int_\Gamma f(z)\,dz$ has the value 0 if Γ is a rectifiable Jordan curve, provided that f is continuous on Γ and has a derivative everywhere in the inner region of Γ. We will prove a weaker version of this theorem in which we assume analyticity of f in a slightly larger region. This added hypothesis makes Cauchy's theorem an immediate consequence of Green's theorem.

16-2 THEOREM (*Cauchy integral theorem*). *Assume that $f = u + iv$ is analytic on an open region S in E_2. Let Γ be a rectifiable Jordan curve such that both Γ and its inner region lie entirely within S. Then*

$$\int_\Gamma f(z)\,dz = 0.$$

Proof. If we write $\int_\Gamma f(z)\,dz = \int_\Gamma (u\,dx - v\,dy) + i\int_\Gamma (v\,dx + u\,dy)$ and apply Green's theorem (Theorem 10-43) to each real-valued line integral, we find

$$\int_\Gamma f(z)\,dz = -\int_R (D_1 v + D_2 u)\,d(x, y) + i\int_R (D_1 u - D_2 v)\,d(x, y),$$

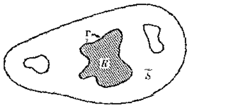

FIG. 16-1. The Cauchy integral theorem.

where $R = I(\Gamma) \cup \Gamma$, $I(\Gamma)$ being the inner region of Γ. (See Fig. 16–1.) The region R is a subset of S (by hypothesis) and the Cauchy-Riemann equations are satisfied on R. (See Theorem 6–24.) Therefore, each double integral is 0 and the proof is complete.

As a kind of converse to Cauchy's theorem, we have:

16–3 THEOREM. *Let $f = u + iv$ be continuous on an open region S in E_2 and suppose u and v have continuous partials D_1u, D_2u, D_1v, D_2v on S. If the contour integral $\int_\Gamma f(z)\,dz$ has the value 0 for every closed polygonal path in S, then f is analytic on S.*

Proof. Let z_1 and z_2 be two arbitrary points of S and let $\Gamma(z_1, z_2)$ be a polygonal curve joining z_1 to z_2. The hypothesis implies that both line integrals $\int_{\Gamma(z_1,z_2)} (u\,dx - v\,dy)$ and $\int_{\Gamma(z_1,z_2)} (v\,dx + u\,dy)$ are independent of the path for all choices of z_1 and z_2 in S. By Theorem 10–38, there exists a function ϕ such that $D_1\phi = u$, $D_2\phi = -v$, and a function ψ such that $D_1\psi = v$, $D_2\psi = u$, everywhere on S. But $D_2u = D_{2,1}\phi$ and $D_1v = -D_{1,2}\phi$. Since D_2u and D_1v are assumed to be continuous, we must have $D_{2,1}\phi = D_{1,2}\phi$, which means that $D_2u = -D_1v$. Similarly, $D_2v = D_1u$. Therefore the Cauchy-Riemann equations are valid in S and (by Theorem 6–25) the derivative $f'(z)$ exists at each point z in S. Continuity of the derivative follows from continuity of the partials of u and v.

16–3 Deformation of the contour. In much of the material that follows we shall have occasion to consider contour integrals taken along rectifiable Jordan curves enclosing one or more isolated points at which the integrand is not analytic. (Such points are called "singularities" of the integrand.) In Chapter 9 we encountered one such example, namely, the integral $\int_{\Gamma[z]} (z - z_0)^{-1}\,dz$, which is used to define the winding number of the path $\Gamma[z]$ with respect to z_0. The integrand in this case is analytic everywhere in E_2 except at z_0, where the derivative fails to exist. In Theorem 9–68 we proved that the winding number is $+1$ or -1 by showing that the curve Γ could be replaced by a circle C with center at z_0 lying entirely inside Γ, the integral along C having the same value as the integral along Γ. By using exactly the same kind of argument that was given in the proof of Theorem 9–68, we can prove the following theorem:

16–4 THEOREM. *Let S be an open region in E_2. Assume that $z_0 \in S$ and suppose there is a neighborhood $N(z_0) \subset S$ such that f is analytic at each point of $S - \overline{N}(z_0)$. Let Γ be a rectifiable Jordan curve, containing $N(z_0)$ in its interior, such that both Γ and its inner region lie within S.*

Let C be a circle with center at z_0 lying entirely inside Γ and also in $S - \overline{N}(z_0)$. Then

$$\int_{\Gamma[z]} f(z)\, dz = \int_{C[w]} f(w)\, dw,$$

provided that the paths $\Gamma[z]$ and $C[w]$ are both positively oriented or both negatively oriented.

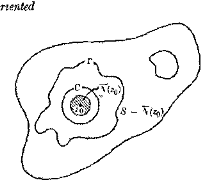

FIG. 16-2 Deformation of the contour

This theorem is sometimes described by saying that the value of a contour integral along a simple closed curve is unaltered by deforming the contour to a circle, provided that the contour remains within the region of analyticity of the integrand during the deformation. (See Fig. 16-2.) The usefulness of the theorem lies in the fact that an integral along a circle is much easier to deal with than an integral along a more general Jordan curve. This is illustrated in the next theorem.

16-4 Cauchy's integral formula.

16-5 THEOREM (*Cauchy's integral formula*). *Assume that f is analytic on an open region S in E_2. Let Γ be a rectifiable Jordan curve such that both Γ and its inner region lie within S. Then, for every point z_0 inside Γ, we have*

$$f(z_0) = \frac{1}{2\pi i} \int_{\Gamma[z]} \frac{f(z)}{z - z_0}\, dz, \tag{1}$$

provided that the path $\Gamma[z]$ is positively oriented.

NOTE. Formula (1) reveals a remarkable phenomenon. It tells us that the values of an analytic function in the interior of a Jordan curve are completely determined by its values on the curve itself.

Proof. Because of Theorem 16–4, it suffices to prove the theorem for the case in which $\Gamma[z]$ is a positively oriented circle with center at z_0. (The radius of the circle may be arbitrary, provided that the circle lies within the original Jordan curve.) Define a new function g on S as follows:

$$g(z) = \begin{cases} \dfrac{f(z) - f(z_0)}{z - z_0} - f'(z_0), & \text{if } z \neq z_0, \\ 0, & \text{if } z = z_0. \end{cases}$$

Then g is continuous everywhere on S and we have

$$f(z) = f(z_0) + (z - z_0)f'(z_0) + (z - z_0)g(z), \tag{2}$$

an equation which holds for every z in S. Since $g(z_0) = 0$, there exists a neighborhood $N(z_0; \delta) \subset S$ such that $|g(z)| < 1$ for each z in this neighborhood. Take the circle Γ so small that its radius R is less than δ. Using (2), we can write

$$\begin{aligned} \int_{\Gamma[z]} \frac{f(z)}{z - z_0}\,dz &= f(z_0)\int_{\Gamma[z]} \frac{1}{z - z_0}\,dz + f'(z_0)\int_{\Gamma[z]} dz + \int_{\Gamma[z]} g(z)\,dz \\ &= 2\pi i f(z_0) + \int_{\Gamma[z]} g(z)\,dz. \end{aligned}$$

Hence we have

$$\left| \int_{\Gamma[z]} \frac{f(z)}{z - z_0}\,dz - 2\pi i f(z_0) \right| = \left| \int_{\Gamma[z]} g(z)\,dz \right| \leq 2\pi R$$

(by Theorem 9–60). The number on the left is independent of R and hence must be 0, since the right-hand side tends to 0 as $R \to 0$.

16–5 The mean value of an analytic function on a circle. Let us apply Cauchy's integral formula to the case where Γ is a circle with center at z_0 described by a function z, where

$$z(t) = z_0 + re^{it}, \qquad \text{if } 0 \leq t \leq 2\pi,$$

r being the radius of the circle. The integral in (1) becomes

$$f(z_0) = \frac{1}{2\pi i}\int_0^{2\pi} \frac{f[z(t)]}{z(t) - z_0}\,dz(t) = \frac{1}{2\pi i}\int_0^{2\pi} \frac{f(z_0 + re^{it})}{z(t) - z_0}\,z'(t)\,dt.$$

Since $z'(t) = ire^{it} = i(z(t) - z_0)$, we obtain

$$f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + re^{it})\, dt$$

This can be interpreted as a *Mean Value Theorem* expressing the value of f at the center of Γ as an "average" of the values on Γ itself. It is valid whenever f is analytic at each point within, and on, Γ.

16–6 Cauchy's integral formula for the derivative of an analytic function. Differentiation under the integral sign in Cauchy's formula (1) (with respect to z_0) yields the integral

$$\frac{1}{2\pi i}\int_{\Gamma[z]} \frac{f(z)}{(z - z_0)^2}\, dz.$$

We will prove that this integral represents the derivative $f'(z_0)$.

16–6 THEOREM. *Assume that f is analytic on an open region S in E_2. Let Γ be a rectifiable Jordan curve such that both Γ and its inner region lie within S. Then, for every point z_0 inside Γ, we have*

$$f'(z_0) = \frac{1}{2\pi i}\int_{\Gamma[z]} \frac{f(z)}{(z - z_0)^2}\, dz, \tag{3}$$

provided that the path $\Gamma[z]$ is positively oriented.

Proof. If z_1 is inside Γ, $z_1 \neq z_0$, we can apply (1) twice to write

$$\begin{aligned}\frac{f(z_1) - f(z_0)}{z_1 - z_0} &= \frac{1}{2\pi i(z_1 - z_0)}\int_{\Gamma[z]} f(z)\left(\frac{1}{z - z_1} - \frac{1}{z - z_0}\right) dz \\ &= \frac{1}{2\pi i}\int_{\Gamma[z]} \frac{f(z)}{(z - z_1)(z - z_0)}\, dz = \frac{1}{2\pi i}\int_{\Gamma[z]} \frac{f(z)}{(z - z_0)^2}\, dz \\ &\quad + \frac{(z_1 - z_0)}{2\pi i}\int_{\Gamma[z]} \frac{f(z)}{(z - z_0)^2(z - z_1)}\, dz\end{aligned}$$

When $z_1 \to z_0$, the left side tends to $f'(z_0)$. Therefore, to prove the theorem, it suffices to show that

$$\lim_{z_1 \to z_0}\, (z_1 - z_0)\int_{\Gamma[z]} \frac{f(z)}{(z - z_0)^2(z - z_1)}\, dz = 0. \tag{4}$$

For this we need only show that the integral multiplying $(z_1 - z_0)$ remains bounded as $z_1 \to z_0$.

Suppose that $|f(z)| \leq M$ for each z on Γ, let $\Lambda(\Gamma)$ denote the length of Γ, and let $\delta = \inf \{|z - z_0| \mid z \in \Gamma\}$. Then $\delta > 0$. If $|z_1 - z_0| < \delta$ and $z \in \Gamma$, we have $|z - z_1| = |(z - z_0) - (z_1 - z_0)| \geq \delta - |z_1 - z_0|$, and hence

$$\left| \int_{\Gamma[z]} \frac{f(z)}{(z - z_0)^2 (z - z_1)} \, dz \right| \leq \frac{M\Lambda(\Gamma)}{\delta^2(\delta - |z_1 - z_0|)} \cdot$$

Since this remains bounded as $z_1 \to z_0$, we get (4) and hence (3).

16–7 The existence of higher derivatives of an analytic function. The method of proof used in the previous theorem can be employed to establish the existence of the second derivative $f''(z_0)$ and, in fact, of all higher derivatives.

16–7 THEOREM. *Assume that f is analytic on an open region S in E_2. Let Γ be a rectifiable Jordan curve such that both Γ and its inner region lie within S. Then, for every point z_0 inside Γ and for every integer $n \geq 1$, the derivative $f^{(n)}(z_0)$ exists and is given by the integral*

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{\Gamma[z]} \frac{f(z)}{(z - z_0)^{n+1}} \, dz, \tag{5}$$

provided that the path $\Gamma[z]$ is positively oriented.

Proof. If z_1 is inside Γ, $z_1 \neq z_0$, we can apply (3) twice to write

$$\frac{f'(z_1) - f'(z_0)}{z_1 - z_0} = \frac{1}{2\pi i(z_1 - z_0)} \int_{\Gamma[z]} f(z) \left\{ \frac{1}{(z - z_1)^2} - \frac{1}{(z - z_0)^2} \right\} dz$$

$$= \frac{2!}{2\pi i} \int_{\Gamma[z]} \frac{f(z)}{(z - z_0)^3} \, dz$$

$$+ \frac{z_1 - z_0}{2\pi i} \int_{\Gamma[z]} f(z) \, \frac{3(z - z_0) - 2(z_1 - z_0)}{(z - z_1)^2 (z - z_0)^3} \, dz.$$

Arguing as in the proof of Theorem 16–6, we can show that the last term tends to 0 as $z_1 \to z_0$. This proves that we have

$$\lim_{z_1 \to z_0} \frac{f'(z_1) - f'(z_0)}{z_1 - z_0} = \frac{2!}{2\pi i} \int_{\Gamma[z]} \frac{f(z)}{(z - z_0)^3} \, dz,$$

the existence of the limit being part of the conclusion. But this means that $f''(z_0)$ exists and is given by formula (5) with $n = 2$. Starting with the formula for $n = 2$, we can show by a similar argument that $f'''(z_0)$ exists and is given by (5) with $n = 3$. The result for general n is proved by induction.

16-8 Power series expansions for analytic functions. In Theorem 13-24 we showed that a power series which converges in a neighborhood $N(z_0)$ defines a function which has derivatives of every order in the neighborhood $N(z_0)$. Such a function is therefore analytic at z_0. Using the integral formulas of the previous section, we can prove the converse of this result.

16-8 THEOREM. *Assume that f is analytic on an open region S in E_2 and let z_0 be a point of S. Then, in every neighborhood $N(z_0)$ such that $N(z_0) \subset S$, f can be represented by the convergent power series*

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n \tag{6}$$

NOTE. This is known as the *Taylor expansion* of f about z_0.

Proof. Let Γ be a circle with center at z_0 and radius $r > 0$, where r is chosen so that both Γ and its interior lie within S. If z_1 is an arbitrary point inside Γ, we can use Cauchy's formula to write

$$f(z_1) = \frac{1}{2\pi i} \int_{\Gamma[z]} \frac{f(z)}{z - z_1} \, dz,$$

where the path $\Gamma[z]$ is positively oriented. Since $|z_1 - z_0| < |z - z_0|$ if $z \in \Gamma$, we have

$$\frac{1}{z - z_1} = \frac{1}{z - z_0} \, \frac{1}{1 - [(z_1 - z_0)/(z - z_0)]} = \sum_{n=0}^{\infty} \frac{(z_1 - z_0)^n}{(z - z_0)^{n+1}},$$

the series being uniformly convergent on Γ. Now f is bounded on Γ and hence we have

$$\frac{f(z)}{z - z_1} = \sum_{n=0}^{\infty} \frac{f(z)}{(z - z_0)^{n+1}} (z_1 - z_0)^n \qquad \text{(uniformly on } \Gamma)$$

By Theorem 13-12, we may integrate term-by-term to get

$$f(z_1) = \sum_{n=0}^{\infty} a_n (z_1 - z_0)^n,$$

where

$$a_n = \frac{1}{2\pi i}\int_{\Gamma[z]} \frac{f(z)}{(z - z_0)^{n+1}}\,dz = \frac{f^{(n)}(z_0)}{n!}.$$

This proves the theorem.

Theorems 16–8 and 13–24 together tell us that a necessary and sufficient condition for a complex-valued function f to be analytic at a point z_0 is that f be representable by a power series in some neighborhood of z_0. When such a power series exists, its radius of convergence is at least as large as the radius of any neighborhood $N(z_0)$ which lies in the region of analyticity of f. Since the circle of convergence cannot contain any points in its interior where f fails to be analytic, it follows that the radius of convergence is exactly equal to the distance from z_0 to the nearest point at which f fails to be analytic.

This observation explains some of the difficulties which arise when we try to find power series expansions for real-valued functions of a real variable. For example, let $f(x) = 1/(1 + x^2)$ if x is real. This function is defined everywhere in E_1 and has derivatives of every order at each point in E_1. Also, it has a power series expansion about the origin, namely,

$$\frac{1}{1 + x^2} = 1 - x^2 + x^4 - x^6 + \cdots.$$

However, this series representation for f is valid only in the open interval $(-1, 1)$. From the standpoint of real variable theory, there is nothing in the behavior of f which explains this. But when we examine the situation in the complex plane, we see at once that the function f defined by $f(z) = 1/(1 + z^2)$ (if $z^2 + 1 \neq 0$) is analytic everywhere in E_2 except at the points $z = \pm i$. Therefore the radius of convergence of the power series expansion about 0 must equal 1.

Examples.

The following power series expansions are valid for all z in E_2:

$$\text{(a)}\quad e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \qquad \text{(b)}\quad \sin z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n + 1)!},$$

$$\text{(c)}\quad \cos z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!}.$$

16-9 Zeros of analytic functions. If f is analytic at z_0 and if $f(z_0) = 0$, the Taylor expansion of f about z_0 has no constant term and hence assumes the following form

$$f(z) = \sum_{n=1}^{\infty} a_n(z - z_0)^n$$

This is valid for each z in some neighborhood $N(z_0)$. If f is identically zero on this neighborhood [that is, if $f(z) = 0$ for every z in $N(z_0)$], then each $a_n = 0$, since $a_n = f^{(n)}(z_0)/n!$. If f is not identically zero on this neighborhood, there will be a first nonzero coefficient a_k in the expansion, in which case the point z_0 is said to be a *zero* of order k. We will prove next that there is a neighborhood of z_0 which contains no further zeros of f.

16-9 THEOREM. *Assume that f is analytic on an open region S in E_2. Suppose $f(z_0) = 0$ for some point z_0 in S and assume that f is not identically zero on any neighborhood of z_0. Then there exists a deleted neighborhood $N'(z_0) \subset S$ on which f does not assume the value 0.*

Proof. The Taylor expansion about z_0 becomes $f(z) = (z - z_0)^k g(z)$, where $k \geq 1$, $g(z) = a_k + a_{k+1}(z - z_0) + \cdots$, and $g(z_0) = a_k \neq 0$. Since g is continuous at z_0, there is a neighborhood $N(z_0) \subset S$ on which g does not vanish. Therefore, f is never zero in the corresponding deleted neighborhood $N'(z_0)$.

This theorem has several important consequences. For example, we can use it to show that a function which is analytic on an open region S cannot be zero on any non-empty open subset of S without being identically zero throughout S.

16-10 THEOREM. *Assume that f is analytic on an open region S in E_2. Let A denote the set of those points z in S for which there exists a neighborhood $N(z)$ on which f is identically zero and let $B = S - A$. Then one of the two sets A or B is empty and the other one is S itself.*

Proof. We have $S = A \cup B$, where A and B are disjoint sets. The set A is open by its very definition. If we prove that B is also open, it will follow from the connectedness of S that at least one of the two sets A or B is empty. (See Definitions 8-29 and 8-39.)

To prove B is open, let z_0 be a point of B and consider the two possibilities $f(z_0) \neq 0$, $f(z_0) = 0$. If $f(z_0) \neq 0$, there is a neighborhood $N(z_0) \subset S$ on which f does not vanish. Each point of this neighborhood must therefore belong to B. Hence, z_0 is an interior point of B if $f(z_0) \neq 0$. But, if $f(z_0) = 0$, Theorem 16-9 provides us with a deleted neighborhood $N'(z_0) \subset S$ on which f does not vanish. This means that $N(z_0) \subset B$. Hence, in either case, z_0 is an interior point of B. Therefore, B is open and one of the two sets A or B must be empty.

16-10 The identity theorem for analytic functions.

16-11 THEOREM. *Assume that f is analytic on an open region S in E_2. Let T be a subset of S having an accumulation point z_0 in S. If $f(z) = 0$ for every z in T, then $f(z) = 0$ for every z in S.*

Proof. There exists an infinite sequence $\{z_n\}$, whose terms are points of T, such that $\lim_{n\to\infty} z_n = z_0$. By continuity, $f(z_0) = \lim_{n\to\infty} f(z_n) = 0$. We will prove next that there is a neighborhood of z_0 on which f is identically zero. Suppose there is no such neighborhood. Then Theorem 16-9 tells us that there must be a deleted neighborhood $N'(z_0)$ on which f does not assume the value 0. But this is impossible, since every deleted neighborhood contains points of T. Therefore there must be a neighborhood of z_0 on which f vanishes identically. Hence the set A of Theorem 16-10 cannot be empty. Therefore, $A = S$, and this means $f(z) = 0$ for every z in S.

As a corollary we have the following important result, sometimes referred to as the *identity theorem for analytic functions:*

16-12 THEOREM. *Let f and g be analytic on an open region S in E_2. If T is a subset of S having an accumulation point z_0 in S and if $f(z) = g(z)$ for every z in T, then $f(z) = g(z)$ for every z in S.*

Proof. Apply Theorem 16-11 to $f - g$.

16-11 Laurent expansions for functions analytic on an annulus.

16-13 DEFINITION. *If $z_0 \in E_2$ and if $0 < r_1 < r_2$, the set*

$$A(z_0; r_1, r_2) = \{z \mid r_1 < |z - z_0| < r_2\}$$

is called an annulus about z_0.

If a function is analytic on an annulus $A(z_0; r_1, r_2)$ but is *not* analytic everywhere in the neighborhood $N(z_0; r_2)$, it cannot be represented by a power series in $z - z_0$. Laurent discovered (in 1843) that such a function can always be represented by a sum of *two* series, one of which is a power series in $z - z_0$ and the other a power series in $(z - z_0)^{-1}$. Laurent's theorem can be stated as follows:

16-14 THEOREM. *Assume that f is analytic on an annulus $A(z_0; r_1, r_2)$. Then, for every point z in this annulus, we have*

$$f(z) = f_1(z) + f_2(z), \tag{7}$$

where

$$f_1(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n \quad \textit{and} \quad f_2(z) = \sum_{n=1}^{\infty} a_{-n}(z - z_0)^{-n}$$

The coefficients are given by the formulas

$$(8) \qquad a_n = \frac{1}{2\pi i}\int_{\Gamma[z]} \frac{f(z)}{(z - z_0)^{n+1}}\,dz \qquad (n = 0, \pm 1, \pm 2, \ldots),$$

the path $\Gamma[z]$ *being any positively oriented circle with center at* z_0 *and radius* r, *where* $r_1 < r < r_2$. *The function* f_1 *(called the regular part of* f *at* z_0*) is analytic on the neighborhood* $N(z_0, r_2)$. *The function* f_2 *(called the principal part of* f *at* z_0*) is analytic on the set* $E_2 - \overline{N}(z_0, r_1)$.

NOTE. The sum $\sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=1}^{\infty} a_{-n}(z - z_0)^{-n}$ is also written as $\sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$ and is referred to as the *Laurent expansion* of f about z_0. Formula (8) shows that a function can have at most one Laurent expansion in a given annulus.

Proof. Suppose $z_1 \in A(z_0, r_1, r_2)$ and let $N(z_1, r)$ be a neighborhood contained within $A(z_0, r_1, r_2)$. Construct a circle C with center at z_0 and radius R_1, where $r_1 < R_1 < |z_1 - z_0| - r$. Let C^+ denote the positively oriented path along C and let C^- denote the corresponding negatively oriented path. Similarly, let K be a circle with center at z_0 and radius R_2, where $|z_1 - z_0| + r < R_2 < r_2$, and let K^+ denote the positively oriented path along K. Then $N(z_1, r) \subset A(z_0, R_1, R_2)$. Choose diametrically opposite points a and b on C, c and d on K (a, b, c, d collinear), such that the line segments $L[b, d]$ and $L[c, a]$ do not intersect the neighborhood $N(z_1, r)$. (See Fig. 16-3.)

FIG. 16-3. Proof of Laurent's theorem.

The paths C^- and K^+, together with the oriented paths along the line segments, determine two positively oriented Jordan curves Γ_1 and Γ_2, exactly one of which, say Γ_1, contains the neighborhood $N(z_1; r)$ in its interior. By Cauchy's integral formula, we have

$$2\pi i f(z_1) = \int_{\Gamma_1} \frac{f(z)}{z - z_1}\, dz.$$

On the other hand, the neighborhood $N(z_1; r)$ is outside Γ_2 and hence

$$\int_{\Gamma_2} \frac{f(z)}{z - z_1}\, dz = 0.$$

Therefore we can write

$$2\pi i f(z_1) = \int_{\Gamma_1} \frac{f(z)}{z - z_1}\, dz + \int_{\Gamma_2} \frac{f(z)}{z - z_1}\, dz. \tag{9}$$

But we have $\int_{\Gamma_1} + \int_{\Gamma_2} = \int_{K^+} + \int_{C^-}$, since the integrals along the line segments cancel out. Therefore, (9) becomes

$$f(z_1) = \frac{1}{2\pi i}\int_{K^+} \frac{f(z)}{z - z_1}\, dz - \frac{1}{2\pi i}\int_{C^+} \frac{f(z)}{z - z_1}\, dz = f_1(z_1) + f_2(z_1), \tag{10}$$

let us say. The integral for $f_1(z_1)$ can be expressed as a power series in $z_1 - z_0$ by arguing exactly as in the proof of Theorem 16–8. Thus we have

$$f_1(z_1) = \sum_{n=0}^{\infty} a_n(z_1 - z_0)^n,$$

where

$$a_n = \frac{1}{2\pi i}\int_{K^+} \frac{f(z)}{(z - z_0)^{n+1}}\, dz \qquad (n = 0, 1, 2, \ldots). \tag{11}$$

Because of Theorem 16–4, the circle K^+ in (11) can be replaced by any other positively oriented circle (with center at z_0) lying within the annulus $A(z_0; r_1, r_2)$. (It is important to realize that we cannot write $a_n = f^{(n)}(z_0)/n!$ in this case, unless f also happens to be analytic everywhere inside K.)

The integral for $f_2(z_1)$ is treated in similar fashion, except that we use the series expansion

$$\frac{1}{z - z_1} = -\frac{1}{z_1 - z_0}\,\frac{1}{1 - [(z - z_0)/(z_1 - z_0)]} = -\sum_{n=1}^{\infty} \frac{(z - z_0)^{n-1}}{(z_1 - z_0)^n},$$

which converges uniformly on C since $|z - z_0| < |z_1 - z_0|$ if $z \in C$.

Multiplying by $f(z)$ and integrating term-by-term, we get

$$-\frac{1}{2\pi i}\int_{C^+}\frac{f(z)}{z-z_1}\,dz = \sum_{n=1}^{\infty} b_n(z_1-z_0)^{-n},$$

where

$$(12) \qquad b_n = \frac{1}{2\pi i}\int_{C^+}\frac{f(z)}{(z-z_0)^{1-n}}\,dz \qquad (n = 1, 2, \quad)$$

The circle C^+ in (12) can be replaced by any other positively oriented concentric circle lying within the annulus $A(z_0, r_1, r_2)$ If we take the same circle in (12) and in (11) and if we write a_{-n} for b_n, both formulas can be combined into one, as indicated in (8) Since z_1 was an arbitrary point of the annulus, this proves (8)

Since the series $\sum_{n=0}^{\infty} a_n(z-z_0)^n$ is a power series in $z-z_0$ which converges for each z in the annulus $A(z_0, r_1, r_2)$, it must have a circle of convergence which includes the whole neighborhood $N(z_0, r_2)$, and hence the function f_1 defined by this series must be analytic on this neighborhood Similarly, we can show that the function f_2 is analytic on the set $E_2 - \overline{N}(z_0; r_1)$ by regarding f_2 as a composite function, say $f_2(z) = g[h(z)]$, where $g(w) = \sum_{n=1}^{\infty} a_{-n}w^n$ and $h(z) = (z-z_0)^{-1}$

16-12 Isolated singularities.

16-15 DEFINITION. *A point z_0 is said to be an isolated singularity of f if*

(a) *f is analytic in a deleted neighborhood of z_0,*

(b) *f is not analytic at z_0*

NOTE f need not be defined at z_0

Because of (a), there exists an annulus $A(z_0, r_1, r_2)$ on which f is analytic and therefore has a (uniquely determined) Laurent expansion, say

$$(13) \qquad f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n + \sum_{n=1}^{\infty} a_{-n}(z-z_0)^{-n}$$

[Since the inner radius r_1 can be arbitrarily small, (13) is valid in the deleted neighborhood $N'(z_0, r_2)$] The singularity z_0 is classified into one of three types (depending on the form of the principal part) as follows

If no negative powers appear in (13), that is, if $a_{-n} = 0$ for every $n = 1, 2, \ldots$, the point z_0 is called a *removable singularity* In this case, the singularity can be removed by defining f at z_0 to have the value $f(z_0) = a_0$ (See Example 1 below)

If $a_{-n} \neq 0$ for some n but $a_{-m} = 0$ for every $m > n$, the point z_0 is said to be a *pole* of order n. In this case, the principal part reduces to a finite sum, namely,

$$\frac{a_{-1}}{z - z_0} + \frac{a_{-2}}{(z - z_0)^2} + \cdots + \frac{a_{-n}}{(z - z_0)^n}.$$

A pole of order 1 is usually called a *simple* pole.

Finally, if $a_{-n} \neq 0$ for infinitely many values of n, the point z_0 is called an *essential singularity*.

EXAMPLE 1. Let $f(z) = \sin z/z$ if $z \neq 0$, $f(0) = 0$. This function is analytic everywhere except at 0. (It is discontinuous at 0, since $\sin z/z \to 1$ as $z \to 0$.) The Laurent expansion about 0 has the form

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - + \cdots.$$

Since no negative powers of z appear, the point 0 is a removable singularity. If we redefine f to have the value 1 at 0, the modified function becomes analytic at 0.

EXAMPLE 2. Let $f(z) = \sin z/z^5$ if $z \neq 0$. The Laurent expansion about 0 is

$$\frac{\sin z}{z^5} = z^{-4} - \frac{1}{3!}z^{-2} + \frac{1}{5!} - \frac{1}{7!}z^2 + \cdots.$$

In this case, the point 0 is a pole of order 4. Note that nothing has been said about the value of f *at* 0.

EXAMPLE 3. Let $f(z) = e^{1/z}$ if $z \neq 0$. The point 0 is an essential singularity, since

$$e^{1/z} = 1 + z^{-1} + \frac{1}{2!}z^{-2} + \cdots + \frac{1}{n!}z^{-n} + \cdots.$$

16–16 THEOREM. *Assume that f is analytic on an open region S in E_2 and define g by the equation $g(z) = 1/f(z)$ if $f(z) \neq 0$. Then f has a zero of order k at a point z_0 in S if, and only if, g has a pole of order k at z_0.*

Proof. If f has a zero of order k at z_0, there is a deleted neighborhood $N'(z_0)$ in which f does not vanish. In the neighborhood $N(z_0)$ we have $f(z) = (z - z_0)^k h(z)$, where $h(z) \neq 0$ if $z \in N(z_0)$. Hence, $1/h$ is analytic in $N(z_0)$ and has an expansion

$$\frac{1}{h(z)} = b_0 + b_1(z - z_0) + \cdots, \qquad \text{where } b_0 = \frac{1}{h(z_0)} \neq 0.$$

Therefore, if $z \in N'(z_0)$, we have

$$g(z) = \frac{1}{(z - z_0)^k h(z)} = \frac{b_0}{(z - z_0)^k} + \frac{b_1}{(z - z_0)^{k-1}} + \cdots,$$

and hence z_0 is a pole of order k for g. The converse is similarly proved.

16–13 The residue of a function at an isolated singular point. If z_0 is an isolated singular point of f, there is a deleted neighborhood $N'(z_0)$ on which f has a Laurent expansion, say

$$(14) \qquad f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=1}^{\infty} a_{-n}(z - z_0)^{-n}.$$

The coefficient a_{-1} which multiplies $(z - z_0)^{-1}$ is called the *residue* of f at z_0 and is denoted by the symbol

$$a_{-1} = \operatorname*{Res}_{z=z_0} f(z).$$

Formula (8) tells us that

$$(15) \qquad \int_\Gamma f(z)\, dz = 2\pi i \operatorname*{Res}_{z=z_0} f(z)$$

if Γ is any positively oriented circle with center at z_0 such that $\Gamma \subset N'(z_0)$.

In many cases it is relatively easy to evaluate the residue at a point without the use of integration. For example, if z_0 is a simple pole, we can use formula (14) to obtain

$$(16) \qquad \operatorname*{Res}_{z=z_0} f(z) = \lim_{z \to z_0} (z - z_0)f(z).$$

Similarly, if z_0 is a pole of order 2, it is easy to show that

$$\operatorname*{Res}_{z=z_0} f(z) = g'(z_0), \qquad \text{where } g(z) = (z - z_0)^2 f(z).$$

In cases like this, where the residue can be computed very easily, equation (15) gives us a powerful method for evaluating contour integrals around closed paths.

Cauchy was the first to realize that there are a great many applications of the theory of residues. (Some of these will be discussed in the later sections.) They are all based on the *Cauchy residue theorem*, which is a generalization of (15) to the case where the path Γ encloses more than one isolated singular point.

16–14 The Cauchy residue theorem.

16–17 THEOREM. *Let f be analytic at all points of an open region S in E_2, with the exception of a finite number of isolated singularities. Let Γ be a rectifiable Jordan curve such that both Γ and its interior lie within S. Assume that Γ contains a certain number of singularities of f in its interior, say $z_1, \ldots, z_n$, but that there are no singularities on Γ itself. Then we have*

$$\int_{\Gamma[z]} f(z)\, dz = 2\pi i \sum_{k=1}^{n} \operatorname*{Res}_{z=z_k} f(z), \tag{17}$$

provided that the path $\Gamma[z]$ is positively oriented.

Proof. When there is only one singularity z_1 inside Γ, Theorem 16–4 allows us to replace the contour Γ by a circle about z_1, and formula (17) reduces to (15).

The general theorem can be proved by induction on the number n of singularities. We shall illustrate the way to derive the case $n = 2$ from the case $n = 1$.

Suppose there are exactly two singularities z_1 and z_2 inside Γ. Then there is a deleted neighborhood $N'(z_1)$ lying inside Γ on which we can write $f(z) = f_1(z) + f_2(z)$, where f_1 is the regular part and f_2 is the principal part of f at z_1. By Laurent's theorem, f_2 is analytic everywhere in E_2 except at z_1, and hence, in particular, f_2 is analytic on $S - \{z_1\}$. Now let g be the function defined on $S - \{z_2\}$ as follows:

$$g(z) = \begin{cases} f(z) - f_2(z), & \text{if } z \in S,\ z \neq z_1,\ z \neq z_2, \\ f_1(z_1), & \text{if } z = z_1. \end{cases}$$

Then g is analytic on $S - \{z_2\}$.

Let us evaluate the integral $\int_\Gamma g(z)\, dz$ in two ways. Since z_1 is not on Γ, we can write

$$\begin{aligned}\int_\Gamma g(z)\, dz &= \int_\Gamma f(z)\, dz - \int_\Gamma f_2(z)\, dz = \int_\Gamma f(z)\, dz - 2\pi i \operatorname*{Res}_{z=z_1} f_2(z) \\ &= \int_\Gamma f(z)\, dz - 2\pi i \operatorname*{Res}_{z=z_1} f(z).\end{aligned}$$

On the other hand, we can replace Γ by a positively oriented circle C with center at z_2, where C lies inside Γ, but such that z_1 is not inside C. We then get

$$\int_\Gamma g(z)\, dz = \int_C g(z)\, dz = 2\pi i \operatorname*{Res}_{z=z_2} g(z) = 2\pi i \operatorname*{Res}_{z=z_2} f(z).$$

Comparing the two expressions for $\int_\Gamma g(z)\,dz$, we obtain

$$\int_\Gamma f(z)\,dz = 2\pi i \sum_{k=1}^{2} \operatorname*{Res}_{z=z_k} f(z)$$

This proves (17) for $n = 2$ The general case is similarly proved

Some of the applications of the Cauchy residue theorem are given in the next few sections

16–15 The difference between the number of zeros and the number of poles inside a closed contour.

16-18 THEOREM *Assume that f is analytic on an open region S in E_2 except, possibly, for a finite number of poles Let Γ be a positively oriented rectifiable Jordan curve such that both Γ and its interior lie within S. Assume further that none of the zeros or poles of f lie on Γ Let $z_1, \ldots, z_q$ be the zeros of f which lie inside Γ and let n_k denote the order of z_k ($k = 1, 2, \ldots, q$) Let $p_1, \ldots, p_r$ be the poles of f which lie inside Γ and let m_k denote the order of p_k ($k = 1, 2, \ldots, r$) Then*

$$\frac{1}{2\pi i}\int_\Gamma \frac{f'(z)}{f(z)}\,dz = \sum_{k=1}^{q} n_k - \sum_{k=1}^{r} m_k \tag{18}$$

NOTE The right member of (18) is the difference between the number of zeros and the number of poles inside Γ, provided that each zero or pole is counted as often as its order indicates

Proof Suppose that in a deleted neighborhood of a point z_0 we have $f(z) = (z - z_0)^n g(z)$, where g is analytic at z_0 and $g(z_0) \neq 0$, n being an integer (positive or negative) Then there is a deleted neighborhood of z_0 on which we can write

$$\frac{f'(z)}{f(z)} = \frac{n}{z - z_0} + \frac{g'(z)}{g(z)},$$

the quotient g'/g being analytic at z_0 This equation tells us that a zero of f of order n is a simple pole of f'/f with residue n Similarly, a pole of f of order n is a simple pole of f'/f with residue $-n$ This fact, used in conjunction with Cauchy's residue theorem, yields (18)

16–16 Evaluation of real-valued integrals by means of residues. Cauchy's residue theorem can sometimes be used to evaluate certain types of real-valued Riemann integrals There are several techniques available, depending on the particular form of the integral to be evaluated. We shall describe briefly two of these methods.

The first method deals with integrals of the form $\int_0^{2\pi} R(\sin t, \cos t)\, dt$, where R is a rational function* of two variables.

16–19 THEOREM. *Let R be a rational function of two variables and let*

$$f(z) = R\left(\frac{z^2 - 1}{2iz}, \frac{z^2 + 1}{2z}\right)$$

whenever the expression on the right is finite. Let $\Gamma[z]$ denote the positively oriented unit circle with center at 0. Then

$$\int_0^{2\pi} R(\sin t, \cos t)\, dt = \int_{\Gamma[z]} \frac{f(z)}{iz}\, dz, \tag{19}$$

provided that f has no poles on Γ.

Proof. One function z which describes $\Gamma[z]$ is given by $z(t) = e^{it}$ if $0 \le t \le 2\pi$. For this function we have

$$z'(t) = iz(t), \qquad \frac{z(t)^2 - 1}{2iz(t)} = \sin t, \qquad \frac{z(t)^2 + 1}{2z(t)} = \cos t,$$

and (19) follows at once from Theorem 9–62.

NOTE. To evaluate the integral on the right of (19), we need only compute the residues of the integrand at those poles which lie inside Γ.

EXAMPLE. Evaluate $I = \int_0^{2\pi} dt/(a + \cos t)$, where a is real, $|a| > 1$. Applying (19), we find

$$I = -2i \int_\Gamma \frac{dz}{z^2 + 2az + 1}\cdot$$

The integrand has simple poles at the roots of the equation $z^2 + 2az + 1 = 0$. These are the points $z_1 = -a + \sqrt{a^2 - 1}$, $z_2 = -a - \sqrt{a^2 - 1}$. The

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* A function P defined on $E_2 \times E_2$ by an equation of the form

$$P(z_1, z_2) = \sum_{m=0}^{p} \sum_{n=0}^{q} a_{m,n}\, z_1^m z_2^n$$

is called a *polynomial in two variables*. The coefficients $a_{m,n}$ may be real or complex. The quotient of two such polynomials is called a *rational function of two variables*.

corresponding residues R_1 and R_2 are given by

$$R_1 = \lim_{z \to z_1} \frac{z - z_1}{z^2 + 2az + 1} = \frac{1}{z_1 - z_2},$$

$$R_2 = \lim_{z \to z_2} \frac{z - z_2}{z^2 + 2az + 1} = \frac{1}{z_2 - z_1}.$$

If $a > 1$, z_1 is inside Γ, z_2 is outside Γ, and $I = 4\pi/(z_1 - z_2) = 2\pi/\sqrt{a^2 - 1}$. If $a < -1$, z_2 is inside Γ, z_1 is outside Γ, and we get $I = -2\pi/\sqrt{a^2 - 1}$.

Many improper integrals can be dealt with by means of the following theorem.

16-20 THEOREM. *Let $T = \{x + iy \mid y \geq 0\}$ denote the upper half-plane. Let S be an open region in E_2 which contains T and suppose f is analytic on S, except, possibly, for a finite number of poles. Suppose further that none of these poles are on the real axis. If*

$$\lim_{R \to +\infty} \int_0^\pi f(Re^{i\theta}) Re^{i\theta} \, d\theta = 0, \tag{20}$$

then the Cauchy principal value of the integral $\int_{-\infty}^{\infty} f(x)\, dx$ exists and equals

$$2\pi i \sum_{k=1}^{n} \operatorname*{Res}_{z=z_k} f(z),$$

where $z_1, \ldots, z_n$ are the poles of f which lie in T.

Proof. Let C be the positively oriented path formed by taking a portion of the real axis from $-R$ to R and a semicircle in T having $[-R, R]$ as its diameter, where R is taken large enough to enclose all the poles $z_1, \ldots, z_n$. Then

$$2\pi i \sum_{k=1}^{n} \operatorname*{Res}_{z=z_k} f(z) = \int_C f(z)\, dz = \int_{-R}^{R} f(x)\, dx + i \int_0^\pi f(Re^{i\theta}) Re^{i\theta}\, d\theta.$$

When $R \to +\infty$, the last integral tends to zero and we obtain

$$\lim_{R \to +\infty} \int_{-R}^{R} f(x)\, dx = 2\pi i \sum_{k=1}^{n} \operatorname*{Res}_{z=z_k} f(z).$$

NOTE. Equation (20) is automatically satisfied if f is the quotient of two polynomials, say $f = P/Q$, provided that the degree of Q exceeds the degree of P by at least 2. (See Exercise 16–34.)

EXAMPLE. To evaluate $\int_{-\infty}^{\infty} dx/(1 + x^4)$, let $f(z) = 1/(z^4 + 1)$. Then $P(z) = 1$, $Q(z) = 1 + z^4$, and hence (20) holds. The poles of f are the roots of the equation $1 + z^4 = 0$. These are z_1, z_2, z_3, z_4, where

$$z_k = e^{(2k-1)\pi i/4} \qquad (k = 1, 2, 3, 4).$$

Of these, only z_1 and z_2 lie in the upper half-plane. The residue at z_1 is

$$\operatorname*{Res}_{z=z_1} f(z) = \lim_{z \to z_1} (z - z_1)f(z) = \frac{1}{(z_1 - z_2)(z_1 - z_3)(z_1 - z_4)} = \frac{e^{-\pi i/4}}{4i}.$$

Similarly, we find $\operatorname{Res}_{z=z_2} f(z) = (1/4i)\, e^{\pi i/4}$. Therefore,

$$\int_{-\infty}^{\infty} \frac{dx}{1 + x^4} = \frac{2\pi i}{4i} (e^{-\pi i/4} + e^{\pi i/4}) = \pi \cos \frac{\pi}{4} = \frac{\pi}{2} \sqrt{2}.$$

16–17 Application of the residue theorem to the inversion formula for Laplace transforms. The following theorem is, in many cases, the easiest method for evaluating the contour integral which appears in the inversion formula for Laplace transforms. (See Theorem 15–34.)

16–21 THEOREM. *Let F be a function which is analytic everywhere in E_2, except, possibly, for a finite number of poles. Suppose that there exist three positive constants M, b, c such that*

$$|F(z)| < \frac{M}{|z|^c}, \qquad \text{whenever } |z| \geq b.$$

Let a be a positive number such that the vertical line $x = a$ contains no poles of F and let $z_1, \ldots, z_n$ denote the poles of F which lie to the left of this line. Then, for each real $t > 0$, we have

$$\lim_{R \to +\infty} \int_{a-iR}^{a+iR} e^{zt} F(z)\, dz = 2\pi i \sum_{k=1}^{n} \operatorname*{Res}_{z=z_k} [e^{zt} F(z)]. \tag{21}$$

Proof. We apply Cauchy's residue theorem to the positively oriented path Γ shown in Fig. 16–4, where the radius R of the circular part is taken large enough to enclose all poles of F which lie to the left of the line $x = a$

and also $R > b$ The residue theorem gives us

(22) $$\int_\Gamma e^{zt}F(z)\,dz = 2\pi i \sum_{k=1}^{n} \operatorname*{Res}_{z=z_k} [e^{zt}F(z)]$$

Now write

$$\int_\Gamma = \int_{\Gamma(A\,B)} + \int_{\Gamma(B\,C)} + \int_{\Gamma(C\,D)} + \int_{\Gamma(D\,E)} + \int_{\Gamma(E\,A)},$$

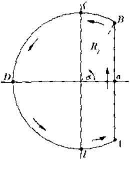

FIGURE 16–4

where A, B, C, D, E are the points indicated in Fig 16–4, and denote these integrals by I_1, I_2, I_3, I_4, I_5 We will prove that $I_k \to 0$ as $R \to +\infty$ when $\lambda > 1$.

First, we have

$$|I_2| < \frac{M}{R^c}\int_\alpha^{\pi/2} e^{tR\cos\theta} R\,d\theta \leq \frac{Me^{at}}{R^{c-1}}\left(\frac{\pi}{2} - \alpha\right) = \frac{Me^{at}}{R^c} R \sin^{-1}\left(\frac{a}{R}\right).$$

Since $R\sin^{-1}(a/R) \to a$ as $R \to +\infty$, it follows that $I_2 \to 0$ as $R \to +\infty$. In the same way we prove $I_5 \to 0$ as $R \to +\infty$.

Next, consider I_3 We have

$$|I_3| < \frac{M}{R^{c-1}}\int_{\pi/2}^{\pi} e^{tR\cos\theta}\,d\theta = \frac{M}{R^{c-1}}\int_0^{\pi/2} e^{-tR\sin\phi}\,d\phi$$

But, $\sin\phi \geq 2/\pi$ if $0 \leq \phi \leq \pi/2$, and hence

$$|I_3| < \frac{M}{R^{c-1}}\int_0^{\pi/2} e^{-2tR\phi/\pi}\,d\phi = \frac{\pi M}{2tR^c}(1 - e^{-tR}) \to 0 \quad \text{as } R \to +\infty.$$

Similarly, we find $I_4 \to 0$ as $R \to +\infty$ But, when $R \to +\infty$ in (22), the right-hand side is independent of R Hence, $\lim_{R\to+\infty} I_1$ exists and has the value indicated in (21)

EXAMPLE Let $F(z) = z/(z^2 + \alpha^2)$, where α is real Then F has simple poles at $\pm i\alpha$ Since $z/(z^2 + \alpha^2) = \frac{1}{2}[1/(z + i\alpha) + 1/(z - i\alpha)]$, we find

$$\operatorname*{Res}_{z=i\alpha} [e^{zt}F(z)] = \tfrac{1}{2}e^{i\alpha t}, \qquad \operatorname*{Res}_{z=-i\alpha} [e^{zt}F(t)] = \tfrac{1}{2}e^{-i\alpha t}$$

Therefore the limit in (21) has the value $2\pi i \cos \alpha t$. From Theorem 15–34, it follows that the function f, continuous on $(0, +\infty)$, whose Laplace transform is F, is given by $f(t) = \cos \alpha t$.

16–18 One-to-one analytic functions. It was pointed out in Chapter 7 that the Jacobian J_f of a complex-valued function f is related to the derivative f' by the equation

$$J_f(z) = |f'(z)|^2$$

at those points where f' is continuous. The results concerning vector-valued functions with nonzero Jacobian apply, therefore, to analytic functions with nonzero derivative. In particular, Theorem 7–4 tells us that if f is analytic on an open set S in E_2 and if $f'(z_0) \neq 0$ for some z_0 in S, then there exists a neighborhood $N(z_0)$ on which f is one-to-one. The Note following Theorem 7–4 shows that this is a local theorem and not a global theorem. We now derive a global condition for one-to-oneness.

16–22 THEOREM. *Assume that f is analytic on an open region S in E_2. Let Γ be a piece-wise smooth Jordan curve such that both Γ and its inner region lie inside S. If f is one-to-one on Γ, then f is also one-to-one on the interior of Γ.*

Proof. Suppose the positively oriented path along Γ is described by a piece-wise smooth function z defined on $[a, b]$. Let $C = f(\Gamma)$ be the image of Γ under f. One-to-oneness of f on Γ shows that C is a Jordan curve. Let w be the composite function given by

$$w(t) = f[z(t)], \qquad \text{if } a \leq t \leq b.$$

Then $w'(t) = f'[z(t)]z'(t)$ at points where $z'(t)$ exists, and hence w is a piece-wise smooth function which describes C. Choose a point w_0 inside C. We will prove that there is one, and only one, point z_0 inside Γ such that $f(z_0) = w_0$.

For this purpose let $g(z) = f(z) - w_0$ and consider the integral

$$\frac{1}{2\pi i}\int_{\Gamma[z]} \frac{g'(z)}{g(z)}\,dz = \frac{1}{2\pi i}\int_{\Gamma[z]} \frac{f'(z)}{f(z) - w_0}\,dz,$$

where $\Gamma[z]$ is positively oriented. The value of this integral is a nonnegative integer N which counts the number of zeros of g inside Γ (since g has no poles inside Γ). The theorem will be proved if we show that $N = 1$.

The integral for N can be expressed as a Riemann integral (by Theorem 9-62) and we get

$$N = \frac{1}{2\pi i}\int_a^b \frac{f'[z(t)]z'(t)}{f[z(t)] - w_0}\,dt = \frac{1}{2\pi i}\int_a^b \frac{w'(t)}{w(t) - w_0}\,dt$$

But the integral for the winding number of the path $C[w]$ with respect to the point w_0 is

$$\frac{1}{2\pi i}\int_{C[w]} \frac{1}{w - w_0}\,dw = \frac{1}{2\pi i}\int_a^b \frac{w'(t)}{w(t) - w_0}\,dt = N$$

By Theorem 9-68, the value of this integral is $+1$ or -1 Since $N \geq 0$, it follows that $N = +1$ and this completes the proof

NOTE In the course of the proof we have shown that the winding number of the path $C[w]$ with respect to w_0 is $+1$ Therefore we conclude that $C[w]$ is positively oriented In other words, f maps positively oriented paths onto positively oriented paths Although we have proved this for piecewise smooth paths only, the result holds for arbitrary rectifiable paths because of Theorem 9-63 This property is sometimes described by saying that f is *sense-preserving*

Next, we derive a necessary condition for global one-to-oneness

16-23 THEOREM *Assume that f is analytic on an open region S in E_2 If f is one-to-one on S, then the derivative $f'(z) \neq 0$ for each z in S*

Proof Suppose $f'(z_0) = 0$ for some z_0 in S We will show that this leads to a contradiction Let $w_0 = f(z_0)$ Then, in some neighborhood $N(z_0) \subset S$, we have a Taylor expansion of the form

$$f(z) - w_0 = \frac{f''(z_0)}{2!}(z - z_0)^2 + \cdots$$

and hence $f - w_0$ has a zero at z_0 of order $N \geq 2$ But, if Γ is a circle about z_0 lying within $N(z_0)$, its image $C = f(\Gamma)$ is a piece-wise smooth Jordan curve containing w_0 in its interior The same argument that was applied in the previous theorem shows that $N = 1$ This contradiction implies $f'(z_0) \neq 0$ for every point z_0 in S

16-19 Conformal mappings. Let Γ_1 and Γ_2 be two smooth Jordan arcs in E_2 intersecting at a point z_0 Let z_1 and z_2 be smooth functions, defined on $[a, b]$ and $[c, d]$, respectively, which describe Γ_1 and Γ_2 Suppose $z_0 = z_1(t_1)$, $z_0 = z_2(t_2)$, $z_1'(t_1) \neq 0$, $z_2'(t_2) \neq 0$ Then the difference $\arg[z_2'(t_2)] - \arg[z_1'(t_1)]$ is called the angle from Γ_1 to Γ_2 at z_0 (See Fig 16-5)

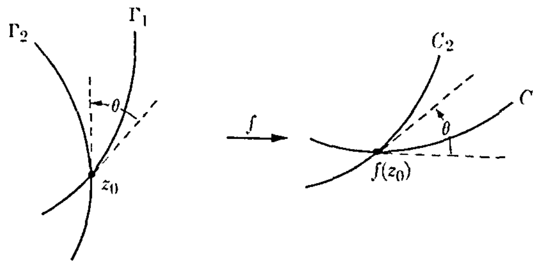

FIG. 16–5. Conformal mapping.

16–24 DEFINITION. *Let f be a complex-valued function defined on an open region S in E_2. Then f is said to be conformal at a point z_0 in S if every pair of smooth Jordan arcs Γ_1 and Γ_2 intersecting at z_0 is mapped by f onto a corresponding pair of smooth Jordan arcs $C_1 = f(\Gamma_1)$ and $C_2 = f(\Gamma_2)$ in such a way that the angle from C_1 to C_2 at $f(z_0)$ differs from the angle from Γ_1 to Γ_2 at z_0 by $2\pi n$, where n is an integer.*

16–25 THEOREM. *Let f be analytic on an open region S in E_2. Then f is conformal at each point z_0 in S at which $f'(z_0) \neq 0$.*

Proof. Let z_1 and z_2 be smooth functions defined on $[a, b]$ and $[c, d]$, respectively, which describe Jordan arcs Γ_1 and Γ_2 intersecting at z_0, where $z_0 = z_1(t_1) = z_2(t_2)$, $z_1'(t_1) \neq 0$, $z_2'(t_2) \neq 0$. Let w_1 and w_2 be the composite functions given by

$$w_1(t) = f[z_1(t)] \qquad \text{if } t \in [a, b], \qquad w_2(t) = f[z_2(t)] \qquad \text{if } t \in [c, d].$$

Then $w_1'(t_1) = f'(z_0)z_1'(t_1) \neq 0$ and $w_2'(t_2) = f'(z_0)z_2'(t_2) \neq 0$. By Theorem 1–31, there exist integers n_1 and n_2 such that

$$\arg [w_1'(t_1)] = \arg [f'(z_0)] + \arg [z_1'(t_1)] + 2\pi n_1,$$

$$\arg [w_2'(t_2)] = \arg [f'(z_0)] + \arg [z_2'(t_2)] + 2\pi n_2.$$

Subtracting these equations gives us the required result.

Angles are not preserved at points where the derivative is zero. For example, if $f(z) = z^2$, a straight line through the origin making an angle α with the real axis is mapped by f onto a straight line making an angle 2α with the real axis. In general, when $f'(z_0) = 0$, the Taylor expansion of f assumes the form

$$f(z) - f(z_0) = (z - z_0)^k[a_k + a_{k+1}(z - z_0) + \cdots],$$

where $k \geq 2$. Using this equation, it is easy to see that angles between curves intersecting at z_0 are multiplied by a factor k under the mapping f.

Among the important examples of conformal mappings are the *Mobius transformations*. These are functions f defined as follows. If a, b, c, d are four complex numbers such that $ad - bc \neq 0$, we define

$$f(z) = \frac{az + b}{cz + d} \tag{23}$$

whenever $cz + d \neq 0$. It is convenient to define f everywhere on the extended plane E_2^* by setting $f(-d/c) = \infty$ and $f(\infty) = a/c$. (If $c = 0$, these last two equations are to be replaced by the single equation $f(\infty) = \infty$.) Now (23) can be solved for z in terms of $f(z)$ to get

$$z = \frac{-df(z) + b}{cf(z) - a}.$$

This means that the inverse function f^{-1} exists and is given by

$$f^{-1}(z) = \frac{-dz + b}{cz - a},$$

with the understanding that $f^{-1}(a/c) = \infty$ and $f^{-1}(\infty) = -d/c$. Thus we see that Mobius transformations are one-to-one mappings of E_2^* onto itself. They are also conformal at each finite $z \neq -d/c$, since

$$f'(z) = \frac{bc - ad}{(cz + d)^2} \neq 0.$$

One of the most important properties of these mappings is that they map circles onto circles (including straight lines as special cases of circles). The proof of this is sketched in Exercise 16–40. Further properties of Mobius transformations are also described in the exercises at the end of the chapter.

Exercises

Cauchy integral formulas; Taylor expansions.

16–1. Assume that f is analytic on a neighborhood $N(z_0; R)$. If $0 < r < R$, let $C(r)$ denote the circle of radius r with z_0 as center and let $M(r)$ denote the maximum of $|f|$ on $C(r)$. Deduce *Cauchy's inequalities:*

$$|f^{(n)}(z_0)| \leq \frac{M(r)n!}{r^n} \qquad (n = 0, 1, 2, \ldots).$$

16–2. A function analytic everywhere on E_2 is called an *entire* function (examples of which are polynomials, the sine and cosine, and the exponential).

(a) Prove *Liouville's theorem: If f is an entire function that is bounded on E_2, then f is a constant.*

(b) Prove the following stronger version of Liouville's theorem: *If f is an entire function such that* $\lim_{z\to\infty} |f(z)/z| = 0$, *then f is a constant.*

(c) What can you conclude about an entire function which satisfies an inequality of the form $|f(z)| \leq M|z|^c$ for every z in E_2, where $c > 0$?

16–3. Let $f = u + iv$ be analytic on a neighborhood $N(z_0; R)$. If $0 < r < R$, show that

$$f'(z_0) = \frac{1}{\pi r}\int_0^{2\pi} u(z_0 + re^{i\theta})e^{-i\theta}\, d\theta.$$

16–4. Assume that f has the Taylor expansion $f(z) = \sum_{n=0}^{\infty} a(n)z^n$, valid in $N(0; R)$. Let

$$g(z) = \frac{1}{p}\sum_{k=0}^{p-1} f(ze^{2\pi ik/p}).$$

Show that g has the Taylor expansion

$$g(z) = \sum_{n=0}^{\infty} a(pn)z^{pn},$$

valid in $N(0; R)$.

16–5. Assume that f has the Taylor expansion $f(z) = \sum_{n=0}^{\infty} a_n z^n$, valid in $N(0; R)$. Let $s_n(z) = \sum_{k=0}^{n} a_k z^k$. If $0 < r < R$ and if $|z_0| < r$, show that

$$s_n(z_0) = \frac{1}{2\pi i}\int_C \frac{f(z)}{z^{n+1}}\, \frac{z^{n+1} - z_0^{n+1}}{z - z_0}\, dz,$$

where C is the positively oriented circle with center at 0 and radius r.

16-6. Assume that f is analytic on the closed disk $\overline{N}(0, 1)$. If $|z_0| < 1$, show that

$$(1 - |z_0|^2)f(z_0) = \frac{1}{2\pi i}\int_C f(z)\,\frac{1 - z\bar{z}_0}{z - z_0}\,dz,$$

where C is the positively oriented unit circle with center at 0. Deduce the inequality

$$(1 - |z_0|^2)|f(z_0)| \leq \frac{1}{2\pi}\int_0^{2\pi} |f(e^{i\theta})|\,d\theta$$

16-7. Given the Taylor expansions $f(z) = \sum_{n=0}^{\infty} a_n z^n$ and $g(z) = \sum_{n=0}^{\infty} b_n z^n$, valid on $\overline{N}(0, R_1)$, $N(0, R_2)$, respectively. Show that

$$\frac{1}{2\pi i}\int_\Gamma \frac{f(z)}{z}\,g\left(\frac{z_0}{z}\right)dz = \sum_{n=0}^{\infty} a_n b_n z_0^n$$

if $|z_0| < R_1 R_2$, where Γ is the positively oriented circle of radius R_1 with center at 0.

16-8. Assume that f is analytic on $N(0, R)$. Let C denote the positively oriented circle with center at 0 and radius r, where $0 < r < R$. If z_0 is inside C, show that

$$f(z_0) = \frac{1}{2\pi i}\int_C f(z)\left\{\frac{1}{z - z_0} - \frac{1}{z - r^2/\bar{z}_0}\right\}dz$$

If $z_0 = Ae^{i\alpha}$, show that this reduces to the formula

$$f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} \frac{(r^2 - A^2)f(re^{i\theta})}{r^2 - 2rA\cos(\alpha - \theta) + A^2}\,d\theta$$

Equating the real parts of this equation gives an expression known as *Poisson's integral formula*.

16-9. Assume that f has the Taylor expansion $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$, valid in $N(z_0, R)$.

(a) If $0 \leq r < R$, deduce *Parseval's identity*

$$\frac{1}{2\pi}\int_0^{2\pi} |f(z_0 + re^{i\theta})|^2\,d\theta = \sum_{n=0}^{\infty} |a_n|^2 r^{2n}$$

(b) Use (a) to deduce the inequality $\sum_{n=0}^{\infty} |a_n|^2 r^{2n} \leq M(r)^2$, where $M(r)$ is the maximum of $|f|$ on the circle $|z - z_0| = r$.

16–10. Prove the *maximum-modulus theorem: Assume that f is analytic on an open region S in E_2 and suppose f is not a constant. Let Γ be a rectifiable Jordan curve such that both Γ and its inner region lie within S. If $|f(z)| \leq M$ on Γ, then $|f(z)| < M$ at all points inside Γ.*

[*Hint:* Assume that $|f(z_0)| = M$ for some interior point. Using the notation of Exercise 16–9, deduce the inequality $\sum_{n=0}^{\infty} |a_n|^2 r^{2n} \leq |a_0|^2$. This implies $a_n = 0$ for every $n \geq 1$, and hence f is constant on a neighborhood of z_0.]

16–11. Let Γ be a rectifiable Jordan curve and let g be a complex-valued function, defined and continuous on Γ. For each z_0 inside Γ, define $f(z_0)$ by the equation

$$f(z_0) = \int_\Gamma \frac{g(z)}{z - z_0}\, dz.$$

Show that f has a derivative at each point z_0 inside Γ given by

$$f'(z_0) = \int_\Gamma \frac{g(z)}{(z - z_0)^2}\, dz.$$

Show also that the second derivative exists inside Γ and that

$$f''(z_0) = 2\int_\Gamma \frac{g(z)}{(z - z_0)^3}\, dz$$

if z_0 is inside Γ. [*Hint:* Argue as in the proof of Theorem 16–6.]

16–12. Prove *Weierstrass' limit theorem: Given a sequence $\{f_n\}$ of functions, each of which is analytic on an open region S in E_2. Let Γ be a rectifiable Jordan curve such that both Γ and its inner region $I(\Gamma)$ lie within S, and suppose $\{f_n\}$ converges uniformly on Γ. Then $\{f_n\}$ converges uniformly on $I(\Gamma)$ and the limit function f is analytic on $I(\Gamma)$.*

Sketch of proof: Uniform convergence of $\{f_n\}$ on $I(\Gamma)$ follows by applying the Cauchy condition (Theorem 13–4) in conjunction with the maximum-modulus theorem. To prove that f is analytic on $I(\Gamma)$, choose z_0 in $I(\Gamma)$ and write

$$f_n(z_0) = \frac{1}{2\pi i}\int_\Gamma \frac{f_n(z)}{z - z_0}\, dz.$$

Because of uniform convergence, we may replace f_n by f in this formula. (See Theorem 13–12.) Now apply Exercise 16–11.

16–13. Prove *Schwarz's lemma: Let f be analytic on the neighborhood $N(0; 1)$. Suppose that $f(0) = 0$ and $|f(z)| \leq 1$ if $|z| < 1$. Then*

$$|f'(0)| \leq 1 \quad \textit{and} \quad |f(z)| \leq |z|, \quad \textit{if } |z| < 1.$$

If $|f'(0)| = 1$ or if $|f(z_0)| = |z_0|$ for at least one z_0 in $N(0, 1)$, then

$$f(z) = e^{i\alpha}z, \qquad \text{where } \alpha \text{ is real}$$

[*Hint* Apply the maximum-modulus theorem to g, where $g(0) = f'(0)$ and $g(z) = f(z)/z$ if $z \neq 0$]

Laurent expansions, singularities, residues

16–14. Let f and g be analytic on an open region S in E_2 Let Γ be a rectifiable Jordan curve such that both Γ and its inner region lie within S Suppose that $|g(z)| < |f(z)|$ for every z on Γ

(a) Show that

$$\frac{1}{2\pi i}\int_\Gamma \frac{f'(z) + g'(z)}{f(z) + g(z)}\,dz = \frac{1}{2\pi i}\int_\Gamma \frac{f'(z)}{f(z)}\,dz$$

[*Hint* Let $m = \inf\{|f(z)| - |g(z)| \mid z \in \Gamma\}$ Then $m > 0$ and hence $|f(z) + tg(z)| \geq m > 0$ for each t in $[0\ 1]$ and each z on Γ Now let

$$\phi(t) = \frac{1}{2\pi i}\int_\Gamma \frac{f'(z) + tg'(z)}{f(z) + tg(z)}\,dz, \qquad \text{if } 0 \leq t \leq 1$$

Then ϕ is continuous, and hence constant, on $[0, 1]$ Thus, $\phi(0) = \phi(1)$]

(b) Use (a) to prove that f and $f + g$ have the same number of zeros inside Γ (*Rouché's theorem*)

16–15. Let p be a polynomial of degree n, say $p(z) = a_0 + a_1 z + \cdots + a_n z^n$, where $a_n \neq 0$ Take $f(z) = a_n z^n$, $g(z) = p(z) - f(z)$ in Rouché's theorem, and prove that p has exactly n zeros in E_2 (*fundamental theorem of algebra*)

16–16. Let f be analytic on the closed disk $\overline{N}(0, 1)$ and suppose $|f(z)| < 1$ if $|z| = 1$ Show that there is one, and only one, point z_0 in $N(0, 1)$ such that $f(z_0) = z_0$

[*Hint* Use Rouché's theorem]

16–17. Let $p_n(z)$ denote the nth partial sum of the Taylor expansion $e^z = \sum_{n=0}^{\infty} z^n/n!$ Using Rouché's theorem (or otherwise), prove that for every $r > 0$ there exists an N (depending on r) such that $n \geq N$ implies $p_n(z) \neq 0$ for every z in $N(0, r)$

16–18. Show that each of the following Laurent expansions is valid in the region indicated

(a) $$\frac{1}{(z-1)(2-z)} = \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} + \sum_{n=1}^{\infty} \frac{1}{z^n}, \qquad \text{if } 1 < |z| < 2$$

(b) $$\frac{1}{(z-1)(2-z)} = \sum_{n=2}^{\infty} \frac{1 - 2^{n-1}}{z^n}, \qquad \text{if } |z| > 2$$

16–19. For each fixed t in E_2, define $J_n(t)$ to be the coefficient of z^n in the Laurent expansion

$$e^{(z-1/z)t/2} = \sum_{n=-\infty}^{\infty} J_n(t)z^n.$$

Show that for $n \geq 0$ we have

$$J_n(t) = \frac{1}{\pi}\int_0^{\pi} \cos(t \sin\theta - n\theta)\, d\theta$$

and that $J_{-n}(t) = (-1)^n J_n(t)$. Deduce the power series expansion

$$J_n(t) = \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{1}{2}t)^{n+2k}}{k!(n+k)!} \qquad (n \geq 0).$$

The function J_n is called the *Bessel function* of order n.

16–20. Prove *Riemann's theorem: If z_0 is an isolated singularity of f and if $|f|$ is bounded on some deleted neighborhood $N'(z_0)$, then z_0 is a removable singularity.*

[*Hint:* Estimate the integrals for the coefficients a_n in the Laurent expansion of f and show that $a_n = 0$ for each $n < 0$.]

16–21. Prove the *Casorati-Weierstrass theorem: Assume that z_0 is an essential singularity of f and let c be an arbitrary complex number. Then, for every $\epsilon > 0$ and every neighborhood $N(z_0)$, there exists a point z in $N(z_0)$ such that $|f(z) - c| < \epsilon$.*

[*Hint:* Assume that the theorem is false and arrive at a contradiction by applying Exercise 16–20 to g, where $g(z) = 1/[f(z) - c]$.

16–22. (*The point at infinity.*) A function f is said to be analytic at ∞ if the function g defined by the equation $g(z) = f(1/z)$ is analytic at the origin. Similarly, we say that f has a zero, a pole, a removable singularity, or an essential singularity at ∞ if g has a zero, a pole, etc., at 0. Liouville's theorem (see Exercise 16–2) states that a function which is analytic everywhere in E_2^* must be a constant. Prove that

(a) f is a polynomial if, and only if, the only singularity of f in E_2^* is a pole at ∞, in which case the order of the pole is equal to the degree of the polynomial.

(b) f is a rational function if, and only if, f has no singularities in E_2^* other than poles.

16–23. Derive the following "short cuts" for computing residues:

(a) If z_0 is a first order pole for f, then

$$\operatorname*{Res}_{z=z_0} f(z) = \lim_{z \to z_0} (z - z_0)f(z).$$

(b) If z_0 is a pole of order 2 for f, then

$$\operatorname*{Res}_{z=z_0} f(z) = g'(z_0), \qquad \text{where } g(z) = (z - z_0)^2 f(z).$$

(c) Suppose f and g are both analytic at z_0, with $f(z_0) \neq 0$ and z_0 a first order zero for g. Show that

$$\operatorname*{Res}_{z=z_0} \frac{f(z)}{g(z)} = \frac{f(z_0)}{g'(z_0)}, \qquad \operatorname*{Res}_{z=z_0} \frac{f(z)}{[g(z)]^2} = \frac{f'(z_0)g'(z_0) - f(z_0)g''(z_0)}{[g'(z_0)]^3}$$

(d) If f and g are as in (c), except that z_0 is a second order zero for g, then

$$\operatorname*{Res}_{z=z_0} \frac{f(z)}{g(z)} = \frac{6f'(z_0)g''(z_0) - 2f(z_0)g'''(z_0)}{3[g''(z_0)]^2}$$

16–24. Compute the residues at the poles of f if

(a) $f(z) = \dfrac{ze^z}{z^2 - 1}$, (b) $f(z) = \dfrac{e^z}{z(z-1)^2}$,

(c) $f(z) = \dfrac{\sin z}{z \cos z}$, (d) $f(z) = \dfrac{1}{1 - e^z}$,

(e) $f(z) = \dfrac{1}{1 - z^n}$ (where n is a positive integer)

16–25. If $C(z_0, r)$ denotes the positively oriented circle with center at z_0 and radius r, show that

(a) $\displaystyle\int_{C(0,4)} \frac{3z - 1}{(z+1)(z-3)}\,dz = 6\pi i$, (b) $\displaystyle\int_{C(0,2)} \frac{2z}{z^2 + 1}\,dz = 4\pi i$,

(c) $\displaystyle\int_{C(0,2)} \frac{z^3}{z^4 - 1}\,dz = 2\pi i$, (d) $\displaystyle\int_{C(2,1)} \frac{e^z}{(z-2)^2}\,dz = 2\pi i e^2$.

Evaluate the integrals in Exercises 16–26 through 16–33 by means of residues

16–26. $\displaystyle\int_0^{2\pi} \frac{dt}{(a + b\cos t)^2} = \frac{2\pi a}{(a^2 - b^2)^{3/2}}$, if $0 < b < a$

16–27. $\displaystyle\int_0^{2\pi} \frac{\cos 2t\,dt}{1 - 2a\cos t + a^2} = \frac{2\pi a^2}{1 - a^2}$, if $a^2 < 1$

16–28. $\displaystyle\int_0^{2\pi} \frac{\cos^3 3t\,dt}{1 - 2a\cos t + a^2} = \frac{\pi(a^2 - a + 1)}{1 - a}$, if $0 < a < 1$.

16–29. $\displaystyle\int_0^{2\pi} \frac{\sin^2 t\,dt}{a + b\cos t} = \frac{2\pi(a - \sqrt{a^2 - b^2})}{b^2}$, if $0 < b < a$.

16–30. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{x^2 + x + 1}\,dx = \frac{2\pi\sqrt{3}}{3}$.

16–31. $\displaystyle\int_{-\infty}^{\infty} \frac{x^6}{(1 + x^4)^2}\,dx = \frac{3\pi\sqrt{2}}{16}$.

16–32. $\displaystyle\int_0^{\infty} \frac{x^2}{(x^2 + 4)^2(x^2 + 9)}\,dx = \frac{\pi}{200}$.

16–33. $\displaystyle\int_0^{\infty} \frac{x^{2m}}{1 + x^{2n}}\,dx = \frac{\pi}{2n}\Big/\sin\left(\frac{2m+1}{2n}\pi\right)$,

if m, n are integers, $0 < m < n$.

16–34. Prove that formula (20) holds if f is the quotient of two polynomials, say $f = P/Q$, where the degree of Q exceeds that of P by 2 or more.

16–35. Prove that formula (20) holds if $f(z) = e^{imz}P(z)/Q(z)$, where $m > 0$ and P and Q are polynomials such that the degree of Q exceeds that of P by 1 or more. This makes it possible to evaluate integrals of the form

$$\int_{-\infty}^{\infty} e^{imx}\frac{P(x)}{Q(x)}\,dx$$

by the method described in Theorem 16–20.

16–36. Use the method suggested in Exercise 16–35 to evaluate the following integrals:

(a) $\displaystyle\int_0^{\infty} \frac{\sin mx}{x(a^2 + x^2)}\,dx = \frac{\pi}{2a^2}(1 - e^{-am})$, if $m \geq 0$, $a > 0$.

(b) $\displaystyle\int_0^{\infty} \frac{\cos mx}{x^4 + a^4}\,dx = \frac{\pi}{2a^3}e^{-ma/\sqrt{2}}\sin\left(\frac{ma}{\sqrt{2}} + \frac{\pi}{4}\right)$, if $m > 0$, $a > 0$.

One-to-one analytic functions.

16–37. Suppose f has the Taylor expansion $f(z) = \sum_{n=0}^{\infty} a_n z^n$, valid in $N(0; r)$. If $|a_1| \geq \sum_{n=2}^{\infty} n|a_n|r^{n-1}$, show that f is one-to-one on $N(0; r)$ if f is not a constant. [*Hint:* Suppose that $f(z_1) = f(z_2)$, where z_1, z_2 are distinct points in $N(0; r)$. Then

$$\sum_{n=1}^{\infty} a_n(z_2^n - z_1^n) = 0.$$

Show that this contradicts the inequality for $|a_1|$ if f is not a constant.]

16-38 If f and g are Möbius transformations, show that the composite function fg is also a Möbius transformation

16-39. Describe geometrically what happens to a point z in E_2 when it is carried into $f(z)$ by the following special Möbius transformations

(a) $f(z) = z + b$ (Translation)
(b) $f(z) = az$, where $a > 0$ (Stretching)
(c) $f(z) = e^{i\alpha}z$, where α is real (Rotation)
(d) $f(z) = 1/z$ (Inversion)

16-40. If $c \neq 0$, we have

$$\frac{az + b}{cz + d} = \frac{a}{c} + \frac{bc - ad}{c(cz + d)}$$

Hence every Mobius transformation can be expressed as a composition of the special cases described in Exercise 16-39. Use this fact to show that Mobius transformations carry circles into circles (where straight lines are considered as special cases of circles)

16-41. (a) Show that all Mobius transformations which map the upper half-plane $T = \{x + iy \mid y \geq 0\}$ onto the closed disk $\bar{N}(0, 1)$ can be expressed in the form $f(z) = e^{i\alpha}(z - a)/(z - \bar{a})$, where α is real and $a \in T$

(b) Show that a and α can always be chosen to map any three given points of the real axis onto any three given points on the unit circle

16-42. Find all Möbius transformations which map the right half-plane $S = \{x + iy \mid x \geq 0\}$ onto the closed disk $\bar{N}(0, 1)$

16-43. Find all Möbius transformations which map the closed disk $\bar{N}(0, 1)$ onto itself

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INDEX OF SPECIAL SYMBOLS

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